

Phenix Overview and General Reference

96 documentation pages

Generated from local Phenix documentation

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Phenix Documentation - version dev-svn

NEW:Phenix documentation for the AI Agent for AI-assisted structure determination, the Phenix chatbot, and AI analysis of results Crystallography Cryo-EM Predicted models How to install GUI introduction FAQs Video tutorials Tutorials & Examples Overview of programs Other links Bibliography Dictionary Index Find a program or regression test Citing Phenix If you use PHENIX please cite: Macromolecular structure determination using X-rays, neutrons and electrons: recent developments in Phenix. D. Liebschner, P. V. Afonine, M. L. Baker, G. Bunkóczy, V. B. Chen, T. I. Croll, B. Hintze, L.-W. Hung, S. Jain, A. J. McCoy, N. W. Moriarty, R. D. Oeffner, B. K. Poon, M. G. Prisant, R. J. Read, J. S. Richardson, D. C. Richardson, M. D. Sammito, O. V. Sobolev, D. H. Stockwell, T. C. Terwilliger, A. G. Urzhumtsev, L. L. Videau, C. J. Williams and P. D. Adams. *Acta Cryst.* (2019). D75, 861-877

Other links

Citing Phenix

If you use PHENIX please cite:

Macromolecular structure determination using X-rays, neutrons and electrons: recent developments in Phenix. D. Liebschner, P. V. Afonine, M. L. Baker, G. Bunkóczy, V. B. Chen, T. I. Croll, B. Hintze, L.-W. Hung, S. Jain, A. J. McCoy, N. W. Moriarty, R. D. Oeffner, B. K. Poon, M. G. Prisant, R. J. Read, J. S. Richardson, D. C. Richardson, M. D. Sammito, O. V. Sobolev, D. H. Stockwell, T. C. Terwilliger, A. G. Urzhumtsev, L. L. Videau, C. J. Williams and P. D. Adams. *Acta Cryst.* (2019). D75, 861-877

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What is PHENIX

[what-is-phenix.html](#)

The PHENIX software suite is a highly automated system for macromolecular structure determination that can rapidly arrive at an initial partial model of a structure without significant human intervention, given moderate resolution and good quality data. This achievement has been made possible by the development of new algorithms for structure determination, maximum-likelihood molecular replacement (PHASER), heavy-atom search (HySS), template and pattern-based automated model-building (RESOLVE), automated macromolecular refinement (phenix.refine), and iterative model-building, density modification and refinement that can operate at moderate resolution (RESOLVE, AutoBuild). These algorithms are based on a highly integrated and comprehensive set of crystallographic libraries that have been built and made available to the community. The algorithms are tightly linked and made easily accessible to users through the PHENIX Wizards and the command line.

There are also a number of tools in PHENIX for handling ligands. Automated fitting of ligands into the electron density is facilitated via the LigandFit wizard. Besides being able to fit a known ligand into a difference map, the LigandFit wizard is capable to identify ligands on the basis of the difference density only. Stereo chemical dictionaries of ligands whose chemical description is not available in the supplied monomer library for the use in restrained macromolecular refinement can be generated with the electronic ligand builder and optimization workbench (eLBOW).

PHENIX builds upon Python, the Boost.Python Library, and C++ to provide an environment for automation and scientific computing. Many of the fundamental crystallographic building blocks, such as data objects and tools for their manipulation are provided by the Computational Crystallography Toolbox (cctbx). The computational tasks which perform complex crystallographic calculations are then built on top of this. Finally, there are a number of different user interfaces available in PHENIX. In order to facilitate automated operation there is the Project Data Storage (PDS) that is used to store and track the results of calculations.

The PHENIX development team consists of members from Lawrence Berkeley Laboratory (Paul Adams's group), Los Alamos National Laboratory (Tom Terwilliger's group), Cambridge University (Randy Read's group) and Duke University (the Richardsons' group). Researchers from Texas A&M University (Tom Ioerger's and Jim Sacchettini's groups) participated in the first five years of PHENIX development.

The development of PHENIX is funded by the National Institutes of Health (General Medicine) under grant P01GM063210, and the PHENIX Industrial Consortium.

Citing PHENIX

If you use PHENIX to solve a structure please cite this publication:

Macromolecular structure determination using X-rays, neutrons and electrons: recent developments in Phenix. D. Liebschner, P. V. Afonine, M. L. Baker, G. Bunkóczi, V. B. Chen, T. I. Croll, B. Hintze, L.-W. Hung, S. Jain, A. J. McCoy, N. W. Moriarty, R. D. Oeffner, B. K. Poon, M. G. Prisant, R. J. Read, J. S. Richardson, D. C. Richardson, M. D. Sammito, O. V. Sobolev, D. H. Stockwell, T. C. Terwilliger, A. G. Urzhumtsev, L. L. Videau, C. J. Williams and P. D. Adams. *Acta Cryst.* (2019). D75, 861-877

Publications

A number of publications describing PHENIX can be found at:

<http://www.phenix-online.org/papers/>

Working with AlphaFold2, RoseTTAFold and other predicted models

[predicted_models_index.html](#)

You can use the predicted models from AlphaFold and other prediction software in Phenix. These models can be very helpful in structure determination because the models can be very accurate over much of their length and the models come with accuracy estimates that allow removal of poorly-predicted regions.

How to use predicted models in Phenix

Phenix has tools for carrying out AlphaFold structure prediction, processing the resulting models, and for using these models in X-ray or cryo-EM structure determination. An in-depth overview with video tutorials describes these steps in detail. You can use the Phenix chatbot to get help on using any Phenix program. Just ask it any question about using Phenix. Here is a summary of the tools in Phenix. You can use `PredictModel` to access the Phenix AlphaFold server and predict the structure of a protein chain. To process predicted models, removing low-confidence regions and converting confidence estimates to B-values, you can use `ProcessPredictedModel`. The `PredictAndBuild` tool will allow you to incorporate predicted models in the structure solution workflow. `PredictAndBuild` generates predicted models and uses them to solve an X-ray structure by MR or to interpret a cryo-EM map. The tool then carries out iterative model rebuilding and prediction to improve the models. The iterative procedure allows creating more accurate predicted models than can be obtained with a simple prediction. `PredictAndBuild` can automatically generate a fairly accurate model starting from just a sequence file and either cryo-EM half-maps or X-ray data. Additionally, it provides morphed versions of unrefined predicted models that can be useful as reference models for refinement. Phenix resources for using predicted models: Overview and video tutorials: AlphaFold and Phenix Predict a structure with AlphaFold Processing a predicted model Solve an X-ray or cryo-EM structure with `PredictAndBuild` Docking a processed predicted model in a cryo-EM map Processing, docking and rebuilding a predicted model in a cryo-EM map Trimming overlapping parts of models

Here is a summary of the tools in Phenix.

You can use `PredictModel` to access the Phenix AlphaFold server and predict the structure of a protein chain.

To process predicted models, removing low-confidence regions and converting confidence estimates to B-values, you can use `ProcessPredictedModel`.

The `PredictAndBuild` tool will allow you to incorporate predicted models in the structure solution workflow. `PredictAndBuild` generates predicted models and uses them to solve an X-ray structure by MR or to interpret a cryo-EM map. The tool then carries out iterative model rebuilding and prediction to improve the models. The iterative procedure allows creating more accurate predicted models than can be obtained with a simple prediction. `PredictAndBuild` can automatically generate a fairly accurate model starting from just a sequence file and either cryo-EM half-maps or X-ray data. Additionally, it provides morphed versions of unrefined predicted models that can be useful as reference models for refinement.

Phenix resources for using predicted models:

- Overview and video tutorials: AlphaFold and Phenix
- Predict a structure with AlphaFold
- Processing a predicted model

- Solve an X-ray or cryo-EM structure with PredictAndBuild
- Docking a processed predicted model in a cryo-EM map
- Processing, docking and rebuilding a predicted model in a cryo-EM map
- Trimming overlapping parts of models

Crystallographic Structure Deposition

xray-structure-deposition.html

Background

Once a crystallographic structure is solved, it is usually deposited in the Worldwide Protein Data Bank (wwPDB). This is necessary for publication as most journals require a PDB ID to be included in any manuscript that describes a new crystallographic structure. The wwPDB is an invaluable resource that contains over 190,000 structures (as of July 2021).

Procedure

The model files (PDB and mmCIF) generated by phenix.refine contain the information required when depositing a structure to the wwPDB. However, the wwPDB currently only accepts the mmCIF (i.e., PDBx/mmCIF) model files for deposition.

To generate a model file suitable for deposition, we recommend a two-stage process:

- By default, phenix.refine will output model files in mmCIF and PDB format. If you turned off the option for mmCIF output, run a final cycle of phenix.refine that writes mmCIF files for model and data. This can be set in the Output section of the GUI.
- You can then process the model file (mmCIF) and a sequence file with mmtbx.prepare_pdb_deposition to create a mmCIF file with the sequence. This program requires the full sequence for the macromolecule to be provided. In the GUI, this program is in the "PDB Deposition" section of tools.

How to use *mmtbx.prepare_pdb_deposition*: [Click here](#)

Related programs

- phenix.get_pdb_validation_report: This program submits your model and data in CIF format to the PDB OneDep server to get a validation report in PDF and XML formats.
- phenix.table_one: This program generates the table of crystallographic statistics required by most scientific journals.

References

Announcing mandatory submission of PDBx/mmCIF format files for crystallographic depositions to the Protein Data Bank (PDB). P.D. Adams, P.V. Afonine, K. Baskaran, H.M. Berman, J. Berrisford, G. Bricogne, D.G. Brown, S.K. Burley, M. Chen, Z. Feng, C. Flensburg, A. Gutmanas, J.C. Hoch, Y. Ikegawa, Y. Kengaku, E. Krissinel, G. Kurisu, Y. Liang, D. Liebschner, L. Mak, J.L. Markley, N.W. Moriarty, G.N. Murshudov, M. Noble, E. Peisach, I. Persikova, B.K. Poon, O.V. Sobolev, E.L. Ulrich, S. Velankar, C. Vonnrhein, J. Westboock, M. Wojdyr, M. Yokochi, and J.Y. Young. Acta Cryst. D75, 451-454 (2019).

- Announcing mandatory submission of PDBx/mmCIF format files for crystallographic depositions to the Protein Data Bank (PDB). P.D. Adams, P.V. Afonine, K. Baskaran, H.M. Berman, J. Berrisford, G. Bricogne, D.G. Brown, S.K. Burley, M. Chen, Z. Feng, C. Flensburg, A. Gutmanas, J.C. Hoch, Y. Ikegawa, Y. Kengaku, E. Krissinel, G. Kurisu, Y. Liang, D. Liebschner, L. Mak, J.L. Markley, N.W. Moriarty, G.N. Murshudov, M. Noble, E. Peisach, I. Persikova, B.K. Poon, O.V. Sobolev, E.L. Ulrich, S. Velankar, C. Vonnrhein, J. Westboock, M. Wojdyr, M. Yokochi, and J.Y. Young. Acta Cryst. D75, 451-454 (2019).

Crystallographic Structure Solution with Phenix

[xray_index.html](#)

X-ray Crystallography

Click on the boxes in the figure to read an overview of each step. Also see the AlphaFold models in Phenix overview and video tutorials for using AlphaFold models in Phenix. You can use the Phenix chatbot to get help on using any Phenix program. Just ask it any question about using Phenix. GUI documentation for most used programs: Predict a structure with AlphaFold using PredictModel Automated structure determination using AlphaFold models with PredictAndBuild Check data quality with Xtriage Experimental phasing with AutoSol Molecular replacement with Phaser Model building with AutoBuild Refine your model with phenix.refine Validation tools in Phenix Fit your ligand with LigandFit How to generate geometry restraints Prepare your model and data for deposition Complete list of programs for each structure solution step Neutron Crystallography Phenix supports neutron scattering lengths and features joint X-N refinement. Microcrystal Electron Diffraction (MicroED) Phenix supports electron scattering factors for neutral atoms.

GUI documentation for most used programs:

- Predict a structure with AlphaFold using PredictModel
- Automated structure determination using AlphaFold models with PredictAndBuild
- Check data quality with Xtriage
- Experimental phasing with AutoSol
- Molecular replacement with Phaser
- Model building with AutoBuild
- Refine your model with phenix.refine
- Validation tools in Phenix
- Fit your ligand with LigandFit
- How to generate geometry restraints
- Prepare your model and data for deposition
- Complete list of programs for each structure solution step

Neutron Crystallography

Phenix supports neutron scattering lengths and features joint X-N refinement.

Microcrystal Electron Diffraction (MicroED)

Phenix supports electron scattering factors for neutral atoms.

AQuaRef: AI-enabled Quantum Refinement of macromolecular models

AQuaRef.html

Authors

Roman Zubatyuk, Malgorzata Biczysko, Kavindri Ranasinghe, Nigel W. Moriarty, Hatice Gokcan, Holger Kruse, Billy K. Poon, Paul D. Adams, Mark P. Waller, Adrian E. Roitberg, Olexandr Isayev, Pavel V. Afonine

Purpose

The purpose of AQuaRef is to refine atomic model coordinates using geometric restraints derived from AI-accelerated quantum-mechanical calculations. Real- and reciprocal-space refinements are supported using crystallographic or cryo-EM experimental data.

Expectations

Atomic models resulting from AQuaRef refinement are expected to have improved geometry, as evaluated by standard validation tools such as MolProbity, Rama-Z, and hydrogen bond skew-kurtosis plots. They should also maintain a similar or improved fit to the experimental data compared to standard classic refinement.

Requirements and Current Limitations

Protein-only (plus water) models are supported at the moment. Static disorder (i.e., alternative conformations) is not supported, except for exchangeable H/D sites in case of refinement against neutron diffraction data.

If a model contains unsupported entity (e.g., a metal ion) or alternative conformations, it is still possible to refine it but unsupported entities need to be excluded using 'exclude' keyword, as shown below in Examples.

The atomic model needs to be nearly final (in the later stages of standard classic refinement) and atom-complete (e.g., all hydrogen atoms are present explicitly, no missing side chains). The model is also expected to have meaningful geometry, such as being devoid of overlapping (clashing) atoms or severely distorted covalent bonds.

Hardware: While this can possibly work on both CPU and GPU computers, currently available builds are specific to Linux machines with CUDA 11 or 12. Additionally, runtimes on CPU-based computers are expected to be much slower compared to GPUs.

No GUI: Command line only.

Availability

AQuaRef is available as part of Phenix or separately from Phenix as part of Q|R refinement project (www.qrefine.com or <https://github.com/qrefine/qrefine>).

Examples

A self-consistent, quick to run, simple example is distributed with the software (Phenix, Q|R) and located in `phenix/modules/qrefine/examples/AQuaRef` folder. Check README file for details. If using AQuaRef for the first time, we suggest to run this example first in order to make sure all works as expected.

Reciprocal space refinement:

```
qr.aquaref model.pdb data.mtz
```

Real space refinement:

```
qr.aquaref model.pdb map.mrc
```

Optimization of model geometry (no experiemntal data used):

```
qr.aquaref model.pdb
```

Reciprocal space refinement excluding Mg and ligand ATP:

```
qr.aquaref model.pdb data.mtz exclude="element MG or name ATP"
```

Literature

AQuaRef: Machine learning accelerated quantum refinement of protein structures

<https://doi.org/10.1101/2024.07.21.604493>

MRage: automated molecular replacement

MRage.html

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MRage is generic molecular replacement program that is designed to run efficiently with a large number of possible models on parallel hardware. It accepts multiple model definitions that are internally processed with integrated tools to search models and used for molecular replacement. Identified solutions can be used to speed up the search even further by model assembly algorithms and limiting search space.

Input

The composition of the unit cell is specified by inputting components, and potential models for the corresponding component in a hierarchical way. It is recommended to specify the sequence of a component, which allows for better handling of partial models (see Composition stage). A stoichiometry of each component in the final assembly is also assigned at this stage. The overall count of assemblies in the unit cell can either be input or determined automatically from the Matthews coefficient. In case no model is known for a component, the list of potential models should be left empty.

Models

Models for a component can be specified stepwise with the amount of preprocessing required. The following options are available:

ensemble. This model is used without modification. model collection. These models will be superposed and optionally trimmed using phenix.ensembl, but no modification of sequence, sidechains, or B-factors will be

performed. template. This model is run through phenix.sculptor with a given set of active protocols (default: use all protocols) to remove unrelated chain segments and solvent, trim sidechains and weight the model according to structure and homology. An alignment can be specified to be used to process the structure; in case this is omitted, phenix.muscle will be used to align the model to the target sequence. homology. Homology search results. Hits will be fetched from the wwPDB, and preprocessed as a template. It is possible to limit the homology search to the top N hits. search. Performs a BLAST search for homologs and processes the results as described above. Requires component sequence.

- ensemble. This model is used without modification.
- model collection. These models will be superposed and optionally trimmed using phenix.enssembler, but no modification of sequence, sidechains, or B-factors will be performed.
- template. This model is run through phenix.sculptor with a given set of active protocols (default: use all protocols) to remove unrelated chain segments and solvent, trim sidechains and weight the model according to structure and homology. An alignment can be specified to be used to process the structure; in case this is omitted, phenix.muscle will be used to align the model to the target sequence.
- homology. Homology search results. Hits will be fetched from the wwPDB, and preprocessed as a template. It is possible to limit the homology search to the top N hits.
- search. Performs a BLAST search for homologs and processes the results as described above. Requires component sequence.

These steps are performed on-demand, i.e. if no clear solution could be found with available ensembles, the program processes the next template, or fetches the next homology search hit, etc.

Symmetry exploration

Specifies the set of space groups to try in the search. Currently, there are three options available:

dataset. The space group as specified on the input or read from the mtz-file. enantiomorph. The space group has been determined from systematic absences. The input space group and its enantiomorph (if exists) will be used in the search. pointgroup. Only the point group of the data has been established. All space groups compatible with the point group will be tried.

- dataset. The space group as specified on the input or read from the mtz-file.
- enantiomorph. The space group has been determined from systematic absences. The input space group and its enantiomorph (if exists) will be used in the search.
- pointgroup. Only the point group of the data has been established. All space groups compatible with the point group will be tried.

Space group exploration is exhaustive and can take several macrocycles to complete. Space groups with clearly inferior results to others are incrementally removed from the search until the correct space group is established.

Search organization

MRage employs a sequential search algorithm, in which molecules are found consecutively. The algorithm consists of several stages, and also allows the exploration of potential space groups.

Composition stage

The composition of each partial structure (located in previous macrocycles) is compared with the composition of the unit cell. Available models that fit in the missing composition are selected for search. In case the modelled sequence for a search model is known, the program can account for the fact that a model may only be a partial model for a component (e.g. one domain only).

Search stage

Calculations are organized into a search tree that is explored according to depth-first traversal. For the underlying calculations, available functions from phaser [PHASER] are used.

For each partial structure and applicable model, a rotation function is calculated. This is then followed up by a translation function calculation. Each translation function peaks are checked for packing clashes, and if accepted, checked whether are significant, probable solutions or not. Calculations are performed in the order of peak score (RF/TF), model quality (sequence identity) or partial structure quality (LLG score). Exploration continues until a significant solution is found (quick mode) or all possibilities has been exhausted (full mode). There is also no thresholding step for rotation (rationale: weak signal in rotation function) and translation function peaks (rationale: packing check fast). As a consequence, more work is performed, but simultaneously this allows weighting translation function results with packing overlap, and also avoids pipe stalls, which would adversely effect scaling in a parallel environment.

In quick mode, if a significant solution is found, conventional search terminates. However, all models that are alternatives of the model that gave this clear solution are superposed onto the solution and refined. This allows the quick evaluation of model quality for a potentially large number of alternative models. Currently, SSM [SSM] is used for superposition.

Parallelisation is done at a function level. Calculations are dispatched until the number of assigned CPUs are filled up. Execution can either take place in a shared memory environment using threads and processes or using a submission queue (currently SGE, LSF and PBS are supported). The degree of parallelisation scales with the complexity of the search, and given unlimited resources, reduces the runtime to a constant. However, in simple cases (e.g. a good model that gives a clear rotation and translation peak), no parallel resources can be used.

Refinement

Peaks above a certain threshold of the best peak and those that are identified as significant solutions are subjected to refinement. After refinement is completed, a second thresholding step is performed. Top peaks will be subjected to post-processing and also carried forward for the next macrocycle. Significant peaks that are below the threshold (e.g. a correct solution for a small domain) are also input to the post-processing step, but will not be propagated to the next cycle.

Post-processing

Significant solutions are analysed for specific features so that the search could be improved.

Amalgamation. Assembly of solutions that come from the same starting point. In case the "starting point" is empty, allowed origin shifts are added to the search. In case the origin is undefined and the space group has floating origin, only the rotation component is extracted. Rotation peak salvage. Propagates significant rotation peaks to the next stage. Regular assembly recognition. Identifies local point group symmetry from the solution. Assembly completion. In case assembly definitions are available (e.g. from regular assembly completion), and a partial assembly has been located, missing molecules are filled in.

- Amalgamation. Assembly of solutions that come from the same starting point. In case the "starting point" is empty, allowed origin shifts are added to the search. In case the origin is undefined and the space group has floating origin, only the rotation component is extracted.
- Rotation peak salvage. Propagates significant rotation peaks to the next stage.
- Regular assembly recognition. Identifies local point group symmetry from the solution.
- Assembly completion. In case assembly definitions are available (e.g. from regular assembly completion), and a partial assembly has been located, missing molecules are filled in.

No actual calculation is performed, only the possibilities are noted. In case the given solution enters the next Search stage (as determined by Composition stage), these possibilities will be scored as a priority.

Space group exploration

Search progress in possible space groups is compared. Space groups having significant solutions or above the threshold (determined by the best score over all space groups) are propagated to the next macrocycle.

Output

The program logs all operations it performs. However, this may be difficult to follow on the screen, especially when multiple space groups are explored, and therefore, after each macrocycle, a summary of all searches performed is printed.

Models generated during the solution process are also saved. This includes structure files fetched from the PDB, but also alignments and homology searches performed.

Solutions

If the search finishes, solutions are written out in all likely space groups. For each space group, the top solution is selected, and all partial solutions above a threshold relative to the top solution is considered. Solutions that are fully "contained" in a higher scoring solution (i.e. all models correspond to a model in the higher scoring solution) are discarded. This way, the top solution is output even if the composition is overestimated, but at the same time removing incomplete solutions (e.g. 14 molecules out of the total 15) that cannot be distinguished based on the final score.

Solutions are written out in an internal compressed format, which is capable of storing a high number of solutions without onerous storage requirements, and can be manipulated with provided tools.

Usage

Graphical interface

As with the generic Phaser GUI, two interfaces are available for running MRage, one designed for single-component searches with minimal customization of input models, and an advanced interface that allows all available features to be used. These are launched from a single button in the main GUI, which will initially pop up an information dialog describing the available options.

The simple GUI is centered around a single input control allowing any combination of files to be dragged in:

Reflections (mandatory): these can be in any file or data format; only amplitudes or intensities are required.
 Sequence: should contain a single polymer sequence, in any format. This is highly recommended, and required if you want to perform an automatic homology search. Template models: unprocessed PDB files,

which will be treated by Sculptor as described above. Homology search: HHPred or BLAST results.

- Reflections (mandatory): these can be in any file or data format; only amplitudes or intensities are required.
- Sequence: should contain a single polymer sequence, in any format. This is highly recommended, and required if you want to perform an automatic homology search.
- Template models: unprocessed PDB files, which will be treated by Sculptor as described above.
- Homology search: HHPred or BLAST results.

In the simplest cases, only the data and sequence are required; you should then select a homology search method in the Options section. "NCBI BLAST" requires internet access; "Local BLAST" requires that BLAST is installed and available on the command line. At this point you can run the program without further input.

Other settings:

Max. number of hits to use controls how many PDB entries from the homology search are processed by the pipeline. Increasing this provides greater coverage, but at the cost of significantly longer runtimes. Resolution cutoff determines the resolution used for running Phaser; this does not need to be changed, but increasing the cutoff will result in a faster but less accurate run. Parallelization method controls how the jobs are distributed across the available computing resources. multiprocessing and threading are essentially equivalent (although in practice they may behave slightly differently), and run locally. The other options require that you have access to one of the supported queuing systems and can submit jobs from the computer where you are running the GUI. Number of CPUs controls how many jobs are run at once. It should not exceed the number of processors available, either on the local system or the queuing system. (Note: if you plan to use your computer for other tasks while MRage is running, you may wish to leave one CPU core free if possible.) Search mode: sets the thoroughness of the search, as describe in the Search stage section. Symmetry exploration determines the space groups to try (Symmetry exploration) Overall count is the number of copies of the component to search for in the asymmetric unit.

- Max. number of hits to use controls how many PDB entries from the homology search are processed by the pipeline. Increasing this provides greater coverage, but at the cost of significantly longer runtimes.
- Resolution cutoff determines the resolution used for running Phaser; this does not need to be changed, but increasing the cutoff will result in a faster but less accurate run.
- Parallelization method controls how the jobs are distributed across the available computing resources. multiprocessing and threading are essentially equivalent (although in practice they may behave slightly differently), and run locally. The other options require that you have access to one of the supported queuing systems and can submit jobs from the computer where you are running the GUI.
- Number of CPUs controls how many jobs are run at once. It should not exceed the number of processors available, either on the local system or the queuing system. (Note: if you plan to use your computer for other tasks while MRage is running, you may wish to leave one CPU core free if possible.)
- Search mode: sets the thoroughness of the search, as describe in the Search stage section.
- Symmetry exploration determines the space groups to try (Symmetry exploration)
- Overall count is the number of copies of the component to search for in the asymmetric unit.

Advanced GUI

This version of the GUI moves the description of the crystal contents and search information to a separate tab, with input options grouped by component. The first tab is essentially similar to the simple GUI, but the only input field is for the reflections file.

The composition tab contains another set of tabs for each unique component. These allow the full range of input files to be specified; some of these have additional parameters (such as the mandatory sequence identity for ensembles and model collections). The "Basic info" tab allows the NCBI BLAST search to be run interactively if desired (the results will be loaded as a homology result). A link to the HHPred server, which does not allow automated queries, is available on the "Homology searches" tab.

You must specify the component stoichiometry, i.e. the number of copies of the component in the overall assembly. This is distinct from the "Overall count" parameter in the main tab: for instance, if you are trying to solve a structure of a hexamer bound to a monomer and expect two copies of the complex in the asymmetric unit of the crystal, the overall count is two, and the stoichiometry of the components is 6 and 1 for the hexamer and monomer respectively.

Command line

phaser.MRage [command-line switches] [PHIL-format parameter files]

Command-line switches:

```
-h, --help show this help message and exit
--show-defaults print PHIL and exit
-i, --stdin read PHIL from stdin as well
-v, --verbosity set verbosity level (DEBUG, INFO, WARNING, VERBOSE)
```

PHIL arguments

Everything not starting with a dash('-') is interpreted as a PHIL argument. This can be a PHIL-format file containing parameters or a command-line assignment.

References

- Phaser crystallographic software. A.J. McCoy, R.W. Grosse-Kunstleve, P.D. Adams, M.D. Winn, L.C. Storoni, and R.J. Read. *J Appl Crystallogr* 40, 658-674 (2007).
- Secondary-structure matching (SSM), a new tool for fast protein structure alignment in three dimensions. E. Krissinel, and K. Henrick. *Acta Crystallogr D Biol Crystallogr* 60, 2256-68 (2004).

List of all keywords

hklin = None labin = None resolution_cutoff = 2.5 no_tncs = False mode = *quick full symmetry_exploration = *dataset enantiomorph pointgroup packing_pool = 20 rotation_peaks_cutoff = 0.75 post_refinement_cutoff = 0.75 final_selection_cutoff = 0.75 b_factor_refinement = True significant_peak_threshold = 7.0 sculptor_protocols = 1 2 3 4 5 6 7 8 9 10 11 *12 13 all minimal template_equivalence = True assembly_acceptance_policy = *always observed never assembly_ensemble_creation_policy = always *observed never pdb_mirror = *rcsb pdbe pdbj exclude_pdb_ids = None Developer option to prevent specific structures from being used as search models, in order to test performance with lower sequence identity structures.crystal_symmetryunit_cell = None space_group = None outputroot = "mrage" max_solutions_to_write = 0 gui_base_dir = None Base directory in which the PHENIX GUI will create a folder containing the results. Ignored when running the program from the command line.save_aniso_data = False job_title = None Job title in PHENIX GUI, not used on command linecompositioncount = None Overall number of copies of the complete molecule (which may be multiple components). If left blank, this will be guessed based on the expected solvent content.componentsequence = None mtype = *protein dna rna Component typestoichiometry = 1 Number of copies of this component relative to the overall macromolecule. This is separate from the overall count parameter in the Basic Options tab, which defines the number of copies of the

complete unit. For instance, if you expect the basic molecule to be two copies of this component and one copy of a second component, the stoichiometry is 2. If you are searching for an unknown number of copies of a single component, you should leave the stoichiometry set to 1.

template_list = None
ensemblealignment = None
Sequence modelled by ensemblecoordinatespdb = None
identity = None
error_translation = *default
rmsd floored_chothia_and_lesk Identity to RMS conversionmodel_collectiontrim = False
coordinatespdb = None
identity = None
error_translation = *default
rmsd floored_chothia_and_lesk Identity to RMS conversiontemplatepdb = None
alignment = None
homologyfile_name = None
max_hits = 3
searchservices = ncbi local Performs a BLAST search of the PDB using either a locally installed version of BLAST and the PDB sequence database (requires separate installation), or the NCBI's BLAST server (requires Internet access).
max_hits = 3
assemblyDefine known assemblyuse_assembled = False Use assembly as model in search or only for completionlocal_searchDefine local search extentsweep = 5
Rotational search radiusextent = 5
Translational search extentcomponentDefine a component of the assemblyfile_name = None
Assembly component structureoperationModel transformation to participate in the assemblyeuler = 0 0 0
Rotationdisplacement = 0 0 0
Translationqueueutechnology = *threading
lsf sge pbs multiprocessing cpus = 1
submission_command = None
qslot_cpus = 1
simple_runParameters for simplified GUI - not intended for command-line use.
enable = False GUI parameter - not intended for command-line use.
component_sequence = None
template_model = None
alignment = None
blast_services = ncbi local Performs a BLAST search of the PDB using either a locally installed version of BLAST and the PDB sequence database (requires separate installation), or the NCBI's BLAST server (requires Internet access).
max_blast_hits = 3

- hklin = None
- labin = None
- resolution_cutoff = 2.5
- no_tncs = False
- mode = *quick full
- symmetry_exploration = *dataset enantiomorph pointgroup
- packing_pool = 20
- rotation_peaks_cutoff = 0.75
- post_refinement_cutoff = 0.75
- final_selection_cutoff = 0.75
- b_factor_refinement = True
- significant_peak_threshold = 7.0
- sculptor_protocols = 1 2 3 4 5 6 7 8 9 10 11 *12 13 all minimal
- template_equivalence = True
- assembly_acceptance_policy = *always observed never
- assembly_ensemble_creation_policy = always *observed never
- pdb_mirror = *rcsb pdbe pdbj
- exclude_pdb_ids = None Developer option to prevent specific structures from being used as search models, in order to test performance with lower sequence identity structures.
- crystal_symmetryunit_cell = None space_group = None
- unit_cell = None
- space_group = None

- outputroot = "mrage" max_solutions_to_write = 0 gui_base_dir = None Base directory in which the PHENIX GUI will create a folder containing the results. Ignored when running the program from the command line. save_aniso_data = False job_title = None Job title in PHENIX GUI, not used on command line
- root = "mrage"
- max_solutions_to_write = 0
- gui_base_dir = None Base directory in which the PHENIX GUI will create a folder containing the results. Ignored when running the program from the command line.
- save_aniso_data = False
- job_title = None Job title in PHENIX GUI, not used on command line
- compositioncount = None Overall number of copies of the complete molecule (which may be multiple components). If left blank, this will be guessed based on the expected solvent content. componentsequence = None mtype = *protein dna rna Component type stoichiometry = 1 Number of copies of this component relative to the overall macromolecule. This is separate from the overall count parameter in the Basic Options tab, which defines the number of copies of the complete unit. For instance, if you expect the basic molecule to be two copies of this component and one copy of a second component, the stoichiometry is 2. If you are searching for an unknown number of copies of a single component, you should leave the stoichiometry set to 1. template_list = None ensemblealignment = None Sequence modelled by ensemble coordinatespdb = None identity = None error_translation = *default rmsd floored_chothia_and_lesk Identity to RMS conversion model_collectiontrim = False coordinatespdb = None identity = None error_translation = *default rmsd floored_chothia_and_lesk Identity to RMS conversion templatepdb = None alignment = None homologyfile_name = None max_hits = 3 searchservices = ncbi local Performs a BLAST search of the PDB using either a locally installed version of BLAST and the PDB sequence database (requires separate installation), or the NCBI's BLAST server (requires Internet access). max_hits = 3
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- sequence = None
- mtype = *protein dna rna Component type
- stoichiometry = 1 Number of copies of this component relative to the overall macromolecule. This is separate from the overall count parameter in the Basic Options tab, which defines the number of copies of the complete unit. For instance, if you expect the basic molecule to be two copies of this component

and one copy of a second component, the stoichiometry is 2. If you are searching for an unknown number of copies of a single component, you should leave the stoichiometry set to 1.

- `template_list = None`
- `ensemblealignment = None` Sequence modelled by ensemble
`coordinatespdb = None` identity = None
`error_translation = *default rmsd floored_chothia_and_lesk` Identity to RMS conversion
- `alignment = None` Sequence modelled by ensemble
- `coordinatespdb = None` identity = None
`error_translation = *default rmsd floored_chothia_and_lesk` Identity to RMS conversion
- `pdb = None`
- `identity = None`
- `error_translation = *default rmsd floored_chothia_and_lesk` Identity to RMS conversion
- `model_collectiontrim = False` `coordinatespdb = None` identity = None
`error_translation = *default rmsd floored_chothia_and_lesk` Identity to RMS conversion
- `trim = False`
- `coordinatespdb = None` identity = None
`error_translation = *default rmsd floored_chothia_and_lesk` Identity to RMS conversion
- `pdb = None`
- `identity = None`
- `error_translation = *default rmsd floored_chothia_and_lesk` Identity to RMS conversion
- `templatepdb = None` `alignment = None`
- `pdb = None`
- `alignment = None`
- `homologyfile_name = None` `max_hits = 3`
- `file_name = None`
- `max_hits = 3`
- `searchservices = ncbi local` Performs a BLAST search of the PDB using either a locally installed version of BLAST and the PDB sequence database (requires separate installation), or the NCBI's BLAST server (requires Internet access).
`max_hits = 3`
- `services = ncbi local` Performs a BLAST search of the PDB using either a locally installed version of BLAST and the PDB sequence database (requires separate installation), or the NCBI's BLAST server (requires Internet access).
- `max_hits = 3`
- `assemblyDefine known assemblyuse_assembled = False` Use assembly as model in search or only for completion
`local_searchDefine local search extentsweep = 5` Rotational search radius
`extent = 5` Translational search extent
`componentDefine a component of the assemblyfile_name = None` Assembly component structure
`operationModel transformation to participate in the assemblyeuler = 0 0 0` Rotation
`displacement = 0 0 0` Translation
- `use_assembled = False` Use assembly as model in search or only for completion
- `local_searchDefine local search extentsweep = 5` Rotational search radius
`extent = 5` Translational search extent
- `sweep = 5` Rotational search radius

- extent = 5 Translational search extent
- componentDefine a component of the assemblyfile_name = None Assembly component structureoperationModel transformation to participate in the assemblyeuler = 0 0 0 Rotationdisplacement = 0 0 0 Translation
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- operationModel transformation to participate in the assemblyeuler = 0 0 0 Rotationdisplacement = 0 0 0 Translation
- euler = 0 0 0 Rotation
- displacement = 0 0 0 Translation
- queueutechnology = *threading lsf sge pbs multiprocessing cpus = 1 submission_command = None qslot_cpus = 1
- technology = *threading lsf sge pbs multiprocessing
- cpus = 1
- submission_command = None
- qslot_cpus = 1
- simple_runParameters for simplified GUI - not intended for command-line use.enable = False GUI parameter - not intended for command-line use.component_sequence = None template_model = None alignment = None blast_services = ncbi local Performs a BLAST search of the PDB using either a locally installed version of BLAST and the PDB sequence database (requires separate installation), or the NCBI's BLAST server (requires Internet access).max_blast_hits = 3
- enable = False GUI parameter - not intended for command-line use.
- component_sequence = None
- template_model = None
- alignment = None
- blast_services = ncbi local Performs a BLAST search of the PDB using either a locally installed version of BLAST and the PDB sequence database (requires separate installation), or the NCBI's BLAST server (requires Internet access).
- max_blast_hits = 3

Running phenix.refine using AFITT gradients for ligands

afitt.html

Contents

- Authors
- Theory and Background
- Running refinement with Phenix and AFITT
- Auxiliary functions
- Literature

Authors

Pawel Janowski and Nigel W. Moriarty

Theory and Background

AFITT refinement in Phenix combines the power and functionality of Phenix crystallographic refinement with OpenEye AFITT's [Wlodek2006] implementation of the MMFF94s [Halgren1996] force field for accurate modeling of small molecule stereochemistry. In combination they provide the user with a cutting-edge crystallographic refinement package while ensuring chemically sensible ligand geometry that significantly reduces conformational strain as compared to traditional refinement. Phenix.refine is a component of the highly regarded Phenix suite of crystallography software. AFITT is OpenEye's comprehensive package for ligands in crystallography. It provides an automated real space ligand fitting process, refinement dictionary generator, implementation of MMFF94, MMFF94s, PM3 and AM1 force fields and the OpenEye's core SZYBKI optimizer.

Phenix.refine employs a maximum likelihood approach to minimize the following residual or energy function:

$$E_{\text{Phenix}} = E_{\text{x-ray}} + E_{\text{geometry}}$$

where, in the presence of ligands, the second term in the above equation can be further broken down into:

$$E_{\text{geometry}} = E_{\text{protein}} + E_{\text{ligand_non-bonded}} + E_{\text{ligand_bonded}}$$

Phenix-AFITT replace Phenix's bonded ligand term with an AFITT term, thus yielding the following final energy function:

$$E_{\text{Phenix-AFITT}} = E_{\text{x-ray}} + E_{\text{geometry}} - E_{\text{ligand_bonded}}^{\text{Phenix}} + E_{\text{ligand_bonded}}^{\text{AFITT}}$$

Phenix's implementation of AFITT is versatile, easy to use and powerful. It can support any ligand for which a cif dictionary has been created. Refinements can include different types of ligands and multiple instances of each ligand type. Support for ligands with full or partial alternate conformations is fully integrated as is refinement of ligands covalently bound to the macromolecule.

Running refinement with Phenix and AFITT

First, the user should make sure that the latest version of Phenix and AFITT are installed. On Linux and OSX, make sure that the OE_EXE environment variable is set and points to the executable (usually flynn). For example:

```
[~:~]$ echo $OE_EXE
```

```
/usr/local/bin/flynn
```

Second prepare the .pdb, .mtz and .cif dictionaries for the ligand in your model. Now running Phenix-AFITT refinement is just a question of running standard Phenix.refine with some additional keywords. For example like this:

```
phenix.refine mymodel.pdb mymodel.mtz myligand.cif use_afitt=True  
afitt.ligand_file=myligand.cif afitt.ligand_names=UNL,BCL
```

The above command will run Phenix.refine with AFITT energy and gradients applied to the ligands UNL and BCL according to the topology provided in the file myligand.cif. The following is the full list of AFITT-related keyword that the user can specify on the command line when running Phenix.refine:

REQUIRED KEYWORDS use_afitt - turn on ligand refinement with AFITT. (default = False)

afitt.ligand_file_name - relative path to cif dictionary file that defines the ligand topology for AFITT. Note that this can be different from the cif file provided to Phenix.refine on the command line without any keyword. The former specifies the topology of the ligand for AFITT. The actual bond and angle parameters in it are irrelevant. The latter specifies the both topology and parameters of non-standard residues for Phenix. Any ligands that are not refined by AFITT (using the "ligand_names" keyword) will be refined by Phenix.refined using the parameters in this file. afitt.ligand_names - three letter ligand name codes. If multiple ligand types are being refined, separate each three letter code by a comma, no spaces. For a given ligand name, AFITT will refine all instances of that ligand as well as alternate conformations of each ligand. OPTIONAL KEYWORDS afitt.ff - AFITT force field to use. Options are: mmff94s, mmff94, am1, pm3. (default=mmff94s) afitt.scale - weight to place on the AFITT ligand energy term. (default=10)

REQUIRED KEYWORDS

use_afitt - turn on ligand refinement with AFITT. (default = False) afitt.ligand_file_name - relative path to cif dictionary file that defines the ligand topology for AFITT. Note that this can be different from the cif file provided to Phenix.refine on the command line without any keyword. The former specifies the topology of the ligand for AFITT. The actual bond and angle parameters in it are irrelevant. The latter specifies the both topology and parameters of non-standard residues for Phenix. Any ligands that are not refined by AFITT (using the "ligand_names" keyword) will be refined by Phenix.refined using the parameters in this file. afitt.ligand_names - three letter ligand name codes. If multiple ligand types are being refined, separate each three letter code by a comma, no spaces. For a given ligand name, AFITT will refine all instances of that ligand as well as alternate conformations of each ligand.

use_afitt - turn on ligand refinement with AFITT. (default = False)

afitt.ligand_file_name - relative path to cif dictionary file that

defines the ligand topology for AFITT. Note that this can be different from the cif file provided to Phenix.refine on the command line without any keyword. The former specifies the topology of the ligand for AFITT. The actual bond and angle parameters in it are irrelevant. The latter specifies the both topology and parameters of non-standard residues for Phenix. Any ligands that are not refined by AFITT (using the "ligand_names" keyword) will be refined by Phenix.refined using the parameters in this file.

afitt.ligand_names - three letter ligand name codes. If multiple

ligand types are being refined, separate each three letter code by a comma, no spaces. For a given ligand name, AFITT will refine all instances of that ligand as well as alternate conformations of each ligand.

OPTIONAL KEYWORDS

afitt.ff - AFITT force field to use. Options are: mmff94s, mmff94, am1, pm3. (default=mmff94s) afitt.scale - weight to place on the AFITT ligand energy term. (default=10)

afitt.ff - AFITT force field to use. Options are: mmff94s, mmff94, am1, pm3. (default=mmff94s)

afitt.scale - weight to place on the AFITT ligand energy term. (default=10)

Auxiliary functions

Two auxiliary command line programs are provided for convenience. mmtbx.afitt returns the AFITT ligand conformational energy given a pdb file (for coordinates) and cif file (for topology) and the three letter names of the ligands the user is querying. Optionally the user can specify the AFITT force field to use using the "-ff" keyword (default=MMFF94s). Example usage:

```
[~/work/afitt:]$ mmtbx.afitt 1cvu.pdb 1cvu.cif BOG,ACD
BOG_702_ AFITT_ENERGY: 2152.1468
BOG_703_ AFITT_ENERGY: 2151.5215
BOG_704_ AFITT_ENERGY: 2137.6938
BOG_2702_ AFITT_ENERGY: 2145.0459
BOG_2704_ AFITT_ENERGY: 2099.0437
ACD_701_ AFITT_ENERGY: 136.1296
ACD_2701_ AFITT_ENERGY: 99.7701
```

mmtbx.afitt_fd runs a finite difference test to check that the Phenix-AFITT energy and gradients are being properly calculated. This is the first test to run if you think something may be wrong with the implementation of the refinement algorithm. mmtbx.afitt_fd requires a pdb file, cif file, ligand names and atom number on whose x-coordinate the finite difference test will be calculated. An optional "-v" keyword provides more verbose output about the individual terms of the energy function. Example usage:

```
[~/work/afitt:]$ mmtbx.afitt_fd 1cvu.pdb 1cvu.cif ACD 9321
-> 0.134306495
-> 0.134306174
TEST PASSES: (analytical - numerical)= -0.000000321
```

As always, running an auxiliary program with the "-h" keyword provides helpful information about the program.

Example ligand conformational energies after traditional refinement and after refinement with Phenix-AFITT. Energies are shown as a percentage of the energy of the ligand conformation deposited in the PDB.

Literature

AI and Phenix

ai.html

Phenix is supplied with several AI tools that can help you decide what to do and to help you solve your structure:

- The Phenix chatbot will answer your questions about Phenix and how to use Phenix tools and how to solve a structure
- The Phenix AI agent can automatically analyze your X-ray or cryo-EM data and solve your structure. It can also run many Phenix tools for you.
- The Phenix AI analysis tool can look at your log file and suggest what to do next.

If you are a beginner, ask the chatbot how to solve your structure if you have X-ray data (or cryo-EM data). Use the AI agent to automatically set up and analyze your data for you, and use the AI analysis tool and the chatbot to interpret your results.

If you are an advanced user, try the AI agent as a tool for quickly screening your data.

If you are either a beginner or an advanced user, also have a look at the overall Phenix documentation for tips and ideas for solving your structure.

Running phenix.refine using Amber gradients

amber.html

Contents

- Authors
- Theory and Background
- Running refinement with Phenix and Amber

Authors

Principal authors: Nigel W. Moriarty, Pawel Janowski, Hai Nguyen & David A. Case

Contributions from: Jason Swails (Python Sander API), Romain Wolf (pdb4amber which is used in AmberPrep)

Theory and Background

Amber refinement in Phenix combines the power and functionality of Phenix crystallographic refinement with the Amber molecular mechanics force field for geometry restraints. Amber restraints replace the conventional set of restraints. In addition to bond, angle and dihedral terms, electrostatics and van der Waals attractive/dispersive interactions are computed. Calls to Amber's energy and gradient calculation engines are seamlessly integrated through internal libraries shared with Phenix. Refinement with Amber molecular mechanics has been shown to improve model quality and reduce overfitting as compared to conventional restraints.

Running refinement with Phenix and Amber

From Phenix 1.16, Amber is distributed with the standard installer for unix and OSX. To run Amber in Phenix, in addition to a PDB file, a topology and a coordinate file must be provided for Amber. From Phenix 1.18, you can allow the phenix.refine GUI to run AmberPrep, automatically creating the files needed. You can also easily create these files by using the provided AmberPrep utility via the AmberPrep GUI or via the commandline:

```
phenix.fetch_pdb 4Lzt --mtz
phenix.AmberPrep 4Lzt.pdb
```

which will generate four files.

```
4amber_4Lzt.prmtop
4amber_4Lzt.rst7
4amber_4Lzt.order
4phenix_4Lzt.pdb
```

The first two are the Amber files and the last one is the PDB file that matches the two Amber files for atom order and must be used in phenix.refine replacing 4Lzt.pdb. Having created the input files, refinement can be run as follows:

```
phenix.refine 4phenix_4Lzt.pdb 4Lzt.mtz use_amber=True topology_file_name=4amber_4Lzt.prmtop coordinate_file_name=4phenix_4Lzt.pdb xray_data.r_free_flags.generate=True
```

Here is list of keywords:

REQUIRED KEYWORDS `use_amber` - turn on refinement with Amber geometry restraints. ("True" or "False"; default = False) `topology_file_name` - relative path to the Amber topology file. This is the prmtop file created by AmberPrep. `coordinate_file_name` - relative path to the Amber coordinate file. This is the rst7 file created by AmberPrep. OPTIONAL KEYWORDS `wxc_factor` - relative weight factor between the x-ray and the Amber geometry restraint terms in refinement. By default this is set to 0.1 which multiplies the standard `wxc_scale` which has a default of 0.5 for the x-ray and E&H restraints. (Float; default=0.1)

REQUIRED KEYWORDS

`use_amber` - turn on refinement with Amber geometry restraints. ("True" or "False"; default = False)
`topology_file_name` - relative path to the Amber topology file. This is the prmtop file created by AmberPrep.
`coordinate_file_name` - relative path to the Amber coordinate file. This is the rst7 file created by AmberPrep.

`use_amber` - turn on refinement with Amber geometry restraints. ("True" or "False"; default = False)

`topology_file_name` - relative path to the Amber topology file.

This is the prmtop file created by AmberPrep.

`coordinate_file_name` - relative path to the Amber coordinate file.

This is the rst7 file created by AmberPrep.

OPTIONAL KEYWORDS

`wxc_factor` - relative weight factor between the x-ray and the Amber geometry restraint terms in refinement. By default this is set to 0.1 which multiplies the standard `wxc_scale` which has a default of 0.5 for the x-ray and E&H restraints. (Float; default=0.1)

`wxc_factor` - relative weight factor between the x-ray and

the Amber geometry restraint terms in refinement. By default this is set to 0.1 which multiplies the standard `wxc_scale` which has a default of 0.5 for the x-ray and E&H restraints. (Float; default=0.1)

ADDITIONAL INFORMATION:

1. We also provide the possibility to energetically optimize model coordinates using Amber's force field rather than the set of Engh & Huber restraints. This leads to models of improved quality. This minimization can be run as follows:

```
phenix.geometry_minimization 4amber_xyz.pdb amber.use=True topology_file_name=4amber_4Lzt.prmtop coordinate_file_name=4amber_4Lzt.rst7
```

2. AmberPrep contains additional options that the user may wish to use. Run:

```
phenix.AmberPrep --show-defaults
```

to view all of the options. Here is a list of the optional keywords:

`minimise` - type of energy minimization to perform. Possible options: `amber_all` - optimize coordinates of all atoms using Amber's minimizer `amber_h` - optimize only hydrogen positions using Amber's minimizer (default) `phenix_all` - optimize coordinates of all atoms using phenix minimizer `off` - do not optimize atoms `clean` - remove temporary files (True/False, default=True) `redq` - use Amber's redq force field instead of the default (currently ff14SB). redq is a version of ff14SB that mitigates the partial charges on charged amino acid side chains. In theory this should account for the electrostatic screening effects of the unmodelled solvent molecules. In practice our tests have not shown any improvement in refinement when using the reduced charge (redq) force field, but users are free to experiment (True/False, default=False)

`minimise` - type of energy minimization to perform. Possible options:

amber_all - optimize coordinates of all atoms using Amber's minimizer
amber_h - optimize only hydrogen positions using Amber's minimizer (default)
phenix_all - optimize coordinates of all atoms using phenix minimizer
off - do not optimize atoms

amber_all - optimize coordinates of all atoms using Amber's minimizer

amber_h - optimize only hydrogen positions using Amber's minimizer (default)

phenix_all - optimize coordinates of all atoms using phenix minimizer

off - do not optimize atoms

clean - remove temporary files (True/False, default=True)

redq - use Amber's redq force field instead of the default (currently

ff14SB). redq is a version of ff14SB that mitigates the partial charges on charged amino acid side chains. In theory this should account for the electrostatic screening effects of the unmodelled solvent molecules. In practice our tests have not shown any improvement in refinement when using the reduced charge (redq) force field, but users are free to experiment (True/False, default=False)

3. Refinement with Amber currently has several limitations. All atoms must be explicitly modelled and there can be no gaps in the peptide or nucleic acid backbone. If hydrogen atoms are missing, they will be added automatically by AmberPrep but missing heavy atoms should first be manually modelled in by the user. Amber does not currently model alternate conformations and AmberPrep will only retain the first alternate conformer of each atom. Solutions to these two limitations are currently in development.

4. Metal-containing ligands are not supported by Amber.

5. Atoms at special positions will most often produce high energy clashes with Amber and usually will not work. If the atom is a solvent molecule, the user may try to refine without it. If the structure contains atoms at special positions, a warning will be issued when running AmberPrep.

FOR ADVANCED USERS:

1. To use MDGX instead of Sander for gradient calculation, use md_engine=mdgx keyword for phenix.refine. The default md_engine is Sander.

2. phenix.refine with Amber also has a hidden wxc_factor parameter. In ordinary refinement the x-ray term is scaled by the ratio of the gradient norms between the x-ray and geometry terms times the wxc_factor (default 0.5 when using E&H; recommended 0.025 when using Amber). If wxc_factor is set, the ratio of the gradient norms will be ignored and the x-ray term will be scaled by wxc_factor directly. This is an experimental feature and has not been shown to improve refinement in any way.

3. The advanced user may wish to change the Amber force field used to impose geometry restraints. By default the force field used is ff14SB. This can easily be changed by modifying AmberPrep script, run_tleap() method, to source a different force field from Amber.

4. phenix.AmberPrep performs several important actions to prepare files for refinement with Phenix and Amber. This includes:

checks for existence of ligands and non-standard small molecules. Derives Amber parameters using eLBOW and Antechamber cleans and modifies PDB file for conformity with Amber runs TLEaP to prepare Amber topology and coordinate files runs Amber's XtalUtilities to prepare Amber inputs that conform to the given crystallographic space group adds missing hydrogen atoms and possibly missing side chain heavy atoms creates a PDB input file for Phenix that corresponds in atom order and content to the Amber input files

- checks for existence of ligands and non-standard small molecules. Derives Amber parameters using eLBOW and Antechamber
- cleans and modifies PDB file for conformity with Amber
- runs TLEaP to prepare Amber topology and coordinate files
- runs Amber's XtalUtilities to prepare Amber inputs that conform to the given crystallographic space group
- adds missing hydrogen atoms and possibly missing side chain heavy atoms
- creates a PDB input file for Phenix that corresponds in atom order and content to the Amber input files

Analyzing alternative conformations matching a map with `analyze_alt_conf`

`analyze_alt_conf.html`

Author(s)

- `analyze_alt_conf`: Tom Terwilliger

Purpose

The purpose of `analyze_alt_conf` is to analyze alternative conformations (focusing on side-chains) for a structure and an optional comparison structure.

The problem that it solves is that if you have multiple conformations present in two structures, identifying which ones match and which ones are unique is tedious.

How `analyze_alt_conf` works:

System Message: WARNING/2 (<string>, line 21)

Title underline too short.

```
How analyze_alt_conf works:
-----
```

This tool first superposes all models that are supplied. Then it analyzes the unique residue conformations in each model that is supplied. Residues that superpose with an rmsd of `min_rotamer_rmsd` (default of 0.25 Å) or better are considered the same conformer; those with a larger rmsd are considered different.

If there are two models supplied, the tool then compares the unique conformers in the first and second models. It identifies which altlocs in the first model (m1) match which altlocs in the second model (m2), and which are unique in one model or the other.

Examples

Standard run of `analyze_alt_conf`:

System Message: WARNING/2 (<string>, line 38)

Title underline too short.

```
Standard run of analyze_alt_conf:
-----
```

Running `analyze_alt_conf` is easy. From the command-line you can type:

```
phenix.analyze_alt_conf model.pdb other_model.pdb
```

where `model.pdb` is the model you would like to use as a starting point and `other_model.pdb` is an optional comparison model.

Possible Problems

The procedure only analyzes alternative side-chain conformers, not main-chain. If you main chain has substantial alternative conformations (not just slight adjustments to match the side chain conformers), you will need to use another approach.

Literature

Additional information

List of all available keywords

job_title = None Job title in PHENIX GUI, not used on command line
input_filesmodel = None Input model
analyze_alt_confmin_rotamer_rmsd = 0.25 Minimum rmsd between residues. If atoms in residue differ by less than this, consider them the same.
match_any_atom = True Ignore atom names when comparing residues. Just take closest pairs between structures
guiGUI-specific parameter required for output directory
output_dir = None

- job_title = None Job title in PHENIX GUI, not used on command line
- input_filesmodel = None Input model
- model = None Input model
- analyze_alt_confmin_rotamer_rmsd = 0.25 Minimum rmsd between residues. If atoms in residue differ by less than this, consider them the same.
- match_any_atom = True Ignore atom names when comparing residues. Just take closest pairs between structures
- min_rotamer_rmsd = 0.25 Minimum rmsd between residues. If atoms in residue differ by less than this, consider them the same.
- guiGUI-specific parameter required for output directory
- output_dir = None

AI analysis of Phenix results

analyze_log.html

Authors

- Nigel Moriarty, Peter Zwart, Thomas C Terwilliger

Purpose

The Phenix AI analysis tool is designed to help you interpret your output files from a run in the Phenix GUI. It analyzes your log file and any output written to the GUI and the names of any output files that are supplied to the GUI and creates a summary of the run. Then it analyzes this summary in the context of the Phenix documentation to describe how this run fits into the framework of structure determination and suggests next steps. The AI analysis tool can be accessed in the Phenix GUI in the Results section of most Phenix GUI runs, next to the Log file button. You can also access it under the Tools menu.

Google and Cohere API Keys

To run the AI analysis, you will need Google and Cohere API keys. The Google key will let you run Gemini from the Phenix server, and the Cohere key will let you run Cohere Langchain tools. Cohere keys are free. Google API keys are free (up to \$300 in usage) for up to 3 months. After that you will be charged, but you can see from your usage in your free trial period how much (or little) you would be charged. You can get the keys from the AI section of the Preferences menu in the Phenix GUI.

You can run a AI analyses without getting your own key as a shared set of keys is supplied, but you should get your own set once you get going so that others can use the shared keys.

How the AI analysis works

The AI analysis is carried out in two steps. In the first step, the log file, along with text written to the GUI and names of output files supplied to the GUI are summarized using a Cohere Langchain analysis. In the second step, the summary from the first step is analyzed in the context of all the Phenix documentation, transcripts of Phenix tutorial videos, and Phenix publications to provide background and to suggest next steps to take.

This type of AI does not save or learn from your questions. However the information in your log file is sent to the Phenix server, and from there, on to Cohere and Google gemini.

What to do with the AI analysis

The purpose of the AI analysis is to help you interpret your output and to suggest ideas for next steps. You should keep in mind that this tools can make mistakes and give you incorrect interpretations and poor suggestions at times, so you want to just take the information as suggestions to think about.

You can combine this AI analysis with the Phenix chatbot . The chatbot can give you interactive answers to your questions using the same database of information as the AI analysis. This allows you to follow up on the AI analysis with questions to the chatbot. You can also paste part of the output from the AI analysis into the chatbot along with a question to get more context.

Limitations of the AI analysis

The AI analysis is limited to the sources that is supplied with, so it only knows about Phenix, and it only knows what is in the documentation, the videos and newsletters, and the papers we have supplied.

AI tools like this one can also just make mistakes and give incorrect answers. This does not seem to happen too often with this tool, but you need to always be on alert when using it. Use the tool as a helper, don't expect it to always be right.

If a detail is missing in the documentation, the AI analysis will not know it.

Privacy in the AI analysis

If you use the AI analysis tool, your log files are sent to the Phenix server, and from there on to Cohere and Gemini. That means the data could potentially be used by Cohere and Google in any way that they use other AI data that is sent to them.

Note that access to Cohere and Gemini is non-commercial only.

Applying NCS to a molecule to generate all NCS copies with `apply_ncs`

`apply_ncs.html`

Author(s)

- `apply_ncs`: Tom Terwilliger

Purpose

If you have a single copy of your molecule placed in the crystal and you have a set of NCS operators defined in an `ncs_info.ncs_spec` file (such as that written by `phenix.find_ncs`), you can apply the operators to your molecule to generate all the NCS copies. This can be useful if you want to work on one molecule in your asymmetric unit and propagate it to the other copies. You can generate your `ncs_info.ncs_spec` file using `phenix.find_ncs` and your original model containing all the NCS copies, then apply it to any one of the individual molecules to regenerate the others.

Usage

How `apply_ncs` works:

`apply_ncs` assumes that you are supplying it with one copy of your molecule, placed in your unit cell, and that you are also supplying a `ncs_info.ncs_spec` file that contains all the NCS operators for your structure along with the approximate molecular centers where each operator applies. `apply_ncs` assumes that the NCS operator with a center closest to the center of the molecule you supply (including space-group symmetry) is the operator corresponding to your input molecule.

`apply_ncs` then calculates the transformation for your molecule to molecule 1, and then uses all the NCS operators to transform molecule 1 to each NCS copy.

Output files from `apply_ncs`

`apply_ncs.pdb`: A PDB file with your input molecule placed at all the NCS-related positions in your structure

Examples

Standard run of `apply_ncs`:

Running `apply_ncs` is easy. From the command-line you can type:

```
phenix.apply_ncs mol1.pdb my_ncs.ncs_spec
```

If `apply_ncs` chooses the wrong NCS copy to match up to your input molecule, you can tell it which one your input molecule is with `match_copy`:

```
phenix.apply_ncs mol1.pdb my_ncs.ncs_spec match_copy=2
```

`apply_ncs` will then assume that you are inputting ncs copy 2.

Possible Problems

Specific limitations and problems:

Literature

Additional information

The documentation for `find_ncs` describes how to find and apply NCS information for helical arrays, point-group symmetry, and non-point-group symmetry.

List of all available keywords

`apply_ncsjob_title` = None Job title in PHENIX GUI, not used on command line
`input_files`
`pdb_in` = None Input PDB file
`ncs_in` = None File with NCS operators to be applied to `pdb_in` This can be a `.ncs_spec` file from `find_ncs`, `simple_ncs_from_pdb` or `find_ncs_from_density` It can also be NCS formatted for resolve, including rotations, translations, and centers NOTE: NCS operators only apply to a limited region within the unit cell. The center of this region for molecule 1 (the one you enter with `pdb_in`) is specified by the NCS center for molecule 1. If your molecule in `pdb_in` is not near this, then it will be brought as close as possible using space-group symmetry (as defined in the CRYST1 record of `pdb_in`) before application of NCS. If your molecule is closer to the center for operator k, then it will instead be brought to that center and then transformed to all other locations
`map_in` = None CCP4 or MRC-style map file (required if `trim_overlapping=True`). Map must cover the entire molecule including NCS
`copies`
`output_file`
`target_output_format` = *None pdb mmCIF Desired output format (if possible). Choices are None (try to use input format), pdb, mmCIF. If output model does not fit in pdb format, mmCIF will be used. Default is pdb.
`pdb_out` = `apply_ncs.pdb` Output PDB file
`log` = `apply_ncs.log` Output log file
`params_out` = `apply_ncs_params.eff` Parameters file to rerun `apply_ncs`
`write_refine_ncs_out` = False Write out `refine_ncs_out` file for refinement
`refine_ncs_out` = `apply_ncs_refine.eff` NCS formatted for refinement. This will put all of the input chains as reference and all of the ncs-related chains in selection records, one for each NCS copy applied.
`directories`
`temp_dir` = "temp_dir" Optional temporary work directory
`output_dir` = "" Output directory where files are to be written
`gui_output_dir` = None
`ncs_trim_overlapping` = False Trim pieces that overlap (including those overlapping by NCS). Requires a full-size (not boxed) map file (ccp4/mrc). Also requires resolution of the map.
`resolution` = None Map resolution (for `trim_overlapping`)
`match_copy` = None Force input model to be associated with NCS copy `match_copy` This can be useful if your molecule center is close to an NCS center other than the one it is associated with
`no_match_copy` = None Do not allow input model to be associated with this NCS copy `no_match_copy`. This can be useful if your molecule center is close to an NCS center other than the one it is associated with
`apply_operators_directly` = False You can apply the NCS matrices directly to an input model, without considering one of the NCS copies to be the identity for this model.
`max_copies` = None You can limit the number of NCS copies to fewer than the actual number
`start_copy` = None You can start the NCS copies with any available copy
`only_start_copy` = None Just use `start_copy` and do not try to arrange chain names
`unique_only` = None Just use unique part of `pdb_in`
`chain_names` = None You can specify the chain names to generate. Note: there must be exactly the same number of chain_names as total chains generated by `apply_ncs`. This includes any chains that are just waters or ligands.
`used_chain_id_list` = None You can specify chain names not to generate
`use_space_group_symmetry` = True Normally `apply_ncs` uses space-group symmetry read from your input PDB file to find the appropriate operator. To disable this say `use_space_group_symmetry=False`. This is what you should do if you are using space group P1 as a dummy space group for example.
`control_verbosity` = False Verbose output
`raise_sorry` = False Raise sorry if problems
`debug` = False Debugging output
`dry_run` = False Just read in and check parameter names
`resolve_command_list` = None You can supply any resolve command here NOTE: for command-line usage you need to enclose the whole set of commands in double

quotes (") and each individual command in single quotes (') like this:
resolve_command_list="'no_build' 'b_overall 23' ";

- apply_ncsjob_title = None Job title in PHENIX GUI, not used on command line
input_files_pdb_in = None Input PDB file
ncs_in = None File with NCS operators to be applied to pdb_in This can be a .ncs_spec file from find_ncs, simple_ncs_from_pdb or find_ncs_from_density It can also be NCS formatted for resolve, including rotations, translations, and centers NOTE: NCS operators only apply to a limited region within the unit cell. The center of this region for molecule 1 (the one you enter with pdb_in) is specified by the NCS center for molecule 1. If your molecule in pdb_in is not near this, then it will be brought as close as possible using space-group symmetry (as defined in the CRYST1 record of pdb_in) before application of NCS. If your molecule is closer to the center for operator k, then it will instead be brought to that center and then transformed to all other locations
map_in = None CCP4 or MRC-style map file (required if trim_overlapping=True). Map must cover the entire molecule including NCS copies.
output_filetarget_output_format = *None pdb mmCIF Desired output format (if possible). Choices are None (try to use input format), pdb, mmCIF. If output model does not fit in pdb format, mmCIF will be used. Default is pdb.
pdb_out = apply_ncs.pdb Output PDB file
log = apply_ncs.log Output log
params_out = apply_ncs.params.eff Parameters file to rerun
apply_ncs.write_refine_ncs_out = False Write out refine_ncs_out file for refinement
refine_ncs_out = apply_ncs.refine.eff NCS formatted for refinement. This will put all of the input chains as reference and all of the ncs-related chains in selection records, one for each NCS copy applied.
directoriestemp_dir = "temp_dir" Optional temporary work directory
output_dir = "" Output directory where files are to be written
gui_output_dir = None
ncstrim_overlapping = False Trim pieces that overlap (including those overlapping by NCS). Requires a full-size (not boxed) map file (ccp4/mrc). Also requires resolution of the map.
resolution = None Map resolution (for trim_overlapping)
match_copy = None Force input model to be associated with NCS copy
match_copy This can be useful if your molecule center is close to an NCS center other than the one it is associated with
no_match_copy = None Do not allow input model to be associated with this NCS copy
no_match_copy. This can be useful if your molecule center is close to an NCS center other than the one it is associated with
apply_operators_directly = False You can apply the NCS matrices directly to an input model, without considering one of the NCS copies to be the identity for this model.
max_copies = None You can limit the number of NCS copies to fewer than the actual number
start_copy = None You can start the NCS copies with any available copy
only_start_copy = None Just use start_copy and do not try to arrange chain names
unique_only = None Just use unique part of pdb_in
chain_names = None You can specify the chain names to generate. Note: there must be exactly the same number of chain_names as total chains generated by apply_ncs. This includes any chains that are just waters or ligands.
used_chain_id_list = None You can specify chain names not to generate
use_space_group_symmetry = True Normally apply_ncs uses space-group symmetry read from your input PDB file to find the appropriate operator. To disable this say use_space_group_symmetry=False. This is what you should do if you are using space group P1 as a dummy space group for example.
controlverbose = False Verbose output
raise_sorry = False Raise sorry if problems
debug = False Debugging output
dry_run = False Just read in and check parameter names
resolve_command_list = None You can supply any resolve command here NOTE: for command-line usage you need to enclose the whole set of commands in double quotes (") and each individual command in single quotes (') like this: resolve_command_list="'no_build' 'b_overall 23' ";
- job_title = None Job title in PHENIX GUI, not used on command line
- input_files_pdb_in = None Input PDB file
ncs_in = None File with NCS operators to be applied to pdb_in This can be a .ncs_spec file from find_ncs, simple_ncs_from_pdb or find_ncs_from_density It can also be NCS formatted for resolve, including rotations, translations, and centers NOTE: NCS operators only apply

to a limited region within the unit cell. The center of this region for molecule 1 (the one you enter with `pdb_in`) is specified by the NCS center for molecule 1. If your molecule in `pdb_in` is not near this, then it will be brought as close as possible using space-group symmetry (as defined in the CRYST1 record of `pdb_in`) before application of NCS. If your molecule is closer to the center for operator k, then it will instead be brought to that center and then transformed to all other locations `map_in = None` CCP4 or MRC-style map file (required if `trim_overlapping=True`). Map must cover the entire molecule including NCS copies.

- `pdb_in = None` Input PDB file
- `ncs_in = None` File with NCS operators to be applied to `pdb_in` This can be a `.ncs_spec` file from `find_ncs`, `simple_ncs_from_pdb` or `find_ncs_from_density` It can also be NCS formatted for resolve, including rotations, translations, and centers NOTE: NCS operators only apply to a limited region within the unit cell. The center of this region for molecule 1 (the one you enter with `pdb_in`) is specified by the NCS center for molecule 1. If your molecule in `pdb_in` is not near this, then it will be brought as close as possible using space-group symmetry (as defined in the CRYST1 record of `pdb_in`) before application of NCS. If your molecule is closer to the center for operator k, then it will instead be brought to that center and then transformed to all other locations
- `map_in = None` CCP4 or MRC-style map file (required if `trim_overlapping=True`). Map must cover the entire molecule including NCS copies.
- `output_filetarget_output_format = *None` pdb mmCIF Desired output format (if possible). Choices are None (try to use input format), pdb, mmCIF. If output model does not fit in pdb format, mmCIF will be used. Default is `pdb.pdb_out = apply_ncs.pdb` Output PDB file `log = apply_ncs.log` Output log file `params_out = apply_ncs_params.eff` Parameters file to rerun `apply_ncs` `write_refine_ncs_out = False` Write out `refine_ncs_out` file for refinement `refine_ncs_out = apply_ncs_refine.eff` NCS formatted for refinement. This will put all of the input chains as reference and all of the ncs-related chains in selection records, one for each NCS copy applied.
- `target_output_format = *None` pdb mmCIF Desired output format (if possible). Choices are None (try to use input format), pdb, mmCIF. If output model does not fit in pdb format, mmCIF will be used. Default is `pdb`.
- `pdb_out = apply_ncs.pdb` Output PDB file
- `log = apply_ncs.log` Output log file
- `params_out = apply_ncs_params.eff` Parameters file to rerun `apply_ncs`
- `write_refine_ncs_out = False` Write out `refine_ncs_out` file for refinement
- `refine_ncs_out = apply_ncs_refine.eff` NCS formatted for refinement. This will put all of the input chains as reference and all of the ncs-related chains in selection records, one for each NCS copy applied.
- `directorytemp_dir = "temp_dir"` Optional temporary work directory `output_dir = ""` Output directory where files are to be written `gui_output_dir = None`
- `temp_dir = "temp_dir"` Optional temporary work directory
- `output_dir = ""` Output directory where files are to be written
- `gui_output_dir = None`
- `ncstrim_overlapping = False` Trim pieces that overlap (including those overlapping by NCS). Requires a full-size (not boxed) map file (ccp4/mrc). Also requires resolution of the map. `resolution = None` Map resolution (for `trim_overlapping`) `match_copy = None` Force input model to be associated with NCS copy `match_copy` This can be useful if your molecule center is close to an NCS center other than the one it is associated with `no_match_copy = None` Do not allow input model to be associated with this NCS copy

no_match_copy. This can be useful if your molecule center is close to an NCS center other than the one it is associated with.

apply_operators_directly = False You can apply the NCS matrices directly to an input model, without considering one of the NCS copies to be the identity for this model.

max_copies = None You can limit the number of NCS copies to fewer than the actual number.

start_copy = None You can start the NCS copies with any available copy.

only_start_copy = None Just use start_copy and do not try to arrange chain names.

unique_only = None Just use unique part of pdb_in.

chain_names = None You can specify the chain names to generate. Note: there must be exactly the same number of chain_names as total chains generated by apply_ncs. This includes any chains that are just waters or ligands.

used_chain_id_list = None You can specify chain names not to generate.

use_space_group_symmetry = True Normally apply_ncs uses space-group symmetry read from your input PDB file to find the appropriate operator. To disable this say use_space_group_symmetry=False. This is what you should do if you are using space group P1 as a dummy space group for example.

- trim_overlapping = False Trim pieces that overlap (including those overlapping by NCS). Requires a full-size (not boxed) map file (ccp4/mrc). Also requires resolution of the map.
- resolution = None Map resolution (for trim_overlapping)
- match_copy = None Force input model to be associated with NCS copy match_copy. This can be useful if your molecule center is close to an NCS center other than the one it is associated with.
- no_match_copy = None Do not allow input model to be associated with this NCS copy no_match_copy. This can be useful if your molecule center is close to an NCS center other than the one it is associated with.
- apply_operators_directly = False You can apply the NCS matrices directly to an input model, without considering one of the NCS copies to be the identity for this model.
- max_copies = None You can limit the number of NCS copies to fewer than the actual number.
- start_copy = None You can start the NCS copies with any available copy.
- only_start_copy = None Just use start_copy and do not try to arrange chain names.
- unique_only = None Just use unique part of pdb_in.
- chain_names = None You can specify the chain names to generate. Note: there must be exactly the same number of chain_names as total chains generated by apply_ncs. This includes any chains that are just waters or ligands.
- used_chain_id_list = None You can specify chain names not to generate.
- use_space_group_symmetry = True Normally apply_ncs uses space-group symmetry read from your input PDB file to find the appropriate operator. To disable this say use_space_group_symmetry=False. This is what you should do if you are using space group P1 as a dummy space group for example.
- controlverbose = False Verbose output.
- raise_sorry = False Raise sorry if problems.
- debug = False Debugging output.
- dry_run = False Just read in and check parameter names.
- resolve_command_list = None You can supply any resolve command here. NOTE: for command-line usage you need to enclose the whole set of commands in double quotes (") and each individual command in single quotes (') like this: resolve_command_list="'no_build' 'b_overall 23' ";
- verbose = False Verbose output.
- raise_sorry = False Raise sorry if problems.
- debug = False Debugging output.
- dry_run = False Just read in and check parameter names.

- `resolve_command_list = None` You can supply any resolve command here NOTE: for command-line usage you need to enclose the whole set of commands in double quotes (") and each individual command in single quotes (') like this: `resolve_command_list="'no_build' 'b_overall 23' "`

Automated molecular replacement with AutoMR

automr.html

Author(s)

- Phaser: Randy J. Read, Airlie J. McCoy and Laurent C. Storoni
- AutoMR Wizard: Tom Terwilliger, Laurent Storoni, Randy Read, and Airlie McCoy
- phenix.xtriage: Peter Zwart

Purpose

Purpose of the AutoMR Wizard

The AutoMR Wizard provides a convenient interface to Phaser molecular replacement and feeds the results of molecular replacement directly into the AutoBuild Wizard for automated model rebuilding.

The AutoMR Wizard begins with datafiles with structure factor amplitudes and uncertainties, a search model or models, and identifies placements of the search models that are compatible with the data.

Usage

The AutoMR wizard has been deprecated and is no longer available in the Phenix GUI, but it may still be run from the command-line, and from parameters files. Both versions are identical except in the way that they take commands from the user. See Using the PHENIX Wizards for details of how to run a Wizard.

Summary of inputs and outputs for AutoMR

Input data file. This file can be in most any format, and must contain either amplitudes or intensities and sigmas. You can specify what resolution to use for molecular replacement and separately what resolution to use for model rebuilding. If you specify "0.0" for resolution (recommended) then defaults will be used for molecular replacement (i.e. use data to 2.5Å if available to solve structure, then carry out rigid body refinement of final solution with all data) and all the data will be used for model rebuilding.

Contents of the asymmetric unit. PHASER needs to know what the total mass in the asymmetric unit is (i.e. not just the mass of the search models). You can define this either by specifying one or more protein or nucleic acid sequence files, or by specifying protein or nucleic acid molecular masses, and telling the Wizard how many copies of each are present.

Space groups to search. You can request that all space groups with the same point group as the one you start out with be searched, and the best one be chosen. If you select this option then the best space group will be used for model rebuilding in AutoBuild.

Ensembles to search for. AutoMR builds up a model by finding a set of good positions and orientations of one "ensemble", and then using each of those placements as starting points for finding the next ensemble, until all the contents of the asymmetric unit are found and a consistent solution is obtained. You can specify any number of different ensembles to search for, and you can search for any number of copies of each ensemble. The order of searching for ensembles makes a difference, but Phaser chooses a sensible default search order based on the size and assumed accuracy of the different ensembles. In difficult cases you could try permuting the search order.

Each ensemble can be specified by a single PDB file or a set of PDB files. The contents of one set of PDB files for an ensemble must all be oriented in the same way, as they will be put together and used as a group always in the molecular replacement process.

You will need to specify how similar you think each input PDB file that is part of an ensemble is to the structure that is in your crystal. You can specify either sequence identity, or expected rmsd. Note that if you use a homology model, you should give the sequence identity of the template from which the model was constructed, not the 100% identity of the model!

Output of AutoMR

Output files from AutoMR

When you run AutoMR the output files will be in a subdirectory with your run number:

AutoMR_run_1_/ # subdirectory with results

- A summary file listing the results of the run and the other files produced:

AutoMR_summary.dat # overall summary

- A warnings file listing any warnings about the run

AutoMR_warnings.dat # any warnings

- A file that lists all parameters and knowledge accumulated by the Wizard during the run (some parts are binary and are not printed)

AutoMR_Facts.dat # all Facts about the run

- Molecular replacement model, structure factors, and map coefficients:

MR.1.pdb

MR.1.mtz

The AutoMR wizard writes out MR.1.pdb and MR.1.mtz as well as output log files. The MR.1.pdb file will contain all the components of your MR solution. If there are multiple PDB files in an ensemble, the model with the lowest estimated rmsd is chosen to represent the whole ensemble and is written to MR.1.pdb. If there are multiple copies of a model, the chains are lettered sequentially A B C... The MR.1.mtz file contains the data from your input file to the full resolution available, as well as sigmaA-weighted 2Fo-Fc map coefficients based on the rigid-body-refined model.

Model rebuilding. After PHASER molecular replacement the AutoMR Wizard loads the AutoBuild Wizard and sets the defaults based on the MR solution that has just been found. You can use the default values, or you may choose to use 2Fo-Fc maps instead of density-modified maps for rebuilding, or you may choose to start the model-rebuilding with the map coefficients from MR.1.mtz.

How to run the AutoMR Wizard

Running the AutoMR Wizard is easy. For example, from the command-line you can type:

```
phenix.automr native.sca search.pdb RMS=0.8 mass=23000 copies=1
```

The AutoMR Wizard will find the best location and orientation of the search model search.pdb in the unit cell based on the data in native.sca, assuming that the RMSD between the correct model and search.pdb is about 0.8 Å, that the molecular mass of the true model is 23000 and that there is 1 copy of this model in the asymmetric unit. Once the AutoMR Wizard has found a solution, it will automatically call the AutoBuild Wizard and rebuild the model.

Components, copies, search models, and ensembles

- Your structure is composed of one or more components such as a 20Kd subunit with sequence seq-of-20Kd-subunit.
- There may be one or more copies of each component in your structure.
- You can search for the location(s) of a component with a search model that consists of a single structure or an ensemble of structures.

What the AutoMR wizard needs to run

In a simple case where you have one search model and are looking for N copies of this model in your structure, you need:

- a datafile name (native.sca or data=native.sca)
- a datafile name (native.sca or data=native.sca)
- a search model (search_model.pdb or coords=search_model.pdb)
- a search model (search_model.pdb or coords=search_model.pdb)
- (3) how similar the search model is to your structure (RMS=0.8 or identity=75)
- (4) information about the contents of the asymmetric unit: (mass=23000 or seq_file=seq.dat) and (copies=1)

It may be advantageous to search using an ensemble of similar structures, rather than a single structure. If you have an ensemble of search models to search for, then specify it as

```
coords="model_1.pdb" coords="model_2.pdb" coords="model_3.pdb"
```

In this case you need to give the RMS or identity for each model: identity='45 40 35'. Each of the models in the ensemble must be in the same orientation as the others, so that the ensemble of models can be placed as a group in the unit cell. You may also use phenix.enssembler to generate a single multi-model PDB file containing the entire ensemble. In this case you should specify a single overall RMS or identity for the ensemble.

Running from a parameters file

You can run phenix.automr from a parameters file. This is often convenient because you can generate a default one with:

```
phenix.automr --show_defaults > my_automr.eff
```

and then you can just edit this file to match your needs and run it with:

```
phenix.automr my_automr.eff
```

If you are searching for more than one ensemble, or if there is more than one component in the a.u., then use a parameters file and specify them like this (put all of this in a file like "my_mr.eff" and run it with "phenix.automr my_mr.eff"):

```
automr {
  ensemble {
    ensembleID = "mol1"
    copies_to_find = 1
    coords = mol1.pdb
    identity = None
    RMS = "0.85"
  }
}
```

```

ensemble {
  ensembleID = "mol2"
  copies_to_find = 1
  coords = mol2.pdb
  identity = None
  RMS = "0.90"
}
component {
  seq_file = "seq1.dat"
  component_type = *protein nucleic_acid
  mass = None
  component_copies = 1
}
component {
  seq_file = "seq2.dat"
  component_type = *protein nucleic_acid
  mass = None
  component_copies = 1
}
}

```

Specifying which columns of data to use from input data files

If one or more of your data files has column names that the Wizard cannot identify automatically, you can specify them yourself. You will need to provide one column "name" for each expected column of data, with "None" for anything that is missing.

For example, if your data file data.mtz has columns F SIGF then you might specify

```

data=data.mtz
input_label_string="F SIGF"

```

You can find out all the possible label strings in a data file that you might use by typing:

```

phenix.autosol display_labels=data.mtz # display all labels for data.mtz

```

You can specify many more parameters as well. See the list of keywords, defaults and descriptions at the end of this page and also general information about running Wizards at Using the PHENIX Wizards for how to do this. Some of the most common parameters are:

```

data=w1.sca # data file
model=coords.pdb # starting model
seq_file=seq.dat # sequence file

```

Examples

Standard AutoMR run with coords.pdb native.sca

Run AutoMR using coords.pdb as search model, native.sca as data, assume RMS between coords.pdb and true model is about 0.85 Å, the sequence of true model is seq.dat and there is 1 copy in the asymmetric unit:

```

phenix.automr coords.pdb native.sca RMS=0.85 seq.dat copies=1 \
  n_cycle_rebuild_max=2 n_cycle_build_max=2

```

Specifying data columns

Run AutoMR as above, but specify the data columns explicitly:

```

phenix.automr coords.pdb RMS=0.85 seq.dat copies=1 \
  data=data.mtz input_label_string="F SIGF" \
  n_cycle_rebuild_max=2 n_cycle_build_max=2

```

Note that the data columns are specified by a string that includes both F and SIGF : "F SIGF". The string must match some set of data labels that can be extracted automatically from your data file. You can find the possible values of this string as described above with

```
phenix.automr display_labels=data.mtz
```

Specifying a refinement file for AutoBuild

Run AutoMR as above, but specify a refinement file that is different from the file used for the MR search:

```
phenix.automr coords.pdb RMS=0.85 seq.dat copies=1 \
data=data.mtz input_label_string="F SIGF" \
input_refinement_file=refinement.mtz \
input_refinement_labels="FP SIGFP FreeR_flag" \
n_cycle_rebuild_max=2 n_cycle_build_max=2
```

Note that the commands `input_refinement_file` and `input_refinement_labels` are in the scope "autobuild_variables" . These commands and others with this prefix are passed on to AutoBuild.

AutoMR searching for 2 components

Run AutoMR on a structure with 2 components. Define the components of the asymmetric unit with sequence files (beta.seq and blip.seq) and number of copies of each component (1). Define the search models with PDB files and estimated RMS from true structures. This is all done by creating a parameters file with all the control information in it. Put all of this in a file like "my_mr.eff" and run it with "phenix.automr my_mr.eff":

```
automr {
  data = "w1.sca"
  build = False
  ensemble {
    ensembleID = "mol1"
    copies_to_find = 1
    coords = mol1.pdb
    identity = None
    RMS = "0.85"
  }
  ensemble {
    ensembleID = "mol2"
    copies_to_find = 1
    coords = mol2.pdb
    identity = None
    RMS = "0.90"
  }
  component {
    seq_file = "seq1.dat"
    component_type = *protein nucleic_acid
    mass = None
    component_copies = 1
  }
  component {
    seq_file = "seq2.dat"
    component_type = *protein nucleic_acid
    mass = None
    component_copies = 1
  }
}
```

Specifying molecular masses of 2 components

Run AutoMR as in the previous example, except specify the components of the asymmetric unit with molecular masses (30000 and 20000), and define the search models with PDB files and percent sequence

identity with the true structures (50% and 60%). This is again all done by creating a parameters file with all the control information in it. Put all of this in a file like "my_mr.eff" and run it with "phenix.automr my_mr.eff":

```
automr {
  data = "w1.sca"
  seq_file = seq.dat
  ensemble {
    ensembleID = "mol1"
    copies_to_find = 1
    coords = mol1.pdb
    identity = 50
  }
  ensemble {
    ensembleID = "mol2"
    copies_to_find = 1
    coords = mol2.pdb
    identity = 60
  }
  component {
    component_type = *protein nucleic_acid
    mass = 30000
    component_copies = 1
  }
  component {
    component_type = *protein nucleic_acid
    mass = 40000
    component_copies = 1
  }
  autobuild_variables{
    n_cycle_rebuild_max = 1
  }
}
```

AutoMR searching for 2 components, but specifying the orientation of one of them

Run AutoMR on a structure with 2 components. Define the components of the asymmetric unit with sequence files (beta.seq and blip.seq) and number of copies of each component (1). Define the search models with PDB files and estimated RMS from true structures. Define the orientation and position of one component. Define the number of copies to find for each component (0 for beta, which is fixed, 1 for blip). This is again all done by creating a parameters file with all the control information in it. Put all of this in a file like "my_mr.eff" and run it with "phenix.automr my_mr.eff":

```
automr {
  data = "w1.sca"
  seq_file = seq.dat
  ensemble {
    ensembleID = "mol1"
    copies_to_find = 1
    coords = mol1.pdb
    identity = 50
  }
  ensemble {
    ensembleID = "mol2"
    copies_to_find = 0
    coords = mol2.pdb
    identity = 60
  }
  component {
    component_type = *protein nucleic_acid
    mass = 30000
    component_copies = 1
  }
  component {
    component_type = *protein nucleic_acid
```

```

mass = 40000
component_copies = 1
}
autobuild_variables{
n_cycle_rebuild_max = 1
}
fixed_ensembles {
fixed_ensembleID_list="mol2"
fixed_euler_list = 199.84 41.535 184.15
fixed_frac_list = -0.49736 -0.15895 -0.28067
}
}

```

Note: you have to define an ensemble for the fixed molecule (mol2 in this example) and that you search for 0 copies of this molecule.

Possible Problems

Specific limitations and problems

The AutoBuild Wizard can build PROTEIN, RNA, or DNA, but it can only build one at a time. If your MR model contains more than one type of chain, then you will need to run AutoBuild separately from AutoMR and when you run AutoBuild, specify one of them with `input_lig_file_list` and the type of chain to build with `chain_type`:

```

input_lig_file_list=ProteinPartofMRmodel.pdb
chain_type=DNA

```

- The keywords "cell" and "sg" have been replaced with "unit_cell" and "space_group" to make the keywords the same as in other phenix applications.
- The syntax for searches with more than one ensemble and more than one component have changed in PHENIX version 1.4.
- If you use an ensemble as a search model, the output structure will contain just the first member of the ensemble, so you may wish to put the member that is likely to be the most similar to the true structure as the first one in your ensemble.
- AutoMR no longer is able to pass arbitrary commands to AutoBuild (this was discontinued to allow for a fixed set of inputs for AutoBuild).
- The AutoMR Wizard can take most settings of most space groups, however it can only use the hexagonal setting of rhombohedral space groups (eg., #146 R3:H or #155 R32:H), and it cannot use space groups 114-119 (not found in macromolecular crystallography) even in the standard setting due to difficulties with the use of `asuset` in the version of `ccp4` libraries used in PHENIX for these settings and space groups.

Literature

Phaser crystallographic software. A.J. McCoy, R.W. Grosse-Kunstleve, P.D. Adams, M.D. Winn, L.C. Storoni, and R.J. Read. *J Appl Crystallogr* 40, 658-674 (2007).

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Likelihood-enhanced fast translation functions. A.J. McCoy, R.W. Grosse-Kunstleve, L.C. Storoni, and R.J. Read. *Acta Crystallogr D Biol Crystallogr* 61, 458-64 (2005).

- Likelihood-enhanced fast translation functions. A.J. McCoy, R.W. Grosse-Kunstleve, L.C. Storoni, and R.J. Read. *Acta Crystallogr D Biol Crystallogr* 61, 458-64 (2005).

Carbohydrate Primer

carbo_primer.html

Introduction

A carbohydrate is an organic compound with the empirical formula $C_m(H_2O)_n$ where m could be different from n . Synonym of saccharide. Saccharides have several characteristics that differentiate them. These include placement of carbonyl group length of the carbon chain sequence of chiral centres handedness of the molecule linear or cyclic

- placement of carbonyl group
- length of the carbon chain
- sequence of chiral centres
- handedness of the molecule
- linear or cyclic

Cyclic saccharides are classified as alpha or beta isomers based on the position of the OH on the anomeric carbon relative to the large group on the ring.

Protein Chemistry

If your protein was produced in an insect or yeast cell line, you should only have NAG and MAN sugars, otherwise you should look at both types of PDB files to see which fits the density better (if you have any density past the main, five sugar core).

MAN

Mannose is a common saccharide in protein chemistry. There are two linear isomers and two cyclic forms of each. The PDB code for each is different because of the different restraints required for each. A leading form of error in protein models with carbohydrates is the use of the three-letter code, MAN, which is for the alpha-D-mannose when that vast majority of units in saccharides are beta-D-mannose or BMA.

NAG

NAG represents the beta form (the most common) and NDG is the alpha form. NAG is interesting because it's a saccharide with a substitute attached to the C2 carbon.

Polysaccharides

Saccharides polymerise into tree like structures that covalently bind to proteins. The links between are generally between the anomeric carbon and an oxygen on the "preceding" units moving from the protein outwards on the chain. Consider the disaccharide, lactose.

Note that the left saccharide is linked with its anomeric carbon via an oxygen to the C4 carbon. The left unit is β -D-galactose and the right is glucose. The link is known as β -(1-4) because the "attaching" unit is β and the it links C1 to C4.

Protein links

The most common link to proteins is the n-link which attaches to ASN in the ASN-X-SER/THR motif.

Using the Phenix chatbot assistant

chatbot.html

Authors

- Nigel Moriarty, Peter Zwart, Thomas C Terwilliger

Purpose

The Phenix chatbot is designed to guide Phenix users through the steps in structure determination, to provide help with running Phenix programs and interpreting their output, to help with debugging problems, and as a reference for the theory behind Phenix tools. The chatbot can be accessed in the Phenix GUI with the Chat button on the toolbar.

How the chatbot works

The Phenix chatbot is based on Google NotebookLM. The notebook was supplied with over 600 pages of Phenix documentation, all the Phenix newsletters, most of the Phenix video tutorials, and many of the Phenix publications.

The chatbot can be started with this chat link. You have to log into a Google account (any Google account should work). This creates a new clean copy of the notebook.

You can ask any questions about how to use Phenix and how to interpret results.

This type of AI does not save or learn from your questions. If you close the chat or hit Refresh, it will be in the same state as it started in. The AI does answer in the context of previous questions or requests, so if you ask it to give you instructions suitable for the GUI, it will try to do that throughout the session.

What kind of questions you can ask the chatbot

You can ask the chatbot almost anything, from big-picture questions to details.

For example:

"How do I solve my structure?"

"I have an mtz file and a sequence file. What do I do next?"

"What is the difference between an mtz file and an mrc file?"

"Can I use AlphaFold with X-ray data?"

"What inputs do I need for autosol?"

"How do I run refinement?"

"Please create an atom selection string that will select all the main-chain atoms of the nucleic acid residues in chain A"

"Please tell me how to run autobuild on the command-line"

"How do I know if my autobuild run worked?"

"How do I tell if I have a good model? Is an R-value of 0.25 ok?"

"What might prevent me from solving my structure by sad?"

Using the chatbot in parallel with other AI tools

A great way to get the most help with Phenix is to use the Phenix chatbot alongside other AI tools. For example if you have a log file from a Phenix run in the GUI and you have enabled AI analysis of log files, there will usually be a button next to the log file button labelled "AI analysis". If you analyze your results with this tool you can then follow up by asking more questions in the chatbot. You can even paste some of the output from the AI analysis into the chatbot and ask the chatbot to help you interpret the results.

You can also go to Perplexity and upload your log file with the paperclip icon. Then you can ask "Is this a good result?" and it will analyze your log file and summarize what it says. You can even ask it to find other log files from the web that are similar. Once you have uploaded your log file, you can ask the AI agent, "What should I do next?" to get ideas about next steps.

Limitations of the chatbot

The chatbot is limited to the sources that it is supplied with, so it only knows about Phenix, and it only knows what is in the documentation, the videos and newsletters, and the papers we have supplied.

AI chatbots like this one can also just make mistakes and give incorrect answers. This does not seem to happen too often with this chatbot, but you need to always be on alert when using it. Use the chatbot as a helper, don't expect it to always be right.

If a detail is missing in the documentation, the chatbot will not know it. For example, the current version of the chatbot does not know that for the `phenix.map_correlations` tool you must supply a resolution if you are comparing a model and a map because that was not stated in the documentation.

While the chatbot can give generally-correct overviews of how to solve a structure, sometimes it can get some aspects mixed up. For example, it might say that density modification improves resolution slightly. While that may be true for cryoEM, density modification does not change the resolution in X-ray crystallography, it only improves the phase information.

Similarly, when the atom selection example above is given to the chatbot, it may give an answer like: "backbone and chain A", instead of "backbone and (not protein) and chain A". Further, the chatbot may suggest testing atom selections with `phenix.pdb_atom_selection`, but can miss the detail that `phenix.pdb_atom_selection` does not recognize the "backbone" smart selection string.

Importing structures from the PDB

cif_as_mtz.html

Unlike most other reflection formats, the CIF files used by the PDB to store experimental data are not supported for direct input to PHENIX. `phenix.cif_as_mtz` is a simple utility for importing these files, and can automatically identify and fix several errors common in the PDB, such as incorrectly labeled intensities in place of amplitudes. Unfortunately, many structures fail to parse properly, so this tool is not guaranteed to work in all cases.

Using the PHENIX GUI

Open the main GUI and click "Import CIF structure factors" under the "Reflection tools" category. Only the CIF file is required, but a PDB file will often be necessary to obtain the correct symmetry information. Click "Use model to guess data type" if you want PHENIX to do a sanity check on the labeled amplitudes (which may not really be amplitudes!).

If you know the PDB ID but haven't downloaded the necessary files yet, you can simply enter the ID into the box in the upper right corner and click "Run", and the PDB and CIF files will be fetched automatically from the RCSB web server.

Command-line use

The output of `phenix.cif_as_mtz` without arguments is show below:

```
Usage: phenix.cif_as_mtz [reflection_cif_file] [options]

Options:
-h, --help show this help message and exit
--unit-cell=10,10,20,90,90,120|FILENAME
External unit cell parameters
--space-group=P212121|FILENAME
External space group symbol
--symmetry=FILENAME External file with symmetry information
--output-file-name=OUTPUT_FILE_NAME
Output mtz file name.
--use-model=USE_MODEL
Use PDB model to make better guess about reflection
data type.
--wavelength-id=WAVELENGTH_ID
Extract data set with given wavelength_id.
--crystal-id=CRYSTAL_ID
Extract data set with given crystal_id.
--show-details-if-error
Show data details for some errors.
--show-log Show some output.
--merge Merge non-unique data (preserves anomalous pairs)
--eliminate_sys_absent
Remove systematic absences
--map_to_asu Map HKL indices to canonical asymmetric unit
--incompatible_flags_to_work_set
When merging place reflections with incompatible
flags into the working set.
```

```
Example: phenix.cif_as_mtz r109ksf.ent --symmetry=pdb109k.ent
```

The `--symmetry` argument is frequently required due to missing symmetry information; in these cases you should use the equivalent PDB file. Most of the other options are usually not necessary, but `--use-model` is

often helpful to get the data type right.

As a shortcut to download the files from the RCSB (or mirror) website, PHENIX includes a tool for fetching data for a given PDB code:

```
phenix.fetch_pdb 2hr0 phenix.fetch_pdb -x 2hr0
```

The argument "-x" specifies that the structure factors should be downloaded instead of the PDB file. (This will only work if the authors deposited their data.)

Output

Assuming that the data were successfully extracted, the program will create an MTZ file with columns FOBS, SIGFOBS (these will be treated as group by most applications in PHENIX), and R-free-flags. In many cases there will be fewer reflections in the output than data lines in the CIF file; this is due to the presence of non-numeric or otherwise unintelligible values.

List of references related to Phenix

[citations.html](#)

Current primary Phenix citation:

Macromolecular structure determination using X-rays, neutrons and electrons: recent developments in Phenix. D. Liebschner, P.V. Afonine, M.L. Baker, G. Bunkóczi, V.B. Chen, T.I. Croll, B. Hintze, L.W. Hung, S. Jain, A.J. McCoy, N.W. Moriarty, R.D. Oeffner, B.K. Poon, M.G. Prisant, R.J. Read, J.S. Richardson, D.C. Richardson, M.D. Sammito, O.V. Sobolev, D.H. Stockwell, T.C. Terwilliger, A.G. Urzhumtsev, L.L. Videau, C.J. Williams, and P.D. Adams. *Acta Cryst. D75*, 861-877 (2019).

- Macromolecular structure determination using X-rays, neutrons and electrons: recent developments in Phenix. D. Liebschner, P.V. Afonine, M.L. Baker, G. Bunkóczi, V.B. Chen, T.I. Croll, B. Hintze, L.W. Hung, S. Jain, A.J. McCoy, N.W. Moriarty, R.D. Oeffner, B.K. Poon, M.G. Prisant, R.J. Read, J.S. Richardson, D.C. Richardson, M.D. Sammito, O.V. Sobolev, D.H. Stockwell, T.C. Terwilliger, A.G. Urzhumtsev, L.L. Videau, C.J. Williams, and P.D. Adams. *Acta Cryst. D75*, 861-877 (2019).

Other articles

The articles below include both papers produced by members of the Phenix collaboration, and works by external groups relating to methods used in Phenix (and elsewhere).

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Putting the best parts of several maps together with `combine_focused_maps`

`combine_focused_maps.html`

Author(s)

- `combine_focused_maps`: Tom Terwilliger

Purpose

The routine `combine_focused_maps` uses rigid-body refinement to place a model in several maps. It then uses the map-model correlation to identify which are the best parts of each map and the relationships among the models in the different maps to superpose the best parts of each map and create a composite map.

Usage

How `combine_focused_maps` works:

Combine-focused-maps uses the fit of chains in a model to each map to identify the correspondence between the maps. Using this approach it is not necessary that the maps are all superimposed or even that they have the same gridding or size.

If the maps all do superimpose (approximately), you can supply a single model. The model will be refined separately against each map with rigid-body refinement. The placements of each chain in each map will then be used to identify the transformations needed to superimpose the parts of each map corresponding to each chain.

If the maps are quite different, then you will need to supply one model for each map, where you have already placed each chain so that it superimposes with the appropriate part of each map. The models should all be the same except that they have been rigid-body-refined against the various maps. You can use the phenix tool `phenix.dock_in_map` to place the chains in each map if you have not done it in some other way.

Once a set of models has been set up to match the maps, the map-model correlation in the region of each chain in each model is calculated. This correlation is calculated in a special way: the B-values of all the atoms in the model are set to zero before map-model correlation is evaluated. This step is important in order to make sure that a perfect high-resolution map always will score better than a perfect low-resolution map. (If B-values are variable then a perfect high-resolution map and a perfect low-resolution map will both have correlations of 1, but with a B-value of zero the high-resolution map will have a higher correlation.)

For each chain in the model, the map-model correlations for each map are then used to identify the weighting on that map.

Alternatively, you can specify what parts of the model are to be grouped together for identifying which map to use in what region. You specify any number of `region_selection` selections (in Phenix atom selection syntax, like `(chain A and resseq 250:360)`). Each one is used to define a region in space and within that region the maps are weighted according to their correlations to the atoms selected with that selection.

An empirical weighting scheme is used. The relative weights of two maps (for this one chain or region) with a difference in `map_model_cc` of `delta_cc` are given by:

$$\exp(-\text{delta_cc}/\text{delta_cc_norm})$$

where `delta_cc_norm` is typically about 0.05. This means a map that has a map-model cc 0.1 less than another map gets a weight of about 1/10 of the better-fitting map. Using this weighting scheme and transformations calculated from the positions of this chain after rigid-body refinement against each map, a weighted average map is created for each chain in the model. These maps are all superimposed on the reference map and masked around the chain that each is to represent.

Finally all the weighted average maps are combined to form a composite map.

Normalization and scaling of input maps

With the default value of the keyword `normalize` (True), maps will be normalized by subtracting their mean value and scaling to yield an rms value of 1.0.

If values of `map_scale` are supplied (one per map; cannot duplicate values so use 0.9999, 1.000, and 1.0001 to if you want the same value of 1 for three maps), the maps will be scaled by these values after normalization.

If you want to just scale and not normalize, set `normalize` to False. (This was the behaviour before phenix version 5705)

Additional output files

If you specify `get_contribution_maps=True` (this is the default), at the end of the procedure you will get one map written out for each input map file, called something like "contribution_map_1.ccp4". This map has values from 0 to 1 showing the contribution of this input map file to the final map. You can use it to color your final composite map, showing how much of the information at each point came from this input map file.

If you specify `get_cc_map=True` (also default) you will get a map file called `cc_map.ccp4` that shows the map correlation between the model and the final composite map. The correlation map will be calculated over whatever regions in the model were used in forming the composite map (i.e. if you used individual chains, then the correlations will be calculated over each chain).

Examples

Standard run of `combine_focused_maps`:

Running `combine_focused_maps` is easy. From the command-line you can type:

```
phenix.combine_focused_maps reference_map.map focused_map_1.map \  
    focused_map_2.map model.pdb resolution=4
```

where `reference_map` is the map (CCP4, mrc or other related format) that will be used as the template to superimpose all other maps will be superimposed, and `focused_map_1.map` and `focused_map_2.map` are maps that are focused on some part of the map (and the remainder of those maps may be poor). The model will be rigid-body refined against all three maps (keeping chains fixed as rigid bodies). The best parts of each map will be selected and combined to create a composite map.

Possible Problems

Specific limitations and problems:

If the density in some of your maps is rather poor, the rigid-body refinement step may not work well. One thing to try in this case is to refine with `rigid_body_refinement_single_unit=True`. This will tend to hold everything together (but it won't give you an adjustment of individual chain positions). Another option is to refine (with rigid-body refinement only) the models before using `combine_focused_maps`. That way you have full control over the refinement process. Then in `combine_focused_maps` use `rigid_body_refinement=False`.

You cannot use `region_selection` with `use_model_symmetry`.

Literature

Additional information

If you supply a model for each map, the models are used to define what parts of each map are included in the analysis. If the model for your target map has chains ABC and one focused map has just chain A, then only chain A from the focused map will be transferred to the target map. If the model for the focused map has chains ABC then weighted versions of the map in the vicinities of ABC will be included.

If you have a model with symmetry (say, chains A B C D are all the same) and just one focused map (say, focused on A), then you can supply your target map, target model with chains ABCD, an one focused map and a model for the focused map that contains just chain A. Then you add the keyword `"use_model_symmetry=True"` and the focused map from chain A is applied to A B C and D in the target map.

If you supply half-maps corresponding to each full map, the half-maps will be processed in the same way as the full map, yielding one pair of half-maps at the end of the procedure. You can use these half-maps to estimate the local resolution of the final composite map (using `phenix.local_resolution`) or for other procedures requiring half maps. Note that the half maps could have discontinuities so they may not be appropriate for all uses of half maps.

List of all available keywords

`job_title = None` Job title in PHENIX GUI, not used on command line
`input_filesmap_file = None` Files with CCP4-style maps. May have origin in any location. All maps will be superimposed on the first map. Normally it is assumed that all the map files are approximately superimposed so that a single `model_file` will approximately match all map files. If the map files do not superimpose, supply one model file per map file, where the model files differ only by rigid-body refinement.
`map_scale = None` Scale to apply to map before use. Must be input in same order as `map_files`. If two values are the same, you need to make them slightly different (like 1 -> 1.000001). Applied after `normalize_maps`.
`model_file = None` Model placed in first map to be used to align maps. Each chain in the model will be used to mark an independent region in each map. Optionally one model can be supplied to match each map. Normally it is assumed that all the map files are approximately superimposed so that a single `model_file` will approximately match all map files. If the map files do not superimpose, supply one model file per map file, where the model files differ only by rigid-body refinement. NOTE: only the parts of maps marked by a chain in the corresponding model file will be included in averaging.
`keep_ligands = False` Allow ligands in density analysis. Note: it is possible that refinement will fail with some ligands. If so, please refine your model against each map individually first, supply those models, and turn off rigid-body refinement.
`half_map_1_file = None` Half-map 1 list matching `map_file` list. If you supply a set of `half_map_1` files and `half_map_2` files, then whatever is done to the files in `map_list` will be applied to both the `half_map_1` files and `half_map_2` files.
`half_map_2_file = None` Half-map 2 list matching `map_file` list. If you supply a set of `half_map_1` files and `half_map_2` files, then whatever is done to the files in `map_list` will be applied to both the `half_map_1` files and `half_map_2` files.
`region_selection = None` Model selection string defining one region to be considered. This is an alternative approach to defining regions (default is to use

each chain as a region). Any number of regions can be supplied. `output_files` `composite_map_file` = `composite_map.ccp4` Output map file with composite map `output_scale_factor` = `None` Scale factor to be applied to output map just before writing. Normally the output map will have a mean of zero and SD of 1. This may lead to the maximum in the map being much greater than 1. You can adjust the output SD with this scale factor. `refined_model_file` = `composite_map.pdb` Output model file after rigid-body refinement `composite_half_map_1_file` = `composite_half_map_1.ccp4` Output map file with composite half-map 1 `composite_half_map_2_file` = `composite_half_map_2.ccp4` Output map file with composite half-map 2 `crystal_info` `resolution` = `None` High-resolution limit for main search. This can be lower resolution than the data. The search is quicker at lower resolution. `short_caption` = `High-resolution limit` `scattering_table` = `n_gaussian` `wk1995` `it1992` *electron neutron Choice of scattering table for structure factor calculations. Standard for X-ray is `n_gaussian`, for cryoEM is `electron`. `superpose_model_symmetry` = `False` If `use_model_symmetry` = `True` then find all the chains in the target model (the first model supplied) that have the same sequence (or use `symmetry_chains` to ID these chains). Any chains in other models that match one of these chains will be matched to all of them. Using this keyword a single focused map can be applied to all corresponding chains in the target map/model. Note: cannot be combined with `region_selection`. `symmetry_chains` = `None` You can specify the chains that are to be considered equivalent. See `use_model_symmetry` for description. Note: cannot be combined with `region_selection`. `rigid_body_refinement` = `True` Run rigid-body refinement on input model vs each map `rigid_body_refinement_single_unit` = `False` Run rigid-bo...

- `job_title` = `None` Job title in PHENIX GUI, not used on command line

- `input_files` `map_file` = `None` Files with CCP4-style maps. May have origin in any location. All maps will be superimposed on the first map. Normally it is assumed that all the map files are approximately superimposed so that a single `model_file` will approximately match all map files. If the map files do not superimpose, supply one model file per map file, where the model files differ only by rigid-body refinement. `map_scale` = `None` Scale to apply to map before use. Must be input in same order as `map_files`. If two values are the same, you need to make them slightly different (like 1 -> 1.000001). Applied after `normalize_maps`. `model_file` = `None` Model placed in first map to be used to align maps. Each chain in the model will be used to mark an independent region in each map. Optionally one model can be supplied to match each map. Normally it is assumed that all the map files are approximately superimposed so that a single `model_file` will approximately match all map files. If the map files do not superimpose, supply one model file per map file, where the model files differ only by rigid-body refinement. NOTE: only the parts of maps marked by a chain in the corresponding model file will be included in averaging. `keep_ligands` = `False` Allow ligands in density analysis. Note: it is possible that refinement will fail with some ligands. If so, please refine your model against each map individually first, supply those models, and turn off rigid-body refinement. `half_map_1_file` = `None` Half-map 1 list matching `map_file` list. If you supply a set of `half_map_1` files and `half_map_2` files, then whatever is done to the files in `map_list` will be applied to both the `half_map_1` files and `half_map_2` files. `half_map_2_file` = `None` Half-map 2 list matching `map_file` list. If you supply a set of `half_map_1` files and `half_map_2` files, then whatever is done to the files in `map_list` will be applied to both the `half_map_1` files and `half_map_2` files. `region_selection` = `None` Model selection string defining one region to be considered. This is an alternative approach to defining regions (default is to use each chain as a region). Any number of regions can be supplied.

- `map_file` = `None` Files with CCP4-style maps. May have origin in any location. All maps will be superimposed on the first map. Normally it is assumed that all the map files are approximately superimposed so that a single `model_file` will approximately match all map files. If the map files do not superimpose, supply one model file per map file, where the model files differ only by rigid-body refinement.

- `map_scale = None` Scale to apply to map before use. Must be input in same order as `map_files`. If two values are the same, you need to make them slightly different (like 1 -> 1.000001). Applied after `normalize_maps`.
- `model_file = None` Model placed in first map to be used to align maps. Each chain in the model will be used to mark an independent region in each map. Optionally one model can be supplied to match each map. Normally it is assumed that all the map files are approximately superimposed so that a single `model_file` will approximately match all map files. If the map files do not superimpose, supply one model file per map file, where the model files differ only by rigid-body refinement. NOTE: only the parts of maps marked by a chain in the corresponding model file will be included in averaging.
- `keep_ligands = False` Allow ligands in density analysis. Note: it is possible that refinement will fail with some ligands. If so, please refine your model against each map individually first, supply those models, and turn off rigid-body refinement.
- `half_map_1_file = None` Half-map 1 list matching `map_file` list. If you supply a set of `half_map_1` files and `half_map_2` files, then whatever is done to the files in `map_list` will be applied to both the `half_map_1` files and `half_map_2` files.
- `half_map_2_file = None` Half-map 2 list matching `map_file` list. If you supply a set of `half_map_1` files and `half_map_2` files, then whatever is done to the files in `map_list` will be applied to both the `half_map_1` files and `half_map_2` files.
- `region_selection = None` Model selection string defining one region to be considered. This is an alternative approach to defining regions (default is to use each chain as a region). Any number of regions can be supplied.
- `output_filescomposite_map_file = composite_map.ccp4` Output map file with composite map
`output_scale_factor = None` Scale factor to be applied to output map just before writing. Normally the output map will have a mean of zero and SD of 1. This may lead to the maximum in the map being much greater than 1. You can adjust the output SD with this scale factor.
`refined_model_file = composite_map.pdb` Output model file after rigid-body refinement
`composite_half_map_1_file = composite_half_map_1.ccp4` Output map file with composite half-map 1
`composite_half_map_2_file = composite_half_map_2.ccp4` Output map file with composite half-map 2
- `composite_map_file = composite_map.ccp4` Output map file with composite map
- `output_scale_factor = None` Scale factor to be applied to output map just before writing. Normally the output map will have a mean of zero and SD of 1. This may lead to the maximum in the map being much greater than 1. You can adjust the output SD with this scale factor.
- `refined_model_file = composite_map.pdb` Output model file after rigid-body refinement
- `composite_half_map_1_file = composite_half_map_1.ccp4` Output map file with composite half-map 1
- `composite_half_map_2_file = composite_half_map_2.ccp4` Output map file with composite half-map 2
- `crystal_inforesolution = None` High-resolution limit for main search. This can be lower resolution than the data. The search is quicker at lower resolution. `.short_caption = High-resolution limit`
`scattering_table = n_gaussian` Choice of scattering table for structure factor calculations. Standard for X-ray is `n_gaussian`, for cryoEM is `electron`.
- `resolution = None` High-resolution limit for main search. This can be lower resolution than the data. The search is quicker at lower resolution. `.short_caption = High-resolution limit`
- `scattering_table = n_gaussian` Choice of scattering table for structure factor calculations. Standard for X-ray is `n_gaussian`, for cryoEM is `electron`.

- `superposeuse_model_symmetry = False` If `use_model_symmetry = True` then find all the chains in the target model (the first model supplied) that have the same sequence (or use `symmetry_chains` to ID these chains). Any chains in other models that match one of these chains will be matched to all of them. Using this keyword a single focused map can be applied to all corresponding chains in the target map/model. Note: cannot be combined with `region_selection.symmetry_chains = None` You can specify the chains that are to be considered equivalent. See `use_model_symmetry` for description. Note: cannot be combined with `region_selection.rigid_body_refinement = True` Run rigid-body refinement on input model vs each map `rigid_body_refinement_single_unit = False` Run rigid-body refinement with just one unit (do not break up into chains) `remove_water = True` Remove waters from input files `local_weighting = False` Use local correlation to weight individual regions of maps. Size of local region determined by `local_residues`. Alternative is one weight for each chain of each map. `local_residues = 10` Number of residues to use in local correlations. `delta_cc_norm = 0.05` When scaling, weight a map or part of a map that has map cc of `delta_cc` less than another map by $\exp(-\text{delta_cc}/\text{delta_cc_norm})$ less than the other map. `mask_atoms_atom_radius = None` Radius for masking atoms when transforming map. Default is $\max(3, \text{resolution})$. `mask_smoothing_radius = None` Radius for smoothing mask. Default is 3 times the resolution `normalize_maps = True` Normalize maps (mean zero, sd 1) before use. Applied before `map_scale` option. `get_contribution_maps = True` Create map-like files showing contribution of each map `get_cc_map = True` Create map-like file showing map-model correlation at end `use_first_map_as_template_only = False` Use first map as template only, do not take any density from it

- `use_model_symmetry = False` If `use_model_symmetry = True` then find all the chains in the target model (the first model supplied) that have the same sequence (or use `symmetry_chains` to ID these chains). Any chains in other models that match one of these chains will be matched to all of them. Using this keyword a single focused map can be applied to all corresponding chains in the target map/model. Note: cannot be combined with `region_selection`.

- `symmetry_chains = None` You can specify the chains that are to be considered equivalent. See `use_model_symmetry` for description. Note: cannot be combined with `region_selection`.

- `rigid_body_refinement = True` Run rigid-body refinement on input model vs each map

- `rigid_body_refinement_single_unit = False` Run rigid-body refinement with just one unit (do not break up into chains)

- `remove_water = True` Remove waters from input files

- `local_weighting = False` Use local correlation to weight individual regions of maps. Size of local region determined by `local_residues`. Alternative is one weight for each chain of each map.

- `local_residues = 10` Number of residues to use in local correlations.

- `delta_cc_norm = 0.05` When scaling, weight a map or part of a map that has map cc of `delta_cc` less than another map by $\exp(-\text{delta_cc}/\text{delta_cc_norm})$ less than the other map.

- `mask_atoms_atom_radius = None` Radius for masking atoms when transforming map. Default is $\max(3, \text{resolution})$.

- `mask_smoothing_radius = None` Radius for smoothing mask. Default is 3 times the resolution

- `normalize_maps = True` Normalize maps (mean zero, sd 1) before use. Applied before `map_scale` option.

- `get_contribution_maps = True` Create map-like files showing contribution of each map

- `get_cc_map = True` Create map-like file showing map-model correlation at end

- `use_first_map_as_template_only = False` Use first map as template only, do not take any density from it

- controlnproc = 1 Number of processors to use random_seed = 171731 Random seed verbose = False Verbose output
- nproc = 1 Number of processors to use
- random_seed = 171731 Random seed
- verbose = False Verbose output
- guiGUI-specific parameter required for output directory output_dir = None restraint_files = None
- output_dir = None
- restraint_files = None

Merging two models with combine_models

combine_models.html

Author(s)

- combine_models: Tom Terwilliger

Purpose

This routine takes a model and finds pieces of a second model file that improve its fit to density when they replace the corresponding pieces in the first model.

Usage

The main uses of phenix.combine_models are:

- Taking the best parts of 2 models and merging them to make a new model.
- Replacing a segment in one model with a corresponding segment from another.

How combine_models works:

The phenix.combine_models tool can combine the best parts of two or more models. You can use it in either of two different ways. One approach keeps segments in the models intact, and the second applies crossovers within segments.

In the first approach (`merge_by_segment_correlation=True`), each fragment in each model is scored based on correlation to the map. The scoring also includes a weight based on the square root of the length of the segment. It also includes weights based on whether a segment has been assigned to the sequence and the number of secondary structure hydrogen bonds in that segment. This approach can combine chains of different types as well. A final increment is added to the score for a segment if it is considered very likely to be correct with different thresholds for chains of different types.

Then the segments are picked in rank order to create a composite model. Any part of a segment that does not overlap an existing part of the composite model is kept. If symmetry is present, overlaps include that symmetry.

In the second approach, (`merge_by_segment_correlation=False`), a working model is created by taking all of the first model supplied and filling in any empty regions with fragments from other models. Then one by one, the segments in the working model are recombined with all other segments. To carry out recombination between two chains, residues that match in the two chains are identified. Then for each segment between matching pairs of residues, whichever chain has the higher correlation to the map is kept to create a composite model.

This second approach to combine_models starts with two input models. The first model is used as the default; if nothing can be found in the second model that is better than what is in the first model, then that part of the first model is kept. The second model is used as a template for improving the first model. Fragments of the second model are considered as alternatives for corresponding segments in the first model.

The fit of the models to density is used to decide which of a pair of fragments is best. In general, the correlation of model density with the map is used as the criterion. In cases where unequal numbers of residues are considered, then this correlation is weighted by the square root of the number of residues in each

case. During the optional `merge_second_model` step, the scoring is optionally based on correlation of density, or by default, based on density at the positions of the main-chain atoms in the model.

If the two input models are not in the same asymmetric unit of the crystal, then `combine_models` will move the pieces from the second model to the corresponding locations in the first model. In this way the final model has all its parts in the same place.

The second approach in `combine_models` has four main steps, each of which is optional:

- Selecting parts of the input models to consider, using atom selections. This step allows you to do things such as removing a segment from the first model, then crossing with the corresponding segment from the second model, effectively replacing that segment in a way that you specify.
- Creating as complete a starting model as possible (`merge_second_model=True`). This is done by cutting the second model into small pieces and reassembling the first model, considering these pieces along with all the segments in the first model as potential pieces of the reassembled model. The result is a new working model.
- Crossing the working model with the second input model, allowing only equal-length crossovers (`matching=True`). In this step, pairs of segments in the working model and the second input model that have overlapping residues are considered, one at a time. The local map correlation is calculated for each residue in each fragment. Then at each position, the coordinates from one fragment or the other are chosen, with the choice crossing over between one fragment and the other only at positions where the main-chain atoms in the residue are within `match_distance` (default 0.5 Å) of each other.
- Crossing the working model with the second input model, allowing only unequal-length crossovers (`non_matching=True`). This step is just like the previous one, except that only unequal-length crossovers are considered. If this method is included, then it may be necessary to reassign the sequence afterwards as the alignment may change
- Checking the sequence. If unequal-length crossovers are made, then the sequence alignment to the model may need to be changed. This step carries that out.

Output files from `combine_models`

`combine_models.pdb`: A PDB file with your combined model.

Examples

Standard run of `combine_models`:

Running `combine_models` is easy. From the command-line you can type:

```
phenix.combine_models pdb_in=first.pdb \
  second_pdb_in=second.pdb \
  seq.dat \
  map_coeffs.mtz
```

This will combine `first.pdb` and `second.pdb` based on fit to the map from `map_coeffs.mtz`, recheck the sequence alignment to `seq.dat`, and write out the resulting model.

Ranking all fragments and picking the best ones:

```
phenix.combine_models pdb_in=model.pdb seq.dat map_coeffs.mtz merge_by_segment_correlation=True
phenix.combine_models pdb_in=model.pdb
```

seq.dat map_coeffs.mtz merge_by_segment_correlation=True

This will score each segment (fragment), then work down from best to worst, keeping any part of any segment that does not overlap with a better-scoring segment.

Selecting pieces from the two models:

To take first.pdb and then see if residues A21-A30 of second.pdb can improve it, you can type:

```
phenix.combine_models pdb_in=first.pdb \  
  second_pdb_in=second.pdb \  
  seq.dat \  
  map_coeffs.mtz \  
  second_pdb_in_atom_selection="(chain A and resid 21:30)" \  

```

Replacing a specific segment:

To take first.pdb and then see if residues A21-A30 and B21-B30 can be improved by replacing them with residues C10-C20 and D10-D20 of second.pdb, you can tell combine_models to ignore residues A22-A29 and B22-B29 and to consider only residues C10-C20 and D10-D20 of second.pdb:

```
phenix.combine_models pdb_in=first.pdb \  
  second_pdb_in=second.pdb \  
  seq.dat \  
  map_coeffs.mtz \  
  pdb_in_atom_selection="(not ( (chain A or chain B) and resid 22:29) )" \  
  second_pdb_in_atom_selection="( (chain C or chain D) and resid 10:20)" \  

```

Crossing two models that have entirely matched residues:

If your first.pdb and second.pdb have exactly the same residues present, and just differ in coordinates, then you might want to preserve all the connectivity by skipping the merge_second_model step, and by skipping the non_matching crossover step, and by skipping the reassignment of sequence. You can type preserve_connectivity=True as a shortcut for this:

```
phenix.combine_models pdb_in=first.pdb \  
  second_pdb_in=second.pdb \  
  seq.dat \  
  map_coeffs.mtz \  
  preserve_connectivity=True
```

Possible Problems

If you are supplying a very big map, running combine_models can require a lot of memory. In this case, you may want to run map_box with your map and a model file that contains all your models. This will box the map around the region where your models are located and make a much smaller map that will require less memory.

Specific limitations and problems:

Literature

Additional information

List of all available keywords

input_files pdb_in = None Input starting PDB file. This model will be the default output model. Parts of the model that are better in the second model be replaced by the second model. pdb_in_atom_selection = None Any selection specified with pdb_in_atom_selection is applied to the pdb_in input model before using it. second_pdb_in = None Input second PDB file. Parts that are better in this model will replace corresponding parts of the first model. second_pdb_in_atom_selection = None Any selection specified with second_pdb_in_atom_selection is applied to the second_pdb_in input model before using it . If multiple second_pdb_in files are supplied, the same number of atom selections are expected. map_in = None CCP4 or MRC-style map file (alternative to mtz_in) mtz_in = None Input MTZ file with map coefficients map_coeff_labels = None If map coefficients cannot be identified automatically from your MTZ file, you can specify the label or labels for them. (Please separate labels with blank space, MTZ columns grouped together separated by commas with no blanks.) You can specify: map_coeff_labels (e.g., FWT,PHIFWT) amplitudes and phases (e.g., FP,SIGFP PHIB) or amplitudes, phases, weights (e.g., FP,SIGFP PHIB FOM) seq_file = None Sequence file for sequence alignment ncs_file = None Input NCS spec file for use with segment_and_split_map. output target_output_format = *None pdb mmCIF Desired output format (if possible). Choices are None (try to use input format), pdb, mmCIF. If output model does not fit in pdb format, mmCIF will be used. Default is pdb. output_files pdb_out = combine_models.pdb Output PDB file log = combine_models.log Output log file params_out = combine_models.params.eff Parameters file to rerun combine_models linkage_output_file = linkage_output.pkl Linkage dictionary output_directory = None Not used (output directory for passing to map_to_model) crystal_info chain_type = *PROTEIN DNA RNA You can specify whether to include protein, DNA, or RNA If you use merge_by_segment_correlation you can include multiple chain types and chain_type is ignored resolution = None High-resolution limit solvent_content = None Solvent fraction of the cell. If this is density cut out from a bigger cell, you can specify the fraction of the volume of this cell that is taken up by the macromolecule. Normally set automatically. Values go from 0 to 1. space_group = None Space group (normally read from the data file) unit_cell = None Unit Cell (normally read from the data file) map_inside_box = True Place centers of all chains inside (0,1). This is required during normal operation. use_sg_symmetry = None Not used directories output_dir = "" Output directory where files are to be written crossing_high_resolution = None Resolution used in map calculation match_distance = 1.5 Residue pairs must have rmsd of match_distance (Å) or lower to be crossed. A value between 0.5 Å and 1.5 Å is generally best. merge_by_segment_correlation = False Split all models into segments (breaking where residue numbers are not sequential), rank segments by correlation to map weighted by number of atoms, take highest-scoring non-overlapping segments. Followed by standard merge with all unused segments. keep_all_of_first_model = None Try to keep all of first model. Applies to merge_by_segment_correlation only. Does not work fully if trim_to_au is used. optimize_ncs = False Optimize NCS after merge_by_segment_correlation. NOTE: assumes that all NCS operators are allowed and may not place the resulting model in density if the NCS is not present in the density. trim_to_au = True Split up any chains that are in multiple NCS au weight_rms = 1 Weight on closest approach of fragments scc_cut = 0.90 Eliminate NCS copies with CC less than cc_cut of max weight_cc = 0.1 Weight on CC in optimizing NCS (typically 10) weight_rad_gyr = 0.1 Weight on radius of gyration in optimize_ncs n_substitutions = 3 Substitutions in optimization n_consider = 20 Substitutions in optimization applied to last n_consider fragments n_random_op...

- input_files pdb_in = None Input starting PDB file. This model will be the default output model. Parts of the model that are better in the second model be replaced by the second model. pdb_in_atom_selection = None Any selection specified with pdb_in_atom_selection is applied to the pdb_in input model before using it. second_pdb_in = None Input second PDB file. Parts that are better in this model will replace corresponding parts of the first model. second_pdb_in_atom_selection = None Any selection specified with second_pdb_in_atom_selection is applied to the second_pdb_in input model before using it . If multiple second_pdb_in files are supplied, the same number of atom selections are expected. map_in = None CCP4 or MRC-style map file (alternative to mtz_in) mtz_in = None Input MTZ file with map

coefficientsmap_coeff_labels = None If map coefficients cannot be identified automatically from your MTZ file, you can specify the label or labels for them. (Please separate labels with blank space, MTZ columns grouped together separated by commas with no blanks.) You can specify: map_coeff_labels (e.g., FWT,PHIFWT) amplitudes and phases (e.g., FP,SIGFP PHIB) or amplitudes, phases, weights (e.g., FP,SIGFP PHIB FOM)seq_file = None Sequence file for sequence alignment ncs_file = None Input NCS spec file for use with segment_and_split_map.

- pdb_in = None Input starting PDB file. This model will be the default output model. Parts of the model that are better in the second model be replaced by the second model.

- pdb_in_atom_selection = None Any selection specified with pdb_in_atom_selection is applied to the pdb_in input model before using it.

- second_pdb_in = None Input second PDB file. Parts that are better in this model will replace corresponding parts of the first model.

- second_pdb_in_atom_selection = None Any selection specified with second_pdb_in_atom_selection is applied to the second_pdb_in input model before using it . If multiple second_pdb_in files are supplied, the same number of atom selections are expected.

- map_in = None CCP4 or MRC-style map file (alternative to mtz_in)

- mtz_in = None Input MTZ file with map coefficients

- map_coeff_labels = None If map coefficients cannot be identified automatically from your MTZ file, you can specify the label or labels for them. (Please separate labels with blank space, MTZ columns grouped together separated by commas with no blanks.) You can specify: map_coeff_labels (e.g., FWT,PHIFWT) amplitudes and phases (e.g., FP,SIGFP PHIB) or amplitudes, phases, weights (e.g., FP,SIGFP PHIB FOM)

- seq_file = None Sequence file for sequence alignment

- ncs_file = None Input NCS spec file for use with segment_and_split_map.

- outputtarget_output_format = *None pdb mmCIF Desired output format (if possible). Choices are None (try to use input format), pdb, mmCIF. If output model does not fit in pdb format, mmCIF will be used. Default is pdb.

- target_output_format = *None pdb mmCIF Desired output format (if possible). Choices are None (try to use input format), pdb, mmCIF. If output model does not fit in pdb format, mmCIF will be used. Default is pdb.

- output_files pdb_out = combine_models.pdb Output PDB file log = combine_models.log Output log file params_out = combine_models_params.eff Parameters file to rerun

- combine_models linkage_output_file = linkage_output.pkl Linkage dictionary output_directory = None Not used (output directory for passing to map_to_model)

- pdb_out = combine_models.pdb Output PDB file

- log = combine_models.log Output log file

- params_out = combine_models_params.eff Parameters file to rerun combine_models

- linkage_output_file = linkage_output.pkl Linkage dictionary

- output_directory = None Not used (output directory for passing to map_to_model)

- crystal_info chain_type = *PROTEIN DNA RNA You can specify whether to include protein, DNA, or RNA If you use merge_by_segment_correlation you can include multiple chain types and chain_type is ignored resolution = None High-resolution limit solvent_content = None Solvent fraction of the cell. If this is density cut out from a bigger cell, you can specify the fraction of the volume of this cell that is taken up by

the macromolecule. Normally set automatically. Values go from 0 to 1. `space_group = None` Space group (normally read from the data file) `unit_cell = None` Unit Cell (normally read from the data file) `map_inside_box = True` Place centers of all chains inside (0,1). This is required during normal operation. `use_sg_symmetry = None` Not used

- `chain_type = *PROTEIN DNA RNA` You can specify whether to include protein, DNA, or RNA If you use `merge_by_segment_correlation` you can include multiple chain types and `chain_type` is ignored
- `resolution = None` High-resolution limit
- `solvent_content = None` Solvent fraction of the cell. If this is density cut out from a bigger cell, you can specify the fraction of the volume of this cell that is taken up by the macromolecule. Normally set automatically. Values go from 0 to 1.
- `space_group = None` Space group (normally read from the data file)
- `unit_cell = None` Unit Cell (normally read from the data file)
- `map_inside_box = True` Place centers of all chains inside (0,1). This is required during normal operation.
- `use_sg_symmetry = None` Not used
- `directoriesoutput_dir = ""` Output directory where files are to be written
- `output_dir = ""` Output directory where files are to be written
- `crossinghigh_resolution = None` Resolution used in map calculation `match_distance = 1.5` Residue pairs must have rmsd of `match_distance` (Å) or lower to be crossed. A value between 0.5 Å and 1.5 Å is generally best. `merge_by_segment_correlation = False` Split all models into segments (breaking where residue numbers are not sequential), rank segments by correlation to map weighted by number of atoms, take highest-scoring non-overlapping segments. Followed by standard merge with all unused segments. `keep_all_of_first_model = None` Try to keep all of first model. Applies to `merge_by_segment_correlation` only. Does not work fully if `trim_to_au` is used. `optimize_ncs = False` Optimize NCS after `merge_by_segment_correlation`. NOTE: assumes that all NCS operators are allowed and may not place the resulting model in density if the NCS is not present in the density. `trim_to_au = True` Split up any chains that are in multiple NCS `auweight_rms = 1` Weight on closest approach of fragments `cc_cut = 0.90` Eliminate NCS copies with CC less than `cc_cut` of `maxweight_cc = 0.1` Weight on CC in optimizing NCS (typically 10) `weight_rad_gyr = 0.1` Weight on radius of gyration in `optimize_ncsn_substitutions = 3` Substitutions in optimization `n_consider = 20` Substitutions in optimization applied to last `n_consider` fragments `sn_random_opt = 0` Try factor `max_optimize_tries = 500` Maximum optimization steps `overlap_tolerance = None` Minimum separation between N/C1'/P atoms in separate chains. Applies to `merge_by_segment_correlation` only. Default is 3 Å for C1'/P and 2 Å for N `ncs_overlap_tolerance_factor = 1.5` Overlap tolerance required for NCS-related chains. Typically larger than for chains in same NCS `au` to ensure they do not overlap `very_good_protein_score = 20.0` Score indicating in combination with `very_good_protein_cc` that an PROTEIN fragment is almost certainly correct `very_good_protein_cc = 0.75` CC indicating in combination with `very_good_protein_score` that an PROTEIN fragment is almost certainly correct `very_good_rna_score = 7.0` Score indicating in combination with `very_good_rna_cc` that an RNA fragment is almost certainly correct `very_good_rna_cc = 0.60` CC indicating in combination with `very_good_rna_score` that an RNA fragment is almost certainly correct `min_fractional = 0.05` Minimum fraction of a chain remaining after removing parts occupied by a better-scoring chain `min_fractional_multiple_chain_types = 0.35` Minimum fraction of a chain remaining after removing parts occupied by a better-scoring chain. Applies if multiple chain types are present (otherwise `min_fractional` applies) `max_residues_in_score = 20` Maximum number of residues to use when increasing score by $\sqrt{n_residues}$ `weight_by_ss = False` Apply extra weight on residues with

H-bondsassignment_weight = None Apply extra weight on residues assigned to sequenceextra_weight_h_bonds = 1. Extra weight on residues with H-bondsmin_absolute = 5 Minimum residues in a chain remaining after removing parts occupied by a better-scoring chainmerge_remainder = True Merge remainder in merge_by_segment_correlationdump_remainder = False If merge_remainder is False and dump_remainder is True, dump the remainder pdb to remainder.pdbncs_in_cell_only = True Only consider NCS-related molecules that are in cellremove_only_if_centers_outside = False Remove outside cell only if centers outside cell. Modifies remove_outside_cell. Must set remove_outside_cell as well to apply.remove_outside_cell = False Remove chains with atoms outside cellmerge_segments = False Identify segments in first and second models For each gap or end in first model, try to insert or attach a segment from the second model. If selected, no other merging is done.min_segment_length = 5 Segments shorter than this will be ignored in merge_segments.merge_second_model = True Cut second model into pieces and merge them into first model. (Useful for filling in gaps in first model.)merge_both_models = False Cut first and second models into pieces and merge them (useful for combining ...

- high_resolution = None Resolution used in map calculation
- match_distance = 1.5 Residue pairs must have rmsd of match_distance (Å) or lower to be crossed. A value between 0.5 Å and 1.5 Å is generally best.
- merge_by_segment_correlation = False Split all models into segments (breaking where residue numbers are not sequential), rank segments by correlation to map weighted by number of atoms, take highest-scoring non-overlapping segments. Followed by standard merge with all unused segments.
- keep_all_of_first_model = None Try to keep all of first model. Applies to merge_by_segment_correlation only. Does not work fully if trim_to_au is used.
- optimize_ncs = False Optimize NCS after merge_by_segment_correlation. NOTE: assumes that all NCS operators are allowed and may not place the resulting model in density if the NCS is not present in the density.
- trim_to_au = True Split up any chains that are in multiple NCS au
- weight_rms = 1 Weight on closest approach of fragments
- cc_cut = 0.90 Eliminate NCS copies with CC less than cc_cut of max
- weight_cc = 0.1 Weight on CC in optimizing NCS (typically 10)
- weight_rad_gyr = 0.1 Weight on radius of gyration in optimize_ncs
- n_substitutions = 3 Substitutions in optimization
- n_consider = 20 Substitutions in optimization applied to last n_consider fragments
- n_random_opt = 0 Try factor
- max_optimize_tries = 500 Maximum optimization steps
- overlap_tolerance = None Minimum separation between N/C1'/P atoms in separate chains. Applies to merge_by_segment_correlation only. Default is 3 Å for C1'/P and 2 Å for N
- ncs_overlap_tolerance_factor = 1.5 Overlap tolerance required for NCS-related chains. Typically larger than for chains in same NCS au to ensure they do not overlap
- very_good_protein_score = 20.0 Score indicating in combination with very_good_protein_cc that an PROTEIN fragment is almost certainly correct
- very_good_protein_cc = 0.75 CC indicating in combination with very_good_protein_score that an PROTEIN fragment is almost certainly correct

- `very_good_rna_score = 7.0` Score indicating in combination with `very_good_rna_cc` that an RNA fragment is almost certainly correct
- `very_good_rna_cc = 0.60` CC indicating in combination with `very_good_rna_score` that an RNA fragment is almost certainly correct
- `min_fractional = 0.05` Minimum fraction of a chain remaining after removing parts occupied by a better-scoring chain
- `min_fractional_multiple_chain_types = 0.35` Minimum fraction of a chain remaining after removing parts occupied by a better-scoring chain. Applies if multiple chain types are present (otherwise `min_fractional` applies)
- `max_residues_in_score = 20` Maximum number of residues to use when increasing score by $\sqrt{n_residues}$
- `weight_by_ss = False` Apply extra weight on residues with H-bonds
- `assignment_weight = None` Apply extra weight on residues assigned to sequence
- `extra_weight_h_bonds = 1` Extra weight on residues with H-bonds
- `min_absolute = 5` Minimum residues in a chain remaining after removing parts occupied by a better-scoring chain
- `merge_remainder = True` Merge remainder in `merge_by_segment_correlation`
- `dump_remainder = False` If `merge_remainder` is False and `dump_remainder` is True, dump the remainder pdb to `remainder.pdb`
- `ncs_in_cell_only = True` Only consider NCS-related molecules that are in cell
- `remove_only_if_centers_outside = False` Remove outside cell only if centers outside cell. Modifies `remove_outside_cell`. Must set `remove_outside_cell` as well to apply.
- `remove_outside_cell = False` Remove chains with atoms outside cell
- `merge_segments = False` Identify segments in first and second models For each gap or end in first model, try to insert or attach a segment from the second model. If selected, no other merging is done.
- `min_segment_length = 5` Segments shorter than this will be ignored in `merge_segments`.
- `merge_second_model = True` Cut second model into pieces and merge them into first model. (Useful for filling in gaps in first model.)
- `merge_both_models = False` Cut first and second models into pieces and merge them (useful for combining all pieces of both models). Alternative to `merge_second_model`.
- `trim_in_merge = False` Trim all the fragments of first and second model back to match density before merging of models. See also `remove_bad_fragments` in `merge` which removes bad fragments (after any trimming is done).
- `remove_bad_fragments_in_merge = True` Remove bad fragments in `merge`. See also `trim_in_merge` which trims back fragments before trying to merge them.
- `extend_in_merge = False` Extend fragments of first and second model during merging of models
- `fragment_length = 10` Length of pieces that second model will be cut down to in `merge_second_model` or `merge_both_models`
- `use_cc_in_combine_extend = None` You can choose to use the correlation of density rather than density at atomic positions to score models in the `merge_second_model` or `merge_both_models` step. This may be useful at lower resolution (> 3 Å)

- `matching = True` Carry out crossover using segments that match (have the same number of residues)
- `non_matching = True` Carry out crossover using segments that do not match (do not have the same number of residues)
- `check_sequence = None` After running all other procedures, redo the sequence assignment. This may be necessary if the models did not have the same sequence assignments or if unassigned pieces are now put together and can be assigned. If `None`, then `check_sequence` will be set to `True` if `non_matching=True`.
- `preserve_connectivity = None` This is a shortcut for turning off `merge_second_model`, `merge_both_models`, `non_matching`, and `check_sequence`. It is useful if your two models have the same residues, just with different coordinates, and you want to maintain the connectivity.
- `merge_only = None` This is a shortcut for turning off everything except `merge_second_model` and `merge_both_models`.
- `standard_merge = None` Merge by reading all chains and running resolve merging.
- `wang_radius = 5` Smoothing radius for solvent identification
- `cc_min = 0.20` Minimum map-model correlation to keep a segment after refinement
- `cc_ratio_min = None` Minimum map-model correlation relative to maximum to keep a segment after refinement
- `combine_models_score_min = None` Minimum score for a chain in `combine_models` (after `helices_strands`). Score is mostly defined by $\sqrt{n_{\text{atoms}}} \times \text{overall map CC}$.
- `cc_min_rna = 0.25` Minimum map-model correlation (RNA building)
- `map_inside_box = None` Map final chains inside unit cell
- `controlverbose = False` Verbose output
`nproc = 1` Number of processors to use
`*multiprocessing sge lsf pbs condor pbspro slurm` Choices are multiprocessing (single machine) or queuing systems
`queue_run_command = None` run command for queue jobs. For example `qsub`.
`raise_sorry = False` Raise sorry if problems
`return_on_failure = False` Just return on failure
`debug = False` Debugging output
`dry_run = False` Just read in and check parameter names
`i_ran_seed = 351219` random seed
`resolve_command_list = None` You can supply any resolve command here NOTE: for command-line usage you need to enclose the whole set of commands in double quotes (") and each individual command in single quotes (') like this: `resolve_command_list="'no_build' 'b_overall 23' "`;
- `verbose = False` Verbose output
- `nproc = 1` Number of processors to use
- `*multiprocessing = *multiprocessing sge lsf pbs condor pbspro slurm` Choices are multiprocessing (single machine) or queuing systems
- `queue_run_command = None` run command for queue jobs. For example `qsub`.
- `raise_sorry = False` Raise sorry if problems
- `return_on_failure = False` Just return on failure
- `debug = False` Debugging output
- `dry_run = False` Just read in and check parameter names
- `i_ran_seed = 351219` random seed
- `resolve_command_list = None` You can supply any resolve command here NOTE: for command-line usage you need to enclose the whole set of commands in double quotes (") and each individual

command in single quotes (') like this: `resolve_command_list="'no_build' 'b_overall 23' "`;

Generating alternative conformations matching a map with `create_alt_conf`

`create_alt_conf.html`

Author(s)

- `create_alt_conf`: Tom Terwilliger, Pavel Afonine, Nigel Moriarty, Dorothee Liebschner

`create_alt_conf`: Tom Terwilliger, Pavel Afonine,

Nigel Moriarty, Dorothee Liebschner

Purpose

The purpose of `create_alt_conf` is to create alternative side-chain conformations for a structure using very high-resolution X-ray data or a very high-resolution cryo-EM map as a guide.

The problem that it solves is that if you have multiple conformations present in a structure, residues that have the same altloc (A or B or C etc) should have plausible relationships, and in particular, they should not clash. Residues with different altloc (A vs B) can have any relationship as they are in different conformers. Finding the assignment of side chain conformations to conformers A, B, C etc that yields good geometry for all the conformers can be difficult.

Note that the model that is produced is not intended to be a final model, rather it is a model that has some plausible alternative conformations. It is recommended that you use the conformers in the output model as suggestions for your model.

How `create_alt_conf` works:

Generating alternative conformations:

The starting point for `create_alt_conf` is normally a model with a single conformation (no A or B altlocs).

The first step is to generate a set of plausible conformations for each side chain based on the current model and the X-ray data or cryo-EM map.

For X-ray data, a 4mFo-3DFc map is created. This map is expected to show conformations that are not represented in the model. For cryo-EM data, the input map is used.

In either case, plausible conformations for at each side chain position are found by testing each rotamer in a rotamer library for the side chain type present at that position and choosing those that fit the density. The conformer that best fits the density and that is different from the current conformer at that position is chosen. If no suitable alternative is found, the original is used. Conformers that clash with all conformers at any other position are discarded. The new side chain conformers are then used to create a new model with side chains that match the density map and that are as different as possible from the starting side chains. This model is refined using the X-ray data or cryo-EM map. The result of this step is a pair of models, the original and one that has alternative side-chain conformations. Each model has just one conformer.

If you want, you can instead supply a model with multiple conformations and it will just split that model into separate models, one for each conformation (each altloc A or B).

Optimizing assignment of side-chain conformers to models:

The key step in `create_alt_conf` is creating a set of models in which each model has good geometry and the models collectively fit the density. The starting point for this step is a set of models that have different side-chain conformations at some or all positions. The challenging part of this step is arranging the side-chain conformers in a way that minimizes clashes but that uses all the supplied side chain conformers.

The procedure used is to generate a diverse set of refined multi-conformer models, to score each model, then to find an optimized model by recombination among the multi-conformer models. The procedure is complicated somewhat by the need to select a small number of test models for scoring, as scoring requires refinement and is therefore a slow process.

Scoring function:

The scoring function used to group side-chain conformers is the Holton geometry validation score. This score is a composite of geometry restraints used in refinement and validation metrics and it is a rather good indication of the overall geometric quality of a model. The scoring function requires a refined model, as unrefined models will generally have very poor geometry.

The scoring function is also calculated on a per-residue basis. This per-residue score is the Holton geometry validation score, only including interactions that involve this residue. Note that the sum of per-residue scores calculated in this way does not equal the total score, both because of the way in which scores are calculated, and because some interactions are only within a residue and others are between residues. Even so, the sum of per-residue scores is closely related to the total score. This allows estimation of the total score from per-residue scores.

Probabilistic estimation of score for a new arrangement of side chains:

To reduce the number of models that need to be fully scored, a procedure for probabilistic estimation of the expected score for a model with a new arrangement of side chains is used. In this estimation process, it is assumed that many of the most significant contacts between residues will be with nearby residues (segments of length `group_length` are considered).

The starting point for this estimation is a set of scored models with varying arrangements of alternative conformations. These arrangements of alternative conformations are set up so that for any stretch of `group_length` residues, there will be models that match at all other positions but that differ in all possible ways at this set of `group_length` residues. The differences in per-residue scores among these models are then used to estimate the expected effect of changing the arrangement of conformers for any one residue, in the context of the conformers surrounding that residue. Additionally, an error estimate in this expected effect is estimated.

With these predictors for the effects of changes in alternative conformations at any one residue in the context of its neighbors, an estimate can be made for the expected score for a model with any arrangement of conformations. The score for the most similar model is taken as a starting value, then the minimal set of changes in arrangement is applied and the expected change in score for each change is noted, along with its uncertainty. This yields a probabilistic estimate of the expected score for this arrangement.

Recombination procedure:

The recombination procedure is targeted (not random recombination). The reason for this is that scoring requires refinement, a relatively slow process. The current set of scored models is used as described above to create a probabilistic estimate of the score that would be found for any new arrangement of side chains as

conformers in a model. A simple recombination and mutation procedure is used to find arrangements that are predicted to have good Holton geometry validation scores. The arrangements with good predicted scores are then generated, refined and scored to identify optimized arrangements that actually do have good scores.

Quick run:

You can skip the extensive optimization procedure and use a shorter one instead with the keyword `quick=True`. This is still not that fast, but it is a lot faster than the full version.

Ligands:

If you supply a single model with no alternate conformations, it may contain ligands. Any ligands are separated out and added in (one copy of each ligand for each conformer created) just before refinement.

Anisotropic refinement:

You can specify `anisotropic_refinement=True` or supply a single model previously refined with anisotropic refinement. This will instruct `create_alt_conf` to use anisotropic refinement.

Examples

Standard run of `create_alt_conf`:

Running `create_alt_conf` is easy. From the command-line you can type:

```
phenix.create_alt_conf model.pdb data.mtz
```

where `model.pdb` is the model you would like to use as a starting point and `data.mtz` is an X-ray data file. The tool will run (for a long time) and create a model named `model_overall_best.pdb` that will contain alternative conformations based on the X-ray data. If you want to use more processors, specify `nproc=32` or whatever number you would like. If you want to create more than two conformers, you can specify the number with `conformers=3` or whatever you like. If you want to use the conformers present in `model.pdb` as starting points, rearranging them as needed but not creating any new ones, you can specify `use_existing_altlocs=True` (or say nothing, this is the default). If you want to throw them away, say `use_existing_altlocs=False`.

Possible Problems

The procedure takes a long time. You would not normally want to run this on a machine with just a few processors. Running with `quick=True` is much faster, but not as comprehensive.

This procedure creates *N* full conformers (altlocs A, B, C etc), with all atoms in the macromolecule present in all the conformers. If you want to only have multiple conformations in a few places, you will need to use `phenix.pdbtools` or another method to remove multiple conformations from the rest of the structure.

This method is only suitable if you have very high-resolution data. Normally 1.5 Å is about the lowest resolution data you would want to use.

The procedure only creates alternative side-chain conformers, not main-chain. If your main chain has substantial alternative conformations (not just slight adjustments to match the side chain conformers), you will need to use another approach.

Literature

Additional information

List of all available keywords

job_title = None Job title in PHENIX GUI, not used on command line
input_filesxray_data = None Data file with experimental data (FP SIGFP or I SIGI).
xray_data_labels = None Optional labels for X-ray data Normally these would be something like I,SIGI or F,SIGF
free_r_labels = None Optional labels for free_r flags. Normally these would be something like FreeR_flags or R_free_flags
free_r_flag_value = 0 FreeR flag value
ncs_file = None File with NCS information (typically point-group NCS with the center specified). Typically in PDB format. Can also be a .ncs_spec file from phenix. Created automatically if symmetry is specified.
map_coeffs = None MTZ file with coefficients for a map
use_map_coeffs_if_optional = False Use map_coeffs (a map) instead of Fobs, sigFobs if both are available
map_coeffs_virtual = None Used internally
map_coeffs_labels = None If map coefficients cannot be identified automatically from your MTZ file, you can specify the label or labels for them. (Please separate labels with blank space, MTZ columns grouped together separated by commas with no blanks.) You can specify: map_coeffs_labels (e.g., FWT,PHIFWT) amplitudes and phases (e.g., FP,SIGFP PHIB) or amplitudes, phases, weights (e.g., FP,SIGFP PHIB FOM)
alt_models = None Set of alternate conformations as individual model files. You can supply alternate conformations with alt_models, or you can supply a single model file with multiple conformations and specify
use_existing_altlocs=True.
fixed_solvent_model = None Model with solvent to be used as is (no refinement). NOTE: set water_and_hydrogens will be set to always if you use a fixed_solvent_model
input_model = None Force this to be input model
restraint_files = None Restraint files for refinement
map_model_full_map = None Input full map file
half_map = None Input half map file
model = None Input model file
output_alt_conf_model = default Model with alternate conformation
overwrite = True Overwrite files with same names
file_name = None Not used
filename = None Not used
serial = None Not used
temp_dir = TEMP_CREATE_ALT_CONF Temporary directory
alt_conf_cycles = 4 Number of overall cycles
use_existing_altlocs = True Use existing conformers. Default is True (use any altlocs in the input model (if one model is supplied). If alt_models are supplied, they are used as a set of conformers.
allow_swap_fixed_solvent = True If fixed_solvent_model is supplied, try swapping conformers at each residue number to optimize geometry score.
score_this_altloc_only = None Score only this altloc if specified
sift_and_rename_waters_only = False Assign waters to altlocs if possible and stop
sift_and_refine_only = False Assign waters to altlocs if possible, refine and stop. Requires fixed_solvent_model
max_sift_cycles_fixed_solvent = None Maximum cycles of sifting with fixed solvent. Default is number of residues in solvent.
final_refinement_fixed_solvent = False Include final low-restraint-weight refinement with fixed solvent. default is False.
water_and_hydrogens = *at_end_of_cycle always never When to add hydrogens and waters in refinement and scoring. Refinement and rebuilding of side chains is done at the same time. If at_end_of_cycle, added at end of each cycle. NOTE: if fixed_solvent_model is set, water_and_hydrogens will be set to always.
max_allowed_residual = 1 Maximum contact energy allowed for adding waters
max_models_per_cycle = 72 Maximum models to consider in any cycle
max_rotamers_to_consider = None Limit the number of rotamers to consider at each site. Default is to consider all possible rotamers at each site.
min_rotamer_rmsd = 0.25 Minimum rmsd between rotamers. If side chain atoms differ by less than this, consider them the same.
conformers = 2 Number of conformers to create. If you supply conformers as alt_models or as a single model (with use_existing_altlocs=True) the number of supplied conformers must match the value of conformers. (This check is to make sure that the number of supplied conformers is what you intend.)
mask_at_large_diffs ...

- job_title = None Job title in PHENIX GUI, not used on command line
- input_filesxray_data = None Data file with experimental data (FP SIGFP or I SIGI).
xray_data_labels = None Optional labels for X-ray data Normally these would be something like I,SIGI or F,SIGF
free_r_labels = None Optional labels for free_r flags. Normally these would be something like

FreeR_flags or R_free_flags free_r_flag_value = 0 FreeR flag value ncs_file = None File with NCS information (typically point-group NCS with the center specified). Typically in PDB format. Can also be a .ncs_spec file from phenix. Created automatically if symmetry is specified. map_coeffs = None MTZ file with coefficients for a map use_map_coeffs_if_optional = False Use map_coeffs (a map) instead of Fobs, sigFobs if both are available map_coeffs_virtual = None Used internally map_coeffs_labels = None If map coefficients cannot be identified automatically from your MTZ file, you can specify the label or labels for them. (Please separate labels with blank space, MTZ columns grouped together separated by commas with no blanks.) You can specify: map_coeffs_labels (e.g., FWT,PHIFWT) amplitudes and phases (e.g., FP,SIGFP PHIB) or amplitudes, phases, weights (e.g., FP,SIGFP PHIB FOM) alt_models = None Set of alternate conformations as individual model files. You can supply alternate conformations with alt_models, or you can supply a single model file with multiple conformations and specify use_existing_altlocs=True. fixed_solvent_model = None Model with solvent to be used as is (no refinement). NOTE: set water_and_hydrogens will be set to always if you use a fixed_solvent_model input_model = None Force this to be input model restraint_files = None Restraint files for refinement map_model full_map = None Input full map file half_map = None Input half map files model = None Input model file

- xray_data = None Data file with experimental data (FP SIGFP or I SIGI).
- xray_data_labels = None Optional labels for X-ray data Normally these would be something like I,SIGI or F,SIGF
- free_r_labels = None Optional labels for free_r flags. Normally these would be something like FreeR_flags or R_free_flags
- free_r_flag_value = 0 FreeR flag value
- ncs_file = None File with NCS information (typically point-group NCS with the center specified). Typically in PDB format. Can also be a .ncs_spec file from phenix. Created automatically if symmetry is specified.
- map_coeffs = None MTZ file with coefficients for a map
- use_map_coeffs_if_optional = False Use map_coeffs (a map) instead of Fobs, sigFobs if both are available
- map_coeffs_virtual = None Used internally
- map_coeffs_labels = None If map coefficients cannot be identified automatically from your MTZ file, you can specify the label or labels for them. (Please separate labels with blank space, MTZ columns grouped together separated by commas with no blanks.) You can specify: map_coeffs_labels (e.g., FWT,PHIFWT) amplitudes and phases (e.g., FP,SIGFP PHIB) or amplitudes, phases, weights (e.g., FP,SIGFP PHIB FOM)
- alt_models = None Set of alternate conformations as individual model files. You can supply alternate conformations with alt_models, or you can supply a single model file with multiple conformations and specify use_existing_altlocs=True.
- fixed_solvent_model = None Model with solvent to be used as is (no refinement). NOTE: set water_and_hydrogens will be set to always if you use a fixed_solvent_model
- input_model = None Force this to be input model
- restraint_files = None Restraint files for refinement
- map_model full_map = None Input full map file half_map = None Input half map files model = None Input model file
- full_map = None Input full map file

- half_map = None Input half map files
- model = None Input model file
- outputalt_conf_model = default Model with alternate conformations
 overwrite = True Overwrite files with same names
 file_name = None Not used
 filename = None Not used
 serial = None Not used
 temp_dir = TEMP_CREATE_ALT_CONF Temporary directory
- alt_conf_model = default Model with alternate conformations
- overwrite = True Overwrite files with same names
- file_name = None Not used
- filename = None Not used
- serial = None Not used
- temp_dir = TEMP_CREATE_ALT_CONF Temporary directory
- alt_confcycles = 4 Number of overall cycles
 use_existing_altlocs = True Use existing conformers. Default is True (use any altlocs in the input model (if one model is supplied). If alt_models are supplied, they are used as a set of conformers.
 allow_swap_fixed_solvent = True If fixed_solvent_model is supplied, try swapping conformers at each residue number to optimize geometry
 score_this_altloc_only = None Score only this altloc if specified
 sift_and_rename_waters_only = False Assign waters to altlocs if possible and stop
 sift_and_refine_only = False Assign waters to altlocs if possible, refine and stop. Requires fixed_solvent_model
 max_sift_cycles_fixed_solvent = None Maximum cycles of sifting with fixed solvent. Default is number of residues in solvent.
 final_refinement_fixed_solvent = False Include final low-restraint-weight refinement with fixed solvent. default is False.
 water_and_hydrogens = *at_end_of_cycle always never When to add hydrogens and waters in refinement and scoring. Refinement and rebuilding of side chains is done at the same time. If at_end_of_cycle, added at end of each cycle. NOTE: if fixed_solvent_model is set, water_and_hydrogens will be set to always.
 max_allowed_residual = 1 Maximum contact energy allowed for adding waters
 max_models_per_cycle = 72 Maximum models to consider in any cycle
 max_rotamers_to_consider = None Limit the number of rotamers to consider at each site. Default is to consider all possible rotamers at each site.
 min_rotamer_rmsd = 0.25 Minimum rmsd between rotamers. If side chain atoms differ by less than this, consider them the same.
 conformers = 2 Number of conformers to create. If you supply conformers as alt_models or as a single model (with use_existing_altlocs=True) the number of supplied conformers must match the value of conformers. (This check is to make sure that the number of supplied conformers is what you intend.)
 mask_at_large_diffs = None Mask all changes to include only residues with large differences between conformations (defined by large_rmsd). Default is False if two conformations, otherwise True
 score_confidence_ratio = 2. Score for a predicted pattern is calculated score plus score_confidence_ratio times the uncertainty. For a score_confidence ratio of 2, this is the score for which there is a 95% probability that the true score is less than this.
 use_edited_model = False Use model edited to remove poor rotamers in next cycle
 rebuild_main_chain = False Find alternate conformations for main chain. Default is False, only look for new side chain positions. Not implemented.
 diff_map_resolution = 2 Resolution for difference map (typically 2 Å)
 macro_cycles = 3 Refinement macro_cycles
 water_and_h_macro_cycles = 5 Refinement macro_cycles when placing waters and hydrogens
 rsr_macro_cycles = 3 Real-space refinement macro_cycles
 mask_atoms_atom_radius = 6 Radius for masking atoms. Set high enough to capture density in alternative conformations not present in starting model.
 large_rmsd = 0.3 Large rmsd between conformer residues indicating need for alternates
 minimum_patterns = 4 Minimum patterns to consider
 group_length = 3 Number of residues in group that is to be varied. Residues in a chain are grouped in sets of group_length and all patterns of side-chain assignments are tested.
 domain_size = 200

Maximum residues to consider when finding clashes `n_swap = 1` Add extra variation of swapping chains, `n_swap` times existing chains added on. Random locations `salt_labels = ABCDEFGHIJKLMNOPQRSTUVWXYZ` Labels to use for alt id. Include enough for all conformations. These are one-character labels only. `geometry_score_type = magnitudes` `gradients` `residue_score` `magnitudes_by_chain` `gradients_by_chain` *`residue_score_by_chain` Score type for geometry evaluation. Both are based on the geometry gradients by residue after refinement. Gradients uses the magnitudes of the mean gradient in a residue. Magnitudes uses the average magnitude of gradients in a residue. `correlated_errors = True` Assume errors in residue ge...

- `cycles = 4` Number of overall cycles
- `use_existing_altlocs = True` Use existing conformers. Default is True (use any altlocs in the input model (if one model is supplied)). If `alt_models` are supplied, they are used as a set of conformers.
- `allow_swap_fixed_solvent = True` If `fixed_solvent_model` is supplied, try swapping conformers at each residue number to optimize geometry score.
- `score_this_altloc_only = None` Score only this altloc if specified
- `sift_and_rename_waters_only = False` Assign waters to altlocs if possible and stop
- `sift_and_refine_only = False` Assign waters to altlocs if possible, refine and stop. Requires `fixed_solvent_model`
- `max_sift_cycles_fixed_solvent = None` Maximum cycles of sifting with fixed solvent. Default is number of residues in solvent.
- `final_refinement_fixed_solvent = False` Include final low-restraint-weight refinement with fixed solvent. default is False.
- `water_and_hydrogens = *at_end_of_cycle` always never When to add hydrogens and waters in refinement and scoring. Refinement and rebuilding of side chains is done at the same time. If `at_end_of_cycle`, added at end of each cycle. NOTE: if `fixed_solvent_model` is set, `water_and_hydrogens` will be set to always.
- `max_allowed_residual = 1` Maximum contact energy allowed for adding waters
- `max_models_per_cycle = 72` Maximum models to consider in any cycle
- `max_rotamers_to_consider = None` Limit the number of rotamers to consider at each site. Default is to consider all possible rotamers at each site.
- `min_rotamer_rmsd = 0.25` Minimum rmsd between rotamers. If side chain atoms differ by less than this, consider them the same.
- `conformers = 2` Number of conformers to create. If you supply conformers as `alt_models` or as a single model (with `use_existing_altlocs=True`) the number of supplied conformers must match the value of `conformers`. (This check is to make sure that the number of supplied conformers is what you intend.)
- `mask_at_large_diffs = None` Mask all changes to include only residues with large differences between conformations (defined by `large_rmsd`). Default is False if two conformations, otherwise True
- `score_confidence_ratio = 2`. Score for a predicted pattern is calculated score plus `score_confidence_ratio` times the uncertainty. For a `score_confidence_ratio` of 2, this is the score for which there is a 95% probability that the true score is less than this.
- `use_edited_model = False` Use model edited to remove poor rotamers in next cycle
- `rebuild_main_chain = False` Find alternate conformations for main chain. Default is False, only look for new side chain positions. Not implemented.
- `diff_map_resolution = 2` Resolution for difference map (typically 2 Å)

- `macro_cycles = 3` Refinement `macro_cycles`
- `water_and_h_macro_cycles = 5` Refinement `macro_cycles` when placing waters and hydrogens
- `rsr_macro_cycles = 3` Real-space refinement `macro_cycles`
- `mask_atoms_atom_radius = 6` Radius for masking atoms. Set high enough to capture density in alternative conformations not present in starting model.
- `large_rmsd = 0.3` Large rmsd between conformer residues indicating need for alternates
- `minimum_patterns = 4` Minimum patterns to consider
- `group_length = 3` Number of residues in group that is to be varied. Residues in a chain are grouped in sets of `group_length` and all patterns of side-chain assignments are tested.
- `domain_size = 200` Maximum residues to consider when finding clashes
- `n_swap = 1` Add extra variation of swapping chains, `n_swap` times existing chains added on. Random locations
- `alt_labels = ABCDEFGHIJKLMNOPQRSTUVWXYZ` Labels to use for alt id. Include enough for all conformations. These are one-character labels only.
- `geometry_score_type = magnitudes` `gradients` `residue_score` `magnitudes_by_chain` `gradients_by_chain` `*residue_score_by_chain` Score type for geometry evaluation. Both are based on the geometry gradients by residue after refinement. Gradients uses the magnitudes of the mean gradient in a residue. Magnitudes uses the average magnitude of gradients in a residue.
- `correlated_errors = True` Assume errors in residue geometry scores are correlated
- `all_single_changes = False` Include all single changes in randomization
- `geometry_mean_or_min = *mean` `min` Geometry mean or minimum. Use the mean value of scores for all examples of a particular pattern of side chains in a set of `group_length` residues.
- `clash_weight = 0` Weight on clashes within a conformer
- `allow_close = NO` `ON` `OO` `NN` atom pairs that can be close (for clashes)
- `close_dist = 2.8` Close distance for `allow_close` atoms
- `other_dist = 4.0` Close distance for non-`allow_close` atoms
- `skip_basic_optimization = False` Skip basic optimization step
- `const_shrink_donor_acceptor = 0` Allow contacts closer by `const_shrink_donor_acceptor` from normal target for H-bonding atoms. NOTE: value of 0.6 matches behavior of Phenix pre-2024.
- `keep_ligands = True` Keep one copy of each input ligand (no altlocs). Add to each model just before refinement. Requires supplying any necessary restraint files.
- `anisotropic_refinement = Auto` Anisotropic refinement. Default is to match input model file.
- `refine_args = None` Any refinement arguments to be added to default arguments. For example, `refinement.refine.occupancies.remove_selection=water` or for example, `refinement.main.number_of_macro_cycles=3`
- `h_selection = None` Selection to apply after adding H atoms
- `fixed_pattern = None` Pattern to apply. Only this pattern is used if specified. Must specify one for each `alt_conf` in this analysis
- `crystal_inforesolution = None` Nominal resolution of mapwrapping = `None` You can specify whether the map is wrapped (can map values outside bounds to inside with cell translations). Always true for crystallographic maps. `scattering_table = *n_gaussian` `wk1995` `it1992` `electron` `neutron` Choice of

scattering table for structure factor calculations. Standard for X-ray is `n_gaussian`, for cryoEM is `electron`. `chain_type = *None` PROTEIN RNA DNA Type of polymer (normally identified by chain automatically).

- `resolution = None` Nominal resolution of map
- `wrapping = None` You can specify whether the map is wrapped (can map values outside bounds to inside with cell translations). Always true for crystallographic maps.
- `scattering_table = *n_gaussian` wk1995 it1992 electron neutron Choice of scattering table for structure factor calculations. Standard for X-ray is `n_gaussian`, for cryoEM is `electron`.
- `chain_type = *None` PROTEIN RNA DNA Type of polymer (normally identified by chain automatically).
- `controlmultiprocessing = *multiprocessing` sge lsf pbs condor pbspro slurm Choices are multiprocessing (single machine) or queuing systems Not implemented `queue_run_command = None` run command for queue jobs. For example `qsub.nproc = 4` Number of processors to use. NOTE: by default multiple processors will only be used in the map-to-model step (this is because multiprocessing requires writing out `nproc` sets of huge files and it can be very slow with distributed queues.). You can override this with `force_nproc = True`. `ignore_symmetry_conflicts = False` You can ignore the symmetry information (CRYST1) from coordinate files. This may be necessary if your model has been placed in a box with `box_map` for example. `use_existing_files = False` Use existing files `random_seed = 771914` random seed `verbose = False` Verbose output `quick = False` Quick run (no optimization with refinement, just find a plausible set of altlocs and finish up) `clean_up = True` Remove temporary directory when done
- `multiprocessing = *multiprocessing` sge lsf pbs condor pbspro slurm Choices are multiprocessing (single machine) or queuing systems Not implemented
- `queue_run_command = None` run command for queue jobs. For example `qsub`.
- `nproc = 4` Number of processors to use. NOTE: by default multiple processors will only be used in the map-to-model step (this is because multiprocessing requires writing out `nproc` sets of huge files and it can be very slow with distributed queues.). You can override this with `force_nproc = True`.
- `ignore_symmetry_conflicts = False` You can ignore the symmetry information (CRYST1) from coordinate files. This may be necessary if your model has been placed in a box with `box_map` for example.
- `use_existing_files = False` Use existing files
- `random_seed = 771914` random seed
- `verbose = False` Verbose output
- `quick = False` Quick run (no optimization with refinement, just find a plausible set of altlocs and finish up)
- `clean_up = True` Remove temporary directory when done
- `gui` GUI-specific parameter required for output directory `output_dir = None`
- `output_dir = None`

Creating maps using the Phenix GUI

create_maps.html

Contents

- The "Create Maps" GUI
- Guidelines for use
- References

Because of the differences in calculation method and input data, there is no single unified interface for creating any electron density maps; instead, separate programs are grouped together in the main GUI. For omit maps, a modified interface to AutoBuild is available; consult the AutoBuild documentation for rules on use. Isomorphous difference maps have a separate GUI, which takes two reflection files as input. For maps using phases from a model, use the "Create Maps" module. All of these programs write output as an MTZ file; the latter two may optionally output XPLOR or CCP4 maps (calculated to cover the PDB file used for phasing) if desired.

The "Create Maps" GUI

This is a frontend to phenix.maps, which shares a common set of parameters with phenix.refine. Many user-defined map types are possible, but the program defaults to the most common ones, 2mFo-DFc and mFo-DFc difference maps. Some additional predefined types are available in the drop-down menus.

The pre-defined maps include all common likelihood-weighted difference maps, anomalous difference map, likelihood-weighted F(obs) or F(calc) maps, and anomalous log-likelihood gradient (LLG) maps generated by Phaser. Custom map types can be defined by typing in an appropriate string in a format similar to those already shown; the map generator will parse the string to determine the components used.

By default the program only generates an MTZ file containing the map coefficients, but you may separately define maps to be generated in X-PLOR or CCP4 format by clicking the button labeled "CCP4 or XPLOR maps". However, if you leave these undefined, the GUI will automatically run the FFT to generate CCP4-format files if you click the "Open in PyMOL" button once map coefficients are available.

Guidelines for use

Map quality may be heavily influenced by reflections that are missing due to errors in data collection or processing. Substituting F(calc) for missing F(obs) increases model bias but often results in a more easily interpreted map; the default maps output by phenix.refine (and REFMAC) are "filled" maps. In phenix.create_maps, the default maps are filled, and we recommend using this option unless you are particularly worried about model bias. Output MTZ column labels must be unique; the predefined map types have associated predefined labels, but you can change these if you wish. Anomalous difference maps require anomalous data (i.e. F₊,F₋ or I₊,I₋); the precalculated anomalous difference DANO generated by some CCP4 programs is not suitable. If you are using neutron data, set the scattering table to "neutron"; otherwise, leave it set to "n_gaussian".

- Map quality may be heavily influenced by reflections that are missing due to errors in data collection or processing. Substituting F(calc) for missing F(obs) increases model bias but often results in a more easily interpreted map; the default maps output by phenix.refine (and REFMAC) are "filled" maps. In phenix.create_maps, the default maps are filled, and we recommend using this option unless you are particularly worried about model bias.

- Output MTZ column labels must be unique; the predefined map types have associated predefined labels, but you can change these if you wish.
- Anomalous difference maps require anomalous data (i.e. F⁺,F⁻ or I⁺,I⁻); the precalculated anomalous difference DANO generated by some CCP4 programs is not suitable.
- If you are using neutron data, set the scattering table to "neutron"; otherwise, leave it set to "n_gaussian".

References

Averaged kick maps: less noise, more signal... and probably less bias. J. Pražnikar, P.V. Afonine, G. Guncar, P.D. Adams, and D. Turk. *Acta Crystallogr D Biol Crystallogr* 65, 921-31 (2009).

- Averaged kick maps: less noise, more signal... and probably less bias. J. Pražnikar, P.V. Afonine, G. Guncar, P.D. Adams, and D. Turk. *Acta Crystallogr D Biol Crystallogr* 65, 921-31 (2009).

Cryo_fit: Fitting Structure to Cryo-EM Map Using MD Simulation

cryo_fit.html

Contents

- Overview
- Introduction
- How to Fit Cryo EM Maps
- Advantages
- How Cryo_fit Works
- How to Run Cryo_fit
- FAQ
- Reference

Overview

Cryo_fit fits biomolecular (protein and nucleic acids) structures to cryo-EM maps using molecular dynamics (MD) simulation.

Introduction

Cryo_fit is an automated cryo-EM density fitting program using MD simulation to guide models into maps. Cryo_fit produces all-atom models highly consistent with the EM density and is being used to optimize models for many functional molecular complexes including one of the first all-atom models of the human ribosome revealing the 'subunit rolling' mechanism unique to eukaryotic ribosomes. Two key advantages of Cryo_fit are: speed and stereochemistry preservation. Its fast speed allows researchers to optimize ribosome structures on a laptop instead of the need of a computing server with thousands of CPU/GPU cores. It uses a reduced model molecular dynamics potential allowing the preservation of stereochemistry information. In addition, secondary and tertiary contact potentials can be specified to place additional restraints on contacts to keep molecules intact during the fitting process.

How to Fit Cryo EM Maps

Advantages

How Cryo_fit Works

How to Run Cryo_fit

See the tutorial notes for cryo_fit

FAQ

See the cryo_fit FAQ

Reference

Journal of Structural Biology, Volume 208, 2019, Pages 1-6, Cryo_fit: Democratization of flexible fitting for cryo-EM

Cryo_fit2: Fitting to a Cryo-EM Map using Phenix Dynamics

cryo_fit2.html

Contents

- Note
- Author
- When cryo_fit1 and cryo_fit2 are useful?
- When cryo_fit2 is useful?
- When cryo_fit2 is more useful than cryo_fit1?
- Theory
- How to install
- How to Run Cryo_fit2
- FAQ
- Limitation

Note

cryo_fit2 is under active development.

Consider to use cryo_fit1 instead.

Author

Pavel Afonine, Dylan Girodat (dgirodat@lanl.gov), Jignesh Prajapati (jprajapati@lanl.gov), Doo Nam Kim (doonam.kim@gmail.com)

When cryo_fit1 and cryo_fit2 are useful?

See the When cryo_fit1 and cryo_fit2 are useful? in FAQ

When cryo_fit2 is useful?

See the When cryo_fit2 is useful? in FAQ

When cryo_fit2 is more useful than cryo_fit1?

See the When cryo_fit2 is more useful than cryo_fit1? in FAQ

Theory

- This program uses phenix.dynamics.
- Simulated annealing is carried out by default to minimize the object function T (= $T_{\text{target_map}} \cdot \text{map_weight} + T_{\text{geom}} \cdot \text{wc}$).

map_weight is cryo-EM map weight and wc is a geometry keeping weight.

- map_weight is cryo-EM map weight and wc is a geometry keeping weight.

How to install

See the installation notes for cryo_fit2

How to Run Cryo_fit2

See the tutorial notes for cryo_fit2

FAQ

See the cryo_fit2 FAQ

Limitation

- Cryo_fit2 does not use electrostatic interaction. Therefore, it does not fully capture all physical forces. See phenix.dynamics for detail. Cryo_fit version 1 also omits electrostatic interaction so that molecular dynamics simulation runs faster.
- See phenix.dynamics for detail.
- Cryo_fit version 1 also omits electrostatic interaction so that molecular dynamics simulation runs faster.

Visualizing reflection data in Phenix

data_viewer.html

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- Overview
- 3D viewer
- 2D viewer

Overview

The program `phenix.data_viewer` is used to explore data in reciprocal space, as a diagnostic and learning tool. Although reciprocal space is largely an imaginary construct, it is a convenient way to describe the geometry of diffraction and data collection. Two different rendering methods are provided: a 3D view of all of reciprocal space at once, and a 2D view of planes of reflections, or "pseudo-precession photographs". These are largely complementary, although certain visualizations may be easier in one or the other view.

`phenix.data_viewer` can be launched from the GUI, under "Reflection tools", where the 2D and 3D versions are listed separately, or directly from the command line, which initially starts the 3D version. OpenGL support (and ideally a reasonably fast graphics card) is required for the 3D graphics; for this reason, we do not recommend running the 3D viewer over an X11 connection. Once the program opens it will ask you to select a reflections file, which depending on contents may prompt you to choose an array of data. Currently any array which includes amplitudes or intensities (including map coefficients) can be viewed, as well as phase angles and R-free flags. However, the examples shown here all use amplitudes or intensities.

3D viewer

For most reflections files, the data will fill some portion of a sphere, whose radius is one divided by the maximum resolution. Depending on space group symmetry and whether the data are symmetry-unique, usually only a wedge of data will be shown. If the data are anomalous this wedge will be mirrored across the origin. For instance, the intensity data for the p9-sad tutorial (included with Phenix) will appear like this:

In this case the space group is $I4$ with anomalous data, so the symmetry-unique reflections fill two 90-degree wedges of reciprocal space. Identifying features in the data is left as an exercise to the user; however, in this case (and many others) the characteristic distribution of intensities seen in the Wilson plot in Xtriage is very obvious: relatively high intensities at low resolution (at the center of the sphere), decreasing rapidly above approximately 10 Angstrom, with a large increase in mean intensity around 4-5 Angstrom due to characteristic inter-atomic distances in organic (especially macromolecular) structures. Clicking on any individual reflection will show the h,k,l index, resolution, and value in the box on the lower-left corner.

The view can be manipulated with the controls on the left-hand side of the window; this includes the option to change the symmetry or generate anomalous data if not are present in the input file. A particularly useful option is to show reflections that are missing. These will appear as white spheres:

Most datasets will have at least a handful of these at high resolution, and often the innermost low-resolution reflections will also be absent due to beamstop overlap. In this case a large contiguous region of missing data is visible for one of the quadrants; this is probably caused by imperfect orientation of the crystal during data collection. For this dataset the number of missing reflections is relatively small and because of the overall excellent quality, no impact on phasing, refinement, or map quality is observed. However, poor completeness can often lead to problems at various stages of refinement, especially if large regions of reciprocal space are

systematically missing.

2D viewer

The 2D view handles symmetry-unique data differently than the 3D version: it will always expand these data to P1 and generate anomalous pairs, covering all of reciprocal space. (For non-unique data, only those reflections present in the input file will be displayed.) This will result in a view something like this (here using the rnase-s tutorial):

In this example both missing reflections (small white open circles) and systematic absences (large purple open circles) are also displayed. The distribution of values is different than observed for the p9-sad example, partly because amplitudes are displayed instead of intensities, and partly because the data were probably not collected or processed to the maximum resolution possible for this crystal.

phenix.den_refine: DEN refinement in Phenix

den_refine.html

Deformable elastic network (DEN) refinement is a refinement protocol that uses a sparse restraint network to maintain local model geometry while allowing for larger global domain motions over the course of several macrocycles of slow-cooling simulated annealing.

The DEN restraint network is a set of atom-atom distance restraints between atoms 3 to 5 Angstrom apart and within 0 and 10 residues in sequence space. The initial target values for these distance restraints may be set to corresponding distances in a reference model (such as a higher resolution structure or homology model) or to the distances in the input model. Over the course of a simulated annealing cycle, the target distances are frequently updated to allow for improved global fit to the diffraction data. DEN restraints have been shown to be particularly useful in improving models with low resolution data.

DEN was originally described by Brunger and colleagues in:

Schroder GF, Levitt M, and Brunger AT. (2010) Super-resolution biomolecular crystallography with low-resolution data. *Nature* 464(7292), 1218-22.

Optimization

The impact of the DEN restraint network is controlled by two variable parameters, gamma, which controls the deformability of the network, and the weight, which controls how large of a contribution the DEN restraints make to the overall refinement energy term (see the above reference for details). To get the most out of a DEN refinement, it is important to run a grid search over combinations of gamma and weight values. This process is very time intensive, and is best carried out on a machine with many processors.

The default behavior is to optimize with six gamma values [0.0, 0.2, 0.4, 0.6, 0.8, 1.0] and five weights [3.0, 10.0, 30.0, 100.0, 300.0], which will run most optimally on a machine with at least 30 processor cores. The number of processors used by Phenix is controlled by the main.nproc parameter, so setting main.nproc=30 will run the optimization step on 30 processors.

Using DEN

How to run the command line version:

```
% phenix.refine model.pdb data.mtz strategy=den
```

See the full list of options below.

There is also a DEN refinement GUI for those who prefer a graphical interface.

phenix.den_refine uses a modified version of the default phenix.refine parameters optimized for DEN refinement.

The DEN-specific subset of refinement parameters:

```
refinement {
  main {
    nproc = Auto
  }
  refine {
    strategy = *individual_sites rigid_body *individual_adp \
    group_adp tls occupancies group_anomalous *den
  }
  den {
```

```

reference_file = None
gamma = 0.5
kappa = 0.1
weight = 30
sigma = 0.44
optimize = True
opt_gamma_values = 0 0.2 0.4 0.6 0.8 1
opt_weight_values = 3 10 30 100 300
num_cycles = 10
kappa_burn_in_cycles = 2
bulk_solvent_and_scale = True
refine_adp = True
final_refinement_cycle = False
verbose = False
annealing_type = *torsion cartesian
output_kinimage = False
restraint_network {
  lower_distance_cutoff = 3
  upper_distance_cutoff = 15
  sequence_separation_low = 0
  sequence_separation_limit = 10
  exclude_hydrogens = True
  ndistance_ratio = 1
  export_den_pairs = False
  den_network_file = None
}
}
tardy {
  mode = *every_macro_cycle second_and_before_last once first first_half
  start_temperature_kelvin = 3300
  final_temperature_kelvin = 300
  number_of_cooling_steps = 1440
  number_of_time_steps = 1
  time_step_pico_seconds = 0.001
  prolsq_repulsion_function_changes {
    c_rep = 100
    k_rep = 1
  }
}
simulated_annealing {
  start_temperature = 3300
  final_temperature = 300
  cool_rate = 50
  number_of_steps = 24
  time_step = 0.001
}
}

```


Processing, docking and rebuilding AlphaFold2 and other predicted models in cryo-EM maps

dock_and_rebuild.html

Author(s)

- dock_and_rebuild: Tom Terwilliger

Purpose

Dock and rebuild combines the functions of processing, docking and rebuilding predicted models produced by AlphaFold, RoseTTAFold and other prediction software into a cryo EM map.

These steps can also be carried out one at a time. There are three steps in working on an AlphaFold or other predicted model with a cryo-EM structure.

The first step is to process the predicted model by trimming off all the uncertain residues in the predicted model and breaking up the remaining structure into a best guess of rigid domains with `phenix.process_predicted_model`.

The next step is to dock each of the domains of the processed model into the map, keeping plausible connectivity. This is done with `phenix.dock_predicted_model`.

The third step is to morph the predicted model onto the docked domains and then to rebuild all the parts of the predicted model using the density in the map.

How dock_and_rebuild works:

The `dock_and_rebuild` procedure is just a way to connect all three steps in processing, docking and rebuilding a predicted model. You may want to use this procedure if you have a simple case, or if you have many models you want to process. If it does not work well, you may want to take its intermediate outputs (i.e., the docked model) and carry out the following steps individually.

The model input to `dock_predicted_model` is your starting predicted model file (i.e., an AlphaFold model)

The map input to `dock_predicted_model` is normally your best sharpened or density-modified cryo-EM map. It can also be a map generated by any other procedure (including crystallography).

If you are able to mask your map, keeping only the part representing the region where this model belongs, that can be very helpful. You can also box the map around this region. If you have a map that has many chains, this masking can greatly shorten the time for docking. If you don't know where in your map the model goes at all, you can supply the entire map. If your map has symmetry this will normally be found automatically.

The three steps described above are then carried out: processing the model to obtain domains representing the accurate parts of the model (`phenix.process_predicted_model`), docking the domains (`phenix.dock_predicted_model`), and morphing and rebuilding the predicted model to yield a rebuilt version of the predicted model.

Examples

Standard run of dock_and_rebuild:

Running dock_and_rebuild is easy. From the command-line you can type:

```
phenix.dock_and_rebuild model=my_model.pdb \  
  full_map=my_map.ccp4 \  
  resolution=3
```

This will carry out all the steps of processing, docking and rebuilding to yield my_model_rebuilt.pdb

Possible Problems

If your map has pseudo-symmetry (like a proteasome) you might need to box one subunit or try ssm_search=False to use a more thorough search in docking.

If your map is inverted (left-handed), docking and model-building will not work properly. You can often tell if your map is inverted because any helices will be left-handed. If you are unsure, you can run MapBox with invert_hand=True to invert the map and then see if docking works. Note that if your map is inverted, you will want to invert all your maps and start everything from the beginning.

Specific limitations and problems:

Literature

Additional information

List of all available keywords

job_title = None Job title in PHENIX GUI, not used on command line
input_files_seq_file = None Input sequence file. Required if no models supplied. Used to create predicted models. Format should be Fasta or simple sequences separated by blank lines. One sequence per chain to be generated. Must match input models if both are supplied. (Fasta format or sequences separated by blank lines)
xray_data_file = None Input X-ray data file (MTZ format).
xray_data_label = None Label specifying which data column to use from xray_data_file
xray_test_flag_label = None Label specifying which test flag column (if any) to use from xray_data_file
truncate_models_to_match_sequences = True If input sequences start after or end before supplied predicted models, trim the predicted models to match sequences
density_select = None If set, trim map to region containing molecule (non-zero region) before use. Recommended for cryo-EM maps that have a large map that is mostly unused. Default is True if map is full-size and False if not.
predicted_model = None One or more input models in one or more files (normally predicted models) to be placed. Normally should start at residue 1 and match sequence file exactly.
b_value_field_is = *plddt rmsd b_value The B-factor field in predicted models can be pLDDT (confidence, 0-1 or 0-100) or rmsd (Å) or a B-factor. Only applies to protein chains (always B-factor for non-protein).
msa_list = None Multiple sequence alignment (MSA) file. Format is a3m only. First sequence in an MSA file must match a sequence in the sequence file exactly.
models_are_already_placed = False You can specify that all models (predicted_model or model) are already placed (docked, placed in unit cell). No docking or MR is done if so. Equivalent to specifying a scaffold_model with the same contents as the model.
model_copies_list = None List of number of copies to find of each model in predicted_model. Default is 1 (must specify all or none of them) . Specify all together like model_copies_list=1,2,1,1
processed_model_file = None Processed model file (e.g., from phenix.process_predicted_model). If supplied, skip the process_predicted_model step and use this file. This model is expected to have one chain for each domain and actual B-values in the B-value field. It is expected that all poorly-predicted residues have been removed.
docked_model_file = None Docked model (e.g., output of dock_processed_model). If supplied, use this file as the docked model. (Skip process_predicted_model

and docking steps This model is expected to have a single chain with gaps for parts of the model that are not accurately known from the prediction. NOTE: You still need to supply the predicted model as the input model for this procedure in addition to this docked model. `morphed_model_file = None` Morphed model (e.g., full model morphed to match output of `dock_processed_model`). If supplied, skip the `process_predicted_model` and docking steps and use this file as the morphed model. This model is expected to have a single chain with no gaps. NOTE: You do not need to supply the predicted model as the input model for this procedure in addition to this morphed model. `previous_model_file = None` Previous model (from a previous run or external). Must match sequence of working model exactly. Used as a source of possible model information and as a hypothesis for docking of working model. If `docked_model_file` is also supplied, `previous_model` is used only as source of model information. `scaffold_model = None` Scaffold model. If supplied, used as target for docking of each chain in `predicted_model` or `model`. Must be similar in sequence to models to be docked. If supplied, `model_copies_list` is ignored and docking is done by superposition onto this structure instead of docking into density or MR. If the scaffold model does not have all chains represented, they are added in by docking. `scaffold_minimum_chain_cc = 0.35` Chains in scaffold model that have `scaffold_minimum_cc` are not re-docked, but those with l...

- `job_title = None` Job title in PHENIX GUI, not used on command line
- `input_filesseq_file = None` Input sequence file. Required if no models supplied. Used to create predicted models. Format should be Fasta or simple sequences separated by blank lines. One sequence per chain to be generated. Must match input models if both are supplied. (Fasta format or sequences separated by blank lines) `xray_data_file = None` Input X-ray data file (MTZ format). `xray_data_label = None` Label specifying which data column to use from `xray_data_file` `xray_test_flag_label = None` Label specifying which test flag column (if any) to use from `xray_data_file` `truncate_models_to_match_sequences = True` If input sequences start after or end before supplied predicted models, trim the predicted models to match sequences `density_select = None` If set, trim map to region containing molecule (non-zero region) before use. Recommended for cryo-EM maps that have a large map that is mostly unused. Default is `True` if map is full-size and `False` if not. `predicted_model = None` One or more input models in one or more files (normally predicted models) to be placed. Normally should start at residue 1 and match sequence file exactly. `b_value_field_is = *plddt rmsd b_value` The B-factor field in predicted models can be pLDDT (confidence, 0-1 or 0-100) or rmsd (Å) or a B-factor. Only applies to protein chains (always B-factor for non-protein). `msa_list = None` Multiple sequence alignment (MSA) file. Format is a3m only. First sequence in an MSA file must match a sequence in the sequence file exactly. `models_are_already_placed = False` You can specify that all models (`predicted_model` or `model`) are already placed (docked, placed in unit cell). No docking or MR is done if so. Equivalent to specifying a `scaffold_model` with the same contents as the model. `model_copies_list = None` List of number of copies to find of each model in `predicted_model`. Default is 1 (must specify all or none of them) . Specify all together like `model_copies_list=1,2,1,1` `processed_model_file = None` Processed model file (e.g., from `phenix.process_predicted_model`). If supplied, skip the `process_predicted_model` step and use this file. This model is expected to have one chain for each domain and actual B-values in the B-value field. It is expected that all poorly-predicted residues have been removed. `docked_model_file = None` Docked model (e.g., output of `dock_processed_model`). If supplied, use this file as the docked model. (Skip `process_predicted_model` and docking steps This model is expected to have a single chain with gaps for parts of the model that are not accurately known from the prediction. NOTE: You still need to supply the predicted model as the input model for this procedure in addition to this docked model. `morphed_model_file = None` Morphed model (e.g., full model morphed to match output of `dock_processed_model`). If supplied, skip the `process_predicted_model` and docking steps and use this file as the morphed model. This model is expected to have a single chain with no gaps. NOTE: You do not need to supply the predicted model as the input model for this procedure in addition to this morphed

`model.previous_model_file` = None Previous model (from a previous run or external). Must match sequence of working model exactly. Used as a source of possible model information and as a hypothesis for docking of working model. If `docked_model_file` is also supplied, `previous_model` is used only as source of model information.
`scaffold_model` = None Scaffold model. If supplied, used as target for docking of each chain in `predicted_model` or `model`. Must be similar in sequence to models to be docked. If supplied, `model_copies_list` is ignored and docking is done by superposition onto this structure instead of docking into density or MR. If the scaffold model does not have all chains represented, they are added in by docking.
`scaffold_minimum_chain_cc` = 0.35 Chains in scaffold model that have `scaffold_minimum_cc` are not re-docked, but those with lower CC are.
`base_scaffold_minimum_chain_cc` = -1 Value of `scaffold_...`

- `seq_file` = None Input sequence file. Required if no models supplied. Used to create predicted models. Format should be Fasta or simple sequences separated by blank lines. One sequence per chain to be generated. Must match input models if both are supplied. (Fasta format or sequences separated by blank lines)
- `xray_data_file` = None Input X-ray data file (MTZ format).
- `xray_data_label` = None Label specifying which data column to use from `xray_data_file`
- `xray_test_flag_label` = None Label specifying which test flag column (if any) to use from `xray_data_file`
- `truncate_models_to_match_sequences` = True If input sequences start after or end before supplied predicted models, trim the predicted models to match sequences
- `density_select` = None If set, trim map to region containing molecule (non-zero region) before use. Recommended for cryo-EM maps that have a large map that is mostly unused. Default is True if map is full-size and False if not.
- `predicted_model` = None One or more input models in one or more files (normally predicted models) to be placed. Normally should start at residue 1 and match sequence file exactly.
- `b_value_field_is` = *plddt rmsd b_value The B-factor field in predicted models can be pLDDT (confidence, 0-1 or 0-100) or rmsd (Å) or a B-factor. Only applies to protein chains (always B-factor for non-protein).
- `msa_list` = None Multiple sequence alignment (MSA) file. Format is a3m only. First sequence in an MSA file must match a sequence in the sequence file exactly.
- `models_are_already_placed` = False You can specify that all models (`predicted_model` or `model`) are already placed (docked, placed in unit cell). No docking or MR is done if so. Equivalent to specifying a `scaffold_model` with the same contents as the model.
- `model_copies_list` = None List of number of copies to find of each model in `predicted_model`. Default is 1 (must specify all or none of them) . Specify all together like `model_copies_list=1,2,1,1`
- `processed_model_file` = None Processed model file (e.g., from `phenix.process_predicted_model`). If supplied, skip the `process_predicted_model` step and use this file. This model is expected to have one chain for each domain and actual B-values in the B-value field. It is expected that all poorly-predicted residues have been removed.
- `docked_model_file` = None Docked model (e.g., output of `dock_processed_model`). If supplied, use this file as the docked model. (Skip `process_predicted_model` and docking steps This model is expected to have a single chain with gaps for parts of the model that are not accurately known from the prediction. NOTE: You still need to supply the predicted model as the input model for this procedure in addition to this docked model.

- `morphed_model_file = None` Morphed model (e.g., full model morphed to match output of `dock_processed_model`). If supplied, skip the `process_predicted_model` and docking steps and use this file as the morphed model. This model is expected to have a single chain with no gaps. NOTE: You do not need to supply the predicted model as the input model for this procedure in addition to this morphed model.
- `previous_model_file = None` Previous model (from a previous run or external). Must match sequence of working model exactly. Used as a source of possible model information and as a hypothesis for docking of working model. If `docked_model_file` is also supplied, `previous_model` is used only as source of model information.
- `scaffold_model = None` Scaffold model. If supplied, used as target for docking of each chain in `predicted_model` or `model`. Must be similar in sequence to models to be docked. If supplied, `model_copies_list` is ignored and docking is done by superposition onto this structure instead of docking into density or MR. If the scaffold model does not have all chains represented, they are added in by docking.
- `scaffold_minimum_chain_cc = 0.35` Chains in scaffold model that have `scaffold_minimum_cc` are not re-docked, but those with lower CC are.
- `base_scaffold_minimum_chain_cc = -1` Value of `scaffold_minimum_chain_cc` in cases where scaffold is known to be good.
- `fragments_model_file = None` Fragments model (e.g., `map_to_model.pdb`). Used as a source of possible model information and as a hypothesis for docking of working model. If `docked_model_file` is also supplied, `fragments_model` is used only as source of model information.
- `model_to_rebuild_file = None` Model to be rebuilt. Must be already in place (already docked). Must match supplied sequences exactly.
- `symmetry_file = None` Symmetry file (.ncs_spec format or MATRX records) with reconstruction symmetry. Used in identification of unique part of map. NOTE: symmetry file applies to map file in original position. NOTE 2: only proper symmetry (point-group, helical) is allowed NOTE 3: Only applies to cryo-EM maps
- `pae_file = None` Optional input json file with matrix of inter-residue estimated errors (pae file)
- `distance_model_file = None` Distance_model_file. A PDB or mmCIF file containing the model corresponding to the PAE matrix. Only needed if `weight_by_ca_ca_distances` is True.
- `search_model_copies = None` Used internally only
- `search_model = None` used internally only
- `map_model_full_map = None` Input map file (for cryo-EM). This can be a boxed or masked map showing just the molecule to dock (best) or a full map with symmetry. If your map has symmetry be sure to set `asymmetric_map = False`. If you have a map with symmetry you can supply a symmetry file if you want. Otherwise symmetry will be automatically determined.
`half_map = None` Input half map files. Usually supply one full map or 2 half maps
`model = None` Input predicted model (e.g., AlphaFold model). Assumed to have pLDDT values in B-value field (or RMSD values). May have multiple chains. Normally use `predicted_model` instead.
- `full_map = None` Input map file (for cryo-EM). This can be a boxed or masked map showing just the molecule to dock (best) or a full map with symmetry. If your map has symmetry be sure to set `asymmetric_map = False`. If you have a map with symmetry you can supply a symmetry file if you want. Otherwise symmetry will be automatically determined.
- `half_map = None` Input half map files. Usually supply one full map or 2 half maps

- `model = None` Input predicted model (e.g., AlphaFold model). Assumed to have pLDDT values in B-value field (or RMSD values). May have multiple chains. Normally use `predicted_model` instead.
- `output_filesoutput_model_prefix = None` Output files with superposed models will begin with this prefix
- `output_seq_file = None` Sequence file (possibly edited) written to this file by PredictAndBuild
- `pdb_out = None` Used internally only
- `temp_dir = None` Temporary directory. Default is `dock_and_build_xx` where `xx` creates new directory
- `crystal_inforesolution = None` High-resolution limit for main search. This can be lower resolution than the data. The search is quicker at lower resolution. If your model is poor, try 2-3 Å lower resolution than your data (i.e, if your data is 2.5 Å, try 5 Å).
- `scattering_table = *None` Choice of scattering table for structure factor calculations. Default for X-ray is `n_gaussian`, for cryoEM is `electron`.
- `wrapping = None` You can specify whether the map is wrapped (can map values outside bounds to inside with cell translations).
- `asymmetric_map = None` Specifies that this is an asymmetric map and no symmetry is to be supplied or found. Alternative to supplying a symmetry file or symmetry. Applies to cryo-EM reconstructions only.
- `solvent_content = None` Solvent fraction (content) of the cell. You can specify the fraction of the volume of this cell that is taken up by the macromolecule. Normally set automatically. Values go from 0 to 1.
- `pdb70_text = None` Database of PDB entries to be used as templates (pdb70 file). Normally used internally only
- `sequence = None` Old-style sequence string (alternative to sequence file).
- `chain_type = *PROTEIN` DNA RNA Chain type

- msa = None MSA as text. Supply MSA as text or as an msa_file. Used internally
- templates_as_string = None Templates as string (single PDB file with one or more chains)
- space_group = None Space group (normally read from the data file, applies to X-ray)
- space_group_alternatives = *HAND ALL NONE Space group alternatives for MR: ALL: all space groups in point group HAND: enantiomer and listed space group LIST: list supplied (not available) NONE: only supplied space group
- unit_cell = None Unit Cell (normally read from the data file, applies to X-ray)
- unique_sequencesFile names and data labels.sequence = "Enter or edit sequence" copies = 1 Copies of this sequencelabel = ""
- sequence = "Enter or edit sequence"
- copies = 1 Copies of this sequence
- label = ""
- process_predicted_modelremove_low_confidence_residues = True Remove low-confidence residues (based on minimum plddt or maximum_rmsd, whichever is specified)continuous_chain = False When removing low-confidence residues, only trim from endsplit_model_by_compact_regions = True Split model into compact regions after removing low-confidence residues.maximum_domains = 3 Maximum domains to obtain. You can use this to merge the closest domains at the end of splitting the model. Make it bigger (and optionally make domain_size smaller) to get more domains. If model is processed in chunks, maximum_domains will apply to each chunk.domain_size = 15 Approximate size of domains to be found (A units). This is the resolution that will be used to make a domain map. If you are getting too many domains, try making domain_size bigger (maximum is 70 A).adjust_domain_size = True If more than maximum_domains are initially found, increase domain_size in increments of 5 A and take the value that gives the smallest number of domains, but at least maximum_domains.minimum_domain_length = None Minimum length of a domain to keep (reject at end if smaller). Default is 10 if no pae matrix or alt_pae_params=False and 20 for pae_matrix with alt_pae_params=Truemaximum_fraction_close = 0.3 Maximum fraction of CA in one domain close to one in another before merging themminimum_sequential_residues = None Minimum length of a short segment to keep (reject at end). Default is 5 if no pae matrix or alt_pae_params=False and 4 for pae_matrix with alt_pae_params=Trueminimum_remainder_sequence_length = 15 used to choose whether the sequence of a removed segment is written to the remainder sequence file.b_value_field_is = *plddt rmsd b_value The B-factor field in predicted models can be pLDDT (confidence, 0-1 or 0-100) or rmsd (A) or a B-factorinput_plddt_is_fractional = None You can specify if the input plddt values (in B-factor field) are fractional (0-1) or not (0-100). By default if all values are between 0 and 1 it is fractional.minimum_plddt = None If low-confidence residues are removed, the cutoff is defined by minimum_plddt or maximum_rmsd, whichever is defined (you cannot define both). A minimum plddt of 0.70 corresponds to a maximum rmsd of 1.5. Minimum plddt values are fractional or not depending on the value of input_plddt_is_fractional.maximum_rmsd = 1.5 If low-confidence residues are removed, the cutoff is defined by minimum_plddt or maximum_rmsd, whichever is defined (you cannot define both). A minimum plddt of 0.70 corresponds to a maximum rmsd of 1.5. Minimum plddt values are fractional or not depending on the value of input_plddt_is_fractional.default_maximum_rmsd = 1.5 Default value of maximum_rmsd, used if maximum_rmsd is not setsubtract_minimum_b = False If set, subtract the lowest B-value from all B-values just before writing out the final files. Does not affect the cutoff for removing low-confidence residues.pae_power = None If PAE matrix (predicted alignment error matrix) is supplied, each edge in the graph will be weighted proportional to (1/pae**pae_power). Use this to try and get the number of domains that you want (try 1, 0.5, 1.5, 2). Default is 1 alt_pae_params=False and 2 for

alt_pae_params=True pae_cutoff = None If PAE matrix (predicted alignment error matrix) is supplied, graph edges will only be created for residue pairs with $pae < pae_cutoff$. Default is 5
alt_pae_params=False and 4 for alt_pae_params=True pae_graph_resolution = None If PAE matrix (predicted alignment error matrix) is supplied, pae_graph_resolution regulates how aggressively the clustering algorithm is. Smaller values lead to larger clusters. Value should be larger than zero, and values larger than 5 are unlikely to be useful, Default is 0.5 alt_pae_params=False and 4 for alt_pae_params=True alt_pae_params = True If PAE matrix is supplied, use alternative set of defaults (minimum_domain_length=20 minimum_sequential_residues=10 pae_power=2 pae_cutoff=4 pae_graph_resolution...

- remove_low_confidence_residues = True Remove low-confidence residues (based on minimum plddt or maximum_rmsd, whichever is specified)
- continuous_chain = False When removing low-confidence residues, only trim from ends
- split_model_by_compact_regions = True Split model into compact regions after removing low-confidence residues.
- maximum_domains = 3 Maximum domains to obtain. You can use this to merge the closest domains at the end of splitting the model. Make it bigger (and optionally make domain_size smaller) to get more domains. If model is processed in chunks, maximum_domains will apply to each chunk.
- domain_size = 15 Approximate size of domains to be found (A units). This is the resolution that will be used to make a domain map. If you are getting too many domains, try making domain_size bigger (maximum is 70 A).
- adjust_domain_size = True If more than maximum_domains are initially found, increase domain_size in increments of 5 A and take the value that gives the smallest number of domains, but at least maximum_domains.
- minimum_domain_length = None Minimum length of a domain to keep (reject at end if smaller). Default is 10 if no pae matrix or alt_pae_params=False and 20 for pae_matrix with alt_pae_params=True
- maximum_fraction_close = 0.3 Maximum fraction of CA in one domain close to one in another before merging them
- minimum_sequential_residues = None Minimum length of a short segment to keep (reject at end). Default is 5 if no pae matrix or alt_pae_params=False and 4 for pae_matrix with alt_pae_params=True
- minimum_remainder_sequence_length = 15 used to choose whether the sequence of a removed segment is written to the remainder sequence file.
- b_value_field_is = *plddt rmsd b_value The B-factor field in predicted models can be pLDDT (confidence, 0-1 or 0-100) or rmsd (A) or a B-factor
- input_plddt_is_fractional = None You can specify if the input plddt values (in B-factor field) are fractional (0-1) or not (0-100). By default if all values are between 0 and 1 it is fractional.
- minimum_plddt = None If low-confidence residues are removed, the cutoff is defined by minimum_plddt or maximum_rmsd, whichever is defined (you cannot define both). A minimum plddt of 0.70 corresponds to a maximum rmsd of 1.5. Minimum plddt values are fractional or not depending on the value of input_plddt_is_fractional.
- maximum_rmsd = 1.5 If low-confidence residues are removed, the cutoff is defined by minimum_plddt or maximum_rmsd, whichever is defined (you cannot define both). A minimum plddt of 0.70 corresponds to a maximum rmsd of 1.5. Minimum plddt values are fractional or not depending on the value of input_plddt_is_fractional.
- default_maximum_rmsd = 1.5 Default value of maximum_rmsd, used if maximum_rmsd is not set

- `subtract_minimum_b = False` If set, subtract the lowest B-value from all B-values just before writing out the final files. Does not affect the cutoff for removing low- confidence residues.
- `pae_power = None` If PAE matrix (predicted alignment error matrix) is supplied, each edge in the graph will be weighted proportional to $(1/pae^{**pae_power})$. Use this to try and get the number of domains that you want (try 1, 0.5, 1.5, 2). Default is 1 `alt_pae_params=False` and 2 for `alt_pae_params=True`
- `pae_cutoff = None` If PAE matrix (predicted alignment error matrix) is supplied, graph edges will only be created for residue pairs with `pae < pae_cutoff`. Default is 5 `alt_pae_params=False` and 4 for `alt_pae_params=True`
- `pae_graph_resolution = None` If PAE matrix (predicted alignment error matrix) is supplied, `pae_graph_resolution` regulates how aggressively the clustering algorithm is. Smaller values lead to larger clusters. Value should be larger than zero, and values larger than 5 are unlikely to be useful, Default is 0.5 `alt_pae_params=False` and 4 for `alt_pae_params=True`
- `alt_pae_params = True` If PAE matrix is supplied, use alternative set of defaults (`minimum_domain_length=20` `minimum_sequential_residues=10` `pae_power=2` `pae_cutoff=4` `pae_graph_resolution=4`). Standard parameters (`alt_pae_params=False`) are: (`minimum_domain_length=10` `minimum_sequential_residues=5` `pae_power=1` `pae_cutoff=5` `pae_graph_resolution=0.5`).
- `weight_by_ca_ca_distance = False` Adjust the edge weighting for each residue pair according to the distance between CA residues. If this is True, then `distance_model` can be provided, otherwise supplied model will be used. See also `distance_power`
- `distance_power = 1` If `weight_by_ca_ca_distance` is True, then edge weights will be multiplied by $1/distance^{**distance_power}$.
- `stop_if_no_residues_obtained = True` Raise Sorry and stop if processing yields no residues
- `keep_all_if_no_residues_obtained = False` Keep everything if processing yields no residues
- `vrms_from_rmsd_intercept = 0.25` Estimate of vrms (error in model) from pLDDT will be based on `vrms_from_rmsd_intercept + vrms_from_rmsd_slope * pLDDT` where mean pLDDT of non-low_confidence_residues is used.
- `vrms_from_rmsd_slope = 1.0` Estimate of vrms (error in model) from pLDDT will be based on `vrms_from_rmsd_intercept + vrms_from_rmsd_slope * pLDDT` where mean pLDDT of non-low_confidence_residues is used.
- `break_into_chunks_if_length_is = 1500` If a sequence is at least `break_into_chunks_if_length_is`, break it into chunks of length `chunk_size` with overlap of `overlap_size` for domain identification using `split_model_by_compact_regions` without a pae matrix
- `chunk_size = 600` If a sequence is at least `break_into_chunks_if_length_is`, break it into chunks of length `chunk_size` with overlap of `overlap_size` for domain identification using `split_model_by_compact_regions` without a pae_matrix
- `overlap_size = 200` If a sequence is at least `break_into_chunks_if_length_is`, break it into chunks of length `chunk_size` with overlap of `overlap_size` for domain identification using `split_model_by_compact_regions` without a pae_matrix
- `searchdock_chains_individually = False` Dock each chain (identified by `chain_id` field) individually. This is useful if the chains might have different orientations in the map compared to the search model, such as in a search model that is a predicted model. Normally used along with `create_unique_chain_at_end=True` for predicted models to put the model back together. `create_unique_chain_at_end = None` Take all docked chains and try to create one chain that has no duplicate residue numbers and that has distances

between ends of fragments consistent with the number of residues between them. If symmetry has been applied, create N copies representing that symmetry. Default is True if dock_chains_individually = True and False otherwise.

minimum_cc_to_keep_domain = 0.2 Do not keep domains in create_unique_chain_at_end if cc is less than minimum_cc_to_keep_domain.

weight_sequential_fragments_by_distance = None Try to dock a chain in a way that minimizes the distance between its first residue and the previously-docked residue with the highest residue number less than this, and similarly for the last residue and the next available placed residue. Default is True if create_unique_chain_at_end is set. Normally use only if you have a model with a single chain and you are docking pieces of that chain.

choose_better_of_individual_and_group_docking = None Dock entire search model (normally one only) and also by chain...pick better-fitting of the two for each chain.

low_res_search = True Try to fit by searching at low resolution.

firstdock_with_mr = True Try using MR to dock.

firstssm_search = None Try to fit by searching for secondary structure. Default is False for dock_in_map and True for dock_and_rebuild if dock_with_mr is not used.

refine_cycles = 3 rigid-body refinement cycles.

ssm_search_min_cc = 0.30 Stop ssm search if this cc achieved.

backup_resolution_cutoff_searches = 2 If initial fitting does not work, try up to this many times increasing resolution by backup_resolution_ratio each time.

backup_resolution_ratio = 1.67 If initial fitting does not work, try up to backup_resolution_cutoff_searches times increasing the resolution by backup_resolution_ratio each time.

resolution_radius_scale = 0.5 Resolution for low-res search will be resolution_radius_scale times the radius of gyration of the search model.

align_moments = False Try to fit by aligning moments of inertia if size of density region and molecule are similar.

max_radius_ratio = 2. Radius of gyration of molecule and density must be within max_radius_ratio of each other.

radius_scale = 1.5 Mask for density and atoms will be radius_scale times the radius of gyration of model.

use_symmetry = True If search_model_copies or search_map_copies are the same for each model and greater than one and allow_symmetry is set, use symmetry in the map to place all copies after the first one.

skip_if_low_cc = True Skip solution before rigid-body refinement if map-model CC is less than half of min_cc.

rigid_body_refinement = True Run rigid-body refinement on final model.

rigid_body_refinement_single_unit = True Run rigid-body refinement with just one unit (do not break up into chains).

rigid_body_refinement_split_method = *chain_id segid When splitting up molecule for rigid-body refinement (if rigid_body_refinement_single_unit=False), use either chain_id or segid to split up molecule.

rigid_body_refinement_resolution = None Run rigid-body refinement at this resolution if specified.

append_to_fixed_model = True Append placed search model to fixed model (if any) after search.

min_cc = 0.4 If quick run, stop if minimum CC is achieved in local search. Also always skip if starting CC_mask is less than 1/4 min_cc.

run_in_boxes = True Run on sub-boxes and combine at the end.

target_box_size = 60 Try to get boxes about this big on a side (grid units).

target_boxes = None Try to get this many boxes. Default is nproc unless this makes box size much smaller than target_box_size.

box_to_run = None Run only this box.

box_overlap_scale = 1 box overlap (ov...

- dock_chains_individually = False Dock each chain (identified by chain_id field) individually. This is useful if the chains might have different orientations in the map compared to the search model, such as in a search model that is a predicted model. Normally used along with create_unique_chain_at_end=True for predicted models to put the model back together.

- create_unique_chain_at_end = None Take all docked chains and try to create one chain that has no duplicate residue numbers and that has distances between ends of fragments consistent with the number of residues between them. If symmetry has been applied, create N copies representing that symmetry. Default is True if dock_chains_individually = True and False otherwise.

- minimum_cc_to_keep_domain = 0.2 Do not keep domains in create_unique_chain_at_end if cc is less than minimum_cc_to_keep_domain.

- `weight_sequential_fragments_by_distance = None` Try to dock a chain in a way that minimizes the distance between its first residue and the previously-docked residue with the highest residue number less than this, and similarly for the last residue and the next available placed residue. Default is True if `create_unique_chain_at_end` is set. Normally use only if you have a model with a single chain and you are docking pieces of that chain.
- `choose_better_of_individual_and_group_docking = None` Dock entire search model (normally one only) and also by chain...pick better-fitting of the two for each chain
- `low_res_search = True` Try to fit by searching at low resolution first
- `dock_with_mr = True` Try using MR to dock first
- `ssm_search = None` Try to fit by searching for secondary structure. Default is False for `dock_in_map` and True for `dock_and_rebuild` if `dock_with_mr` is not used.
- `refine_cycles = 3` rigid-body refinement cycles
- `ssm_search_min_cc = 0.30` Stop ssm search if this cc achieved
- `backup_resolution_cutoff_searches = 2` If initial fitting does not work, try up to this many times increasing resolution by `backup_resolution_ratio` each time
- `backup_resolution_ratio = 1.67` If initial fitting does not work, try up to `backup_resolution_cutoff_searches` times increasing the resolution by `backup_resolution_ratio` each time
- `resolution_radius_scale = 0.5` Resolution for low-res search will be `resolution_radius_scale` times the radius of gyration of the search model.
- `align_moments = False` Try to fit by aligning moments of inertia if size of density region and molecule are similar
- `max_radius_ratio = 2`. Radius of gyration of molecule and density must be within `max_radius_ratio` of each other
- `radius_scale = 1.5` Mask for density and atoms will be `radius_scale` times the radius of gyration of model
- `use_symmetry = True` If `search_model_copies` or `search_map_copies` are the same for each model and greater than one and `allow_symmetry` is set, use symmetry in the map to place all copies after the first one.
- `skip_if_low_cc = True` Skip solution before rigid-body refinement if map-model CC is less than half of `min_cc`.
- `rigid_body_refinement = True` Run rigid-body refinement on final model
- `rigid_body_refinement_single_unit = True` Run rigid-body refinement with just one unit (do not break up into chains)
- `rigid_body_refinement_split_method = *chain_id segid` When splitting up molecule for rigid-body refinement (if `rigid_body_refinement_single_unit=False`), use either `chain_id` or `segid` to split up molecule
- `rigid_body_refinement_resolution = None` Run rigid-body refinement at this resolution if specified
- `append_to_fixed_model = True` Append placed search model to fixed model (if any) after search
- `min_cc = 0.4` If quick run, stop if minimum CC is achieved in local search. Also always skip if starting `CC_mask` is less than $1/4$ `min_cc`.
- `run_in_boxes = True` Run on sub-boxes and combine at the end
- `target_box_size = 60` Try to get boxes about this big on a side (grid units)

- `target_boxes = None` Try to get this many boxes. Default is `nproc` unless this makes box size much smaller than `target_box_size`
- `box_to_run = None` Run only this box
- `box_overlap_scale = 1` box overlap (overlap of boxes) will be `box_overlap_scale` times the density radius
- `edge_ratio = 10` box edge `box_overlap` times `edge_ratio`
- `density_radius = None` Radius for density to be cut out and compared. Default is 6 times the resolution.
- `model_radius = 3` Radius for removing density near `fixed_model`
- `zero_value = 0` Value to set map in regions overlapping fixed model
- `density_peaks = 20` Number of NCS-related peaks of density to check
- `delta_phi = 20` Angular spacing of search
- `max_rot = None` Maximum rotations to try
- `rotz_only = None` Rotate only around Z
- `single_positions_to_try = 10` Number of offset positions to try in optimizing orientation. Positions along the chain are selected as centers and a local fit near each position is carried out. The resulting offsets relative to the original placement are used to optimize the overall orientation and position.
- `max_position_shift_frac = 0.05` Maximum fractional positional shift in `single_positions` run
- `min_relative_cc = 0.67` Minimum local CC relative to original CC to keep a local search. This is a way to reject local searches that are completely wrong.
- `sieve_fit = None` Use `sieve_fit` fraction of single positions in fitting. If `None`, use all
- `ncs_copies_max = None` Maximum number of matching models to write. If more than one they will be written as MODEL records in the output PDB file. You can get them individually afterwards with `phenix.pdbtools placed_model.pdb keep="model 1"` etc.
- `start_rot = None` Three numbers `rotx`, `rotz`, `roty` defining the starting rotation of the search model. Normally used along with `delta_phi=1000` or `max_rot=1` to generate exactly one defined rotation.
- `search_center = None` Optional coordinates in search model for centering search. Note this is different from `target_search_center` which is the location xyz in the map to look.
- `search_center_selection = None` Optional selection defining coordinates in search model for centering search.
- `target_search_center = None` Optional coordinates in reference map where `search_center` should be approximately located after superimposing maps. Used to eliminate possible superpositions that place the search center elsewhere. Overlap scores decreased based on distance to `target_search_center/density_radius`.
- `model_search_position = None` Optional coordinates (usually part of search model) for matching to `target_search_position` after transformation. These can be specified in addition to `target_search_center`. Can have multiple `model_search_position` and `target_search_position` pairs by specifying each multiple times in order. NOTE: Not compatible with map search
- `target_search_position = None` Target positions for `model_search_position` after transformation.
- `search_position_radius = None` Radius for comparison of `target_search_position` and transformed `search_position` values. Default is `density_radius`. If specified, must be a single value or the same number of values as entries in `model_search_position` and `target_search_position`

- `rot_id_n` = None Number of rotation groups. Along with `rot_id_group`, allows defining groups of rotations to be carried out in one run. `.short_caption` = Number of rotation groups
- `rot_id_group` = None rotation group to include. See `rot_id_n`.
- `map_box` = True Run `map_box` to extract useful part of map before search
- `fix_search_position` = False You can choose to not move your model center of mass to the origin by fixing the search position
- `search_box_size` = None You can choose the size of the search box for local searches. Normally set by default corresponding to 3 times `density_radius`
- `lower_bounds` = None You can select a part of your map for analysis with `lower_bounds` and `upper_bounds`.
- `upper_bounds` = None You can select a part of your map for analysis with `lower_bounds` and `upper_bounds`.
- `keep_search_order` = None Keep search order as input
- `remove_water` = False Remove waters and other hetero atoms from input files
- `iterationcycles` = None Cycles of prediction and rebuilding (default is 10 for Thorough, 3 for Standard and 1 for Quickcycle = 1 Iteration cycle
- `cycles` = None Cycles of prediction and rebuilding (default is 10 for Thorough, 3 for Standard and 1 for Quick
- `cycle` = 1 Iteration cycle
- `buildrebuild_strategy` = Thorough Standard Quick Rebuilding strategy. Standard is up to 3 cycles, refining and replacing poorly-fitting loops in predicted models each cycle, using autobuild for density modification (if Xray). Quick is refinement only, one cycle. Thorough is up to 10 cycles, extensive rebuilding.`refine_only` = None Refine only, no rebuilding steps (set automatically with Quick)`refine_only_resolution` = 3.5 Default cutoff for using Refine only`run_fit_loops` = True Run standard fit_loops`run_iterative_morph` = True Run iterative morphing`run_trace_loops_through_density` = True Run loop fitting with trace_through_density algorithm`run_refine` = True Refine morphed model. Required for `run_fit_loops` or `run_trace_loops_through_density``run_iterative_resolution_refine` = True Refine morphed model with iterative resolution method`run_extend` = True Extend ends of morphed model`extract_unique` = True Extract unique part of map. Applies after `density_select` (if set).`use_symmetry_in_extract_unique` = True Use symmetry from symmetry file (if available) when extracting unique part of map.`acceptable_docking_cc` = 0.5 Acceptable docking CC for a chain`minimum_docking_cc` = 0.15 Minimum docking CC for a chain`minimum_cutoff` = 0.1 Minimum cutoff for estimating density in better parts of model`reasonable_cc_ratio` = 0.80 Acceptable CC (ratio) for a chain relative to average. Used to make sure a chain entirely in very bad density is not kept`reasonable_cc_diff` = 0.15 Acceptable CC (difference) for a chain relative to average. Used to make sure a chain entirely in very bad density is not kept`shift_field_distance` = None Shift field characteristic distance (default = 10 Å) for morphing`cc_sd_ratio` = 3. Keep residues with CC at least within `cc_sd_ratio` of the mean CC for good residues`cc_sd_ratio_end` = 2. Keep residues with CC at least within `cc_sd_ratio_end` of the mean CC for good residues (applies to end of fragments)`cc_sd_ratio_ok` = 2. Residues with with CC at least within `cc_sd_ratio_ok` of the mean CC for good residues are ok (do not delete on the basis of `plddt`)`max_gap_ratio` = 3. Allow CA-CA distance to be up to `max_gap_ratio` times expected`maximum_connectivity_deviation` = 15 Maximum connectivity deviation. Reject a solution with bigger deviation than this plus 2 \* `resolution`.`keep_fraction_of_best` = 0.5 Acceptable CC to keep as ratio to best found. Applies if best found is at least `acceptable_docking_cc`.`keep_maximum_entries` = 10 Maximum dock positions to keep`rmsd_for_similar_placement` = None RMSD value indicating that two placements are similar. default

is resolution of maprigid_body_refine_cycles = 1 Refinement cycles for rigid-body
 refinementoverlap_ca_ca_distance = 3 Overlap distance for CA-CA atoms (or P-P)ca_ca_distance = 3.8
 CA-CA distance (or P-P)allowed_fraction_overlapping = 0.10 Allowed fraction
 overlappingmaximum_combinations = 100000 Maximum combinations of placements to
 considerproceed_with_any_symmetry = False Run even with symmetry that is not point-group or helical.
 Not recommended as symmetry may not work properlybox_cushion = 20 Size of buffer around model
 when boxingrefine_cycles = 3 Refinement cyclesloop_refine_cycles = 5 Refinement cycles for
 loopsloop_backup_residues = 3 Number of tries removing one residue at a time from each end of existing
 ends of loop if no loop is found with initial gapresidues_to_trim = 5 Residues to trim on each end of all
 trimmed fragmentsfind_ncs_from_model = True Find NCS (symmetry) from working models and apply in
 density modification. Also turns on finding NCS in any autobuild density modification (including
 density-based search)allow_split_more_than_one_chain_for_mr = False Allow splitting chains into
 domains for mr (X-ray only) even if there are multiple chains. Normally only split if a single chain as
 reassembly may not work with split multiple chains.acceptable_cc_ratio = 0.8 Keep segments with CC at
 least equal to average of base segments times minimum_cc_ratio minus difference between segment
 confide...

- rebuild_strategy = Thorough Standard Quick Rebuilding strategy. Standard is up to 3 cycles, refining and replacing poorly-fitting loops in predicted models each cycle, using autobuild for density modification (if Xray). Quick is refinement only, one cycle. Thorough is up to 10 cycles, extensive rebuilding.
- refine_only = None Refine only, no rebuilding steps (set automatically with Quick)
- refine_only_resolution = 3.5 Default cutoff for using Refine only
- run_fit_loops = True Run standard fit_loops
- run_iterative_morph = True Run iterative morphing
- run_trace_loops_through_density = True Run loop fitting with trace_through_density algorithm
- run_refine = True Refine morphed model. Required for run_fit_loops or run_trace_loops_through_density
- run_iterative_resolution_refine = True Refine morphed model with iterative resolution method
- run_extend = True Extend ends of morphed model
- extract_unique = True Extract unique part of map. Applies after density_select (if set).
- use_symmetry_in_extract_unique = True Use symmetry from symmetry file (if available) when extracting unique part of map.
- acceptable_docking_cc = 0.5 Acceptable docking CC for a chain
- minimum_docking_cc = 0.15 Minimum docking CC for a chain
- minimum_cutoff = 0.1 Minimum cutoff for estimating density in better parts of model
- reasonable_cc_ratio = 0.80 Acceptable CC (ratio) for a chain relative to average. Used to make sure a chain entirely in very bad density is not kept
- reasonable_cc_diff = 0.15 Acceptable CC (difference) for a chain relative to average. Used to make sure a chain entirely in very bad density is not kept
- shift_field_distance = None Shift field characteristic distance (default = 10 Å) for morphing
- cc_sd_ratio = 3. Keep residues with CC at least within cc_sd_ratio of the mean CC for good residues
- cc_sd_ratio_end = 2. Keep residues with CC at least within cc_sd_ratio_end of the mean CC for good residues (applies to end of fragments)

- `cc_sd_ratio_ok = 2`. Residues with with CC at least within `cc_sd_ratio_ok` of the mean CC for good residues are ok (do not delete on the basis of `plddt`)
- `max_gap_ratio = 3`. Allow CA-CA distance to be up to `max_gap_ratio` times expected
- `maximum_connectivity_deviation = 15` Maximum connectivity deviation. Reject a solution with bigger deviation than this plus $2 * \text{resolution}$.
- `keep_fraction_of_best = 0.5` Acceptable CC to keep as ratio to best found. Applies if best found is at least `acceptable_docking_cc`.
- `keep_maximum_entries = 10` Maximum dock positions to keep
- `rmsd_for_similar_placement = None` RMSD value indicating that two placements are similar. default is resolution of map
- `rigid_body_refine_cycles = 1` Refinement cycles for rigid-body refinement
- `overlap_ca_ca_distance = 3` Overlap distance for CA-CA atoms (or P-P)
- `ca_ca_distance = 3.8` CA-CA distance (or P-P)
- `allowed_fraction_overlapping = 0.10` Allowed fraction overlapping
- `maximum_combinations = 100000` Maximum combinations of placements to consider
- `proceed_with_any_symmetry = False` Run even with symmetry that is not point-group or helical. Not recommended as symmetry may not work properly
- `box_cushion = 20` Size of buffer around model when boxing
- `refine_cycles = 3` Refinement cycles
- `loop_refine_cycles = 5` Refinement cycles for loops
- `loop_backup_residues = 3` Number of tries removing one residue at a time from each end of existing ends of loop if no loop is found with initial gap
- `residues_to_trim = 5` Residues to trim on each end of all trimmed fragments
- `find_ncs_from_model = True` Find NCS (symmetry) from working models and apply in density modification. Also turns on finding NCS in any autobuild density modification (including density-based search)
- `allow_split_more_than_one_chain_for_mr = False` Allow splitting chains into domains for mr (X-ray only) even if there are multiple chains. Normally only split if a single chain as reassembly may not work with split multiple chains.
- `acceptable_cc_ratio = 0.8` Keep segments with CC at least equal to average of base segments times `minimum_cc_ratio` minus difference between segment confidence (`plDDT`) and confidence cutoff (typically 0.7)
- `low_res_if_multiple_solutions = 3.5` Try phaser MR at increasing lower resolutions up to this value if multiple solutions are found
- `delta_low_res = 0.5` Resolution increment for `low_res_if_multiple_solutions`
- `predictionprediction_method = *alphafold` Prediction method to use
`template_search_method = *mmseqs2` structure_search Method to identify templates from PDB and generate MSAs. If `structure_search` is set, you can specify what PDB databases to use (same as in `phenix.structure_search`). If `structure_search` is set, you must supply your own MSA file with the keyword `upload_msa_file=True` (`structure_search` does not generate MSAs).
`starting_alphafold_model = None` Starting AlphaFold model. You can supply an AlphaFold model and skip the initial AlphaFold step. This is equivalent to setting up an `output_directory` with just an AlphaFold model named as the expected first

AlphaFold model and specifying `carry_on=True`. `input_directory` = ColabInputs Input directory containing density map. The map filename must start with the same characters as the jobname (only including characters before the first underscore). If you are supplying an MSA file it goes here as well. `output_directory` = ColabOutputs Output directory. Copy outputs to `output_directory`. Used to restart with `carry on`. `save_outputs_in_google_drive` = False If run on Colab, copy outputs to `output_directory` in Google drive. `content_dir` = None Content directory. Default is working directory. `maxit_path` = None Path to maxit (pdb to cif) converter. Optional. `data_dir` = /mnt Data directory (location of AlphaFold parameters). `upload_file_with_jobname_resolution_sequence_lines` = None Upload a file with a set of jobs (Colab only). Each line in the file is a jobname, resolution, and sequence. `maximum_cycles` = 10 Maximum cycles to carry out. `cycle_rmsd_to_resolution_ratio` = 0.25 Stop iteration if rmsd between subsequent AlphaFold models is less than `cycle_rmsd_to_resolution_ratio` times the resolution for two cycles in a row. `significant_increase_in_residues` = 5 Significant increase in residues in model. `password` = None Phenix download password (Colab only). The password used to download Phenix at your institution. Updated weekly, so you may need to request a new one frequently. `version` = dev-4502 Version of Phenix to run (Colab only). `query_sequence` = None Sequence. `resolution` = None Resolution of map (Å). Internal use only. Normally set instead `crystal_info.resolution`. `jobname` = None Name of this job. The first characters before any underscore must be unique and will define the first characters of the corresponding map file in the input directory. The job name will normally also be the name of the working directory. Used internally only. `use_msa` = True Use multiple sequence alignments at some points. `skip_all_msa_after_first_cycle` = False Skip multiple sequence alignments after first cycle. `include_templates_from_pdb` = True Include templates from PDB. Note special behavior when running `predict_and_build`: applies only on first cycle, and if PhenixServer or Colab used prediction will be run with and without templates and PDB and highest plddt model will be kept. `maximum_templates_from_pdb` = 20 Maximum templates from PDB to include. `release_date` = None release date for templates from pdb (only use up to this date. Format is 2020-05-14). `upload_msa_file` = False Supply MSA directly (.a3m format). You can supply the MSA as a file in your `input_directory` and it will be used instead of using `mmseqs2` to generate an MSA. Your file name must end in .a3m. The format is two lines per sequence, the first starts with a greater-than sign and is ignored, the second is a sequence of letters or minus signs. All sequence lines must have the same length. The first sequence must be the target. `upload_manual_templates` = False Supply templates for AlphaFold prediction. Used in the same way as templates from the PDB unless `uploaded_templates_are_map_to_model` is set. May be .cif or .pdb files. Templates must start with characters matching the first characters in the jobname (before the first underscore), the remainder of the file name can be anything but just end in .cif or .pdb. The files must be in the `input_directory` or be uploaded (in Colab only). `uploaded_tem...`

- `prediction_method` = *alphafold Prediction method to use
- `template_search_method` = *mmseqs2 structure_search Method to identify templates from PDB and generate MSAs. If `structure_search` is set, you can specify what PDB databases to use (same as in `phenix.structure_search`). If `structure_search` is set, you must supply your own MSA file with the keyword `upload_msa_file=True` (`structure_search` does not generate MSAs).
- `starting_alphafold_model` = None Starting AlphaFold model. You can supply an AlphaFold model and skip the initial AlphaFold step. This is equivalent to setting up an `output_directory` with just an AlphaFold model named as the expected first AlphaFold model and specifying `carry_on=True`.
- `input_directory` = ColabInputs Input directory containing density map. The map filename must start with the same characters as the jobname (only including characters before the first underscore). If you are supplying an MSA file it goes here as well.
- `output_directory` = ColabOutputs Output directory. Copy outputs to `output_directory`. Used to restart with `carry on`.

- `save_outputs_in_google_drive` = False If run on Colab, copy outputs to `output_directory` in Google drive
- `content_dir` = None Content directory. Default is working directory
- `maxit_path` = None Path to maxit (pdb to cif) converter. Optional.
- `data_dir` = /mnt Data directory (location of AlphaFold parameters)
- `upload_file_with_jobname_resolution_sequence_lines` = None Upload a file with a set of jobs (Colab only). Each line in the file is a jobname, resolution, and sequence
- `maximum_cycles` = 10 Maximum cycles to carry out
- `cycle_rmsd_to_resolution_ratio` = 0.25 Stop iteration if rmsd between subsequent AlphaFold models is less than `cycle_rmsd_to_resolution_ratio` times the resolution for two cycles in a row
- `significant_increase_in_residues` = 5 Significant increase in residues in model
- `password` = None Phenix download password (Colab only). The password used to download Phenix at your institution. Updated weekly, so you may need to request a new one frequently.
- `version` = dev-4502 Version of Phenix to run (Colab only)
- `query_sequence` = None Sequence
- `resolution` = None Resolution of map (Å). Internal use only. Normally set instead `crystal_info.resolution`.
- `jobname` = None Name of this job. The first characters before any underscore must be unique and will define the first characters of the corresponding map file in the input directory. The job name will normally also be the name of the working directory. Used internally only
- `use_msa` = True Use multiple sequence alignments at some point
- `skip_all_msa_after_first_cycle` = False Skip multiple sequence alignments after first cycle
- `include_templates_from_pdb` = True Include templates from PDB. Note special behavior when running `predict_and_build`: applies only on first cycle, and if PhenixServer or Colab used prediction will be run with and without templates and PDB and highest plddt model will be kept
- `maximum_templates_from_pdb` = 20 Maximum templates from PDB to include
- `release_date` = None release date for templates from pdb (only use up to this date. Format is 2020-05-14.
- `upload_msa_file` = False Supply MSA directly (.a3m format). You can supply the MSA as a file in your `input_directory` and it will be used instead of using `mmseqs2` to generate an MSA. Your file name must end in .a3m. The format is two lines per sequence, the first starts with a greater-than sign and is ignored, the second is a sequence of letters or minus signs. All sequence lines must have the same length. The first sequence must be the target.
- `upload_manual_templates` = False Supply templates for AlphaFold prediction. Used in the same way as templates from the PDB unless `uploaded_templates_are_map_to_model` is set. May be .cif or .pdb files. Templates must start with characters matching the first characters in the jobname (before the first underscore), the remainder of the file name can be anything but just end in .cif or .pdb. The files must be in the `input_directory` or be uploaded (in Colab only).
- `uploaded_templates_are_map_to_model` = False The manual templates are models that may or may not have the sequence of the alphafold models. These are used only as suggestions for placement of the main chain.
- `upload_maps` = True Use maps (required to be True)
- `random_seed` = 7231771 Random seed. Used internally

- `random_seed_iterations = 1` Random seed iterations of AlphaFold in first cycle. The model with the highest pLDDT will be used. If running `predict_and_build` and `include_templates_from_pdb` is True this many iterations will be carried out with and without templates. In `predict_and_build` and `predict_chain` set `number_of_models` instead.
- `minimum_random_seed_iterations = 1` Random seed iterations of AlphaFold after first cycle. The model with the highest pLDDT will be used
- `big_improvement = 10` How much improvement in pLDDT is worth going through all randomization cycles
- `good_enough_plddt = 80` Value of pLDDT that is good enough to not make any more models
- `nproc = 4` Number of processors to use. Internal use only. Normally set instead `control.nproc`
- `debug = False` Debugging run (print traceback error messages)
- `get_msa_only = False` Just get MSA and save it on server, no prediction. Does not return the MSA
- `carry_on = False` Carry on from where previous run ended. Used (usually in Colab) to go on after a crash or timeout. Requires that files are saved in the output directory
- `cif_dir = None` Location of templates (normally set automatically)
- `template_hit_list = None` List of templates (normally set automatically)
- `jobnames = None` List of jobnames (normally set automatically)
- `resolutions = None` List of resolutions (normally set automatically)
- `include_templates_from_pdb_list = None` List of `include_templates_from_pdb` ((normally set automatically)
- `include_side_in_templates_list = None` List of `include_side_in_templates_list` ((normally set automatically)
- `map_filename_dict = None` Map filename dict (used internally only)
- `msa_filename_dict = None` MSA filename dict (used internally only)
- `cif_filename_dict = None` CIF filename dict (used internally only)
- `query_sequences = None` List of query sequences (one per jobname) (used internally only)
- `manual_templates_uploaded = None` List of manual templates (used internally only)
- `upload_dir = None` Upload directory (used internally only)
- `working_directory = None` Working directory (used internally only)
- `maps_uploaded = None` List of maps (used internally only)
- `msas_uploaded = None` List of msas (used internally only)
- `num_models = 1` Number of models (used internally only)
- `homooligomer = 1` Number of copies (used internally only)
- `cycle = None` Cycle number (internal use only)
- `host_url = https://api.colabfold.com` Host url for mmseqs
- `use_env = None` Use env (internal use only)
- `use_custom_msa = None` Use custom msa (internal use only)
- `use_templates = None` Use templates (internal use only)
- `template_paths = None` Template paths (internal use only)

- `mtm_file_name` = None map_to_model file name (internal use only)
- `cycle_model_file_name` = None cycle_model_file_name (internal use only)
- `previous_final_model_name` = None previous_final_model_name (internal use only)
- `msa` = None MSA (internal use only)
- `msa_is_msa_object` = None MSA info (internal use only)
- `deletion_matrix` = None Deletion matrix (internal use only)
- `structure_search_params` structure_searchpdb_file = None Enter a PDB file name sequence = None Optional Fasta sequence file. Only needed for a quick sequence search against RCSB without a PDB. output_prefix = 'output' Provide an output prefix if needed blastpath = None Enter path to blastall executable sequence_only = False Do a Blast search against PDBaa sequence instead of doing a Ramachandran-based structure search structure_only = False Do only a Ramachandran-based structure search. db_used = 'rcsb' structure database used in search. rcsb, scop95, or AF2db = 'rcsb' Database used in search. rcsb, scop95, or AF2. get_ligand = False Use get_ligand=True to retrieve ligands. get_ramcode_only = False Generate Rama code for input pdb/cif only. This is for developers only. get_xml_only = False Get BLAST XML output returned as a string object. No coordinate superposition will be performed. Developers only. use_pdb100aa = False Use PDB100 sequence database for sequence search. use_custom_db = False Use custom database specified by custom_db_files/custom_db_dir. custom_db_dir = None The directory of pdb/cif files to make custom database. Default is current directory custom_db_files = None Filenames of the pdb/cif files separated by spaces for database. If none specified, all pdb/cif in the custom_db_dir will be collected atom_selection = 'all' Choose part of the pdb used in the search (default=all). for example: chain B, resseq 113:219, ... etc. get_pdb = 10 get_pdb=N will collect and superpose the top N homologous pdbs. Use get_pdb=0 to disable this option. deposited_before = 0 Specify the latest year of matching structures to be considered for scoring. Pdb's deposited after this year will be discarded. deposited_after = 0 Specify the earliest year of matching structures to be considered for scoring. Pdb's deposited before this year will be discarded. batch_size = 0 Process the pdbs in batch of <batch_size> until <min_match> hits are identified or until all <get_pdb> pdbs are processed min_match = 0 Finish structure_search when <min_match> matches are found. Usually uses with <trim_ends> to exit the search once find suitable pdbs. keep_all_pdb = False Keep all the PDB files, including full PDB, PDB_Chain and superposed PDB_Chain. Default is False which will keep only superposed PDB_Chain files in the directory specified in the output message. trim_ends = False Remove terminal residues of hit pdbs extending beyond those of the target pdb. write_pdb = True Set to False if no output pdb file is needed. Sometimes useful if use Structure_Search within another program and only want to pass pdb objects. write_results = True Set to False if no output results/log files is needed. Useful when calling Structure_Search within another program and only want to pass pdb objects. trim_hit_pdb = False Remove extra domains, extended loops, and unfit portions of hit pdbs after superposed to the target pdb. pickle_hits = False Pickle blast hit results from xml output. coot_display = False (default) Display output pdb files in coot. ask_coot = True prompt for coot display options PDB_MIRROR_DIR = None Enter the top directory of local RCSB PDB mirror. The program will try to retrieve PDBs and/or structure factors from this mirror first. Note this assumes the directory trees under it follows those in RCSB -- pdb files as 'pdb####.ent.gz' in PDB_MIRROR_DIR/data/structures/divided/pdb directory. If you use PDB's rsync script, this variable would be the same as the \$MIRROR_DIR set in the script PDB_MIRROR_MMCIF = None Enter the parent directory of the mmCIF files in the local PDB mirror. MMCIFs will be retrieved from subdirectory ## where ## are the second and third letters in the PDB id. This keyword should be \$PDB_MIRROR_DIR/data/structures/divided/mmcif directory. PDB_MIRROR_PDB = None Enter the parent directory of the PDB files in the local PDB mirror. PDBs will be retrieved from subdirectory ##

where ## are the second and...

- `structure_searchpdb_file` = None Enter a PDB file name sequence = None Optional Fasta sequence file. Only needed for a quick sequence search against RCSB without a PDB. `output_prefix` = 'output' Provide an output prefix if needed `blastpath` = None Enter path to blastall executable `sequence_only` = False Do a Blast search against PDBaa sequence instead of doing a Ramachandran-based structure search `structure_only` = False Do only a Ramachandran-based structure search. `db_used` = 'rcsb' structure database used in search. `rcsb`, `scop95`, or `AF2db` = 'rcsb' Database used in search. `rcsb`, `scop95`, or `AF2`. `get_ligand` = False Use `get_ligand=True` to retrieve ligands. `get_ramcode_only` = False Generate Rama code for input pdb/cif only. This is for developers only. `get_xml_only` = False Get BLAST XML output returned as a string object. No coordinate superposition will be performed. Developers only. `use_pdb100aa` = False Use PDB100 sequence database for sequence search. `use_custom_db` = False Use custom database specified by `custom_db_files/custom_db_dir`. `custom_db_dir` = None The directory of pdb/cif files to make custom database. Default is current directory `custom_db_files` = None Filenames of the pdb/cif files separated by spaces for database. If none specified, all pdb/cif in the `custom_db_dir` will be collected `atom_selection` = 'all' Choose part of the pdb used in the search (default=all). for example: chain B, resseq 113:219, ... etc. `get_pdb` = 10 `get_pdb=N` will collect and superpose the top N homologous pdbs. Use `get_pdb=0` to disable this option. `deposited_before` = 0 Specify the latest year of matching structures to be considered for scoring. Pdb's deposited after this year will be discarded. `deposited_after` = 0 Specify the earliest year of matching structures to be considered for scoring. Pdb's deposited before this year will be discarded. `batch_size` = 0 Process the pdbs in batch of <batch_size> until <min_match> hits are identified or until all <get_pdb> pdbs are processed `min_match` = 0 Finish structure_search when <min_match> matches are found. Usually uses with <trim_ends> to exit the search once find suitable pdbs. `keep_all_pdb` = False Keep all the PDB files, including full PDB, PDB_Chain and superposed PDB_Chain. Default is False which will keep only superposed PDB_Chain files in the directory specified in the output message. `trim_ends` = False Remove terminal residues of hit pdbs extending beyond those of the target pdb. `write_pdb` = True Set to False if no output pdb file is needed. Sometimes useful if use Structure_Search within another program and only want to pass pdb objects. `write_results` = True Set to False if no output results/log files is needed. Useful when calling Structure_Search within another program and only want to pass pdb objects. `trim_hit_pdb` = False Remove extra domains, extended loops, and unfit portions of hit pdbs after superposed to the target pdb. `pickle_hits` = False Pickle blast hit results from xml output. `coot_display` = False (default) Display output pdb files in coot. `ask_coot` = True prompt for coot display options `PDB_MIRROR_DIR` = None Enter the top directory of local RCSB PDB mirror. The program will try to retrieve PDBs and/or structure factors from this mirror first. Note this assumes the directory trees under it follows those in RCSB -- pdb files as 'pdb####.ent.gz' in `PDB_MIRROR_DIR/data/structures/divided/pdb` directory. If you use PDB's rsync script, this variable would be the same as the `$MIRROR_DIR` set in the script `PDB_MIRROR_MMCIFF` = None Enter the parent directory of the mmCIF files in the local PDB mirror. MMCIFFs will be retrieved from subdirectory ## where ## are the second and third letters in the PDB id. This keyword should be `$PDB_MIRROR_DIR/data/structures/divided/mmCIF` directory. `PDB_MIRROR_PDB` = None Enter the parent directory of the PDB files in the local PDB mirror. PDBs will be retrieved from subdirectory ## where ## are the second and third letters in the ...
- `pdb_file` = None Enter a PDB file name
- `sequence` = None Optional Fasta sequence file. Only needed for a quick sequence search against RCSB without a PDB.
- `output_prefix` = 'output' Provide an output prefix if needed
- `blastpath` = None Enter path to blastall executable

- `sequence_only = False` Do a Blast search against PDBaa sequence instead of doing a Ramachandran-based structure search
- `structure_only = False` Do only a Ramachandran-based structure search.
- `db_used = 'rcsb'` structure database used in search. rcsb, scop95, or AF2
- `db = 'rcsb'` Database used in search. rcsb, scop95, or AF2.
- `get_ligand = False` Use `get_ligand=True` to retrieve ligands.
- `get_ramcode_only = False` Generate Rama code for input pdb/cif only. This is for developers only.
- `get_xml_only = False` Get BLAST XML output returned as a string object. No coordinate superposition will be performed. Developers only.
- `use_pdb100aa = False` Use PDB100 sequence database for sequence search.
- `use_custom_db = False` Use custom database specified by `custom_db_files/custom_db_dir`.
- `custom_db_dir = None` The directory of pdb/cif files to make custom database. Default is current directory
- `custom_db_files = None` Filenames of the pdb/cif files separated by spaces for database. If none specified, all pdb/cif in the `custom_db_dir` will be collected
- `atom_selection = 'all'` Choose part of the pdb used in the search (default=all). for example: chain B, resseq 113:219, ... etc.
- `get_pdb = 10` `get_pdb=N` will collect and superpose the top N homologous pdbs. Use `get_pdb=0` to disable this option.
- `deposited_before = 0` Specify the latest year of matching structures to be considered for scoring. Pdb's deposited after this year will be discarded.
- `deposited_after = 0` Specify the earliest year of matching structures to be considered for scoring. Pdb's deposited before this year will be discarded.
- `batch_size = 0` Process the pdbs in batch of `<batch_size>` until `<min_match>` hits are identified or until all `<get_pdb>` pdbs are processed
- `min_match = 0` Finish structure_search when `<min_match>` matches are found. Usually uses with `<trim_ends>` to exit the search once find suitable pdbs.
- `keep_all_pdb = False` Keep all the PDB files, including full PDB, PDB_Chain and superposed PDB_Chain. Default is False which will keep only superposed PDB_Chain files in the directory specified in the output message.
- `trim_ends = False` Remove terminal residues of hit pdbs extending beyond those of the the target pdb.
- `write_pdb = True` Set to False if no output pdb file is needed. Sometimes useful if use Structure_Search within another program and only want to pass pdb objects.
- `write_results = True` Set to False if no output results/log files is needed. Useful when calling Structure_Search within another program and only want to pass pdb objects.
- `trim_hit_pdb = False` Remove extra domains, extended loops, and unfit portions of hit pdbs after superposed to the target pdb.
- `pickle_hits = False` Pickle blast hit results from xml output.
- `coot_display = False` (default) Display output pdb files in coot.
- `ask_coot = True` prompt for coot display options

- `PDB_MIRRORDIR = None` Enter the top directory of local RCSB PDB mirror. The program will try to retrieve PDBs and/or structure factors from this mirror first. Note this assumes the directory trees under it follows those in RCSB -- pdb files as 'pdb####.ent.gz' in `PDB_MIRRORDIR/data/structures/divided/pdb` directory. If you use PDB's rsync script, this variable would be the same as the `$MIRRORDIR` set in the script
- `PDB_MIRROR_MMCIF = None` Enter the parent directory of the mmcif files in the local PDB mirror. MMCIFs will be retrieved from subdirectory `##` where `##` are the second and third letters in the PDB id. This keyword should be `$PDB_MIRRORDIR/data/structures/divided/mmcif` directory.
- `PDB_MIRROR_PDB = None` Enter the parent directory of the PDB files in the local PDB mirror. PDBs will be retrieved from subdirectory `##` where `##` are the second and third letters in the PDB id. This keyword should be `$PDB_MIRRORDIR/data/structures/divided/pdb` directory. We recommend setting `PDB_MIRRORDIR` and it will take care of both `PDB_MIRROR_PDB` and others together. However, users may choose to specify `PDB_MIRROR_PDB` directly
- `PDB_MIRROR_STRUCTURE_FACTORS = None` Enter the parent directory of the PDB files in the local PDB mirror. structure factors s will be retrieved from subdirectory `##` where `##` are the second and third letters in the PDB id. This keyword should be the same as the `$PDB_MIRRORDIR/data/structures/divided/structure_factors` directory. We recommend setting `PDB_MIRRORDIR` and it will take care of both `PDB_MIRROR_PDB` and `DB_MIRROR_STRUCTURE_FACTORS` together. However, users may choose to specify `PDB_MIRROR_STRUCTURE_FACTORS` directly
- `local_pdb_dir = None` Enter the path directly to your local PDB repository.
- `verbose = False` verbose output
- `debug = False` debugging output
- `job_title = None` Job title in PHENIX GUI, not used on command line
- guiGUI-specific parameter required for output directory `output_dir = None`
- `output_dir = None`
- `controlstop_after_dock = None` Stop after docking step `stop_after_morph = False` Stop after morphing step `read_files = False` Read existing output files and use them if present `write_files = True` Write output files `nproc = 1` Number of processors to use on your local machine `ignore_symmetry_conflicts = True` You can ignore the symmetry information (CRYST1) from coordinate files. This may be necessary if your model has been placed in a box with `box_map` for example. `random_seed = 171731` Random seed `max_dirs = 1000` Maximum number of directories (dock_and_build_xxx) `verbose = False` Verbose output `quick = True` Run quickly
- `stop_after_dock = None` Stop after docking step
- `stop_after_morph = False` Stop after morphing step
- `read_files = False` Read existing output files and use them if present
- `write_files = True` Write output files
- `nproc = 1` Number of processors to use on your local machine
- `ignore_symmetry_conflicts = True` You can ignore the symmetry information (CRYST1) from coordinate files. This may be necessary if your model has been placed in a box with `box_map` for example.
- `random_seed = 171731` Random seed
- `max_dirs = 1000` Maximum number of directories (dock_and_build_xxx)
- `verbose = False` Verbose output

- quick = True Run quickly
- guiGUI-specific parameter required for output directoryoutput_dir = None
- output_dir = None

Docking processed AlphaFold2 and other predicted models in cryo-EM maps

dock_predicted_model.html

Author(s)

- dock_predicted_model: Tom Terwilliger

Purpose

Dock predicted model docks the domains from a model produced by AlphaFold, RoseTTAFold and other prediction software into a cryo EM map. It uses the connectivity of the model as a restraint in the docking process so that the docked domains normally are in a reasonable arrangement. It can take map symmetry into account.

The dock_predicted_model program is normally the second step in working on an AlphaFold or other predicted model with a cryo-EM structure.

The first step is to process the predicted model by trimming off all the uncertain residues in the predicted model and breaking up the remaining structure into a best guess of rigid domains with phenix.process_predicted_model .

The next step is to dock each of the domains of the processed model into the map, keeping plausible connectivity. This is done with phenix.dock_predicted_model .

The third step is to morph the predicted model onto the docked domains and then to rebuild all the parts of the predicted model using the density in the map.

Note that you can run all three steps sequentially with phenix.dock_and_rebuild .

How dock_predicted_model works:

The model input to dock_predicted_model is a model file that comes from your starting predicted model (AlphaFold model) but which has been split so that each chain represents one domain (fixed rigid unit) of the predicted model, and from which all uncertain residues are removed. This is normally the output of phenix.process_predicted_model .

Note that you can work with one chain at a time even in a cryo-EM map that has many chains.

The map input to dock_predicted model is normally your best sharpened or density-modified cryo-EM map. It can also be a map generated by any other procedure (including crystallography).

If you are able to mask your map, keeping only the part representing the region where this model (one chain) belongs, that can be very helpful. You can also box the map around this region. If you have a map that has many chains, this masking can greatly shorten the time for docking. If you don't know where in your map the model goes at all, you can supply the entire map. If your map has symmetry this will normally be found automatically.

The first step in docking the predicted model is to extract the unique part of the map. If the map is asymmetric, this is the region where there is density. If it is symmetric, it is the unique part of the density, chosen to make a compact unit. Note that this step may not be perfect, particularly in the case of a very asymmetric molecule. You can possibly do much better by segmenting the map by hand and selecting just the region representing

your molecule if you have the time.

The next step is to sort the domains in the processed model by size and dock them in the map. Each domain is docked and refined by rigid-body refinement. Then the transformation obtained for that domain is applied to all the other domains as a quick guess as to how they should be placed. If they match the density, they are refined as well. If a complete model can be obtained by connecting these pieces (with distances between residues in one domain and the next residues in another domain compatible with the number of residues in between that are missing), then the docking is complete. If it is not, additional domains are docked, until all have been tried.

When all domains are placed, all the residues in the domains are sorted and a single clean chain with chain ID from the original predicted model and gaps corresponding to the processed model is produced.

Examples

Standard run of dock_predicted_model:

Running dock_predicted_model is easy. From the command-line you can type:

```
phenix.dock_predicted_model model=my_model.pdb \
  processed_model_file=my_model_processed.pdb map_file=my_map.ccp4 \
  resolution=3
```

This will dock the domains (one for each chain ID) from my_model_processed.pdb into my_map.ccp4. It will rename all the chains to match the chain in my_model.pdb and arrange all the residues in order.

Possible Problems

For complicated molecules, this procedure may not work well because it does not examine every possible docking position of every domain, and a complex structure might have more than one copy of a particular chain. You can improve the success a lot if you are able to mask the map around one copy of the chain you are looking for.

If that doesn't work, you can try to dock the domains from your AlphaFold model using phenix.dock_in_map and asking for multiple copies of the domains, then you could manually try to choose which domains go together. You can then create one PDB file with one copy of each domain (you will need to create a PDB file that has just one chain ID and all residues arranged sequentially, but with gaps allowed) and go on to phenix.dock_and_rebuild with that as your docked model.

Specific limitations and problems:

Literature

Additional information

List of all available keywords

job_title = None Job title in PHENIX GUI, not used on command line
input_filesseq_file = None Input sequence file. Required if no models supplied. Used to create predicted models. Format should be Fasta or simple sequences separated by blank lines. One sequence per chain to be generated. Must match input models if both are supplied. (Fasta format or sequences separated by blank lines)
xray_data_file = None Input X-ray

data file (MTZ format).xray_data_label = None Label specifying which data column to use from
 xray_data_filexray_test_flag_label = None Label specifying which test flag column (if any) to use from
 xray_data_filetruncate_models_to_match_sequences = True If input sequences start after or end before
 supplied predicted models, trim the predicted models to match sequencesdensity_select = None If set, trim
 map to region containing molecule (non-zero region) before use. Recommended for cryo-EM maps that have
 a large map that is mostly unused. Default is True if map is full-size and False if not.predicted_model = None
 One or more input models in one or more files (normally predicted models) to be placed. Normally should start
 at residue 1 and match sequence file exactly.b_value_field_is = *plddt rmsd b_value The B-factor field in
 predicted models can be pLDDT (confidence, 0-1 or 0-100) or rmsd (Å) or a B-factor. Only applies to protein
 chains (always B-factor for non-protein).msa_list = None Multiple sequence alignment (MSA) file. Format is
 a3m only. First sequence in an MSA file must match a sequence in the sequence file
 exactly.models_are_already_placed = False You can specify that all models (predicted_model or model) are
 already placed (docked, placed in unit cell). No docking or MR is done if so. Equivalent to specifying a
 scaffold_model with the same contents as the model.model_copies_list = None List of number of copies to
 find of each model in predicted_model. Default is 1 (must specify all or none of them) . Specify all together like
 model_copies_list=1,2,1,1processed_model_file = None Processed model file (e.g., from
 phenix.process_predicted_model). If supplied, skip the process_predicted_model step and use this file. This
 model is expected to have one chain for each domain and actual B-values in the B-value field. It is expected
 that all poorly-predicted residues have been removed.docked_model_file = None Docked model (e.g., output
 of dock_processed_model). If supplied, use this file as the docked model. (Skip process_predicted_model
 and docking steps This model is expected to have a single chain with gaps for parts of the model that are not
 accurately known from the prediction. NOTE: You still need to supply the predicted model as the input model
 for this procedure in addition to this docked model.morphed_model_file = None Morphed model (e.g., full
 model morphed to match output of dock_processed_model). If supplied, skip the process_predicted_model
 and docking steps and use this file as the morphed model. This model is expected to have a single chain with
 no gaps. NOTE: You do not need to supply the predicted model as the input model for this procedure in
 addition to this morphed model.previous_model_file = None Previous model (from a previous run or external).
 Must match sequence of working model exactly. Used as a source of possible model information and as a
 hypothesis for docking of working model. If docked_model_file is also supplied, previous_model is used only
 as source of model information.scaffold_model = None Scaffold model. If supplied, used as target for docking
 of each chain in predicted_model or model. Must be similar in sequence to models to be docked. If supplied,
 model_copies_list is ignored and docking is done by superposition onto this structure instead of docking into
 density or MR. If the scaffold model does not have all chains represented, they are added in by
 docking.scaffold_minimum_chain_cc = 0.35 Chains in scaffold model that have scaffold_minimum_cc are not
 re-docked, but those with l...

- job_title = None Job title in PHENIX GUI, not used on command line

- input_filesseq_file = None Input sequence file. Required if no models supplied. Used to create predicted models. Format should be Fasta or simple sequences separated by blank lines. One sequence per chain to be generated. Must match input models if both are supplied. (Fasta format or sequences separated by blank lines)xray_data_file = None Input X-ray data file (MTZ format).xray_data_label = None Label specifying which data column to use from xray_data_filexray_test_flag_label = None Label specifying which test flag column (if any) to use from xray_data_filetruncate_models_to_match_sequences = True If input sequences start after or end before supplied predicted models, trim the predicted models to match sequencesdensity_select = None If set, trim map to region containing molecule (non-zero region) before use. Recommended for cryo-EM maps that have a large map that is mostly unused. Default is True if map is full-size and False if not.predicted_model = None One or more input models in one or more files (normally predicted models) to be placed. Normally should start at residue 1 and match sequence file

exactly.b_value_field_is = *plddt rmsd b_value The B-factor field in predicted models can be pLDDT (confidence, 0-1 or 0-100) or rmsd (A) or a B-factor. Only applies to protein chains (always B-factor for non-protein).msa_list = None Multiple sequence alignment (MSA) file. Format is a3m only. First sequence in an MSA file must match a sequence in the sequence file exactly.models_are_already_placed = False You can specify that all models (predicted_model or model) are already placed (docked, placed in unit cell). No docking or MR is done if so. Equivalent to specifying a scaffold_model with the same contents as the model.model_copies_list = None List of number of copies to find of each model in predicted_model. Default is 1 (must specify all or none of them) . Specify all together like model_copies_list=1,2,1,1processed_model_file = None Processed model file (e.g., from phenix.process_predicted_model). If supplied, skip the process_predicted_model step and use this file. This model is expected to have one chain for each domain and actual B-values in the B-value field. It is expected that all poorly-predicted residues have been removed.docked_model_file = None Docked model (e.g., output of dock_processed_model). If supplied, use this file as the docked model. (Skip process_predicted_model and docking steps This model is expected to have a single chain with gaps for parts of the model that are not accurately known from the prediction. NOTE: You still need to supply the predicted model as the input model for this procedure in addition to this docked model.morphed_model_file = None Morphed model (e.g., full model morphed to match output of dock_processed_model). If supplied, skip the process_predicted_model and docking steps and use this file as the morphed model. This model is expected to have a single chain with no gaps. NOTE: You do not need to supply the predicted model as the input model for this procedure in addition to this morphed model.previous_model_file = None Previous model (from a previous run or external). Must match sequence of working model exactly. Used as a source of possible model information and as a hypothesis for docking of working model. If docked_model_file is also supplied, previous_model is used only as source of model information.scaffold_model = None Scaffold model. If supplied, used as target for docking of each chain in predicted_model or model. Must be similar in sequence to models to be docked. If supplied, model_copies_list is ignored and docking is done by superposition onto this structure instead of docking into density or MR. If the scaffold model does not have all chains represented, they are added in by docking.scaffold_minimum_chain_cc = 0.35 Chains in scaffold model that have scaffold_minimum_cc are not re-docked, but those with lower CC are.base_scaffold_minimum_chain_cc = -1 Value of scaffold_...

- seq_file = None Input sequence file. Required if no models supplied. Used to create predicted models. Format should be Fasta or simple sequences separated by blank lines. One sequence per chain to be generated. Must match input models if both are supplied. (Fasta format or sequences separated by blank lines)
- xray_data_file = None Input X-ray data file (MTZ format).
- xray_data_label = None Label specifying which data column to use from xray_data_file
- xray_test_flag_label = None Label specifying which test flag column (if any) to use from xray_data_file
- truncate_models_to_match_sequences = True If input sequences start after or end before supplied predicted models, trim the predicted models to match sequences
- density_select = None If set, trim map to region containing molecule (non-zero region) before use. Recommended for cryo-EM maps that have a large map that is mostly unused. Default is True if map is full-size and False if not.
- predicted_model = None One or more input models in one or more files (normally predicted models) to be placed. Normally should start at residue 1 and match sequence file exactly.
- b_value_field_is = *plddt rmsd b_value The B-factor field in predicted models can be pLDDT (confidence, 0-1 or 0-100) or rmsd (A) or a B-factor. Only applies to protein chains (always B-factor for

non-protein).

- `msa_list` = None Multiple sequence alignment (MSA) file. Format is a3m only. First sequence in an MSA file must match a sequence in the sequence file exactly.
- `models_are_already_placed` = False You can specify that all models (`predicted_model` or `model`) are already placed (docked, placed in unit cell). No docking or MR is done if so. Equivalent to specifying a `scaffold_model` with the same contents as the model.
- `model_copies_list` = None List of number of copies to find of each model in `predicted_model`. Default is 1 (must specify all or none of them) . Specify all together like `model_copies_list=1,2,1,1`
- `processed_model_file` = None Processed model file (e.g., from `phenix.process_predicted_model`). If supplied, skip the `process_predicted_model` step and use this file. This model is expected to have one chain for each domain and actual B-values in the B-value field. It is expected that all poorly-predicted residues have been removed.
- `docked_model_file` = None Docked model (e.g., output of `dock_processed_model`). If supplied, use this file as the docked model. (Skip `process_predicted_model` and docking steps This model is expected to have a single chain with gaps for parts of the model that are not accurately known from the prediction. NOTE: You still need to supply the predicted model as the input model for this procedure in addition to this docked model.
- `morphed_model_file` = None Morphed model (e.g., full model morphed to match output of `dock_processed_model`). If supplied, skip the `process_predicted_model` and docking steps and use this file as the morphed model. This model is expected to have a single chain with no gaps. NOTE: You do not need to supply the predicted model as the input model for this procedure in addition to this morphed model.
- `previous_model_file` = None Previous model (from a previous run or external). Must match sequence of working model exactly. Used as a source of possible model information and as a hypothesis for docking of working model. If `docked_model_file` is also supplied, `previous_model` is used only as source of model information.
- `scaffold_model` = None Scaffold model. If supplied, used as target for docking of each chain in `predicted_model` or `model`. Must be similar in sequence to models to be docked. If supplied, `model_copies_list` is ignored and docking is done by superposition onto this structure instead of docking into density or MR. If the scaffold model does not have all chains represented, they are added in by docking.
- `scaffold_minimum_chain_cc` = 0.35 Chains in scaffold model that have `scaffold_minimum_cc` are not re-docked, but those with lower CC are.
- `base_scaffold_minimum_chain_cc` = -1 Value of `scaffold_minimum_chain_cc` in cases where scaffold is known to be good.
- `fragments_model_file` = None Fragments model (e.g., `map_to_model.pdb`). Used as a source of possible model information and as a hypothesis for docking of working model. If `docked_model_file` is also supplied, `fragments_model` is used only as source of model information.
- `model_to_rebuild_file` = None Model to be rebuilt. Must be already in place (already docked). Must match supplied sequences exactly.
- `symmetry_file` = None Symmetry file (.ncs_spec format or MATRX records) with reconstruction symmetry. Used in identification of unique part of map. NOTE: symmetry file applies to map file in original position. NOTE 2: only proper symmetry (point-group, helical) is allowed NOTE 3: Only applies to cryo-EM maps

- `pae_file` = None Optional input json file with matrix of inter-residue estimated errors (pae file)
- `distance_model_file` = None Distance_model_file. A PDB or mmCIF file containing the model corresponding to the PAE matrix. Only needed if `weight_by_ca_ca_distances` is True.
- `search_model_copies` = None Used internally only
- `search_model` = None used internally only
- `map_model` = None Input map file (for cryo-EM). This can be a boxed or masked map showing just the molecule to dock (best) or a full map with symmetry. If your map has symmetry be sure to set `asymmetric_map` = False. If you have a map with symmetry you can supply a symmetry file if you want. Otherwise symmetry will be automatically determined.
- `half_map` = None Input half map files. Usually supply one full map or 2 half maps
- `model` = None Input predicted model (e.g., AlphaFold model). Assumed to have pLDDT values in B-value field (or RMSD values). May have multiple chains. Normally use `predicted_model` instead.
- `output_files` = None Output files with superposed models will begin with this prefix
- `output_model_prefix` = None Output files with superposed models will begin with this prefix
- `output_seq_file` = None Sequence file (possibly edited) written to this file by PredictAndBuild
- `pdb_out` = None Used internally only
- `temp_dir` = None Temporary directory. Default is `dock_and_build_xx` where xx creates new directory
- `crystal_info` = None High-resolution limit for main search. This can be lower resolution than the data. The search is quicker at lower resolution. If your model is poor, try 2-3 A lower resolution than your data (i.e, if your data is 2.5 A, try 5 A).
- `scattering_table` = *None n_gaussian wk1995 it1992 electron neutron Choice of scattering table for structure factor calculations. Default for X-ray is n_gaussian, for cryoEM is electron.
- `wrapping` = None You can specify whether the map is wrapped (can map values outside bounds to inside with cell translations).
- `asymmetric_map` = None Specifies that this is an asymmetric map and no symmetry is to be supplied or found. Alternative to supplying a symmetry file or symmetry. Applies to cryo-EM reconstructions only.
- `solvent_content` = None Solvent fraction (content) of the cell. You can specify the fraction of the volume of this cell that is taken up by the macromolecule. Normally set automatically. Values go from 0 to 1.
- `pdb70_text` = None Database of PDB entries to be used as templates (pdb70 file). Normally used internally only
- `sequence` = None Old-style sequence string (alternative to sequence file).
- `chain_type` = *PROTEIN DNA RNA Chain type
- `msa` = None MSA as text. Supply MSA as text or as an `msa_file`. Used internally
- `templates_as_string` = None Templates as string (single PDB file with one or more chains)
- `space_group` = None Space group (normally read from the data file, applies to X-ray)
- `space_group_alternatives` = *HAND ALL NONE Space group alternatives for MR: ALL: all space groups in point group HAND: enantiomer and listed space group LIST: list supplied (not available) NONE: only supplied space group
- `unit_cell` = None Unit Cell (normally read from the data file,

applies to X-ray)unique_sequencesFile names and data labels.sequence = "Enter or edit sequence"
copies = 1 Copies of this sequencelabel = ""

- resolution = None High-resolution limit for main search. This can be lower resolution than the data. The search is quicker at lower resolution. If your model is poor, try 2-3 Å lower resolution than your data (i.e, if your data is 2.5 Å, try 5 Å).
- scattering_table = *None n_gaussian wk1995 it1992 electron neutron Choice of scattering table for structure factor calculations. Default for X-ray is n_gaussian, for cryoEM is electron.
- wrapping = None You can specify whether the map is wrapped (can map values outside bounds to inside with cell translations).
- asymmetric_map = None Specifies that this is an asymmetric map and no symmetry is to be supplied or found. Alternative to supplying a symmetry file or symmetry. Applies to cryo-EM reconstructions only.
- solvent_content = None Solvent fraction (content) of the cell. You can specify the fraction of the volume of this cell that is taken up by the macromolecule. Normally set automatically. Values go from 0 to 1.
- pdb70_text = None Database of PDB entries to be used as templates (pdb70 file). Normally used internally only
- sequence = None Old-style sequence string (alternative to sequence file).
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- msa = None MSA as text. Supply MSA as text or as an msa_file. Used internally
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- unit_cell = None Unit Cell (normally read from the data file, applies to X-ray)
- unique_sequencesFile names and data labels.sequence = "Enter or edit sequence" copies = 1 Copies of this sequencelabel = ""
- sequence = "Enter or edit sequence"
- copies = 1 Copies of this sequence
- label = ""
- process_predicted_modelremove_low_confidence_residues = True Remove low-confidence residues (based on minimum plddt or maximum_rmsd, whichever is specified)continuous_chain = False When removing low-confidence residues, only trim from ends
split_model_by_compact_regions = True Split model into compact regions after removing low-confidence residues.
maximum_domains = 3 Maximum domains to obtain. You can use this to merge the closest domains at the end of splitting the model. Make it bigger (and optionally make domain_size smaller) to get more domains. If model is processed in chunks, maximum_domains will apply to each chunk.
domain_size = 15 Approximate size of domains to be found (Å units). This is the resolution that will be used to make a domain map. If you are getting too many domains, try making domain_size bigger (maximum is 70 Å).
adjust_domain_size = True If more than maximum_domains are initially found, increase domain_size in increments of 5 Å and take the value that gives the smallest number of domains, but at least maximum_domains.
minimum_domain_length = None Minimum length of a domain to keep (reject at end if smaller). Default is 10 if no pae matrix or alt_pae_params=False and 20 for pae_matrix with alt_pae_params=True
maximum_fraction_close = 0.3 Maximum fraction of CA in one domain close to one in another before merging

themminimum_sequential_residues = None Minimum length of a short segment to keep (reject at end).
 Default is 5 if no pae matrix or alt_pae_params=False and 4 for pae_matrix with
 alt_pae_params=Trueminimum_remainder_sequence_length = 15 used to choose whether the sequence
 of a removed segment is written to the remainder sequence file.b_value_field_is = *plddt rmsd b_value
 The B-factor field in predicted models can be pLDDT (confidence, 0-1 or 0-100) or rmsd (A) or a
 B-factorinput_plddt_is_fractional = None You can specify if the input plddt values (in B-factor field) are
 fractional (0-1) or not (0-100). By default if all values are between 0 and 1 it is fractional.minimum_plddt =
 None If low-confidence residues are removed, the cutoff is defined by minimum_plddt or maximum_rmsd,
 whichever is defined (you cannot define both). A minimum plddt of 0.70 corresponds to a maximum rmsd
 of 1.5. Minimum plddt values are fractional or not depending on the value of
 input_plddt_is_fractional.maximum_rmsd = 1.5 If low-confidence residues are removed, the cutoff is
 defined by minimum_plddt or maximum_rmsd, whichever is defined (you cannot define both). A minimum
 plddt of 0.70 corresponds to a maximum rmsd of 1.5. Minimum plddt values are fractional or not
 depending on the value of input_plddt_is_fractional.default_maximum_rmsd = 1.5 Default value of
 maximum_rmsd, used if maximum_rmsd is not setsubtract_minimum_b = False If set, subtract the lowest
 B-value from all B-values just before writing out the final files. Does not affect the cutoff for removing low-
 confidence residues.pae_power = None If PAE matrix (predicted alignment error matrix) is supplied, each
 edge in the graph will be weighted proportional to $(1/\text{pae}^{\text{pae_power}})$. Use this to try and get the number
 of domains that you want (try 1, 0.5, 1.5, 2). Default is 1 alt_pae_params=False and 2 for
 alt_pae_params=Truepae_cutoff = None If PAE matrix (predicted alignment error matrix) is supplied,
 graph edges will only be created for residue pairs with $\text{pae} < \text{pae_cutoff}$. Default is 5
 alt_pae_params=False and 4 for alt_pae_params=Truepae_graph_resolution = None If PAE matrix
 (predicted alignment error matrix) is supplied, pae_graph_resolution regulates how aggressively the
 clustering algorithm is. Smaller values lead to larger clusters. Value should be larger than zero, and
 values larger than 5 are unlikely to be useful, Default is 0.5 alt_pae_params=False and 4 for
 alt_pae_params=Truealt_pae_params = True If PAE matrix is supplied, use alternative set of defaults
 (minimum_domain_length=20 minimum_sequential_residues=10 pae_power=2 pae_cutoff=4
 pae_graph_resolution...

- remove_low_confidence_residues = True Remove low-confidence residues (based on minimum plddt or maximum_rmsd, whichever is specified)
- continuous_chain = False When removing low-confidence residues, only trim from ends
- split_model_by_compact_regions = True Split model into compact regions after removing low-confidence residues.
- maximum_domains = 3 Maximum domains to obtain. You can use this to merge the closest domains at the end of splitting the model. Make it bigger (and optionally make domain_size smaller) to get more domains. If model is processed in chunks, maximum_domains will apply to each chunk.
- domain_size = 15 Approximate size of domains to be found (A units). This is the resolution that will be used to make a domain map. If you are getting too many domains, try making domain_size bigger (maximum is 70 A).
- adjust_domain_size = True If more that maximum_domains are initially found, increase domain_size in increments of 5 A and take the value that gives the smallest number of domains, but at least maximum_domains.
- minimum_domain_length = None Minimum length of a domain to keep (reject at end if smaller). Default is 10 if no pae matrix or alt_pae_params=False and 20 for pae_matrix with alt_pae_params=True
- maximum_fraction_close = 0.3 Maximum fraction of CA in one domain close to one in another before merging them

- `minimum_sequential_residues = None` Minimum length of a short segment to keep (reject at end). Default is 5 if no pae matrix or `alt_pae_params=False` and 4 for pae_matrix with `alt_pae_params=True`
- `minimum_remainder_sequence_length = 15` used to choose whether the sequence of a removed segment is written to the remainder sequence file.
- `b_value_field_is = *plddt rmsd b_value` The B-factor field in predicted models can be pLDDT (confidence, 0-1 or 0-100) or rmsd (A) or a B-factor
- `input_plddt_is_fractional = None` You can specify if the input plddt values (in B-factor field) are fractional (0-1) or not (0-100). By default if all values are between 0 and 1 it is fractional.
- `minimum_plddt = None` If low-confidence residues are removed, the cutoff is defined by `minimum_plddt` or `maximum_rmsd`, whichever is defined (you cannot define both). A minimum plddt of 0.70 corresponds to a maximum rmsd of 1.5. Minimum plddt values are fractional or not depending on the value of `input_plddt_is_fractional`.
- `maximum_rmsd = 1.5` If low-confidence residues are removed, the cutoff is defined by `minimum_plddt` or `maximum_rmsd`, whichever is defined (you cannot define both). A minimum plddt of 0.70 corresponds to a maximum rmsd of 1.5. Minimum plddt values are fractional or not depending on the value of `input_plddt_is_fractional`.
- `default_maximum_rmsd = 1.5` Default value of `maximum_rmsd`, used if `maximum_rmsd` is not set
- `subtract_minimum_b = False` If set, subtract the lowest B-value from all B-values just before writing out the final files. Does not affect the cutoff for removing low- confidence residues.
- `pae_power = None` If PAE matrix (predicted alignment error matrix) is supplied, each edge in the graph will be weighted proportional to $(1/\text{pae}^{\text{pae_power}})$. Use this to try and get the number of domains that you want (try 1, 0.5, 1.5, 2). Default is 1 `alt_pae_params=False` and 2 for `alt_pae_params=True`
- `pae_cutoff = None` If PAE matrix (predicted alignment error matrix) is supplied, graph edges will only be created for residue pairs with `pae < pae_cutoff`. Default is 5 `alt_pae_params=False` and 4 for `alt_pae_params=True`
- `pae_graph_resolution = None` If PAE matrix (predicted alignment error matrix) is supplied, `pae_graph_resolution` regulates how aggressively the clustering algorithm is. Smaller values lead to larger clusters. Value should be larger than zero, and values larger than 5 are unlikely to be useful, Default is 0.5 `alt_pae_params=False` and 4 for `alt_pae_params=True`
- `alt_pae_params = True` If PAE matrix is supplied, use alternative set of defaults (`minimum_domain_length=20` `minimum_sequential_residues=10` `pae_power=2` `pae_cutoff=4` `pae_graph_resolution=4`). Standard parameters (`alt_pae_params=False`) are: (`minimum_domain_length=10` `minimum_sequential_residues=5` `pae_power=1` `pae_cutoff=5` `pae_graph_resolution=0.5`).
- `weight_by_ca_ca_distance = False` Adjust the edge weighting for each residue pair according to the distance between CA residues. If this is True, then `distance_model` can be provided, otherwise supplied model will be used. See also `distance_power`
- `distance_power = 1` If `weight_by_ca_ca_distance` is True, then edge weights will be multiplied by $1/\text{distance}^{\text{distance_power}}$.
- `stop_if_no_residues_obtained = True` Raise Sorry and stop if processing yields no residues
- `keep_all_if_no_residues_obtained = False` Keep everything if processing yields no residues
- `vrms_from_rmsd_intercept = 0.25` Estimate of vrms (error in model) from pLDDT will be based on `vrms_from_rmsd_intercept + vrms_from_rmsd_slope * pLDDT` where mean pLDDT of

non-low_confidence_residues is used.

- `vrms_from_rmsd_slope = 1.0` Estimate of vrms (error in model) from pLDDT will be based on $\text{vrms_from_rmsd_intercept} + \text{vrms_from_rmsd_slope} * \text{pLDDT}$ where mean pLDDT of non-low_confidence_residues is used.

- `break_into_chunks_if_length_is = 1500` If a sequence is at least `break_into_chunks_if_length_is`, break it into chunks of length `chunk_size` with overlap of `overlap_size` for domain identification using `split_model_by_compact_regions` without a `pae_matrix`

- `chunk_size = 600` If a sequence is at least `break_into_chunks_if_length_is`, break it into chunks of length `chunk_size` with overlap of `overlap_size` for domain identification using `split_model_by_compact_regions` without a `pae_matrix`

- `overlap_size = 200` If a sequence is at least `break_into_chunks_if_length_is`, break it into chunks of length `chunk_size` with overlap of `overlap_size` for domain identification using `split_model_by_compact_regions` without a `pae_matrix`

- `searchdock_chains_individually = False` Dock each chain (identified by `chain_id` field) individually. This is useful if the chains might have different orientations in the map compared to the search model, such as in a search model that is a predicted model. Normally used along with `create_unique_chain_at_end=True` for predicted models to put the model back together. `create_unique_chain_at_end = None` Take all docked chains and try to create one chain that has no duplicate residue numbers and that has distances between ends of fragments consistent with the number of residues between them. If symmetry has been applied, create N copies representing that symmetry. Default is True if `dock_chains_individually = True` and False otherwise. `minimum_cc_to_keep_domain = 0.2` Do not keep domains in `create_unique_chain_at_end` if cc is less than

`minimum_cc_to_keep_domain`. `weight_sequential_fragments_by_distance = None` Try to dock a chain in a way that minimizes the distance between its first residue and the previously-docked residue with the highest residue number less than this, and similarly for the last residue and the next available placed residue. Default is True if `create_unique_chain_at_end` is set. Normally use only if you have a model with a single chain and you are docking pieces of that chain. `choose_better_of_individual_and_group_docking = None` Dock entire search model (normally one only) and also by chain...pick better-fitting of the two for each chain. `low_res_search = True` Try to fit by searching at low resolution first. `dock_with_mr = True` Try using MR to dock first. `ssm_search = None` Try to fit by searching for secondary structure. Default is False for `dock_in_map` and True for `dock_and_rebuild` if `dock_with_mr` is not used. `refine_cycles = 3` rigid-body refinement cycles. `ssm_search_min_cc = 0.30` Stop ssm search if this cc

achieved. `backup_resolution_cutoff_searches = 2` If initial fitting does not work, try up to this many times increasing resolution by `backup_resolution_ratio` each time. `backup_resolution_ratio = 1.67` If initial fitting does not work, try up to `backup_resolution_cutoff_searches` times increasing the resolution by

`backup_resolution_ratio` each time. `resolution_radius_scale = 0.5` Resolution for low-res search will be `resolution_radius_scale` times the radius of gyration of the search model. `align_moments = False` Try to fit by aligning moments of inertia if size of density region and molecule are similar. `max_radius_ratio = 2`.

Radius of gyration of molecule and density must be within `max_radius_ratio` of each other. `radius_scale = 1.5` Mask for density and atoms will be `radius_scale` times the radius of gyration of model. `use_symmetry = True` If `search_model_copies` or `search_map_copies` are the same for each model and greater than one and `allow_symmetry` is set, use symmetry in the map to place all copies after the first one. `skip_if_low_cc = True` Skip solution before rigid-body refinement if map-model CC is less than half of

`min_cc`. `rigid_body_refinement = True` Run rigid-body refinement on final

model. `rigid_body_refinement_single_unit = True` Run rigid-body refinement with just one unit (do not break up into chains). `rigid_body_refinement_split_method = *chain_id segid` When splitting up molecule for rigid-body refinement (if `rigid_body_refinement_single_unit=False`), use either `chain_id` or `segid` to

split up molecule rigid_body_refinement_resolution = None Run rigid-body refinement at this resolution if specified
 append_to_fixed_model = True Append placed search model to fixed model (if any) after search
 min_cc = 0.4 If quick run, stop if minimum CC is achieved in local search. Also always skip if starting CC_mask is less than 1/4 min_cc.
 run_in_boxes = True Run on sub-boxes and combine at the end
 target_box_size = 60 Try to get boxes about this big on a side (grid units)
 target_boxes = None Try to get this many boxes. Default is nproc unless this makes box size much smaller than target_box_size
 box_to_run = None Run only this box
 box_overlap_scale = 1 box overlap (overlap)

- dock_chains_individually = False Dock each chain (identified by chain_id field) individually. This is useful if the chains might have different orientations in the map compared to the search model, such as in a search model that is a predicted model. Normally used along with create_unique_chain_at_end=True for predicted models to put the model back together.

- create_unique_chain_at_end = None Take all docked chains and try to create one chain that has no duplicate residue numbers and that has distances between ends of fragments consistent with the number of residues between them. If symmetry has been applied, create N copies representing that symmetry. Default is True if dock_chains_individually = True and False otherwise.

- minimum_cc_to_keep_domain = 0.2 Do not keep domains in create_unique_chain_at_end if cc is less than minimum_cc_to_keep_domain.

- weight_sequential_fragments_by_distance = None Try to dock a chain in a way that minimizes the distance between its first residue and the previously-docked residue with the highest residue number less than this, and similarly for the last residue and the next available placed residue. Default is True if create_unique_chain_at_end is set. Normally use only if you have a model with a single chain and you are docking pieces of that chain.

- choose_better_of_individual_and_group_docking = None Dock entire search model (normally one only) and also by chain...pick better-fitting of the two for each chain

- low_res_search = True Try to fit by searching at low resolution first

- dock_with_mr = True Try using MR to dock first

- ssm_search = None Try to fit by searching for secondary structure. Default is False for dock_in_map and True for dock_and_rebuild if dock_with_mr is not used.

- refine_cycles = 3 rigid-body refinement cycles

- ssm_search_min_cc = 0.30 Stop ssm search if this cc achieved

- backup_resolution_cutoff_searches = 2 If initial fitting does not work, try up to this many times increasing resolution by backup_resolution_ratio each time

- backup_resolution_ratio = 1.67 If initial fitting does not work, try up to backup_resolution_cutoff_searches times increasing the resolution by backup_resolution_ratio each time

- resolution_radius_scale = 0.5 Resolution for low-res search will be resolution_radius_scale times the radius of gyration of the search model.

- align_moments = False Try to fit by aligning moments of inertia if size of density region and molecule are similar

- max_radius_ratio = 2. Radius of gyration of molecule and density must be within max_radius_ratio of each other

- radius_scale = 1.5 Mask for density and atoms will be radius_scale times the radius of gyration of model

- use_symmetry = True If search_model_copies or search_map_copies are the same for each model and greater than one and allow_symmetry is set, use symmetry in the map to place all copies after the first

one.

- `skip_if_low_cc = True` Skip solution before rigid-body refinement if map-model CC is less than half of `min_cc`.
- `rigid_body_refinement = True` Run rigid-body refinement on final model
- `rigid_body_refinement_single_unit = True` Run rigid-body refinement with just one unit (do not break up into chains)
- `rigid_body_refinement_split_method = *chain_id segid` When splitting up molecule for rigid-body refinement (if `rigid_body_refinement_single_unit=False`), use either `chain_id` or `segid` to split up molecule
- `rigid_body_refinement_resolution = None` Run rigid-body refinement at this resolution if specified
- `append_to_fixed_model = True` Append placed search model to fixed model (if any) after search
- `min_cc = 0.4` If quick run, stop if minimum CC is achieved in local search. Also always skip if starting `CC_mask` is less than $1/4$ `min_cc`.
- `run_in_boxes = True` Run on sub-boxes and combine at the end
- `target_box_size = 60` Try to get boxes about this big on a side (grid units)
- `target_boxes = None` Try to get this many boxes. Default is `nproc` unless this makes box size much smaller than `target_box_size`
- `box_to_run = None` Run only this box
- `box_overlap_scale = 1` box overlap (overlap of boxes) will be `box_overlap_scale` times the density radius
- `edge_ratio = 10` box edge `box_overlap` times `edge_ratio`
- `density_radius = None` Radius for density to be cut out and compared. Default is 6 times the resolution.
- `model_radius = 3` Radius for removing density near `fixed_model`
- `zero_value = 0` Value to set map in regions overlapping fixed model
- `density_peaks = 20` Number of NCS-related peaks of density to check
- `delta_phi = 20` Angular spacing of search
- `max_rot = None` Maximum rotations to try
- `rotz_only = None` Rotate only around Z
- `single_positions_to_try = 10` Number of offset positions to try in optimizing orientation. Positions along the chain are selected as centers and a local fit near each position is carried out. The resulting offsets relative to the original placement are used to optimize the overall orientation and position.
- `max_position_shift_frac = 0.05` Maximum fractional positional shift in `single_positions` run
- `min_relative_cc = 0.67` Minimum local CC relative to original CC to keep a local search. This is a way to reject local searches that are completely wrong.
- `sieve_fit = None` Use `sieve_fit` fraction of single positions in fitting. If `None`, use all
- `ncs_copies_max = None` Maximum number of matching models to write. If more than one they will be written as MODEL records in the output PDB file. You can get them individually afterwards with `phenix.pdbtools placed_model.pdb keep="model 1"` etc.
- `start_rot = None` Three numbers `rotx`, `rotz`, `roty` defining the starting rotation of the search model. Normally used along with `delta_phi=1000` or `max_rot=1` to generate exactly one defined rotation.

- `search_center` = None Optional coordinates in search model for centering search. Note this is different from `target_search_center` which is the location xyz in the map to look.
- `search_center_selection` = None Optional selection defining coordinates in search model for centering search.
- `target_search_center` = None Optional coordinates in reference map where `search_center` should be approximately located after superimposing maps. Used to eliminate possible superpositions that place the search center elsewhere. Overlap scores decreased based on distance to `target_search_center/density_radius`.
- `model_search_position` = None Optional coordinates (usually part of search model) for matching to `target_search_position` after transformation. These can be specified in addition to `target_search_center`. Can have multiple `model_search_position` and `target_search_position` pairs by specifying each multiple times in order. NOTE: Not compatible with map search
- `target_search_position` = None Target positions for `model_search_position` after transformation.
- `search_position_radius` = None Radius for comparison of `target_search_position` and transformed `search_position` values. Default is `density_radius`. If specified, must be a single value or the same number of values as entries in `model_search_position` and `target_search_position`
- `rot_id_n` = None Number of rotation groups. Along with `rot_id_group`, allows defining groups of rotations to be carried out in one run. `short_caption` = Number of rotation groups
- `rot_id_group` = None rotation group to include. See `rot_id_n`.
- `map_box` = True Run `map_box` to extract useful part of map before search
- `fix_search_position` = False You can choose to not move your model center of mass to the origin by fixing the search position
- `search_box_size` = None You can choose the size of the search box for local searches. Normally set by default corresponding to 3 times `density_radius`
- `lower_bounds` = None You can select a part of your map for analysis with `lower_bounds` and `upper_bounds`.
- `upper_bounds` = None You can select a part of your map for analysis with `lower_bounds` and `upper_bounds`.
- `keep_search_order` = None Keep search order as input
- `remove_water` = False Remove waters and other hetero atoms from input files
- `buildrebuild_strategy` = Thorough Standard Quick Rebuilding strategy. Standard is up to 3 cycles, refining and replacing poorly-fitting loops in predicted models each cycle, using autobuild for density modification (if Xray). Quick is refinement only, one cycle. Thorough is up to 10 cycles, extensive rebuilding.
`refine_only` = None Refine only, no rebuilding steps (set automatically with Quick)
`refine_only_resolution` = 3.5 Default cutoff for using Refine only
`run_fit_loops` = True Run standard fit loops
`run_iterative_morph` = True Run iterative morphing
`run_trace_loops_through_density` = True Run loop fitting with trace_through_density algorithm
`run_refine` = True Refine morphed model. Required for `run_fit_loops` or `run_trace_loops_through_density`
`run_iterative_resolution_refine` = True Refine morphed model with iterative resolution method
`run_extend` = True Extend ends of morphed model
`extract_unique` = True Extract unique part of map. Applies after `density_select` (if set).
`use_symmetry_in_extract_unique` = True Use symmetry from symmetry file (if available) when extracting unique part of map.
`acceptable_docking_cc` = 0.5 Acceptable docking CC for a chain
`minimum_docking_cc` = 0.15 Minimum docking CC for a chain
`minimum_cutoff` = 0.1 Minimum cutoff for estimating density in better parts of model
`reasonable_cc_ratio` = 0.80 Acceptable CC (ratio) for a chain relative to average. Used to

make sure a chain entirely in very bad density is not kept
 reasonable_cc_diff = 0.15 Acceptable CC (difference) for a chain relative to average. Used to make sure a chain entirely in very bad density is not kept
 shift_field_distance = None Shift field characteristic distance (default = 10 Å) for morphing
 cc_sd_ratio = 3. Keep residues with CC at least within cc_sd_ratio of the mean CC for good residues
 cc_sd_ratio_end = 2. Keep residues with CC at least within cc_sd_ratio_end of the mean CC for good residues (applies to end of fragments)
 cc_sd_ratio_ok = 2. Residues with CC at least within cc_sd_ratio_ok of the mean CC for good residues are ok (do not delete on the basis of plddt)
 max_gap_ratio = 3. Allow CA-CA distance to be up to max_gap_ratio times expected
 maximum_connectivity_deviation = 15 Maximum connectivity deviation. Reject a solution with bigger deviation than this plus 2 * resolution.
 keep_fraction_of_best = 0.5 Acceptable CC to keep as ratio to best found. Applies if best found is at least acceptable_docking_cc.
 keep_maximum_entries = 10 Maximum dock positions to keep
 rmsd_for_similar_placement = None RMSD value indicating that two placements are similar. default is resolution of map
 rigid_body_refine_cycles = 1 Refinement cycles for rigid-body refinement
 overlap_ca_ca_distance = 3 Overlap distance for CA-CA atoms (or P-P)
 ca_ca_distance = 3.8 CA-CA distance (or P-P)
 allowed_fraction_overlapping = 0.10 Allowed fraction overlapping
 maximum_combinations = 100000 Maximum combinations of placements to consider
 proceed_with_any_symmetry = False Run even with symmetry that is not point-group or helical. Not recommended as symmetry may not work properly
 box_cushion = 20 Size of buffer around model when boxing
 refine_cycles = 3 Refinement cycles
 loop_refine_cycles = 5 Refinement cycles for loops
 loop_backup_residues = 3 Number of tries removing one residue at a time from each end of existing ends of loop if no loop is found with initial gap
 residues_to_trim = 5 Residues to trim on each end of all trimmed fragments
 find_ncs_from_model = True Find NCS (symmetry) from working models and apply in density modification. Also turns on finding NCS in any autobuild density modification (including density-based search)
 allow_split_more_than_one_chain_for_mr = False Allow splitting chains into domains for mr (X-ray only) even if there are multiple chains. Normally only split if a single chain as reassembly may not work with split
 multiple_chains.acceptable_cc_ratio = 0.8 Keep segments with CC at least equal to average of base segments times minimum_cc_ratio minus difference between segment confide...

- rebuild_strategy = Thorough Standard Quick Rebuilding strategy. Standard is up to 3 cycles, refining and replacing poorly-fitting loops in predicted models each cycle, using autobuild for density modification (if X-ray). Quick is refinement only, one cycle. Thorough is up to 10 cycles, extensive rebuilding.
- refine_only = None Refine only, no rebuilding steps (set automatically with Quick)
- refine_only_resolution = 3.5 Default cutoff for using Refine only
- run_fit_loops = True Run standard fit_loops
- run_iterative_morph = True Run iterative morphing
- run_trace_loops_through_density = True Run loop fitting with trace_through_density algorithm
- run_refine = True Refine morphed model. Required for run_fit_loops or run_trace_loops_through_density
- run_iterative_resolution_refine = True Refine morphed model with iterative resolution method
- run_extend = True Extend ends of morphed model
- extract_unique = True Extract unique part of map. Applies after density_select (if set).
- use_symmetry_in_extract_unique = True Use symmetry from symmetry file (if available) when extracting unique part of map.
- acceptable_docking_cc = 0.5 Acceptable docking CC for a chain

- `minimum_docking_cc` = 0.15 Minimum docking CC for a chain
- `minimum_cutoff` = 0.1 Minimum cutoff for estimating density in better parts of model
- `reasonable_cc_ratio` = 0.80 Acceptable CC (ratio) for a chain relative to average. Used to make sure a chain entirely in very bad density is not kept
- `reasonable_cc_diff` = 0.15 Acceptable CC (difference) for a chain relative to average. Used to make sure a chain entirely in very bad density is not kept
- `shift_field_distance` = None Shift field characteristic distance (default = 10 Å) for morphing
- `cc_sd_ratio` = 3. Keep residues with CC at least within `cc_sd_ratio` of the mean CC for good residues
- `cc_sd_ratio_end` = 2. Keep residues with CC at least within `cc_sd_ratio_end` of the mean CC for good residues (applies to end of fragments)
- `cc_sd_ratio_ok` = 2. Residues with CC at least within `cc_sd_ratio_ok` of the mean CC for good residues are ok (do not delete on the basis of pLDDT)
- `max_gap_ratio` = 3. Allow CA-CA distance to be up to `max_gap_ratio` times expected
- `maximum_connectivity_deviation` = 15 Maximum connectivity deviation. Reject a solution with bigger deviation than this plus $2 * \text{resolution}$.
- `keep_fraction_of_best` = 0.5 Acceptable CC to keep as ratio to best found. Applies if best found is at least `acceptable_docking_cc`.
- `keep_maximum_entries` = 10 Maximum dock positions to keep
- `rmsd_for_similar_placement` = None RMSD value indicating that two placements are similar. default is resolution of map
- `rigid_body_refine_cycles` = 1 Refinement cycles for rigid-body refinement
- `overlap_ca_ca_distance` = 3 Overlap distance for CA-CA atoms (or P-P)
- `ca_ca_distance` = 3.8 CA-CA distance (or P-P)
- `allowed_fraction_overlapping` = 0.10 Allowed fraction overlapping
- `maximum_combinations` = 100000 Maximum combinations of placements to consider
- `proceed_with_any_symmetry` = False Run even with symmetry that is not point-group or helical. Not recommended as symmetry may not work properly
- `box_cushion` = 20 Size of buffer around model when boxing
- `refine_cycles` = 3 Refinement cycles
- `loop_refine_cycles` = 5 Refinement cycles for loops
- `loop_backup_residues` = 3 Number of tries removing one residue at a time from each end of existing ends of loop if no loop is found with initial gap
- `residues_to_trim` = 5 Residues to trim on each end of all trimmed fragments
- `find_ncs_from_model` = True Find NCS (symmetry) from working models and apply in density modification. Also turns on finding NCS in any autobuild density modification (including density-based search)
- `allow_split_more_than_one_chain_for_mr` = False Allow splitting chains into domains for mr (X-ray only) even if there are multiple chains. Normally only split if a single chain as reassembly may not work with split multiple chains.
- `acceptable_cc_ratio` = 0.8 Keep segments with CC at least equal to average of base segments times `minimum_cc_ratio` minus difference between segment confidence (pLDDT) and confidence cutoff

(typically 0.7)

- `low_res_if_multiple_solutions = 3.5` Try phaser MR at increasing lower resolutions up to this value if multiple solutions are found
- `delta_low_res = 0.5` Resolution increment for `low_res_if_multiple_solutions`
- `controlstop_after_dock = True` Stop after docking step
`stop_after_morph = False` Stop after morphing
- `stepread_files = False` Read existing output files and use them if present
`write_files = True` Write output files
- `nproc = 1` Number of processors to use on your local machine
- `ignore_symmetry_conflicts = True` You can ignore the symmetry information (CRYST1) from coordinate files. This may be necessary if your model has been placed in a box with `box_map` for example.
- `random_seed = 171731` Random seed
- `max_dirs = 1000` Maximum number of directories (`dock_and_build_xxx`)
- `verbose = False` Verbose output
- `quick = True` Run quickly
- `gui` GUI-specific parameter required for output directory
`output_dir = None`
- `output_dir = None`

Build water into cryo-EM map using phenix.douse

douse.html

phenix.douse is a tool to build water into cryo-EM maps.

Facts about the procedure

- 1) The water is built into density map also ensuring each water is linked to the protein via H-bond, directly or indirectly (via another water).
- 2) Atomic model of the macro-molecule is assumed to be as complete and accurate as possible (for example, B factors are refined individually).
- 3) Water is looked for in a band of 6Å width extending out from the protein surface. This allows to build about two solvation layers around protein. Any unmodeled density in this region is treated as potential water.
- 4) Minimal (dist_min) and maximal (dist_max) distances between any water and protein or other water are 2 and 3.2Å, correspondingly.
- 5) Not all protein atoms are used to assess contacts with prospective water molecules but only those that can participate in H-bonds. These atoms are additionally filtered to leave a subset of atoms that are confidently placed in the map. The confidence is quantified by CCmask (cc_mask_threshold_interacting_atoms) - the correlation between model-calculated and experimental maps for the atom in question. The default is set to 0.5. The CCmask is calculated in a sphere of 2Å radius (atom_radius) around the atom in question.
- 6) Only strong enough density peaks are considered as potential water-candidates. "Strong enough" is defined as the average of map density values interpolated at atomic centers of protein atoms times the attenuation scale factor map_threshold_scale which is set to 0.25 by default.
- 7) The remaining peaks are filtered by two additional criteria.

One is CCmask (cc_mask_filter) with the cutoff value of 0.5 (cc_mask_filter_threshold). The CCmask is calculated similarly to "5)". Since CCmask strongly depends on atomic B factors and B factors of newly placed water-candidates are not known, unrestrained isotropic B factor refinement is performed for newly placed water prior to CCmask calculation.

Another selection criterion is peak sphericity (scc). The peak sphericity is evaluated as minimal correlation between any combination of two sets of map values interpolated along vectors going out from the peak center and having length of 1Å. For a perfectly spherical peak the correlation between any "pair of vectors" (any pair of sets of map values along these vectors) must be 1. The water is rejected if scc is less than 0.5.

Comments

- 1) Geometric (water-protein or water-water distance) and chemical (types of interacting atoms) criteria are rather strong and rigid criteria and do not allow much room for variation. The set default values are taken from standard literature.
- 2) Map-based criteria, such as various correlations, are more heuristics and rule-of-thumb based. For correlations, 1 means perfect similarity, 0 is no similarity and -1 is anti-similarity. So a choice like 0.5 means some similarity, half-way between none and perfect. Choosing a larger value will mean choosing more confident waters.

3) Map sphericity reflects the observation that water density appears to be approximately spherical most of the time. This observation breaks down in a number of special cases that need a special treatment and that can be a subject of future improvement of the tool. For example, at high resolution water with high directional mobility typically yield elliptical peak shapes. Water coordinated by heavier ions may not even resolved as individual peak or even if resolved as individual peak its sphericity can be largely perturbed.

4) Peak height threshold for selecting peak candidates is another rather ad-hoc criterion. It uses peak heights of confidently placed protein atoms as a guide. However, since water molecules are more mobile and often partially occupied, their peaks are expected to be weaker than peaks of better resolved protein atoms. By how much weaker? That is unknown. Thus some arbitrary attenuation scale is used.

5) Building 'single-atom' entities, such as water or ions, is a much more difficult task to do compared to building more feature-rich molecules such as proteins or multi-atom ligands. This is because it is much easier to confuse a true peak with a noise peak, while a constellation of many peaks (corresponding to protein chain, for example) contains very important information about shape and connectivity, making interpretation such peaks in terms of atomic model easier. Therefore, water building procedure is unlikely to be ever fully automated.

Availability

Command line, GUI.

Required inputs

Atomic model (PDB or mmCIF format) and map (MRC format).

resolution is required if `cc_mask_filter` is set to True.

Optional inputs

`mode = whole or per_chain`: work on the whole model or per chain. Working on the whole model may require a lot of memory.

`sphericity_filter = True`: use peak sphericity criteria. Disable to add more water.

`scc = 0.5`: map peak sphericity characteristic. Decrease to add more water.

`keep_input_water = False`: keep or remove existing water.

`dist_min = 2.0`: minimal peak-peak or peak-atom distance. Decrease to add more water.

`dist_max = 3.2`: maximal peak-peak or peak-atom distance. Increase to add more water.

`step = 0.3`: map re-sampling step.

`map_threshold_scale = 0.5`: map threshold for peak selection = (mean map value at atom centers) * `map_threshold_scale` Decrease to add more water.

`cc_mask_filter = True`: reject water with `CC_mask < cc_mask_filter_threshold`.

`cc_mask_filter_threshold = 0.5`: `CC_mask` threshold for rejecting water.

`resolution`: map resolution. Required if `cc_mask_filter=True`

Limitations

P1 symmetry is the only supported symmetry.

Assumptions

Macromolecular part of the atomic model is as complete as possible (no missing chains or other unmodelled parts present). The model is fully refined including individual B-factors.

Map resolution is sufficiently high to expect water oxygens be visible in the map.

Example

```
phenix.douse model.pdb map.mrc resolution=2.5
```

resolution is required if `cc_mask_filter` is set to `True`.

More information

https://phenix-online.org/phenixwebsite_static/mainsite/files/presentations/water.pdf

This document reflects the procedure at the time of making the document. The procedure in most current version of Phenix can be (slightly) different.

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phenix.dynamics: simple model perturbation

dynamics.html

This program performs very crude molecular dynamics on a model, usually for the purpose of removing any bias in R-free due to previous refinement against a different test set. It is not suitable for generating physically realistic trajectories, due to the lack of attractive forces such as electrostatics, and lengthy runs will result in the model falling apart. However, it is very effective for de-biasing while maintaining approximately correct geometry, and is also very fast (unlike simulated annealing in phenix.refine).

The inputs for the program are very similar to those for phenix.geometry_minimization; the only required files are a model and any necessary restraints (CIF) files for non-standard ligands. By default the dynamics is run at approximately room temperature; higher temperatures will result in faster movement. Note however that unlike simulated annealing, the model will tend to unfold rapidly at high temperatures.

Alternatives and complementary programs

Simulated annealing has some advantages due to the ability to simultaneously de-bias and optimize the model against the data, which prevents the structure from unfolding at high temperatures. The main disadvantage is the much longer runtime. phenix.pdbtools can be used to reset B-factors, or shake coordinates without geometry restraints. The latter is redundant if dynamics is run, but setting the B-factors to suitably low starting values (approximately 10 for most resolutions) will help remove bias. Both of the above steps can also be run at the start of phenix.refine; click the "Modify start model" button in the GUI to access options. To perturb structures for use as molecular replacement models, there is a separate program providing an interface to the normal modes method in Phaser (also available in the GUI under "Model tools"). This does a better job conserving local structure.

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List of all available keywords

silent = False write_geo_file = True file_name = None show_states = False restraints = None
restraints_directory = None output_file_name_prefix = None directory = None job_title = None Job title in
PHENIX GUI, not used on command line fix_rotamer_outliers = True Remove outliers allow_allowed_rotamers
= True More strict fixing outliers stop_for_unknowns = True dynamics_type = *cartesian stop_at_diff = None
stop after reaching specified cutoff value pdb_interpretationsort_atoms = True use_ncs_to_build_restraints =
True show_restraints_histograms = True flip_symmetric_amino_acids = True superpose_ideal_ligand =
*None all SF4 F3S DVT disable_uc_volume_vs_n_atoms_check = False
allow_polymer_cross_special_position = False correct_hydrogens = True c_beta_restraints = True

use_neutron_distances = False Use neutron X-H distances (which are longer than X-ray ones)
 disulfide_bond_exclusions_selection_string = None exclusion_distance_cutoff = 3 If SG of CYS forming SS bond is closer than this distance to an atom that it may coordinate then this SG is excluded from SS bond.
 link_distance_cutoff = 3 Length of link between the linked residues
 disulfide_distance_cutoff = 3
 add_angle_and_dihedral_restraints_for_disulfides = True dihedral_function_type = *determined_by_sign_of_periodicity all_sinusoidal all_harmonic chir_volume_esd = 0.2
 max_reasonable_bond_distance = 50.0 nonbonded_distance_cutoff = None default_vdw_distance = 1
 min_vdw_distance = 1 nonbonded_buffer = 1 **EXPERIMENTAL, developers only**nonbonded_weight = None
 Weighting of nonbonded restraints term. By default, this will be set to 16 if explicit hydrogens are used (this was the default in earlier versions of Phenix), or 100 if hydrogens are missing.
 const_shrink_donor_acceptor = 0 **EXPERIMENTAL, developers only**vdw_1_4_factor = 0.8
 min_distance_sym_equiv = 0.5 custom_nonbonded_symmetry_exclusions = None
 translate_cns_dna_rna_residue_names = None proceed_with_excessive_length_bonds = False
 restraints_librarycdl = True Use Conformation Dependent Library (CDL) for geometry
 restraintsinclude_modified_amino_acid_in_cdl = False mcl = True Use Metal Coordination Library (MCL) for tetrahedral Zn⁺⁺ and iron-sulfur clusters SF₄, FES, F₃S, ...cis_pro_eh99 = True omega_cdl = False Use Omega Conformation Dependent Library (omega-CDL) for geometry
 restraintscdl_svl = False rdl = False rdl_selection = *all TRP hpd1 = False cdl_nucleotidesenable = False Use RestraintsLib for DNA and RNA for geometry
 restraintsesd = *phenix csd factor = 2.0 user_suppliedpath = None Root directory of user supplied restraints
 action = *pre post Choice to look here before GeoStd, the default, or after
 secondary_structuress_by_chain = True Only applies if search_method = from_ca. Find secondary structure only within individual chains. Alternative is to allow H-bonds between chains. Can be much slower with ss_by_chain=False. If your model is complete use ss_by_chain=True. If your model is many fragments, use ss_by_chain=False.
 from_ca_conservative = False various parameters changed to make from_ca method more conservative, hopefully to closer resemble ksdssp.
 max_rmsd = 1 Only applies if search_method = from_ca. Maximum rmsd to consider two chains with identical sequences as the same for ss identification
 use_representative_chains = True Only applies if search_method = from_ca. Use a representative of all chains with the same sequence. Alternative is to examine each chain individually. Can be much slower with use_representative_of_chain=False if there are many symmetry copies. Ignored unless ss_by_chain is True.
 max_representative_chains = 100 Only applies if search_method = from_ca. Maximum number of representative chains
 enabled = False Turn on secondary structure restraints (main switch)
 proteinenabled = True Turn on secondary structure restraints for protein
 search_method = *ksdssp from_ca cablam Particular method to search protein secondary structure.
 distance_ideal_n_o = 2.9 Target length for N-O hydrogen bond
 distance_cut_n_o...

- silent = False
- write_geo_file = True
- file_name = None
- show_states = False
- restraints = None
- restraints_directory = None
- output_file_name_prefix = None
- directory = None
- job_title = None Job title in PHENIX GUI, not used on command line
- fix_rotamer_outliers = True Remove outliers
- allow_allowed_rotamers = True More strict fixing outliers

- stop_for_unknowns = True
- dynamics_type = *cartesian
- stop_at_diff = None stop after reaching specified cutoff value
- pdb_interpretationsort_atoms = True use_ncs_to_build_restraints = True show_restraints_histograms = True flip_symmetric_amino_acids = True superpose_ideal_ligand = *None all SF4 F3S DVT
 disable_uc_volume_vs_n_atoms_check = False allow_polymer_cross_special_position = False
 correct_hydrogens = True c_beta_restraints = True use_neutron_distances = False Use neutron X-H
 distances (which are longer than X-ray ones)disulfide_bond_exclusions_selection_string = None
 exclusion_distance_cutoff = 3 If SG of CYS forming SS bond is closer than this distance to an atom that it
 may coordinate then this SG is excluded from SS bond.link_distance_cutoff = 3 Length of link between
 the linked residuesdisulfide_distance_cutoff = 3 add_angle_and_dihedral_restraints_for_disulfides =
 True dihedral_function_type = *determined_by_sign_of_periodicity all_sinusoidal all_harmonic
 chir_volume_esd = 0.2 max_reasonable_bond_distance = 50.0 nonbonded_distance_cutoff = None
 default_vdw_distance = 1 min_vdw_distance = 1 nonbonded_buffer = 1 **EXPERIMENTAL, developers
 only**nonbonded_weight = None Weighting of nonbonded restraints term. By default, this will be set to 16
 if explicit hydrogens are used (this was the default in earlier versions of Phenix), or 100 if hydrogens are
 missing.const_shrink_donor_acceptor = 0 **EXPERIMENTAL, developers only**vdw_1_4_factor = 0.8
 min_distance_sym_equiv = 0.5 custom_nonbonded_symmetry_exclusions = None
 translate_cns_dna_rna_residue_names = None proceed_with_excessive_length_bonds = False
 restraints_librarycdl = True Use Conformation Dependent Library (CDL) for geometry
 restraintsinclude_modified_amino_acid_in_cdl = False mcl = True Use Metal Coordination Library (MCL)
 for tetrahedral Zn++ and iron-sulfur clusters SF4, FES, F3S, ...cis_pro_eh99 = True omega_cdl = False
 Use Omega Conformation Dependent Library (omega-CDL) for geometry restraintscdl_svl = False rdl =
 False rdl_selection = *all TRP hpdl = False cdl_nucleotidesenable = False Use RestraintsLib for DNA and
 RNA for geometry restraintsesd = *phenix csd factor = 2.0 user_suppliedpath = None Root directory of
 user supplied restraintsaction = *pre post Choice to look here before GeoStd, the default, or
 aftersecondary_structuress_by_chain = True Only applies if search_method = from_ca. Find secondary
 structure only within individual chains. Alternative is to allow H-bonds between chains. Can be much
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 Can be much slower with use_representative_of_chain=False if there are many symmetry copies.
 Ignored unless ss_by_chain is True.max_representative_chains = 100 Only applies if search_method =
 from_ca. Maximum number of representative chainsenabled = False Turn on secondary structure
 restraints (main switch)proteinenabled = True Turn on secondary structure restraints for
 proteinsearch_method = *ksdssp from_ca cablam Particular method to search protein secondary
 structure.distance_ideal_n_o = 2.9 Target length for N-O hydrogen bonddistance_cut_n_o = 3.5
 Hydrogen bond with length exceeding this value will not be establishedremove_outliers = True If true,
 h-bonds exceeding distance_cut_n_o length will not be establishedrestrain_hbond_angles = True
 helixserial_number = None helix_identifier = None enabled = True Restraining this particular helixselection =
 None helix_type = *alpha pi 3_10 unknown Type of helix, defaults to alpha. Only alpha, pi, and 3_10
 helices are used for hydrogen-bond res...
- sort_atoms = True
- use_ncs_to_build_restraints = True

- show_restraints_histograms = True
- flip_symmetric_amino_acids = True
- superpose_ideal_ligand = *None all SF4 F3S DVT
- disable_uc_volume_vs_n_atoms_check = False
- allow_polymer_cross_special_position = False
- correct_hydrogens = True
- c_beta_restraints = True
- use_neutron_distances = False Use neutron X-H distances (which are longer than X-ray ones)
- disulfide_bond_exclusions_selection_string = None
- exclusion_distance_cutoff = 3 If SG of CYS forming SS bond is closer than this distance to an atom that it may coordinate then this SG is excluded from SS bond.
- link_distance_cutoff = 3 Length of link between the linked residues
- disulfide_distance_cutoff = 3
- add_angle_and_dihedral_restraints_for_disulfides = True
- dihedral_function_type = *determined_by_sign_of_periodicity all_sinusoidal all_harmonic
- chir_volume_esd = 0.2
- max_reasonable_bond_distance = 50.0
- nonbonded_distance_cutoff = None
- default_vdw_distance = 1
- min_vdw_distance = 1
- nonbonded_buffer = 1 **EXPERIMENTAL, developers only**
- nonbonded_weight = None Weighting of nonbonded restraints term. By default, this will be set to 16 if explicit hydrogens are used (this was the default in earlier versions of Phenix), or 100 if hydrogens are missing.
- const_shrink_donor_acceptor = 0 **EXPERIMENTAL, developers only**
- vdw_1_4_factor = 0.8
- min_distance_sym_equiv = 0.5
- custom_nonbonded_symmetry_exclusions = None
- translate_cns_dna_rna_residue_names = None
- proceed_with_excessive_length_bonds = False
- restraints_librarycdl = True Use Conformation Dependent Library (CDL) for geometry
- restraintsinclude_modified_amino_acid_in_cdl = False mcl = True Use Metal Coordination Library (MCL) for tetrahedral Zn++ and iron-sulfur clusters SF4, FES, F3S, ...cis_pro_eh99 = True omega_cdl = False Use Omega Conformation Dependent Library (omega-CDL) for geometry restraintscdl_svl = False rdl = False rdl_selection = *all TRP hpdI = False cdl_nucleotidesenable = False Use RestraintsLib for DNA and RNA for geometry restraintsesd = *phenix csd factor = 2.0 user_suppliedpath = None Root directory of user supplied restraintsaction = *pre post Choice to look here before GeoStd, the default, or after
- cdl = True Use Conformation Dependent Library (CDL) for geometry restraints
- include_modified_amino_acid_in_cdl = False

- mcl = True Use Metal Coordination Library (MCL) for tetrahedral Zn⁺⁺ and iron-sulfur clusters SF₄, FES, F3S, ...
- cis_pro_eh99 = True
- omega_cdl = False Use Omega Conformation Dependent Library (omega-CDL) for geometry restraints
- cdl_svl = False
- rdl = False
- rdl_selection = *all TRP
- hpdl = False
- cdl_nucleotidesenable = False Use RestraintsLib for DNA and RNA for geometry restraintsesd = *phenix csd factor = 2.0
- enable = False Use RestraintsLib for DNA and RNA for geometry restraints
- esd = *phenix csd
- factor = 2.0
- user_suppliedpath = None Root directory of user supplied restraintsaction = *pre post Choice to look here before GeoStd, the default, or after
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- action = *pre post Choice to look here before GeoStd, the default, or after
- secondary_structuresss_by_chain = True Only applies if search_method = from_ca. Find secondary structure only within individual chains. Alternative is to allow H-bonds between chains. Can be much slower with ss_by_chain=False. If your model is complete use ss_by_chain=True. If your model is many fragments, use ss_by_chain=False.from_ca_conservative = False various parameters changed to make from_ca method more conservative, hopefully to closer resemble ksdssp.max_rmsd = 1 Only applies if search_method = from_ca. Maximum rmsd to consider two chains with identical sequences as the same for ss identificationuse_representative_chains = True Only applies if search_method = from_ca. Use a representative of all chains with the same sequence. Alternative is to examine each chain individually. Can be much slower with use_representative_of_chain=False if there are many symmetry copies. Ignored unless ss_by_chain is True.max_representative_chains = 100 Only applies if search_method = from_ca. Maximum number of representative chainsenabled = False Turn on secondary structure restraints (main switch)proteinenabled = True Turn on secondary structure restraints for proteinsearch_method = *ksdssp from_ca cablam Particular method to search protein secondary structure.distance_ideal_n_o = 2.9 Target length for N-O hydrogen bonddistance_cut_n_o = 3.5 Hydrogen bond with length exceeding this value will not be establishedremove_outliers = True If true, h-bonds exceeding distance_cut_n_o length will not be establishedrestrain_hbond_angles = True helixserial_number = None helix_identifier = None enabled = True Restraining this particular helixselection = None helix_type = *alpha pi 3_10 unknown Type of helix, defaults to alpha. Only alpha, pi, and 3_10 helices are used for hydrogen-bond restraints.sigma = 0.05 slack = 0 top_out = False angle_sigma_scale = 1 Multiply sigmas for h-bond angles by this value. Original sigmas range from 5 to 10.angle_sigma_set = None Use this parameter to set sigmas for h-bond angles to a particular valuehbonddonor = None acceptor = None sheetenabled = True Restraining this particular sheetfirst_strand = None sheet_id = None sigma = 0.05 slack = 0 top_out = False angle_sigma_scale = 1 Multiply sigmas for h-bond angles by this value. Original sigmas range from 5 to 10.angle_sigma_set = None Use this parameter to set sigmas for h-bond angles to a particular valuestrandselection = None sense = parallel antiparallel *unknown bond_start_current = None bond_start_previous = None hbonddonor = None acceptor = None nucleic_acidenabled = True Turn on secondary structure restraints for nucleic

acidshbond_distance_cutoff = 3.4 Hydrogen bonds with length exceeding this limit will not be establishedscale_bonds_sigma = 1. All sigmas for h-bond length will be multiplied by this number. The smaller number is tighter restraints.base_pairenabled = True Restrain this particular base-pairbase1 = None Selection string selecting at least one atom in the desired residuebase2 = None Selection string selecting at least one atom in the desired residuesaenger_class = 0 Type of base-pairingrestrain_planarity = False Apply planarity restraint to this base-pairplanarity_sigma = 0.176 restrain_hbonds = True Restrain hydrogen bonds length for this base-pairrestrain_hb_angles = True Restrain angles around hydrogen bonds for this base-pairrestrain_parallelity = True Apply parallelity restraint to this base-pairparallelity_target = 0 parallelity_sigma = 0.0335 stacking_pairenabled = True Restrain this particular base-pairbase1 = None Selection string selecting at least one atom in the desired residuebase2 = None Selection string selecting at least one atom in the desired residueangle = 0 sigma = 0.027

- ss_by_chain = True Only applies if search_method = from_ca. Find secondary structure only within individual chains. Alternative is to allow H-bonds between chains. Can be much slower with ss_by_chain=False. If your model is complete use ss_by_chain=True. If your model is many fragments, use ss_by_chain=False.
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- max_rmsd = 1 Only applies if search_method = from_ca. Maximum rmsd to consider two chains with identical sequences as the same for ss identification
- use_representative_chains = True Only applies if search_method = from_ca. Use a representative of all chains with the same sequence. Alternative is to examine each chain individually. Can be much slower with use_representative_of_chain=False if there are many symmetry copies. Ignored unless ss_by_chain is True.
- max_representative_chains = 100 Only applies if search_method = from_ca. Maximum number of representative chains
- enabled = False Turn on secondary structure restraints (main switch)
- proteinenabled = True Turn on secondary structure restraints for proteinsearch_method = *ksdssp from_ca cablam Particular method to search protein secondary structure.distance_ideal_n_o = 2.9 Target length for N-O hydrogen bonddistance_cut_n_o = 3.5 Hydrogen bond with length exceeding this value will not be establishedremove_outliers = True If true, h-bonds exceeding distance_cut_n_o length will not be establishedrestrain_hbond_angles = True helixserial_number = None helix_identifier = None enabled = True Restrain this particular helixselection = None helix_type = *alpha pi 3_10 unknown Type of helix, defaults to alpha. Only alpha, pi, and 3_10 helices are used for hydrogen-bond restraints.sigma = 0.05 slack = 0 top_out = False angle_sigma_scale = 1 Multiply sigmas for h-bond angles by this value. Original sigmas range from 5 to 10.angle_sigma_set = None Use this parameter to set sigmas for h-bond angles to a particular valuehbonddonor = None acceptor = None sheetenabled = True Restrain this particular sheetfirst_strand = None sheet_id = None sigma = 0.05 slack = 0 top_out = False angle_sigma_scale = 1 Multiply sigmas for h-bond angles by this value. Original sigmas range from 5 to 10.angle_sigma_set = None Use this parameter to set sigmas for h-bond angles to a particular valuestrandselection = None sense = parallel antiparallel *unknown bond_start_current = None bond_start_previous = None hbonddonor = None acceptor = None
- enabled = True Turn on secondary structure restraints for protein
- search_method = *ksdssp from_ca cablam Particular method to search protein secondary structure.
- distance_ideal_n_o = 2.9 Target length for N-O hydrogen bond

- distance_cut_n_o = 3.5 Hydrogen bond with length exceeding this value will not be established
- remove_outliers = True If true, h-bonds exceeding distance_cut_n_o length will not be established
- restrain_hbond_angles = True
- helixserial_number = None helix_identifier = None enabled = True Restrain this particular helixselection = None helix_type = *alpha pi 3_10 unknown Type of helix, defaults to alpha. Only alpha, pi, and 3_10 helices are used for hydrogen-bond restraints.sigma = 0.05 slack = 0 top_out = False angle_sigma_scale = 1 Multiply sigmas for h-bond angles by this value. Original sigmas range from 5 to 10.angle_sigma_set = None Use this parameter to set sigmas for h-bond angles to a particular valuehbonddonor = None acceptor = None
- serial_number = None
- helix_identifier = None
- enabled = True Restrain this particular helix
- selection = None
- helix_type = *alpha pi 3_10 unknown Type of helix, defaults to alpha. Only alpha, pi, and 3_10 helices are used for hydrogen-bond restraints.
- sigma = 0.05
- slack = 0
- top_out = False
- angle_sigma_scale = 1 Multiply sigmas for h-bond angles by this value. Original sigmas range from 5 to 10.
- angle_sigma_set = None Use this parameter to set sigmas for h-bond angles to a particular value
- hbonddonor = None acceptor = None
- donor = None
- acceptor = None
- sheetenabled = True Restrain this particular sheetfirst_strand = None sheet_id = None sigma = 0.05 slack = 0 top_out = False angle_sigma_scale = 1 Multiply sigmas for h-bond angles by this value. Original sigmas range from 5 to 10.angle_sigma_set = None Use this parameter to set sigmas for h-bond angles to a particular valuestrandselection = None sense = parallel antiparallel *unknown bond_start_current = None bond_start_previous = None hbonddonor = None acceptor = None
- enabled = True Restrain this particular sheet
- first_strand = None
- sheet_id = None
- sigma = 0.05
- slack = 0
- top_out = False
- angle_sigma_scale = 1 Multiply sigmas for h-bond angles by this value. Original sigmas range from 5 to 10.
- angle_sigma_set = None Use this parameter to set sigmas for h-bond angles to a particular value
- strandselection = None sense = parallel antiparallel *unknown bond_start_current = None bond_start_previous = None

- selection = None
- sense = parallel antiparallel *unknown
- bond_start_current = None
- bond_start_previous = None
- hbonddonor = None acceptor = None
- donor = None
- acceptor = None
- nucleic_acidenabled = True Turn on secondary structure restraints for nucleic acids
- hbond_distance_cutoff = 3.4 Hydrogen bonds with length exceeding this limit will not be established
- scale_bonds_sigma = 1. All sigmas for h-bond length will be multiplied by this number. The smaller number is tighter restraints.
- base_pairenable = True Restraint this particular base-pairbase1 = None Selection string selecting at least one atom in the desired residuebase2 = None Selection string selecting at least one atom in the desired residue
- saenger_class = 0 Type of base-pairing
- restrain_planarity = False Apply planarity restraint to this base-pairplanarity_sigma = 0.176
- restrain_hbonds = True Restrain hydrogen bonds length for this base-pairrestrain_hb_angles = True
- restrain_hb_angles = True Restrain angles around hydrogen bonds for this base-pairrestrain_parallelity = True Apply parallelity restraint to this base-pairparallelity_target = 0
- parallelity_sigma = 0.0335 stacking_pairenable = True
- Restraint this particular base-pairbase1 = None Selection string selecting at least one atom in the desired residuebase2 = None Selection string selecting at least one atom in the desired residueangle = 0 sigma = 0.027
- enabled = True Turn on secondary structure restraints for nucleic acids
- hbond_distance_cutoff = 3.4 Hydrogen bonds with length exceeding this limit will not be established
- scale_bonds_sigma = 1. All sigmas for h-bond length will be multiplied by this number. The smaller number is tighter restraints.
- base_pairenable = True Restraint this particular base-pairbase1 = None Selection string selecting at least one atom in the desired residuebase2 = None Selection string selecting at least one atom in the desired residue
- saenger_class = 0 Type of base-pairing
- restrain_planarity = False Apply planarity restraint to this base-pairplanarity_sigma = 0.176
- restrain_hbonds = True Restrain hydrogen bonds length for this base-pairrestrain_hb_angles = True
- restrain_hb_angles = True Restrain angles around hydrogen bonds for this base-pairrestrain_parallelity = True Apply parallelity restraint to this base-pairparallelity_target = 0
- parallelity_sigma = 0.0335

- `stacking_pairenable` = True Restraint this particular base-pair
- `base1` = None Selection string selecting at least one atom in the desired residue
- `base2` = None Selection string selecting at least one atom in the desired residue
- `angle` = 0
- `sigma` = 0.027
- `reference_coordinate_restraints` Restrains coordinates in Cartesian space to stay near their starting positions. This is intended for use in generating simulated annealing omit maps, to prevent refined atoms from collapsing in on the region missing atoms. For conserving geometry quality at low resolution, the more flexible reference model restraints should be used instead.
- `enabled` = False
- `exclude_outliers` = True
- `selection` = all
- `sigma` = 0.2
- `limit` = 1.0
- `top_out` = False
- `reference_is_average_alt_confs` = False Sets the reference coordinate to the average of two alt. confs. Ignores selection option.
- `automatic_restraint` `side_chain_parallelity` = False
- `side_chain_parallelity` = False
- `automatic_linking` `link_all` = False If True, bond restraints will be generated for any appropriate ligand-protein or ligand-nucleic acid covalent bonds. This includes sugars, amino acid modifications, and other prosthetic groups.
- `link_none` = False
- `link_metals` = Auto
- `link_residues` = True
- `link_amino_acid_rna_dna` = False
- `link_carbohydrates` = True
- `link_ligands` = True
- `link_small_molecules` = False

- metal_coordination_cutoff = 3.0
- amino_acid_bond_cutoff = 1.9
- inter_residue_bond_cutoff = 2.2
- buffer_for_second_row_elements = 0.5
- carbohydrate_bond_cutoff = 1.99
- ligand_bond_cutoff = 1.99
- small_molecule_bond_cutoff = 1.98
- exclude_hydrogens_from_bonding_decisions = False
- include_in_automatic_linkingselection_1 = None selection_2 = None bond_cutoff = 4.5
- selection_1 = None
- selection_2 = None
- bond_cutoff = 4.5
- exclude_from_automatic_linkingselection_1 = None selection_2 = None
- selection_1 = None
- selection_2 = None
- apply_cis_trans_specificationcis_trans_mod = cis *trans residue_selection = None Residues containing C-alpha atom of omega dihedral
- cis_trans_mod = cis *trans
- residue_selection = None Residues containing C-alpha atom of omega dihedral
- apply_cif_restraintsrestraints_file_name = None residue_selection = None
- restraints_file_name = None
- residue_selection = None
- apply_cif_modificationdata_mod = None residue_selection = None
- data_mod = None
- residue_selection = None
- apply_cif_linkdata_link = None residue_selection_1 = None residue_selection_2 = None
- data_link = None
- residue_selection_1 = None
- residue_selection_2 = None
- peptide_linkramachandran_restraints = False !!! OBSOLETE. Kept for backward compatibility only !!!
 Restrains peptide backbone to fall within allowed regions of Ramachandran plot. Although it does not eliminate outliers, it can significantly improve the percent favored and percent outliers at low resolution. Probably not useful (and maybe even harmful) at resolutions much higher than 3.5Å.cis_threshold = 45
 apply_all_trans = False discard_omega = False discard_psi_phi = True apply_peptide_plane = False
 omega_esd_override_value = None rama_weight = 1.0 scale_allowed = 1.0 rama_potential = *oldfield
 emsley rama_selection = None Selection of part of the model for which Ramachandran restraints will be set up.restrain_rama_outliers = True Apply restraints to Ramachandran outliersrestrain_rama_allowed = True Apply restraints to residues in allowed region on Ramachandran
 plotrestrain_allowed_outliers_with_emsley = False In case of restrain_rama_outliers=True and/or
 restrain_rama_allowed=True still restrain these residues with emsley. Make sense only in case of using

oldfield potential.oldfieldesd = 10.0 weight_scale = 1.0 dist_weight_max = 10.0 weight = None plot_cutoff = 0.027

- ramachandran_restraints = False !!! OBSOLETE. Kept for backward compatibility only !!! Restrains peptide backbone to fall within allowed regions of Ramachandran plot. Although it does not eliminate outliers, it can significantly improve the percent favored and percent outliers at low resolution. Probably not useful (and maybe even harmful) at resolutions much higher than 3.5Å.
- cis_threshold = 45
- apply_all_trans = False
- discard_omega = False
- discard_psi_phi = True
- apply_peptide_plane = False
- omega_esd_override_value = None
- rama_weight = 1.0
- scale_allowed = 1.0
- rama_potential = *oldfield emsley
- rama_selection = None Selection of part of the model for which Ramachandran restraints will be set up.
- restrain_rama_outliers = True Apply restraints to Ramachandran outliers
- restrain_rama_allowed = True Apply restraints to residues in allowed region on Ramachandran plot
- restrain_allowed_outliers_with_emsley = False In case of restrain_rama_outliers=True and/or restrain_rama_allowed=True still restrain these residues with emsley. Make sense only in case of using oldfield potential.
- oldfieldesd = 10.0 weight_scale = 1.0 dist_weight_max = 10.0 weight = None plot_cutoff = 0.027
- esd = 10.0
- weight_scale = 1.0
- dist_weight_max = 10.0
- weight = None
- plot_cutoff = 0.027
- ramachandran_plot_restraintsenabled = False favored = *oldfield emsley emsley8k phi_psi_2 allowed = *oldfield emsley emsley8k phi_psi_2 outlier = *oldfield emsley emsley8k phi_psi_2 selection = None Selection of part of the model for which Ramachandran restraints will be set up.
inject_emsley8k_into_oldfield_favored = True Backdoor to disable temporary dirty hack to use botholdfieldweight = 0. Direct weight value. If 0 the weight will be calculated as following: $(w, op.esd, op.dist_weight_max, 2.0, op.weight_scale) \cdot 1 / esd^2 \cdot \max(2.0, \min(current_distance_to_allowed, dist_weight_max)) \cdot weight_scale \cdot \max(2.0, current_distance_to_allowed) \cdot 1 / esd^2 \cdot weight_scale \cdot \max(distance_to_allowed_cutoff, current_distance_to_allowed) \cdot weight_scale (=0.01) \cdot \max(distance_weight_min(=2.), \min(distance_weight_max(=10.), current_distance_to_allowed))$
weight_scale = 0.01 distance_weight_min = 2.0 minimum coefficient when scaling depending on how far the residue is from allowed region.
distance_weight_max = 10.0 maximum coefficient when scaling depending on how far the residue is from allowed region.
plot_cutoff = 0.027
emsleyweight = 1.0 scale_allowed = 1.0 emsley8kweight_favored = 5.0 weight_allowed = 10.0
weight_outlier = 10.0 phi_psi_2favored_strategy = *closest highest_probability random weighted_random
allowed_strategy = *closest highest_probability random weighted_random outlier_strategy = *closest

highest_probability random weighted_random

- enabled = False
- favored = *oldfield emsley emsley8k phi_psi_2
- allowed = *oldfield emsley emsley8k phi_psi_2
- outlier = *oldfield emsley emsley8k phi_psi_2
- selection = None Selection of part of the model for which Ramachandran restraints will be set up.
- inject_emsley8k_into_oldfield_favored = True Backdoor to disable temporary dirty hack to use both
- oldfieldweight = 0. Direct weight value. If 0 the weight will be calculated as following: $(w, op.esd, op.dist_weight_max, 2.0, op.weight_scale) \cdot 1 / esd^2 \cdot \max(2.0, \min(current_distance_to_allowed, dist_weight_max)) \cdot weight_scale \cdot \max(2.0, current_distance_to_allowed) \cdot 1 / esd^2 \cdot weight_scale \cdot \max(distance_to_allowed_cutoff, current_distance_to_allowed) \cdot weight_scale (=0.01) \cdot \max(distance_weight_min(=2.), \min(distance_weight_max(=10.), current_distance_to_allowed))$ weight_scale = 0.01 distance_weight_min = 2.0 minimum coefficient when scaling depending on how far the residue is from allowed region. distance_weight_max = 10.0 maximum coefficient when scaling depending on how far the residue is from allowed region. plot_cutoff = 0.027
- weight = 0. Direct weight value. If 0 the weight will be calculated as following: $(w, op.esd, op.dist_weight_max, 2.0, op.weight_scale) \cdot 1 / esd^2 \cdot \max(2.0, \min(current_distance_to_allowed, dist_weight_max)) \cdot weight_scale \cdot \max(2.0, current_distance_to_allowed) \cdot 1 / esd^2 \cdot weight_scale \cdot \max(distance_to_allowed_cutoff, current_distance_to_allowed) \cdot weight_scale (=0.01) \cdot \max(distance_weight_min(=2.), \min(distance_weight_max(=10.), current_distance_to_allowed))$
- weight_scale = 0.01
- distance_weight_min = 2.0 minimum coefficient when scaling depending on how far the residue is from allowed region.
- distance_weight_max = 10.0 maximum coefficient when scaling depending on how far the residue is from allowed region.
- plot_cutoff = 0.027
- emsleyweight = 1.0 scale_allowed = 1.0
- weight = 1.0
- scale_allowed = 1.0
- emsley8kweight_favored = 5.0 weight_allowed = 10.0 weight_outlier = 10.0
- weight_favored = 5.0
- weight_allowed = 10.0
- weight_outlier = 10.0
- phi_psi_2favored_strategy = *closest highest_probability random weighted_random allowed_strategy = *closest highest_probability random weighted_random outlier_strategy = *closest highest_probability random weighted_random
- favored_strategy = *closest highest_probability random weighted_random
- allowed_strategy = *closest highest_probability random weighted_random
- outlier_strategy = *closest highest_probability random weighted_random
- rna_sugar_pucker_analysisbond_min_distance = 1.2 bond_max_distance = 1.8 epsilon_range_min = 155.0 epsilon_range_max = 310.0 delta_range_2p_min = 129.0 delta_range_2p_max = 162.0

delta_range_3p_min = 65.0 delta_range_3p_max = 104.0 p_distance_c1p_outbound_line_2p_max = 2.9
o3p_distance_c1p_outbound_line_2p_max = 2.4 bond_detection_distance_tolerance = 0.5 enable =
True

- bond_min_distance = 1.2
- bond_max_distance = 1.8
- epsilon_range_min = 155.0
- epsilon_range_max = 310.0
- delta_range_2p_min = 129.0
- delta_range_2p_max = 162.0
- delta_range_3p_min = 65.0
- delta_range_3p_max = 104.0
- p_distance_c1p_outbound_line_2p_max = 2.9
- o3p_distance_c1p_outbound_line_2p_max = 2.4
- bond_detection_distance_tolerance = 0.5
- enable = True
- show_histogram_slotsbond_lengths = 5 nonbonded_interaction_distances = 5
bond_angle_deviations_from_ideal = 5 dihedral_angle_deviations_from_ideal = 5
chiral_volume_deviations_from_ideal = 5
- bond_lengths = 5
- nonbonded_interaction_distances = 5
- bond_angle_deviations_from_ideal = 5
- dihedral_angle_deviations_from_ideal = 5
- chiral_volume_deviations_from_ideal = 5
- show_max_itemsnot_linked = 5 bond_restraints_sorted_by_residual = 5
nonbonded_interactions_sorted_by_model_distance = 5 bond_angle_restraints_sorted_by_residual = 5
dihedral_angle_restraints_sorted_by_residual = 3 chirality_restraints_sorted_by_residual = 3
planarity_restraints_sorted_by_residual = 3 residues_with_excluded_nonbonded_symmetry_interactions
= 12 fatal_problem_max_lines = 10
- not_linked = 5
- bond_restraints_sorted_by_residual = 5
- nonbonded_interactions_sorted_by_model_distance = 5
- bond_angle_restraints_sorted_by_residual = 5
- dihedral_angle_restraints_sorted_by_residual = 3
- chirality_restraints_sorted_by_residual = 3
- planarity_restraints_sorted_by_residual = 3
- residues_with_excluded_nonbonded_symmetry_interactions = 12
- fatal_problem_max_lines = 10
- ncs_groupThe definition of one NCS group. Note, that almost always in refinement programs they will
be checked and filtered if needed.reference = None Residue selection string for the complete master
NCS copyselection = None Residue selection string for each NCS copy location in ASU

- reference = None Residue selection string for the complete master NCS copy
- selection = None Residue selection string for each NCS copy location in ASU
- ncs_searchSet of parameters for NCS search procedure. Some of them also used for filtering user-supplied ncs_group.enabled = False Enable NCS restraints or constraints in refinement (in some cases may be switched on inside refinement program).exclude_selection = "element H or element D or water" Atoms selected by this selection will be excluded from the model before any NCS search and/or filtering procedures. There is no way atoms defined by this selection will be in NCS.chain_similarity_threshold = 0.85 Threshold for sequence similarity between matching chains. A smaller value may cause more chains to be grouped together and can lower the number of common residueschain_max_rmsd = 2. limit of rms difference between chains to be considered as copiesresidue_match_radius = 4.0 Maximum allowed distance difference between pairs of matching atoms of two residuestry_shortcuts = False Try very quick check to speed up the search when chains are identical. If failed, regular search will be performed automatically.minimum_number_of_atoms_in_copy = 3 Do not create ncs groups where master and copies would contain less than specified amount of atomsvalidate_user_supplied_groups = True Enable validation of user-supplied ncs_group. Need to exercise a lot of caution turning this off. This option is for developers only.
- enabled = False Enable NCS restraints or constraints in refinement (in some cases may be switched on inside refinement program).
- exclude_selection = "element H or element D or water" Atoms selected by this selection will be excluded from the model before any NCS search and/or filtering procedures. There is no way atoms defined by this selection will be in NCS.
- chain_similarity_threshold = 0.85 Threshold for sequence similarity between matching chains. A smaller value may cause more chains to be grouped together and can lower the number of common residues
- chain_max_rmsd = 2. limit of rms difference between chains to be considered as copies
- residue_match_radius = 4.0 Maximum allowed distance difference between pairs of matching atoms of two residues
- try_shortcuts = False Try very quick check to speed up the search when chains are identical. If failed, regular search will be performed automatically.
- minimum_number_of_atoms_in_copy = 3 Do not create ncs groups where master and copies would contain less than specified amount of atoms
- validate_user_supplied_groups = True Enable validation of user-supplied ncs_group. Need to exercise a lot of caution turning this off. This option is for developers only.
- clash_guardnonbonded_distance_threshold = 0.5 max_number_of_distances_below_threshold = 100 max_fraction_of_distances_below_threshold = 0.1
- nonbonded_distance_threshold = 0.5
- max_number_of_distances_below_threshold = 100
- max_fraction_of_distances_below_threshold = 0.1
- geometry_restraintseditsexcessive_bond_distance_limit = 10 bondaction = *add delete change atom_selection_1 = None atom_selection_2 = None symmetry_operation = None The bond is between atom_1 and symmetry_operation * atom_2, with atom_1 and atom_2 given in fractional coordinates. Example: symmetry_operation = -x-1,-y,zdistance_ideal = None sigma = None slack = None limit = -1.0 top_out = False angleaction = *add delete change atom_selection_1 = None atom_selection_2 = None atom_selection_3 = None angle_ideal = None sigma = None dihedralaction = *add delete change atom_selection_1 = None atom_selection_2 = None atom_selection_3 = None atom_selection_4 = None

angle_ideal = None alt_angle_ideals = None sigma = None periodicity = 1 planarityaction = *add delete
change atom_selection = None sigma = None parallelityaction = *add delete change atom_selection_1 =
None atom_selection_2 = None sigma = 0.027 target_angle_deg = 0 removeangles = None dihedrals =
None chiralities = None planarities = None parallelities = None

- editsexcessive_bond_distance_limit = 10 bondaction = *add delete change atom_selection_1 = None
atom_selection_2 = None symmetry_operation = None The bond is between atom_1 and
symmetry_operation * atom_2, with atom_1 and atom_2 given in fractional coordinates. Example:
symmetry_operation = -x-1,-y,zdistance_ideal = None sigma = None slack = None limit = -1.0 top_out =
False angleaction = *add delete change atom_selection_1 = None atom_selection_2 = None
atom_selection_3 = None angle_ideal = None sigma = None dihedralaction = *add delete change
atom_selection_1 = None atom_selection_2 = None atom_selection_3 = None atom_selection_4 = None
angle_ideal = None alt_angle_ideals = None sigma = None periodicity = 1 planarityaction = *add delete
change atom_selection = None sigma = None parallelityaction = *add delete change atom_selection_1 =
None atom_selection_2 = None sigma = 0.027 target_angle_deg = 0

- excessive_bond_distance_limit = 10

- bondaction = *add delete change atom_selection_1 = None atom_selection_2 = None
symmetry_operation = None The bond is between atom_1 and symmetry_operation * atom_2, with
atom_1 and atom_2 given in fractional coordinates. Example: symmetry_operation =
-x-1,-y,zdistance_ideal = None sigma = None slack = None limit = -1.0 top_out = False

- action = *add delete change

- atom_selection_1 = None

- atom_selection_2 = None

- symmetry_operation = None The bond is between atom_1 and symmetry_operation * atom_2, with
atom_1 and atom_2 given in fractional coordinates. Example: symmetry_operation = -x-1,-y,z

- distance_ideal = None

- sigma = None

- slack = None

- limit = -1.0

- top_out = False

- angleaction = *add delete change atom_selection_1 = None atom_selection_2 = None
atom_selection_3 = None angle_ideal = None sigma = None

- action = *add delete change

- atom_selection_1 = None

- atom_selection_2 = None

- atom_selection_3 = None

- angle_ideal = None

- sigma = None

- dihedralaction = *add delete change atom_selection_1 = None atom_selection_2 = None
atom_selection_3 = None atom_selection_4 = None angle_ideal = None alt_angle_ideals = None sigma
= None periodicity = 1

- action = *add delete change

- atom_selection_1 = None

- atom_selection_2 = None
- atom_selection_3 = None
- atom_selection_4 = None
- angle_ideal = None
- alt_angle_ideals = None
- sigma = None
- periodicity = 1
- planarityaction = *add delete change atom_selection = None sigma = None
- action = *add delete change
- atom_selection = None
- sigma = None
- parallelityaction = *add delete change atom_selection_1 = None atom_selection_2 = None sigma = 0.027 target_angle_deg = 0
- action = *add delete change
- atom_selection_1 = None
- atom_selection_2 = None
- sigma = 0.027
- target_angle_deg = 0
- removeangles = None dihedrals = None chiralities = None planarities = None parallelities = None
- angles = None
- dihedrals = None
- chiralities = None
- planarities = None
- parallelities = None
- reference_modelenabled = False Restrains the dihedral angles to a high-resolution reference structure to reduce overfitting at low resolution. You will need to specify a reference PDB file (in the input list in the main window) to use this option.file = None use_starting_model_as_reference = False sigma = 1.0 limit = 15.0 hydrogens = False Include dihedrals with hydrogen atomsmain_chain = True Include dihedrals formed by main chain atomsside_chain = True Include dihedrals formed by side chain atomsfix_outliers = True Try to fix rotamer outliers in refined modelstrict_rotamer_matching = False Make sure that rotamers in refinement model matches those in reference model even when they are not outliersauto_shutoff_for_ncs = False Do not apply to parts of structure covered by NCS restraintssecondary_structure_only = False Only apply reference model restraints to secondary structure elements (helices and sheets)reference_groupreference = None selection = None file_name = None this is to used internally to disambiguate cases where multiple reference models contain the same chain ID. This normally does not need to be set by the usersearch_optionsSet of parameters for NCS search procedure. Some of them also used for filtering user-supplied ncs_group.exclude_selection = "element H or element D or water" Atoms selected by this selection will be excluded from the model before any NCS search and/or filtering procedures. There is no way atoms defined by this selection will be in NCS.chain_similarity_threshold = 0.85 Threshold for sequence similarity between matching chains. A smaller value may cause more chains to be grouped together and can lower the number of common

residueschain_max_rmsd = 100. limit of rms difference between chains to be considered as copies
residue_match_radius = 1000 Maximum allowed distance difference between pairs of matching atoms of two residues
try_shortcuts = False Try very quick check to speed up the search when chains are identical. If failed, regular search will be performed automatically.
minimum_number_of_atoms_in_copy = 3 Do not create ncs groups where master and copies would contain less than specified amount of atoms
validate_user_supplied_groups = True Enable validation of user-supplied ncs_group. Need to exercise a lot of caution turning this off. This option is for developers only.

- enabled = False Restrains the dihedral angles to a high-resolution reference structure to reduce overfitting at low resolution. You will need to specify a reference PDB file (in the input list in the main window) to use this option.

- file = None

- use_starting_model_as_reference = False

- sigma = 1.0

- limit = 15.0

- hydrogens = False Include dihedrals with hydrogen atoms

- main_chain = True Include dihedrals formed by main chain atoms

- side_chain = True Include dihedrals formed by side chain atoms

- fix_outliers = True Try to fix rotamer outliers in refined model

- strict_rotamer_matching = False Make sure that rotamers in refinement model matches those in reference model even when they are not outliers

- auto_shutoff_for_ncs = False Do not apply to parts of structure covered by NCS restraints

- secondary_structure_only = False Only apply reference model restraints to secondary structure elements (helices and sheets)

- reference_groupreference = None selection = None file_name = None this is to used internally to disambiguate cases where multiple reference models contain the same chain ID. This normally does not need to be set by the user

- reference = None

- selection = None

- file_name = None this is to used internally to disambiguate cases where multiple reference models contain the same chain ID. This normally does not need to be set by the user

- search_optionsSet of parameters for NCS search procedure. Some of them also used for filtering user-supplied ncs_group.
exclude_selection = "element H or element D or water" Atoms selected by this selection will be excluded from the model before any NCS search and/or filtering procedures. There is no way atoms defined by this selection will be in NCS.
chain_similarity_threshold = 0.85 Threshold for sequence similarity between matching chains. A smaller value may cause more chains to be grouped together and can lower the number of common residues
chain_max_rmsd = 100. limit of rms difference between chains to be considered as copies
residue_match_radius = 1000 Maximum allowed distance difference between pairs of matching atoms of two residues
try_shortcuts = False Try very quick check to speed up the search when chains are identical. If failed, regular search will be performed automatically.
minimum_number_of_atoms_in_copy = 3 Do not create ncs groups where master and copies would contain less than specified amount of atoms
validate_user_supplied_groups = True Enable validation of user-supplied ncs_group. Need to exercise a lot of caution turning this off. This option is for developers only.

- `exclude_selection = "element H or element D or water"` Atoms selected by this selection will be excluded from the model before any NCS search and/or filtering procedures. There is no way atoms defined by this selection will be in NCS.
- `chain_similarity_threshold = 0.85` Threshold for sequence similarity between matching chains. A smaller value may cause more chains to be grouped together and can lower the number of common residues
- `chain_max_rmsd = 100`. limit of rms difference between chains to be considered as copies
- `residue_match_radius = 1000` Maximum allowed distance difference between pairs of matching atoms of two residues
- `try_shortcuts = False` Try very quick check to speed up the search when chains are identical. If failed, regular search will be performed automatically.
- `minimum_number_of_atoms_in_copy = 3` Do not create ncs groups where master and copies would contain less than specified amount of atoms
- `validate_user_supplied_groups = True` Enable validation of user-supplied `ncs_group`. Need to exercise a lot of caution turning this off. This option is for developers only.
- `cartesian_dynamicstemperture = 300` `number_of_steps = 200` `time_step = 0.0005`
`initial_velocities_zero_fraction = 0` `n_print = 100` `verbose = -1` `random_seed = None` `n_collect = 10`
`stop_cm_motion = True`
- `temperature = 300`
- `number_of_steps = 200`
- `time_step = 0.0005`
- `initial_velocities_zero_fraction = 0`
- `n_print = 100`
- `verbose = -1`
- `random_seed = None`
- `n_collect = 10`
- `stop_cm_motion = True`

Docking a model into a cryo-EM map with em_placement

em_placement.html

Authors

- Randy J Read, Claudia Millán, Airlie J McCoy, Thomas C Terwilliger

Purpose

The em_placement tool docks the input model in a cryo-EM reconstruction, using the half-maps in an adaptive search strategy.

How em_placement works

The half-maps are compared and analysed to determine signal and noise power as smooth functions in Fourier space. This information is used to predict whether or not a rotation search with a structural model over the full map volume is likely to succeed. If not, the maps are divided into sub-volumes small enough that the noise will be reduced to a level where signal from the rotation search can be detected. Following this, translation searches in the full map volume or the sub-volumes are carried out. Finally, rigid-body refinement is carried out using spherical sub-volumes centered on potential solutions to give the final solutions and confidence indicators.

A log-likelihood-gain (LLG) score greater than 60 suggests that a solution is significantly better than random.

NB: the statistical analysis requires the half-maps to correspond to the full volume of the cryo-EM reconstruction, so cut-out boxes cannot be used. Half-maps are required for a search over the whole reconstruction volume, whereas just a single full map can be used optionally for a search over a local volume.

Running em_placement

There is an EMplacement GUI to run em_placement, but it can also be controlled by an input script using the phil syntax. A basic phil script would look like the following, in which a model derived from the membrane domain of PDB entry 4cof is docked into the reconstruction of the GABA receptor in EMDB entry 11657.

```
em_placement
{
  remove_phaser_tng_folder = True

  map_model
  {
    half_map = emd_11657_half_map_1.map
    half_map = emd_11657_half_map_2.map
    best_resolution = 1.7
    point_group_symmetry = C5
    sequence_composition = 7a5v.fa
  }

  biological_unit {
    molecule
    {
      molecule_name = 4cofA_membrane
      model_file = 4cofA_membrane.pdb
      starting_model_vrms = 1.2
    }
  }
}
```

```
}
```

If the phil script is in a file named `docking_script.phil`, the search would be run from the command-line as follows.

```
phenix.voyager.em_placement docking_script.phil
```

Most parameters specified in this script have been named in a way intended to convey the purpose of that parameter.

The `remove_phasertng_folder` parameter is activated to clean up the graph database produced by phasertng, which could be used in other circumstances as part of a larger automation framework or for debugging.

If the `point_group_symmetry` is provided, the symmetry detected in the reconstruction is applied to generate a full assembly from a single copy.

The `sequence_composition` parameter specifies the name of a file containing the sequences of all the components in the reconstruction.

By default, `em_placement` carries out an adaptive search, which is normally recommended. Aspects of this search can be over-ridden with expert user keywords specified below, for example to force a full 6D search or to force only a single rotation search over the whole ordered volume regardless of the estimated signal.

Literature

Likelihood-based signal and noise analysis for docking of models into cryo-EM maps. R.J. Read, C. Millán, A.J. McCoy, and T.C. Terwilliger. *Acta Cryst.* D79, 271-280 (2023). Likelihood-based docking of models into cryo-EM maps. C. Millán, A.J. McCoy, T.C. Terwilliger, and R.J. Read. *Acta Cryst.* D79, 281-289 (2023).

- Likelihood-based signal and noise analysis for docking of models into cryo-EM maps. R.J. Read, C. Millán, A.J. McCoy, and T.C. Terwilliger. *Acta Cryst.* D79, 271-280 (2023).
- Likelihood-based docking of models into cryo-EM maps. C. Millán, A.J. McCoy, T.C. Terwilliger, and R.J. Read. *Acta Cryst.* D79, 281-289 (2023).

List of all available keywords

`em_placementremove_phasertng_folder = True` Removes the phasertng folder with the databases at the end of the run
`level = process concise *summary logfile verbose testing` Set the verbosity level
`job_title = None` Job title in PHENIX GUI, not used on command line
`map_modelfull_map = None` Input full map file as alternative to half-map
`half_map = None` Input half map files
`point_group_symmetry = C1` Schonflies symbol plus order (e.g. D2, C7, I) or helical. Default is C1, no symmetry
`best_resolution = None` Best resolution (smallest d-spacing) found in input map. If zero it will be set automatically
`cell_rescale = None` Rescaling factor for map cell or magnification factor
`sequence_composition = None` Sequence file for entire EM reconstruction. Ignored if `fixed_scattering_ratio` is set, in which case total composition is set to model composition times `fixed_scattering_ratio`
`fixed_scattering_ratio = None` If set, total composition is model composition times this factor
`region_approach = *automatic manual rigid_body` Method for specifying search regions
`region_for_dockingManual` specification for region(s)
`name = None` A name for the region
`radius = None` radius for the sphere for anisotropy correction and docking
`centerx = None` x coord center of sphere for docking search
`chy = None` y coord center of sphere for docking search
`chz = None` z coord center of sphere for docking search
`dockingfrf_rescoring = True` use frfr after frf or not
`frf_clustering = True` use clustering or not for frf (or frfr if `frf_rescoring`)
`frf_target_ellg = 225`. ideal target eLLG for rotation search with strong signal
`minimum_frf_ellg = 7.5` minimum target eLLG for rotation search: controls sub-volume size
`refine_scale = False` flag for whether or not to refine cell scale
`top_files = 5` number of top potential solutions to save for

output and focused dockingsigma_cut = 7. threshold for carrying out focused docking is top score - sigma_cut*sqrt(top score)save_output_folder = True create output folder to save docked models and statisticsoutput_folder = None non-default path for output folderbrute_approach = *search six_dimensional_full six_dimensional_local search strategy (frf or brute)angle_aroundalpha = None alpha for brute rotation function anglebeta = None beta for brute rotation function anglegamma = None gamma for brute rotation function anglerange = None angular range for brute rotation functionbiological_unitmoleculename = model A name for the search modelmodel_file = None Coordinate in .pdb or .cifstarting_model_vrms = 1.2 Starting RMSD estimate for the modelmolecule_fixedfixed_molecule_name = "fixed" A name for the fixed modelfixed_model_file = None Fixed model coordinates in .pdb or .cifguiGUI-specific parameter required for output directoryoutput_dir = None

- em_placementremove_phasertng_folder = True Removes the phasertng folder with the databases at the end of the runlevel = process concise *summary logfile verbose testing Set the verbosity leveljob_title = None Job title in PHENIX GUI, not used on command linemap_modelfull_map = None Input full map file as alternative to half-mapshalf_map = None Input half map filespoint_group_symmetry = C1 Schonflies symbol plus order (e.g. D2, C7, I) or helical. Default is C1, no symmetrybest_resolution = None Best resolution (smallest d-spacing) found in input map. If zero it will be set automaticallycell_rescale = None Rescaling factor for map cell or magnification factorsequence_composition = None Sequence file for entire EM reconstruction. Ignored if fixed_scattering_ratio is set, in which case total composition is set to model composition times fixed_scattering_ratio.fixed_scattering_ratio = None If set, total composition is model composition times this factorregion_approach = *automatic manual rigid_body Method for specifying search regionsregion_for_dockingManual specification for region(s)name = None A name for the regionradius = None radius for the sphere for anisotropy correction and dockingcenterx = None x coord center of sphere for docking searchy = None y coord center of sphere for docking searchz = None z coord center of sphere for docking searchdockingfrf_rescoring = True use frf after frf or notfrf_clustering = True use clustering or not for frf (or frf if frf_rescoring)frf_target_ellg = 225. ideal target eLLG for rotation search with strong signalminimum_frf_ellg = 7.5 minimum target eLLG for rotation search: controls sub-volume sizefine_scale = False flag for whether or not to refine cell scaletop_files = 5 number of top potential solutions to save for output and focused dockingsigma_cut = 7. threshold for carrying out focused docking is top score - sigma_cut*sqrt(top score)save_output_folder = True create output folder to save docked models and statisticsoutput_folder = None non-default path for output folderbrute_approach = *search six_dimensional_full six_dimensional_local search strategy (frf or brute)angle_aroundalpha = None alpha for brute rotation function anglebeta = None beta for brute rotation function anglegamma = None gamma for brute rotation function anglerange = None angular range for brute rotation functionbiological_unitmoleculename = model A name for the search modelmodel_file = None Coordinate in .pdb or .cifstarting_model_vrms = 1.2 Starting RMSD estimate for the modelmolecule_fixedfixed_molecule_name = "fixed" A name for the fixed modelfixed_model_file = None Fixed model coordinates in .pdb or .cifguiGUI-specific parameter required for output directoryoutput_dir = None

- remove_phasertng_folder = True Removes the phasertng folder with the databases at the end of the run

- level = process concise *summary logfile verbose testing Set the verbosity level

- job_title = None Job title in PHENIX GUI, not used on command line

- map_modelfull_map = None Input full map file as alternative to half-mapshalf_map = None Input half map filespoint_group_symmetry = C1 Schonflies symbol plus order (e.g. D2, C7, I) or helical. Default is

C1, no symmetry
 best_resolution = None Best resolution (smallest d-spacing) found in input map. If zero it will be set automatically
 cell_rescale = None Rescaling factor for map cell or magnification
 factorsequence_composition = None Sequence file for entire EM reconstruction. Ignored if fixed_scattering_ratio is set, in which case total composition is set to model composition times fixed_scattering_ratio.
 fixed_scattering_ratio = None If set, total composition is model composition times this factor
 region_approach = *automatic manual rigid_body Method for specifying search regions
 region_for_dockingManual specification for region(s)
 name = None A name for the region
 radius = None radius for the sphere for anisotropy correction and docking
 centerx = None x coord center of sphere for docking search
 y = None y coord center of sphere for docking search
 z = None z coord center of sphere for docking search

- full_map = None Input full map file as alternative to half-maps
- half_map = None Input half map files
- point_group_symmetry = C1 Schonflies symbol plus order (e.g. D2, C7, I) or helical. Default is C1, no symmetry
- best_resolution = None Best resolution (smallest d-spacing) found in input map. If zero it will be set automatically
- cell_rescale = None Rescaling factor for map cell or magnification factor
- sequence_composition = None Sequence file for entire EM reconstruction. Ignored if fixed_scattering_ratio is set, in which case total composition is set to model composition times fixed_scattering_ratio.
- fixed_scattering_ratio = None If set, total composition is model composition times this factor
- region_approach = *automatic manual rigid_body Method for specifying search regions
- region_for_dockingManual specification for region(s)
 name = None A name for the region
 radius = None radius for the sphere for anisotropy correction and docking
 centerx = None x coord center of sphere for docking search
 y = None y coord center of sphere for docking search
 z = None z coord center of sphere for docking search
- name = None A name for the region
- radius = None radius for the sphere for anisotropy correction and docking
- centerx = None x coord center of sphere for docking search
 y = None y coord center of sphere for docking search
 z = None z coord center of sphere for docking search
- x = None x coord center of sphere for docking search
- y = None y coord center of sphere for docking search
- z = None z coord center of sphere for docking search
- dockingfrf_rescoring = True use frfr after frf or not
 frf_clustering = True use clustering or not for frf (or frfr if frf_rescoring)
 frf_target_ellg = 225. ideal target eLLG for rotation search with strong signal
 minimum_frf_ellg = 7.5 minimum target eLLG for rotation search: controls sub-volume size
 refine_scale = False flag for whether or not to refine cell scale
 top_files = 5 number of top potential solutions to save for output and focused dockings
 sigma_cut = 7. threshold for carrying out focused docking is top score - sigma_cut*sqrt(top score)
 save_output_folder = True create output folder to save docked models and statistics
 output_folder = None non-default path for output folder
 brute_approach = *search six_dimensional_full six_dimensional_local search strategy (frf or brute)
 angle_aroundalpha = None alpha for brute rotation function
 anglebeta = None beta for brute rotation function
 anglegamma = None gamma for brute rotation function
 anglerange = None angular range for brute rotation function

- frf_rescoring = True use frfr after frf or not
- frf_clustering = True use clustering or not for frf (or frfr if frf_rescoring)
- frf_target_ellg = 225. ideal target eLLG for rotation search with strong signal
- minimum_frf_ellg = 7.5 minimum target eLLG for rotation search: controls sub-volume size
- refine_scale = False flag for whether or not to refine cell scale
- top_files = 5 number of top potential solutions to save for output and focused docking
- sigma_cut = 7. threshold for carrying out focused docking is top score - sigma_cut*sqrt(top score)
- save_output_folder = True create output folder to save docked models and statistics
- output_folder = None non-default path for output folder
- brute_approach = *search six_dimensional_full six_dimensional_local search strategy (frf or brute)
- angle_aroundalpha = None alpha for brute rotation function anglebeta = None beta for brute rotation function anglegamma = None gamma for brute rotation function anglerange = None angular range for brute rotation function
- alpha = None alpha for brute rotation function angle
- beta = None beta for brute rotation function angle
- gamma = None gamma for brute rotation function angle
- range = None angular range for brute rotation function
- biological_unitmoleculename = model A name for the search modelmodel_file = None Coordinate in .pdb or .cifstarting_model_vrms = 1.2 Starting RMSD estimate for the modelmolecule_fixedfixed_molecule_name = "fixed" A name for the fixed modelfixed_model_file = None Fixed model coordinates in .pdb or .cif
- moleculename = model A name for the search modelmodel_file = None Coordinate in .pdb or .cifstarting_model_vrms = 1.2 Starting RMSD estimate for the model
- molecule_name = model A name for the search model
- model_file = None Coordinate in .pdb or .cif
- starting_model_vrms = 1.2 Starting RMSD estimate for the model
- molecule_fixedfixed_molecule_name = "fixed" A name for the fixed modelfixed_model_file = None Fixed model coordinates in .pdb or .cif
- fixed_molecule_name = "fixed" A name for the fixed model
- fixed_model_file = None Fixed model coordinates in .pdb or .cif
- guiGUI-specific parameter required for output directoryoutput_dir = None
- output_dir = None

Euclidian Model Matching

emma.html

phenix.emma

EMMA stands for Euclidean Model Matching and is the algorithm used by HySS to superimpose two putative solutions and to derive the consensus model. The same algorithm is also available through the external phenix.emma command-line interface. Enter phenix.emma without arguments to obtain the help page:

```
usage: phenix.emma [options] reference_coordinates reference_coordinates other_coordinates
```

options:

```
-h, --help show this help message and exit
--unit_cell=10,10,20,90,90,120|FILENAME
External unit cell parameters
--space_group=P212121|FILENAME
External space group symbol
--symmetry=FILENAME External file with symmetry information
--tolerance=FLOAT match tolerance
--diffraction_index_equivalent
Use only if models are diffraction-index equivalent.
```

Example: phenix.emma model1.pdb model2.sdb

The command takes two coordinate files in various formats (.pdb, CNS .sdb, SOLVE output, SHELX .ins) and compares the structures taking the space group symmetry, the allowed origin shifts and the hand ambiguity into account. The output is similar to the Match summary shown above in the example HySS output.

The match tolerance defaults to 3 Angstrom. For structures obtained with very low resolution data it may be necessary to specify a different tolerance, e.g. --tolerance=5.

The --symmetry option works just like it does for phenix.hyss. It can be used to extract symmetry information from external files such as input files for other programs (CNS, SHELX, SOLVE, ...) or reflection files.

However, the --symmetry option is only required if the information about the unit cell and the space group is missing in both coordinate files given to phenix.emma.

phenix.emma conducts an exhaustive search and, in contrast to HySS, displays all possible matches. The match with the largest number of matching sites is shown first, the match with the smallest number of matching sites is shown last (often just one site). Therefore you have to look at the beginning of the output to see the best match. I.e. if the output goes to the screen don't let yourself get distracted if you see a large number of Singles near the end of the output. Scroll back to see the best match.

Emma is also available via a web interface.

Docking a model into a local region of density with `emplace_local`

`emplace_local.html`

Authors

- Randy J Read, Claudia Millán, Airlie J McCoy, Thomas C Terwilliger
- Eric F Petterson, Tristan I Croll, Billy K Poon, Elaine C Meng, Dorothee Liebschner, Paul D Adams (ChimeraX interface)

Purpose

The `emplace_local` tool docks the input model within a sphere of density centered on a defined point in the reconstruction, returning the docked model, a local weighted map, and quality of fit statistics.

How `emplace_local` works

Ideally, half-maps are provided, then the chosen region of density is analysed for local signal and noise (both of which are resolution and direction-dependent) to define a likelihood target for rotation and translation searches, followed by rigid-body refinement.

If half-maps are unavailable, just a full map can be provided and then the likelihood target is defined by assuming that signal-to-noise drops off isotropically to the stated resolution limit.

The quality of fit can be judged by three factors:

- 1) whether there is a unique solution;
- 2) the log-likelihood-gain (LLG) score, which should preferably be greater than 60;
- 3) the map correlation coefficient to the weighted map used for docking.

The basic inputs are:

```
model file
two unmasked half-map files (preferred) or full-map file
nominal resolution (optional if half-maps provided, but recommended)
specification of center of search region
```

Note that the search center can be specified explicitly with the `sphere_center` keyword (the usual approach) or by translating the search model to the desired search region and not providing a `sphere_center` keyword.

It is also possible to provide a fixed model as prior information, accounting for the contributions of the fixed model to the sphere.

If the symmetry of a symmetric reconstruction is specified, the symmetry is applied to the output structure. Note that in the ChimeraX interface, ChimeraX functions are used to apply the symmetry.

Standard run of `emplace_local`

```
phenix.voyager.emplace_local map1=half1.map map2=half2.map
d_min=3.3 model_file=model.pdb sphere_center=100,140,170
```

Literature

Likelihood-based signal and noise analysis for docking of models into cryo-EM maps. R.J. Read, C. Millán, A.J. McCoy, and T.C. Terwilliger. *Acta Cryst.* D79, 271-280 (2023). Likelihood-based docking of models into cryo-EM maps. C. Millán, A.J. McCoy, T.C. Terwilliger, and R.J. Read. *Acta Cryst.* D79, 281-289 (2023). Likelihood-based interactive local docking into cryo-EM maps in ChimeraX. R.J. Read, E.F. Pettersen, A.J. McCoy, T.I. Croll, T.C. Terwilliger, B.K. Poon, E.C. Meng, D. Liebschner, and P.D. Adams. *Acta Cryst.* D80, 588-598 (2024).

- Likelihood-based signal and noise analysis for docking of models into cryo-EM maps. R.J. Read, C. Millán, A.J. McCoy, and T.C. Terwilliger. *Acta Cryst.* D79, 271-280 (2023).
- Likelihood-based docking of models into cryo-EM maps. C. Millán, A.J. McCoy, T.C. Terwilliger, and R.J. Read. *Acta Cryst.* D79, 281-289 (2023).
- Likelihood-based interactive local docking into cryo-EM maps in ChimeraX. R.J. Read, E.F. Pettersen, A.J. McCoy, T.I. Croll, T.C. Terwilliger, B.K. Poon, E.C. Meng, D. Liebschner, and P.D. Adams. *Acta Cryst.* D80, 588-598 (2024).

List of all available keywords

model_file = None Search model file
 fixed_model_file = None Fixed model file
 sequence_composition = None Optional sequence file with composition for full reconstruction, overrides fixed_scattering_ratio
 map = None Map file for full map (optional, not used if half-maps provided)
 map1 = None Map file for half-map 1 (optional if full map is provided)
 map2 = None Map file for half-map 2 (optional if full map is provided)
 point_group_symmetry = C1 Schonflies symbol plus order (e.g. D2, C7, I) or helical. Default is C1, no symmetry
 d_min = None Nominal local resolution. Preferred but optional if half-maps provided, required otherwise
 sphere_center = None Centre of sphere for docking
 model_vrms = 1.5 estimated RMSD for model
 fixed_scattering_ratio = 1.5 Estimated scattering ratio between sphere and model
 brute_rotation = False Test all orientations instead of fast rotation function
 refine_scale = False Turn on cell scale (voxel size) refinement
 cell_rescale = None Rescaling factor for map cell or magnification factor
 level = *process concise summary logfile verbose testing Output level from process (minimum output) to testing
 top_files = 5 Maximum number of top docking solutions to save and report
 outputfileroot = top_model Manually set file root

- model_file = None Search model file
- fixed_model_file = None Fixed model file
- sequence_composition = None Optional sequence file with composition for full reconstruction, overrides fixed_scattering_ratio
- map = None Map file for full map (optional, not used if half-maps provided)
- map1 = None Map file for half-map 1 (optional if full map is provided)
- map2 = None Map file for half-map 2 (optional if full map is provided)
- point_group_symmetry = C1 Schonflies symbol plus order (e.g. D2, C7, I) or helical. Default is C1, no symmetry
- d_min = None Nominal local resolution. Preferred but optional if half-maps provided, required otherwise
- sphere_center = None Centre of sphere for docking
- model_vrms = 1.5 estimated RMSD for model
- fixed_scattering_ratio = 1.5 Estimated scattering ratio between sphere and model
- brute_rotation = False Test all orientations instead of fast rotation function
- refine_scale = False Turn on cell scale (voxel size) refinement

- cell_rescale = None Rescaling factor for map cell or magnification factor
- level = *process concise summary logfile verbose testing Output level from process (minimum output) to testing
- top_files = 5 Maximum number of top docking solutions to save and report
- outputfileroot = top_model Manually set file root
- fileroot = top_model Manually set file root

Explore Metric Symmetry

explore_metric_symmetry.html

Purpose

iotbx.explore_metric_symmetry is a program that allows a user to quickly determine the symmetry of the lattice, given a unit cell, and determine relations between various possible point groups.

Another use of iotbx.explore_metric_symmetry is in the comparison of related unit cells, that are related by a linear recombination of their basis vectors.

Keywords

A list of keywords and concise help can be obtained by typing: iotbx.explore_metric_symmetry

```
options:
-h, --help show this help message and exit
--unit_cell=10,10,20,90,90,120|FILENAME
External unit cell parameters
--space_group=P212121|FILENAME
External space group symbol
--symmetry=FILENAME External file with symmetry information
--max_delta=FLOAT Maximum delta/obliquity used in determining the
lattice symmetry, using a modified Le-Page algorithm.
Default is 5.0 degrees
--start_from_p1 Reduce to Niggli cell and forget the input space group
before higher metric symmetry is sought.
--graph=GRAPH A graphical representation of the graph will be
written out. Requires Graphviz to be installed and in
path.
--centring_type=CENTRING_TYPE
Centring type, choose from P,A,B,C,I,R,F
--other_unit_cell=10,20,30,90,103.7,90
Other unit cell, for unit cell comparison
--other_space_group=OTHER_SPACE_GROUP
space group for other_unit_cell, for unit cell
comparison
--other_centring_type=OTHER_CENTRING_TYPE
Centring type, choose from P,A,B,C,I,R,F
--no_point_group_graph
Do not carry out the construction of a point group
graph.
--relative_length_tolerance=FLOAT
Tolerance for unit cell lengths to be considered
equal-ish.
--absolute_angle_tolerance=FLOAT
Angular tolerance in unit cell comparison
--max_order=INT Maximum volume change for target cell
```

Explore Metric Symmetry. A list of possible unit cells and space groups is given for the given specified unit cell and space group combination.

The keywords unit_cell, space_group (or centring_type) define the crystal symmetry for which a point group graph is constructed.

The keyword max_delta sets the tolerance used in the determination of the lattice symmetry.

the keyword start_from_p1 in combination with the space group is equivalent to specification of the centring_type only.

If graphviz is installed, an a png file with the point group graph can be constructed by specifying the filename of the png graph with the keyword graph.

If a second crystal is specified by the keywords other_unit_cell, other_space_group (or other_centring_type) the unit cells will be compared. Using linear combinations of the smallest unit cell, possible matches for the large unit cell are sought. If desired, the larger unit cell can be expanded as well using the keyword max_order.

The tolerances in the unit cell comparison can be changed from their defaults (10% on the lengths and 20 degrees on the angles) using the keywords relative_length_tolerance and absolute_angle_tolerance.

Construction of a point group graph can be skipped using the key no_point_group_graph.

Examples

Constructing a point group graph given some basic information:

```
iotbx.explore_metric_symmetry --unit_cell="20,30,40,90,90,90" \  
--centring_type=P
```

All point groups between P 1 and P 2 2 2 will be listed

Comparing two related unit cells can be done using:

```
iotbx.explore_metric_symmetry --unit_cell="20,30,40,90,90,90" \  
--centring_type=P --other_unit_cell="40,80,60,90,90,90" \  
--other_centring_type=F
```

phenix.f000: given PDB file and mean bulk-solvent density estimate F000 reflection

f000.html

Description

The program reads PDB model and optionally provided mean bulk-solvent density value (default is 0.35).
Output: estimate of F000 value and solvent content

The program reads PDB model and optionally provided mean bulk-solvent density value (default is 0.35).
Output: estimate of F000 value and solvent content

Command line usage examples:

```
% phenix.f000 model.pdb  
% phenix.f000 model.pdb mean_solvent_density=0.44
```


phenix.find_alt_orig_sym_mate: Identify equivalent MR solutions of the same dataset irrespective of origin and symmetry operation

find_alt_orig_sym_mate.html

Contents

- Author(s)
- Purpose
- Usage
- Using from the command-line
- Examples of PHIL input
- Output
- Do the MR solutions match?
- Algorithm
- Also move HETATM (hetero atoms)
- Debug mode
- Caveats
- Changes
- Literature
- List of all available keywords

Author(s)

- phenix.find_alt_orig_sym_mate: Robert D. Oeffner
- PHENIX GUI: Nat Echols

Purpose

phenix.find_alt_orig_sym_mate (or phenix.famos) attempts to find the best superposition of two different molecular replacement solutions of the same dataset termed moving.pdb and fixed.pdb with respect to all symmetry operations and alternate origin shifts permitted by the spacegroup of the crystal. If either of the pdb files contain more chains each chain will be tested for their best match against chains in the other pdb file. It does so by calculating a score value, MLAD (see Algorithm), for all possible symmetry operations and alternate origin shifts of each chain in moving.pdb compared with chains in fixed.pdb. The transformation with the smallest MLAD is retained for that particular pair of chains.

Usage

phenix.find_alt_orig_sym_mate can be run from the GUI by clicking on the Find alternate-origin symmetry mates button under the Model tools category on the right hand side of the main Phenix window.

Once the run button has been pressed the program will execute for a minute or two depending on the size of the molecules and space group. Afterwards the run status page features a table of the best matches between

pairs of chains in the two files. This includes the MLAD values, chain IDs, symmetry transformations and alternate origins for the two structures. If the same origin shift is not applied to all pairs of chains a warning will be printed. If the spacegroup of the crystal has a floating origin this generally results in an origin offset between chains that is not a rational number.

In the case illustrated above the chains A, B, C in the file 1tfx.pdb all have unique good matches with chains H, A, I in the file test42.pdb. This is stated in the log file as well as printed with green text on the GUI. One pair of chains does not make a good match. A warning is then printed with red text. The second column in the table in the end of the log file contains the RMSDs of the pairs of matched chains. From the results tab the superposed structure can be visually inspected in Coot or Pymol. Matches with low MLAD scores will have correspondingly good superpositions.

Using from the command-line

phenix.find_alt_orig_sym_mate can also be run from the command-line with PHIL input in either a text file or as keywords like:

```
phenix.famos moving.pdb=pdbfile1 fixed.pdb=pdbfile2
```

or:

```
phenix.find_alt_orig_sym_mate my_phil_input.txt.
```

The PHIL input scopes, moving and fixed, specifies the MR solutions. Both scopes are programmatically equivalent and must be non-empty. This means that a scope should either specify the parameter xyzfname or the sub-scope mrsolution or the sub-scope pickle.solution.

Examples:

- Use the pdb file from one of the MR solutions in a scope. This is useful for simple cases such as when the solution comes from the PHENIX MRage GUI or when it is obtained from another MR program than Phaser.
- Specify a Phaser MR solution by assigning the .sol file name of the solution to the mrsolution.solfname parameter of the sub-scope as well as assigning the IDs of the ensembles and the corresponding MR models to the parameters mrsolution.ensembles.name and mrsolution.ensembles.xyzfname. Multiple search components are specified as multiple ensembles. This way of specifying solutions is useful for solution files produced when Phaser is run from the command line and it produces several solutions of which one is the correct solution.
- Use a solution from the Phaser-MR GUI in PHENIX. In that case the parameter pickle.solution.pkifname in the sub-scope should be assigned to the solution file that is produced by PHENIX after an MR calculation. The pickle.solution.philfname should then be assigned to the input file for the MR calculation. This way of specifying solutions is useful when MR solutions are available from having run Phaser through the PHENIX interface.
- If space group and the unit cell dimensions is not available from any of the input files then these need to be specified by assigning the parameter, spacegroupname, either to a PDB file with a CRYST1 record or to an MTZ file with that information. This should be the data file used for the molecular replacement calculation.

Examples of PHIL input

Unless the input just constitutes of two PDB files a PHIL file is the easiest way to enter input. A few examples of PHIL for phenix.find_alt_orig_sym_mate are given below.

Testing a command-line Phaser MR solution file against a solution specified as a PDB file:

```
AltOrigSymMates.fixed.mrsolution
{
  solfname = "testdata/MR_3ECI_A0_2P82_A0.sol"
  ensembles
  {
    name = "MR_2P82_A0"
    xyzfname = "testdata/sculpt_2P82_A0.pdb"
  }
  ensembles
  {
    name = "MR_3ECI_A0"
    xyzfname = "testdata/sculpt_3ECI_A0.pdb"
  }
}
AltOrigSymMates.moving.pdb="testdata/2z0d.pdb"
```

Testing two set of solution files from the PHENIX Phaser-MR GUI against one another:

```
AltOrigSymMates.fixed.pickle_solution
{
  philfname = "testdata/phaser_mr_13.eff"
  pklfname = "testdata/phaser_mr_13.pkl"
}

AltOrigSymMates.moving.pickle_solution
{
  philfname = "testdata/phaser_mr_11.eff"
  pklfname = "testdata/phaser_mr_11.pkl"
}
```

Testing two solution files from the command-line version of Phaser against one another:

```
AltOrigSymMates.fixed.mrsolution
{
  solfname = "testdata/MR_3ECI_A0_2P82_A0.sol"
  ensembles
  {
    name = "MR_2P82_A0"
    xyzfname = "testdata/sculpt_2P82_A0.pdb"
  }
  ensembles
  {
    name = "MR_3ECI_A0"
    xyzfname = "testdata/sculpt_3ECI_A0.pdb"
  }
}

AltOrigSymMates.moving.mrsolution
{
  solfname = "testdata/MR_2ZPN_A0_2P82_A0.sol"
  ensembles
  {
    name = "MR_2P82_A0"
    xyzfname = "testdata/sculpt_2P82_A0.pdb"
  }
  ensembles
  {
    name = "MR_2ZPN_A0"
    xyzfname = "testdata/sculpt_2ZPN_A0.pdb"
  }
}
AltOrigSymMates.spacegroupfname = "testdata/2Z0D.mtz"
```

For more information on the PHIL input see the bottom of this page.

Output

The closest match between chains in the moving section to chains in the fixed section is saved with the name of the pdb file in the fixed scope, concatenated with the name of the pdb file in the moving scope but prepended with "MinMLAD_". A log file with the name of the pdb file in the fixed scope, concatenated with the name of the pdb file in the moving scope but prepended with "AltOrigSymMLAD_" is written containing standard output. All files are saved in the current working directory. If MR solutions specified in the PHIL contains multiple solutions then phenix.find_alt_orig_sym_mate will output multiple log files corresponding to each MR solution.

Do the MR solutions match?

A good match between two chains usually have a MLAD value below 1.5 whereas a bad match usually have a value above 2.0. The RMSD value for each match is also computed using the sequence alignment that the SSM superpose program uses. But without superposing the chains. If a match is good the RMSD will be small just like the MLAD value. It is advisable to visually inspect that the chains of structures have been transformed to the same origins in a molecular graphics viewing program.

Algorithm

phenix.find_alt_orig_sym_mate computes configurations by looping over all symmetry operations and alternative origin shifts. An alignment between C-alpha atoms from moving.pdb and fixed.pdb is computed using secondary structure matching (SSM). If that fails or if the MLAD score achieved with SSM is larger than 2.0 an alignment is computed using MMTBX alignment functions which is part of the CCTBX. To estimate the best match a distance measure between the aligned C-alpha atoms is computed for each configuration. The mean log absolute deviation (MLAD) is defined as:

$$\text{MLAD}(\text{dR}) = \Sigma(\log(\text{dr} \cdot \text{dr} / (|\text{dr}| + 0.9) + \max(0.9, \min(\text{dr} \cdot \text{dr}, 1))) - \log(0.9)),$$

where dr is the difference vector between a pair of aligned C-alpha atoms and the sum is taken over all atom pairs in the alignment. The factor log(0.9) is subtracted to ensure that MLAD(0) = 0.0, i.e. that two identical structures produces the value zero.

MLAD can loosely be interpreted as a distance measure between structures. But it is not a metric in a strict mathematical sense since the triangle inequality is not fulfilled. Unlike a plain root mean square deviation the logarithm in the MLAD formula will downplay contributions of atom pairs where the atoms are spatially distant. If an RMSD were employed such atom pairs would contribute on an equal footing with those groups of atom pairs that can be superposed perfectly. This in turn may lead to non-optimal superpositions when the structures tested against one another consists of multiple domains where one domain has undergone a domain motion, i.e. where a subset of atoms in one chain are bound to be spatially distant from the atoms in the other chain they have been paired with.

phenix.find_alt_orig_sym_mate.py will for each chain in the fixed scope find the smallest MLAD with a copy of each chain in the moving scope for a given symmetry operation and alternative origin. When all chains in the fixed scope have been tested these copies will be saved to a file. For spacegroups with floating origin the minimum MLAD is found by doing a Golden Sectioning minimization along the polar axis for each copy of chains in moving.pdb.

Also move HETATM (hetero atoms)

Invoking this flag will move hetero atoms (ligands, waters, metals, etc.) in conjunction with their associated peptide chain. The program first invokes phenix.sort_hetatms as to associate hetero atoms sensibly with adjacent peptide chains.

After having identified the transformation for a chain yielding the smallest MLAD score it then subjects the associated hetero atoms to the same transformation.

Debug mode

Invoking the debug flag produces individual pdb files of the C-alpha atoms used for each SSM alignment of the moving scope for each permitted symmetry operation and alternative origin. A gold atom is placed at the centroid of the C-alpha atoms. Similar files are produced for the fixed scope. These files are stored in a subfolder named "AltOrigSymMatesFiles".

If a floating origin is present in the space group a table of MLAD values is produced by sliding a copy of the chains in the moving scope along the polar axis in the fractional interval [-0.5, 0.5] for each permitted symmetry operation and alternative origin.

Caveats

The program tests all solutions present in solution files entered as fixed scope against all solutions present in solution files entered as moving scope. Consequently the execution time is proportional to the number of solutions in fixed scope times the number of solutions in moving scope.

The execution time is proportional to the number of SSM alignments being tested; if SSM identifies 12 alignments the program will take 12 times as long.

Changes

The command-line syntax mentioned in the Computational Crystallography Newsletter 2012 January has been replaced by PHIL syntax.

Literature

Algorithms for deriving crystallographic space-group information R.W. Große-Kunstleve Acta Cryst. A55, 383-395 (1999)

phenix.find_alt_orig_sym_mate Robert D. Oeffner, Gábor Bunkóczi and Randy J. Read Computational Crystallography Newsletter 2012 January, 5-10 (2012)

Secondary-structure matching (SSM), a new tool for fast protein structure alignment in three dimensions. E. Krissinel and K. Henrick Acta Cryst. D60, 2256-2268 (2004)

List of all available keywords

AltOrigSymMatespacegroupfname = None A model file or a reflection data file providing the space group and cell dimension. This is needed in the uncommon situation both moving and fixed model files do not already contain these, (CRYST1 record in a pdb file or _cell and _symmetry blocks in a cif file). Must be based on the data set used for the MR calculation.chosenorigin = "Any" By default all allowed origin shifts are probed. Specifying '2/3, 1/3, 1/2' tells the program to only explore symmetry operations for the unit cells displaced at fractional positions (2/3, 1/3, 1/2)use_all_SSM = False By default only the best sequence alignment from SSM is used. Setting this value to True will include other suboptimal alignments for

testing.no_symmetry_operations = False When set to False all symmetry operations are probed during the computation.movehetatms = False Move hetatoms close to a chain in conjunction with that chain. Hetatoms are first associated with individual chains with phenix.sort_hetatms before undergoing the same spatial transformation as the chains.writepdbfiles = True Save final transformed pdb file and any intermediate pdb files realised from MR solution filesoutputfileprefix = None Alternative prefix for output files .pdb, .log and .sol instead of AltOrigSymMLAD_*.nproc = 1 Number of CPU cores to use when submitting MR solution files with multiple solutionsquiet = False Set to True for silencing output when using multiple cpu coresdebug = False verbose = False write_cif = False goodmatch = 1.5 Weed out poor matches with MLAD values larger than goodmatch.strictly_protein = True Relax the condition that structures must only be protein chains with conventional residue names This might help solving cases where one or both pdb files are flagged as having no chainsworkdir = None Specify if output should be written elsewhere than the current working directory.gui_output_dir = None For PHENIX GUI only.job_title = None Job title in PHENIX GUI, not used on command linemovingSpecify either a PDB file, an MR solution or a pickle_solution.pdb = None Deprecated, use model in the future. Path to a PDB file from an MR solution. Constituent chains will be moved separately.model = None Path to a PDB file from an MR solution. Constituent chains will be moved separately.mrsolutionMR solution produced by command-line version of Phasersolfname = None path to the solution file, i.e. the .sol fileensemblesEnsembles constituent in this MR solution. Typically these are chains in the PDB file.name = None ID for ensemble in the MR solution, i.e. the name after the 'SOLU 6DIM ENSE' string in the .sol file.xyzfname = None Deprecated, use model in the future Path to the PDB file with atoms representing this ensemble This is the search model used by Phaser for this ensemble prior to solving the structure. It is not the solved structure.model = None Path to the PDB file with atoms representing this ensemble. This is the search model used by Phaser for this ensemble prior to solving the structure. It is not the solved structure.fixedSpecify either a PDB file, an MR solution or a pickle_solution.pdb = None Deprecated, use model in the future. Path to a PDB file from an MR solution. Constituent chains will remain fixed.model = None Path to a PDB file from an MR solution. Constituent chains will remain fixed.mrsolutionMR solution produced by command-line version of Phasersolfname = None path to the solution file, i.e. the .sol fileensemblesEnsembles constituent in this MR solution. Typically these are chains in the PDB file.name = None ID for ensemble in the MR solution, i.e. the name after the 'SOLU 6DIM ENSE' string in the .sol file.xyzfname = None Deprecated, use model in the future. Path to the PDB file with atoms representing this ensemble. This is the search model used by Phaser for this ensemble prior to solving the structure. It is not the solved structure.model = None Path to the PDB file with atoms representing this ensembl...

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- spacegroupfname = None A model file or a reflection data file providing the space group and cell dimension. This is needed in the uncommon situation both moving and fixed model files do not already contain these, (CRYST1 record in a pdb file or _cell and _symmetry blocks in a cif file). Must be based on the data set used for the MR calculation.
- chosenorigin = "Any" By default all allowed origin shifts are probed. Specifying '2/3, 1/3, 1/2' tells the program to only explore symmetry operations for the unit cells displaced at fractional positions (2/3, 1/3, 1/2)
- use_all_SSM = False By default only the best sequence alignment from SSM is used. Setting this value to True will include other suboptimal alignments for testing.
- no_symmetry_operations = False When set to False all symmetry operations are probed during the computation.
- movehetatms = False Move hetatoms close to a chain in conjunction with that chain. Hetatoms are first associated with individual chains with phenix.sort_hetatms before undergoing the same spatial transformation as the chains.
- writepdbfiles = True Save final transformed pdb file and any intermediate pdb files realised from MR solution files
- outputfileprefix = None Alternative prefix for output files .pdb, .log and .sol instead of AltOrigSymMLAD_*.*
- nproc = 1 Number of CPU cores to use when submitting MR solution files with multiple solutions
- quiet = False Set to True for silencing output when using multiple cpu cores
- debug = False
- verbose = False

- `write_cif = False`
- `goodmatch = 1.5` Weed out poor matches with MLAD values larger than `goodmatch`.
- `strictly_protein = True` Relax the condition that structures must only be protein chains with conventional residue names This might help solving cases where one or both `pdb` files are flagged as having no chains
- `workdir = None` Specify if output should be written elsewhere than the current working directory.
- `gui_output_dir = None` For PHENIX GUI only.
- `job_title = None` Job title in PHENIX GUI, not used on command line
- `movingSpecify` either a PDB file, an MR solution or a `pickle_solution.pdb = None` Deprecated, use `model` in the future. Path to a PDB file from an MR solution. Constituent chains will be moved separately.
- `model = None` Path to a PDB file from an MR solution. Constituent chains will be moved separately.
- `mrsolution` MR solution produced by command-line version of Phaser
- `solfname = None` path to the solution file, i.e. the `.sol` file
- `ensembles` Ensembles constituent in this MR solution. Typically these are chains in the PDB file.
- `name = None` ID for ensemble in the MR solution, i.e. the name after the `"SOLU 6DIM ENSE"` string in the `.sol` file.
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- `pdb = None` Deprecated, use `model` in the future. Path to a PDB file from an MR solution. Constituent chains will be moved separately.
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- `fixedSpecify` either a PDB file, an MR solution or a `pickle_solution.pdb = None` Deprecated, use `model` in the future. Path to a PDB file from an MR solution. Constituent chains will remain fixed.
`model = None` Path to a PDB file from an MR solution. Constituent chains will remain fixed.
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Rapid fitting of secondary structure to a map with `find_helices_strands`

`find_helices_strands.html`

Contents

- Author(s)
- Purpose
- Running the program
- Methods
- Using `find_helices_strands` to bootstrap `phenix.autobuild`
- Command-line use
- Specific limitations and problems
- References
- List of all available keywords

Author(s)

`find_helices_strands`: Tom Terwilliger PULCHRA: Piotr Rotkiewicz

- `find_helices_strands`: Tom Terwilliger
- PULCHRA: Piotr Rotkiewicz

Purpose

`find_helices_strands` is a command line tool for finding helices and strands in a map and building an model of the parts of a structure that have regular secondary structure. It can be used for protein, RNA, and DNA. An option is to use a rapid chain-tracing algorithm to build CA of proteins, followed by reconstruction of a full model.

Running the program

The program is available in the Phenix GUI under "Model building". The only required input is an electron density map, such as the output of density modification. A sequence file is optional.

The default behavior is to search for protein helices and strands, but you may instead look for nucleic acid helices, or use the `trace_chain` method (describe below) to find C-alpha positions independently of secondary structure motifs. Using the default options without a sequence, the model will be a poly-alanine backbone; with the `trace_chain` method it will be a C-alpha trace, but you may optionally have PULCHRA (Rotkiewicz & Skolnick 2008) complete the backbone.

The program usually only requires a few minutes at most to run. Output will be to a new directory, with a single PDB file containing a partial model.

Methods

How `find_helices_strands` finds helices and strands in maps:

`find_helices_strands` first identifies helical segments as rods of density at 5-8 Å. Then it identifies helices at higher resolution keeping the overall locations of the helices fixed. Then it identifies the directions and CA positions of helices by noting the helical pattern of high-density points offset slightly along the helix axis from the main helical density (as used in "O" to identify helix direction). Finally model helices are fit to the density using the positions and orientations identified in the earlier steps. A similar procedure is used to identify strands. Then the helices and strands are combined into a single model.

How `find_helices_strands` finds RNA and DNA helices in maps:

`find_helices_strands` finds RNA and DNA helices differently than it finds helices in proteins. It uses a convolution search to find places in the asymmetric unit where an A-form RNA or B-form DNA helix can be placed. These are assembled into contiguous helical segments if possible. The resolution of this search is 4.5 Å if you have resolution beyond 4.5 Å, and the resolution of your data otherwise.

How `trace_chain` finds CA positions in maps:

The RESOLVE `trace_chain` algorithm places dummy atoms down the middle of all the tubes of density in a map, then it attempts to find sets of these atoms that may be CA atoms, where the atoms are spaced by 3.8 Å and where there is strong density between each pair. This yields segments represented by CA atoms. Next PULCHRA (Rotkiewicz & Skolnick 2008) can optionally be used to reconstruct a full main-chain model. Finally RESOLVE is used to assemble all the resulting fragments into a model.

Using `find_helices_strands` to bootstrap `phenix.autobuild`

If you run `phenix.autobuild` at low resolution (3.5 Å or lower) then your model may have strands built instead of helices. You can use `find_helices_strands` to help bootstrap autobuild model-building by providing the helical model from `find_helices_strands` to `phenix.autobuild`. Just run `phenix.find_helices_strands` with your best map `map_coeffs.mtz`. Then take the helical model pass it to `phenix.autobuild` with the keyword (in addition to your usual keywords for autobuild):

```
consider_main_chain_list=map_coeffs.mtz_helices.pdb
```

Then the AutoBuild wizard will treat your helical model just like one of the models that it builds, and merge it into the model as it is being assembled.

Command-line use

From the command-line you can type:

```
phenix.find_helices_strands map_coeffs.mtz quick=True
```

If you want a more thorough run, then skip the "quick=True" flag. If you want (or need) to specify the column names from your mtz file, you will need to tell `find_helices_strands` what FP and PHIB are, in this format:

```
phenix.find_helices_strands map_coeffs.mtz \  
  labin="LABIN FP=2FOFCWT PHIB=PH2FOFCWT"
```

If you want to specify a sequence file, then in the last step `find_helices_strands` will try to align your sequence with the map and model:

```
phenix.find_helices_strands map_coeffs.mtz seq_file=seq.dat
```

If you want to use the `trace_chain` algorithm, then specify:

```
phenix.find_helices_strands map_coeffs.mtz seq_file=seq.dat trace_chain=True
```

Here is an example using data from the PHENIX examples library:

```
phenix.find_helices_strands $PHENIX/phenix_examples/p9-build/p9-resolve.mtz \
  labin="FP=FP PHIB=PHIM FOM=FOMM" trace_chain=True
```

That should build a model using sample data in a few seconds.

Specific limitations and problems

References

Fast procedure for reconstruction of full-atom protein models from reduced representations. P. Rotkiewicz, and J. Skolnick. J Comput Chem 29, 1460-5 (2008).

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Rapid model building of alpha-helices in electron-density maps. T.C. Terwilliger. Acta Crystallogr D Biol Crystallogr 66, 268-75 (2010).

- Rapid model building of alpha-helices in electron-density maps. T.C. Terwilliger. Acta Crystallogr D Biol Crystallogr 66, 268-75 (2010).

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- Rapid chain tracing of polypeptide backbones in electron-density maps. T.C. Terwilliger. Acta Crystallogr D Biol Crystallogr 66, 285-94 (2010).

List of all available keywords

use_pdb_in_directly = False Use PDB input model directly
output_dir = None Output directory
seq_file = None Sequence file for sequence alignment
compare_file = None PDB file for comparison only
res_convolution = 4.5 high-resolution limit for convolution calculation. (Applies to nucleic acids only)
temp_dir = "temp_dir" Optional temporary work directory
helices_only = False Find only helices (Applies only if trace_chain=False)
strands_only = False Find only strands (Applies only if trace_chain=False)
trace_chain = False Use resolve trace_chain algorithm
helices_before_trace = None Find helices before running resolve trace_chain algorithm
strands_before_trace = None Find strands before running resolve trace_chain algorithm
pulchra = False Use PULCHRA to create all-atom model from CA model
reverse_if_better = True Reverse chain direction in trace_chain if pulchra=True and reversed chain has improved geometry.
remove_side = False Try to remove density for side chains in trace_chain
rho_cut_min = None Minimum rho/sigma at potential CA positions
rat_pair_min = None Target minimum ratio of rho/sigma at midpoint between potential CA positions to the average at the two CA positions
minimum_length_angstroms_helices_strands = None Minimum length (A CA start - CA end) for helix/strand to keep
dist_ca_target = 3.8 Target CA-CA distance
dist_ca_sep = 4.0 Max separation of CA atoms
dist_ca_tol_max = None Maximum tolerance for CA-CA distances. Normally 0.8 A for medium and 1.3 A for quick
dist_ca_start = None Guess of tolerance for CA-CA distances. Set automatically by default.
dist_ca_tol = None Tolerance for CA-CA distances. Normally set automatically Typical values are 0.3-1.0 A A high number can be used to force a more thorough search. Compare with target_p_ratio which adjusts CA-CA tolerance to achieve a targeted ratio of nonamers to atoms.
cutoff_trace = 0.000 The top

cutoff_trace fraction of peaks in trace_chain will be assumed to be non-protein, and all peaks near them will be ignored. (Default = 0.00)ncut_trace_min = 0 The top ncut_trace_min peaks in trace_chain will be assumed to be non-protein, and all peaks near them will be ignored. (Default = 0)target_p_ratio = None Target ratio of nonamers found to atoms in a.u default=3 for quick, 4 otherwise. This can be used as an alternate method to adjust the thoroughness of trace_chain searches. Differs from setting dist_ca_tol by adjusting CA-CA tolerance to achieve the desired target ratio, while dist_ca_tol=tol sets the tolerance directly to toltrace_ratio_long = 0.5 When setting tolerances for CA-CA distances in trace_chain the upper bound will increase trace_ratio_long as fast as the lower bound. (Default=0.5)ratio_trace_extra = None Minimum ratio of dist between extra atoms added in trace_chain to rad_sep_trace Default is 1.25 for standard and 1.5 for quickrad_sep_trace = None Dummy atom separation in trace_chain Default is 0.6 A for standard and 0.75 for quick Increased if resolution is greater than 3 A Value of rad_mask_trace in resolve will be rad_sep_trace*2atom_target_ratio = 1.0 Ratio of target number of atoms to estimated actual number_n_atoms_total_scale = 1.0 Ratio of estimated atoms in au to standard estimatefill_gaps = True Try to fill in gaps in trace_chaintarget_angle = None Target angle for CA-CA-CA or P-P-P use_any_side = False Use any side chain that fits density in assemblycc_helix_min = None Minimum CC of low-res helical density to map to keep. group_ca_length = None Minimum length of a segment of helix or strand to keep. (only applies if trace_chain=False) cc_strand_min = None Minimum CC of strand density to map to keep. (only applies if trace_chain=False) quick = False Try to find structure quicklytarget_time_for_trace_chain = 15 Target length of time to work on trace_chain (sec)recycle = False Recycle CA positions in trace_chainoptimize = True Try to optimize CA-CA tolerance in trace_chain to obtain nonamer ratio equal to target_p_ratioassemble = False Assemble model wi...

- use_pdb_in_directly = False Use PDB input model directly
- output_dir = None Output directory
- seq_file = None Sequence file for sequence alignment
- compare_file = None PDB file for comparison only
- res_convolution = 4.5 high-resolution limit for convolution calculation. (Applies to nucleic acids only)
- temp_dir = "temp_dir" Optional temporary work directory
- helices_only = False Find only helices (Applies only if trace_chain=False)
- strands_only = False Find only strands (Applies only if trace_chain=False)
- trace_chain = False Use resolve trace_chain algorithm
- helices_before_trace = None Find helices before running resolve trace_chain algorithm
- strands_before_trace = None Find strands before running resolve trace_chain algorithm
- pulchra = False Use PULCHRA to create all-atom model from CA model
- reverse_if_better = True Reverse chain direction in trace_chain if pulchra=True and reversed chain has improved geometry.
- remove_side = False Try to remove density for side chains in trace_chain
- rho_cut_min = None Minimum rho/sigma at potential CA positions
- rat_pair_min = None Target minimum ratio of rho/sigma at midpoint between potential CA positions to the average at the two CA positions
- minimum_length_angstroms_helices_strands = None Minimum length (A CA start - CA end) for helix/strand to keep
- dist_ca_target = 3.8 Target CA-CA distance

- `dist_ca_sep` = 4.0 Max separation of CA atoms
- `dist_ca_tol_max` = None Maximum tolerance for CA-CA distances. Normally 0.8 Å for medium and 1.3 Å for quick
- `dist_ca_start` = None Guess of tolerance for CA-CA distances. Set automatically by default.
- `dist_ca_tol` = None Tolerance for CA-CA distances. Normally set automatically Typical values are 0.3-1.0 Å A high number can be used to force a more thorough search. Compare with `target_p_ratio` which adjusts CA-CA tolerance to achieve a targeted ratio of nonamers to atoms.
- `cutoff_trace` = 0.000 The top `cutoff_trace` fraction of peaks in `trace_chain` will be assumed to be non-protein, and all peaks near them will be ignored. (Default = 0.00)
- `ncut_trace_min` = 0 The top `ncut_trace_min` peaks in `trace_chain` will be assumed to be non-protein, and all peaks near them will be ignored. (Default = 0)
- `target_p_ratio` = None Target ratio of nonamers found to atoms in a.u default=3 for quick, 4 otherwise. This can be used as an alternate method to adjust the thoroughness of `trace_chain` searches. Differs from setting `dist_ca_tol` by adjusting CA-CA tolerance to achieve the desired target ratio, while `dist_ca_tol=tol` sets the tolerance directly to `tol`
- `trace_ratio_long` = 0.5 When setting tolerances for CA-CA distances in `trace_chain` the upper bound will increase `trace_ratio_long` as fast as the lower bound. (Default=0.5)
- `ratio_trace_extra` = None Minimum ratio of dist between extra atoms added in `trace_chain` to `rad_sep_trace` Default is 1.25 for standard and 1.5 for quick
- `rad_sep_trace` = None Dummy atom separation in `trace_chain` Default is 0.6 Å for standard and 0.75 for quick Increased if resolution is greater than 3 Å Value of `rad_mask_trace` in `resolve` will be `rad_sep_trace*2`
- `atom_target_ratio` = 1.0 Ratio of target number of atoms to estimated actual number
- `n_atoms_total_scale` = 1.0 Ratio of estimated atoms in au to standard estimate
- `fill_gaps` = True Try to fill in gaps in `trace_chain`
- `target_angle` = None Target angle for CA-CA-CA or P-P-P
- `use_any_side` = False Use any side chain that fits density in assembly
- `cc_helix_min` = None Minimum CC of low-res helical density to map to keep.
- `group_ca_length` = None Minimum length of a segment of helix or strand to keep. (only applies if `trace_chain=False`)
- `cc_strand_min` = None Minimum CC of strand density to map to keep. (only applies if `trace_chain=False`)
- `quick` = False Try to find structure quickly
- `target_time_for_trace_chain` = 15 Target length of time to work on `trace_chain` (sec)
- `recycle` = False Recycle CA positions in `trace_chain`
- `optimize` = True Try to optimize CA-CA tolerance in `trace_chain` to obtain nonamer ratio equal to `target_p_ratio`
- `assemble` = False Assemble model with `resolve` after `trace_chain`
- `resolve_size` = 12 Size of `resolve` to use. You may need a bigger size than in other `resolve` applications
- `coarse_grid` = False `Coarse_grid` allows the use of a smaller `resolve` size
- `verbose` = False Verbose output

- `raise_sorry` = False Raise sorry if problems
- `debug` = False Debugging output
- `dry_run` = False Just read in and check parameter names
- `mtz_label_prefix` = None Prefix for column names, used by Coot. The GUI will detect and set this automatically. It should not be changed directly by users.
- `delete_tmp_dir` = False GUI setting, does not apply to command-line version
- `job_title` = None Job title in PHENIX GUI, not used on command line
- `input_filesmtz_in` = None MTZ file with coefficients for a map
`map_coeff_labels` = None If map coefficients cannot be identified automatically from your MTZ file, you can specify the label or labels for them. (Please separate labels with blank space, MTZ columns grouped together separated by commas with no blanks.) You can specify: `map_coeff_labels` (e.g., FWT,PHIFWT) amplitudes and phases (e.g., FP,SIGFP PHIB) or amplitudes, phases, weights (e.g., FP,SIGFP PHIB FOM)
`labin` = "" Labin line for MTZ file with map coefficients. Normally use instead `map_coeff_labels`. This is available for backward compatibility. You can specify: LABIN FP=myFP PHIB=myPHI FOM=myFOM where myFP is your column label for FP
`map_in` = None CCP4 or MRC-style map file
`pdb_in` = None Optional PDB file to be used for seeding trace_chain with CA
- `mtz_in` = None MTZ file with coefficients for a map
- `map_coeff_labels` = None If map coefficients cannot be identified automatically from your MTZ file, you can specify the label or labels for them. (Please separate labels with blank space, MTZ columns grouped together separated by commas with no blanks.) You can specify: `map_coeff_labels` (e.g., FWT,PHIFWT) amplitudes and phases (e.g., FP,SIGFP PHIB) or amplitudes, phases, weights (e.g., FP,SIGFP PHIB FOM)
- `labin` = "" Labin line for MTZ file with map coefficients. Normally use instead `map_coeff_labels`. This is available for backward compatibility. You can specify: LABIN FP=myFP PHIB=myPHI FOM=myFOM where myFP is your column label for FP
- `map_in` = None CCP4 or MRC-style map file
- `pdb_in` = None Optional PDB file to be used for seeding trace_chain with CA
- `output_filetarget_output_format` = *None pdb mmCIF Desired output format (if possible). Choices are None (try to use input format), pdb, mmCIF. If output model does not fit in pdb format, mmCIF will be used. Default is pdb.
`output_model` = None Output PDB file
`output_log` = None Output log file name. If you want to specify a directory to put this file in then please use "`output_dir=myoutput_dir`"
`gui_output_dir` = None GUI parameter only
- `target_output_format` = *None pdb mmCIF Desired output format (if possible). Choices are None (try to use input format), pdb, mmCIF. If output model does not fit in pdb format, mmCIF will be used. Default is pdb.
- `output_model` = None Output PDB file
- `output_log` = None Output log file name. If you want to specify a directory to put this file in then please use "`output_dir=myoutput_dir`"
- `gui_output_dir` = None GUI parameter only
- `crystal_inforesolution` = None high-resolution limit for map calculation
`chain_type` = *PROTEIN DNA RNA Chain type (for identifying main-chain and side-chain atoms)
`solvent_fraction` = 0.5 You can specify the solvent fraction
- `resolution` = None high-resolution limit for map calculation

- chain_type = *PROTEIN DNA RNA Chain type (for identifying main-chain and side-chain atoms)
- solvent_fraction = 0.5 You can specify the solvent fraction

Finding and analyzing NCS from heavy-atom sites or a model with `find_ncs`

`find_ncs.html`

Contents

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- How `find_ncs` works
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- References
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Authors:

- `find_ncs`: Tom Terwilliger
- `simple_ncs_from_pdb` : Tom Terwilliger
- Phil command interpreter: Ralf W. Grosse-Kunstleve
- `find_domain`: Peter Zwart

Purpose

The `find_ncs` method identifies NCS in (a) the chains in a PDB file or (b) a set of heavy-atom sites, or (c) from the density in a map and writes out the NCS operators in forms suitable for phenix.refine, resolve, and the AutoSol and AutoBuild Wizards, or (d) from an existing symmetry file (`.ncs_spec`), or (e) it can generate NCS for a helical array from helical parameters.

How `find_ncs` works

The basic steps that the `find_ncs` carries out are:

- Decide whether to use `simple_ncs_from_pdb` (used if the input file contains chains from a PDB file) or RESOLVE NCS identification (used if the input file contains heavy-atom sites) or `find_ncs_from_density` (used if a map is supplied and sites are not found from either of the above)
- call either `simple_ncs_from_pdb` or RESOLVE or `find_ncs_from_density` to identify NCS

- Evaluate the NCS by calculating the correlation of NCS-related electron density based on the input map coefficients mtz file.
- Report the NCS operators and correlations

Output files from find_ncs

The output files that are produced are:

- NCS operators written in format for phenix.refine: find_ncs.ncs
- NCS operators written in format for the PHENIX Wizards: find_ncs.ncs_spec

What find_ncs needs

- find_ncs needs a file with map coefficients and optionally a second file containing NCS information.

The file with NCS information can be...

- a PDB file with a model (find_ncs will call simple_ncs_from_pdb to extract NCS operators from the chains in your model)
- a PDB file with heavy-atom sites (find_ncs will call RESOLVE to find NCS operators from your heavy-atom sites)
- an NCS definitions file written by a PHENIX wizard (e.g., AutoSol_1.ncs_spec, produced by AutoSol)
- a RESOLVE log file containing formatted NCS operators

Map coefficients file

The file with map coefficients can be any MTZ file with coefficients for a map. If find_ncs does not choose the correct columns automatically, then you can specify them with a command like:

```
labin="labin FP=FP PHIB=PHIB FOM=FOM "
```

If you have no map coefficients yet (you just have some sites and want to get operators, for example), you can tell find_ncs to ignore the map with:

```
ncs_parameters.force_ncs=True
```

You still need to supply a map coefficients file (the map itself will be ignored though).

Format of NCS operators

The NCS operators look like this:

```
rota_matrix 0.7955 -0.5660 0.2164
rota_matrix -0.5511 -0.8242 -0.1300
rota_matrix 0.2520 -0.0159 -0.9676
tran_orth 18.4021 5.3569 -23.3621

center_orth 15.9430 11.8822 0.6609
```

rota_matrix and tran_orth are the rotation and translation information, and center_orth is (approximately) the center of the region where the operator applies.

The standard in the .ncs_spec files is that the NCS operator is applied to copy j to map copy j on to copy 1. If you want the inverse operators you can specify invert_matrices=True and then you will get operators that map copy 1 on to copy j.

Examples

Standard run of find_ncs

If you have just a map coefficients file you can say:

```
phenix.find_ncs mlt.mtz
```

and find_ncs will look for NCS-related density in mlt.mtz (see find_ncs_from_density for more details on how this works)

If you have a model and a map file, from the command-line you can type:

```
phenix.find_ncs anb.pdb mlt.mtz
```

This will produce the following output:

```
Getting column labels from mlt.mtz for input map file
FILE TYPE: ccp4_mtz
All labels: ['FP', 'SIGFP', 'PHIC', 'FOM']
Labin line will be: labin FP=FP PHIB=PHIC FOM=FOM
To change it modify this: params.ncs.labin="labin FP=FP PHIB=PHIC FOM=FOM "
This is the map that will be used to evaluate NCS

Reading NCS information from: anb.pdb
Copying mlt.mtz to temp_dir/mlt.mtz
This PDB file contains 2 chains and 636 total residues
and 636 C-alpha or P atoms and 4740 total atoms
NCS will be found using the chains in this PDB file
Chains in this PDB file: ['M', 'Z']
Two chains were found in the file anb.pdb, chain M and chain Z

GROUPS BASED ON QUICK COMPARISON: []
Looking for invariant domains for ...: ['M', 'Z'] [[[2, 138], [193, 373]], [[2, 138], [193, 373]]]
Residues 2-138, 193-373, matched between the two chains

Copying mlt.mtz to temp_dir/mlt.mtz
Copying temp_dir/NCS_correlation.log to NCS_correlation.log
Log file for NCS correlation is in NCS_correlation.log
List of refined NCS correlations: [1.0, 0.80000000000000004]
There were two separate groups of residues that had different NCS
relationships. Residues 193-373 of each chain were in one group, and
residues 2-138 in each chain were in the other group.

The electron density map had a correlation between the two NCS-related
chains of 1.0 for the first group, and 0.8 for the second

The NCS operators for each are listed.

GROUP 1
Summary of NCS group with 2 operators:
ID of chain/residue where these apply: [['M', 'Z'], [[[193, 373]], [[193, 373]]]]
RMSD (A) from chain M: 0.0 0.0
Number of residues matching chain M:[181, 181]
Source of NCS info: anb.pdb
Correlation of NCS: 1.0

OPERATOR 1
CENTER: 69.1058 -9.5443 59.4674

ROTA 1: 1.0000 0.0000 0.0000
ROTA 2: 0.0000 1.0000 0.0000
ROTA 3: 0.0000 0.0000 1.0000
TRANS: 0.0000 0.0000 0.0000

OPERATOR 2
```

CENTER: 37.5004 -37.0709 -62.5441

ROTA 1: 0.7751 -0.6211 -0.1162
ROTA 2: -0.3607 -0.5859 0.7256
ROTA 3: -0.5188 -0.5205 -0.6782
TRANS: 9.7485 27.6460 17.2076

GROUP 2

Summary of NCS group with 2 operators:

ID of chain/residue where these apply: [['M', 'Z'], [[2, 138]], [[2, 138]]]

RMSD (A) from chain M: 0.0 0.0

Number of residues matching chain M:[137, 137]

Source of NCS info: anb.pdb

Correlation of NCS: 0.8

OPERATOR 1

CENTER: 66.6943 -13.3128 21.6769

ROTA 1: 1.0000 0.0000 0.0000
ROTA 2: 0.0000 1.0000 0.0000
ROTA 3: 0.0000 0.0000 1.0000
TRANS: 0.0000 0.0000 0.0000

OPERATOR 2

CENTER: 39.0126 -53.7392 -13.4457

ROTA 1: 0.3702 -0.9275 -0.0516
ROTA 2: -0.8933 -0.3402 -0.2938
ROTA 3: 0.2549 0.1548 -0.9545
TRANS: 1.7147 -0.6936 7.2172

Creating a model containing unique set of chains and nearby NCS-related chains

If you want to create a model that surrounds a unique set of chains with all the NCS-related ones, for now you need to carry out a few extra steps. In this example the symmetry is helical with 5 unique chains; a similar approach can be used for point-group symmetry.

For helical assemblies, NCS is not always simple to apply. The general idea is to generate more than enough copies, pare them down to the ones you want, get the NCS file for these copies, and apply it to your original 5 unique copies

- Make a PDB file with one set of the unique 5 copies
- Make a PDB file with a single monomer from these 5
- use `apply_ncs` on the file with one chain to generate more copies
- if you didn't get enough copies to cover all the locations you want, extend your NCS operators:

```
phenix.find_ncs my_helical_ncs.ncs_spec helix_extend_range=30
```

This will extend the operators +/-30 Å and write out a new `.ncs_spec` file

- Once you get enough copies from step 2, edit that PDB file to remove all the chains that you don't want.
- Run `find_ncs` on the PDB file with your desired chains to get the NCS relationships

6. Now (finally) run `apply_ncs` with the new `.ncs_spec` file and your PDB file with one set of 5 copies

Additional information

References

List of all available keywords

find_ncsjob_title = None Job title in PHENIX GUI, not used on command line
input_filesncs_in_type = *None chains sites ncs_file Type of ncs information. Choices are: chains: a PDB file with two or more chains that have a consistent residue-numbering system. sites: a PDB file or fractional-coordinate file with atomic positions of heavy-atoms that show NCS ncs_file: an ncs object file from PHENIX. ncs_in = None File with NCS information (PDB file with heavy-atom sites or with NCS-related chains) or NCS object
mtz_in = None MTZ file with coefficients for a map that can be used to assess NCS. Required for finding NCS from heavy-atom sites
labin = "" Labin line for MTZ file with map coefficients. This is optional if find_ncs can guess the correct coefficients for FP PHI and FOM. Otherwise specify: LABIN FP=myFP PHIB=myPHI FOM=myFOM where myFP is your column label for FP
map_in = None CCP4 or MRC-style map file (instead of mtz_in)
output_filesparams_out = find_ncs_params.eff Parameters file to rerun find_ncs
ncs_out = find_ncs.ncs_spec Output file with NCS information. If set, then files will also be written for resolve and phenix.refine and as BIOMT records
ncs_format = *ncs_spec biomt Format of output NCS information. ncs_spec is standard phenix ncs format. biomt is BIOMT records.
ncs_mask_file = None If defined and mtz_in is defined, map coefficients will be written to this file that show the asymmetric unit of NCS. By default the file contains map coefficients. To write a CCP4-style map. Use ncs_mask_as_mtz=False to write map coefficients for the mask.
ncs_mask_as_mtz = True If True, the ncs_mask_file will be map coefficients. If False a CCP4-style map will be written.
ncs_au_file = None If defined, map coefficients showing the asymmetric unit of NCS will be written to this file.
crystal_inforesolution = None high-resolution limit for map calculation
directorytemp_dir = "temp_dir" Temporary work directory
output_dir = "" Output directory where files are to be written
gui_output_dir = None control
verbose = True Verbose output
debug = False Debugging output
raise_sorry = False Raise sorry if problems
dry_run = False Just read in and check parameter names
require_nonzero = True Require non-zero values in data columns to consider reading in.
ncs_parametersinvert_matrices = False Matrices normally describe how to map each operator on to the first one. To get operators for mapping the first one on the others, set invert_matrices=True
chain_max_rmsd = 2 limit of rms difference between chains to be considered as copies
ncs_restrict = 0 You can specify the number of NCS operators to look for
force_ncs = False You can tell find_ncs to ignore the map. This is useful if you only have FP but no phases yet...
optimize_ncs = False You can tell find_ncs to optimize the NCS by making as compact a molecule as possible.
no_ncs_sg_mismatch = True If one or more NCS operators is the same as a SG operator, skip the NCS
stry_density_after_ha = True If search for NCS using ha sites does not yield NCS correlation of at least target_ncs_cc or the number of operators is less than target_ncs_operators then try finding NCS directly from density
target_ncs_operators = None Number of expected NCS operators (used to decide whether to try density search)
ncs_copies_max = None Max number of allowed NCS operators in density search
target_ncs_cc = 0.80 Expected NCS correlation
minimum_ncs_cc = 0.30 Minimum NCS correlation . If the overall NCS correlation is less than minimum_ncs_cc it will be ignored. NOTE: This is applied at the very end of NCS identification. The parameter minimum_ncs_cc_from_density is applied in finding NCS from density as well.
fraction_ncs_min = 0.05 Minimum fraction of the asymmetric unit covered by NCS. NCS ignored if less than this.
helix_extend_range = None Range in z to extend helical operators (A)
helix_trans_along_z = None Along with helix_theta, defines helical parameters. Note: matrices are for mapping position k on to identity, this may be the ...

- find_ncsjob_title = None Job title in PHENIX GUI, not used on command line
input_filesncs_in_type = *None chains sites ncs_file Type of ncs information. Choices are: chains: a PDB file with two or more chains that have a consistent residue-numbering system. sites: a PDB file or fractional-coordinate file with atomic positions of heavy-atoms that show NCS ncs_file: an ncs object file from PHENIX. ncs_in = None File with NCS information (PDB file with heavy-atom sites or with NCS-related chains) or NCS object
mtz_in = None MTZ file with coefficients for a map that can be used to assess NCS. Required for finding NCS from heavy-atom sites
labin = "" Labin line for MTZ file with map coefficients. This is optional

if find_ncs can guess the correct coefficients for FP PHI and FOM. Otherwise specify: LABIN FP=myFP PHIB=myPHI FOM=myFOM where myFP is your column label for FP map_in = None CCP4 or MRC-style map file (instead of mtz_in)output_filesparams_out = find_ncs_params.eff Parameters file to rerun find_ncsncs_out = find_ncs.ncs_spec Output file with NCS information. If set, then files will also be written for resolve and phenix.refine and as BIOMT recordsncs_format = *ncs_spec biomt Format of output NCS information. ncs_spec is standard phenix ncs format. biomt is BIOMT records.ncs_mask_file = None If defined and mtz_in is defined, map coefficients will be written to this file that show the asymmetric unit of NCS. By default the file contains map coefficients. To write a CCP4-style map. Use ncs_mask_as_mtz=False to write map coefficients for the mask.ncs_mask_as_mtz = True If True, the ncs_mask_file will be map coefficients. If False a CCP4-style map will be written.ncs_au_file = None If defined, map coefficients showing the asymmetric unit of NCS will be written to this file.crystal_inresolution = None high-resolution limit for map calculationdirectorytemp_dir = "temp_dir" Temporary work directoryoutput_dir = "" Output directory where files are to be writtengui_output_dir = None controlverbose = True Verbose outputdebug = False Debugging outputraise_sorry = False Raise sorry if problemsdry_run = False Just read in and check parameter namesrequire_nonzero = True Require non-zero values in data columns to consider reading in.ncs_parametersinvert_matrices = False Matrices normally describe how to map each operator on to the first one. To get operators for mapping the first one on the others, set invert_matrices=Truechain_max_rmsd = 2 limit of rms difference between chains to be considered as copiesncs_restrict = 0 You can specify the number of NCS operators to look forforce_ncs = False You can tell find_ncs to ignore the map. This is useful if you only have FP but no phases yet...optimize_ncs = False You can tell find_ncs to optimize the NCS by making as compact a molecule as possible. no_ncs_sg_mismatch = True If one or more NCS operators is the same as a SG operator, skip the NCstry_density_after_ha = True If search for NCS using ha sites does not yield NCS correlation of at least target_ncs_cc or the number of operators is less than target_ncs_operators then try finding NCS directly from densitytarget_ncs_operators = None Number of expected NCS operators (used to decide whether to try density search)ncs_copies_max = None Max number of allowed NCS operators in density search target_ncs_cc = 0.80 Expected NCS correlationminimum_ncs_cc = 0.30 Minimum NCS correlation . If the overall NCS correlation is less than minimum_ncs_cc it will be ignored. NOTE: This is applied at the very end of NCS identification. The parameter minimum_ncs_cc_from_density is applied in finding NCS from density as well.fraction_ncs_min = 0.05 Minimum fraction of the asymmetric unit covered by NCS. NCS ignored if less than this.helix_extend_range = None Range in z to extend helical operators (A)helix_trans_along_z = None Along with helix_theta, defines helical parameters. Note: matrices are for mapping position k on to identity, this may be the ...

- job_title = None Job title in PHENIX GUI, not used on command line

- input_filesncs_in_type = *None chains sites ncs_file Type of ncs information. Choices are: chains: a PDB file with two or more chains that have a consistent residue-numbering system. sites: a PDB file or fractional-coordinate file with atomic positions of heavy-atoms that show NCS ncs_file: an ncs object file from PHENIX. ncs_in = None File with NCS information (PDB file with heavy-atom sites or with NCS-related chains) or NCS objectmtz_in = None MTZ file with coefficients for a map that can be used to assess NCS. Required for finding NCS from heavy-atom siteslabin = "" Labin line for MTZ file with map coefficients. This is optional if find_ncs can guess the correct coefficients for FP PHI and FOM. Otherwise specify: LABIN FP=myFP PHIB=myPHI FOM=myFOM where myFP is your column label for FP map_in = None CCP4 or MRC-style map file (instead of mtz_in)

- ncs_in_type = *None chains sites ncs_file Type of ncs information. Choices are: chains: a PDB file with two or more chains that have a consistent residue-numbering system. sites: a PDB file or fractional-coordinate file with atomic positions of heavy-atoms that show NCS ncs_file: an ncs object file from PHENIX.

- `ncs_in` = None File with NCS information (PDB file with heavy-atom sites or with NCS-related chains) or NCS object
- `mtz_in` = None MTZ file with coefficients for a map that can be used to assess NCS. Required for finding NCS from heavy-atom sites
- `labin` = "" Labin line for MTZ file with map coefficients. This is optional if `find_ncs` can guess the correct coefficients for FP PHI and FOM. Otherwise specify: LABIN FP=myFP PHIB=myPHI FOM=myFOM where myFP is your column label for FP
- `map_in` = None CCP4 or MRC-style map file (instead of `mtz_in`)
- `output_filesparams_out` = `find_ncs_params.eff` Parameters file to rerun `find_ncs`
`ncs_out` = `find_ncs.ncs_spec` Output file with NCS information. If set, then files will also be written for `resolve` and `phenix.refine` and as BIOMT records
`ncs_format` = `*ncs_spec biomt` Format of output NCS information. `ncs_spec` is standard phenix ncs format. `biomt` is BIOMT records.
`ncs_mask_file` = None If defined and `mtz_in` is defined, map coefficients will be written to this file that show the asymmetric unit of NCS. By default the file contains map coefficients. To write a CCP4-style map. Use `ncs_mask_as_mtz=False` to write map coefficients for the mask.
`ncs_mask_as_mtz` = True If True, the `ncs_mask_file` will be map coefficients. If False a CCP4-style map will be written.
`ncs_au_file` = None If defined, map coefficients showing the asymmetric unit of NCS will be written to this file.
- `params_out` = `find_ncs_params.eff` Parameters file to rerun `find_ncs`
- `ncs_out` = `find_ncs.ncs_spec` Output file with NCS information. If set, then files will also be written for `resolve` and `phenix.refine` and as BIOMT records
- `ncs_format` = `*ncs_spec biomt` Format of output NCS information. `ncs_spec` is standard phenix ncs format. `biomt` is BIOMT records.
- `ncs_mask_file` = None If defined and `mtz_in` is defined, map coefficients will be written to this file that show the asymmetric unit of NCS. By default the file contains map coefficients. To write a CCP4-style map. Use `ncs_mask_as_mtz=False` to write map coefficients for the mask.
- `ncs_mask_as_mtz` = True If True, the `ncs_mask_file` will be map coefficients. If False a CCP4-style map will be written.
- `ncs_au_file` = None If defined, map coefficients showing the asymmetric unit of NCS will be written to this file.
- `crystal_inforesolution` = None high-resolution limit for map calculation
- `resolution` = None high-resolution limit for map calculation
- `directories`
`temp_dir` = "temp_dir" Temporary work directory
`output_dir` = "" Output directory where files are to be written
`gui_output_dir` = None
- `temp_dir` = "temp_dir" Temporary work directory
- `output_dir` = "" Output directory where files are to be written
- `gui_output_dir` = None
- `control`
`verbose` = True Verbose output
`debug` = False Debugging output
`raise_sorry` = False Raise sorry if problems
`sorry_run` = False Just read in and check parameter names
`require_nonzero` = True Require non-zero values in data columns to consider reading in.
- `verbose` = True Verbose output
- `debug` = False Debugging output
- `raise_sorry` = False Raise sorry if problems

- `dry_run = False` Just read in and check parameter names
- `require_nonzero = True` Require non-zero values in data columns to consider reading in.
- `ncs_parameters`
 - `invert_matrices = False` Matrices normally describe how to map each operator on to the first one. To get operators for mapping the first one on the others, set `invert_matrices=True`
 - `chain_max_rmsd = 2` limit of rms difference between chains to be considered as copies
 - `ncs_restrict = 0` You can specify the number of NCS operators to look for
 - `force_ncs = False` You can tell `find_ncs` to ignore the map. This is useful if you only have FP but no phases yet...
 - `optimize_ncs = False` You can tell `find_ncs` to optimize the NCS by making as compact a molecule as possible.
 - `no_ncs_sg_mismatch = True` If one or more NCS operators is the same as a SG operator, skip the NCS
 - `try_density_after_ha = True` If search for NCS using ha sites does not yield NCS correlation of at least `target_ncs_cc` or the number of operators is less than `target_ncs_operators` then try finding NCS directly from density
 - `target_ncs_operators = None` Number of expected NCS operators (used to decide whether to try density search)
 - `ncs_copies_max = None` Max number of allowed NCS operators in density search
 - `target_ncs_cc = 0.80` Expected NCS correlation
 - `minimum_ncs_cc = 0.30` Minimum NCS correlation . If the overall NCS correlation is less than `minimum_ncs_cc` it will be ignored. NOTE: This is applied at the very end of NCS identification. The parameter `minimum_ncs_cc_from_density` is applied in finding NCS from density as well.
 - `fraction_ncs_min = 0.05` Minimum fraction of the asymmetric unit covered by NCS. NCS ignored if less than this.
 - `helix_extend_range = None` Range in z to extend helical operators (A)
 - `helix_trans_along_z = None` Along with `helix_theta`, defines helical parameters. Note: matrices are for mapping position k on to identity, this may be the opposite of what you would expect.
 - `helix_theta = None` Along with `helix_trans_along_z`, defines helical parameters. Note: matrices are for mapping position k on to identity, this may be the opposite of what you would expect.
 - `n_try_ncs = 3` Number of tries to find ncs from heavy-atom sites
 - `ncs_thorough = 8` ha_ds_window = None Used to specify window of atom-atom lengths in heavy-atom search. None uses default
 - `sha_window_max = 30` Used to specify max atom-atom length in heavy-atom search. None uses default. Specify a longer distance to look more thoroughly, a shorter distance to speed up the process
 - `dist_cut_ncs = None` Thoroughness for looking for heavy-atom sites (high=more thorough)
 - `coordinate_offset = None` Origin offset to apply to `ncs_intol_r`
 - `intol_r = 0.02` tolerance in rotations for point group or helical symmetry
 - `abs_tol_t = 2` tolerance in translations (A) for point group or helical symmetry
 - `rel_tol_t = 0.05` tolerance in translations (fractional) for point group or helical symmetry
- `invert_matrices = False` Matrices normally describe how to map each operator on to the first one. To get operators for mapping the first one on the others, set `invert_matrices=True`
- `chain_max_rmsd = 2` limit of rms difference between chains to be considered as copies
- `ncs_restrict = 0` You can specify the number of NCS operators to look for
- `force_ncs = False` You can tell `find_ncs` to ignore the map. This is useful if you only have FP but no phases yet...
- `optimize_ncs = False` You can tell `find_ncs` to optimize the NCS by making as compact a molecule as possible.
- `no_ncs_sg_mismatch = True` If one or more NCS operators is the same as a SG operator, skip the NCS
- `try_density_after_ha = True` If search for NCS using ha sites does not yield NCS correlation of at least `target_ncs_cc` or the number of operators is less than `target_ncs_operators` then try finding NCS directly from density
- `target_ncs_operators = None` Number of expected NCS operators (used to decide whether to try density search)
- `ncs_copies_max = None` Max number of allowed NCS operators in density search

- `target_ncs_cc = 0.80` Expected NCS correlation
- `minimum_ncs_cc = 0.30` Minimum NCS correlation . If the overall NCS correlation is less than `minimum_ncs_cc` it will be ignored. NOTE: This is applied at the very end of NCS identification. The parameter `minimum_ncs_cc_from_density` is applied in finding NCS from density as well.
- `fraction_ncs_min = 0.05` Minimum fraction of the asymmetric unit covered by NCS. NCS ignored if less than this.
- `helix_extend_range = None` Range in z to extend helical operators (A)
- `helix_trans_along_z = None` Along with `helix_theta`, defines helical parameters. Note: matrices are for mapping position k on to identity, this may be the opposite of what you would expect.
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- `n_try_ncs = 3` Number of tries to find ncs from heavy-atom sites
- `ncs_thorough = 8`
- `ha_ds_window = None` Used to specify window of atom-atom lengths in heavy-atom search. None uses defaults
- `ha_window_max = 30` Used to specify max atom-atom length in heavy-atom search. None uses defaults. Specify a longer distance to look more thoroughly, a shorter distance to speed up the process
- `dist_cut_ncs = None` Thoroughness for looking for heavy-atom sites (high=more thorough)
- `coordinate_offset = None` Origin offset to apply to `ncs_in`
- `tol_r = 0.02` tolerance in rotations for point group or helical symmetry
- `abs_tol_t = 2` tolerance in translations (A) for point group or helical symmetry
- `rel_tol_t = 0.05` tolerance in translations (fractional) for point group or helical symmetry
- `find_ncs_from_densityinput_filesmtz_in = None` MTZ file with coefficients for a map `labin = ""` Labin line for MTZ file with map coefficients. This is optional if `find_ncs_from_density` can guess the correct coefficients for FP PHI and FOM. Otherwise specify: `LABIN FP=myFP PHIB=myPHI FOM=myFOM` where `myFP` is your column label for FP `center_pdb_in = None` Optional PDB file with list of centers of density to be used as search model `density_mtz_in = None` MTZ file with coefficients for density around center defined by first atom in `center_pdb_in` (required) NOTE: Must be in space group p1 with identical cell to `mtz_in` `density_labin = ""` Labin line for MTZ file `density_mtz_in`. This is optional if `find_ncs_from_density` can guess the correct coefficients for FP PHI and FOM. Otherwise specify: `LABIN FP=myFP PHIB=myPHI FOM=myFOM` where `myFP` is your column label for FP
- `output_filescenters_pdb_out = guess_molecular_centers.pdb` Output PDB file with coords of centers
- `log = find_ncs_from_density.log` Output log file
- `ncs_spec_file = 'find_ncs_from_density.ncs_spec'` NCS specification file with all NCS information
- `params_out = find_ncs_from_density_params.eff` Parameters file to rerun `find_ncs_from_density`
- `ncs_mask_file = None` If defined, map coefficients will be written to this file that show the asymmetric unit of NCS.
- `directorytemp_dir = "temp_dir"` Temporary work directory
- `output_dir = ""` Output directory where files are to be written
- `density_searchdensity_radius = 10.` Radius for density to be cut out and compared
- `peak_separation = 15` Minimum distance between centers. Use about $1.5 \times \text{density_radius}$
- `density_peaks = 20` Number of NCS-related peaks of density to output
- `delta_phi = 20` Angular spacing of search
- `rotz_only = False` Rotate around Z only
- `ncs_copies_max = None` Maximum number of NCS copies to look for
- `min_ratio_to_top_cc = 0.75` Peaks will be kept up to `ncs_copies_max`, or `min_ratio_to_top_cc * best cc`, whichever comes first
- `minimum_ncs_cc_from_density = 0.30` Overall NCS CC at full resolution must be at least this high
- `fraction_ncs_min_from_density = 0.05` Minimum fraction of the asymmetric unit covered by NCS. NCS ignored if less than

`this.dump_ncs_density = False` You can dump 1 mtz file for each possible NCS copy so that you can see if the NCS is real Each file has the local map transformed to the orientation and position of copy 1 so they should all superimpose The files are written to the `temp_dir` (e.g., `temp_dir/ncs_1_map_coeffs_001.mtz`)
`map_operators_inside_unit_cell = False` If set then put the operators inside the unit cell. This removes the use of space-group symmetry for SG P1 and is used normally for non-crystal
`data.find_centers`
`smoothing_radius = None` Radius for smoothing squared density to find centers Choose a smaller value to get more center guesses. Default is same value as `density_radius`.
`n_center_find = None` Target number of centers to find Smoothing radius will be varied from 2 to $2 \times \text{smoothing_radius}$ and the value giving the number of peaks closest to `n_center_find` will be used. Normally leave this set to
`Nonen_center_use = 20` Number of locations to consider as molecular centers in density search. Try increasing this number to look harder for NCS.
`try_all_centers = False` If set, try all `n_center_use` centers. Normally stop if a good solution is found.
`ok_ncs_cc = 0.6` NCS correlation good enough to terminate search
`crystal_info.resolution = 4` High-resolution limit for map calculation It is useful to cut the resolution at 3-5 Å If you use higher resolution, use a finer `delta`
`phisolvent_fraction = 0.5` solvent fraction
`control.verbose = True` Verbose output
`debug = False` Debugging output
`raise_sorry = False` Raise sorry if problems
`dry_run = False` Just read in and check parameter names
`resolve_command_list = None` You can supply any resolve command here NOTE: for command-line usage you need to enclose the whole set of commands in double quotes (") and each individual command in single quotes (') like this: `resolve_command_list="'no_build...'`

- `input_files.mtz_in = None` MTZ file with coefficients for a map `labin = ""` Labin line for MTZ file with map coefficients. This is optional if `find_ncs_from_density` can guess the correct coefficients for FP PHI and FOM. Otherwise specify: `LABIN FP=myFP PHIB=myPHI FOM=myFOM` where `myFP` is your column label for FP
`center_pdb_in = None` Optional PDB file with list of centers of density to be used as search model
`density.mtz_in = None` MTZ file with coefficients for density around center defined by first atom in `center_pdb_in` (required) NOTE: Must be in space group p1 with identical cell to `mtz_in`
`labin = ""` Labin line for MTZ file `density.mtz_in`. This is optional if `find_ncs_from_density` can guess the correct coefficients for FP PHI and FOM. Otherwise specify: `LABIN FP=myFP PHIB=myPHI FOM=myFOM` where `myFP` is your column label for FP
- `mtz_in = None` MTZ file with coefficients for a map
- `labin = ""` Labin line for MTZ file with map coefficients. This is optional if `find_ncs_from_density` can guess the correct coefficients for FP PHI and FOM. Otherwise specify: `LABIN FP=myFP PHIB=myPHI FOM=myFOM` where `myFP` is your column label for FP
- `center_pdb_in = None` Optional PDB file with list of centers of density to be used as search model
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- `output_files.centers_pdb_out = guess_molecular_centers.pdb` Output PDB file with coords of centers
`log = find_ncs_from_density.log` Output log file
`ncs_spec_file = 'find_ncs_from_density.ncs_spec'` NCS specification file with all NCS information
`params_out = find_ncs_from_density.params.eff` Parameters file to rerun `find_ncs_from_density`
`ncs_mask_file = None` If defined, map coefficients will be written to this file that show the asymmetric unit of NCS.
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- `ncs_spec_file = 'find_ncs_from_density.ncs_spec'` NCS specification file with all NCS information
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- `directoriestemp_dir = "temp_dir"` Temporary work directory `output_dir = ""` Output directory where files are to be written
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- `density_searchdensity_radius = 10` Radius for density to be cut out and compared `peak_separation = 15` Minimum distance between centers. Use about $1.5 \times \text{density_radius}$ `density_peaks = 20` Number of NCS-related peaks of density to output `delta_phi = 20` Angular spacing of search `rotz_only = False` Rotate around Z only `ncs_copies_max = None` Maximum number of NCS copies to look for `min_ratio_to_top_cc = 0.75` Peaks will be kept up to `ncs_copies_max`, or `min_ratio_to_top_cc * best cc`, whichever comes first `minimum_ncs_cc_from_density = 0.30` Overall NCS CC at full resolution must be at least this high `fraction_ncs_min_from_density = 0.05` Minimum fraction of the asymmetric unit covered by NCS. NCS ignored if less than this. `dump_ncs_density = False` You can dump 1 mtz file for each possible NCS copy so that you can see if the NCS is real Each file has the local map transformed to the orientation and position of copy 1 so they should all superimpose The files are written to the `temp_dir` (e.g., `temp_dir/ncs_1_map_coeffs_001.mtz`) `map_operators_inside_unit_cell = False` If set then put the operators inside the unit cell. This removes the use of space-group symmetry for SG P1 and is used normally for non-crystal data.
- `density_radius = 10` Radius for density to be cut out and compared
- `peak_separation = 15` Minimum distance between centers. Use about $1.5 \times \text{density_radius}$
- `density_peaks = 20` Number of NCS-related peaks of density to output
- `delta_phi = 20` Angular spacing of search
- `rotz_only = False` Rotate around Z only
- `ncs_copies_max = None` Maximum number of NCS copies to look for
- `min_ratio_to_top_cc = 0.75` Peaks will be kept up to `ncs_copies_max`, or `min_ratio_to_top_cc * best cc`, whichever comes first
- `minimum_ncs_cc_from_density = 0.30` Overall NCS CC at full resolution must be at least this high
- `fraction_ncs_min_from_density = 0.05` Minimum fraction of the asymmetric unit covered by NCS. NCS ignored if less than this.
- `dump_ncs_density = False` You can dump 1 mtz file for each possible NCS copy so that you can see if the NCS is real Each file has the local map transformed to the orientation and position of copy 1 so they should all superimpose The files are written to the `temp_dir` (e.g., `temp_dir/ncs_1_map_coeffs_001.mtz`)
- `map_operators_inside_unit_cell = False` If set then put the operators inside the unit cell. This removes the use of space-group symmetry for SG P1 and is used normally for non-crystal data.
- `find_centerssmoothing_radius = None` Radius for smoothing squared density to find centers Choose a smaller value to get more center guesses. Default is same value as `density_radius`. `n_center_find = None` Target number of centers to find Smoothing radius will be varied from 2 to $2 \times \text{smoothing_radius}$ and the value giving the number of peaks closest to `n_center_find` will be used. Normally leave this set to `None` `n_center_use = 20` Number of locations to consider as molecular centers in density search. Try increasing this number to look harder for NCS. `try_all_centers = False` If set, try all `n_center_use` centers.

Normally stop if a good solution is found.
`ok_ncs_cc = 0.6` NCS correlation good enough to terminate search

- `smoothing_radius = None` Radius for smoothing squared density to find centers Choose a smaller value to get more center guesses. Default is same value as `density_radius`.

- `n_center_find = None` Target number of centers to find Smoothing radius will be varied from 2 to $2 \times \text{smoothing_radius}$ and the value giving the number of peaks closest to `n_center_find` will be used. Normally leave this set to None

- `n_center_use = 20` Number of locations to consider as molecular centers in density search. Try increasing this number to look harder for NCS.

- `try_all_centers = False` If set, try all `n_center_use` centers. Normally stop if a good solution is found.

- `ok_ncs_cc = 0.6` NCS correlation good enough to terminate search

- `crystal_inforesolution = 4`. High-resolution limit for map calculation It is useful to cut the resolution at 3-5 Å If you use higher resolution, use a finer `delta_phisolvant_fraction = 0.5` solvent fraction

- `resolution = 4`. High-resolution limit for map calculation It is useful to cut the resolution at 3-5 Å If you use higher resolution, use a finer `delta_phi`

- `solvent_fraction = 0.5` solvent fraction

- `controlverbose = True` Verbose output
`debug = False` Debugging output
`raise_sorry = False` Raise sorry if problems
`dry_run = False` Just read in and check parameter names
`resolve_command_list = None` You can supply any resolve command here NOTE: for command-line usage you need to enclose the whole set of commands in double quotes (") and each individual command in single quotes (') like this:
`resolve_command_list="'no_build' 'b_overall 23' "`

- `verbose = True` Verbose output

- `debug = False` Debugging output

- `raise_sorry = False` Raise sorry if problems

- `dry_run = False` Just read in and check parameter names

- `resolve_command_list = None` You can supply any resolve command here NOTE: for command-line usage you need to enclose the whole set of commands in double quotes (") and each individual command in single quotes (') like this: `resolve_command_list="'no_build' 'b_overall 23' "`

Finding and analyzing difference map peaks

find_peaks_holes.html

Contents

- Overview
- Configuration
- Output
- List of all available keywords

Overview

phenix.find_peaks_holes is a utility for identifying points of interest in difference maps, typically at the end of refinement. It will calculate an mFo-DFc map and locate peaks and holes, plus anomalous map peaks (if anomalous data are provided) and flagging of waters with suspiciously high mFo-DFc values. This information may be used to improve the model, especially in the addition of solvent and small ions.

Configuration

The program requires only a PDB file and data (including R-free flags). The default options should be sufficient for most cases, although for very large structures or especially noisy maps, you may find it helpful to increase the sigma cutoff.

By default, peaks which overlap with the input atoms (within a cutoff distance) will not be included. If you want to find all peaks, set the minimum peak-model distance to zero. Additional options are available by clicking the "Peak search" button.

On the command line, the minimal input is simply the paths to your model and data files (column labels may need to be specified if the choice is ambiguous). Running the program without arguments will print out a summary of available options, which are also listed below.

Output

A summary tab will appear at the end of the run with peak counts, plus output files and buttons to view them. By default the program will write a PDB file containing up to five "chains", all with atoms labeled UNK:

A: mFo-DFc peaks (positive). B: mFo-DFc holes (negative) C: anomalous map peaks D: waters in mFo-DFc peaks E: waters in anomalous peaks

- A: mFo-DFc peaks (positive).
- B: mFo-DFc holes (negative)
- C: anomalous map peaks
- D: waters in mFo-DFc peaks
- E: waters in anomalous peaks

The program will also write the 2mFo-DFc, mFo-DFc, and anomalous (if available) maps. Additional tabs will display each type of result in a list; clicking these items will recenter the view in Coot and/or PyMOL on the selected point.

List of all available keywords

map_cutoff = 3.0 anom_map_cutoff = 3.0 include_peaks_near_model = False By default, the program will only display peaks that map to points outside of the current model, ignoring those that overlap with atoms. Setting this option to True is equivalent to specifying a distance cutoff of zero for the filtering step. wavelength = None Optional parameter, if defined this will cause all atoms to be treated as anomalous scatterers using the standard Sasaki table to obtain theoretical fp and fpp values. Only really useful if the Phaser LLG map is being used for the anomalous map. filter_peaks_by_2fofc = None If this is set, peaks outside 2mFo-DFc density at the cutoff will be discarded. (This does not apply to the analysis of solvent atoms.) Holes will not be changed. use_phaser_if_available = True If True, and Phaser is installed and configured, an anomalous LLG map will be used in place of the simple anomalous difference map. The wavelength should be specified for this to be maximally useful. write_pdb = True write_maps = True output_file_prefix = peaks_holes job_title = None Job title in PHENIX GUI, not used on command line. input_sequence = None scattering_table = wk1995 it1992 *n_gaussian neutron electron wavelength = None energy = None use_experimental_phases = False skip_twin_detection = False twin_law = Auto Enter twin law if known. xray_dataScope of X-ray data and free-R flags. file_name = None labels = None high_resolution = None low_resolution = None twin_law = None outliers_rejection = True Remove basic wilson outliers, extreme wilson outliers, and beamstop shadow outliers. french_wilson_scale = True sigma_fobs_rejection_criterion = None sigma_iobs_rejection_criterion = None ignore_all_zeros = True force_anomalous_flag_to_be_equal_to = None convert_to_non_anomalous_if_ratio_pairs_lone_less_than_threshold = 0.5 french_wilson_max_bins = 60 Maximum number of resolution bins. min_bin_size = 40 Minimum number of reflections per bin. free_flags. file_name = None This is normally the same as the file containing Fobs and is usually selected automatically. label = None required = True Specify if free-r flags are must be present (or else generated). test_flag_value = None This value is usually selected automatically - do not change unless you really know what you're doing! ignore_r_free_flags = False Use all reflections in refinement (work and test). disable_suitability_test = False ignore_pdb_hexdigest = False If True, disables safety check based on MD5 hexdigests stored in PDB files produced by previous runs. generate = False Generate R-free flags (if not available in input files). fraction = 0.1 max_free = 2000 lattice_symmetry_max_delta = 5 use_lattice_symmetry = True use_dataman_shells = False Used to avoid biasing of the test set by certain types of non-crystallographic symmetry. n_shells = 20 pdbfile_name = None Model file(s) name (PDB). monomersfile_name = None Monomer file(s) name (CIF). mapsmap_file_name = None A CCP4-formatted map. d_min = None Resolution of map. map_coefficients_file_name = None MTZ file containing map. map_coefficients_label = None Data label for complex map coefficients in MTZ. file. find_peaks. use_sigma_scaled_maps = True Default is sigma scaled map, map in absolute scale is used otherwise. grid_step = 0.6 Grid step for map sampling. max_number_of_peaks = None map_next_to_model. min_model_peak_dist = 1.8 max_model_peak_dist = 3.2 min_peak_peak_dist = 1.8 use_hydrogens = False peak_search. peak_search_level = 1 max_peaks = 0 interpolate = True min_distance_sym_equiv = None general_positions_only = False min_cross_distance = 1.8 min_cubicle_edge = 5

- map_cutoff = 3.0
- anom_map_cutoff = 3.0
- include_peaks_near_model = False By default, the program will only display peaks that map to points outside of the current model, ignoring those that overlap with atoms. Setting this option to True is equivalent to specifying a distance cutoff of zero for the filtering step.
- wavelength = None Optional parameter, if defined this will cause all atoms to be treated as anomalous scatterers using the standard Sasaki table to obtain theoretical fp and fpp values. Only really useful if the Phaser LLG map is being used for the anomalous map.

- `filter_peaks_by_2focf` = None If this is set, peaks outside 2mFo-DFc density at the cutoff will be discarded. (This does not apply to the analysis of solvent atoms.) Holes will not be changed.
- `use Phaser if available` = True If True, and Phaser is installed and configured, an anomalous LLG map will be used in place of the simple anomalous difference map. The wavelength should be specified for this to be maximally useful.
- `write_pdb` = True
- `write_maps` = True
- `output_file_prefix` = peaks_holes
- `job_title` = None Job title in PHENIX GUI, not used on command line
- `inputsequence` = None `scattering_table` = wk1995 it1992 *n_gaussian neutron electron wavelength = None `energy` = None `use_experimental_phases` = False `skip_twin_detection` = False `twin_law` = Auto Enter twin law if known. `xray_dataScope` of X-ray data and free-R flags `file_name` = None `labels` = None `high_resolution` = None `low_resolution` = None `twin_law` = None `outliers_rejection` = True Remove basic wilson outliers , extreme wilson outliers , and beamstop shadow outliers `french_wilson_scale` = True `sigma_fobs_rejection_criterion` = None `sigma_jobs_rejection_criterion` = None `ignore_all_zeros` = True `force_anomalous_flag_to_be_equal_to` = None `convert_to_non_anomalous_if_ratio_pairs_lone_less_than_threshold` = 0.5 `french_wilsonmax_bins` = 60 Maximum number of resolution bins `min_bin_size` = 40 Minimum number of reflections per `binr_free_flagsfile_name` = None This is normally the same as the file containing Fobs and is usually selected automatically. `label` = None `required` = True Specify if free-r flags are must be present (or else generated) `test_flag_value` = None This value is usually selected automatically - do not change unless you really know what you're doing! `ignore_r_free_flags` = False Use all reflections in refinement (work and test) `disable_suitability_test` = False `ignore_pdb_hexdigest` = False If True, disables safety check based on MD5 hexdigests stored in PDB files produced by previous runs. `generate` = False Generate R-free flags (if not available in input files) `fraction` = 0.1 `max_free` = 2000 `lattice_symmetry_max_delta` = 5 `use_lattice_symmetry` = True `use_dataman_shells` = False Used to avoid biasing of the test set by certain types of non-crystallographic symmetry. `n_shells` = 20 `pdbfile_name` = None Model file(s) name (PDB) `monomersfile_name` = None Monomer file(s) name (CIF) `mapsmap_file_name` = None A CCP4-formatted map `d_min` = None Resolution of map `map_coefficients_file_name` = None MTZ file containing map `map_coefficients_label` = None Data label for complex map coefficients in MTZ file
- `sequence` = None
- `scattering_table` = wk1995 it1992 *n_gaussian neutron electron
- `wavelength` = None
- `energy` = None
- `use_experimental_phases` = False
- `skip_twin_detection` = False
- `twin_law` = Auto Enter twin law if known.
- `xray_dataScope` of X-ray data and free-R flags `file_name` = None `labels` = None `high_resolution` = None `low_resolution` = None `twin_law` = None `outliers_rejection` = True Remove basic wilson outliers , extreme wilson outliers , and beamstop shadow outliers `french_wilson_scale` = True `sigma_fobs_rejection_criterion` = None `sigma_jobs_rejection_criterion` = None `ignore_all_zeros` = True `force_anomalous_flag_to_be_equal_to` = None `convert_to_non_anomalous_if_ratio_pairs_lone_less_than_threshold` = 0.5 `french_wilsonmax_bins` = 60 Maximum number of resolution bins `min_bin_size` = 40 Minimum number of reflections per

binr_free_flagsfile_name = None This is normally the same as the file containing Fobs and is usually selected automatically.label = None required = True Specify if free-r flags are must be present (or else generated)test_flag_value = None This value is usually selected automatically - do not change unless you really know what you're doing!ignore_r_free_flags = False Use all reflections in refinement (work and test)disable_suitability_test = False ignore_pdb_hexdigest = False If True, disables safety check based on MD5 hexdigests stored in PDB files produced by previous runs.generate = False Generate R-free flags (if not available in input files)fraction = 0.1 max_free = 2000 lattice_symmetry_max_delta = 5 use_lattice_symmetry = True use_dataman_shells = False Used to avoid biasing of the test set by certain types of non-crystallographic symmetry.n_shells = 20

- file_name = None
- labels = None
- high_resolution = None
- low_resolution = None
- twin_law = None
- outliers_rejection = True Remove basic wilson outliers , extreme wilson outliers , and beamstop shadow outliers
- french_wilson_scale = True
- sigma_fobs_rejection_criterion = None
- sigma_iobs_rejection_criterion = None
- ignore_all_zeros = True
- force_anomalous_flag_to_be_equal_to = None
- convert_to_non_anomalous_if_ratio_pairs_lone_less_than_threshold = 0.5
- french_wilsonmax_bins = 60 Maximum number of resolution binsmin_bin_size = 40 Minimum number of reflections per bin
- max_bins = 60 Maximum number of resolution bins
- min_bin_size = 40 Minimum number of reflections per bin
- r_free_flagsfile_name = None This is normally the same as the file containing Fobs and is usually selected automatically.label = None required = True Specify if free-r flags are must be present (or else generated)test_flag_value = None This value is usually selected automatically - do not change unless you really know what you're doing!ignore_r_free_flags = False Use all reflections in refinement (work and test)disable_suitability_test = False ignore_pdb_hexdigest = False If True, disables safety check based on MD5 hexdigests stored in PDB files produced by previous runs.generate = False Generate R-free flags (if not available in input files)fraction = 0.1 max_free = 2000 lattice_symmetry_max_delta = 5 use_lattice_symmetry = True use_dataman_shells = False Used to avoid biasing of the test set by certain types of non-crystallographic symmetry.n_shells = 20
- file_name = None This is normally the same as the file containing Fobs and is usually selected automatically.
- label = None
- required = True Specify if free-r flags are must be present (or else generated)
- test_flag_value = None This value is usually selected automatically - do not change unless you really know what you're doing!
- ignore_r_free_flags = False Use all reflections in refinement (work and test)

- `disable_suitability_test = False`
- `ignore_pdb_hexdigest = False` If True, disables safety check based on MD5 hexdigests stored in PDB files produced by previous runs.
- `generate = False` Generate R-free flags (if not available in input files)
- `fraction = 0.1`
- `max_free = 2000`
- `lattice_symmetry_max_delta = 5`
- `use_lattice_symmetry = True`
- `use_dataman_shells = False` Used to avoid biasing of the test set by certain types of non-crystallographic symmetry.
- `n_shells = 20`
- `pdbfile_name = None` Model file(s) name (PDB)
- `file_name = None` Model file(s) name (PDB)
- `monomersfile_name = None` Monomer file(s) name (CIF)
- `file_name = None` Monomer file(s) name (CIF)
- `mapsmmap_file_name = None` A CCP4-formatted map
- `mapmap_min = None` Resolution of map
- `mapmap_coefficients_file_name = None` MTZ file containing map
- `mapmap_coefficients_label = None` Data label for complex map coefficients in MTZ file
- `map_file_name = None` A CCP4-formatted map
- `d_min = None` Resolution of map
- `map_coefficients_file_name = None` MTZ file containing map
- `map_coefficients_label = None` Data label for complex map coefficients in MTZ file
- `find_peaksuse_sigma_scaled_maps = True` Default is sigma scaled map, map in absolute scale is used otherwise.
- `grid_step = 0.6` Grid step for map sampling
- `max_number_of_peaks = None`
- `map_next_to_modelmin_model_peak_dist = 1.8` max_model_peak_dist = 3.2 min_peak_peak_dist = 1.8
- `use_hydrogens = False` peak_searchpeak_search_level = 1 max_peaks = 0 interpolate = True
- `min_distance_sym_equiv = None` general_positions_only = False min_cross_distance = 1.8
- `min_cubicle_edge = 5`
- `use_sigma_scaled_maps = True` Default is sigma scaled map, map in absolute scale is used otherwise.
- `grid_step = 0.6` Grid step for map sampling
- `max_number_of_peaks = None`
- `map_next_to_modelmin_model_peak_dist = 1.8` max_model_peak_dist = 3.2 min_peak_peak_dist = 1.8
- `use_hydrogens = False`
- `min_model_peak_dist = 1.8`
- `max_model_peak_dist = 3.2`
- `min_peak_peak_dist = 1.8`
- `use_hydrogens = False`
- `peak_searchpeak_search_level = 1` max_peaks = 0 interpolate = True min_distance_sym_equiv = None
- `general_positions_only = False` min_cross_distance = 1.8 min_cubicle_edge = 5
- `peak_search_level = 1`

- max_peaks = 0
- interpolate = True
- min_distance_sym_equiv = None
- general_positions_only = False
- min_cross_distance = 1.8
- min_cubicle_edge = 5

Finding a program or test using find_program

find_program.html

Author(s)

- find_program: Tom Terwilliger

Purpose

The routine find_program will show all the programs that match supplied text (either in program name or in program descriptions). The locations of the programs (which tab in the GUI or command-line name) are shown.

Alternatively, find_program can search for a regression test based on words in the name of the test or based on the name of a module that is tested by that regression test.

Lastly, find_program can find all the Python programs that have changed since last checking in programs to github and list all the regression tests that test those programs.

Usage

How find_program works:

Searches for programs:

Find_program uses the list of programs and short descriptions in the file \$PHENIX/modules/phenix/phenix/utilities/phenix_list_help.py as a list of program names. Additionally it compiles program names and descriptions from \$PHENIX/modules/phenix/wxGUI2/Programs/__init__.params and \$PHENIX/modules/phenix/wxGUI2/Home/Tools.py.

Find_program then makes a list of all programs, descriptions, and locations.

This list is then searched with the search_text values provided.

Searches for regression tests:

Find_program can also look for and display regression tests. A database (test_and_module_info.pkl) of regression tests and all the modules called by each test is created using the tool phenix_regression.get_test_and_module_info. This database is read by find_program and can be searched either by words from the name of a regression test (providing the full path of the regression test), or by words from the name of a module (providing the full path of regression tests that use that module).

The search for regression tests by module is a way to find a test that will run a module that you are interested in.

Search for regression tests affecting all changed programs:

Find_program can find all the Python programs that have changed since the last check-in to github (it uses git to do this). Then for each of these programs, it finds all the regression tests that are affected. Then it lists all

the regression tests it finds.

This search is a way to find all the tests that could possibly fail due to any changed Python module. Note that this search will not find C++ code that is affected. It will also not find code that is accessed in tests using the `easy_run` command which uses a system call that does not pass back traceback information.

Running regression tests

The standard framework for running regression tests is described in [Running Phenix regression tests](#) . This approach uses `find_program` if you use the `git` or `sort` options.

Standard run of `find_program`:

You can use `find_program` to find a program like this:

```
phenix.find_program search_text=autosol
```

You can also use it to try and find a regression test that exercises a routine with the word `anomalous` in it (anywhere) and also the word `signal` (anywhere):

```
phenix.find_program search_type=tests search_tests_by=function_called search_text=anomalous search_text=signal
```

Setting up the database for `find_program`:

You can update the database for `find_program` by running the tool

```
phenix_regression.get_test_and_module_info nproc=64
```

in an empty directory. This will take a while (perhaps an hour or so with `nproc=64`) and produce a file called:

```
test_and_module_info.pkl
```

You can copy this file to its official location at:

```
$PHENIX/modules/phenix_regression/misc/test_and_module_info.pkl
```

and then `find_program` will use this new version. You can also specify where this `.pkl` file is located with the keyword:

```
phenix_test_database_file=xxx.pkl
```

Possible Problems

Specific limitations and problems:

The `search_tests` and `git_affected_tests` features of `find_program` do not find C++ code that is affected by a test (only Python code).

Literature

Additional information

List of all available keywords

`all_tests = False` Find all regression tests
`git_affected_tests = False` Find regression tests to cover all changed Python files in
`githubdirectories_to_search = phenix cctbx_project solve_resolve` Directories to search to find

changed filessearch_type = *programs tests You can search for programs (real_space_refine) or tests (tst_ncs.py). For programs, your search text must match part of the program name or its description. For tests you can search based on the name of the test or based on the name of any function that is accessed by the test.

search_text = None Text to search for. If multiple, any match is displayed. If search_text=ANY (must be capitalized) or None (empty) then searches will match any text. Each entry of search_text counts as one search term. So if there are two entries of search text like < density modification > and <autosol> , then if all_search_terms_required=True, the phrase < density modification > and the word < autosol > must both appear somewhere in the name or description to be selected.

case_sensitive_search = False Normally searches are case-insensitive. You can force a case-sensitive search with this keyword.

all_search_terms_required = True All search terms required (a AND b AND c). Alternative is to match any search terms (a OR b OR c). Each entry of search_text counts as one search term. So if there are two entries of search text like < density modification > and <autosol> , then if all_search_terms_required=True, the phrase < density modification > and the word < autosol > must both appear somewhere in the name or description to be selected.

try_splitting_text = True If no programs are found, try splitting search text entries into individual words and search for all the words in any order.

ignore_hyphens = True Ignore hyphens in searches.

number_to_display = 20 Number of programs or tests to display. Up to this number of entries will be shown.

show_tests_by_module = True Show tests by module.

find_programs_without_tests = None Find programs without tests.

job_title = None Job title in PHENIX GUI, not used on command line.

input_filesprogram_data_file_name = None Input data file name with program information.

read_program_data_if_available = True Read program data.

phenix_test_database_file = None Input database file (.pkl) with information about all Phenix tests and modules that are called by those tests. This is normally found automatically.

output_filesprogram_data_file_name = None Output data file name with program information.

files_with_tests_file_name = files_with_tests.txt Output data file name with list of files with tests.

files_with_no_tests_file_name = files_with_no_tests.txt Output data file name with list of files with no tests.

modules_with_tests_file_name = modules_with_tests.txt Output data file name with list of modules with tests.

modules_with_no_tests_file_name = modules_with_no_tests.txt Output data file name with list of modules with no tests.

programsinclude_program_description = True Search in documentation.

stringtestssearch_tests_by = *test_name function_called For tests you can search based on the name of any function that is called by the test. This is a way to find a test that will run a function that you are interested in. The alternative is to search by the name of the test.

guiGUI-specific parameter required for output directoryoutput_dir = None

- all_tests = False Find all regression tests
- git_affected_tests = False Find regression tests to cover all changed Python files in github
- directories_to_search = phenix cctbx_project solve_resolve Directories to search to find changed files
- search_type = *programs tests You can search for programs (real_space_refine) or tests (tst_ncs.py). For programs, your search text must match part of the program name or its description. For tests you can search based on the name of the test or based on the name of any function that is accessed by the test.
- search_text = None Text to search for. If multiple, any match is displayed. If search_text=ANY (must be capitalized) or None (empty) then searches will match any text. Each entry of search_text counts as one search term. So if there are two entries of search text like < density modification > and <autosol> , then if all_search_terms_required=True, the phrase < density modification > and the word < autosol > must both appear somewhere in the name or description to be selected.
- case_sensitive_search = False Normally searches are case-insensitive. You can force a case-sensitive search with this keyword.
- all_search_terms_required = True All search terms required (a AND b AND c). Alternative is to match any search terms (a OR b OR c). Each entry of search_text counts as one search term. So if there are

two entries of search text like < density modification > and <autosol> , then if all_search_terms_required=True, the phrase < density modification > and the word < autosol > must both appear somewhere in the name or description to be selected.

- try_splitting_text = True If no programs are found, try splitting search text entries into individual words and search for all the words in any order.
- ignore_hyphens = True Ignore hyphens in searches
- number_to_display = 20 Number of programs or tests to display. Up to this number of entries will be shown.
- show_tests_by_module = True Show tests by module
- find_programs_without_tests = None Find programs without tests
- job_title = None Job title in PHENIX GUI, not used on command line
- input_filesprogram_data_file_name = None Input data file name with program information
- read_program_data_if_available = True Read program data
- phenix_test_database_file = None Input database file (.pkl) with information about all Phenix tests and modules that are called by those tests. This is normally found automatically
- program_data_file_name = None Input data file name with program information
- read_program_data_if_available = True Read program data
- phenix_test_database_file = None Input database file (.pkl) with information about all Phenix tests and modules that are called by those tests. This is normally found automatically
- output_filesprogram_data_file_name = None Output data file name with program information
- files_with_tests_file_name = files_with_tests.txt Output data file name with list of files with tests
- files_with_no_tests_file_name = files_with_no_tests.txt Output data file name with list of files with no tests
- modules_with_tests_file_name = modules_with_tests.txt Output data file name with list of modules with tests
- modules_with_no_tests_file_name = modules_with_no_tests.txt Output data file name with list of modules with no tests
- programsinclude_program_description = True Search in documentation string
- include_program_description = True Search in documentation string
- testssearch_tests_by = *test_name function_called For tests you can search based on the name of any function that is called by the test. This is a way to find a test that will run a function that you are interested in. The alternative is to search by the name of the test.
- search_tests_by = *test_name function_called For tests you can search based on the name of any function that is called by the test. This is a way to find a test that will run a function that you are interested in. The alternative is to search by the name of the test.
- guiGUI-specific parameter required for output directoryoutput_dir = None

- output_dir = None

Fitting loops to fill in gaps in models with fit_loops

fit_loops.html

Contents

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- Using a loop library
- Using a loop library to connect all segments
- Specifying a gap to fill in
- Possible Problems
- Specific limitations and problems:
- Literature
- Additional information
- List of all available keywords

Author(s)

- fit_loops: Tom Terwilliger

Purpose

fit_loops is a tool for building a loop into density to connect existing chain ends. You supply a model with a gap and a sequence file and coefficients for an electron density map, and you specify the first and last residues to be built. Then fit_loops will attempt to build the loop that you specify. One loop can be done at a time with fit_loops (but if you have multiple identical chains, you can fit them all at once).

You can use either of two methods to fit the loops. By default fit_loops uses resolve chain extension to try to trace residues from the ends of segments in your input PDB file. If it can connect the segments, it writes out the connecting loops. Alternatively you can use a loop library supplied with PHENIX to connect ends of segments from your input PDB file.

If you want a more complete model-building process, then you will want to use phenix.autobuild .

fit_loops can be run from the command line or from the PHENIX GUI.

Usage

How fit_loops works:

fit_loops calculates a map based on the supplied map coefficients, then tries to extend the ends of the supplied model into the gap region, following the electron density in the map.

Output files from fit_loops

model_with_loops.pdb: The output from fit_loops is a new PDB file containing your input model with the newly-built loop inserted into it (if a loop could be found).

Examples

Standard run of fit_loops:

A typical command-line input would be:

```
phenix.fit_loops pdb_in=nsf_gap.pdb mtz_in=map_coeffs.mtz \  
seq_file=nsf.seq start=37 end=43 chain_id=None
```

This will fit a loop starting with residue 37 and ending with residue 43 in nsf_gap.pdb. phenix.fit_loops will expect that your existing nsf_gap.pdb model has a chain ending at residue 36 and another starting at residue 44. As chain_id=None in this example, if there are multiple chains A,B,C in nsf_gap.pdb then all 3 will be filled in.

Specifying column names

If you want (or need) to specify the column names from your mtz file, you will need to tell fit_loops what FP and PHIB (and optionally FOM) are, in this format:

```
phenix.fit_loops pdb_in=nsf_gap.pdb mtz_in=map_coeffs.mtz \  
seq_file=nsf.seq start=37 end=43 chain_id=None \  
labin="FP=2F0FCWT PHIB=PH2F0FCWT"
```

Fitting a loop with poor density

If you want to try and fit a loop with poor density, you might want to lower the threshold for the correlation of density in the loop (default minimum correlation is 0.2):

```
phenix.fit_loops pdb_in=nsf_gap.pdb mtz_in=map_coeffs.mtz \  
seq_file=nsf.seq start=37 end=43 chain_id=None \  
loop_cc_min=0.1
```

Using a loop library

To use the loop library in PHENIX, use the keyword loop_lib:

```
phenix.fit_loops pdb_in=nsf_gap.pdb mtz_in=map_coeffs.mtz \  
seq_file=nsf.seq start=37 end=39 chain_id=None loop_lib=True
```

This will fit a loop starting with residue 37 and ending with residue 39. The maximum current length in the loop library is 3 residues.

Using a loop library to connect all segments

To use the loop library in PHENIX and try to connect any pair of segments that have an geometrical relationship, use the keyword connect_all:

```
phenix.fit_loops pdb_in=nsf_gap.pdb mtz_in=map_coeffs.mtz \  
seq_file=nsf.seq connect_all=True
```

This will go through all pairs of segments, trying to connect them with a loop from the PHENIX loop library. Note that this is a last-resort approach, normally instead use the default and let fit_loops connect segments that are close in sequence.

Specifying a gap to fill in

To specify a particular gap for loop fitting, you can say

```
phenix.fit_loops pdb_in=nsf_gap.pdb mtz_in=map_coeffs.mtz \  
seq_file=nsf.seq start=37 end=39 chain_id=A
```

Note the residue number for start and end are the first and last in the gap, not the residues flanking the gap that are already present. This command fits a gap present at residues 37-39. Note: if any of the residues from 37-39 are still present in your model, specify remove_loop=True and fit_loops will remove them (otherwise it will just fit the existing gap).

Possible Problems

One of the most common problems is that the residues just before and after a gap are themselves poorly fitted. A solution is to remove the last couple residues before a loop and the first couple residues after the loop and to try and fit the resulting larger gap. To do this you can say something like:

```
phenix.fit_loops pdb_in=nsf_gap.pdb mtz_in=map_coeffs.mtz \  
seq_file=nsf.seq start=35 end=41 chain_id=A replace_residues=True
```

Here the replace_residues=True command is required to tell fit_loops to remove residues 35 and 36 and residues 40 and 41 before trying to fit residues 35-41.

You can also accomplish this by leaving the parameter trim_ends_if_needed=True, which will do this automatically (trying several possible trimmings of the ends).

Specific limitations and problems:

Literature

Additional information

List of all available keywords

input_files_pdb_in = None Model with gap to fill. mtz_in = None MTZ file with coefficients for a map
mtz_in_virtual = None Used internally map_coeff_labels = None If map coefficients cannot be identified automatically from your MTZ file, you can specify the label or labels for them. (Please separate labels with blank space, MTZ columns grouped together separated by commas with no blanks.) You can specify: map_coeff_labels (e.g., FWT,PHIFWT) amplitudes and phases (e.g., FP,SIGFP PHIB) or amplitudes, phases, weights (e.g., FP,SIGFP PHIB FOM) labin = "" Labin line for MTZ file with map coefficients. Normally use instead map_coeff_labels. This is available for backward compatibility. You can specify: LABIN FP=myFP PHIB=myPHI FOM=myFOM where myFP is your column label for FP map_in = None CCP4 or MRC-style map file seq_file = None sequence file (1-letter code, one copy of each chain) seq_prob_file = None File seq_prob.dat from resolve sequence alignment output_files_pdb_out = connect.pdb Output PDB file (will be missing if no result). results_as_json = None Output result summary as JSON log = None Output logfile

params_out = fit_loops_params.eff Parameters file to rerun fit_loops
 fitting_min_acceptable_prob = None Minimum loop probability to consider
 refine_loops = True Refine fitted loops in loop_lib
 chain_id = None Chain ID for chain containing missing loops. If None any chain ID is allowed. All missing segments matching chain_id, start and end will be fit. If chain_id is specified without start and end, all gaps in that chain will be fit. If no chain_id and no start and end are specified, all gaps are fit.
 start = None Starting residue number of loop(s) to be fit. NOTE: This is the residue number of the first residue that is NOT present in your model already (one more than the residue number of the residue just before the gap). If None any start is allowed. All missing segments matching chain_id, start and end will be fit.
 end = None Ending residue number of loop(s) to be fit. NOTE: This is the residue number of the last residue that is NOT present in your model already (one fewer than the residue number of the residue just after the gap). If None any end is allowed. All missing segments matching chain_id, start and end will be fit.
 insert_or_delete_residues = None Try to insert or delete this many residues in the loop (ignore loop sequence if so and use any side chain).
 remove_loops = True Remove existing residues and replace with new loop. All segments matching chain_id, start and end will be fit.
 skip_trim = True If skip_trim=True (default) then model with added loops will be written out without checking for overlaps with non-sequence-aligned residues. If skip_trim=False then this check will be carried out. Note that skip_trim=False can cause some residues to be de-assigned from sequence if they cannot be successfully matched to density. In such cases you might try skip_trim=True.
 score_min = 1.0 Minimum connection score for connect_all_segments.
 min_dist = 3. Minimum distance between connections for connect_all_segments.
 min_log_prob = -5. Minimum log(P) for sequence to consider locations.
 skip_seq_prob = False Skip sequence probability calculations.
 save_acceptable_loops = False Just return a file with possible loops.
 sn_random_loop = 200 Number of tries at building loops.
 connect_all_segments = False Try to connect all segments to each other, regardless of sequence and residue numbers. Note: this is a last-resort approach. Normally just use the default and let fit_loops connect segments that are close in sequence.
 sequential_only = False Only connect adjacent segments in connect_all_segments.
 dist_max = 15 Maximum CA-CA distance in connect_all_segments.
 ignore_sequence_register = False Ignore the input sequence register.
 all_assigned = True Assume all residues in model can be assigned to sequence. If you set all_assigned=False then any residues with residue number greater than the number of residues in the longest sequence in the sequence file are...

- input_files
 pdb_in = None Model with gap to fill.
 mtz_in = None MTZ file with coefficients for a map.
 mtz_in_virtual = None Used internally.
 map_coeff_labels = None If map coefficients cannot be identified automatically from your MTZ file, you can specify the label or labels for them. (Please separate labels with blank space, MTZ columns grouped together separated by commas with no blanks.) You can specify: map_coeff_labels (e.g., FWT,PHIFWT) amplitudes and phases (e.g., FP,SIGFP PHIB) or amplitudes, phases, weights (e.g., FP,SIGFP PHIB FOM).
 labin = "" Labin line for MTZ file with map coefficients. Normally use instead map_coeff_labels. This is available for backward compatibility. You can specify: LABIN FP=myFP PHIB=myPHI FOM=myFOM where myFP is your column label for FP.
 map_in = None CCP4 or MRC-style map file.
 seq_file = None sequence file (1-letter code, one copy of each chain).
 seq_prob_file = None File seq_prob.dat from resolve sequence alignment.
- pdb_in = None Model with gap to fill.
- mtz_in = None MTZ file with coefficients for a map
- mtz_in_virtual = None Used internally
- map_coeff_labels = None If map coefficients cannot be identified automatically from your MTZ file, you can specify the label or labels for them. (Please separate labels with blank space, MTZ columns grouped together separated by commas with no blanks.) You can specify: map_coeff_labels (e.g., FWT,PHIFWT) amplitudes and phases (e.g., FP,SIGFP PHIB) or amplitudes, phases, weights (e.g., FP,SIGFP PHIB FOM)

- `labin = ""` Labin line for MTZ file with map coefficients. Normally use instead `map_coeff_labels`. This is available for backward compatibility. You can specify: `LABIN FP=myFP PHIB=myPHI FOM=myFOM` where `myFP` is your column label for FP
- `map_in = None` CCP4 or MRC-style map file
- `seq_file = None` sequence file (1-letter code, one copy of each chain)
- `seq_prob_file = None` File `seq_prob.dat` from resolve sequence alignment
- `output_files`
 - `pdb_out = connect.pdb` Output PDB file (will be missing if no result).
 - `results_as_json = None` Output result summary as JSON
 - `log = None` Output logfile
 - `params_out = fit_loops_params.eff` Parameters file to rerun `fit_loops`
- `fittingmin_acceptable_prob = None` Minimum loop probability to consider
 - `refine_loops = True` Refine fitted loops in `loop_libchain_id = None` Chain ID for chain containing missing loops. If `None` any chain ID is allowed. All missing segments matching `chain_id`, start and end will be fit. If `chain_id` is specified without start and end, all gaps in that chain will be fit. If no `chain_id` and no start and end are specified, all gaps are fit.
 - `start = None` Starting residue number of loop(s) to be fit. NOTE: This is the residue number of the first residue that is NOT present in your model already (one more than the residue number of the residue just before the gap). If `None` any start is allowed. All missing segments matching `chain_id`, start and end will be fit.
 - `end = None` Ending residue number of loop(s) to be fit. NOTE: This is the residue number of the last residue that is NOT present in your model already (one fewer than the residue number of the residue just after the gap). If `None` any end is allowed. All missing segments matching `chain_id`, start and end will be fit.
 - `insert_or_delete_residues = None` Try to insert or delete this many residues in the loop (ignore loop sequence if so and use any side chain)
 - `remove_loops = True` Remove existing residues and replace with new loop. All segments matching `chain_id`, start and end will be fit.
 - `skip_trim = True` If `skip_trim=True` (default) then model with added loops will be written out without checking for overlaps with non-sequence-aligned residues. If `skip_trim=False` then this check will be carried out. Note that `skip_trim=False` can cause some residues to be de-assigned from sequence if they cannot be successfully matched to density. In such cases you might try `skip_trim=True`.
 - `score_min = 1.0` Minimum connection score for `connect_all_segments`
 - `min_dist = 3` Minimum distance between connections for `connect_all_segments`
 - `min_log_prob = -5` Minimum log(P) for sequence to consider
 - `locationskip_seq_prob = False` Skip sequence probability calculations
 - `save_acceptable_loops = False` Just return a file with possible loops
 - `sn_random_loop = 200` Number of tries at building loops
 - `connect_all_segments = False` Try to connect all segments to each other, regardless of sequence and residue numbers. Note: this is a last-resort approach. Normally just use the default and let `fit_loops` connect segments that are close in sequence.
 - `sequential_only = False` Only connect adjacent segments in `connect_all_segments`
 - `dist_max = 15` Maximum CA-CA distance in `connect_all_segments`
 - `ignore_sequence_register = False` Ignore the input sequence register
 - `all_assigned = True` Assume all residues in model can be assigned to sequence. If you set `all_assigned=False` then any residues with residue number greater than the number of residues in the longest sequence in the sequence file are assumed not to be assigned to sequence. This is useful if you are using resolve model-building as resolve will identify non-sequence-aligned residues with high residue numbers.
 - `sequence_offset = None` You can specify that `fit_loops` should offset input sequence file residue numbers by `sequence_offset` before use (same effect as adding `sequence_offset*X` to the beginning of

the sequence file.) Note: number of of sequence_offset values must match number of chains in sequence file
 loop_cc_min = 0.2 Minimum loop map-model correlation
 aggressive = False Aggressive loop building (risky)
 target_insert = None You can try to force the number of residues to insert with trace_loops. If none, try to fill in the gap based on the number of missing residues. If set and greater than 0, only accept the number of residues given. If zero, take any length. Not supported.
 time_per_residue = None You can specify how long to try in fitting (sec/residue)
 trim_ends_if_needed = True If a single loop is specified and no loop is found, try trimming back the ends.
 loop_backup_residues = None Number of tries removing one residue at a time from each end of existing ends of loop if no loop is found with initial ...

- min_acceptable_prob = None Minimum loop probability to consider
- refine_loops = True Refine fitted loops in loop_lib
- chain_id = None Chain ID for chain containing missing loops. If None any chain ID is allowed. All missing segments matching chain_id, start and end will be fit. If chain_id is specified without start and end, all gaps in that chain will be fit. If no chain_id and no start and end are specified, all gaps are fit.
- start = None Starting residue number of loop(s) to be fit. NOTE: This is the residue number of the first residue that is NOT present in your model already (one more than the residue number of the residue just before the gap). if None any start is allowed. All missing segments matching chain_id, start and end will be fit
- end = None Ending residue number of loop(s) to be fit. NOTE: This is the residue number of the last residue that is NOT present in your model already (one fewer than the residue number of the residue just after the gap). If None any end is allowed. All missing segments matching chain_id, start and end will be fit
- insert_or_delete_residues = None Try to insert or delete this many residues in the loop (ignore loop sequence if so and use any side chain)
- remove_loops = True Remove existing residues and replace with new loop All segments matching chain_id, start and end will be fit
- skip_trim = True If skip_trim=True (default) then model with added loops will be written out without checking for overlaps with non-sequence-aligned residues. If skip_trim=False then this check will be carried out. Note that skip_trim=False can cause some residues to be de-assigned from sequence if they cannot be successfully matched to density. In such cases you might try skip_trim=True."
- score_min = 1.0 Minimum connection score for connect_all_segments
- min_dist = 3. Minimum distance between connections for connect_all_segments"
- min_log_prob = -5. Minimum log(P) for sequence to consider location
- skip_seq_prob = False Skip sequence probability calculation
- save_acceptable_loops = False Just return a file with possible loops
- n_random_loop = 200 Number of tries at building loops
- connect_all_segments = False Try to connect all segments to each other, regardless of sequence and residue numbers. Note: this is a last-resort approach. Normally just use the default and let fit_loops connect segments that are close in sequence.
- sequential_only = False Only connect adjacent segments in connect_all_segments
- dist_max = 15 Maximum CA-CA distance in connect_all_segments
- ignore_sequence_register = False Ignore the input sequence register
- all_assigned = True Assume all residues in model can be assigned to sequence If you set all_assigned=False then any residues with residue number greater than the number of residues in the

longest sequence in the sequence file are assumed not to be assigned to sequence. This is useful if you are using resolve model-building as resolve will identify non-sequence-aligned residues with high residue numbers.

- `sequence_offset = None` You can specify that `fit_loops` should offset input sequence file residue numbers by `sequence_offset` before use (same effect as adding `sequence_offset*X` to the beginning of the sequence file.) Note: number of `sequence_offset` values must match number of chains in sequence file
- `loop_cc_min = 0.2` Minimum loop map-model correlation
- `aggressive = False` Aggressive loop building (risky)
- `target_insert = None` You can try to force the number of residues to insert with `trace_loops`. If none, try to fill in the gap based on the number of missing residues. If set and greater than 0, only accept the number of residues given. If zero, take any length. Not supported.
- `time_per_residue = None` You can specify how long to try in fitting (sec/residue)
- `trim_ends_if_needed = True` If a single loop is specified and no loop is found, try trimming back the ends.
- `loop_backup_residues = None` Number of tries removing one residue at a time from each end of existing ends of loop if no loop is found with initial gap Default is 4 (1 if quick)
- `split_by_gap = True` Split up work by gaps and run individually (normally True)
- `loop_lib = False` Use loop library to fit loops Only applicable for `chain_type=PROTEIN`
- `standard_loops = True` Use standard loop fitting
- `trace_loops = False` Use loop tracing to fit loops Only applicable for `chain_type=PROTEIN`
- `refine_trace_loops = True` Refine loops (real-space) after `trace_loops`
- `density_of_points = None` Packing density of points to consider as as possible CA atoms in `trace_loops`. Try 1.0 for a quick run, up to 5 for much more thorough run If None, try value depending on value of quick.
- `a_cut_min = None` Minimum density (relative to SD of map, normalized for solvent content) for `trace_loop` points
- `max_density_of_points = None` Maximum packing density of points to consider as as possible CA atoms in `trace_loops`.
- `cutout_model_radius = None` Radius to cut out density for `trace_loops` If None, guess based on length of loop
- `max_cutout_model_radius = 20`. Maximum value of `cutout_model_radius` to try
- `padding = 1`. Padding for cut out density in `trace_loops`
- `cut_out_density = True` Cut out density for `trace_loops`
- `max_span = 30` Maximum length of a gap to try to fill
- `max_c_ca_dist = None` Maximum C-CA distance for a connection
- `max_overlap_rmsd = 2`. Maximum rmsd for 3 residues on each end of loop-lib fit
- `max_overlap = None` Maximum number of residues from ends to start with. (1=use existing ends, 2=one in from ends etc) If None, set based on value of quick.
- `min_overlap = None` Minimum number of residues from ends to start with. (1=use existing ends, 2=one in from ends etc)

- `crystal_inforesolution = 0`. high-resolution limit for map calculation
- `chain_type = *PROTEIN DNA RNA` Chain type (for identifying main-chain and side-chain atoms)
- `final_solvent_content = None` Solvent content (after cutout if `trace_loops` used) . Normally determined automatically.
- `resolution = 0`. high-resolution limit for map calculation
- `chain_type = *PROTEIN DNA RNA` Chain type (for identifying main-chain and side-chain atoms)
- `final_solvent_content = None` Solvent content (after cutout if `trace_loops` used) . Normally determined automatically.
- `directories`
`temp_dir = "temp_dir"` Temporary work directory
`create_temp_dir_if_missing = True` Create `temp_dir` if not present
`output_dir = None` Output directory where files are to be written
`gui_output_dir = None` GUI use only - does not apply to command line version
- `temp_dir = "temp_dir"` Temporary work directory
- `create_temp_dir_if_missing = True` Create `temp_dir` if not present
- `output_dir = None` Output directory where files are to be written
- `gui_output_dir = None` GUI use only - does not apply to command line version
- `control`
`verbose = False` Verbose output
`quick = False` Try to run quickly
`raise_sorry = False` Raise sorry if problems
`debug = False` Debugging output
`dry_run = False` Just read in and check parameter names
`coarse_grid = False` Use coarse grid (saves on memory)
`i_ran_seed = None` random seed
`resolve_command_list = None` Commands for resolve. One per line in the form: keyword value value can be optional Examples: `coarse_grid resolution 200 2.0 hkl in test.mtz` NOTE: for command-line usage you need to enclose the whole set of commands in double quotes (") and each individual command in single quotes (') like this: `resolve_command_list="'no_build' 'b_overall 23' "`
`write_run_directory_to_file = None` The working directory name is written to this file
`pickled_arg_dict = None` Pickled keywords to `__init__` are in this file
`nproc = 1` You can specify the number of processors to use
`max_wait_time = 1.0` You can specify the length of time (seconds) to wait when looking for a file. If you have a cluster where jobs do not start right away you may need a longer time to wait. The symptom of too short a wait time is 'File not found'
`wait_between_submit_time = 1.0` You can specify the length of time (seconds) to wait between each job that is submitted when running sub-processes. This can be helpful on NFS-mounted systems when running with multiple processors to avoid file conflicts. The symptom of too short a wait_time is File exists:....
`background = None` run jobs in background or not (if `nproc` is greater than 1) Usually set automatically. If `run_command` is `sh` or `csh`,
`run_command = "sh "` Command for running jobs (e.g., `sh` or `qsub`)
- `verbose = False` Verbose output
- `quick = False` Try to run quickly
- `raise_sorry = False` Raise sorry if problems
- `debug = False` Debugging output
- `dry_run = False` Just read in and check parameter names
- `coarse_grid = False` Use coarse grid (saves on memory)
- `i_ran_seed = None` random seed
- `resolve_command_list = None` Commands for resolve. One per line in the form: keyword value value can be optional Examples: `coarse_grid resolution 200 2.0 hkl in test.mtz` NOTE: for command-line usage you need to enclose the whole set of commands in double quotes (") and each individual command in single quotes (') like this: `resolve_command_list="'no_build' 'b_overall 23' "`
- `write_run_directory_to_file = None` The working directory name is written to this file

- pickled_arg_dict = None Pickled keywords to __init__ are in this file
- nproc = 1 You can specify the number of processors to use
- max_wait_time = 1.0 You can specify the length of time (seconds) to wait when looking for a file. If you have a cluster where jobs do not start right away you may need a longer time to wait. The symptom of too short a wait time is 'File not found'
- wait_between_submit_time = 1.0 You can specify the length of time (seconds) to wait between each job that is submitted when running sub-processes. This can be helpful on NFS-mounted systems when running with multiple processors to avoid file conflicts. The symptom of too short a wait_between_submit_time is File exists:....
- background = None run jobs in background or not (if nproc is greater than 1) Usually set automatically. If run_command is sh or csh, True
- run_command = "sh " Command for running jobs (e.g., sh or qsub)
- non_user_paramsprint_citations = True Print citation information at end of run
- print_citations = True Print citation information at end of run
- guiGUI-specific parameters, not used on command lineresult_file = None job_title = None Job title in PHENIX GUI, not used on command line
- result_file = None
- job_title = None Job title in PHENIX GUI, not used on command line
- outputDuplicate of output scope, needed because fit_loops constructs its own parameters in __init__ with params=self.process_paramstarget_output_format = *None mmCIF pdb
- target_output_format = *None mmCIF pdb

Fixing register errors in a model with `fix_insertions_deletions`

`fix_insertions_deletions.html`

Author(s)

- `fix_insertions_deletions`: Tom Terwilliger

Purpose

The routine `fix_insertions_deletions` is a tool for fixing register errors in a model by comparing the density at side-chain positions to a sequence file.

Usage

Normally you will access the functionality of `fix_insertions_deletions` by running the Phenix `map_to_model` tool in the Phenix GUI. However you can run it directly as well (there is no GUI for `fix_insertions_deletions`).

How `fix_insertions_deletions` works:

The `fix_insertions_deletions` tool examines the density in the supplied map at the position of each side chain in the supplied model and creates a table of side-chain probabilities corresponding to each segment in the model.

These side-chain probabilities are used to generate a map-based sequence for the model and map. This map-based sequence is then compared to the actual sequence to identify positions where the sequence register is likely to be incorrect, and what changes in register are needed to fix it.

Each place a register shift is needed is used as a target for main-chain rebuilding. During rebuilding the specified insertion or deletion is enforced so that only models with the desired changes are obtained (if possible).

Additional rebuilding of the worst-fitting regions is also carried out.

Using `fix_insertions_deletions`:

The tool `fix_insertions_deletions` is usually run automatically as part of `trace_and_build`. However you can run it yourself to try and fix up a model.

Input map file: The map file should cover the model you supply.

Resolution: Specify the resolution of your map (usually the resolution defined by your half-dataset Fourier shell correlation)

Model: Supply a model that you want to fix. Only the main-chain will matter.

Sequence: Supply a sequence file that covers at least the part of the model that is supplied

Examples

Standard run of `fix_insertions_deletions`:

You can use `fix_insertions_deletions` to fix register shifts in a model based on a cryo-EM map:

```
phenix.fix_insertions_deletions my_map.mrc resolution=3 my_model.pdb seq.dat
```

Possible Problems

Specific limitations and problems:

Literature

Additional information

List of all available keywords

job_title = None Job title in PHENIX GUI, not used on command line
input_filesmap_file = None File with CCP4-style map. May have origin in any location.
model_file = None Input PDB file with chains to be adjusted.
seq_file = None Optional sequence file
placement_pickle_file = None Read placements from this file
output_filetarget_output_format = *None pdb mmCIF Desired output format (if possible). Choices are None (try to use input format), pdb, mmCIF. If output model does not fit in pdb format, mmCIF will be used. Default is pdb.
pdb_out = fix_insertions_deletions.pdb Output rebuilt model
output_placement_pickle_file = None Write placements to this file
temp_dir = None Temporary directory. Default is fix_insertions_deletions_xx where xx creates new directory
crystal_inforesolution = None High-resolution limit for map analysis.
scattering_table = n_gaussian wk1995 it1992 *electron neutron Choice of scattering table for structure factor calculations. Standard for X-ray is n_gaussian, for cryoEM is electron.
chain_type = *PROTEIN Chain type. Must be PROTEIN
solvent_content = None Solvent fraction of the cell. If this is density cut out from a bigger cell, you can specify the fraction of the volume of this cell that is taken up by the macromolecule. Normally set automatically. Values go from 0 to 1.
solvent_content_iterations = 3 Iterations of solvent fraction estimation
use_mask_if_present = True If map is masked, use the mask as solvent content
sequence = None Sequences
origin_cart = None Origin (cartesian coordinates, overrides value based on map)
strategyfirst_cycle_for_fixing = 1 First macro-cycle where insertions/deletions should be tested
loop_method = *trace_chain *extend_only *split_loop *rebuild Method for loop building. None means try everything. trace_chain is finding CA positions to trace chain. extend-only is using resolve model-building to build loop. split_loop is cut loop in the middle and refine. rebuild is rebuild
loopsmask_secondary_structure_in_split_loop = True Mask out mask secondary structure in split
loopresidues_to_skip_on_ends = -2 Skip residues_to_skip_on_ends at ends of secondary structure in masking. Negative means keep that many residues off ends of secondary
structurebuildingfix_insertions_deletions = False Fix insertions and deletions using sequence_from_map to adjust alignment to match sequence that is supplied. NOTE: Use instead split_with_sequence which works better.
restrain_ends_to_original = None Restrict this many residues at each end to original
fix_insertions_deletions_only = None Only fix clear insertions and deletions
stake_all_insertions_deletions = None Accept insertion/deletion fixes without scoring (default quick is True)
ratio_for_sequence_register = 1.0 Accept fixes for insertions/deletions if score is above ratio_for_sequence_register times previous score
try_as_is = None Try rebuilding without insertions/deletions
try_insertions = None Try insertions
try_deletions = None Try deletions
refine = None Refine models at start of procedure. Default is True unless quick is set.
refine_cycles = 1 Refinement cycles (except final cycles)
refine_b = None Refine B-values
good_enough_cc = None If all residues have this CC, don't bother rebuilding them. Default is 0.7 (0.6 if quick=True)
rebuild_length_worst = 15 Longest rebuild length to try
max_insert_or_delete = 1 Maximum residues to insert or delete in on rebuild
stageaverage_rebuild_length = True Choose rebuild length as average of minimum and optimal if optimal is

longer than rebuild_length_worst. Alternative is use minimum if optimal is too long.time_per_residue = 1 How long to try in fitting loops (sec/residue)max_rebuild_cycles = None Maximum rebuilding cycles per macro cycle. Default is 20 (1 if quick is set and sequence is present)macro_cycles = None Macro cycles of rebuilding and refinement. Default is 4 (1 if quick is set)start_rebuild = None Starting residue to rebuild. If specified, this is all that is doneend_rebuild = None Ending residue to rebuild. If s...

- job_title = None Job title in PHENIX GUI, not used on command line
- input_filesmap_file = None File with CCP4-style map. May have origin in any location.model_file = None Input PDB file with chains to be adjusted.seq_file = None Optional sequence fileplacement_pickle_file = None Read placements from this file
- map_file = None File with CCP4-style map. May have origin in any location.
- model_file = None Input PDB file with chains to be adjusted.
- seq_file = None Optional sequence file
- placement_pickle_file = None Read placements from this file
- output_filetarget_output_format = *None pdb mmCIF Desired output format (if possible). Choices are None (try to use input format), pdb, mmCIF. If output model does not fit in pdb format, mmCIF will be used. Default is pdb.pdb_out = fix_insertions_deletions.pdb Output rebuilt modeloutput_placement_pickle_file = None Write placements to this filetemp_dir = None Temporary directory. Default is fix_insertions_deletions_xx where xx creates new directory
- target_output_format = *None pdb mmCIF Desired output format (if possible). Choices are None (try to use input format), pdb, mmCIF. If output model does not fit in pdb format, mmCIF will be used. Default is pdb.
- pdb_out = fix_insertions_deletions.pdb Output rebuilt model
- output_placement_pickle_file = None Write placements to this file
- temp_dir = None Temporary directory. Default is fix_insertions_deletions_xx where xx creates new directory
- crystal_inforesolution = None High-resolution limit for map analysis.scattering_table = n_gaussian wk1995 it1992 *electron neutron Choice of scattering table for structure factor calculations. Standard for X-ray is n_gaussian, for cryoEM is electron.chain_type = *PROTEIN Chain type. Must be PROTEINsolvent_content = None Solvent fraction of the cell. If this is density cut out from a bigger cell, you can specify the fraction of the volume of this cell that is taken up by the macromolecule. Normally set automatically. Values go from 0 to 1.solvent_content_iterations = 3 Iterations of solvent fraction estimationuse_mask_if_present = True If map is masked, use the mask as solvent contentsequence = None Sequencesorigin_cart = None Origin (cartesian coordinates, overrides value based on map)
- resolution = None High-resolution limit for map analysis.
- scattering_table = n_gaussian wk1995 it1992 *electron neutron Choice of scattering table for structure factor calculations. Standard for X-ray is n_gaussian, for cryoEM is electron.
- chain_type = *PROTEIN Chain type. Must be PROTEIN
- solvent_content = None Solvent fraction of the cell. If this is density cut out from a bigger cell, you can specify the fraction of the volume of this cell that is taken up by the macromolecule. Normally set automatically. Values go from 0 to 1.
- solvent_content_iterations = 3 Iterations of solvent fraction estimation
- use_mask_if_present = True If map is masked, use the mask as solvent content

- sequence = None Sequences
- origin_cart = None Origin (cartesian coordinates, overrides value based on map)
- strategyfirst_cycle_for_fixing = 1 First macro-cycle where insertions/deletions should be tested
 loop_method = *trace_chain *extend_only *split_loop *rebuild Method for loop building. None means try everything. trace_chain is finding CA positions to trace chain. extend-only is using resolve model-building to build loop. split_loop is cut loop in the middle and refine. rebuild is rebuild
 loopsmask_secondary_structure_in_split_loop = True Mask out mask secondary structure in split
 loopresidues_to_skip_on_ends = -2 Skip residues_to_skip_on_ends at ends of secondary structure in masking. Negative means keep that many residues off ends of secondary structure
- first_cycle_for_fixing = 1 First macro-cycle where insertions/deletions should be tested
- loop_method = *trace_chain *extend_only *split_loop *rebuild Method for loop building. None means try everything. trace_chain is finding CA positions to trace chain. extend-only is using resolve model-building to build loop. split_loop is cut loop in the middle and refine. rebuild is rebuild loops
- mask_secondary_structure_in_split_loop = True Mask out mask secondary structure in split loop
- residues_to_skip_on_ends = -2 Skip residues_to_skip_on_ends at ends of secondary structure in masking. Negative means keep that many residues off ends of secondary structure
- rebuildingfix_insertions_deletions = False Fix insertions and deletions using sequence_from_map to adjust alignment to match sequence that is supplied. NOTE: Use instead split_with_sequence which works better.
 restrain_ends_to_original = None Restrain this many residues at each end to
 originalfix_insertions_deletions_only = None Only fix clear insertions and
 deletionstake_all_insertions_deletions = None Accept insertion/deletion fixes without scoring (default quick is True)
 ratio_for_sequence_register = 1.0 Accept fixes for insertions/deletions if score is above ratio_for_sequence_register times previous score
 try_as_is = None Try rebuilding without insertions/deletion
 try_insertions = None Try insertion
 try_deletions = None Try deletions
 refine = None Refine models at start of procedure. Default is True unless quick is set.
 refine_cycles = 1 Refinement cycles (except final cycles)
 refine_b = None Refine B-values
 good_enough_cc = None If all residues have this CC, don't bother rebuilding them. Default is 0.7 (0.6 if quick=True)
 rebuild_length_worst = 15 Longest rebuild length to try
 max_insert_or_delete = 1 Maximum residues to insert or delete in on rebuild
 stageaverage_rebuild_length = True Choose rebuild length as average of minimum and optimal if optimal is longer than rebuild_length_worst. Alternative is use minimum if optimal is too long.
 time_per_residue = 1 How long to try in fitting loops (sec/residue)
 max_rebuild_cycles = None Maximum rebuilding cycles per macro cycle. Default is 20 (1 if quick is set and sequence is present)
 macro_cycles = None Macro cycles of rebuilding and refinement. Default is 4 (1 if quick is set)
 start_rebuild = None Starting residue to rebuild. If specified, this is all that is done
 end_rebuild = None Ending residue to rebuild. If specified, this is all that is done
 rebuild_segment = 1 Segment to rebuild from start_rebuild to end_rebuild. None means rebuild all segments from start_rebuild to end_rebuild.
 minimum_contact_distance = 3 Minimum distance between CA atoms not immediately connected
 split_with_sequence = False Use sequence assignment to identify sequence register errors. Runs sequence_from_map to split and fix assignment and then fit_all_loops to fill in gaps.
 keep_connectivity_in_split_with_sequence = True Keep connectivity in
 split_with_sequencefirst_cycle_for_split = 2 First macro-cycle where splitting should be tested
 split_input_model = True Input model will be split into segments
- fix_insertions_deletions = False Fix insertions and deletions using sequence_from_map to adjust alignment to match sequence that is supplied. NOTE: Use instead split_with_sequence which works better.
- restrain_ends_to_original = None Restrain this many residues at each end to original

- `fix_insertions_deletions_only` = None Only fix clear insertions and deletions
- `take_all_insertions_deletions` = None Accept insertion/deletion fixes without scoring (default quick is True)
- `ratio_for_sequence_register` = 1.0 Accept fixes for insertions/deletions if score is above `ratio_for_sequence_register` times previous score
- `try_as_is` = None Try rebuilding without insertions/deletions
- `try_insertions` = None Try insertions
- `try_deletions` = None Try deletions
- `refine` = None Refine models at start of procedure. Default is True unless quick is set.
- `refine_cycles` = 1 Refinement cycles (except final cycles)
- `refine_b` = None Refine B-values
- `good_enough_cc` = None If all residues have this CC, don't bother rebuilding them. Default is 0.7 (0.6 if quick=True)
- `rebuild_length_worst` = 15 Longest rebuild length to try
- `max_insert_or_delete` = 1 Maximum residues to insert or delete in on rebuild stage
- `average_rebuild_length` = True Choose rebuild length as average of minimum and optimal if optimal is longer than `rebuild_length_worst`. Alternative is use minimum if optimal is too long.
- `time_per_residue` = 1 How long to try in fitting loops (sec/residue)
- `max_rebuild_cycles` = None Maximum rebuilding cycles per macro cycle. Default is 20 (1 if quick is set and sequence is present)
- `macro_cycles` = None Macro cycles of rebuilding and refinement. Default is 4 (1 if quick is set)
- `start_rebuild` = None Starting residue to rebuild. If specified, this is all that is done
- `end_rebuild` = None Ending residue to rebuild. If specified, this is all that is done
- `rebuild_segment` = 1 Segment to rebuild from `start_rebuild` to `end_rebuild`. None means rebuild all segments from `start_rebuild` to `end_rebuild`.
- `minimum_contact_distance` = 3 Minimum distance between CA atoms not immediately connected
- `split_with_sequence` = False Use sequence assignment to identify sequence register errors. Runs `sequence_from_map` to split and fix assignment and then `fit_all_loops` to fill in gaps.
- `keep_connectivity_in_split_with_sequence` = True Keep connectivity in `split_with_sequence`
- `first_cycle_for_split` = 2 First macro-cycle where splitting should be tested
- `split_input_model` = True Input model will be split into segments
- `weight_vdw` = 10 weight on very close CA-CA contacts
- `weight_ca_ca_dist` = 1. weight on CA-CA distance
- `weight_proximity_to_known_position` = 0.5 weight on proximity to well-identified CA position
- `weight_density` = 1.0 weight on density at mid-points between CA atoms
- `weight_cc_mask_score` = 1.0 Weight on map correlation in evaluating chain direction
- `weight_seq_score` = 1.0 Weight on sequence-map match in evaluating chain direction
- `weight_x_gly_score` = 1.0 Weight on excess Gly and X residues in evaluating chain direction

- `weight_density = 1.0` weight on density at mid-points between CA atoms
- `weight_cc_mask_score = 1.0` Weight on map correlation in evaluating chain direction
- `weight_seq_score = 1.0` Weight on sequence-map matchin evaluating chain direction
- `weight_x_gly_score = 1.0` Weight on excess Gly and X residues in evaluating chain direction
- `sequencingrandom_sequences = 100` Number of random sequences of each length to use as baseline
- `positive_gap_penalty = 1` Penalty for missing residues is $\text{positive_gap_penalty} * \text{gap}^{**2}$
- `negative_gap_penalty = 2` Gap penalty for extra residues is $\text{negative_gap_penalty} * \text{gap}^{**2}$
- `max_gap_length = 2` Maximum gap length for adjacent alignments
- `minimum_alignment_length = 5` Minimum length of an alignment
- `score_by_residue_groups = None` Use residue groups in sequence alignment and listing of optimal sequences. Default is True unless no sequence is supplied.
- `random_sequences = 100` Number of random sequences of each length to use as baseline
- `positive_gap_penalty = 1` Penalty for missing residues is $\text{positive_gap_penalty} * \text{gap}^{**2}$
- `negative_gap_penalty = 2` Gap penalty for extra residues is $\text{negative_gap_penalty} * \text{gap}^{**2}$
- `max_gap_length = 2` Maximum gap length for adjacent alignments
- `minimum_alignment_length = 5` Minimum length of an alignment
- `score_by_residue_groups = None` Use residue groups in sequence alignment and listing of optimal sequences. Default is True unless no sequence is supplied.
- `trace_chainhelices_strands_cc_min = 0.5` Minimum map CC for helices/strands
- `helices_strands_cc_min = 0.5` Minimum map CC for helices/strands
- `n_random_frag = 100`
- `controlmultiprocessing = *` multiprocessing sge lsf pbs condor pbspro slurm Choices are multiprocessing (single machine) or queuing systems
- `queue_run_command = None` run command for queue jobs. For example qsub.
- `nproc = 1` Number of processors to use
- `random_seed = 171731` Random seed
- `verbose = False` Verbose output
- `skip_temp_dir = True` Skip temp_dir when scoring
- `quick = True` Quick run
- `superquick = False` Very quick run
- `ignore_symmetry_conflicts = False` You can ignore the symmetry information (CRYST1) from coordinate files. This may be necessary if your model has been placed in a box with box_map for

example.

- `max_dirs = 1000` Maximum number of directories (`fix_insertions_deletions_xxxx`)
- `resolve_size = None` Size of resolve to use.
- `coarse_grid = None` Use a coarse grid in RESOLVE (saves on memory)
- `em_side_density = False` Use EM side chain density. Alternative is to use standard x-ray side chain density in sequence templates.
- `gui` GUI-specific parameter required for output directory `output_dir = None`
- `output_dir = None`

<string>

fmodel.html

phenix.fmodel: a tool to compute structure factors, Fmodel:

$$F_{\text{model}} = \text{scale} * \exp(\text{AnisoScale}) * (F_{\text{calc}} + k_{\text{sol}} * \exp(-b_{\text{sol}} * s^2/4) * F_{\text{mask}})$$
 where: Fmodel - total model structure factor (complex value) AnisoScale = $-\text{ht} * A(-1) * b_{\text{cart}} * A(-1)^{\text{th}}/4$ h - column vector with Miller indices A - orthogonalization matrix b_cart - anisotropic scale matrix t and (-1) denotes transposition and inversion operations scale - overall scale factor Fcalc - structure factors calculated from atomic model k_sol and b_sol - Flat Bulk solvent model parameters Fmask - structure factors calculated from bulk solvent mask

$$F_{\text{model}} = \text{scale} * \exp(\text{AnisoScale}) * (F_{\text{calc}} + k_{\text{sol}} * \exp(-b_{\text{sol}} * s^2/4) * F_{\text{mask}})$$

where:

- Fmodel - total model structure factor (complex value)
- AnisoScale = $-\text{ht} * A(-1) * b_{\text{cart}} * A(-1)^{\text{th}}/4$
- h - column vector with Miller indices
- A - orthogonalization matrix
- b_cart - anisotropic scale matrix
- t and (-1) denotes transposition and inversion operations
- scale - overall scale factor
- Fcalc - structure factors calculated from atomic model
- k_sol and b_sol - Flat Bulk solvent model parameters
- Fmask - structure factors calculated from bulk solvent mask

Usage examples:

phenix.fmodel model.pdb high_resolution=1.5 will result in a file containing complete set of Fmodel = Fcalc computed from atomic model up to 1.5Å resolution. phenix.fmodel model.pdb scale=2 k_sol=0.35 b_sol=50 b_cart="1 2 3 0 4 7" high_res=1.5 low_res=10 will result in a file containing complete set of Fmodel computed using the above formula in resolution range 1.5-20.0Å. phenix.fmodel model.pdb high_resolution=1.5 algorithm=direct is similar to "1)" but the Fcalc are computed using direct summation algorithm. phenix.fmodel model.pdb high_res=1.5 format=cns label=FOBS type=real r_free=0.1 will result in CNS formatted file containing complete set of amplitudes of Fmodel = Fcalc computed up to 1.5Å resolution, labelled as FOBS, and free-R flags with 10% of test reflections. This is a typical command to simulate Fobs. phenix.fmodel model.pdb high_res=1.5 scattering_table=neutron will result in a file containing complete set of Fmodel = Fcalc computed from atomic model up to 1.5Å resolution using neutron scattering table. phenix.fmodel model.pdb parameters.txt will result in a structure factor file, where Fmodel were computed using parameters defined in parameters.txt file. The parameters.txt file can contain all or any subset of parameters listed below. Note, that each { must have a matching one }. phenix.fmodel model.pdb reflection_data.mtz will result in a file containing a set of Fmodel = Fcalc that will match the set of Miller indices of the data in reflection_data.mtz file. phenix.fmodel model.pdb reflection_data.mtz data_column_label="FOBS,SIGMA" similar to "7)", where the specific data array is selected. phenix.fmodel model.pdb reflection_data.mtz twin_law="l,-k,h" twin_fraction=0.3 generates twin data set (real type) with given twin law and fraction.

- phenix.fmodel model.pdb high_resolution=1.5 will result in a file containing complete set of Fmodel = Fcalc computed from atomic model up to 1.5Å resolution.

phenix.fmodel model.pdb high_resolution=1.5

will result in a file containing complete set of Fmodel = Fcalc computed from atomic model up to 1.5A resolution.

- phenix.fmodel model.pdb scale=2 k_sol=0.35 b_sol=50 b_cart="1 2 3 0 4 7" high_res=1.5 low_res=10 will result in a file containing complete set of Fmodel computed using the above formula in resolution range 1.5-20.0A.

phenix.fmodel model.pdb scale=2 k_sol=0.35 b_sol=50 b_cart="1 2 3 0 4 7" high_res=1.5 low_res=10

will result in a file containing complete set of Fmodel computed using the above formula in resolution range 1.5-20.0A.

- phenix.fmodel model.pdb high_resolution=1.5 algorithm=direct is similar to "1)" but the Fcalc are computed using direct summation algorithm.

phenix.fmodel model.pdb high_resolution=1.5 algorithm=direct

is similar to "1)" but the Fcalc are computed using direct summation algorithm.

- phenix.fmodel model.pdb high_res=1.5 format=cns label=FOBS type=real r_free=0.1 will result in CNS formatted file containing complete set of amplitudes of Fmodel = Fcalc computed up to 1.5A resolution, labelled as FOBS, and free-R flags with 10% of test reflections. This is a typical command to simulate Fobs.

phenix.fmodel model.pdb high_res=1.5 format=cns label=FOBS type=real r_free=0.1

will result in CNS formatted file containing complete set of amplitudes of Fmodel = Fcalc computed up to 1.5A resolution, labelled as FOBS, and free-R flags with 10% of test reflections. This is a typical command to simulate Fobs.

- phenix.fmodel model.pdb high_res=1.5 scattering_table=neutron will result in a file containing complete set of Fmodel = Fcalc computed from atomic model up to 1.5A resolution using neutron scattering table.

phenix.fmodel model.pdb high_res=1.5 scattering_table=neutron

will result in a file containing complete set of Fmodel = Fcalc computed from atomic model up to 1.5A resolution using neutron scattering table.

- phenix.fmodel model.pdb parameters.txt will result in a structure factor file, where Fmodel were computed using parameters defined in parameters.txt file. The parameters.txt file can contain all or any subset of parameters listed below. Note, that each { must have a matching one }.

phenix.fmodel model.pdb parameters.txt

will result in a structure factor file, where Fmodel were computed using parameters defined in parameters.txt file. The parameters.txt file can contain all or any subset of parameters listed below. Note, that each { must have a matching one }.

- phenix.fmodel model.pdb reflection_data.mtz will result in a file containing a set of Fmodel = Fcalc that will match the set of Miller indices of the data in reflection_data.mtz file.

phenix.fmodel model.pdb reflection_data.mtz

will result in a file containing a set of Fmodel = Fcalc that will match the set of Miller indices of the data in reflection_data.mtz file.

- phenix.fmodel model.pdb reflection_data.mtz data_column_label="FOBS,SIGMA" similar to "7)", where the specific data array is selected.

```
phenix.fmodel model.pdb reflection_data.mtz data_column_label="FOBS,SIGMA"
```

similar to "7)", where the specific data array is selected.

- `phenix.fmodel model.pdb reflection_data.mtz twin_law="l,-k,h" twin_fraction=0.3` generates twin data set (real type) with given twin law and fraction.

```
phenix.fmodel model.pdb reflection_data.mtz twin_law="l,-k,h" twin_fraction=0.3
```

generates twin data set (real type) with given twin law and fraction.

See below for complete list of available parameters.

Find matching fragments in the PDB (fragment_search)

fragment_search.html

Author(s)

- fragment_search: Tom Terwilliger

Purpose

Extract a segment from the PDB that is similar to a segment in a supplied model.

This tool can be used in early stages of model-building to try and improve a segment in a preliminary model by replacing it with a similar fragment from a deposited structure.

Note that the fragment from the PDB may not actually have good geometry because it is morphed (distorted) to match the supplied model.

Fragment search differs from `replace_with_fragments_from_pdb` in that `fragment_search` is looking for a single fragment from the PDB that matches a target segment in your model, while `replace_with_fragments_from_pdb` is trying to completely replace your model with a set of fragments.

Usage

`fragment_search` can find a fragment in the PDB similar to a supplied fragment (protein chains only)

You can adjust the maximum number of insertions/deletions in the replacement fragment and whether to morph the fragment.

You can also choose to exclude or include specific PDB entries or exclude all PDB entries with sequences similar to a particular PDB entry.

You can choose to create a replacement for just a selected part of your model (by supplying a selection string that specifies what to select).

How fragment_search works:

Fragment searching is done using a quality-filtered set of chains from the PDB, filtered at 90% sequence identity (`pdb2018_90`). The curated subset of the PDB consist of non-redundant (<90% identity), high-resolution structures (< 2 Å) that have very good geometry. The main-chain coordinates of these structures are supplied in a database as part of Phenix.

Potential matching fragments are found by comparing end-to-end distances for the target with all fragments of the desired length in structures in the database. Fragments with the desired end-to-end distance are then examined for matching of distances from end to several internal positions in the target fragment. Finally those that match all these are superimposed on the target fragment, morphed to match the target fragment, and the rmsd of CA positions are used to evaluate the match.

Morphing includes two steps. In the first, the fragment is distorted with an offset that changes linearly for each residue along the chain and that allows the first and last CA positions of the matching fragment to superimpose on those of the target fragment.

In the second step, a distortion field is applied to the matching fragment to make it more similar to the target fragment. This morphing consists of calculation of a real-space distortion field that changes over a typical

distance of about 10 Å (set by the parameter `distortion_field_length`), and applying this distortion field to all the atoms in the matching fragment.

Examples

Standard run of `fragment_search`:

Running `fragment_search` is easy. From the command-line you can type:

```
phenix.fragment_search model.pdb nproc=8 selection="chain A and resseq 1:20"
```

This will find fragments in the PDB similar to residues 1-20 of `model.pdb`.

Possible Problems

Specific limitations and problems:

Literature

Additional information

List of all available keywords

`job_title` = None Job title in PHENIX GUI, not used on command line
`input_filesmodel` = None Input PDB file with segments to try and match
`selection` = None If specified, match selected part of model
`output_files`superposed_model_prefix = None Output files with superposed models will begin with this prefix
`fragment_searchtrim` = True Trim models down to match the target fragment. Otherwise return full
`chainn_res` = None Number of residues to get. If None, use number of residues in input fragment
`max_indel` = 0 Maximum insertions/deletions relative to `n_resn_half_window` = 5 Number of residues over which to average morphing shifts
`max_gap` = 20 Maximum gap in match to keep. Interpolate morphing values over gaps up to `max_gap`. Trim gaps longer than `max_gap`
`max_segment_matching_ca_ca_deviation` = 0.8 Maximum rmsd between segments to allow a match
`max_models` = 1 Number of models to return
`morph` = True Morph output models after superposition and before trimming
`pdb_include_list` = None List of PDB ID or pdb ID and chain_id to include (2DY1 or 2DY1_A)
`pdb_exclude_list` = None List of PDB ID or pdb ID and chain_id to exclude (2DY1 or 2DY1_A)
`exclude_similar_to_this_pdb_id` = None Exclude all PDB chains with sequence identity > `maximum_percent_identity_to_this_pdb_id` to `exclude_similar_to_this_pdb_id`
`maximum_percent_identity_to_this_pdb_id` = 30 Maximum percent identity to supplied PDB chain to include. Exclude all PDB chains with sequence identity > `maximum_percent_identity_to_this_pdb_id` to `exclude_similar_to_this_pdb_id`
`controlnproc` = 1 Number of processors to use. If None, uses all available
`write_files` = True Write output files
GUI-specific parameter required for output directory
`output_dir` = None

- `job_title` = None Job title in PHENIX GUI, not used on command line
- `input_filesmodel` = None Input PDB file with segments to try and match
`selection` = None If specified, match selected part of model
- `model` = None Input PDB file with segments to try and match
- `selection` = None If specified, match selected part of model

- `output_filesuperposed_model_prefix = None` Output files with superposed models will begin with this prefix
- `superposed_model_prefix = None` Output files with superposed models will begin with this prefix
- `fragment_searchtrim = True` Trim models down to match the target fragment. Otherwise return full chain
- `n_res = None` Number of residues to get. If None, use number of residues in input fragment
- `max_indel = 0` Maximum insertions/deletions relative to `n_res`
- `n_half_window = 5` Number of residues over which to average morphing shifts
- `max_gap = 20` Maximum gap in match to keep. Interpolate morphing values over gaps up to `max_gap`. Trim gaps longer than `max_gap`
- `max_segment_matching_ca_ca_deviation = 0.8` Maximum rmsd between segments to allow a match
- `max_models = 1` Number of models to return
- `morph = True` Morph output models after superposition and before trimming
- `pdb_include_list = None` List of PDB ID or pdb ID and chain_id to include (2DY1 or 2DY1_A)
- `pdb_exclude_list = None` List of PDB ID or pdb ID and chain_id to exclude (2DY1 or 2DY1_A)
- `exclude_similar_to_this_pdb_id = None` Exclude all PDB chains with sequence identity > `maximum_percent_identity_to_this_pdb_id` to exclude_similar_to_this_pdb_id
- `maximum_percent_identity_to_this_pdb_id = 30` Maximum percent identity to supplied PDB chain to include. Exclude all PDB chains with sequence identity > `maximum_percent_identity_to_this_pdb_id` to exclude_similar_to_this_pdb_id
- `controlnproc = 1` Number of processors to use. If None, uses all available
- `write_files = True` Write output files
- `nproc = 1` Number of processors to use. If None, uses all available
- `write_files = True` Write output files
- `gui` GUI-specific parameter required for output directory
- `output_dir = None`

Correlation of two maps (deprecated, use MapCorrelations)

`get_cc_mtz_mtz.html`

Author(s)

- `get_cc_mtz_mtz`: Tom Terwilliger

Purpose

`get_cc_mtz_mtz` is a command line tool for adjusting the origin of a map so that the map superimposes on another map, and obtaining the correlation of the two maps. The maps are calculated from map coefficients supplied by the user in two mtz files.

Usage

How `get_cc_mtz_mtz` works:

`get_cc_mtz_mtz` calculates maps based on the supplied mtz files, then uses RESOLVE to find the origin shift compatible with space-group symmetry that maximizes the correlation of the two maps. This shift is applied to the second map and the correlation of the maps is calculated.

Several parameters can be set by the user to define how the correlations are calculated. By default, maps are calculated using all the reflections present (to the specified high-resolution limit, if any) in each mtz file.

If you specify

```
use_only_refl_present_in_mtz_1=True
```

Then the map calculated using your second mtz file will only include reflections that were present in your first mtz file. This removes the effects of missing reflections on the correlation.

If you specify

```
scale=True
```

then `get_cc_mtz_mtz` scales the amplitudes from the second input mtz file to those in the first input mtz, including an overall B factor and a scale factor. This reduces effects of differences in overall B factors between the two mtz files on the correlation.

If you specify

```
keep_f_mag=False
```

then `get_cc_mtz_mtz` uses amplitudes from the first input mtz file and phases and figure of merit from both to do the correlation. This has the effect of removing effects due to differences in amplitudes on the correlation, and focusing on differences in phases and figures of merit.

Output files from `get_cc_mtz_mtz`

`offset.log`: Log file for correlation calculation.

Examples

Standard run of get_cc_mtz_mtz:

Running the get_cc_mtz_mtz is easy. From the command-line you can type:

```
phenix.get_cc_mtz_mtz map_coeffs_1.mtz map_coeffs_2.mtz
```

If you want (or need) to specify the column names from your mtz file, you will need to tell get_cc_mtz_mtz what FP and PHIB (and optionally FOM) are, in this format:

```
phenix.get_cc_mtz_mtz map_coeffs_1.mtz map_coeffs_2.mtz \  
labin_1="FP=2FOFCWT PHIB=PH2FOFCWT" labin_2="FP=2FOFCWT PHIB=PH2FOFCWT"
```

Possible Problems

Specific limitations and problems:

- Versions of phenix.get_cc_mtz_mtz up to 1.3-final used a different set of defaults, with the values,

```
scale=True  
keep_f_mag=False  
use_only_refl_present_in_mtz_1=True
```

These defaults were changed after version 1.3-final in order to make the results independent of the order of the mtz files and to make the default be to get the correlation of maps without manipulation.

List of all available keywords

get_cc_mtz_mtzmtz_1 = None MTZ file 1 with coefficients for a map mtz_2 = None MTZ file 2 with coefficients for a map labin_1 = "" Labin line for MTZ file 1 with map coefficients. This is optional if get_cc_mtz_mtz can guess the correct coefficients for FP PHI and FOM. Otherwise specify: LABIN FP=myFP PHIB=myPHI FOM=myFOM where myFP is your column label for FP labin_2 = "" Labin line for MTZ file 2 with map coefficients. This is optional if get_cc_mtz_mtz can guess the correct coefficients for FP PHI and FOM. Otherwise specify: LABIN FP=myFP PHIB=myPHI FOM=myFOM where myFP is your column label for FP map_in_1 = None CCP4 or MRC-style map file for file 1 (instead of mtz_1)map_in_2 = None CCP4 or MRC-style map file for file 2 (instead of mtz_2)resolution = None high-resolution limit for map calculationlow_resolution = None low-resolution limit for map calculationoffset_suffix = offset Suffix for output version of mtz file 2 or map_in_2, offset to maximize correlation with mtz file 1temp_dir = "temp_dir" Temporary work directoryoutput_dir = "" Output directory where files are to be writtengui_output_dir = None keep_f_mag = True If you set keep_f_mag=False then get_cc_mtz_mtz uses amplitudes from the first input mtz file and phases and fom from both to do the correlation. If you specify keep_f_mag=True then the amplitudes from both files are included.scale = False If you set scale=True then get_cc_mtz_mtz scales the amplitudes from the second input mtz file to those in the first input mtz, including an overall B factor and a scale factor. use_only_refl_present_in_mtz_1 = False You can specify that only reflections present in your first mtz file are used in the comparison. Note that this means that the order of the files will have an effect on the correlation coefficientany_offset = False You can search for a match with any offset even though this is not allowed by space-group symmetryfix_xyz = False You can get map-map correlation with no translations Does not apply to mtz-mtz comparisonsverbose = True Verbose outputraise_sorry = False Raise sorry if problemsdebug = False Debugging outputdry_run = False Just read in and check parameter namesresolve_command_list = None You can supply any resolve command here NOTE: for command-line usage you need to enclose the whole set of commands in double quotes (") and each individual command in single quotes (') like this: resolve_command_list="'no_build' 'b_overall 23' "; job_title = None Job title in PHENIX GUI, not used on command line

• `get_cc_mtz_mtz` `mtz_1` = None MTZ file 1 with coefficients for a map `mtz_2` = None MTZ file 2 with coefficients for a map `labin_1` = "" Labin line for MTZ file 1 with map coefficients. This is optional if `get_cc_mtz_mtz` can guess the correct coefficients for FP PHI and FOM. Otherwise specify: LABIN FP=`myFP` PHIB=`myPHI` FOM=`myFOM` where `myFP` is your column label for FP `labin_2` = "" Labin line for MTZ file 2 with map coefficients. This is optional if `get_cc_mtz_mtz` can guess the correct coefficients for FP PHI and FOM. Otherwise specify: LABIN FP=`myFP` PHIB=`myPHI` FOM=`myFOM` where `myFP` is your column label for FP `map_in_1` = None CCP4 or MRC-style map file for file 1 (instead of `mtz_1`) `map_in_2` = None CCP4 or MRC-style map file for file 2 (instead of `mtz_2`) `resolution` = None high-resolution limit for map calculation `low_resolution` = None low-resolution limit for map calculation `offset_suffix` = offset Suffix for output version of `mtz` file 2 or `map_in_2`, offset to maximize correlation with `mtz` file 1 `temp_dir` = "temp_dir" Temporary work directory `output_dir` = "" Output directory where files are to be written `gui_output_dir` = None `keep_f_mag` = True If you set `keep_f_mag`=False then `get_cc_mtz_mtz` uses amplitudes from the first input `mtz` file and phases and fom from both to do the correlation. If you specify `keep_f_mag`=True then the amplitudes from both files are included. `scale` = False If you set `scale`=True then `get_cc_mtz_mtz` scales the amplitudes from the second input `mtz` file to those in the first input `mtz`, including an overall B factor and a scale factor. `use_only_refl_present_in_mtz_1` = False You can specify that only reflections present in your first `mtz` file are used in the comparison. Note that this means that the order of the files will have an effect on the correlation coefficient `any_offset` = False You can search for a match with any offset even though this is not allowed by space-group symmetry `fix_xyz` = False You can get map-map correlation with no translations Does not apply to `mtz-mtz` comparisons `verbose` = True Verbose output `raise_sorry` = False Raise sorry if problems `debug` = False Debugging output `dry_run` = False Just read in and check parameter names `resolve_command_list` = None You can supply any resolve command here NOTE: for command-line usage you need to enclose the whole set of commands in double quotes (") and each individual command in single quotes (') like this: `resolve_command_list`="'no_build' 'b_overall 23' "; `job_title` = None Job title in PHENIX GUI, not used on command line

- `mtz_1` = None MTZ file 1 with coefficients for a map
- `mtz_2` = None MTZ file 2 with coefficients for a map
- `labin_1` = "" Labin line for MTZ file 1 with map coefficients. This is optional if `get_cc_mtz_mtz` can guess the correct coefficients for FP PHI and FOM. Otherwise specify: LABIN FP=`myFP` PHIB=`myPHI` FOM=`myFOM` where `myFP` is your column label for FP
- `labin_2` = "" Labin line for MTZ file 2 with map coefficients. This is optional if `get_cc_mtz_mtz` can guess the correct coefficients for FP PHI and FOM. Otherwise specify: LABIN FP=`myFP` PHIB=`myPHI` FOM=`myFOM` where `myFP` is your column label for FP
- `map_in_1` = None CCP4 or MRC-style map file for file 1 (instead of `mtz_1`)
- `map_in_2` = None CCP4 or MRC-style map file for file 2 (instead of `mtz_2`)
- `resolution` = None high-resolution limit for map calculation
- `low_resolution` = None low-resolution limit for map calculation
- `offset_suffix` = offset Suffix for output version of `mtz` file 2 or `map_in_2`, offset to maximize correlation with `mtz` file 1
- `temp_dir` = "temp_dir" Temporary work directory
- `output_dir` = "" Output directory where files are to be written
- `gui_output_dir` = None

- `keep_f_mag = True` If you set `keep_f_mag=False` then `get_cc_mtz_mtz` uses amplitudes from the first input mtz file and phases and fom from both to do the correlation. If you specify `keep_f_mag=True` then the amplitudes from both files are included.
- `scale = False` If you set `scale=True` then `get_cc_mtz_mtz` scales the amplitudes from the second input mtz file to those in the first input mtz, including an overall B factor and a scale factor.
- `use_only_refl_present_in_mtz_1 = False` You can specify that only reflections present in your first mtz file are used in the comparison. Note that this means that the order of the files will have an effect on the correlation coefficient
- `any_offset = False` You can search for a match with any offset even though this is not allowed by space-group symmetry
- `fix_xyz = False` You can get map-map correlation with no translations Does not apply to mtz-mtz comparisons
- `verbose = True` Verbose output
- `raise_sorry = False` Raise sorry if problems
- `debug = False` Debugging output
- `dry_run = False` Just read in and check parameter names
- `resolve_command_list = None` You can supply any resolve command here NOTE: for command-line usage you need to enclose the whole set of commands in double quotes (") and each individual command in single quotes (') like this: `resolve_command_list="'no_build' 'b_overall 23' "`
- `job_title = None` Job title in PHENIX GUI, not used on command line

Correlation of map and model (deprecated, use MapCorrelations)

[get_cc_mtz_pdb.html](#)

Author(s)

- `get_cc_mtz_pdb`: Tom Terwilliger

Purpose

`get_cc_mtz_pdb` is a command line tool for adjusting the origin of a PDB file using space-group symmetry so that the PDB file superimposes on a map, obtaining the correlation of model and map, and analyzing the correlation for each residue.

Usage

How `get_cc_mtz_pdb` works:

`get_cc_mtz_pdb` calculates a model map based on the supplied PDB file, then uses RESOLVE to find the origin shift (using space-group symmetry) that maximizes the correlation of this model map with a map calculated with the supplied map coefficients in an mtz file. This shift is applied to the atoms in the PDB file to create `offset.pdb` and then the correlation, residue-by-residue of `offset.pdb` with the map is analyzed. Atoms and residues that are out of density or are in weak density are flagged.

You can set several parameters to define how the correlations are calculated.

By default model density is calculated using the atom types, occupancies and isotropic thermal factors (B-values) supplied in the PDB file. If you specify

```
scale=True
```

then an overall B as well as an increment in B-values for each atom beyond CB (for proteins) will be added to the values in the PDB file, after adjusting these parameters to maximize the map correlation.

If you specify

```
use_only_refl_present_in_mtz=True
```

then the model-based map will be calculated using the same set of reflections as the map calculated from your input mtz file. This reduces the contribution of missing reflections on the calculation (but the correlation is no longer the actual map-model correlation).

In the calculation of the map correlation in the region of the model, the region where the model is located is defined as all points within a distance `rad_max` of an atom in the model. The value of `rad_max` is adjusted in each case to maximize this correlation. Its value is typically similar to the high-resolution limit of the map.

Output files from `get_cc_mtz_pdb`

`offset.pdb`: A PDB file offset to match the origin in the mtz file.

Examples

Standard run of get_cc_mtz_pdb:

Running the get_cc_mtz_pdb is easy. From the command-line you can type:

```
phenix.get_cc_mtz_pdb map_coeffs.mtz coords.pdb
```

If you want (or need) to specify the column names from your mtz file, you will need to tell get_cc_mtz_pdb what FP and PHIB (and optionally FOM) are, in this format:

```
phenix.get_cc_mtz_pdb map_coeffs.mtz coords.pdb \
  labin="FP=2FOFCWT PHIB=PH2FOFCWT"
```

Possible Problems

Specific limitations and problems:

The option to use atom selections in get_cc_mtz_pdb can cause some confusion, because the CC values obtained can depend on the atom selections. The way this works is that the PDB file (after atom selections) is used to calculate model density. This model density is compared to the map from your input map coefficients in the region surrounding the atoms selected. This model density can depend on the presence of nearby atoms (where the density extends a ways away from those atoms).

In versions of PHENIX up to 1.3-final, defaults were set to maximize the correlation coefficient rather than to give the correlation using the existing thermal parameters and including only the reflections present in the mtz file. These previous defaults were equivalent to using the values:

```
scale=True
use_only_refl_present_in_mtz=True
```

These defaults were changed so that the correlation values obtained by default in a case where no origin shifts are needed would correspond to those obtained by simply calculating (1) a map using the input map coefficients and (2) a map from the PBB file and then determining the correlation between these maps.

List of all available keywords

get_cc_mtz_pdbpdb_in = None PDB file with coordinates to evaluateatom_selection = None Any selection specified with atom_selection is applied to input model (pdb_in) before using the model. NOTE: this option can result in confusing output because the model density near an atom and therefore the local CC value depends not just on the atoms being considered but also nearby atoms. Therefore if you remove some atoms next to a residue with atom_selection then the CC for that residue may change.mtz_in = None MTZ file with coefficients for a map map_in = None CCP4 or MRC-style map filelabin = "" Labin line for MTZ file with map coefficients. This is optional if get_cc_mtz_pdb can guess the correct coefficients for FP PHI and FOM. Otherwise specify: LABIN FP=myFP PHIB=myPHI FOM=myFOM where myFP is your column label for FP offset_pdb_suffix = offset Suffix for output version of pdb file, (offset to maximize correlation with mtz file if fix_xyz is not set)resolution = None high-resolution limit for map calculationuse_only_refl_present_in_mtz = False You can specify that only reflections present in your mtz file are used in the comparison. scale = False If you set scale=True then get_cc_mtz_pdb applies an overall B factor and a delta_b for each atom beyond CB. split_conformers = False If you want to have A and B conformers analyzed separately you can say split_conformers=True. Note that this requires that the entire A conformer for a residue must be before (or after) the entire B conformer for that residue.use_cc_mask = None Use cc_mask calculation to match validation methods Default is True for map_in and False for mtz_infix_xyz = False If you want your PDB file compared to the map from your mtz file with no offsets at all (fixed position) then specify fix_xyz=Truefix_rad_max = False If you want to use a fixed radius around all atoms for calculation of the

correlation use fix_rad=True. To set the value, use rad_max=xxx, otherwise it is set automatically. If rad_max is set, fix_rad_max is set to True. rad_max = None If you want to use a fixed radius around all atoms for calculation of the correlation use fix_rad=True. To set the value, use rad_max=2.5, otherwise it is set automatically. any_offset = False You can search for a match with any offset even though this is not allowed by space-group symmetry. chain_type = *PROTEIN DNA RNA Chain type (for identifying main-chain and side-chain atoms). temp_dir = "temp_dir" Temporary work directory. output_dir = "" Output directory where files are to be written. gui_output_dir = None verbose = True Verbose output. quick = False Skip the residue-by-residue correlations for a quick run. raise_sorry = False Raise sorry if problems. debug = False Debugging output. dry_run = False Just read in and check parameter names. resolve_command_list = None You can supply any resolve command here. NOTE: for command-line usage you need to enclose the whole set of commands in double quotes (") and each individual command in single quotes (') like this: resolve_command_list="'no_build' 'b_overall 23' "; job_title = None Job title in PHENIX GUI, not used on command line. output_file_target_output_format = *None pdb mmCIF Desired output format (if possible). Choices are None (try to use input format), pdb, mmCIF. If output model does not fit in pdb format, mmCIF will be used. Default is pdb.

- get_cc_mtz_pdb_pdb_in = None PDB file with coordinates to evaluate. atom_selection = None Any selection specified with atom_selection is applied to input model (pdb_in) before using the model. NOTE: this option can result in confusing output because the model density near an atom and therefore the local CC value depends not just on the atoms being considered but also nearby atoms. Therefore if you remove some atoms next to a residue with atom_selection then the CC for that residue may change. mtz_in = None MTZ file with coefficients for a map. map_in = None CCP4 or MRC-style map file. labin = "" Labin line for MTZ file with map coefficients. This is optional if get_cc_mtz_pdb can guess the correct coefficients for FP PHI and FOM. Otherwise specify: LABIN FP=myFP PHIB=myPHI FOM=myFOM where myFP is your column label for FP. offset_pdb_suffix = offset Suffix for output version of pdb file, (offset to maximize correlation with mtz file if fix_xyz is not set). resolution = None high-resolution limit for map calculation. use_only_refl_present_in_mtz = False You can specify that only reflections present in your mtz file are used in the comparison. scale = False If you set scale=True then get_cc_mtz_pdb applies an overall B factor and a delta_b for each atom beyond CB. split_conformers = False If you want to have A and B conformers analyzed separately you can say split_conformers=True. Note that this requires that the entire A conformer for a residue must be before (or after) the entire B conformer for that residue. use_cc_mask = None Use cc_mask calculation to match validation methods. Default is True for map_in and False for mtz_in. fix_xyz = False If you want your PDB file compared to the map from your mtz file with no offsets at all (fixed position) then specify fix_xyz=True. fix_rad_max = False If you want to use a fixed radius around all atoms for calculation of the correlation use fix_rad=True. To set the value, use rad_max=xxx, otherwise it is set automatically. If rad_max is set, fix_rad_max is set to True. rad_max = None If you want to use a fixed radius around all atoms for calculation of the correlation use fix_rad=True. To set the value, use rad_max=2.5, otherwise it is set automatically. any_offset = False You can search for a match with any offset even though this is not allowed by space-group symmetry. chain_type = *PROTEIN DNA RNA Chain type (for identifying main-chain and side-chain atoms). temp_dir = "temp_dir" Temporary work directory. output_dir = "" Output directory where files are to be written. gui_output_dir = None verbose = True Verbose output. quick = False Skip the residue-by-residue correlations for a quick run. raise_sorry = False Raise sorry if problems. debug = False Debugging output. dry_run = False Just read in and check parameter names. resolve_command_list = None You can supply any resolve command here. NOTE: for command-line usage you need to enclose the whole set of commands in double quotes (") and each individual command in single quotes (') like this: resolve_command_list="'no_build' 'b_overall 23' "; job_title = None Job title in PHENIX GUI, not used on command line. output_file_target_output_format = *None pdb mmCIF Desired output format (if possible). Choices are None (try to use input format), pdb, mmCIF. If output model does

not fit in pdb format, mmCIF will be used. Default is pdb.

- `pdb_in` = None PDB file with coordinates to evaluate
- `atom_selection` = None Any selection specified with `atom_selection` is applied to input model (`pdb_in`) before using the model. NOTE: this option can result in confusing output because the model density near an atom and therefore the local CC value depends not just on the atoms being considered but also nearby atoms. Therefore if you remove some atoms next to a residue with `atom_selection` then the CC for that residue may change.
- `mtz_in` = None MTZ file with coefficients for a map
- `map_in` = None CCP4 or MRC-style map file
- `labin` = "" Labin line for MTZ file with map coefficients. This is optional if `get_cc_mtz_pdb` can guess the correct coefficients for FP PHI and FOM. Otherwise specify: `LABIN FP=myFP PHIB=myPHI FOM=myFOM` where `myFP` is your column label for FP
- `offset_pdb_suffix` = offset Suffix for output version of pdb file, (offset to maximize correlation with mtz file if `fix_xyz` is not set)
- `resolution` = None high-resolution limit for map calculation
- `use_only_refl_present_in_mtz` = False You can specify that only reflections present in your mtz file are used in the comparison.
- `scale` = False If you set `scale=True` then `get_cc_mtz_pdb` applies an overall B factor and a `delta_b` for each atom beyond CB.
- `split_conformers` = False If you want to have A and B conformers analyzed separately you can say `split_conformers=True`. Note that this requires that the entire A conformer for a residue must be before (or after) the entire B conformer for that residue.
- `use_cc_mask` = None Use `cc_mask` calculation to match validation methods Default is True for `map_in` and False for `mtz_in`
- `fix_xyz` = False If you want your PDB file compared to the map from your mtz file with no offsets at all (fixed position) then specify `fix_xyz=True`
- `fix_rad_max` = False If you want to use a fixed radius around all atoms for calculation of the correlation use `fix_rad=True`. To set the value, use `rad_max=xxx`, otherwise it is set automatically. If `rad_max` is set, `fix_rad_max` is set to True
- `rad_max` = None If you want to use a fixed radius around all atoms for calculation of the correlation use `fix_rad=True`. To set the value, use `rad_max=2.5`, otherwise it is set automatically
- `any_offset` = False You can search for a match with any offset even though this is not allowed by space-group symmetry
- `chain_type` = *PROTEIN DNA RNA Chain type (for identifying main-chain and side-chain atoms)
- `temp_dir` = "temp_dir" Temporary work directory
- `output_dir` = "" Output directory where files are to be written
- `gui_output_dir` = None
- `verbose` = True Verbose output
- `quick` = False Skip the residue-by-residue correlations for a quick run
- `raise_sorry` = False Raise sorry if problems
- `debug` = False Debugging output

- `dry_run = False` Just read in and check parameter names
- `resolve_command_list = None` You can supply any resolve command here NOTE: for command-line usage you need to enclose the whole set of commands in double quotes (",) and each individual command in single quotes (') like this: `resolve_command_list="'no_build' 'b_overall 23' "`
- `job_title = None` Job title in PHENIX GUI, not used on command line
- `output_filetarget_output_format = *None` pdb mmCIF Desired output format (if possible). Choices are None (try to use input format), pdb, mmCIF. If output model does not fit in pdb format, mmCIF will be used. Default is pdb.
- `target_output_format = *None` pdb mmCIF Desired output format (if possible). Choices are None (try to use input format), pdb, mmCIF. If output model does not fit in pdb format, mmCIF will be used. Default is pdb.

Guess location of molecular centers in a map with guess_molecular_centers

[guess_molecular_centers.html](#)

Author(s)

- guess_molecular_centers: Tom Terwilliger

Purpose

guess_molecular_centers is a tool to identify potential locations of molecules in an electron density map.

Usage

How guess_molecular_centers works:

guess_molecular_centers identifies centers of macromolecules in the density map by finding local maxima of the local RMS density. You can specify the radius of the sphere used to estimate local RMS density.

Output files from guess_molecular_centers

guess_molecular_centers.pdb: A PDB file with pseudo-atoms marking the locations of centers of your molecule(s)

Examples

Standard run of guess_molecular_centers:

Running guess_molecular_centers is easy. From the command-line you can type:

```
phenix.guess_molecular_centers map_coeffs.mtz
```

Possible Problems

If your molecule is large, you may wish to increase the radius used for smoothing (smoothing_radius). If you cannot find the centers you want, you can try looking for more by increasing n_centers_find.

Literature

Finding non-crystallographic symmetry in density maps of macromolecular structures. T.C. Terwilliger. J Struct Funct Genomics 14, 91-5 (2013).

- Finding non-crystallographic symmetry in density maps of macromolecular structures. T.C. Terwilliger. J Struct Funct Genomics 14, 91-5 (2013).

List of all available keywords

guess_molecular_centersinput_filesmtz_in = None MTZ file with coefficients for a map labin = "" Labin line for MTZ file with map coefficients. This is optional if guess_molecular_centers can guess the correct coefficients

for FP PHI and FOM. Otherwise specify: LABIN FP=myFP PHIB=myPHI FOM=myFOM where myFP is your column label for FP output_filetarget_output_format = *None pdb mmCIF Desired output format (if possible). Choices are None (try to use input format), pdb, mmCIF. If output model does not fit in pdb format, mmCIF will be used. Default is pdb.pdb_out = guess_molecular_centers.pdb Output PDB filelog = guess_molecular_centers.log Output log fileparams_out = guess_molecular_centers_params.eff Parameters file to rerun guess_molecular_centersdirectoriestemp_dir = "temp_dir" Optional temporary work directory output_dir = "" Output directory where files are to be writtenfind_centersssmoothing_radius = 10. Radius for smoothing squared density to find centers Choose a smaller value to get more center guessesn_centers = 1 Report up to n_centers centersn_center_find = None Target number of centers to find Smoothing radius will be varied from 2 to 2*smoothing_radius and the value giving the number of peaks closest to n_center_find will be usedcrystal_inresolution = 0. high-resolution limit for map calculationsolvent_fraction = 0.5 solvent fractioncontrolverbose = True Verbose outputraise_sorry = False Raise sorry if problemsdebug = False Debugging outputdry_run = False Just read in and check parameter namesresolve_command_list = None You can supply any resolve command here NOTE: for command-line usage you need to enclose the whole set of commands in double quotes (") and each individual command in single quotes (') like this: resolve_command_list="'no_build' 'b_overall 23' ";

- guess_molecular_centersinput_filesmtz_in = None MTZ file with coefficients for a map labin = "" Labin line for MTZ file with map coefficients. This is optional if guess_molecular_centers can guess the correct coefficients for FP PHI and FOM. Otherwise specify: LABIN FP=myFP PHIB=myPHI FOM=myFOM where myFP is your column label for FP output_filetarget_output_format = *None pdb mmCIF Desired output format (if possible). Choices are None (try to use input format), pdb, mmCIF. If output model does not fit in pdb format, mmCIF will be used. Default is pdb.pdb_out = guess_molecular_centers.pdb Output PDB filelog = guess_molecular_centers.log Output log fileparams_out = guess_molecular_centers_params.eff Parameters file to rerun guess_molecular_centersdirectoriestemp_dir = "temp_dir" Optional temporary work directory output_dir = "" Output directory where files are to be writtenfind_centersssmoothing_radius = 10. Radius for smoothing squared density to find centers Choose a smaller value to get more center guessesn_centers = 1 Report up to n_centers centersn_center_find = None Target number of centers to find Smoothing radius will be varied from 2 to 2*smoothing_radius and the value giving the number of peaks closest to n_center_find will be usedcrystal_inresolution = 0. high-resolution limit for map calculationsolvent_fraction = 0.5 solvent fractioncontrolverbose = True Verbose outputraise_sorry = False Raise sorry if problemsdebug = False Debugging outputdry_run = False Just read in and check parameter namesresolve_command_list = None You can supply any resolve command here NOTE: for command-line usage you need to enclose the whole set of commands in double quotes (") and each individual command in single quotes (') like this: resolve_command_list="'no_build' 'b_overall 23' ";
- input_filesmtz_in = None MTZ file with coefficients for a map labin = "" Labin line for MTZ file with map coefficients. This is optional if guess_molecular_centers can guess the correct coefficients for FP PHI and FOM. Otherwise specify: LABIN FP=myFP PHIB=myPHI FOM=myFOM where myFP is your column label for FP
- mtz_in = None MTZ file with coefficients for a map
- labin = "" Labin line for MTZ file with map coefficients. This is optional if guess_molecular_centers can guess the correct coefficients for FP PHI and FOM. Otherwise specify: LABIN FP=myFP PHIB=myPHI FOM=myFOM where myFP is your column label for FP
- output_filetarget_output_format = *None pdb mmCIF Desired output format (if possible). Choices are None (try to use input format), pdb, mmCIF. If output model does not fit in pdb format, mmCIF will be used. Default is pdb.pdb_out = guess_molecular_centers.pdb Output PDB filelog =

guess_molecular_centers.log Output log file
params_out = guess_molecular_centers_params.eff
Parameters file to rerun guess_molecular_centers

- target_output_format = *None pdb mmCIF Desired output format (if possible). Choices are None (try to use input format), pdb, mmCIF. If output model does not fit in pdb format, mmCIF will be used. Default is pdb.
- pdb_out = guess_molecular_centers.pdb Output PDB file
- log = guess_molecular_centers.log Output log file
- params_out = guess_molecular_centers_params.eff Parameters file to rerun guess_molecular_centers
- directorytemp_dir = "temp_dir" Optional temporary work directory output_dir = "" Output directory where files are to be written
- temp_dir = "temp_dir" Optional temporary work directory
- output_dir = "" Output directory where files are to be written
- find_centerssmoothing_radius = 10. Radius for smoothing squared density to find centers Choose a smaller value to get more center guessesn_centers = 1 Report up to n_centers centersn_center_find = None Target number of centers to find Smoothing radius will be varied from 2 to 2*smoothing_radius and the value giving the number of peaks closest to n_center_find will be used
- smoothing_radius = 10. Radius for smoothing squared density to find centers Choose a smaller value to get more center guesses
- n_centers = 1 Report up to n_centers centers
- n_center_find = None Target number of centers to find Smoothing radius will be varied from 2 to 2*smoothing_radius and the value giving the number of peaks closest to n_center_find will be used
- crystal_inforesolution = 0. high-resolution limit for map calculationsolvent_fraction = 0.5 solvent fraction
- resolution = 0. high-resolution limit for map calculation
- solvent_fraction = 0.5 solvent fraction
- controlverbose = True Verbose outputraise_sorry = False Raise sorry if problemsdebug = False Debugging outputdry_run = False Just read in and check parameter namesresolve_command_list = None You can supply any resolve command here NOTE: for command-line usage you need to enclose the whole set of commands in double quotes (") and each individual command in single quotes (') like this: resolve_command_list="'no_build' 'b_overall 23' ";
- verbose = True Verbose output
- raise_sorry = False Raise sorry if problems
- debug = False Debugging output
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phenix.reduce: tool for adding hydrogens to a PDB model

hydrogens.html

Purpose

phenix.reduce is a command line tool for adding hydrogens to a PDB structure file. Hydrogens are added in standardized geometry with optimization of the orientations of OH, SH, NH₃⁺, Met methyls, Asn and Gln sidechain amides, and His rings. Both proteins and nucleic acids can be processed. HET groups can also be processed as long as the atom connectivity is provided. The program is described in Word, et al.(1999). J. Mol. Biol. 285, 1735-1747. For more information visit: <http://kinemage.biochem.duke.edu/software/reduce.php>

How to run

phenix.reduce is run from the command line:

```
% phenix.reduce [pdb_file] [options]
```

To get information about command line options type:

```
% phenix.reduce
```

or for a longer list:

```
% phenix.reduce -h
```

Hydrogens in refinement

Please refer to phenix.refine documentation to see how hydrogen atoms are used in structure refinement.

Phenix Documentation - version dev-svn

index.html

Phenix help chatbot, AI agent, and AI analysis tool

You can use the Phenix chatbot to get help on using any Phenix program. Just ask it any question about using Phenix.

You can use the Phenix AI Agent to help you run Phenix tools with your data in the GUI or from the command line.

You can also use the Phenix AI analysis tool to summarize the results of a run in the GUI and suggest next steps.

- Instructions for using the chatbot
- Phenix AI tool overview
- Phenix AI analysis tool
- Phenix AI Agent

Introduction to Phenix

- What is Phenix
- How to install and run Phenix
- The Phenix graphical interface
- The main modules in Phenix
- Phenix atom selection syntax
- FAQS: Frequently asked questions
- Dictionary of common terms
- Overview of file formats
- Index for Phenix documentation
- Tutorials and Examples
- Finding a program or regression test in Phenix

Tools for analysing and manipulating experimental data in Phenix

- Data quality assessment with phenix.xtriage (GUI version)
- Planning a SAD experiment with phenix.plan_sad_experiment
- Scaling unmerged anomalous data or multiple datasets with phenix.scale_and_merge
- Analyzing anomalous signal in a SAD experiment with phenix.anomalous_signal
- Data quality assessment with phenix.reflection_statistics
- Structure factor file inspection and conversions
- Graphical editor for MTZ files
- Manipulating reflection data with phenix.xmanip

- Exploring the symmetry of your crystal with `phenix.explore_metric_symmetry`
- Importing data from the PDB with `phenix.cif_as_mtz`
- Visualizing reciprocal-space data
- Using unmerged data in Phenix
- Showing information about a map file
- Calculate local resolution from two cryo-EM half-maps

Experimental phasing

- Automated Structure Solution using AutoSol (GUI version)
- SAD phasing in the Phaser GUI
- Substructure determination with `phenix.hyss`
- Comparison of substructure sites with `phenix.emma`
- Setting up SAD data from the PDB with `phenix.sad_data_from_pdb`
- Frequently asked questions about experimental phasing

Molecular replacement

- Overview of molecular replacement in Phenix
- Molecular replacement in the Phaser GUI
- MRage: automated molecular replacement
- MR and model building with AutoMR, AutoBuild and Rosetta
- Morphing a poor model into electron density
- Documentation for the Phaser program
- Ensembler: create ensemble of models for molecular replacement
- Sculptor: modify input model for molecular replacement
- Frequently asked questions about molecular replacement

Model-building

- Solve an X-ray structure or interpret a cryo-EM map with `PredictAndBuild`
- Automated Model Building and Rebuilding using AutoBuild (GUI version)
- Rapid cryo-EM model-building with `trace_and_build`
- Dock chains in Cryo-EM map with `dock_in_map`
- Likelihood-based single model docking in Cryo-EM map with `em_placement`
- Dock AlphaFold model with `dock_predicted_model`
- Dock and rebuild AlphaFold model with `dock_and_rebuild`
- Cryo-EM model rebuilding with `rebuild_model`
- Cryo-EM model-building with Pathwalker
- Low-resolution/cryo-EM model-building with `map_to_model`
- Rapid model-building with `find_helices_strands`

- Fitting loops with `fit_loops`
- Building starting with a very poor map with `parallel_autobuild`
- Rapid iterative map improvement and model-building with `phase_and_build`
- Rapid model-building with `build_one_model`
- Assigning sequence to a model with `assign_sequence`
- Replace side chains with rotamers using `replace_side_chains`
- Combine the best parts of two models with `combine_models`
- Remove incorrect protein from complex with `remove_poor_fragments`
- Fixing register errors with `fix_insertions_deletions`
- Adjusting a CA/CB model with `refine_ca_model`
- Frequently asked questions about model-building
- Build water into cryo-EM map
- Create a model with alternative side-chain conformations with `phenix.create_alt_conf`
- Analyze a model with alternative side-chain conformations with `phenix.analyze_alt_conf`

Structure refinement and restraint generation

- AQuaRef: AI-enabled Quantum Refinement of macromolecular models <AQuaRef.html>
- Structure refinement with `phenix.refine` (GUI version)
- Determining non-crystallographic symmetry (NCS) from a PDB file with `phenix.simple_ncs_from_pdb`
- Finding and analyzing NCS from heavy-atom sites or a model with `phenix.find_ncs`
- Generating ligand coordinates and restraints using eLBOW (GUI version)
- Editing ligand restraints from eLBOW using REEL
- Add hydrogens; generate metal coordination files; introduce neutron exchange sites using ReadySet!
- Adding secondary structure restraints
- Mode idealization before refinement
- Generating hydrogen atoms for refinement using `phenix.reduce`
- Refinement in real space with `phenix.real_space_refine`
- Ensemble refinement
- Variability refinement
- DEN refinement with `phenix.den_refine`
- Hybrid Rosetta/Phenix refinement of protein X-ray crystal structures
- Hybrid Phenix/Amber refinement for protein and RNA/DNA X-ray crystal structure
- Hybrid Phenix/AFITT refinement for ligands in X-ray crystal structure
- Frequently asked questions (FAQs) about `phenix.refine`

Validation

- Validation in the Phenix GUI
- `phenix.model_vs_data`: compute model, data and model-to-data fit statistics

- Low-resolution secondary structure validation with CaBLAM
- Map statistics (resolution, etc) with Mtriage
- `phenix.get_pdb_validation_report`: get validation report from RCSB via OneDep API
- Model geometry evaluation with `phenix.holton_geometry_validation`

Ligand fitting and restraints

- Quantum Interface (QI)
- Automated Ligand Fitting using LigandFit (GUI version)
- Automated Ligand Fitting using Guided Ligand Replacement (GLR)
- Automated ligand identification in Phenix
- Finding all the ligands in a map with `phenix.find_all_ligands`
- Generating ligand coordinates and restraints using eLBOW (GUI version)
- Editing ligand restraints from eLBOW using REEL
- Automated protein-ligand structure determination with `phenix.ligand_pipeline`
- Frequently asked questions about ligand fitting and restraints

Carbohydrates

- Carbohydrate Primer
- Simple Carbohydrate Line-Entry System (SCaLES)

Map utilities

- `phenix.map_symmetry`: Map symmetry analysis
- Generating electron density maps in the GUI
- `phenix.maps`: tool to compute various maps and save them in most of known formats
- Generating composite omit maps
- `phenix.map_value_at_point`: compute map value at a given point
- `phenix.mtz2map`: FFT-ing map coefficients
- `phenix.ncs_average`: Simple NCS averaging
- `phenix.polder`: compute omit map excluding bulk solvent
- Correlation of maps or map and model with `map_correlations`
- Cut out density from a crystallographic map with `cut_out_density`
- Cut out density from a cryo-EM or crystallographic map with `map_box`
- Segment a cryo-EM map with `segment_and_split_map`
- Compare CA positions in two models with `chain_comparison`
- Optimize an Xray or cryo-EM map with `map_sharpening`
- Combine cryo-EM maps with `combine_focused_maps`
- Density modify a cryo-EM map with `resolve_cryo_em`
- Remove test set reflections from a map using `remove_free_from_map`

- Guessing molecular centers in a map with `guess_molecular_centers`
- Density modification with `phenix.density_modification`
- Density modification with multi-crystal averaging with `phenix.multi_crystal_average`
- Find NCS in a map with `find_ncs_from_density`
- Map superposition with `phenix.superpose_maps`
- Modify map coefficients to remove density near a model
- `phenix.fem`: compute Feature Enhanced Map (FEM)
- Residual difference map analysis with `phenix.find_peaks_holes`
- Convert map into structure factors `phenix.map_to_structure_factors`

Model utilities

- Predict AlphaFold2 models with `phenix.predict_model`
- Process AlphaFold2 and other predicted models with `phenix.process_predicted_model`
- Idealize model geometry with `phenix.geometry_minimization`
- Superimpose models with `phenix.superpose_models`
- Generating NCS copies with `apply_ncs`
- PDB model: statistics, manipulations and more with `phenix.pdbtools`
- `phenix.fmodel`: structure factors from a PDB file
- `phenix.f000`: given PDB file and mean bulk-solvent density estimate $F(0,0,0)$
- Map one PDB file close to another using SG symmetry with `phenix.map_to_object`
- SCEDS - identify domains with normal mode analysis
- `structural_domains_search` - identify domains using a graph search
- Find symmetry equivalent molecular replacement solution on alternative origin with `phenix.find_alt_orig_sym_mate`
- Building biological assembly and crystallographic asymmetric unit
- Calculating Fragment Antigen-Binding (Fab) elbow angle
- Find Similar Structures in PDB using phi-psi angles and sequence
- Find Structures in PDB using SSM matching and morphing
- Find fragments in PDB matching a target fragment
- Replace model with fragments from the PDB
- Shaking a structure with simple dynamics
- Sorting ligands and waters to group with polymer chains
- Trim one model to avoid atoms in another with `trim_overlapping`

Other tools in Phenix

- Setting up GUI tests in Phenix
- Finding a program or regression test in Phenix
- Running regression tests in Phenix

- Characterizing raw data and model with LABELIT
- Running SOLVE/RESOLVE in Phenix
- Parallel validation and analysis of related structures
- phenix.reciprocal_space_arrays: compute various arrays such as Fcalc, Fmask, Fbulk, HL ...
- Generating Table 1 for publication

Useful tools outside of Phenix

- Manual model inspection and building with Coot
- Molecular graphics with PyMOL
- MolProbity - An **Active** Validation Tool

Appendix

- Hints on writing scripts with AI tools
- Generating code documentation strings with with AI tools
- Phenix html documentation generation procedures
- Phenix Windows build procedures
- Bundling base components as zip files for the Windows build
- Creating a Windows installer for PHENIX

Optimizing a map with local_aniso_sharpen (deprecated, use MapSharpening)

[local_aniso_sharpen.html](#)

Optimizing a map with local_aniso_sharpen (deprecated, use MapSharpening)

Author(s)

- local_aniso_sharpen: Tom Terwilliger

Purpose

The routine local_aniso_sharpen is a tool for optimizing a map by applying a local, anisotropic, resolution-dependent scaling factor.

How local_aniso_sharpen works:

Local anisotropic sharpening can be carried out locally, as the name suggests, or on a map as a whole.

The basis for sharpening is an analysis of the resolution-dependent fall-off of amplitudes of Fourier coefficients for the map, along with an analysis of the resolution-dependent fall-off of correlation between two half-maps, or between a map and a model-based map.

When carried out locally, a full map is divided into small boxes. The density near the edges of each box is masked so that it gradually diminishes to zero at the edges. Each small box is treated as a full map to identify its optimal sharpening. Then the optimal sharpening parameters from the small boxes are applied to the full map in a way that has no edge effects and smoothly varies from one place to another in the map.

To identify the optimal anisotropic sharpening of a map based on the information in two half-maps, two analyses are done. The first is an analysis of the resolution-dependent fall-off of rms amplitudes of Fourier coefficients representing the map. This is examined as a function of direction in reciprocal space, and is similar to the calculation normally done to apply an anisotropy correction to a map. This analysis shows the anisotropy of the map itself.

The second is an analysis of the correlation between Fourier coefficients for the two half-maps. This is also done as a function of resolution and direction in reciprocal space. This analysis shows the anisotropy of the errors in the map.

For purposes of this analysis, the optimal map is the one that has the maximal expected correlation to an idealized version of the true map. This idealized map is a map that would be obtained from a model where all the atoms are point atoms (B values of about zero).

For a map with zero error (all correlations at all resolutions and directions equal to 1), the optimal map will be one that has no anisotropy and the same resolution dependence as the idealized map. For such a map, first the anisotropy in the map is removed, then an overall resolution-dependence matching that of the idealized map is imposed by simple multiplication with a resolution_dependent scale factor.

For a map with errors, the map coefficients obtained from the previous step are modified by a local scale factor that reflects the expected signal-to-noise in that map coefficient. The scale factor for a particular map coefficient is given by $1/(1 + E^2)$, where E is the normalized expected error in that map coefficient. This

scale factor will ordinarily be anisotropic and resolution-dependent.

Examples

You can use `local_aniso_sharpen` with either two half-maps or a map and a model.

Standard run of `local_aniso_sharpen` with two half-maps:

To run `local_aniso_sharpen` with two half maps, you can say:

```
phenix.local_aniso_sharpen half_map_A.mrc half_map_B.mrc seq_file=seq.fa
```

If you wish, you can specify a nominal resolution. The sequence file is to let the program figure out the volume of the molecule that is present.

To run `local_aniso_sharpen` with a map and model, you can say:

```
phenix.local_aniso_sharpen map.mrc model.pdb resolution=3 seq_file=seq.fa
```

The resolution is again optional.

You can specify whether anisotropy or local sharpening are to be applied:

```
local_sharpen=True
anisotropic_sharpen=True
```

Output

`Local_aniso_sharpen` will provide a summary of the anisotropy of your map if you leave `anisotropic_sharpen=True`. Here are some of the values that are calculated for a test case, with annotations:

```
NOTE: All values apply after removal of overall U of:
      (0.81, 0.84, 0.78, -0.03, -0.01, 0.04)

>>> This overall U value has large and similar values for the first 3
>>> numbers, and small values for the last 3. This indicates that this
>>> map is relatively isotropic.
>>> In this output, the overall U listed above is removed from all other
>>> U values before printing them out.

>>> All values are reported in normalized units (U values). See
>>> https://www.iucr.org/\_\_data/iucr/cifdic\_html/1/cif\_core.dic/Iatom\_site\_U\_iso\_or\_equiv.html

      Estimated anisotropic fall-off of the data relative to ideal
      (Positive means amplitudes fall off more in this direction)
      ( X, Y, Z, XY, XZ, YZ)
      (0.002, 0.089, 0.037, 0.031, 0.042, -0.002)

>>> This is the fall-off of the data, after removal of the overall U values,
>>> relative to a standard of values calculated from a beta-galactosidase
>>> model without a bulk solvent correction (available from the phenix
>>> python module cctbx.development.approx_amplitude_vs_resolution)

      Anisotropy of the data
      (Positive means amplitudes fall off more in this direction)
      ( X, Y, Z, XY, XZ, YZ)
      (-0.119, 0.167, -0.133, -0.013, 0.031, 0.064)

>>> This is the anisotropy of the data, after removal of overall U values

      Anisotropy of the uncertainties
      (Positive means uncertainties decrease more in this direction)
      ( X, Y, Z, XY, XZ, YZ)
      (0.160, -0.032, -0.035, 0.031, 0.054, -0.010)
```

>> This is the anisotropy of the uncertainties in scale factors,
>> after removal of overall U values

Anisotropy of the scale factors
(Positive means scale factors increase more in this direction)
(X, Y, Z, XY, XZ, YZ)
(-0.022, -0.121, -0.088, -0.050, -0.040, -0.019)

>> This is the anisotropy of the scale factors,
>> after removal of overall U values

Possible Problems

If the half-maps are not actually independent the procedure will not work well

If the model is very poor the procedure will not work well

For model-based sharpening, if local sharpening is used, the sharpening is only applied in the region of the model

Specific limitations and problems:

Literature

Additional information

List of all available keywords

job_title = None Job title in PHENIX GUI, not used on command line
input_filesseq_file = None Sequence file. Include all copies of each sequence so that the full contents of the map can be estimated. (Fasta format or sequences separated by blank lines)
map_modelfull_map = None Input full map file
half_map = None Input half map files
model = None Input model file
outputsharpened_map_file = default Sharpened map file
namesharpened_map_file_1 = default Sharpened half map 1 file
namesharpened_map_file_2 = default Sharpened half map 2 file
name_local_resolution_map_file = default Local resolution map file
nameoverwrite = True Overwrite files with same names
file_name = None Not used
filename = None Not used
serial = None Not used
output_scale_factor = None Scale factor to be applied to output map just before writing. Normally the output map will have a mean of zero and SD of 1. This may lead to the maximum in the map being much greater than 1. You can adjust the output SD with this scale factor.
sharpeninglocal_sharpen = False Sharpen locally (alternative is global sharpening). Note: can take a long time
anisotropic_sharpen = True Use anisotropic sharpening. Can be combined with local sharpening
model_sharpen = None Model sharpening (default if model is supplied and only one map is supplied)
n_bins = None Number of bins for sharpening (default 200 overall and 20 local)
n_boxes = None Number of boxes
box_size_grid_units = None Size of core region of boxes (not including region where mask is applied, in grid units)
sharpen_all_maps = True Sharpen and write out half-maps in addition to the full map
crystal_inforesolution = None Nominal resolution of map
wrapping = None You can specify whether the map is wrapped (can map values outside bounds to inside with cell translations).
controlmultiprocessing = *multiprocessing sge lsf pbs condor pbspro slurm Choices are multiprocessing (single machine) or queuing systems Not implemented
queue_run_command = None run command for queue jobs. For example qsub. Not implemented
nproc = 1 Number of processors to use. NOTE: by default multiple processors will only be used in the map-to-model step (this is because multiprocessing requires writing out nproc sets of huge files and it can be very slow with distributed queues.). You can override this with force_nproc = True.
ignore_symmetry_conflicts = False You can ignore the symmetry information (CRYST1) from coordinate files. This may be necessary if your model has been placed in a box with box_map

for example.verbose = False Verbose outputguiGUI-specific parameter required for output directoryoutput_dir = None

- job_title = None Job title in PHENIX GUI, not used on command line
- input_filesseq_file = None Sequence file. Include all copies of each sequence so that the full contents of the map can be estimated. (Fasta format or sequences separated by blank lines)map_modelfull_map = None Input full map filehalf_map = None Input half map filesmodel = None Input model file
- seq_file = None Sequence file. Include all copies of each sequence so that the full contents of the map can be estimated. (Fasta format or sequences separated by blank lines)
- map_modelfull_map = None Input full map filehalf_map = None Input half map filesmodel = None Input model file
- full_map = None Input full map file
- half_map = None Input half map files
- model = None Input model file
- outputsharpened_map_file = default Sharpened map file namesharpened_map_file_1 = default Sharpened half map 1 file namesharpened_map_file_2 = default Sharpened half map 2 file name
local_resolution_map_file = default Local resolution map file name
overwrite = True Overwrite files with same names
file_name = None Not used
filename = None Not used
serial = None Not used
output_scale_factor = None Scale factor to be applied to output map just before writing. Normally the output map will have a mean of zero and SD of 1. This may lead to the maximum in the map being much greater than 1. You can adjust the output SD with this scale factor.
- sharpened_map_file = default Sharpened map file name
- sharpened_map_file_1 = default Sharpened half map 1 file name
- sharpened_map_file_2 = default Sharpened half map 2 file name
- local_resolution_map_file = default Local resolution map file name
- overwrite = True Overwrite files with same names
- file_name = None Not used
- filename = None Not used
- serial = None Not used
- output_scale_factor = None Scale factor to be applied to output map just before writing. Normally the output map will have a mean of zero and SD of 1. This may lead to the maximum in the map being much greater than 1. You can adjust the output SD with this scale factor.
- sharpeninglocal_sharpen = False Sharpen locally (alternative is global sharpening). Note: can take a long time
anisotropic_sharpen = True Use anisotropic sharpening. Can be combined with local sharpening
model_sharpen = None Model sharpening (default if model is supplied and only one map is supplied)
n_bins = None Number of bins for sharpening (default 200 overall and 20 local)
n_boxes = None Number of boxes
box_size_grid_units = None Size of core region of boxes (not including region where mask is applied, in grid units)
sharpen_all_maps = True Sharpen and write out half-maps in addition to the full map
- local_sharpen = False Sharpen locally (alternative is global sharpening). Note: can take a long time
- anisotropic_sharpen = True Use anisotropic sharpening. Can be combined with local sharpening
- model_sharpen = None Model sharpening (default if model is supplied and only one map is supplied)
- n_bins = None Number of bins for sharpening (default 200 overall and 20 local)

- `n_boxes` = None Number of boxes
- `box_size_grid_units` = None Size of core region of boxes (not including region where mask is applied, in grid units)
- `sharpen_all_maps` = True Sharpen and write out half-maps in addition to the full map
- `crystal_inforesolution` = None Nominal resolution of mapwrapping = None You can specify whether the map is wrapped (can map values outside bounds to inside with cell translations).
- `resolution` = None Nominal resolution of map
- `wrapping` = None You can specify whether the map is wrapped (can map values outside bounds to inside with cell translations).
- `controlmultiprocessing` = *multiprocessing sge lsf pbs condor pbspro slurm Choices are multiprocessing (single machine) or queuing systems Not implemented
- `queue_run_command` = None run command for queue jobs. For example qsub. Not implemented
- `nproc` = 1 Number of processors to use. NOTE: by default multiple processors will only be used in the map-to-model step (this is because multiprocessing requires writing out nproc sets of huge files and it can be very slow with distributed queues.). You can override this with `force_nproc` = True.
- `ignore_symmetry_conflicts` = False You can ignore the symmetry information (CRYST1) from coordinate files. This may be necessary if your model has been placed in a box with `box_map` for example.
- `verbose` = False Verbose output
- `gui` GUI-specific parameter required for output directory
- `output_dir` = None

phenix.model_idealization

model_idealization.html

Description

The program for geometry idealization of protein secondary structure.

Program reads model from .pdb file, extracts information from HELIX/SHEET records if available and replaces these helices and sheets with geometrically ideal ones while preserving initial atom positions as much as possible.

If HELIX/SHEET records are not available, it will try to find them automatically. Correct secondary structure annotation is extremely important for proper functioning of this program. Incorrect annotations most likely will lead to distorted model. Please refer to SS documentation for more details.

If input .pdb file contains unknown ligands, their description should be supplied. This description can be obtained using eLBOW (GUI version)

The model should not contain non-standard protein residues inside secondary structure elements. The presence of unknown (UNK) residues in model (outside SS elements) is not advised because appropriate geometry restraints will not be generated for them and therefore such residues may be significantly distorted.

Command line usage examples

```
% phenix.model_idealization model.pdb
```

GUI

Not available

List of all available keywords

model_file_name = None map_file_name = None User-provided map that will be used as
reference hkl_file_name = None User-provided X-ray data to generate 2mFo-DFc map that will be used as
referencedata_labels = None Labels for experimental data.r_free_flags_labels = None Labels for free
reflections.ligands_file_name = None User-provided ligand restraintsmask_and_he_map = False Mask and
histogram equalization of the input maptrim_alternative_conformations = False Leave only atoms with empty
altlocadditionally_fix_rotamer_outliers = True At the late stage if rotamer is still outlier choose another one
with minimal clash with surrounding atomsuse_ss_restraints = True Use Secondary Structure
restraintsuse_starting_model_for_final_gm = False Use supplied model for final geometry minimization.
Otherwise just use self.output_prefix = None output_pkl = False output_model_h = True
use_map_for_reference = True run_minimization_first = True run_minimization_last = True
use_hydrogens_in_minimization = False reference_map_resolution = 5 number_of_refinement_cycles = 5
cycles_to_converge = 2 Number of cycles of geometry optimization without considerable stat change before
stoppingignore_ncs = False Don't use NCS even if it is present in model.filter_input_ss = True Filter input
annotationdebug = False Output all intermediate filesverbose = False More output to lognonbonded_weight =
10000 apply_all_trans = True rama_weight = 1.0 scale_allowed = 1.0 rama_potential = *oldfield emsley
rama_selection = None Selection of part of the model for which Ramachandran restraints will be set
up.restrain_rama_outliers = True Apply restraints to Ramachandran outliersrestrain_rama_allowed = True
Apply restraints to residues in allowed region on Ramachandran plotrestrain_allowed_outliers_with_emsley =

False In case of restrain_rama_outliers=True and/or restrain_rama_allowed=True still restrain these residues with emsley. Make sense only in case of using oldfield potential.ss_idealizationenabled = True Enable secondary structure idealizationfile_name_before_regularization = None skip_empty_ss_elements = True skip SS element if there is no atoms in the structure skip_good_ss_elements = False Skip SS elements without wrong/outlier Ramachandran angles and with reasonable hydrogen bonds lengthrestrain_torsion_angles = False Restraine torsion anglesfix_rotamer_outliers = True Fix rotamer outliers after replacing SS elementssigma_on_reference_non_ss = 1 Weight on original model coordinates restraints where no ss present. Keeps loops close to initial model. (bigger number gives lighter restraints)sigma_on_reference_helix = 1 Weight on original model coordinates restraints where helix present. Bends helices a bit according to initial model. (bigger number gives lighter restraints)sigma_on_reference_sheet = 0.5 Weight on original model coordinates restraints where sheet present. Bends helices a bit according to initial model. (bigger number gives lighter restraints)sigma_on_torsion_ss = 5 Weight on torsion angles restraints where ss present. Keeps helices torsion angles close to ideal. (bigger number gives lighter restraints)sigma_on_torsion_nonss = 5 sigma_on_ramachandran = 1 sigma_on_cbeta = 2.5 n_macro = Auto n_iter = 300 nproc = 1 verbose = False oldfieldesd = 10.0 weight_scale = 1.0 dist_weight_max = 10.0 weight = None plot_cutoff = 0.027 secondary_structuress_by_chain = True Only applies if search_method = from_ca. Find secondary structure only within individual chains. Alternative is to allow H-bonds between chains. Can be much slower with ss_by_chain=False. If your model is complete use ss_by_chain=True. If your model is many fragments, use ss_by_chain=False.from_ca_conservative = False various parameters changed to make from_ca method more conservative, hopefully to closer resemble ksdssp.max_rmsd = 1 Only applies if search_method = from_ca. Maximum rmsd to consider two chains with identical sequences as the same for ss identificationuse_representative_chains = True...

- model_file_name = None
- map_file_name = None User-provided map that will be used as reference
- hkl_file_name = None User-provided X-ray data to generate 2mFo-DFc map that will be used as reference
- data_labels = None Labels for experimental data.
- r_free_flags_labels = None Labels for free reflections.
- ligands_file_name = None User-provided ligand restraints
- mask_and_he_map = False Mask and histogram equalization of the input map
- trim_alternative_conformations = False Leave only atoms with empty altloc
- additionally_fix_rotamer_outliers = True At the late stage if rotamer is still outlier choose another one with minimal clash with surrounding atoms
- use_ss_restraints = True Use Secondary Structure restraints
- use_starting_model_for_final_gm = False Use supplied model for final geometry minimization. Otherwise just use self.
- output_prefix = None
- output_pkl = False
- output_model_h = True
- use_map_for_reference = True
- run_minimization_first = True
- run_minimization_last = True

- use_hydrogens_in_minimization = False
- reference_map_resolution = 5
- number_of_refinement_cycles = 5
- cycles_to_converge = 2 Number of cycles of geometry optimization without considerable stat change before stopping
- ignore_ncs = False Don't use NCS even if it is present in model.
- filter_input_ss = True Filter input annotation
- debug = False Output all intermediate files
- verbose = False More output to log
- nonbonded_weight = 10000
- apply_all_trans = True
- rama_weight = 1.0
- scale_allowed = 1.0
- rama_potential = *oldfield emsley
- rama_selection = None Selection of part of the model for which Ramachandran restraints will be set up.
- restrain_rama_outliers = True Apply restraints to Ramachandran outliers
- restrain_rama_allowed = True Apply restraints to residues in allowed region on Ramachandran plot
- restrain_allowed_outliers_with_emsley = False In case of restrain_rama_outliers=True and/or restrain_rama_allowed=True still restrain these residues with emsley. Make sense only in case of using oldfield potential.
- ss_idealizationenabled = True Enable secondary structure idealizationfile_name_before_regularization = None skip_empty_ss_elements = True skip SS element if there is no atoms in the structureskip_good_ss_elements = False Skip SS elements without wrong/outlier Ramachandran angles and with reasonable hydrogen bonds lengthrestrain_torsion_angles = False Restraining torsion anglesfix_rotamer_outliers = True Fix rotamer outliers after replacing SS elementssigma_on_reference_non_ss = 1 Weight on original model coordinates restraints where no ss present. Keeps loops close to initial model. (bigger number gives lighter restraints)sigma_on_reference_helix = 1 Weight on original model coordinates restraints where helix present. Bends helices a bit according to initial model. (bigger number gives lighter restraints)sigma_on_reference_sheet = 0.5 Weight on original model coordinates restraints where sheet present. Bends helices a bit according to initial model. (bigger number gives lighter restraints)sigma_on_torsion_ss = 5 Weight on torsion angles restraints where ss present. Keeps helices torsion angles close to ideal. (bigger number gives lighter restraints)sigma_on_torsion_nonss = 5 sigma_on_ramachandran = 1 sigma_on_cbeta = 2.5 n_macro = Auto n_iter = 300 nproc = 1 verbose = False
- enabled = True Enable secondary structure idealization
- file_name_before_regularization = None
- skip_empty_ss_elements = True skip SS element if there is no atoms in the structure
- skip_good_ss_elements = False Skip SS elements without wrong/outlier Ramachandran angles and with reasonable hydrogen bonds length
- restrain_torsion_angles = False Restraining torsion angles

- `fix_rotamer_outliers = True` Fix rotamer outliers after replacing SS elements
- `sigma_on_reference_non_ss = 1` Weight on original model coordinates restraints where no ss present. Keeps loops close to initial model. (bigger number gives lighter restraints)
- `sigma_on_reference_helix = 1` Weight on original model coordinates restraints where helix present. Bends helices a bit according to initial model. (bigger number gives lighter restraints)
- `sigma_on_reference_sheet = 0.5` Weight on original model coordinates restraints where sheet present. Bends helices a bit according to initial model. (bigger number gives lighter restraints)
- `sigma_on_torsion_ss = 5` Weight on torsion angles restraints where ss present. Keeps helices torsion angles close to ideal. (bigger number gives lighter restraints)
- `sigma_on_torsion_nonss = 5`
- `sigma_on_ramachandran = 1`
- `sigma_on_cbeta = 2.5`
- `n_macro = Auto`
- `n_iter = 300`
- `nproc = 1`
- `verbose = False`
- `oldfieldesd = 10.0` `weight_scale = 1.0` `dist_weight_max = 10.0` `weight = None` `plot_cutoff = 0.027`
- `esd = 10.0`
- `weight_scale = 1.0`
- `dist_weight_max = 10.0`
- `weight = None`
- `plot_cutoff = 0.027`
- `secondary_structuress_by_chain = True` Only applies if `search_method = from_ca`. Find secondary structure only within individual chains. Alternative is to allow H-bonds between chains. Can be much slower with `ss_by_chain=False`. If your model is complete use `ss_by_chain=True`. If your model is many fragments, use `ss_by_chain=False`. `from_ca_conservative = False` various parameters changed to make `from_ca` method more conservative, hopefully to closer resemble `ksdssp`. `max_rmsd = 1` Only applies if `search_method = from_ca`. Maximum rmsd to consider two chains with identical sequences as the same for ss identification. `use_representative_chains = True` Only applies if `search_method = from_ca`. Use a representative of all chains with the same sequence. Alternative is to examine each chain individually. Can be much slower with `use_representative_of_chain=False` if there are many symmetry copies. Ignored unless `ss_by_chain` is `True`. `max_representative_chains = 100` Only applies if `search_method = from_ca`. Maximum number of representative chains. `enabled = False` Turn on secondary structure restraints (main switch). `protein_enabled = True` Turn on secondary structure restraints for protein. `search_method = *ksdssp from_ca cablam` Particular method to search protein secondary structure. `distance_ideal_n_o = 2.9` Target length for N-O hydrogen bond. `distance_cut_n_o = 3.5` Hydrogen bond with length exceeding this value will not be established. `remove_outliers = True` If true, h-bonds exceeding `distance_cut_n_o` length will not be established. `restrain_hbond_angles = True` helix. `serial_number = None` `helix_identifier = None` `enabled = True` Restrict this particular helix. `selection = None` `helix_type = *alpha pi 3_10 unknown` Type of helix, defaults to alpha. Only alpha, pi, and 3_10 helices are used for hydrogen-bond restraints. `sigma = 0.05` `slack = 0` `top_out = False` `angle_sigma_scale = 1` Multiply sigmas for h-bond angles by this value. Original sigmas range from 5 to 10. `angle_sigma_set = None` Use this parameter to set sigmas for h-bond angles to a particular value. `hbond_donor = None`

acceptor = None sheetenabled = True Restraining this particular sheetfirst_strand = None sheet_id = None
sigma = 0.05 slack = 0 top_out = False angle_sigma_scale = 1 Multiply sigmas for h-bond angles by this
value. Original sigmas range from 5 to 10.angle_sigma_set = None Use this parameter to set sigmas for
h-bond angles to a particular valuestrandselection = None sense = parallel antiparallel *unknown
bond_start_current = None bond_start_previous = None hbonddonor = None acceptor = None
nucleic_acidenabled = True Turn on secondary structure restraints for nucleic
acidshbond_distance_cutoff = 3.4 Hydrogen bonds with length exceeding this limit will not be
establishedscale_bonds_sigma = 1. All sigmas for h-bond length will be multiplied by this number. The
smaller number is tighter restraints.base_pairenabled = True Restraining this particular base-pairbase1 =
None Selection string selecting at least one atom in the desired residuebase2 = None Selection string
selecting at least one atom in the desired residuesaenger_class = 0 Type of
base-pairingrestrain_planarity = False Apply planarity restraint to this base-pairplanarity_sigma = 0.176
restrain_hbonds = True Restraining hydrogen bonds length for this base-pairrestrain_hb_angles = True
Restraining angles around hydrogen bonds for this base-pairrestrain_parallelity = True Apply parallelity
restraint to this base-pairparallelity_target = 0 parallelity_sigma = 0.0335 stacking_pairenabled = True
Restraining this particular base-pairbase1 = None Selection string selecting at least one atom in the desired
residuebase2 = None Selection string selecting at least one atom in the desired residueangle = 0 sigma =
0.027

- ss_by_chain = True Only applies if search_method = from_ca. Find secondary structure only within individual chains. Alternative is to allow H-bonds between chains. Can be much slower with ss_by_chain=False. If your model is complete use ss_by_chain=True. If your model is many fragments, use ss_by_chain=False.
- from_ca_conservative = False various parameters changed to make from_ca method more conservative, hopefully to closer resemble ksdssp.
- max_rmsd = 1 Only applies if search_method = from_ca. Maximum rmsd to consider two chains with identical sequences as the same for ss identification
- use_representative_chains = True Only applies if search_method = from_ca. Use a representative of all chains with the same sequence. Alternative is to examine each chain individually. Can be much slower with use_representative_of_chain=False if there are many symmetry copies. Ignored unless ss_by_chain is True.
- max_representative_chains = 100 Only applies if search_method = from_ca. Maximum number of representative chains
- enabled = False Turn on secondary structure restraints (main switch)
- proteinenabled = True Turn on secondary structure restraints for proteinsearch_method = *ksdssp from_ca cablam Particular method to search protein secondary structure.distance_ideal_n_o = 2.9 Target length for N-O hydrogen bonddistance_cut_n_o = 3.5 Hydrogen bond with length exceeding this value will not be establishedremove_outliers = True If true, h-bonds exceeding distance_cut_n_o length will not be establishedrestrain_hbond_angles = True helixserial_number = None helix_identifier = None enabled = True Restraining this particular helixselection = None helix_type = *alpha pi 3_10 unknown Type of helix, defaults to alpha. Only alpha, pi, and 3_10 helices are used for hydrogen-bond restraints.sigma = 0.05 slack = 0 top_out = False angle_sigma_scale = 1 Multiply sigmas for h-bond angles by this value. Original sigmas range from 5 to 10.angle_sigma_set = None Use this parameter to set sigmas for h-bond angles to a particular valuehbonddonor = None acceptor = None sheetenabled = True Restraining this particular sheetfirst_strand = None sheet_id = None sigma = 0.05 slack = 0 top_out = False angle_sigma_scale = 1 Multiply sigmas for h-bond angles by this value. Original sigmas range from 5 to 10.angle_sigma_set = None Use this parameter to set sigmas for h-bond angles to a particular valuestrandselection = None

sense = parallel antiparallel *unknown bond_start_current = None bond_start_previous = None
hbonddonor = None acceptor = None

- enabled = True Turn on secondary structure restraints for protein
- search_method = *ksdssp from_ca cablam Particular method to search protein secondary structure.
- distance_ideal_n_o = 2.9 Target length for N-O hydrogen bond
- distance_cut_n_o = 3.5 Hydrogen bond with length exceeding this value will not be established
- remove_outliers = True If true, h-bonds exceeding distance_cut_n_o length will not be established
- restrain_hbond_angles = True
- helixserial_number = None helix_identifier = None enabled = True Restrain this particular helixselection = None helix_type = *alpha pi 3_10 unknown Type of helix, defaults to alpha. Only alpha, pi, and 3_10 helices are used for hydrogen-bond restraints.sigma = 0.05 slack = 0 top_out = False angle_sigma_scale = 1 Multiply sigmas for h-bond angles by this value. Original sigmas range from 5 to 10.angle_sigma_set = None Use this parameter to set sigmas for h-bond angles to a particular valuehbonddonor = None acceptor = None
- serial_number = None
- helix_identifier = None
- enabled = True Restrain this particular helix
- selection = None
- helix_type = *alpha pi 3_10 unknown Type of helix, defaults to alpha. Only alpha, pi, and 3_10 helices are used for hydrogen-bond restraints.
- sigma = 0.05
- slack = 0
- top_out = False
- angle_sigma_scale = 1 Multiply sigmas for h-bond angles by this value. Original sigmas range from 5 to 10.
- angle_sigma_set = None Use this parameter to set sigmas for h-bond angles to a particular value
- hbonddonor = None acceptor = None
- donor = None
- acceptor = None
- sheetenabled = True Restrain this particular sheetfirst_strand = None sheet_id = None sigma = 0.05 slack = 0 top_out = False angle_sigma_scale = 1 Multiply sigmas for h-bond angles by this value. Original sigmas range from 5 to 10.angle_sigma_set = None Use this parameter to set sigmas for h-bond angles to a particular valuestrandselection = None sense = parallel antiparallel *unknown bond_start_current = None bond_start_previous = None hbonddonor = None acceptor = None
- enabled = True Restrain this particular sheet
- first_strand = None
- sheet_id = None
- sigma = 0.05
- slack = 0
- top_out = False

- angle_sigma_scale = 1 Multiply sigmas for h-bond angles by this value. Original sigmas range from 5 to 10.
- angle_sigma_set = None Use this parameter to set sigmas for h-bond angles to a particular value
- strandselection = None sense = parallel antiparallel *unknown bond_start_current = None bond_start_previous = None
- selection = None
- sense = parallel antiparallel *unknown
- bond_start_current = None
- bond_start_previous = None
- hbonddonor = None acceptor = None
- donor = None
- acceptor = None
- nucleic_acidenabled = True Turn on secondary structure restraints for nucleic acidshbond_distance_cutoff = 3.4 Hydrogen bonds with length exceeding this limit will not be establishedscale_bonds_sigma = 1. All sigmas for h-bond length will be multiplied by this number. The smaller number is tighter restraints.base_pairenable = True Restrict this particular base-pairbase1 = None Selection string selecting at least one atom in the desired residuebase2 = None Selection string selecting at least one atom in the desired residueaenger_class = 0 Type of base-pairingrestrain_planarity = False Apply planarity restraint to this base-pairplanarity_sigma = 0.176 restrain_hbonds = True Restrict hydrogen bonds length for this base-pairrestrain_hb_angles = True Restrict angles around hydrogen bonds for this base-pairrestrain_parallelity = True Apply parallelity restraint to this base-pairparallelity_target = 0 parallelity_sigma = 0.0335 stacking_pairenable = True Restrict this particular base-pairbase1 = None Selection string selecting at least one atom in the desired residuebase2 = None Selection string selecting at least one atom in the desired residueangle = 0 sigma = 0.027
- enabled = True Turn on secondary structure restraints for nucleic acids
- hbond_distance_cutoff = 3.4 Hydrogen bonds with length exceeding this limit will not be established
- scale_bonds_sigma = 1. All sigmas for h-bond length will be multiplied by this number. The smaller number is tighter restraints.
- base_pairenable = True Restrict this particular base-pairbase1 = None Selection string selecting at least one atom in the desired residuebase2 = None Selection string selecting at least one atom in the desired residueaenger_class = 0 Type of base-pairingrestrain_planarity = False Apply planarity restraint to this base-pairplanarity_sigma = 0.176 restrain_hbonds = True Restrict hydrogen bonds length for this base-pairrestrain_hb_angles = True Restrict angles around hydrogen bonds for this base-pairrestrain_parallelity = True Apply parallelity restraint to this base-pairparallelity_target = 0 parallelity_sigma = 0.0335
- enabled = True Restrict this particular base-pair
- base1 = None Selection string selecting at least one atom in the desired residue
- base2 = None Selection string selecting at least one atom in the desired residue
- saenger_class = 0 Type of base-pairing
- restrain_planarity = False Apply planarity restraint to this base-pair
- planarity_sigma = 0.176

- `restrain_hbonds = True` Restrain hydrogen bonds length for this base-pair
- `restrain_hb_angles = True` Restrain angles around hydrogen bonds for this base-pair
- `restrain_parallelity = True` Apply parallelity restraint to this base-pair
- `parallelity_target = 0`
- `parallelity_sigma = 0.0335`
- `stacking_pairenabled = True` Restraint this particular base-pair
`base1 = None` Selection string selecting at least one atom in the desired residue
`base2 = None` Selection string selecting at least one atom in the desired residue
`angle = 0`
`sigma = 0.027`
- `loop_idealizationenabled = True` `output_prefix = None` `minimize_whole = True` `fix_rotamers = True` Fix rotamers before minimization
`force_rama_fixes = False` If true, the procedure will pick and apply the best variant even if all of them are above thresholds to be picked straight away. Alternatively, when False, the procedure will accept failure and leave a ramachandran outlier intact.
`include_allowed = False` Try to fix residues in allowed Ramachandran region with CCD
`avoid_allowed_region = False` If true, CCD will not let residues go to allowed region on Ramachandran plot
`make_all_trans = True` If true, procedure will try to get rid of all twisted and cis-peptides making all of them trans.
`save_states = False` Save states of CCD. Generates a states file for every model. Warning! Significantly slower!
`number_of_ccd_trials = 3` How many times we are trying to fix outliers in the same chain
`variant_search_level = 2` how thoroughly variants will be explored (1-3)
`variant_number_cutoff = 50` how many first variants to take from generated
`variant_deviation_accept_level = low * med high very_high` what deviation from starting model is acceptable when screening variants. med is fine for real-space, low is for reciprocal space.
`expert_level = 2` `debug = False` Output of intermediate files
- `enabled = True`
- `output_prefix = None`
- `minimize_whole = True`
- `fix_rotamers = True` Fix rotamers before minimization
- `force_rama_fixes = False` If true, the procedure will pick and apply the best variant even if all of them are above thresholds to be picked straight away. Alternatively, when False, the procedure will accept failure and leave a ramachandran outlier intact.
- `include_allowed = False` Try to fix residues in allowed Ramachandran region with CCD
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- `number_of_ccd_trials = 3` How many times we are trying to fix outliers in the same chain
- `variant_search_level = 2` how thoroughly variants will be explored (1-3)

- variant_number_cutoff = 50 how many first variants to take from generated
- variant_deviation_accept_level = low *med high very_high what deviation from starting model is acceptable when screening variants. med is fine for real-space, low is for reciprocal space.
- expert_level = 2
- debug = False Output of intermediate files
- cablam_idealizationenabled = True nproc = 1 Parallelization is not implementedrotation_angle = 30 angle to rotate the oxygen while searching for better conformationrequire_h_bond = True require presence of (new) h-bond after rotationdo_gm = False Run geometry minimization after rotationfind_ss_after_fixes = True re-evaluate SS after fixing Cablam outliers. May be helpful to identify new or extend previous SS elementssave_intermediates = False Save all cablam rotation for particular residue in separate file
- enabled = True
- nproc = 1 Parallelization is not implemented
- rotation_angle = 30 angle to rotate the oxygen while searching for better conformation
- require_h_bond = True require presence of (new) h-bond after rotation
- do_gm = False Run geometry minimization after rotation
- find_ss_after_fixes = True re-evaluate SS after fixing Cablam outliers. May be helpful to identify new or extend previous SS elements
- save_intermediates = False Save all cablam rotation for particular residue in separate file

FFT-ing map coefficients

mtz2map.html

If you already have map coefficients in an MTZ file and just want to run an FFT to produce X-plor or CCP4 map files, use `phenix.fft` or `phenix.mtz2map` (these are synonyms for the same program). In most cases this can be run automatically by passing the name of the MTZ file. MTZ files generated by `phenix.maps`, `phenix.refine`, `SOLVE/RESOLVE`, and `REFMAC` are supported, and additional combinations of map coefficients are possible if defined by the user. By default, output files will take the base name of the MTZ file, with the appropriate map type appended if it can be guessed from the column labels, otherwise a unique integer, followed by the file type extension (`.ccp4` or `.xplor`).

Other map types

For isomorphous difference maps (Fobs-Fobs), use `phenix.fobs_minus_fobs_map` (documentation to be added). For omit maps, including composite omit maps and iterative build omit maps, use the AutoBuild wizard. You can also generate a simple simulated-annealing omit map by deleting the relevant part of the molecule and running `phenix.refine` with `simulated_annealing=True`. (This is essentially the same as what is done by AutoBuild, however.)

- For isomorphous difference maps (Fobs-Fobs), use `phenix.fobs_minus_fobs_map` (documentation to be added).
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List of all available keywords

`mtz_file` = None MTZ file containing map coefficients
`pdb_file` = None Model file around which to draw the map. If not supplied, the map will fill the unit cell.
`labels` = None Map column labels. Common examples are 2FOFCWT, PH2FOFCWT and FP, SIGFP PHIM FOMM . If left blank, all maps present in the input file will be used.
`buffer` = 5.0 Extra padding (in Angstroms) around the selected region (or unit cell)
`selection` = None Atom selection around which to draw map (plus buffer). If left blank, the entire model file will be used.
`d_min` = None High-resolution cutoff
`d_max` = None Low-resolution cutoff
`grid_resolution_factor` = 0.25 Grid spacing (multiplied by the high-resolution limit)
`gridding` = None Gridding
`scale` = *sigma volume Scaling method for map values
`include_fmodel` = False
`show_maps` = False
`output_directory` = None Output directory (defaults to current)
`prefix` = None Output file prefix (defaults to the MTZ file name base)
`job_title` = None Job title in PHENIX GUI, not used on command line
`format` = xplor *ccp4 dsn6 extension = *Auto ccp4 xplor map dsn6
`r_free_flags_remove` = False
`file_name` = None
`label` = None
`test_flag_value` = None

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- `pdb_file` = None Model file around which to draw the map. If not supplied, the map will fill the unit cell.
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- d_min = None High-resolution cutoff
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- grid_resolution_factor = 0.25 Grid spacing (multiplied by the high-resolution limit)
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- scale = *sigma volume Scaling method for map values
- include_fmodel = False
- show_maps = False
- outputdirectory = None Output directory (defaults to current)prefix = None Output file prefix (defaults to the MTZ file name base)job_title = None Job title in PHENIX GUI, not used on command lineformat = xplor *ccp4 dsn6 extension = *Auto ccp4 xplor map dsn6
- directory = None Output directory (defaults to current)
- prefix = None Output file prefix (defaults to the MTZ file name base)
- job_title = None Job title in PHENIX GUI, not used on command line
- format = xplor *ccp4 dsn6
- extension = *Auto ccp4 xplor map dsn6
- r_free_flagsremove = False file_name = None label = None test_flag_value = None
- remove = False
- file_name = None
- label = None
- test_flag_value = None

Density modification with multi-crystal averaging with `phenix.multi_crystal_average`

`multi_crystal_average.html`

Contents

- Author(s)
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- How `phenix.multi_crystal_average` works:
- Output files from `phenix.multi_crystal_average`
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- Standard run of `phenix.multi_crystal_average`:
- Run of `phenix.multi_crystal_average` with multiple domains:
- Run of `phenix.multi_crystal_average` using PDB files to define the NCS asymmetric unit:
- Possible Problems
- Specific limitations and problems:
- Literature
- Additional information
- List of all available keywords

Author(s)

- `phenix.multi_crystal_average`: Tom Terwilliger

Purpose

`phenix.multi_crystal_average` is a tool for carrying out density modification, including NCS symmetry within a crystal and electron density from multiple crystals.

Usage

How `phenix.multi_crystal_average` works:

The inputs to `phenix.multi_crystal_average` are a set of PDB files that define the NCS within each crystal and the relationships of density between crystals, structure factor amplitudes (and optional phases, FOM and HL coefficients) for each crystal, and starting electron density maps for one or more crystals.

The PDB files should each be composed of the exact same chains, placed in a different position and orientation for each NCS asymmetric unit of each crystal. You might create these PDB files by molecular replacement starting with the same search model for each crystal. You should not refine these MR solutions; they are only used to get the NCS relationships and the NCS will be more reliably found if the models for all NCS asymmetric units are identical. You can break the NCS asymmetric unit into domains and place them

independently. You can specify the domains by giving them unique chain IDs, (or you can use the routine `edit_chains.py` to do this for you, see below). A separate NCS group will be created for each domain. Additionally if your NCS asymmetric unit consists of more than one chain (A+B for example) then each chain will always be treated as a separate NCS group.

`phenix.multi_crystal_average` first uses the supplied PDB files to calculate NCS operators relating the NCS asymmetric unit in each crystal to all other NCS asymmetric units in that crystal and in other crystals. This is done by adding the unique chains in one crystal to each PDB file in turn, finding all the NCS relationships from all chains in that composite PDB file, and removing duplicate identity transformations. For example, suppose the NCS asymmetric unit is one chain (A,B,C,...). Then to relate all NCS asymmetric units to the NCS asymmetric unit of crystal 0, `phenix.multi_crystal_average` will compare all chains in the PDB file for each crystal to the unique chain in the PDB file for crystal 0, generating one NCS operator for each chain in each crystal. In this process the unique chain (in this case the NCS asymmetric unit of crystal 0) is renamed to a unique name (usually "***") and a composite PDB file is created with this chain along with all the chains in the PDB file for the crystal being considered, and `phenix.simple_ncs_from_pdb` is used to find the NCS operators. The centroids of the chains defining NCS are used as centers of the regions where the NCS operator is to be applied.

If the supplied PDB files have more than one domain or chain in each NCS asymmetric unit, then the domains or chains are grouped into separate NCS groups.

Once NCS operators have been identified, density modification is carried out sequentially on data from each crystal. During density modification for one crystal, the current electron density maps from all other crystals are used in generating target density for density modification in exactly the same way as NCS-related density is normally used when only a single crystal is available.

First the asymmetric unit of NCS is defined, in this case including the density in all NCS copies within the crystal being density modified as well as the density in all NCS copies in all other crystals. The asymmetric unit of NCS is the region over which the NCS operators apply. It is assumed to be identical for all NCS copies for all crystals, with orientation and position identified by the NCS operators. It is identified as the region over which all NCS copies have correlated density. If a mask for the protein/solvent boundary is supplied (by specifying "`use_model_mask`"), then the asymmetric unit of NCS is constrained to be within the non-solvent region of the map. Alternatively, if you request that the domains provided in your PDB files be used to define the NCS asymmetric unit (by specifying "`write_ncs_domain_pdb`") then the NCS asymmetric unit (for each NCS group) is limited to the region occupied by the corresponding chains in your PDB files.

Then a target density map is created for the crystal being density modified. For each NCS copy in this crystal, the average density for all other NCS copies in this and other crystals is used as a target.

Finally, statistical density modification is carried out using histograms of expected density, solvent flattening, and the NCS-based target density for this crystal. The process is then repeated for all other crystals. For those crystals for which no starting phases were available, one additional step is carried out in which the target density map is used by itself to calculate a starting electron density map (using RESOLVE map-based phasing).

This entire process is carried out several times, leading to electron density maps for all crystals that typically have a high level of correlation of density within all NCS copies in each crystal and between the corresponding NCS regions in different crystals.

Output files from `phenix.multi_crystal_average`

denmod_cycle_1_xl_0.mtz: Density-modified map coefficients for crystal 0, cycle 1. Crystal 0 is the first crystal specified in your pdb_list, map_coeff_list, etc.

denmod_cycle_5_xl_1.mtz: Density-modified map coefficients for crystal 1, cycle 5. These map coefficients are suitable for model-building. They also contain HL coefficients that can optionally be used in refinement. As the HL coefficients contain information from all crystals they may in some cases be useful in refinement (normally you would only use experimental HL phase information in refinement as the NCS-based information would come from your NCS restraints in refinement).

Graphical interface

The GUI is listed in the "Maps" category of the main interface. The configuration tab contains a set of tabs for the crystal forms: The label strings are automatically filled in when an MTZ file is loaded; these normally do not need to be further edited. At least one crystal form must have map coefficients, and all are required to have PDB and data files, as well as the estimated solvent content. All other options are on the second tab of the configuration panel: At the end of the run, the final density-modified and averaged map coefficients will be listed in a new tab. Selecting one of these and clicking the Coot or PyMOL buttons below will load the map coefficients and corresponding model for that crystal form.

Examples

Standard run of phenix.multi_crystal_average:

Running phenix.multi_crystal_average from the command line is easy. Usually you will want to edit a small parameter file (run_multi.eff) to contain your commands like this:

```
# run_multi.eff commands for running phenix.multi_crystal_average
# use: "phenix.multi_crystal_average run_multi.eff"
multi {
  crystal {
    pdb_file= "gene-5.pdb"
    map_coeffs = "resolve_1_offset.mtz"
    datafile = "phaser_1_offset.mtz"
    datafile_labin = "FP=F SIGFP=SIGF PHIB=PHI FOM=FOM HLA=HLA HLB=HLB HLC=HLC HLD=HLD"
    solvent_content = 0.43
  }
  crystal {
    pdb_file= "multi.pdb"
    map_coeffs = "None"
    datafile = "multi.pdb.mtz"
    datafile_labin = "FP=FP"
    solvent_content = 0.50
  }
}
```

Then you can run this with the command:

```
phenix.multi_crystal_average run_multi.eff
```

In this example we have 2 crystals. Crystal 1 has starting map coefficients in crystal_1_map_coeffs.mtz and data for FP in crystal_1_data.mtz. The contents of this crystal are represented by crystal_1.pdb. The second crystal has no starting map, has data for FP as well as PHI and HL coefficients in crystal_2_data.mtz, and the contents of this crystal are represented by crystal_2.pdb. The solvent contents of the 2 crystals are 0.43 and 0.50, and 5 overall cycles are to be done.

NOTE: Usually you will want to supply coefficients for a density-modified map (if available, and perhaps with NCS as well) for your starting map coefficients, and you will want to supply experimental phases and HL

coefficients for your datfiles.

The column label strings like "FP=FP" are optional...if you say instead "None" then phenix.multi_crystal_average will guess them for you. NOTE: You need to put quotes around the None. Also note: the order of entering pdb_list, map_coeff_list, data_file_list, solvent_content_list etc matters. They must all match.

Run of phenix.multi_crystal_average with multiple domains:

If your PDB files have more than one NCS domain within a chain, then you may want to split the chains up into sub-chains representing the individual NCS domains. This will provide a better definition of the NCS operators when the PDB files are analyzed.

You can use the jiffy "edit_chains.py" to do this. This jiffy splits your chains up into sub-chains based on the domains that you specify in "edit_chains.dat".

NOTE: edit_chains.py only works if your chains have single-letter ID's. (It simply adds another character to your chain ID's to make new ones.) If you have two-letter chain ID's, then you'll have to do this another way. To use it, type:

```
phenix.python $PHENIX/phenix/phenix/autosol/edit_chains.py file.pdb edited_file.pdb
```

The file edit_chains.dat is required and should look like:

```
A 1 321
A 322 597
A 598 750
A 751 902
A 903 1082
B 1 58
B 424 425
B 59 101
B 343 423
B 102 342
```

where the letter and residue range is your chain ID and residue range for a particular domain. You should specify these for ALL chains in your PDB files (not just the unique ones).

Run of phenix.multi_crystal_average using PDB files to define the NCS asymmetric unit:

If you specify the parameter write_ncs_domain_pdb=True, then phenix.multi_crystal_average will write out domain-specific PDB files for each domain in your model (based on its analysis of NCS, one for each NCS group). Then it will use those domain-specific PDB files to define the region over which the corresponding set of NCS operators apply. This is generally a good idea if you have multiple domains in your structure.

Possible Problems

Specific limitations and problems:

- The format for the parameters file for phenix.multi_crystal_average has changed (twice) (from a list of inputs for pdb_list etc, to several groups of inputs, one for each crystal.)

The format for the parameters file for phenix.multi_crystal_average has changed (twice) (from a list of inputs for pdb_list etc, to several groups of inputs, one for each crystal.)

- If the NCS asymmetric unit of your crystal contains more than one type of chain, `phenix.multi_crystal_average` will consider it to have more than one domain. This limitation comes from `phenix.simple_ncs_from_pdb`, which assigns one NCS group to each unique chain in the NCS asymmetric unit. If you would like `phenix.multi_crystal_average` to consider several chains as a single NCS group, then you would need to rename your chains and residues so that all the residues in a single NCS group have the same chain name and so that residue numbers are not duplicated. Normally you not need to do this, but if you want to use `phenix.multi_crystal_average` to generate phases for one crystal from another and you have more than one chain in the NCS asymmetric unit you would have to do this.

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- If your NCS asymmetric unit has more than one domain (more than one type of chain chain, or else multiple domains within a chain that have different arrangements in different NCS asymmetric units) then `phenix.multi_crystal_average` requires that you provide map coefficients for all crystals. This is because `phenix.multi_crystal_average` cannot use the PDB files you provide to generate the NCS asymmetric unit directly at this point (i.e., it cannot use `pdb_domain` in `RESOLVE`.) Therefore if you don't provide map coefficients for one crystal then it does not have a way to individually identify the region occupied by each domain in the NCS asymmetric unit for that crystal. This isn't a problem if there are not multiple domains or types of chains in the NCS asymmetric unit because the automatic method for generation of the NCS asymmetric unit can be used.

If your NCS asymmetric unit has more than one domain (more than one type of chain chain, or else multiple domains within a chain that have different arrangements in different NCS asymmetric units) then `phenix.multi_crystal_average` requires that you provide map coefficients for all crystals. This is because `phenix.multi_crystal_average` cannot use the PDB files you provide to generate the NCS asymmetric unit directly at this point (i.e., it cannot use `pdb_domain` in `RESOLVE`.) Therefore if you don't provide map coefficients for one crystal then it does not have a way to individually identify the region occupied by each domain in the NCS asymmetric unit for that crystal. This isn't a problem if there are not multiple domains or types of chains in the NCS asymmetric unit because the automatic method for generation of the NCS asymmetric unit can be used.

- Normally you should supply PDB files defining the NCS in your crystals in which all the chains have identical sequences and conformations within each NCS copy. This is not absolutely required, however. If your PDB file contains chains that are not identical then NCS will be estimated from the chains you provide. It may be necessary to set the parameter:
`simple_ncs_from_pdb.maximize_size_of_groups=True` to get this to work if the chains have insertions, deletions, or sequence differences.

Normally you should supply PDB files defining the NCS in your crystals in which all the chains have identical sequences and conformations within each NCS copy. This is not absolutely required, however. If your PDB file contains chains that are not identical then NCS will be estimated from the chains you provide. It may be necessary to set the parameter:

```
simple_ncs_from_pdb.maximize_size_of_groups=True
```

to get this to work if the chains have insertions, deletions, or sequence differences.

- The size of the asymmetric unit in the SOLVE/RESOLVE portion of phenix.multi_crystal_average is limited by the memory in your computer and the binaries used. The Wizard is supplied with regular-size ("", size=6), giant ("_giant", size=12), huge ("_huge", size=18) and extra_huge ("_extra_huge", size=36). Larger-size versions can be obtained on request.

The size of the asymmetric unit in the SOLVE/RESOLVE portion of phenix.multi_crystal_average is limited by the memory in your computer and the binaries used. The Wizard is supplied with regular-size ("", size=6), giant ("_giant", size=12), huge ("_huge", size=18) and extra_huge ("_extra_huge", size=36). Larger-size versions can be obtained on request.

Literature

Rapid automatic NCS identification using heavy-atom substructures. T.C. Terwilliger. Acta Crystallogr D Biol Crystallogr 58, 2213-5 (2002).

- Rapid automatic NCS identification using heavy-atom substructures. T.C. Terwilliger. Acta Crystallogr D Biol Crystallogr 58, 2213-5 (2002).

Statistical density modification with non-crystallographic symmetry. T.C. Terwilliger. Acta Crystallogr D Biol Crystallogr 58, 2082-6 (2002).

- Statistical density modification with non-crystallographic symmetry. T.C. Terwilliger. Acta Crystallogr D Biol Crystallogr 58, 2082-6 (2002).

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- Map-likelihood phasing. T.C. Terwilliger. Acta Crystallogr D Biol Crystallogr 57, 1763-75 (2001).

Additional information

List of all available keywords

job_title = None Job title in PHENIX GUI, not used on command line
 crystal_ncs_domain_file = None PDB file for identifying region occupied by ncs copies. If specified, region defined by this file is used for ncs domains.
 pdb_file = None PDB files, one for each crystal (One pdb_file=xxx.pdb per crystal) These should be in the same order as datafiles and map files. They are used to identify the NCS within each crystal and between crystals. You should create these by placing the unique set of atoms (the NCS asymmetric unit) in each NCS asymmetric unit of each unit cell. Normally you would do this by carrying out molecular replacement on each crystal with the same search model.
 map_coeffs = None Mtz files with map coefficients. (One map_coeffs=xxx.mtz per crystal). At least one crystal must have map coefficients. Use "None" with quotes for any crystals that do not have starting maps. NOTE: If you have multiple NCS groups then you need map coefficients for all crystals.
 map_coeffs_labin = None Optional labin lines for mtz files with map coefficients. (One map_coeffs_labin=my_labin per crystal, or none at all) They look like map_coeffs_labin="FP=FP PHIB=PHIM FOM=FOMM" Put each set of labin values inside single quotes, and the whole list inside double quotes. You can leave out a labin statement for a file by putting "None" and the routine will guess the column labels.
 datafile = None Mtz files with structure factors and optional phases and FOM and optional HL coefficients. One datafile for each crystal to be included (One

datafile=xxx.mtz for each crystal). datafile_labin = None Optional labin line for mtz file (In same order as mtz file). It can contain FP SIGFP [PHIB FOM] [HLA HLB HLC HLD]. It looks like this: datafile_labin='FP=FP SIGFP=SIGFP PHIB=PHIM FOM=FOMM' You can leave out a labin statement for a file by putting in "None" and the routine will guess the column labels NOTE: If you supply HL coefficients they will be used in phase recombination. If you supply PHIB or PHIB and FOM and not HL coefficients, then HL coefficients will be derived from your PHIB and FOM and used in phase recombination. solvent_content = None Solvent content (0 to 1, typically 0.5) for each crystal (one solvent_content=xxx for each crystal). ha_file = None Optional file, normally containing heavy atom sites or other coordinates that are present in this structure, but not in the others. If supplied, the density near (within 2 Å) of atoms in this file will not be transferred to other crystals. perfect_map_coeffs = None Optional mtz files with perfect map coefficients for comparison. perfect_map_coeffs_labin = None Optional labin lines for mtz files with perfect map coefficients. For comparison and checking only. Not normally used. update_map = True Update the map for this crystal on each cycle. Set to False if this crystal represents a high-resolution model or is otherwise expected to be already as good as it can be. averaging_cycles = 5 Number of cycles of density modification resolution = None high-resolution limit for map calculation Default is use all data fill = True Fill in all missing reflections to resolution res_fill.res_fill = None Resolution for filling in missing data (default = highest resolution of any datafile) Default is use all data temp_dir = "temp_dir" Optional temporary work directory output_dir = "" Output directory where files are to be written write_target_map_and_sigma = False Write target map and sigma use_model_mask = False You can use the ncs_domain_pdb PDB files you input to define the solvent boundary if you wish. These will partially define the NCS asymmetric unit (by limiting it to the non-solvent region) but the exact NCS asymmetric unit will always be defined automatically (by the overlap of NCS-related density). Note that this is different than the command write_ncs_domain_pdb which defines individual regi...

- job_title = None Job title in PHENIX GUI, not used on command line

- crystal_ncs_domain_file = None PDB file for identifying region occupied by ncs copies. If specified, region defined by this file is used for ncs domains. pdb_file = None PDB files, one for each crystal (One pdb_file=xxx.pdb per crystal) These should be in the same order as datafiles and map files. They are used to identify the NCS within each crystal and between crystals. You should create these by placing the unique set of atoms (the NCS asymmetric unit) in each NCS asymmetric unit of each unit cell. Normally you would do this by carrying out molecular replacement on each crystal with the same search model. map_coeffs = None Mtz files with map coefficients. (One map_coeffs=xxx.mtz per crystal). At least one crystal must have map coefficients. Use "None" with quotes for any crystals that do not have starting maps. NOTE: If you have multiple NCS groups then you need map coefficients for all crystals. map_coeffs_labin = None Optional labin lines for mtz files with map coefficients. (One map_coeffs_labin=my_labin per crystal, or none at all) They look like map_coeffs_labin=" 'FP=FP PHIB=PHIM FOM=FOMM'" Put each set of labin values inside single quotes, and the whole list inside double quotes. You can leave out a labin statement for a file by putting in "None" and the routine will guess the column labels datafile = None Mtz files with structure factors and optional phases and FOM and optional HL coefficients. One datafile for each crystal to be included (One datafile=xxx.mtz for each crystal). datafile_labin = None Optional labin line for mtz file (In same order as mtz file). It can contain FP SIGFP [PHIB FOM] [HLA HLB HLC HLD]. It looks like this: datafile_labin='FP=FP SIGFP=SIGFP PHIB=PHIM FOM=FOMM' You can leave out a labin statement for a file by putting in "None" and the routine will guess the column labels NOTE: If you supply HL coefficients they will be used in phase recombination. If you supply PHIB or PHIB and FOM and not HL coefficients, then HL coefficients will be derived from your PHIB and FOM and used in phase recombination. solvent_content = None Solvent content (0 to 1, typically 0.5) for each crystal (one solvent_content=xxx for each crystal). ha_file = None Optional file, normally containing heavy atom sites or other coordinates that are present in this structure, but not in the others. If supplied, the density near (within 2 Å) of atoms in

this file will not be transferred to other crystals. perfect_map_coeffs = None Optional mtz files with perfect map coefficients for comparison. perfect_map_coeffs_labin = None Optional labin lines for mtz files with perfect map coefficients. For comparison and checking only. Not normally used. update_map = True Update the map for this crystal on each cycle. Set to False if this crystal represents a high-resolution model or is otherwise expected to be already as good as it can be.

- ncs_domain_file = None PDB file for identifying region occupied by ncs copies. If specified, region defined by this file is used for ncs domains.

- pdb_file = None PDB files, one for each crystal (One pdb_file=xxx.pdb per crystal) These should be in the same order as datafiles and map files. They are used to identify the NCS within each crystal and between crystals. You should create these by placing the unique set of atoms (the NCS asymmetric unit) in each NCS asymmetric unit of each unit cell. Normally you would do this by carrying out molecular replacement on each crystal with the same search model.

- map_coeffs = None Mtz files with map coefficients. (One map_coeffs=xxx.mtz per crystal). At least one crystal must have map coefficients. Use "None" with quotes for any crystals that do not have starting maps. NOTE: If you have multiple NCS groups then you need map coefficients for all crystals.

- map_coeffs_labin = None Optional labin lines for mtz files with map coefficients. (One map_coeffs_labin=my_labin per crystal, or none at all) They look like map_coeffs_labin="FP=FP PHIB=PHIM FOM=FOMM" Put each set of labin values inside single quotes, and the whole list inside double quotes. You can leave out a labin statement for a file by putting in "None" and the routine will guess the column labels

- datafile = None Mtz files with structure factors and optional phases and FOM and optional HL coefficients. One datafile for each crystal to be included (One datafile=xxx.mtz for each crystal).

- datafile_labin = None Optional labin line for mtz file (In same order as mtz file). It can contain FP SIGFP [PHIB FOM] [HLA HLB HLC HLD]. It looks like this: datafile_labin='FP=FP SIGFP=SIGFP PHIB=PHIM FOM=FOMM' You can leave out a labin statement for a file by putting in "None" and the routine will guess the column labels NOTE: If you supply HL coefficients they will be used in phase recombination. If you supply PHIB or PHIB and FOM and not HL coefficients, then HL coefficients will be derived from your PHIB and FOM and used in phase recombination.

- solvent_content = None Solvent content (0 to 1, typically 0.5) for each crystal (one solvent_content=xxx for each crystal).

- ha_file = None Optional file, normally containing heavy atom sites or other coordinates that are present in this structure, but not in the others. If supplied, the density near (within 2 Å) of atoms in this file will not be transferred to other crystals.

- perfect_map_coeffs = None Optional mtz files with perfect map coefficients for comparison.

- perfect_map_coeffs_labin = None Optional labin lines for mtz files with perfect map coefficients. For comparison and checking only. Not normally used.

- update_map = True Update the map for this crystal on each cycle. Set to False if this crystal represents a high-resolution model or is otherwise expected to be already as good as it can be.

- averagingcycles = 5 Number of cycles of density modification resolution = None high-resolution limit for map calculation Default is use all datafill = True Fill in all missing reflections to resolution res_fill.res_fill = None Resolution for filling in missing data (default = highest resolution of any datafile) Default is use all datatemp_dir = "temp_dir" Optional temporary work directoryoutput_dir = "" Output directory where files are to be writtenwrite_target_map_and_sigma = False Write target map and sigmause_model_mask = False You can use the ncs_domain_pdb PDB files you input to define the solvent boundary if you wish. These will partially define the NCS asymmetric unit (by limiting it to the non-solvent region) but the exact

NCS asymmetric unit will always be defined automatically (by the overlap of NCS-related density). Note that this is different than the command `write_ncs_domain_pdb` which defines individual regions where NCS applies for each domain. `sharpen = False` You can sharpen the maps or not in the density-modification process. (They are unsharpened at the end of the process if so). Not normally used, as an anisotropy correction with sharpening is normally applied to all the data. `equal_ncs_weight = False` You can fix the NCS weighting to equally weight all copies. `weight_ncs = None` You can set the weighting on NCS symmetry (and cross-crystal averaging) `write_ncs_domain_pdb = False` You can use the input PDB files to define NCS boundaries. The atoms in the PDB files will be grouped into domains during the analysis of NCS and written out to domain-specific PDB files. (If there is only one domain or NCS group then there will be only one domain-specific PDB file and it will be the same as the starting PDB file.) Then the domain-specific PDB files will be used to define the regions over which the corresponding NCS operators apply. Note that this is different than the command `use_model_mask` which only defines the overall solvent boundary with your model. `mask_cycles = 1` Number of mask cycles in each cycle of density modification `map_phasing_ab = False` Use map phasing (A,B instead of r,phi) `use_FP_PHIM = False` Use FP and PHIM as labels for all intermediate map coeffs. Used to get around problem with labels when they vary. `no_recombination = False` Do not recombine with original data. Use just map phases and amplitudes. `output_filetarget_output_format = *None` `pdb` `mmCIF` Desired output format (if possible). Choices are `None` (try to use input format), `pdb`, `mmCIF`. If output model does not fit in `pdb` format, `mmCIF` will be used. Default is `pdb`.

- `cycles = 5` Number of cycles of density modification
- `resolution = None` high-resolution limit for map calculation Default is use all data
- `fill = True` Fill in all missing reflections to resolution `res_fill`.
- `res_fill = None` Resolution for filling in missing data (default = highest resolution of any datafile) Default is use all data
- `temp_dir = "temp_dir"` Optional temporary work directory
- `output_dir = ""` Output directory where files are to be written
- `write_target_map_and_sigma = False` Write target map and sigma
- `use_model_mask = False` You can use the `ncs_domain_pdb` PDB files you input to define the solvent boundary if you wish. These will partially define the NCS asymmetric unit (by limiting it to the non-solvent region) but the exact NCS asymmetric unit will always be defined automatically (by the overlap of NCS-related density). Note that this is different than the command `write_ncs_domain_pdb` which defines individual regions where NCS applies for each domain.
- `sharpen = False` You can sharpen the maps or not in the density-modification process. (They are unsharpened at the end of the process if so). Not normally used, as an anisotropy correction with sharpening is normally applied to all the data.
- `equal_ncs_weight = False` You can fix the NCS weighting to equally weight all copies.
- `weight_ncs = None` You can set the weighting on NCS symmetry (and cross-crystal averaging)
- `write_ncs_domain_pdb = False` You can use the input PDB files to define NCS boundaries. The atoms in the PDB files will be grouped into domains during the analysis of NCS and written out to domain-specific PDB files. (If there is only one domain or NCS group then there will be only one domain-specific PDB file and it will be the same as the starting PDB file.) Then the domain-specific PDB files will be used to define the regions over which the corresponding NCS operators apply. Note that this is different than the command `use_model_mask` which only defines the overall solvent boundary with your model.

- `mask_cycles = 1` Number of mask cycles in each cycle of density modification
- `map_phasing_ab = False` Use map phasing (A,B instead of r,phi)
- `use_FP_PHIM = False` Use FP and PHIM as labels for all intermediate map coeffs . Used to get around problem with labels when they vary.
- `no_recombination = False` Do not recombine with original data. Use just map phases and amplitudes.
- `output_filetarget_output_format = *None` pdb mmCIF Desired output format (if possible). Choices are None (try to use input format), pdb, mmCIF. If output model does not fit in pdb format, mmCIF will be used. Default is pdb.
- `target_output_format = *None` pdb mmCIF Desired output format (if possible). Choices are None (try to use input format), pdb, mmCIF. If output model does not fit in pdb format, mmCIF will be used. Default is pdb.
- `anisoremove_aniso = True` Remove anisotropy from data files before use Note: map files are assumed to be already corrected
`b_iso = None` Target overall B value for anisotropy correction. Ignored if `remove_aniso = False`. If None, default is minimum of (`max_b_iso`, lowest B of datasets, `target_b_ratio*resolution`)
`max_b_iso = 40`. Default maximum overall B value for anisotropy correction. Ignored if `remove_aniso = False`. Ignored if `b_iso` is set. If used, default is minimum of (`max_b_iso`, lowest B of datasets, `target_b_ratio*resolution`)
`target_b_ratio = 10`. Default ratio of target B value to resolution for anisotropy correction. Ignored if `remove_aniso = False`. Ignored if `b_iso` is set. If used, default is minimum of (`max_b_iso`, lowest B of datasets, `target_b_ratio*resolution`)
- `remove_aniso = True` Remove anisotropy from data files before use Note: map files are assumed to be already corrected
- `b_iso = None` Target overall B value for anisotropy correction. Ignored if `remove_aniso = False`. If None, default is minimum of (`max_b_iso`, lowest B of datasets, `target_b_ratio*resolution`)
- `max_b_iso = 40`. Default maximum overall B value for anisotropy correction. Ignored if `remove_aniso = False`. Ignored if `b_iso` is set. If used, default is minimum of (`max_b_iso`, lowest B of datasets, `target_b_ratio*resolution`)
- `target_b_ratio = 10`. Default ratio of target B value to resolution for anisotropy correction. Ignored if `remove_aniso = False`. Ignored if `b_iso` is set. If used, default is minimum of (`max_b_iso`, lowest B of datasets, `target_b_ratio*resolution`)
- `controlverbose = True` verbose output
`debug = False` debugging output
`raise_sorry = False` Raise sorry if problems
`sn_xyz = None` gridding for maps
`coarse_grid = False` You can set `coarse_grid` in `resolve`
`resolve_size = 12` Size for solve/resolve ('giant', 'huge', 'extra_huge' or a number where 12=giant, 18=huge)
`resolve_command_list = None` Commands for resolve. One per line in the form: keyword value value can be optional Examples: `coarse_grid resolution 200 2.0 hklIn test.mtz` NOTE: for command-line usage you need to enclose the whole set of commands in double quotes (") and each individual command in single quotes (') like this: `resolve_command_list="no_build 'b_overall 23' "`
`dry_run = False` Just read in and check parameter names
`base_gui_dir = None` GUI parameter only
- `verbose = True` verbose output
- `debug = False` debugging output
- `raise_sorry = False` Raise sorry if problems
- `sn_xyz = None` gridding for maps
- `coarse_grid = False` You can set `coarse_grid` in `resolve`

- `size = 12` Size for solve/resolve ("", "giant", "huge", "extra_huge", or a number where 12=giant 18=huge)
- `resolve_command_list = None` Commands for resolve. One per line in the form: keyword value value can be optional Examples: `coarse_grid resolution 200 2.0 hklIn test.mtz` NOTE: for command-line usage you need to enclose the whole set of commands in double quotes (") and each individual command in single quotes (') like this: `resolve_command_list="no_build 'b_overall 23' "`
- `dry_run = False` Just read in and check parameter names
- `base_gui_dir = None` GUI parameter only
- `simple_ncs_from_pdbpdb_in = None` Input PDB file to be used to identify `ncstemp_dir = ""` temporary directory (`ncs_domain_pdb` will be written there)
`ncs_domain_pdb_stem = None` NCS domains will be written to `ncs_domain_pdb_stem+"group_"+nn`
`write_ncs_domain_pdb = False` You can write out PDB files representing NCS domains for density modification if you want
`write_ncs_phil = False` Write out phil parameters for the NCS groups to file in working directory
`show_summary = False` Show NCS information summary
`write_spec_files = False` Write .ncs_spec and .resolve representation of NCS groups
`ncs_search` Set of parameters for NCS search procedure. Some of them also used for filtering user-supplied `ncs_group`.
`enabled = False` Enable NCS restraints or constraints in refinement (in some cases may be switched on inside refinement program).
`exclude_selection = "element H or element D or water"` Atoms selected by this selection will be excluded from the model before any NCS search and/or filtering procedures. There is no way atoms defined by this selection will be in
`ncs.chain_similarity_threshold = 0.85` Threshold for sequence similarity between matching chains. A smaller value may cause more chains to be grouped together and can lower the number of common residues
`chain_max_rmsd = 2` limit of rms difference between chains to be considered as copies
`residue_match_radius = 4.0` Maximum allowed distance difference between pairs of matching atoms of two residues
`try_shortcuts = False` Try very quick check to speed up the search when chains are identical. If failed, regular search will be performed automatically.
`minimum_number_of_atoms_in_copy = 3` Do not create ncs groups where master and copies would contain less than specified amount of atoms
`validate_user_supplied_groups = True` Enable validation of user-supplied `ncs_group`. Need to exercise a lot of caution turning this off. This option is for developers only.
`ncs_group` The definition of one NCS group. Note, that almost always in refinement programs they will be checked and filtered if needed.
`reference = None` Residue selection string for the complete master NCS copy
`selection = None` Residue selection string for each NCS copy location in ASU
- `pdb_in = None` Input PDB file to be used to identify ncs
- `temp_dir = ""` temporary directory (`ncs_domain_pdb` will be written there)
- `ncs_domain_pdb_stem = None` NCS domains will be written to `ncs_domain_pdb_stem+"group_"+nn`
- `write_ncs_domain_pdb = False` You can write out PDB files representing NCS domains for density modification if you want
- `write_ncs_phil = False` Write out phil parameters for the NCS groups to file in working directory
- `show_summary = False` Show NCS information summary
- `write_spec_files = False` Write .ncs_spec and .resolve representation of NCS groups
- `ncs_search` Set of parameters for NCS search procedure. Some of them also used for filtering user-supplied `ncs_group`.
`enabled = False` Enable NCS restraints or constraints in refinement (in some cases may be switched on inside refinement program).
`exclude_selection = "element H or element D or water"` Atoms selected by this selection will be excluded from the model before any NCS search and/or filtering procedures. There is no way atoms defined by this selection will be in
`ncs.chain_similarity_threshold = 0.85` Threshold for sequence similarity between matching chains. A

smaller value may cause more chains to be grouped together and can lower the number of common residues
`chain_max_rmsd = 2`. limit of rms difference between chains to be considered as copies
`residue_match_radius = 4.0` Maximum allowed distance difference between pairs of matching atoms of two residues
`try_shortcuts = False` Try very quick check to speed up the search when chains are identical. If failed, regular search will be performed automatically.
`minimum_number_of_atoms_in_copy = 3` Do not create ncs groups where master and copies would contain less than specified amount of atoms
`validate_user_supplied_groups = True` Enable validation of user-supplied ncs_group. Need to exercise a lot of caution turning this off. This option is for developers only.

- `enabled = False` Enable NCS restraints or constraints in refinement (in some cases may be switched on inside refinement program).

- `exclude_selection = "element H or element D or water"` Atoms selected by this selection will be excluded from the model before any NCS search and/or filtering procedures. There is no way atoms defined by this selection will be in NCS.

- `chain_similarity_threshold = 0.85` Threshold for sequence similarity between matching chains. A smaller value may cause more chains to be grouped together and can lower the number of common residues

- `chain_max_rmsd = 2`. limit of rms difference between chains to be considered as copies

- `residue_match_radius = 4.0` Maximum allowed distance difference between pairs of matching atoms of two residues

- `try_shortcuts = False` Try very quick check to speed up the search when chains are identical. If failed, regular search will be performed automatically.

- `minimum_number_of_atoms_in_copy = 3` Do not create ncs groups where master and copies would contain less than specified amount of atoms

- `validate_user_supplied_groups = True` Enable validation of user-supplied ncs_group. Need to exercise a lot of caution turning this off. This option is for developers only.

- `ncs_group` The definition of one NCS group. Note, that almost always in refinement programs they will be checked and filtered if needed.
`reference = None` Residue selection string for the complete master NCS copy
`selection = None` Residue selection string for each NCS copy location in ASU

- `reference = None` Residue selection string for the complete master NCS copy

- `selection = None` Residue selection string for each NCS copy location in ASU

Building biological assembly and crystallographic asymmetric unit from a PDB file

multimer_reconstruction.html

Introduction

In PDB files, not all the atoms locations of a biological assembly and the crystallographic asymmetric unit are given. The transformations information, rotational and translational (for both crystallographic and non-crystallographic), needed to reconstruct the multimer, is given in REMARK 350 and MTRIX line of the pdb file.

Phenix provide a commands to create a pdb file of the reconstructed a multimer for both the biological assembly and the crystallographic asymmetric unit.

The bioassembly_reconstruction command function applies all the transformations in the the pdb file REMARK 350 BIOMT, to each of the chains, produces new chains with new coordinates and adds them to the current model. The crystall_asym_unit_reconstruction does the same, using the MTRIX data.

Command-line use

Biological Assembly

Usage: phenix.pdb.biomt_reconstruction [pdf_file] [pdf_file] [options]

Options:
--help, -h

Example:

```
>>> phenix.pdb.biomt_reconstruction 'input_file_name.pdb'
bio-assembly reconstructed pdb file was added to your current directory
/path/biological_assembly_input_input_file_name.pdb
```

```
>>> phenix.pdb.biomt_reconstruction 'input_file_name.pdb' 'output_file_name.pdb'
bio-assembly reconstructed pdb file was added to your current directory
/path/output_file_name.pdb
```

Crystallographic Asymmetric Unit

Usage: phenix.pdb.mtrix_reconstruction [pdf_file] [pdf_file] [options]

Options:
--help, -h

Example:

```
>>> phenix.pdb.mtrix_reconstruction 'input_file_name.pdb'
bio-assembly reconstructed pdb file was added to your current directory
/path/crystall_asymmetric_unit_input_file_name.pdb
```

```
>>> phenix.pdb.mtrix_reconstruction 'input_file_name.pdb' 'output_file_name.pdb'
bio-assembly reconstructed pdb file was added to your current directory
/path/output_file_name.pdb
```

Output

For both commands the output is a pdb file in the current working directory

Calculating a simple NCS averaged map

[ncs_average.html](#)

If you already have map coefficients in an MTZ file and have identified your NCS with `phenix.find_ncs` and just want to calculate the average density in all your NCS copies, you can use `phenix.ncs_average`. In most cases this can be run automatically by passing the name of the MTZ file and the `ncs_spec` file with NCS information.

The output of `phenix.ncs_average` is a CCP4-style map file.

These simple NCS averaged maps can be accurate maps for cases with high NCS symmetry and they are useful for evaluating what an NCS density modified map should be expected to look like.

If you want to see the region that is being averaged, you can run `phenix.ncs_average` like this (notice the double and single quotes, both are necessary):

```
phenix.ncs_average ncs_map_coeffs.mtz ncs.ncs_spec resolve_command_list=" 'ncs_mask_file  
ncs_mask.ccp4' "
```

If you want to specify the region to be averaged, you can do this by supplying a PDB file and the mask will be around atoms in that file. Note: the mask you supply is an overall mask around ALL the areas to be averaged, not just a single asymmetric unit of NCS.

```
phenix.ncs_average ncs_map_coeffs.mtz ncs.ncs_spec resolve_command_list=" 'ncs_mask_file  
ncs_mask.ccp4' 'model model_defining_mask.pdb' 'use_model_mask' 'rad_mask 5' "
```

List of all available keywords

`input_filesmtz_in` = None Input MTZ file with map coefficients
`map_coeff_labels` = None If map coefficients cannot be identified automatically from your MTZ file, you can specify the label or labels for them. (Please separate labels with blank space, MTZ columns grouped together separated by commas with no blanks.) You can specify: `map_coeff_labels` (e.g., FWT,PHIFWT) amplitudes and phases (e.g., FP,SIGFP PHIB) or amplitudes, phases, weights (e.g., FP,SIGFP PHIB FOM)
`map_file` = None Optional input CCP4-style map file. No `mtz_in` is needed if a map file is supplied
`ncs_in` = None File with NCS operators to be applied to `ncs_in`. This can be a `.ncs_spec` file from `find_ncs`, `simple_ncs_from_pdb` or `find_ncs_from_density`. It can also be NCS formatted for `resolve`, including rotations, translations, and centers. Note: you can alternatively supply a PDB file with `pdb_in`
`pdb_in` = None Optional PDB file containing NCS to apply
`output_filesccp4_map_out` = `ncs_average.ccp4` Output CCP4-style map file with NCS-averaged density
`ncs_mask_file` = `ncs_mask.ccp4` Output CCP4-style map file with NCS-mask
`map_coeffs_out` = `ncs_average.mtz` Output MTZ file with map coefficients representing NCS-averaged density. Note: only applicable if input map is represented as map coefficients information with `mtz_in`. not applicable if `map_in` is used.
`log` = `ncs_average.log` Output log file
`params_out` = `ncs_average_params.eff` Parameters file to rerun `ncs_averaging`
`directories`
`temp_dir` = "temp_dir" Optional temporary work directory
`output_dir` = "" Output directory where files are to be written
`crystal_info`
`high_resolution` = None high-resolution limit for map calculations
`solvent_content` = 0.5 solvent fraction
`wang_radius` = 5 Smoothing radius for identification of solvent boundary
`use_pdb_in_as_mask` = False Use `pdb_in` file as mask for NCS (if supplied)
`rad_mask` = 4 mask radius around atoms in `pdb_in`
`high_resolution_scale` = 0.95 Resolution of output map coeffs will be `high_resolution_scale`* resolution of input map coeffs if `high_resolution` is None
`fraction_ncs_min` = 0.05 If the fraction of the asymmetric unit occupied by the NCS related molecules is less than `fraction_ncs_min` it is ignored
`refine_ncs_operators` = False Refine NCS operators
`control`
`verbose` = False Verbose output
`raise_sorry` = False Raise sorry if problems
`debug` = False Debugging output
`dry_run` = False Just read in and check parameter

namescoarse_grid = False Use coarse grid (saves on memory)resolve_command_list = None You can supply any resolve command here NOTE: for command-line usage you need to enclose the whole set of commands in double quotes (") and each individual command in single quotes (') like this:
 resolve_command_list="'no_build' 'b_overall 23' ";

- input_filesmtz_in = None Input MTZ file with map coefficientsmap_coeff_labels = None If map coefficients cannot be identified automatically from your MTZ file, you can specify the label or labels for them. (Please separate labels with blank space, MTZ columns grouped together separated by commas with no blanks.) You can specify: map_coeff_labels (e.g., FWT,PHIFWT) amplitudes and phases (e.g., FP,SIGFP PHIB) or amplitudes, phases, weights (e.g., FP,SIGFP PHIB FOM)map_file = None Optional input CCP4-style map file. No mtz_in is needed if a map file is suppliedncs_in = None File with NCS operators to be applied to ncs_in This can be a .ncs_spec file from find_ncs, simple_ncs_from_pdb or find_ncs_from_density It can also be NCS formatted for resolve, including rotations, translations, and centers Note: you can alternatively supply a PDB file with pdb_inpdb_in = None Optional PDB file containing NCS to apply
- mtz_in = None Input MTZ file with map coefficients
- map_coeff_labels = None If map coefficients cannot be identified automatically from your MTZ file, you can specify the label or labels for them. (Please separate labels with blank space, MTZ columns grouped together separated by commas with no blanks.) You can specify: map_coeff_labels (e.g., FWT,PHIFWT) amplitudes and phases (e.g., FP,SIGFP PHIB) or amplitudes, phases, weights (e.g., FP,SIGFP PHIB FOM)
- map_file = None Optional input CCP4-style map file. No mtz_in is needed if a map file is supplied
- ncs_in = None File with NCS operators to be applied to ncs_in This can be a .ncs_spec file from find_ncs, simple_ncs_from_pdb or find_ncs_from_density It can also be NCS formatted for resolve, including rotations, translations, and centers Note: you can alternatively supply a PDB file with pdb_in
- pdb_in = None Optional PDB file containing NCS to apply
- output_filesccp4_map_out = ncs_average.ccp4 Output CCP4-style map file with NCS-averaged densityncs_mask_file = ncs_mask.ccp4 Output CCP4-style map file with NCS-maskmap_coeffs_out = ncs_average.mtz Output MTZ file with map coefficients representing NCS-averaged density. Note: only applicable if input map is represented as map coefficients information with mtz_in. not applicable if map_in is used.log = ncs_average.log Output log fileparams_out = ncs_average_params.eff Parameters file to rerun ncs averaging
- ccp4_map_out = ncs_average.ccp4 Output CCP4-style map file with NCS-averaged density
- ncs_mask_file = ncs_mask.ccp4 Output CCP4-style map file with NCS-mask
- map_coeffs_out = ncs_average.mtz Output MTZ file with map coefficients representing NCS-averaged density. Note: only applicable if input map is represented as map coefficients information with mtz_in. not applicable if map_in is used.
- log = ncs_average.log Output log file
- params_out = ncs_average_params.eff Parameters file to rerun ncs averaging
- directoriestemp_dir = "temp_dir" Optional temporary work directory output_dir = "" Output directory where files are to be written
- temp_dir = "temp_dir" Optional temporary work directory
- output_dir = "" Output directory where files are to be written

- `crystal_info` `high_resolution = None` high-resolution limit for map calculation `solvent_content = 0.5` solvent fraction `wang_radius = 5` Smoothing radius for identification of solvent boundary `use_pdb_in_as_mask = False` Use `pdb_in` file as mask for NCS (if supplied) `rad_mask = 4` mask radius around atoms in `pdb_in`. `high_resolution_scale = 0.95` Resolution of output map coeffs will be `high_resolution_scale * resolution` of input map coeffs if `high_resolution` is `None` `fraction_ncs_min = 0.05` If the fraction of the asymmetric unit occupied by the NCS related molecules is less than `fraction_ncs_min` it is ignored `refine_ncs_operators = False` Refine NCS operators
- `high_resolution = None` high-resolution limit for map calculation
- `solvent_content = 0.5` solvent fraction
- `wang_radius = 5` Smoothing radius for identification of solvent boundary
- `use_pdb_in_as_mask = False` Use `pdb_in` file as mask for NCS (if supplied)
- `rad_mask = 4` mask radius around atoms in `pdb_in`.
- `high_resolution_scale = 0.95` Resolution of output map coeffs will be `high_resolution_scale * resolution` of input map coeffs if `high_resolution` is `None`
- `fraction_ncs_min = 0.05` If the fraction of the asymmetric unit occupied by the NCS related molecules is less than `fraction_ncs_min` it is ignored
- `refine_ncs_operators = False` Refine NCS operators
- `control` `verbose = False` Verbose output `raise_sorry = False` Raise sorry if problems `debug = False` Debugging output `dry_run = False` Just read in and check parameter names `coarse_grid = False` Use coarse grid (saves on memory) `resolve_command_list = None` You can supply any resolve command here NOTE: for command-line usage you need to enclose the whole set of commands in double quotes (") and each individual command in single quotes (') like this:
`resolve_command_list="'no_build' 'b_overall 23' "`
- `verbose = False` Verbose output
- `raise_sorry = False` Raise sorry if problems
- `debug = False` Debugging output
- `dry_run = False` Just read in and check parameter names
- `coarse_grid = False` Use coarse grid (saves on memory)
- `resolve_command_list = None` You can supply any resolve command here NOTE: for command-line usage you need to enclose the whole set of commands in double quotes (") and each individual command in single quotes (') like this: `resolve_command_list="'no_build' 'b_overall 23' "`

Model building with Pathwalker

pathwalker.html

Author(s)

- pathwalker: Matthew Baker

Purpose

Build a model with path walking

How pathwalker works:

Examples

Standard run of pathwalker:

Running pathwalker is easy. From the command-line you can type:

```
phenix.pathwalker my_map.mrc resolution=3 threshold=0.2 nres=150
```

This will build a model based on the map my_map.mrc at a resolution of 3 Å using a threshold of 0.2 and looking for 150 CA atoms

Possible Problems

Specific limitations and problems:

Literature

Additional information

List of all available keywords

job_title = None Job title in PHENIX GUI, not used on command line
input_filesmap_file = None File with CCP4-style map. May have origin in any location.
default_paramsthreshold = None Threshold used for populating pseudatoms.
nres = None Number of pseudoatoms to populate the density with
strategyseq_file = None Sequence file
pa_file = None Optional input pseudoatoms to be used as starting point for pathwalker to be run. This model should be in the same reference frame as the input map unless starting_model_is_from_box_map_au is set.
start_pt = 0 If a pa_file is used, does the user want to specify a start PA. Must be used in conjunction with end_pt.
end_pt = 0 If a pa_file is used, does the user want to specify a end PA. Must be used in conjunction with start_pt.
verbose = False Verbose turns on path evaluation.
tsp = *ortools LKH TSP solver to be used.
pa_type = *kmeans sc ac ms gmm birch PA type used for populating pseudatoms. (kmeans is default)
noise = 0 Add noise to atom positions (float)
map_weight = False Weighting distances by map value
filter = False Apply a filter to the map. Default is 4.5Å, if turned on. This may help with PA placement
filter_resolution = 4.5 Resolution to apply a filter. The default is 4.5Å if turned on.
or_time = 30 Time (in second) to allow OR tools to refine pathwalker trace Default is 30

secondslocal_search_strategy = *GUIDED_LOCAL_SEARCH AUTOMATIC GREEDY_DESCENT
SIMULATED_ANNEALING TABU_SEARCH OBJECTIVE_TABU_SEARCH Local search strategies (also
called metaheuristics) for the OR-Tools TSP solver.prob_model = False compute a probabilistic model based
on multiple pathwalker paths.all_atom = False Convert the final pathwalker model into an all-atom model
using Pulchra.reverse = False Generate the all-atom reverse path with Pulchra.refine_resolution = 0 Add a
map resolution value to run real-space refinementrefine_cycles = 5 The number of real space refinement
cycles that are to be run.bracket = [] This handles the options for iterative PA generation and tsp solver. Three
values are required: Threshold minimum Threshold maximum Threshold step They should be entered like
bracket=0.22,0.24,0.01tsp_runs = [0] Number of TSP runs and the amount of noise to apply to each run. for
example: tsp_runs=0,0,0,1,0 will actually create 5 models, where runs 1-3 and 5 have no additive noise,
while run 4 has noise equivalent to 1 sigma added to the original PAsaverage_model = False if multiple TSP
runs are set, an average model can be generated with this flagoutput_files pw_out_pdb = pw-threshold.pdb
Search PDB file, transformed to superpose on the target map file.pts_out_pdb = pts-threshold.pdb Search
PDB file, transformed to superpose on the target map file.pts_out = pts.txt Pseudoatom points generated
(coordinates)guiGUI-specific parameter required for output directoryoutput_dir = None

- job_title = None Job title in PHENIX GUI, not used on command line
- input_filesmap_file = None File with CCP4-style map. May have origin in any location.
- map_file = None File with CCP4-style map. May have origin in any location.
- default_paramsthreshold = None Threshold used for populating pseudatoms.nres = None Number of
pseudoatoms to populate the density with
- threshold = None Threshold used for populating pseudatoms.
- nres = None Number of pseudoatoms to populate the density with
- strategyseq_file = None Sequence filepa_file = None Optional input pseudoatoms to be used as starting
point for pathwalker to be run. This model should be in the same reference frame as the input map unless
starting_model_is_from_box_map_au is set.start_pt = 0 If a pa_file is used, does the user want to specify
a start PA. Must be used in conjunction with end_pt.end_pt = 0 If a pa_file is used, does the user want to
specify a end PA. Must be used in conjunction with start_pt.verbose = False Verbose turns on path
evaluation.tsp = *ortools LKH TSP solver to be used.pa_type = *kmeans sc ac ms gmm birch PA type
used for populating pseudatoms. (kmeans is default)noise = 0 Add noise to atom positions
(float)map_weight = False Weighting distances by map valuefilter = False Apply a filter to the map.
Default is 4.5A, if turned on. This may help with PA placementfilter_resolution = 4.5 Resolution to apply a
filter. The default is 4.5A if turned on.or_time = 30 Time (in second) to allow OR tools to refine pathwalker
trace Default is 30 secondslocal_search_strategy = *GUIDED_LOCAL_SEARCH AUTOMATIC
GREEDY_DESCENT SIMULATED_ANNEALING TABU_SEARCH OBJECTIVE_TABU_SEARCH Local
search strategies (also called metaheuristics) for the OR-Tools TSP solver.prob_model = False compute
a probabilistic model based on multiple pathwalker paths.all_atom = False Convert the final pathwalker
model into an all-atom model using Pulchra.reverse = False Generate the all-atom reverse path with
Pulchra.refine_resolution = 0 Add a map resolution value to run real-space refinementrefine_cycles = 5
The number of real space refinement cycles that are to be run.bracket = [] This handles the options for
iterative PA generation and tsp solver. Three values are required: Threshold minimum Threshold
maximum Threshold step They should be entered like bracket=0.22,0.24,0.01tsp_runs = [0] Number of
TSP runs and the amount of noise to apply to each run. for example: tsp_runs=0,0,0,1,0 will actually
create 5 models, where runs 1-3 and 5 have no additive noise, while run 4 has noise equivalent to 1
sigma added to the original PAsaverage_model = False if multiple TSP runs are set, an average model
can be generated with this flag
- seq_file = None Sequence file

- `pa_file = None` Optional input pseudoatoms to be used as starting point for pathwalker to be run. This model should be in the same reference frame as the input map unless `starting_model_is_from_box_map_au` is set.
- `start_pt = 0` If a `pa_file` is used, does the user want to specify a start PA. Must be used in conjunction with `end_pt`.
- `end_pt = 0` If a `pa_file` is used, does the user want to specify a end PA. Must be used in conjunction with `start_pt`.
- `verbose = False` Verbose turns on path evaluation.
- `tsp = *ortools` LKH TSP solver to be used.
- `pa_type = *kmeans` sc ac ms gmm birch PA type used for populating pseudatoms. (kmeans is default)
- `noise = 0` Add noise to atom positions (float)
- `map_weight = False` Weighting distances by map value
- `filter = False` Apply a filter to the map. Default is 4.5Å, if turned on. This may help with PA placement
- `filter_resolution = 4.5` Resolution to apply a filter. The default is 4.5Å if turned on.
- `or_time = 30` Time (in second) to allow OR tools to refine pathwalker trace Default is 30 seconds
- `local_search_strategy = *GUIDED_LOCAL_SEARCH` AUTOMATIC GREEDY_DESCENT SIMULATED_ANNEALING TABU_SEARCH OBJECTIVE_TABU_SEARCH Local search strategies (also called metaheuristics) for the OR-Tools TSP solver.
- `prob_model = False` compute a probabilistic model based on multiple pathwalker paths.
- `all_atom = False` Convert the final pathwalker model into an all-atom model using Pulchra.
- `reverse = False` Generate the all-atom reverse path with Pulchra
- `refine_resolution = 0` Add a map resolution value to run real-space refinement
- `refine_cycles = 5` The number of real space refinement cycles that are to be run.
- `bracket = []` This handles the options for iterative PA generation and tsp solver. Three values are required: Threshold minimum Threshold maximum Threshold step They should be entered like `bracket=0.22,0.24,0.01`
- `tsp_runs = [0]` Number of TSP runs and the amount of noise to apply to each run. for example: `tsp_runs=0,0,0,1,0` will actually create 5 models, where runs 1-3 and 5 have no additive noise, while run 4 has noise equivalent to 1 sigma added to the original PAs
- `average_model = False` if multiple TSP runs are set, an average model can be generated with this flag
- `output_files` `pw_out_pdb = pw-threshold.pdb` Search PDB file, transformed to superpose on the target map file. `pts_out_pdb = pts-threshold.pdb` Search PDB file, transformed to superpose on the target map file. `pts_out = pts.txt` Pseudoatom points generated (coordinates)
- `pw_out_pdb = pw-threshold.pdb` Search PDB file, transformed to superpose on the target map file.
- `pts_out_pdb = pts-threshold.pdb` Search PDB file, transformed to superpose on the target map file.
- `pts_out = pts.txt` Pseudoatom points generated (coordinates)
- `gui` GUI-specific parameter required for output directory `output_dir = None`
- `output_dir = None`

Rapid phase improvement and model-building using phase_and_build

phase_and_build.html

Contents

- Author(s)
- Purpose
- Usage
- How phase_and_build works:
- Output files from phase_and_build
- Parameters files in phase_and_build
- Examples
- Standard run of phase_and_build:
- Possible Problems
- Specific limitations and problems:
- Literature
- Additional information

Author(s)

- phase_and_build: Tom Terwilliger

Purpose

phase_and_build is a new and rapid method for improving the quality of your map and building a model. The approach is to carry out an iterative process of building a model as rapidly as possible and using this model in density modification to improve the map. This approach is related to the older phenix.autobuild approach. The difference is that in phenix.autobuild much effort was spent on building the best possible model at each stage before carrying out density modification, while in phenix.phase_and_build speed of model-building is optimized. The result is that phenix.phase_and_build is 10 times faster than phenix.autobuild, yet it produces nearly as good a model in the end. The phenix.phase_and_build approach will also find NCS from your starting map and apply it during density modification.

Usage

How phase_and_build works:

phase_and_build first identifies a free set of reflections if they are not supplied. A map without the test set reflections is created for real-space refinement. A set of data for density modification is created which contains anisotropy-corrected data (if supplied).

- Next phase_and_build estimates the solvent fraction (if not supplied) from the sequence file and cell parameters and the maximum and most likely number of NCS copies.

- A starting map is obtained by density modification of the input data (unless a map coefficients file is supplied).
- NCS is identified from the starting map (if the maximum number of NCS copies is greater than 1). If a heavy-atom file is supplied, NCS will first be identified from the heavy-atom sites, and the map will be used if the expected number of sites is not obtained.
- Next one or more cycles of phase improvement are carried out. In each cycle a model is built using `phenix.build_one_model`, the model is refined in real-space and/or in reciprocal space, and then the model is included in density modification to create a new map.
- Finally, a full model-building cycle is carried out using the most recent map. One or more models are built with `phenix.build_one_model` and are refined, then are combined, again with `phenix.build_one_model`. The resulting model is refined and sequence assignment and fitting of short loops is carried out with `phenix.assign_sequence`. Then longer loops are fit with `phenix.fit_loops` and the final model is refined and written out.

Output files from `phase_and_build`

`phase_and_build.pdb`: A PDB file with the resulting model
`phase_and_build_map_coeffs.mtz`: An MTZ file with optimized phases
`phase_and_build_refine_data.mtz`: An MTZ file with data and test set used in refinement

Parameters files in `phase_and_build`

When you run `phenix.phase_and_build` it will write out a `phase_and_build_params.eff` parameter file that can be used to re-run `phenix.phase_and_build` (just as for essentially all PHENIX methods). In addition, `phenix.phase_and_build` will write out the parameters files for the intermediate methods used as part of `phenix.phase_and_build` to the temporary directory used in building. You can run these with:

```
phenix.find_ncs temp_dir/find_ncs_params.eff # runs NCS identification
phenix.autobuild temp_dir/AutoBuild_run_1/autobuild.eff # runs first cycle of density modification
phenix.build_one_model temp_dir/build_one_model_params.eff # runs most recent model-building
phenix.assign_sequence temp_dir/assign_sequence_params.eff # runs sequence assignment and filling short gaps
phenix.fit_loops temp_dir/fit_loops_params.eff # runs loop fitting
```

This gives you control of all the steps in map improvement and model-building in addition to letting you run them all together with `phenix.phase_and_build`

Examples

Standard run of `phase_and_build`:

Running `phase_and_build` is easy. From the command-line you can type:

```
phenix.phase_and_build exptl_fobs_phases_freeR_flags.mtz sequence.dat
```

If you want to supply a file with anisotropy-corrected data to use in density modification you can do so:

```
phenix.phase_and_build data=exptl_fobs_phases_freeR_flags.mtz \
seq_file=sequence.dat \
aniso_corrected_data=solve_1.mtz
```

where `solve_1.mtz` is anisotropy-corrected (the amplitudes are not measured amplitudes, but rather are corrected with an anisotropic B-factor), and `exptl_fobs_phases_freeR_flags.mtz` contains experimental amplitudes. These two files normally will contain the same phase information. (Usually these files will come

from phenix.autosol.)

You can also add a starting model or a starting map to phenix.phase_and_build. This means that you can run it once, get a new model and map, then run it again to further improve your model and map.

Possible Problems

Specific limitations and problems:

phenix.phase_and_build does not have the full flexibility of phenix.autobuild, so you may want to get a nearly-complete model with phenix.phase_and_build and then use phenix.autobuild to increase the completeness and quality.

Literature

Additional information

<string>

phenix_maps.html

phenix.maps: a command line tool to compute various maps and save them in most of known formats.

How to run the command line version:

Run phenix.maps without any arguments: just type phenix.maps in the command line and hit Enter. This will create a parameter file called maps.params, which can be renamed if desired. Edit maps.params file to specify input/output file names, data labels and the desired maps. It is possible to request as many maps as desired. By default, the file maps.params specifies 5 maps to be created: 2mFo-DFc, 2mFo-DFc with missing Fobs filled with DFcalc, mFo-DFc and anomalous difference maps will be output in MTZ format, and one 2mFo-DFc map will be output in CCP4 format. NOTE: the anomalous difference map will only be created if the input reflection data file contains Bijvoet maps (F+/F- or I+/I-). Run this command to compute requested maps: phenix.maps maps.params

- Run phenix.maps without any arguments: just type phenix.maps in the command line and hit Enter. This will create a parameter file called maps.params, which can be renamed if desired.
- Edit maps.params file to specify input/output file names, data labels and the desired maps. It is possible to request as many maps as desired. By default, the file maps.params specifies 5 maps to be created: 2mFo-DFc, 2mFo-DFc with missing Fobs filled with DFcalc, mFo-DFc and anomalous difference maps will be output in MTZ format, and one 2mFo-DFc map will be output in CCP4 format. NOTE: the anomalous difference map will only be created if the input reflection data file contains Bijvoet maps (F+/F- or I+/I-).
- Run this command to compute requested maps: phenix.maps maps.params

Alternately, you may specify input files (and additional parameters) directly on the command line:

```
% phenix.maps model.pdb data.mtz
```

and it will automatically generate the default maps as described above.

Important Facts:

phenix.maps is available in PHENIX GUI. The scope of parameters 'map_coefficients' defines the map that will be output as Fourier map coefficients. The scope of parameters 'map' defines the maps that will be output as CCP4 or X-plor format. To create several maps: duplicate either 'map_coefficients' or 'map' or both scopes of parameters as many times as many maps is desired. Then edit each of them to define the maps. A map is defined by specifying a map type using 'map_type' keyword available within each scope of parameters: 'map_coefficients' or 'map'. The general supported format for 'map_type' is: [p][m]Fo+[q][D]Fc[_filled]. For example: 2Fo-Fc, 2mFobs-DFcalc, 3Fobs-2Fmodel, Fo-Fc, mFobs-DFcalc, anom, llg. The 'map_type' parser will automatically recognize which map is requested. The program creates as many files with CCP4 or X-plor formatted maps as is requested, and it creates only one MTZ formatted file with all Fourier map coefficients in it. The CCP4 or X-plor formatted maps can be computed in the entire unit cell or around selected atoms only. Twinning (if detected) will be accounted for automatically. This can be disabled by using "skip_twin_detection=True" keyword. All arrays used in map calculation, for example: Fobs, Fmodel, Fcalc, Fmask, m, D, etc., can be output into a CNS or MTZ formatted reflection file. For those who likes to experiment: bulk solvent correction and anisotropic scaling can be turned off, the data can be filtered by sigma and resolution. For some map types certain 'map_coefficients' or 'map' scope parameters may not be

applicable. For example, for "map_type=anomalous" the keywords "fill_missing_f_obs" and some other are not applicable. For LLG map calculation, if you specify the wavelength any existing heavy atoms (P or heavier) will be modeled as anomalous scatterers using the theoretical values of f' and f'' .

- phenix.maps is available in PHENIX GUI.
- The scope of parameters 'map_coefficients' defines the map that will be output as Fourier map coefficients. The scope of parameters 'map' defines the maps that will be output as CCP4 or X-plor format.
- To create several maps: duplicate either 'map_coefficients' or 'map' or both scopes of parameters as many times as many maps is desired. Then edit each of them to define the maps.
- A map is defined by specifying a map type using 'map_type' keyword available within each scope of parameters: 'map_coefficients' or 'map'. The general supported format for 'map_type' is: [p][m]Fo+[q][D]Fc[_filled]. For example: 2Fo-Fc, 2mFobs-DFcalc, 3Fobs-2Fmodel, Fo-Fc, mfobs-Dfcalc, anom, llg. The 'map_type' parser will automatically recognize which map is requested.
- The program creates as many files with CCP4 or X-plor formatted maps as is requested, and it creates only one MTZ formatted file with all Fourier map coefficients in it.
- The CCP4 or X-plor formatted maps can be computed in the entire unit cell or around selected atoms only.
- Twinning (if detected) will be accounted for automatically. This can be disabled by using "skip_twin_detection=True" keyword.
- All arrays used in map calculation, for example: Fobs, Fmodel, Fcalc, Fmask, m, D, etc., can be output into a CNS or MTZ formatted reflection file.
- For those who likes to experiment: bulk solvent correction and anisotropic scaling can be turned off, the data can be filtered by sigma and resolution.
- For some map types certain 'map_coefficients' or 'map' scope parameters may not be applicable. For example, for "map_type=anomalous" the keywords "fill_missing_f_obs" and some other are not applicable.
- For LLG map calculation, if you specify the wavelength any existing heavy atoms (P or heavier) will be modeled as anomalous scatterers using the theoretical values of f' and f'' .

All phenix.maps parameters:

```
maps {
  input {
    pdb_file_name = None
    reflection_data {
      file_name = None
      labels = None
      high_resolution = None
      low_resolution = None
      twin_law = None
      outliers_rejection = True
      french_wilson_scale = True
      french_wilson {
        max_bins = 60
        min_bin_size = 40
      }
      sigma_fobs_rejection_criterion = None
      sigma_iobs_rejection_criterion = None
      r_free_flags {
        file_name = None
        label = None
        required = True
      }
    }
  }
}
```

```

test_flag_value = None
ignore_r_free_flags = False
}
}
}
output {
  directory = None
  prefix = None
  job_title = None
  fmodel_data_file_format = mtz
  include_r_free_flags = False
}
scattering_table = wk1995 it1992 *n_gaussian neutron electron
wavelength = None
bulk_solvent_correction = True
anisotropic_scaling = True
skip_twin_detection = False
omit {
  method = *simple
  selection = None
}
map_coefficients {
  map_type = 2mFo-DFc
  format = *mtz phs
  mtz_label_amplitudes = 2FOFCWT
  mtz_label_phases = PH2FOFCWT
  fill_missing_f_obs = False
  sharpening = False
  sharpening_b_factor = None
  exclude_free_r_reflections = False
  isotropize = True
}
map_coefficients {
  map_type = 2mFo-DFc
  format = *mtz phs
  mtz_label_amplitudes = 2FOFCWT_fill
  mtz_label_phases = PH2FOFCWT_fill
  fill_missing_f_obs = True
  sharpening = False
  sharpening_b_factor = None
  exclude_free_r_reflections = False
  isotropize = True
}
map_coefficients {
  map_type = mFo-DFc
  format = *mtz phs
  mtz_label_amplitudes = FOFCWT
  mtz_label_phases = PHFOFCWT
  fill_missing_f_obs = False
  sharpening = False
  sharpening_b_factor = None
  exclude_free_r_reflections = False
  isotropize = True
}
map_coefficients {
  map_type = anomalous
  format = *mtz phs
  mtz_label_amplitudes = ANOM
  mtz_label_phases = PHANOM
  fill_missing_f_obs = False
  sharpening = False
  sharpening_b_factor = None
  exclude_free_r_reflections = False
  isotropize = True
}
}
map {
  map_type = 2mFo-DFc

```

```
format = xplor *ccp4
file_name = None
fill_missing_f_obs = False
grid_resolution_factor = 1/4.
region = *selection cell
atom_selection = None
atom_selection_buffer = 3
sharpening = False
sharpening_b_factor = None
exclude_free_r_reflections = False
isotropize = True
}
}
```

Planning a SAD experiment with `plan_sad_experiment`

`plan_sad_experiment.html`

Author(s)

- `plan_sad_experiment`: Tom Terwilliger

Purpose

`plan_sad_experiment` is a tool for estimating the anomalous signal that you might get from your SAD experiment and for predicting whether this signal would be sufficient to solve the structure.

`plan_sad_experiment` is normally used along with `scale_and_merge` and `anomalous_signal` to plan a SAD experiment, scale the data, and analyze the anomalous signal before solving the structure.

Usage

How `plan_sad_experiment` works:

- You supply `plan_sad_experiment` with a sequence file, the anomalously-scattering atom you plan to use for the experiment, and the wavelength for data collection.
- `plan_sad_experiment` will estimate the necessary I/σ_I of your dataset to provide enough anomalous signal to solve the structure.
- `plan_sad_experiment` will try various values of I/σ_I for your dataset at each of several resolutions. For each I/σ_I it will estimate the half-dataset anomalous correlation that would result along with the likely true correlation between your anomalous differences and those that would be calculated from a final model of your structure (cc^*_ano). From this anomalous correlation (cc^*_ano), `plan_sad_experiment` will estimate the anomalous signal (related to cc^*_ano by the square root of the number of reflections divided by the square root of the number of sites). Then `plan_sad_experiment` will choose a value of I/σ_I that gives an anomalous signal of about 30 (if achievable with the maximum I/σ_I you specify).
- The way that `plan_sad_experiment` and `anomalous_signal` estimate the probability that you can solve your dataset is to compare the anomalous signal in this dataset with the anomalous signal in other datasets at the same resolution. Then the fraction of similar datasets that can be solved by HySS is used as the probability that the anomalous substructure for your dataset will also be found.
- Similarly, the mean figure of merit for datasets with an estimated anomalous correlation (cc^*_ano) similar to that for your data is used as an estimate of the figure of merit that you would obtain if the substructure is found for your crystal.

Output from `plan_sad_experiment`

`plan_sad_experiment` provides a summary of the scattering expected from your crystal and a summary of the anomalous signal expected if you are able to measure your data with the suggested overall I/σ_I . You can set the maximum I/σ_I to look for. Here is an example setting `max_i_over_sigma=30`:

```
-----Dataset overall I/sigma required to solve a structure-----  
  
Dataset characteristics:  
  Target anomalous signal: 30.0  
  Residues: 325
```

Chain-type: PROTEIN
 Solvent_fraction: 0.50
 Atoms: 2642
 Anomalous-scattering atom: se
 Wavelength: 0.9792 Å
 Sites: 7
 f-double-prime: 3.84

Target anomalous scatterer:
 Atom: se f": 3.84 n: 7 rmsF: 10.2

Other anomalous scatterers in the structure:

Atom: C f": 0.00 n: 1674 rmsF: 0.1
 Atom: N f": 0.01 n: 445 rmsF: 0.1
 Atom: O f": 0.01 n: 514 rmsF: 0.3
 Atom: S f": 0.23 n: 10 rmsF: 0.7

Normalized anomalous scattering:

From target anomalous atoms $\text{rms}(x^2)/\text{rms}(F^2)$: 2.97
 From other anomalous atoms $\text{rms}(e^2)/\text{rms}(F^2)$: 0.24
 Correlation of useful to total anomalous scattering: 1.00

-----Dataset <I>/<sigma>; needed for anomalous signal of 15-30-----

-----Targets for entire dataset----- -----Likely outcome-----

Anomalous Useful Useful
 Half-dataset Anom CC Anomalous
 Dmin N I/sigI sigF/F CC (cc*_anom) Signal P(Substr) FOM
 (%) (%)

6.00	852	29	3.0	0.58	0.64	7	51	0.22
5.00	1473	29	3.0	0.62	0.66	9	79	0.15
3.00	6821	29	3.0	0.64	0.66	19	89	0.22
2.50	11787	29	3.0	0.70	0.68	25	96	0.19
2.00	23021	28	3.2	0.62	0.66	29	97	0.17
1.50	54569	13	6.7	0.18	0.42	29	97	0.15

Note: Target anomalous signal not achievable with tested I/sigma (up to 30)
 for resolutions of 2.50 Å and lower. I/sigma shown is value
 of $\max_i \frac{i}{\sigma}$.

This table says that if you collect your data to a resolution of 2.0 Å with
 an overall <I>/<sigma> of about 28 then the half-dataset anomalous
 correlation should be about 0.62 (typically within a factor of 2). This
 should lead to a correlation of your anomalous data to true anomalous
 differences (CC*_ano) of about 0.66, and a useful anomalous signal around
 29 (again within a factor of about two). With this value of estimated
 anomalous signal the probability of finding the anomalous substructure is
 about 96% (based on estimated anomalous signal and actual outcomes for
 real structures.), and the estimated figure of merit of phasing is 0.17.

The value of sigF/F (actually $\text{rms}(\text{sigF})/\text{rms}(F)$) is approximately the inverse
 of I/sigma. The calculations are based on $\text{rms}(\text{sigF})/\text{rms}(F)$.

Note that these values assume data measured with little radiation damage or at
 least with anomalous pairs measured close in time. The values also assume that
 the anomalously-scattering atoms are nearly as well-ordered as other atoms.
 If your crystal does not fit these assumptions it may be necessary to collect
 data with even higher I/sigma than indicated here.

Note also that anomalous signal is roughly proportional to the anomalous
 structure factors at a given resolution. That means that if you have 50%
 occupancy of your anomalous atoms, the signal will be 50% of what it otherwise
 would be. Also it means that if your anomalously scattering atoms only
 contribute to 5 Å, you should only consider data to 5 Å in this analysis.

What to do next:

1. Collect your data, trying to obtain a value of I/σ for the whole dataset at least as high as your target.
2. Scale and analyze your unmerged data with `phenix.scale\_and\_merge` to get accurate scaled and merged data as well as two half-dataset data files that can be used to estimate the quality of your data.
3. Analyze your anomalous data (the scaled merged data and the two half-dataset data files) with `phenix.anomalous\_signal` to estimate the anomalous signal in your data. This tool will again guess the fraction of the substructure that can be obtained with your data, this time with knowledge of the actual anomalous signal. It will also estimate the figure of merit of phasing that you can obtain once you solve the substructure.
4. Compare the anomalous signal in your measured data with the estimated values in the table above. If they are lower than expected you may need to collect more data to obtain the target anomalous signal.

Literature

List of all available keywords

`job_title` = None Job title in PHENIX GUI, not used on command line
`i_over_sigma` = None Optional I/σ . If supplied, the expected values of half-dataset correlation and `cc*_ano` based on this I/σ be calculated.
`max_i_over_sigma` = 100 Limit search of necessary I/σ to less than this value. You might increase this if you plan to do a very careful or very high-multiplicity experiment.
`i_over_sigma_range_low` = None If you set `i_over_sigma_range_low` and `i_over_sigma_range_high` then the value of `i_over_sigma` will be varied between these limits
`i_over_sigma_range_high` = None If you set `i_over_sigma_range_low` and `i_over_sigma_range_high` then the value of `i_over_sigma` will be varied between these limits
`steps` = 20 Number of steps for sampling ranges (i.e., `i_over_sigma_low` to `i_over_sigma_high`)
`min_in_bin` = 50 Minimum data points per bin in Bayesian estimation. Higher values smooth the predictor.
`target_signal` = 30. The anomalous signal that you would like to obtain. The value of I/σ will be adjusted to obtain this signal. Typically you will need a signal of 15-30 so solve the substructure.
`min_cc_ano` = 0.15 You can set the target minimum (true) anomalous correlation (`CC*_ano`). This value affects the phasing accuracy after the substructure is determined.
`ideal_cc_anom` = 0.75 The `ideal_cc_anom` is the expected anomalous correlation between an accurate model with isotropic anomalous scatterers and perfectly-measured data. The `ideal_cc_anom` is determined empirically. It is typically not unity because anomalous scatterers may have multiple locations with low occupancy or may be non-isotropic. A value of about 0.75 is a reasonable guess.
`include_weak_anomalous_scattering` = Auto At longer wavelengths the scattering of C, N, and O become significant relative to S. Default is to consider the scattering from C, N, O as noise. Additionally, (see `intrinsic_scatterers_as_noise`) if intrinsic anomalous scatterers (P and S) are weak, they will be counted as noise. This weak anomalous scattering is effectively noise and has the same effect as the `ideal_cc_anom` but it can be calculated from the composition. Its effects are added to those modeled by the `ideal_cc_anom` parameter. Default is to include weak anomalous scattering if a sequence file or the number of sulfurs is provided.
`intrinsic_scatterers_as_noise` = None Applies if `include_weak_anomalous_scattering`=True. You can choose to treat any intrinsic scatterers (S for protein, P for nucleic acid) as noise, just like any contributions from C, N, or O atoms. This is default if anomalous scattering (f-double-prime) from these atoms is less than half that of your specified anomalous scatterer. Alternatively these atoms are excluded from the noise calculation and are assumed to be included in the number of sites you specify.
`bayesian_updates` = False Use Bayesian updates of half-dataset CC and signal. First predict these values using standard approach, then use empirical half-dataset CC and signal for a training set of datasets to re-estimate these values. This helps correct for typical errors in measurement and typical resolution-dependent effects. Note that if you use `bayesian_updates`=True then the predictions may not vary smoothly with resolution or changes in parameters.
`input_filesdata` = None Data file (I or I+ and I- or F or F+ and F-). Any standard format is

fine.data_labels = None Optional label specifying which columns of anomalous data to use. Not necessary if your input file has only one set of anomalous data.
 crystal_inforesolution = None High-resolution limit. Either a high-resolution limit or a (Wilson) b_value or both is required
 b_value = None Estimated Wilson B-value for the dataset. Either a high-resolution limit or a (Wilson) b_value or both is required
 b_value_anomalous = None Estimated Wilson B-value for the anomalously-scattering atoms. Normally leave as None and it will be estimated from b_value.
 seq_file = None Optional sequence file (1-letter code). Separate chains with a blank line or line ...

- job_title = None Job title in PHENIX GUI, not used on command line
- i_over_sigma = None Optional I/sigI. If supplied, the expected values of half-dataset correlation and cc^*_ano based on this I/sigI be calculated.
- max_i_over_sigma = 100 Limit search of necessary I/sigI to less than this value. You might increase this if you plan to do a very careful or very high-multiplicity experiment.
- i_over_sigma_range_low = None If you set i_over_sigma_range_low and i_over_sigma_range_high then the value of i_over_sigma will be varied between these limits
- i_over_sigma_range_high = None If you set i_over_sigma_range_low and i_over_sigma_range_high then the value of i_over_sigma will be varied between these limits
- steps = 20 Number of steps for sampling ranges (i.e., i_over_sigma_low to i_over_sigma_high)
- min_in_bin = 50 Minimum data points per bin in Bayesian estimation. Higher values smooth the predictor.
- target_signal = 30. The anomalous signal that you would like to obtain. The value of I/sigma will be adjusted to obtain this signal. Typically you will need a signal of 15-30 so solve the substructure.
- min_cc_ano = 0.15 You can set the target minimum (true) anomalous correlation (CC^*_ano). This value affects the phasing accuracy after the substructure is determined.
- ideal_cc_anom = 0.75 The ideal_cc_anom is the expected anomalous correlation between an accurate model with isotropic anomalous scatterers and perfectly-measured data. The ideal_cc_anom is determined empirically. It is typically not unity because anomalous scatterers may have multiple locations with low occupancy or may be non-isotropic. A value of about 0.75 is a reasonable guess.
- include_weak_anomalous_scattering = Auto At longer wavelengths the scattering of C, N, and O become significant relative to S. Default is to consider the scattering from C, N, O as noise. Additionally, (see intrinsic_scatterers_as_noise) if intrinsic anomalous scatterers (P and S) are weak, they will be counted as noise. This weak anomalous scattering is effectively noise and has the same effect as the ideal_cc_anom but it can be calculated from the composition. Its effects are added to those modeled by the ideal_cc_anom parameter. Default is to include weak anomalous scattering if a sequence file or the number of sulfurs is provided
- intrinsic_scatterers_as_noise = None Applies if include_weak_anomalous_scattering=True. You can choose to treat any intrinsic scatterers (S for protein, P for nucleic acid) as noise, just like any contributions from C, N, or O atoms. This is default if anomalous scattering (f-double-prime) from these atoms is less than half that of your specified anomalous scatterer. Alternatively these atoms are excluded from the noise calculation and are assumed to be included in the number of sites you specify.
- bayesian_updates = False Use Bayesian updates of half-dataset CC and signal. First predict these values using standard approach, then use empirical half-dataset CC and signal for a training set of datasets to re-estimate these values. This helps correct for typical errors in measurement and typical resolution-dependent effects. Note that if you use bayesian_updates=True then the predictions may not vary smoothly with resolution or changes in parameters.

- `input_filesdata` = None Data file (I or I+ and I- or F or F+ and F-). Any standard format is fine.`data_labels` = None Optional label specifying which columns of anomalous data to use. Not necessary if your input file has only one set of anomalous data.

- `data` = None Data file (I or I+ and I- or F or F+ and F-). Any standard format is fine.

- `data_labels` = None Optional label specifying which columns of anomalous data to use. Not necessary if your input file has only one set of anomalous data.

- `crystal_inforesolution` = None High-resolution limit. Either a high-resolution limit or a (Wilson) `b_value` or both is required.`b_value` = None Estimated Wilson B-value for the dataset. Either a high-resolution limit or a (Wilson) `b_value` or both is required.`b_value_anomalous` = None Estimated Wilson B-value for the anomalously-scattering atoms. Normally leave as None and it will be estimated from `b_value`.`seq_file` = None Optional sequence file (1-letter code). Separate chains with a blank line or line starting with `>`.`chain_type` = *PROTEIN RNA DNA Chain type (PROTEIN RNA DNA). This is used to estimate the number of atoms from the number of residues.`ncs_copies` = None Optional estimate of NCS copies in your crystals (only used if a data file is supplied).`solvent_fraction` = None Optional estimate of solvent fraction in your crystals (0 to 1).`residues` = None The number of residues in the molecule or asymmetric unit. Note that it is the ratio of residues to anomalously-scattering atoms that matters.`atom_type` = None Optional name of anomalously-scattering atom. If supplied, you also need to supply the wavelength for X-ray data collection. If not supplied, then you need to supply a value for `f_double_prime`.`number_of_s` = None You can specify the number of S atoms in the asymmetric unit. Only used if `include_weak_anomalous_scattering=True`. If not set, the number is guessed from the sequence file if present.`f_double_prime` = None F-double-prime value for the anomalously-scattering atom. Alternatively you can specify the atom type and wavelength.`wavelength` = None Wavelength for X-ray data collection. If supplied, also specify the `atom_type`. Alternatively you can specify the value of `f_double_prime`.`sites` = None The number of anomalously-scattering atoms in the molecule or asymmetric unit. Note that it is the ratio of residues to anomalously-scattering atoms that matters.`sites_min` = None If you set `sites_min` and `sites_max` and not `sites` the sites will be varied from `sites_min` to `sites_max`.`sites_max` = None If you set `sites_min` and `sites_max` and not `sites` the sites will be varied from `sites_min` to `sites_max`.`occupancy` = 1 Estimate of occupancy of anomalously-scattering atoms

- `resolution` = None High-resolution limit. Either a high-resolution limit or a (Wilson) `b_value` or both is required

- `b_value` = None Estimated Wilson B-value for the dataset. Either a high-resolution limit or a (Wilson) `b_value` or both is required

- `b_value_anomalous` = None Estimated Wilson B-value for the anomalously-scattering atoms. Normally leave as None and it will be estimated from `b_value`.

- `seq_file` = None Optional sequence file (1-letter code). Separate chains with a blank line or line starting with `>`.

- `chain_type` = *PROTEIN RNA DNA Chain type (PROTEIN RNA DNA). This is used to estimate the number of atoms from the number of residues

- `ncs_copies` = None Optional estimate of NCS copies in your crystals (only used if a data file is supplied).

- `solvent_fraction` = None Optional estimate of solvent fraction in your crystals (0 to 1)

- `residues` = None The number of residues in the molecule or asymmetric unit. Note that it is the ratio of residues to anomalously-scattering atoms that matters.

- `atom_type` = None Optional name of anomalously-scattering atom. If supplied, you also need to supply the wavelength for X-ray data collection. If not supplied, then you need to supply a value for `f_double_prime`.

- `number_of_s = None` You can specify the number of S atoms in the asymmetric unit. Only used if `include_weak_anomalous_scattering=True`. If not set, the number is guessed from the sequence file if present.
- `f_double_prime = None` F-double-prime value for the anomalously-scattering atom. Alternatively you can specify the atom type and wavelength.
- `wavelength = None` Wavelength for X-ray data collection. If supplied, also specify the `atom_type`. Alternatively you can specify the value of `f_double_prime`
- `sites = None` The number of anomalously-scattering atoms in the molecule or asymmetric unit. Note that it is the ratio of residues to anomalously-scattering atoms that matters.
- `sites_min = None` If you set `sites_min` and `sites_max` and not `sites` the sites will be varied from `sites_min` to `sites_max`
- `sites_max = None` If you set `sites_min` and `sites_max` and not `sites` the sites will be varied from `sites_min` to `sites_max`
- `occupancy = 1` Estimate of occupancy of anomalously-scattering atoms
- `controlfixed_resolution = False` Only run calculation at high_resolution limit
`show_summary = False` Show summary only
`verbose = False` Verbose output
- `fixed_resolution = False` Only run calculation at high_resolution limit
- `show_summary = False` Show summary only
- `verbose = False` Verbose output

PredictModel: Predict structures of chains in a sequence file

predict_model.html

Author(s)

- predict_model: Tom Terwilliger

Tutorial video

A tutorial video for AlphaFold prediction is available on the Phenix YouTube channel and covers the following topics:

- basic overview of AlphaFold model prediction
- how to run PredictModel with the Phenix AlphaFold server using the Phenix GUI
- how to interpret the output

Purpose

Predict model can be used to generate predicted models from a sequence file. One predicted model is generated for each unique sequence. It does not make predictions of a complex.

You can choose whether to use a multiple sequence alignment (MSA), whether go use templates from the PDB, and whether to supply your own template to be used as a guide in prediction.

Sequence file

The sequence file that you supply specifies what is going to be predicted and how many copies of each chain are present in the structure (one for every copy in the sequence file). All models that are created and input will be associated with one or more of the chains in your sequence file based on sequence identity (normally every model should match a chain in the sequence file exactly).

Model prediction

Predict model normally uses the Phenix server notebook to carry out AlphaFold prediction of one chain at a time.

You can specify what inputs the AlphaFold prediction should use. These always include a sequence file, but it can include an optional multiple sequence alignment file, optional templates, keywords for model prediction such as the number of models to generate, random seed, and whether to use multiple sequence alignment.

When prediction is being carried out, the Phenix GUI waits for the predicted models to appear in the working directory, then it writes out the resulting models.

Using the Phenix server for model prediction

Prediction is fully automated with the Phenix server through the Phenix GUI, so all you have to do is let it run. The Phenix GUI will use the Phenix server to carry out the prediction and put it in the working directory.

MSA calculation vs model prediction

The calculation of multiple sequence alignments (MSAs) is a separate step from model prediction. The Phenix GUI on your computer sends a request to the mmseqs2 server, which creates an MSA and sends it back to your computer. Then your computer uploads the MSA to the Phenix server, which uses it in an AlphaFold prediction.

If you want, you can supply your own MSAs. The key requirement is that the sequence of the first entry in your MSA must exactly match the sequence to be predicted.

You can also skip the use of MSAs. This can be useful if you supply a template and you want AlphaFold to rebuild your template instead of doing a new prediction.

Number of models

AlphaFold can carry out multiple predictions for a sequence. You can specify how many of these to carry out. The PredictModel tool will choose the one with the highest value of pLDDT (predicted local difference distance test, Mariani et al., 2013). Normally the default (5 models) is a reasonable compromise between optimal prediction and the length of time needed to carry out the prediction. Note that the Phenix server will stop carrying out predictions if it appears unlikely that further predictions will significantly improve the pLDDT (the estimate is based on the variability of pLDDT values so far.)

Using templates from the PDB

You can request that templates from the PDB be used in prediction. If you use this feature, models will be predicted both with and without the templates, and the model with the highest pLDDT will be saved.

Using supplied templates

You can supply your own templates. As for templates from the PDB, if you use this feature, models will be predicted both with and without the templates, and the model with the highest pLDDT will be saved.

Predicting very long sequences in chunks and reassembling

Sequences longer than about 1500 residues take quite a long time to run. By default if your sequence is more than 1500 residues it will be split into overlapping chunks of length 600 with an overlap of 200. These chunks will be run individually, then reassembled at the end. To improve the contacts between domains in different chunks, a contact map is created and domains that are in contact but in different chunks are re-predicted together. During assembly the domain pairs are used to adjust the positions of domains, and also positions of domains are adjusted to minimize overlaps. The repositioned domains are then used as a template to morph the full reassembled prediction. The result of this procedure is to assemble a full-length prediction from overlapping chunk predictions and to distort this assembled prediction to minimize overlaps and make domain-domain contacts present between chunks.

Note that if you have a sequence longer than the threshold for breaking into chunks then you can only include one sequence in a PredictModel run.

Precalculated MSAs and models save time if you rerun with the same sequence

When you run a prediction on the Phenix server, it saves the MSA and the predicted model. Then if you run the same request again (same number of AlphaFold models, same sequence, same choice of including templates from the PDB, no supplied template), the server will return the result from the original run. You can turn this off with the precalculated results and precalculated MSA keywords if you want. Note that the server

does not save your sequence or parameters. It just makes a hash string from them so that if the same request is made again it can detect it. The results are simply saved in a file that has the hash string as a name.

Precalculated MSAs and models allow you go get your result even if disconnected

If you get disconnected from the server during a prediction job, you can just wait for the job to finish, then submit a new request with the same parameters. As the server saves the predictions, it will return the result to you right away (if it is finished).

Using AlphaFold to improve a model you already have

As AlphaFold often produces models with quite good geometry, you can use it as a procedure for geometry optimization. You supply your working model and your sequence, and you turn off the use of MSAs and the use of templates from the PDB. Of course AlphaFold does not know about your density map, so it could move the model away from the density. Normally you would use refinement to help with this.

Using AlphaFold to build parts of a model that are missing

If you want AlphaFold to try and build the missing parts of a chain that you have partially built, supply your working partial model and your sequence, and allow the use of MSAs (and optionally, the use of templates from the PDB). Your working model will be used as a template for the part of the structure that you have already built. Note that you have to have the correct sequence for your partially-built model.

Notes on using the Phenix server from the Phenix GUI for model prediction

The Phenix GUI can automatically send a request to the Phenix server for an AlphaFold prediction. This requires an internet connection, the Phenix server to be up, and a token that is supplied with Phenix.

If the Phenix GUI successfully communicates with the server, it will wait for the server to carry out the prediction, showing the status of the prediction while it is being carried out. The status will usually start out as waiting in the queue for a place on the GPU. If this status lasts more than a few seconds, an estimated waiting time (before starting processing) is displayed in the Phenix GUI. The waiting time is very approximate. It is based on how long previous jobs waited to start and how many jobs are ahead of yours. If the currently-running jobs happen to take a very short or long time, the estimate could be off in either direction. If you submit more than one job at a time, the first job you submit will get higher priority and may move nearer the front of the queue than the others.

Once your job starts, the status in the Phenix GUI will change to reflect that it is running. There won't be an estimated run time shown, but most of the time jobs with a few hundred residues take 10-20 min, and longer sequences take more or less proportionally longer. If you include templates from the PDB, it takes longer to run. If you specify that the number of AlphaFold models to use is more than the default (5), it will also take longer.

If your prediction is successful, the Phenix GUI will list the estimated confidence (pLDDT) and the output data files. The output files include the AlphaFold model (named something like PredictAndBuild_10_ALPHAFOLD_A.pdb), the multiple sequence alignment used (PredictAndBuild_10_ALPHAFOLD_A_MSA.a3m), and the Predicted Aligned Error (PAE) file (PredictAndBuild_10_ALPHAFOLD_A_PAE.jsn).

Privacy of your data when using the Phenix server

When you run the Phenix server to get an AlphaFold prediction, your data are uploaded by the Phenix GUI to the Phenix server, the prediction is carried out, and then the results are downloaded.

Normally your sequence is also sent to the mmseqs2 MSA server to obtain a multiple sequence alignment. You can also supply your own MSA and then the mmseqs2 server is not used. The MSA and the hashed sequence are sent to the Phenix server and are saved unless you specify `allow_precalculated_msa_file=False`.

When the job is completed and Phenix downloads the data, all data is deleted if you specify `allow_precalculated_result_file=False`. If you specify `allow_precalculated_result_file=True` (default), a hashed version of the inputs (sequence, number of models, etc) is saved along with the result (the AlphaFold model). In this way it is not possible to see the inputs, but if you run the exact same job again, the server will see that the hash of the inputs matches and it will return the model directly.

Here are some details on the privacy of your data when using the Phenix server:

- Accessible only from Phenix

- Password-protected (password supplied by Phenix)

- Server provides Phenix with a passkey when data are uploaded. That passkey allows download of the data. All this is done within Phenix.

- Phenix sends your IP address to the server. This is used as a unique identifier in the queueing system. The IP address is deleted when the data are downloaded.

- Server is accessible in two modes (user can choose, default is Mode 1 (save hashed inputs and results)).

- A. Mode 1: hashed inputs and results saved on server
 - a. in this mode if anyone submits the identical job (identical sequence, identical parameters) later, they get back the result model from this job
 - b. result model does not contain any information about the user
 - c. inputs and log files are deleted after data are downloaded
 - d. results removed from server after 1 year
- B. Mode 2: nothing saved. Input and results deleted from server after data are downloaded

- Server can only carry out pre-specified commands (i.e., model prediction), and only allows json text to be uploaded.

- Data is accessible to the Phenix developers while running. The developers will only access this data to debug the system.

The output predicted model

The predicted model produced by AlphaFold with Phenix will consist of a single chain with exactly the sequence that is supplied. It will be in the standard PDB (version 3.2) format. It will contain all (non-hydrogen) atoms for all the residues. The coordinates of the atoms have the usual meaning (locations of atoms in 3D space), but the orientation and position of the chain are arbitrary. The B-value field (normally atomic displacement parameters) will contain the values of pLDDT (confidence), with one value per residue. This means that some processing of this model is necessary before use (this is normally done by `phenix.process_predicted_model`).

The pLDDT confidence measure

AlphaFold provides residue-level estimates of model accuracy in the form of predicted values of the LDDT (Local Distance Difference Test, Mariani et al., 2013). These predicted values are referred to as pLDDT values (predicted Local Distance Difference Test). The LDDT measure is a number from zero to one that reflects the similarity of CA positions in two structures.

The LDDT is not a simple measure of accuracy, but rather a composite that is based on the CA distances between all pairs of CA atoms in one model and how similar these inter-atomic distances are to those in the other model. It is a local test, where only inter-atomic distances that are less than 15 Å are considered.

The LDDT is a composite that is generated from the fraction of CA-CA distances that are very accurate (less than 0.5 Å off), the fraction that are quite accurate (less than 1 Å off), moderately accurate (less than 2 Å off), and somewhat accurate (less than 4 Å off). If all CA-CA distances are within 0.5 Å, the resulting score will be 1, if all are more than 4 Å off, the score is zero. Note that the LDDT is related to RMSD, but not in a simple way.

The pLDDT values (predicted Local Distance Difference Test) provided by AlphaFold are estimates of the LDDT values. The pLDDT values are essentially unbiased (just as likely to be too low as too high), and reasonably accurate. The correlation between pLDDT values and actual LDDT values calculated using AlphaFold models and models in the PDB is about 0.7 - 0.75, which means that the values of pLDDT you get are useful indicators of model quality but also that sometimes AlphaFold will have high confidence in an incorrect prediction or low confidence in a correct prediction.

As the pLDDT metric is an estimate of local error, it does not tell you that much about long-range distortions or domain movements between the predicted and actual structure. The PAE matrix does have some information about these features because it is not restricted to short distances.

The output MSA file

The MSA file that was used in AlphaFold prediction is available if the Phenix server is used. This MSA file can be used in subsequent predictions (if the sequence is identical). Normally there is no need to supply an MSA file because the MSA is saved by the Phenix server and it will just be re-used if the same sequence is supplied. However if you want to edit this MSA file before running an AlphaFold prediction, you can do that.

The output PAE file

AlphaFold provides a Predicted Aligned Error matrix in the form of a PAE file. The error that it is describing reflects the estimated error in CA positions in the predicted model compared to the (unknown) true structure.

Suppose we had both the prediction and the true structure. Then suppose we superpose the predicted model on the true structure, basing the superposition on the residues near residue *i*. Using that superposition, we can now look at the location of the CA atom of residue *j* in the predicted model and in the true structure. We could in particular note the distance between the predicted and actual position of the CA atom of residue *j*, when the structures are superposed using residues near residue *i*. We could do this for every residue pair (*i,j*), and the result would be the Aligned Error matrix.

The Predicted Aligned Error matrix is an estimate of the Aligned Error matrix. It is generated by AlphaFold. Basically, AlphaFold predicts the model and also it generates an error estimate in the prediction.

The PAE matrix is useful for identifying domains within a predicted chain that are accurately predicted, along with connecting regions that are more uncertain. In the Phenix GUI the PAE matrix is plotted with dark green indicating low PAE values (relationships between pairs of residues that are predicted to be accurate). A domain that is accurately predicted can be seen in such a plot as a solid green square along the diagonal. If your plot has just one big green square, it is probably all quite accurate. If it looks instead like a few squares

along the diagonal, probably the individual domains are accurately predicted, but their relationship is not known.

Relationship between the pLDDT values and the PAE matrix

The PAE matrix is related to the pLDDT values, but not in a simple way.

The pLDDT value for a residue describes the expected errors in the distances between that CA atom and nearby CA atoms (compared to the true structure) on a scale of 0 to 1, where 1 means all CA-CA distances are expected to match within 0.5 Å.

The value of the PAE matrix for a pair of residues is the estimated error in Å in the position of the CA atom for the second residue if the model is superposed on the true structure using CA atoms near the first CA atom.

Both measures tell you something about the uncertainties in your model, but neither tells you exactly how well the model would superpose on the true structure.

Rough guide as to expected RMSD for a residue if you have the pLDDT

Here is a table that gives you an idea of the RMSD of CA atoms in an AlphaFold prediction to a superposed deposited model (see Terwilliger et al., 2023):

pLDDT Median prediction error % with error over 2 Å

pLDDT > 90	0.6 Å	10%
pLDDT 80-90	1.1 Å	22%
pLDDT 70-80	1.5 Å	33%
pLDDT < 70	3.5 Å	77%

This table shows that high-confidence residues (pLDDT >90) are generally incredibly accurate, but a few (10%) still are off by more than 2 Å. Low-confidence residues (pLDDT < 70) are usually pretty far off (median of 3.5 Å) but still almost a quarter are better than 2 Å.

Examples

Standard run of predict_model

Running predict_model is easy. From the command-line you can type:

```
phenix.predict_and_build jobname=myjob seq_file=seq.dat \
  prediction_server=PhenixServer \
  stop_after_predict=True
```

Common questions

How long will it take?

A typical protein chain with 200 residues will take about 5-10 minutes to run, once it has started on the Phenix server.

Timing is more or less proportional to chain length. If you specify that more than 5 models are to be built, it will typically take longer.

If your chain has already been predicted for you or for someone else, you normally should get a result (a copy of that prediction) in about 30 seconds or less.

If the Phenix server is full (typically 6 jobs can run at once), you can use the Server button on the Phenix GUI to see the expected wait time to start a job.

What to do next after PredictModel

Normally the next step after PredictModel is running ProcessPredictedModel to trim off low-confidence residues, identify compact domains, and convert pLDDT values to pseudo-B values. If you have cryo-EM data the next step after ProcessPredictedModel is usually docking the trimmed model into your map with DockInMap or EMPlacement or automated docking and rebuilding with PredictAndBuild. If you have X-ray data, the next step after ProcessPredictedModel is usually molecular replacement with Phaser or automated molecular replacement and rebuilding with PredictAndBuild.

Note that a model produced by PredictModel is not yet aligned with your density map for cryo-EM or for your crystal unit cell in crystallography. This means you must carry out some step for placing and orienting your PredictModel model before carrying out refinement or any steps that involve comparing your model to data.

Server problems

The most common problem in running Predict model is that the Phenix server is not working as expected. Normally the first thing to try is just let the program retry (it will do this for a while normally). If that does not work, you can take the files in the packaged .tgz file (listed in the GUI output), use them to get your own prediction with any server, and put the resulting predicted models in the place specified in the GUI or program output.

Specific limitations and problems:

Literature

Accelerating crystal structure determination with iterative AlphaFold prediction. T.C. Terwilliger, P.V. Afonine, Liebschner, D., T.I. Croll, McCoy, AJ, Oeffner, RD., Williams, CJ, Poon, BK, J.S. Richardson, R.J. Read, and P.D. Adams. *Acta Cryst D* 79, 234-244 (2023). Improved AlphaFold modeling with implicit experimental information. T.C. Terwilliger, B.K. Poon, P.V. Afonine, C.J. Schlicksup, T.I. Croll, C. Millán, J.S. Richardson, R.J. Read, and P.D. Adams. *Nature Methods* 19, 1376-1382 (2022). AlphaFold predictions: great hypotheses but no match for experiment. T.C. Terwilliger, P.V. Afonine, Liebschner, D., T.I. Croll, McCoy, AJ, Oeffner, RD., Williams, CJ, Poon, BK, J.S. Richardson, R.J. Read, and P.D. Adams. *BiorXiv* (2023). Making protein folding accessible to all. M. Mirdita, K. Schütze, Y. Moriwaki, L. Heo, S. Ovchinnikov, and M. Steinegger. *Nature Methods* 19, 679–682 (2022). Highly accurate protein structure prediction with AlphaFold. J. Jumper, R. Evans, A. Pritzel, T. Green, M. Figurnov, O. Ronneberger, K. Tunyasuvunakool, R. Bates, A. Žídek, A. Potapenko, A. Bridgland, C. Meyer, S.A.A. Kohl, A.J. Ballard, A. Cowie, B. Romera-Paredes, S. Nikolov, R. Jain, J. Adler, T. Back, S. Petersen, D. Reiman, E. Clancy, M. Zielinski, M. Steinegger, M. Pacholska, T. Berghammer, S. Bodenstein, D. Silver, O. Vinyals, A.W. Senior, K. Kavukcuoglu, P. Kohli, and D. Hassabis. *Nature* 596, 583-589 (2021). MMseqs2 enables sensitive protein sequence searching for the analysis of massive data sets. M. Steinegger, and J. Söding. *Nature biotechnology* 35, 1026-1028 (2017). IDDT: a local superposition-free score for comparing protein structures and models using distance difference tests. V. Mariani, M. Biasini, A. Barbato, and T.. Schwede. *Bioinformatics* 29, 2722-2728 (2013).

- Accelerating crystal structure determination with iterative AlphaFold prediction. T.C. Terwilliger, P.V. Afonine, Liebschner, D., T.I. Croll, McCoy, AJ, Oeffner, RD., Williams, CJ, Poon, BK, J.S. Richardson, R.J. Read, and P.D. Adams. *Acta Cryst D* 79, 234-244 (2023).

- Improved AlphaFold modeling with implicit experimental information. T.C. Terwilliger, B.K. Poon, P.V. Afonine, C.J. Schlicksup, T.I. Croll, C. Millán, J.S. Richardson, R.J. Read, and P.D. Adams. *Nature Methods* 19, 1376-1382 (2022).
- AlphaFold predictions: great hypotheses but no match for experiment. T.C. Terwilliger, P.V. Afonine, Liebschner, D., T.I. Croll, McCoy, AJ, Oeffner, RD., Williams, CJ, Poon, BK, J.S. Richardson, R.J. Read, and P.D. Adams. *BiorXiv* (2023).
- Making protein folding accessible to all. M. Mirdita, K. Schütze, Y. Moriwaki, L. Heo, S. Ovchinnikov, and M. Steinegger. *Nature Methods* 19, 679–682 (2022).
- Highly accurate protein structure prediction with AlphaFold. J. Jumper, R. Evans, A. Pritzel, T. Green, M. Figurnov, O. Ronneberger, K. Tunyasuvunakool, R. Bates, A. Žídek, A. Potapenko, A. Bridgland, C. Meyer, S.A.A. Kohl, A.J. Ballard, A. Cowie, B. Romera-Paredes, S. Nikolov, R. Jain, J. Adler, T. Back, S. Petersen, D. Reiman, E. Clancy, M. Zielinski, M. Steinegger, M. Pacholska, T. Berghammer, S. Bodenstein, D. Silver, O. Vinyals, A.W. Senior, K. Kavukcuoglu, P. Kohli, and D. Hassabis. *Nature* 596, 583-589 (2021).
- MMseqs2 enables sensitive protein sequence searching for the analysis of massive data sets. M. Steinegger, and J. Söding. *Nature biotechnology* 35, 1026-1028 (2017).
- IDDT: a local superposition-free score for comparing protein structures and models using distance difference tests. V. Mariani, M. Biasini, A. Barbato, and T.. Schwede. *Bioinformatics* 29, 2722-2728 (2013).

Additional information

List of all available keywords (same as predict_and_build keywords)

jobname = " Name of this job. Used to create a directory that will hold all the working files for the run. Normally set automatically by the GUI to match the directory where the job is run: directory: PredictAndBuild_30, jobname: PredictAndBuild_30. When run on command line you can specify any 4-or-more character jobname if you want. carry_on_directory = None Name of carry-on directory (stores intermediate and final results). You can continue with a previous run by specifying this directory from a previous run. To rerun or carry on from the job in /Users/Documents/PredictAndBuild_30, you specify carry_on_directory= /Users/Documents/PredictAndBuild_30/PredictAndBuild_30_CarryOn. Then if you also specify carry_on=True, files will be read from this directory and copied to a new carry_on_directory inside the working directory. Normally set automatically by the GUI in the directory containing your job: directory: PredictAndBuild_30, jobname: PredictAndBuild_30 carry_on_directory: PredictAndBuild_30_CarryOn. NOTE: Cannot be the the current directory. NOTE: if you specify a CarryOn directory you must specify carry_on=True if you want it to be used. upload_download_directory = /Users/terwill/Downloads/ Directory to use for uploading and downloading files from Colab. Normally use your standard Downloads folder. alphafold_model_suffix = AlphaFold_cycle_1.pdb Name of AlphaFold model is <jobname>_<alphafold_model_suffix> when coming from Colab server. pae_suffix = PAE_cycle_1.jsn Name of PAE file is <jobname>_<pae_suffix> when coming from Colab server. carry_on = False Carry on with a job from where it left off. Read in all files that were created instead of making new ones. To use, run the job you want exactly as you did the first time except set carry_on=True and specify carry_on_directory=xxx where xxx is the carry_on_directory from the run you want to continue. If your previous run was /Users/Documents/PredictAndBuild_30, specify: carry_on_directory= /Users/Documents/PredictAndBuild_30/PredictAndBuild_30_CarryOn. NOTE: if you specify a CarryOn directory you must also specify carry_on=True (False is the default) if you want this to be used. prediction_server = *PhenixServer Colab ManualPrediction LocalServer Choose where prediction

calculations will be done. Normally use the Phenix server (PhenixServer). As a backup you can use Colab (you will need a Google account.) Use ManualPrediction if you are going to do the prediction off-line and copy the result to the location specified by PredictAndBuild. LocalServer is used internally only.
`run_directly = False` Run directly. Used internally only
`max_len_at_server = 1500` Maximum length of sequence at server. Used internally only
`dummy_run = False` Write dummy file. Used internally only
`run_server_test = False` Run test of server and stop (does nothing else and does not require any inputs)
`copy_of_final_model = None` Copy final model here. Used internally and by predict_model. (Also copy map files to same directory). If None, copy to working directory. Use `copy_of_final_model=null` to not write these files at all.
`copy_of_msa = None` Copy final msa here (only if there is just one). Used internally
`copy_of_pae = None` Copy final pae here (only if there is just one). Used internally
`predicted_model_prefix = None` Predicted models will have this prefix. Used by predict_model.
`job_title = None` Job title in PHENIX GUI, not used on command line
`predict_and_build_max_rmsd_to_predict_together = 2` Two models are considered similar (and not worth predicting separately) if they have an rmsd after superposition of this value or less (A)
`template_file = None` Template model to be used in prediction.
`get_msa_with_mmseqs2 = True` Run the mmseqs2 server from this computer to get MSA if not already available. Alternatives are to supply and MSA or to run the mmseqs2 server from the Phenix server or Colab.
`combine_rebuilt_and_autobuild = True` Combine rebuilt and autobuild models and use if better Rfree (X-ray only)....

- `jobname = "` Name of this job. Used to create a directory that will hold all the working files for the run. Normally set automatically by the GUI to match the directory where the job is run: directory: PredictAndBuild_30, jobname: PredictAndBuild_30. When run on command line you can specify any 4-or-more character jobname if you want.
- `carry_on_directory = None` Name of carry-on directory (stores intermediate and final results). You can continue with a previous run by specifying this directory from a previous run. To rerun or carry on from the job in /Users/Documents/PredictAndBuild_30, you specify `carry_on_directory= /Users/Documents/PredictAndBuild_30/PredictAndBuild_30_CarryOn`. Then if you also specify `carry_on=True`, files will be read from this directory and copied to a new `carry_on_directory` inside the working directory. Normally set automatically by the GUI in the directory containing your job: directory: PredictAndBuild_30, jobname: PredictAndBuild_30 `carry_on_directory: PredictAndBuild_30_CarryOn`. NOTE: Cannot be the the current directory. NOTE: if you specify a CarryOn directory you must specify `carry_on=True` if you want it to be used.
- `upload_download_directory = /Users/terwill/Downloads/` Directory to use for uploading and downloading files from Colab. Normally use your standard Downloads folder.
- `alphafold_model_suffix = AlphaFold_cycle_1.pdb` Name of AlphaFold model is <jobname>_<alphafold_model_suffix> when coming from Colab server.
- `pae_suffix = PAE_cycle_1.jsn` Name of PAE file is <jobname>_<pae_suffix> when coming from Colab server.
- `carry_on = False` Carry on with a job from where it left off. Read in all files that were created instead of making new ones. To use, run the job you want exactly as you did the first time except set `carry_on=True` and specify `carry_on_directory=xxx` where xxx is the `carry_on_directory` from the run you want to continue. If your previous run was /Users/Documents/PredictAndBuild_30, specify: `carry_on_directory= /Users/Documents/PredictAndBuild_30/PredictAndBuild_30_CarryOn`. NOTE: if you specify a CarryOn directory you must also specify `carry_on=True` (False is the default) if you want this to be used.
- `prediction_server = *PhenixServer` Colab ManualPrediction LocalServer Choose where prediction calculations will be done. Normally use the Phenix server (PhenixServer). As a backup you can use Colab (you will need a Google account.) Use ManualPrediction if you are going to do the prediction off-line and copy the result to the location specified by PredictAndBuild. LocalServer is used internally

only.

- `run_directly` = False Run directly. Used internally only
- `max_len_at_server` = 1500 Maximum length of sequence at server. Used internally only
- `dummy_run` = False Write dummy file. Used internally only
- `run_server_test` = False Run test of server and stop (does nothing else and does not require any inputs)
- `copy_of_final_model` = None Copy final model here. Used internally and by `predict_model`. (Also copy map files to same directory). If None, copy to working directory. Use `copy_of_final_model=null` to not write these files at all.
- `copy_of_msa` = None Copy final msa here (only if there is just one). Used internally
- `copy_of_pae` = None Copy final pae here (only if there is just one). Used internally
- `predicted_model_prefix` = None Predicted models will have this prefix. Used by `predict_model`.
- `job_title` = None Job title in PHENIX GUI, not used on command line
- `predict_and_buildmax_rmsd_to_predict_together` = 2 Two models are considered similar (and not worth predicting separately) if they have an rmsd after superposition of this value or less (A)
- `template_file` = None Template model to be used in prediction.
- `get_msa_with_mmseqs2` = True Run the mmseqs2 server from this computer to get MSA if not already available. Alternatives are to supply an MSA or to run the mmseqs2 server from the Phenix server or Colab.
- `combine_rebuilt_and_autobuild` = True Combine rebuilt and autobuild models and use if better Rfree (X-ray only).
- `include_side_in_templates` = False Include side-chains beyond CB in templates as well as just main and CB (two runs).
- `only_include_side_in_templates` = None Include side-chains beyond CB in templates only (one run)
- `use_templates_in_prediction` = True Use templates if available (i.e., after first cycle)
- `required_free_r_to_keep` = 0.49 Skip refinements if R is worse than this value
- `autobuild_resolution` = 3.5 Worst resolution where autobuild density modification may be useful.
- `autobuild_satisfactory_r_value` = 0.25 Skip autobuild if R is this value or better
- `autobuild_for_xray_density_modification` = None Use autobuild rebuilding for density modification. Default is True if resolution is `autobuild_resolution` or better.
- `autobuild_n_cycle_rebuild_max` = 3 Maximum autobuild rebuild cycles
- `autobuild_n_cycle_build_max` = 1 Maximum autobuild build cycles
- `autobuild_n_cycle_refine` = 3 AutoBuild refinement cycles
- `autobuild_quick` = False Run AutoBuild in quick mode
- `autobuild_min_resolution` = 2 Minimum resolution for autobuilding. Resolution will be cut off at this value if it is better (smaller number).
- `density_modify_at_start` = None Density modify cryo-EM maps before analysis. Requires two half maps. Default is True if half maps are supplied and no full map is supplied, otherwise False. Works best if maps contain entire structure with boundary (at least 5 Å) or are original full-sized maps.
- `density_modify_with_model` = None Density modify using rebuilt model each cycle. default is True. Requires two half-maps for cryo-EM. Works best for cryo-EM if maps contain entire structure with boundary (at least 5 Å) or are original full-sized maps. Default is True for X-ray and False for cryo-EM
- `vrms_value` = None Value of VRMS to use in molecular replacement. This value will be used for all chains if set. Default is to estimate this value for each chain from the average pLDDT value after trimming.
- `large_sequence` = 1500 Length of sequence that requires `job_size=large`
- `pause_after_mr` = False Pause after MR and refinement (X-ray only). You can go on after checking your results.
- `pause_after_docking` = False Pause after docking and refinement (Cryo-EM only). You can go on after checking your results.
- `stop_after_predict` = False Stop after prediction. No map file needed if set.
- `stop_after_mr` = False Stop after MR.
- `parallel_predictions` = 1 Number of predictions to run at once. If more than one, no feedback is provided until the end. If there are waiting jobs in the queue running more than one at a time does not help (your first job may go near the front of the line but the others stay at the end until the first one finishes).
- `write_prediction_files_and_wait` = False Write prediction files in working

directory and wait for result to appear. Only used for servers.put_files_for_prediction_here = None
 Program will put prediction files (pdb, msa) in this directory for the rest server to read (may look like
 /app/xxx/xxx)files_for_prediction_will_be_transferred_here = None Program will expect prediction files
 (pdb, msa) from the put_files_for_prediction_here directory will be copied to this directory so the rest
 server can read them. Your system has to automatically do this transfer.precalculated_msas_dir =
 /app/results/precaculated_msas Location of precalculated msas (usually on
 server)precaculated_results_dir = /app/results/precaculated_results Location of precalculated results
 (usually on server)precaculated_pae_dir = /app/results/precac...

- max_rmsd_to_predict_together = 2 Two models are considered similar (and not worth predicting separately) if they have an rmsd after superposition of this value or less (A)
- template_file = None Template model to be used in prediction.
- get_msa_with_mmseqs2 = True Run the mmseqs2 server from this computer to get MSA if not already available. Alternatives are to supply and MSA or to run the mmseqs2 server from the Phenix server or Colab.
- combine_rebuilt_and_autobuild = True Combine rebuilt and autobuild models and use if better Rfree (X-ray only).
- include_side_in_templates = False Include side-chains beyond CB in templates as well as just main and CB (two runs).
- only_include_side_in_templates = None Include side-chains beyond CB in templates only (one run)
- use_templates_in_prediction = True Use templates if available (i.e., after first cycle)
- required_free_r_to_keep = 0.49 Skip refinements if R is worse than this value
- autobuild_resolution = 3.5 Worst resolution where autobuild density modification may be useful.
- autobuild_satisfactory_r_value = 0.25 Skip autobuild if R is this value or better
- autobuild_for_xray_density_modification = None Use autobuild rebuilding for density modification. Default is True if resolution is autobuild_resolution or better.
- autobuild_n_cycle_rebuild_max = 3 Maximum autobuild rebuild cycles
- autobuild_n_cycle_build_max = 1 Maximum autobuild build cycles
- autobuild_n_cycle_refine = 3 AutoBuild refinement cycles
- autobuild_quick = False Run AutoBuild in quick mode
- autobuild_min_resolution = 2 Minimum resolution for autobuilding. Resolution will be cut off at this value if it is better (smaller number).
- density_modify_at_start = None Density modify cryo-EM maps before analysis. Requires two half maps. Default is True if half maps are supplied and no full map is supplied, otherwise False. Works best if maps contain entire structure with boundary (at least 5 A) or are original full-sized maps.
- density_modify_with_model = None Density modify using rebuilt model each cycle. default is True. Requires two half-maps for cryo-EM. Works best for cryo-EM if maps contain entire structure with boundary (at least 5 A) or are original full-sized maps. Default is True for X-ray and False for cryo_EM
- vrms_value = None Value of VRMS to use in molecular replacement. This value will be used for all chains if set. Default is to estimate this value for each chain from the average pLDDT value after trimming.
- large_sequence = 1500 Length of sequence that requires job_size=large

- `pause_after_mr = False` Pause after MR and refinement (X-ray only). You can go on after checking your results.
- `pause_after_docking = False` Pause after docking and refinement (Cryo-EM only). You can go on after checking your results.
- `stop_after_predict = False` Stop after prediction. No map file needed if set.
- `stop_after_mr = False` Stop after MR.
- `parallel_predictions = 1` Number of predictions to run at once. If more than one, no feedback is provided until the end. If there are waiting jobs in the queue running more than one at a time does not help (your first job may go near the front of the line but the others stay at the end until the first one finishes).
- `write_prediction_files_and_wait = False` Write prediction files in working directory and wait for result to appear. Only used for servers.
- `put_files_for_prediction_here = None` Program will put prediction files (pdb, msa) in this directory for the rest server to read (may look like `/app/xxx/xxx`)
- `files_for_prediction_will_be_transferred_here = None` Program will expect prediction files (pdb, msa) from the `put_files_for_prediction_here` directory will be copied to this directory so the rest server can read them. Your system has to automatically do this transfer.
- `precalculated_msas_dir = /app/results/precalculated_msas` Location of precalculated msas (usually on server)
- `precalculated_results_dir = /app/results/precalculated_results` Location of precalculated results (usually on server)
- `precalculated_pae_dir = /app/results/precalculated_pae` Location of precalculated pae files (usually on server)
- `allow_precalculated_result_file = True` Allow using precalculated results if inputs are identical
- `allow_precalculated_msa_file = True` Allow using precalculated msas if inputs are identical
- `files_from_prediction_may_be_found_here = None` Look for output from prediction here (in case full path is not available)
- `rest_server_incoming = None` Location on REST server where files should be placed
- `max_wait_time = 100000` Maximum wait time for `write_prediction_files_and_wait` (sec)
- `stop_if_internet_not_available = True` Stop if attempting to predict models online and internet is not available
- `number_of_models = 5` Number of models to generate for each chain. Normal way to set the value of `random_seed_iterations`. If a good pLDDT is found, no more models are created, where good is `good_enough_plddt` (typically 80)
- `use_supplied_random_seed_in_prediction = False` Use random seed to set up prediction (default is to use a standard random seed).
- `update_unique_sequences = True` Update unique sequences based on MR or docking results
- `minimum_residues = 20` Minimum residues in a chain
- `maximum_residues = 10000` Maximum residues in a sequence that can be used for prediction
- `break_into_chunks_if_length_is = 1500` If a sequence is less than `maximum_residues` but at least `break_into_chunks_if_length_is`, break it into chunks of length `chunk_size` with overlap of `overlap_size`, then reassemble using overlap window of `overlap_match_size`.

- `chunk_size = 600` If a sequence is less than `maximum_residues` but at least `break_into_chunks_if_length_is`, break it into chunks of length `chunk_size` with overlap of `overlap_size`, then reassemble using overlap window of `overlap_match_size`.
- `overlap_size = 200` If a sequence is less than `maximum_residues` but at least `break_into_chunks_if_length_is`, break it into chunks of length `chunk_size` with overlap of `overlap_size`, then reassemble using overlap window of `overlap_match_size`.
- `overlap_match_size = 50` If a sequence is less than `maximum_residues` but at least `break_into_chunks_if_length_is`, break it into chunks of length `chunk_size` with overlap of `overlap_size`, then reassemble using overlap window of `overlap_match_size`.
- `morph_reassembled_model = True` If `break_into_chunks_if_length_is` is applied, after reassembling identify close domains that were in different chunks and repredict them to get their relationships. Then adjust the reassembly to minimize overlaps and match pair relationships. NOTE: This reduces overlaps and can improve domain relationships but can yield distortions of the predicted model.
- `minimum_cycles = 2` Minimum cycles to carry out
- `use_input_scaffold_if_supplied = True` If input scaffold is provided, use it throughout (all cycles)
- `default_maxit_dir = /app/results/rest/` Default place to look for maxit if not supplied. This directory should contain the whole maxit-v11.100-prod-src
- `skip_copying_files = False` Skip copying output files to GUI directory
- `refinementmacro_cycles = 3` Refinement macro_cycles
- `macro_cycles = 3` Refinement macro_cycles
- `rest_serverurl = None` The URL for the Phenix REST server Normally set automatically
`url_type = *prediction ai` Type of server url (prediction or ai). Normally set automatically
`port = None` The port for interacting with the Phenix REST server Normally set automatically
`token = None` Authentication token for accessing the Phenix REST server. Normally set automatically
`timeout = 5` Time to keep trying to connect to a server
`quick_check_interval = 1` Time in seconds between job status checks
`check_interval = 60` Time in seconds between job status checks
`max_tries = 10` Number of tries to get result from server
`max_tries_on_availability = 2` Number of tries to see if server is up
`stop_if_server_not_available = True` Stop if server is not available (wrong url or down)
`job_size = small *medium large` Size of job (small, medium, large)
`requires_gpu = True` Job requires GPU
`running_server_test = False` This is a test
`verbose = False` Verbose output
- `url = None` The URL for the Phenix REST server Normally set automatically
- `url_type = *prediction ai` Type of server url (prediction or ai). Normally set automatically
- `port = None` The port for interacting with the Phenix REST server Normally set automatically
- `token = None` Authentication token for accessing the Phenix REST server. Normally set automatically
- `timeout = 5` Time to keep trying to connect to a server
- `quick_check_interval = 1` Time in seconds between job status checks
- `check_interval = 60` Time in seconds between job status checks
- `max_tries = 10` Number of tries to get result from server
- `max_tries_on_availability = 2` Number of tries to see if server is up
- `stop_if_server_not_available = True` Stop if server is not available (wrong url or down)
- `job_size = small *medium large` Size of job (small, medium, large)
- `requires_gpu = True` Job requires GPU

- `running_server_test = False` This is a test job

- `verbose = False` Verbose output

- `input_filesseq_file = None` Input sequence file. Required if no models supplied. Used to create predicted models. Format should be Fasta or simple sequences separated by blank lines. One sequence per chain to be generated. Must match input models if both are supplied. (Fasta format or sequences separated by blank lines)
`xray_data_file = None` Input X-ray data file (MTZ format).
`xray_data_label = None` Label specifying which data column to use from `xray_data_file`
`xray_test_flag_label = None` Label specifying which test flag column (if any) to use from `xray_data_file`
`truncate_models_to_match_sequences = True` If input sequences start after or end before supplied predicted models, trim the predicted models to match sequences
`density_select = None` If set, trim map to region containing molecule (non-zero region) before use. Recommended for cryo-EM maps that have a large map that is mostly unused. Default is `True` if map is full-size and `False` if not.
`predicted_model = None` One or more input models in one or more files (normally predicted models) to be placed. Normally should start at residue 1 and match sequence file exactly.
`b_value_field_is = *plddt rmsd b_value` The B-factor field in predicted models can be pLDDT (confidence, 0-1 or 0-100) or rmsd (Å) or a B-factor. Only applies to protein chains (always B-factor for non-protein).
`msa_list = None` Multiple sequence alignment (MSA) file. Format is a3m only. First sequence in an MSA file must match a sequence in the sequence file exactly.
`models_are_already_placed = False` You can specify that all models (predicted_model or model) are already placed (docked, placed in unit cell). No docking or MR is done if so. Equivalent to specifying a scaffold_model with the same contents as the model.
`model_copies_list = None` List of number of copies to find of each model in predicted_model. Default is 1 (must specify all or none of them) . Specify all together like `model_copies_list=1,2,1,1`
`processed_model_file = None` Processed model file (e.g., from `phenix.process_predicted_model`). If supplied, skip the `process_predicted_model` step and use this file. This model is expected to have one chain for each domain and actual B-values in the B-value field. It is expected that all poorly-predicted residues have been removed.
`docked_model_file = None` Docked model (e.g., output of `dock_processed_model`). Not available in `predict_and_build` GUI. If supplied, use this file as the docked model. (Skip `process_predicted_model` and docking steps This model is expected to have a single chain with gaps for parts of the model that are not accurately known from the prediction. NOTE: You still need to supply the predicted model as the input model for this procedure in addition to this docked model.
`morphed_model_file = None` Morphed model (e.g., full model morphed to match output of `dock_processed_model`). If supplied, skip the `process_predicted_model` and docking steps and use this file as the morphed model. This model is expected to have a single chain with no gaps. NOTE: You do not need to supply the predicted model as the input model for this procedure in addition to this morphed model.
`previous_model_file = None` Previous model (from a previous run or external). Must match sequence of working model exactly. Used as a source of possible model information and as a hypothesis for docking of working model. If `docked_model_file` is also supplied, `previous_model` is used only as source of model information.
`scaffold_model = None` Scaffold model. If supplied, used as target for docking of each chain in predicted_model or model. Must be similar in sequence to models to be docked. If supplied, `model_copies_list` is ignored and docking is done by superposition onto this structure instead of docking into density or MR. If the scaffold model does not have all chains represented, they are added in by docking.
`scaffold_minimum_chain_cc = 0.35` Chains in scaffold model that have `scaffold_minimum_cc` are not re-docked, but those with lower CC are.
`base_scaffold_...`

- `seq_file = None` Input sequence file. Required if no models supplied. Used to create predicted models. Format should be Fasta or simple sequences separated by blank lines. One sequence per chain to be generated. Must match input models if both are supplied. (Fasta format or sequences separated by blank lines)

- `xray_data_file = None` Input X-ray data file (MTZ format).

- `xray_data_label` = None Label specifying which data column to use from `xray_data_file`
- `xray_test_flag_label` = None Label specifying which test flag column (if any) to use from `xray_data_file`
- `truncate_models_to_match_sequences` = True If input sequences start after or end before supplied predicted models, trim the predicted models to match sequences
- `density_select` = None If set, trim map to region containing molecule (non-zero region) before use. Recommended for cryo-EM maps that have a large map that is mostly unused. Default is True if map is full-size and False if not.
- `predicted_model` = None One or more input models in one or more files (normally predicted models) to be placed. Normally should start at residue 1 and match sequence file exactly.
- `b_value_field_is` = *plddt rmsd b_value The B-factor field in predicted models can be pLDDT (confidence, 0-1 or 0-100) or rmsd (Å) or a B-factor. Only applies to protein chains (always B-factor for non-protein).
- `msa_list` = None Multiple sequence alignment (MSA) file. Format is a3m only. First sequence in an MSA file must match a sequence in the sequence file exactly.
- `models_are_already_placed` = False You can specify that all models (`predicted_model` or `model`) are already placed (docked, placed in unit cell). No docking or MR is done if so. Equivalent to specifying a `scaffold_model` with the same contents as the model.
- `model_copies_list` = None List of number of copies to find of each model in `predicted_model`. Default is 1 (must specify all or none of them) . Specify all together like `model_copies_list=1,2,1,1`
- `processed_model_file` = None Processed model file (e.g., from `phenix.process_predicted_model`). If supplied, skip the `process_predicted_model` step and use this file. This model is expected to have one chain for each domain and actual B-values in the B-value field. It is expected that all poorly-predicted residues have been removed.
- `docked_model_file` = None Docked model (e.g., output of `dock_processed_model`). Not available in `predict_and_build` GUI. If supplied, use this file as the docked model. (Skip `process_predicted_model` and docking steps This model is expected to have a single chain with gaps for parts of the model that are not accurately known from the prediction. NOTE: You still need to supply the predicted model as the input model for this procedure in addition to this docked model.
- `morphed_model_file` = None Morphed model (e.g., full model morphed to match output of `dock_processed_model`). If supplied, skip the `process_predicted_model` and docking steps and use this file as the morphed model. This model is expected to have a single chain with no gaps. NOTE: You do not need to supply the predicted model as the input model for this procedure in addition to this morphed model.
- `previous_model_file` = None Previous model (from a previous run or external). Must match sequence of working model exactly. Used as a source of possible model information and as a hypothesis for docking of working model. If `docked_model_file` is also supplied, `previous_model` is used only as source of model information.
- `scaffold_model` = None Scaffold model. If supplied, used as target for docking of each chain in `predicted_model` or `model`. Must be similar in sequence to models to be docked. If supplied, `model_copies_list` is ignored and docking is done by superposition onto this structure instead of docking into density or MR. If the scaffold model does not have all chains represented, they are added in by docking.
- `scaffold_minimum_chain_cc` = 0.35 Chains in scaffold model that have `scaffold_minimum_cc` are not re-docked, but those with lower CC are.

- `base_scaffold_minimum_chain_cc = -1` Value of `scaffold_minimum_chain_cc` in cases where scaffold is known to be good.
- `fragments_model_file = None` Fragments model (e.g., `map_to_model.pdb`). Used as a source of possible model information and as a hypothesis for docking of working model. If `docked_model_file` is also supplied, `fragments_model` is used only as source of model information.
- `model_to_rebuild_file = None` Model to be rebuilt. Must be already in place (already docked). Must match supplied sequences exactly.
- `symmetry_file = None` Symmetry file (.ncs_spec format or MATRX records) with reconstruction symmetry. Used in identification of unique part of map. NOTE: symmetry file applies to map file in original position. NOTE 2: only proper symmetry (point-group, helical) is allowed NOTE 3: Only applies to cryo-EM maps
- `pae_file = None` Optional input json file with matrix of inter-residue estimated errors (pae file)
- `distance_model_file = None` Distance_model_file. A PDB or mmCIF file containing the model corresponding to the PAE matrix. Only needed if `weight_by_ca_ca_distances` is True.
- `search_model_copies = None` Used internally only
- `search_model = None` used internally only
- `map_modelfull_map = None` Input map file (for cryo-EM). This can be a boxed or masked map showing just the molecule to dock (best) or a full map with symmetry. If your map has symmetry be sure to set `asymmetric_map = False`. If you have a map with symmetry you can supply a symmetry file if you want. Otherwise symmetry will be automatically determined.
`half_map = None` Input half map files. Usually supply one full map or 2 half maps
`model = None` Input predicted model (e.g., AlphaFold model). Assumed to have pLDDT values in B-value field (or RMSD values). May have multiple chains. Normally use `predicted_model` instead.
- `full_map = None` Input map file (for cryo-EM). This can be a boxed or masked map showing just the molecule to dock (best) or a full map with symmetry. If your map has symmetry be sure to set `asymmetric_map = False`. If you have a map with symmetry you can supply a symmetry file if you want. Otherwise symmetry will be automatically determined.
- `half_map = None` Input half map files. Usually supply one full map or 2 half maps
- `model = None` Input predicted model (e.g., AlphaFold model). Assumed to have pLDDT values in B-value field (or RMSD values). May have multiple chains. Normally use `predicted_model` instead.
- `output_filesoutput_model_prefix = None` Output files with superposed models will begin with this prefix
`output_seq_file = None` Sequence file (possibly edited) written to this file by PredictAndBuild
`pdb_out = None` Used internally only
`temp_dir = None` Temporary directory. Default is `dock_and_build_xx` where xx creates new directory
- `output_model_prefix = None` Output files with superposed models will begin with this prefix
- `output_seq_file = None` Sequence file (possibly edited) written to this file by PredictAndBuild
- `pdb_out = None` Used internally only
- `temp_dir = None` Temporary directory. Default is `dock_and_build_xx` where xx creates new directory
- `crystal_inforesolution = None` High-resolution limit for main search. This can be lower resolution than the data. The search is quicker at lower resolution. If your model is poor, try 2-3 A lower resolution than your data (i.e, if your data is 2.5 A, try 5 A).
`scattering_table = *None` n_gaussian wk1995 it1992 electron neutron Choice of scattering table for structure factor calculations. Default for X-ray is n_gaussian, for cryoEM is electron.
`wrapping = None` You can specify whether the map is wrapped (can map values

outside bounds to inside with cell translations).asymmetric_map = None Specifies that this is an asymmetric map and no symmetry is to be supplied or found. Alternative to supplying a symmetry file or symmetry. Applies to cryo-EM reconstructions only.solvent_content = None Solvent fraction (content) of the cell. You can specify the fraction of the volume of this cell that is taken up by the macromolecule. Normally set automatically. Values go from 0 to 1.pdb70_text = None Database of PDB entries to be used as templates (pdb70 file). Normally used internally onlysequence = None Old-style sequence string (alternative to sequence file).chain_type = *PROTEIN DNA RNA Chain typemsa = None MSA as text. Supply MSA as text or as an msa_file. Used internallytemplates_as_string = None Templates as string (single PDB file with one or more chains)space_group = None Space group (normally read from the data file, applies to X-ray)space_group_alternatives = *HAND ALL NONE Space group alternatives for MR: ALL: all space groups in point group HAND: enantiomer and listed space group LIST: list supplied (not available) NONE: only supplied space groupunit_cell = None Unit Cell (normally read from the data file, applies to X-ray)unique_sequencesFile names and data labels.sequence = "Enter or edit sequence" copies = 1 Copies of this sequencelabel = ""

- resolution = None High-resolution limit for main search. This can be lower resolution than the data. The search is quicker at lower resolution. If your model is poor, try 2-3 Å lower resolution than your data (i.e, if your data is 2.5 Å, try 5 Å).
- scattering_table = *None n_gaussian wk1995 it1992 electron neutron Choice of scattering table for structure factor calculations. Default for X-ray is n_gaussian, for cryoEM is electron.
- wrapping = None You can specify whether the map is wrapped (can map values outside bounds to inside with cell translations).
- asymmetric_map = None Specifies that this is an asymmetric map and no symmetry is to be supplied or found. Alternative to supplying a symmetry file or symmetry. Applies to cryo-EM reconstructions only.
- solvent_content = None Solvent fraction (content) of the cell. You can specify the fraction of the volume of this cell that is taken up by the macromolecule. Normally set automatically. Values go from 0 to 1.
- pdb70_text = None Database of PDB entries to be used as templates (pdb70 file). Normally used internally only
- sequence = None Old-style sequence string (alternative to sequence file).
- chain_type = *PROTEIN DNA RNA Chain type
- msa = None MSA as text. Supply MSA as text or as an msa_file. Used internally
- templates_as_string = None Templates as string (single PDB file with one or more chains)
- space_group = None Space group (normally read from the data file, applies to X-ray)
- space_group_alternatives = *HAND ALL NONE Space group alternatives for MR: ALL: all space groups in point group HAND: enantiomer and listed space group LIST: list supplied (not available) NONE: only supplied space group
- unit_cell = None Unit Cell (normally read from the data file, applies to X-ray)
- unique_sequencesFile names and data labels.sequence = "Enter or edit sequence" copies = 1 Copies of this sequencelabel = ""
- sequence = "Enter or edit sequence"
- copies = 1 Copies of this sequence
- label = ""
- iterationcycles = None Cycles of prediction and rebuilding (default is 10 for Thorough, 3 for Standard and 1 for Quickcycle = 1 Iteration cycle

- cycles = None Cycles of prediction and rebuilding (default is 10 for Thorough, 3 for Standard and 1 for Quick
- cycle = 1 Iteration cycle
- process_predicted_modelremove_low_confidence_residues = True Remove low-confidence residues (based on minimum plddt or maximum_rmsd, whichever is specified)continuous_chain = False When removing low-confidence residues, only trim from endssplit_model_by_compact_regions = True Split model into compact regions after removing low-confidence residues.maximum_domains = 3 Maximum domains to obtain. You can use this to merge the closest domains at the end of splitting the model. Make it bigger (and optionally make domain_size smaller) to get more domains. If model is processed in chunks, maximum_domains will apply to each chunk.domain_size = 15 Approximate size of domains to be found (A units). This is the resolution that will be used to make a domain map. If you are getting too many domains, try making domain_size bigger (maximum is 70 A).adjust_domain_size = True If more than maximum_domains are initially found, increase domain_size in increments of 5 A and take the value that gives the smallest number of domains, but at least maximum_domains.minimum_domain_length = None Minimum length of a domain to keep (reject at end if smaller). Default is 10 if no pae matrix or alt_pae_params=False and 20 for pae_matrix with alt_pae_params=Truemaximum_fraction_close = 0.3 Maximum fraction of CA in one domain close to one in another before mergingthemminimum_sequential_residues = None Minimum length of a short segment to keep (reject at end). Default is 5 if no pae matrix or alt_pae_params=False and 4 for pae_matrix with alt_pae_params=Trueminimum_remainder_sequence_length = 15 used to choose whether the sequence of a removed segment is written to the remainder sequence file.b_value_field_is = *plddt rmsd b_value The B-factor field in predicted models can be pLDDT (confidence, 0-1 or 0-100) or rmsd (A) or a B-factorinput_plddt_is_fractional = None You can specify if the input plddt values (in B-factor field) are fractional (0-1) or not (0-100). By default if all values are between 0 and 1 it is fractional.minimum_plddt = None If low-confidence residues are removed, the cutoff is defined by minimum_plddt or maximum_rmsd, whichever is defined (you cannot define both). A minimum plddt of 0.70 corresponds to a maximum rmsd of 1.5. Minimum plddt values are fractional or not depending on the value of input_plddt_is_fractional.maximum_rmsd = 1.5 If low-confidence residues are removed, the cutoff is defined by minimum_plddt or maximum_rmsd, whichever is defined (you cannot define both). A minimum plddt of 0.70 corresponds to a maximum rmsd of 1.5. Minimum plddt values are fractional or not depending on the value of input_plddt_is_fractional.default_maximum_rmsd = 1.5 Default value of maximum_rmsd, used if maximum_rmsd is not setsubtract_minimum_b = False If set, subtract the lowest B-value from all B-values just before writing out the final files. Does not affect the cutoff for removing low-confidence residues.pae_power = None If PAE matrix (predicted alignment error matrix) is supplied, each edge in the graph will be weighted proportional to $(1/\text{pae}^{\text{pae_power}})$. Use this to try and get the number of domains that you want (try 1, 0.5, 1.5, 2). Default is 1 alt_pae_params=False and 2 for alt_pae_params=Truepae_cutoff = None If PAE matrix (predicted alignment error matrix) is supplied, graph edges will only be created for residue pairs with $\text{pae} < \text{pae_cutoff}$. Default is 5 alt_pae_params=False and 4 for alt_pae_params=Truepae_graph_resolution = None If PAE matrix (predicted alignment error matrix) is supplied, pae_graph_resolution regulates how aggressively the clustering algorithm is. Smaller values lead to larger clusters. Value should be larger than zero, and values larger than 5 are unlikely to be useful, Default is 0.5 alt_pae_params=False and 4 for alt_pae_params=Truealt_pae_params = False If PAE matrix is supplied, use alternative set of defaults (minimum_domain_length=20 minimum_sequential_residues=10 pae_power=2 pae_cutoff=4 pae_graph_resolutio...
- remove_low_confidence_residues = True Remove low-confidence residues (based on minimum plddt or maximum_rmsd, whichever is specified)

- `continuous_chain = False` When removing low-confidence residues, only trim from ends
- `split_model_by_compact_regions = True` Split model into compact regions after removing low-confidence residues.
- `maximum_domains = 3` Maximum domains to obtain. You can use this to merge the closest domains at the end of splitting the model. Make it bigger (and optionally make `domain_size` smaller) to get more domains. If model is processed in chunks, `maximum_domains` will apply to each chunk.
- `domain_size = 15` Approximate size of domains to be found (Å units). This is the resolution that will be used to make a domain map. If you are getting too many domains, try making `domain_size` bigger (maximum is 70 Å).
- `adjust_domain_size = True` If more than `maximum_domains` are initially found, increase `domain_size` in increments of 5 Å and take the value that gives the smallest number of domains, but at least `maximum_domains`.
- `minimum_domain_length = None` Minimum length of a domain to keep (reject at end if smaller). Default is 10 if no pae matrix or `alt_pae_params=False` and 20 for pae_matrix with `alt_pae_params=True`
- `maximum_fraction_close = 0.3` Maximum fraction of CA in one domain close to one in another before merging them
- `minimum_sequential_residues = None` Minimum length of a short segment to keep (reject at end). Default is 5 if no pae matrix or `alt_pae_params=False` and 4 for pae_matrix with `alt_pae_params=True`
- `minimum_remainder_sequence_length = 15` used to choose whether the sequence of a removed segment is written to the remainder sequence file.
- `b_value_field_is = *plddt rmsd b_value` The B-factor field in predicted models can be pLDDT (confidence, 0-1 or 0-100) or rmsd (Å) or a B-factor
- `input_plddt_is_fractional = None` You can specify if the input plddt values (in B-factor field) are fractional (0-1) or not (0-100). By default if all values are between 0 and 1 it is fractional.
- `minimum_plddt = None` If low-confidence residues are removed, the cutoff is defined by `minimum_plddt` or `maximum_rmsd`, whichever is defined (you cannot define both). A minimum plddt of 0.70 corresponds to a maximum rmsd of 1.5. Minimum plddt values are fractional or not depending on the value of `input_plddt_is_fractional`.
- `maximum_rmsd = 1.5` If low-confidence residues are removed, the cutoff is defined by `minimum_plddt` or `maximum_rmsd`, whichever is defined (you cannot define both). A minimum plddt of 0.70 corresponds to a maximum rmsd of 1.5. Minimum plddt values are fractional or not depending on the value of `input_plddt_is_fractional`.
- `default_maximum_rmsd = 1.5` Default value of `maximum_rmsd`, used if `maximum_rmsd` is not set
- `subtract_minimum_b = False` If set, subtract the lowest B-value from all B-values just before writing out the final files. Does not affect the cutoff for removing low- confidence residues.
- `pae_power = None` If PAE matrix (predicted alignment error matrix) is supplied, each edge in the graph will be weighted proportional to $(1/\text{pae}^{\text{pae_power}})$. Use this to try and get the number of domains that you want (try 1, 0.5, 1.5, 2). Default is 1 `alt_pae_params=False` and 2 for `alt_pae_params=True`
- `pae_cutoff = None` If PAE matrix (predicted alignment error matrix) is supplied, graph edges will only be created for residue pairs with `pae < pae_cutoff`. Default is 5 `alt_pae_params=False` and 4 for `alt_pae_params=True`
- `pae_graph_resolution = None` If PAE matrix (predicted alignment error matrix) is supplied, `pae_graph_resolution` regulates how aggressively the clustering algorithm is. Smaller values lead to

larger clusters. Value should be larger than zero, and values larger than 5 are unlikely to be useful, Default is 0.5 alt_pae_params=False and 4 for alt_pae_params=True

- alt_pae_params = False If PAE matrix is supplied, use alternative set of defaults (minimum_domain_length=20 minimum_sequential_residues=10 pae_power=2 pae_cutoff=4 pae_graph_resolution=4). Standard parameters (alt_pae_params=False) are: (minimum_domain_length=10 minimum_sequential_residues=5 pae_power=1 pae_cutoff=5 pae_graph_resolution=0.5).
- weight_by_ca_ca_distance = False Adjust the edge weighting for each residue pair according to the distance between CA residues. If this is True, then distance_model can be provided, otherwise supplied model will be used. See also distance_power
- distance_power = 1 If weight_by_ca_ca_distance is True, then edge weights will be multiplied by $1/\text{distance}^{**\text{distance_power}}$.
- stop_if_no_residues_obtained = True Raise Sorry and stop if processing yields no residues
- keep_all_if_no_residues_obtained = False Keep everything if processing yields no residues
- vrms_from_rmsd_intercept = 0.25 Estimate of vrms (error in model) from pLDDT will be based on $\text{vrms_from_rmsd_intercept} + \text{vrms_from_rmsd_slope} * \text{pLDDT}$ where mean pLDDT of non-low_confidence_residues is used.
- vrms_from_rmsd_slope = 1.0 Estimate of vrms (error in model) from pLDDT will be based on $\text{vrms_from_rmsd_intercept} + \text{vrms_from_rmsd_slope} * \text{pLDDT}$ where mean pLDDT of non-low_confidence_residues is used.
- break_into_chunks_if_length_is = 1500 If a sequence is at least break_into_chunks_if_length_is, break it into chunks of length chunk_size with overlap of overlap_size for domain identification using split_model_by_compact_regions without a pae matrix
- chunk_size = 600 If a sequence is at least break_into_chunks_if_length_is, break it into chunks of length chunk_size with overlap of overlap_size for domain identification using split_model_by_compact_regions without a pae_matrix
- overlap_size = 200 If a sequence is at least break_into_chunks_if_length_is, break it into chunks of length chunk_size with overlap of overlap_size for domain identification using split_model_by_compact_regions without a pae_matrix
- searchdock_chains_individually = False Dock each chain (identified by chain_id field) individually. This is useful if the chains might have different orientations in the map compared to the search model, such as in a search model that is a predicted model. Normally used along with create_unique_chain_at_end=True for predicted models to put the model back together. create_unique_chain_at_end = None Take all docked chains and try to create one chain that has no duplicate residue numbers and that has distances between ends of fragments consistent with the number of residues between them. If symmetry has been applied, create N copies representing that symmetry. Default is True if dock_chains_individually = True and False otherwise. minimum_cc_to_keep_domain = 0.2 Do not keep domains in create_unique_chain_at_end if cc is less than minimum_cc_to_keep_domain. weight_sequential_fragments_by_distance = None Try to dock a chain in a way that minimizes the distance between its first residue and the previously-docked residue with the highest residue number less than this, and similarly for the last residue and the next available placed residue. Default is True if create_unique_chain_at_end is set. Normally use only if you have a model with a single chain and you are docking pieces of that chain. choose_better_of_individual_and_group_docking = None Dock entire search model (normally one only) and also by chain...pick better-fitting of the two for each chain low_res_search = True Try to fit by searching at low resolution first dock_with_mr = True Try

using MR to dock firstssm_search = None Try to fit by searching for secondary structure. Default is False for dock_in_map and True for dock_and_rebuild if dock_with_mr is not used.refine_cycles = 3 rigid-body refinement cyclesssm_search_min_cc = 0.30 Stop ssm search if this cc achievedbackup_resolution_cutoff_searches = 2 If initial fitting does not work, try up to this many times increasing resolution by backup_resolution_ratio each timebackup_resolution_ratio = 1.67 If initial fitting does not work, try up to backup_resolution_cutoff_searches times increasing the resolution by backup_resolution_ratio each timeresolution_radius_scale = 0.5 Resolution for low-res search will be resolution_radius_scale times the radius of gyration of the search model.align_moments = False Try to fit by aligning moments of inertia if size of density region and molecule are similarmax_radius_ratio = 2. Radius of gyration of molecule and density must be within max_radius_ratio of each otherradius_scale = 1.5 Mask for density and atoms will be radius_scale times the radius of gyration of modeluse_symmetry = True If search_model_copies or search_map_copies are the same for each model and greater than one and allow_symmetry is set, use symmetry in the map to place all copies after the first one.skip_if_low_cc = True Skip solution before rigid-body refinement if map-model CC is less than half of min_cc.rigid_body_refinement = True Run rigid-body refinement on final modelrigid_body_refinement_single_unit = True Run rigid-body refinement with just one unit (do not break up into chains)rigid_body_refinement_split_method = *chain_id segid When splitting up molecule for rigid-body refinement (if rigid_body_refinement_single_unit=False), use either chain_id or segid to split up moleculeigid_body_refinement_resolution = None Run rigid-body refinement at this resolution if specifiedappend_to_fixed_model = True Append placed search model to fixed model (if any) after searchmin_cc = 0.4 If quick run, stop if minimum CC is achieved in local search. Also always skip if starting CC_mask is less than 1/4 min_cc.run_in_boxes = True Run on sub-boxes and combine at the endtarget_box_size = 60 Try to get boxes about this big on a side (grid units)target_boxes = None Try to get this many boxes. Default is nproc unless this makes box size much smaller than target_box_sizebox_to_run = None Run only this boxbox_overlap_scale = 1 box overlap (ov...

- dock_chains_individually = False Dock each chain (identified by chain_id field) individually. This is useful if the chains might have different orientations in the map compared to the search model, such as in a search model that is a predicted model. Normally used along with create_unique_chain_at_end=True for predicted models to put the model back together.

- create_unique_chain_at_end = None Take all docked chains and try to create one chain that has no duplicate residue numbers and that has distances between ends of fragments consistent with the number of residues between them. If symmetry has been applied, create N copies representing that symmetry. Default is True if dock_chains_individually = True and False otherwise.

- minimum_cc_to_keep_domain = 0.2 Do not keep domains in create_unique_chain_at_end if cc is less than minimum_cc_to_keep_domain.

- weight_sequential_fragments_by_distance = None Try to dock a chain in a way that minimizes the distance between its first residue and the previously-docked residue with the highest residue number less than this, and similarly for the last residue and the next available placed residue. Default is True if create_unique_chain_at_end is set. Normally use only if you have a model with a single chain and you are docking pieces of that chain.

- choose_better_of_individual_and_group_docking = None Dock entire search model (normally one only) and also by chain...pick better-fitting of the two for each chain

- low_res_search = True Try to fit by searching at low resolution first

- dock_with_mr = True Try using MR to dock first

- ssm_search = None Try to fit by searching for secondary structure. Default is False for dock_in_map and True for dock_and_rebuild if dock_with_mr is not used.

- `refine_cycles` = 3 rigid-body refinement cycles
- `ssm_search_min_cc` = 0.30 Stop ssm search if this cc achieved
- `backup_resolution_cutoff_searches` = 2 If initial fitting does not work, try up to this many times increasing resolution by `backup_resolution_ratio` each time
- `backup_resolution_ratio` = 1.67 If initial fitting does not work, try up to `backup_resolution_cutoff_searches` times increasing the resolution by `backup_resolution_ratio` each time
- `resolution_radius_scale` = 0.5 Resolution for low-res search will be `resolution_radius_scale` times the radius of gyration of the search model.
- `align_moments` = False Try to fit by aligning moments of inertia if size of density region and molecule are similar
- `max_radius_ratio` = 2. Radius of gyration of molecule and density must be within `max_radius_ratio` of each other
- `radius_scale` = 1.5 Mask for density and atoms will be `radius_scale` times the radius of gyration of model
- `use_symmetry` = True If `search_model_copies` or `search_map_copies` are the same for each model and greater than one and `allow_symmetry` is set, use symmetry in the map to place all copies after the first one.
- `skip_if_low_cc` = True Skip solution before rigid-body refinement if map-model CC is less than half of `min_cc`.
- `rigid_body_refinement` = True Run rigid-body refinement on final model
- `rigid_body_refinement_single_unit` = True Run rigid-body refinement with just one unit (do not break up into chains)
- `rigid_body_refinement_split_method` = *chain_id segid When splitting up molecule for rigid-body refinement (if `rigid_body_refinement_single_unit`=False), use either `chain_id` or `segid` to split up molecule
- `rigid_body_refinement_resolution` = None Run rigid-body refinement at this resolution if specified
- `append_to_fixed_model` = True Append placed search model to fixed model (if any) after search
- `min_cc` = 0.4 If quick run, stop if minimum CC is achieved in local search. Also always skip if starting `CC_mask` is less than 1/4 `min_cc`.
- `run_in_boxes` = True Run on sub-boxes and combine at the end
- `target_box_size` = 60 Try to get boxes about this big on a side (grid units)
- `target_boxes` = None Try to get this many boxes. Default is `nproc` unless this makes box size much smaller than `target_box_size`
- `box_to_run` = None Run only this box
- `box_overlap_scale` = 1 box overlap (overlap of boxes) will be `box_overlap_scale` times the density radius
- `edge_ratio` = 10 box edge `box_overlap` times `edge_ratio`
- `density_radius` = None Radius for density to be cut out and compared. Default is 6 times the resolution.
- `model_radius` = 3 Radius for removing density near `fixed_model`
- `zero_value` = 0 Value to set map in regions overlapping fixed model
- `density_peaks` = 20 Number of NCS-related peaks of density to check
- `delta_phi` = 20 Angular spacing of search

- `max_rot` = None Maximum rotations to try
- `rotz_only` = None Rotate only around Z
- `single_positions_to_try` = 10 Number of offset positions to try in optimizing orientation. Positions along the chain are selected as centers and a local fit near each position is carried out. The resulting offsets relative to the original placement are used to optimize the overall orientation and position.
- `max_position_shift_frac` = 0.05 Maximum fractional positional shift in `single_positions` run
- `min_relative_cc` = 0.67 Minimum local CC relative to original CC to keep a local search. This is a way to reject local searches that are completely wrong.
- `sieve_fit` = None Use `sieve_fit` fraction of single positions in fitting. If None, use all
- `ncs_copies_max` = None Maximum number of matching models to write. If more than one they will be written as MODEL records in the output PDB file. You can get them individually afterwards with `phenix.pdbtools placed_model.pdb keep="model 1"` etc.
- `start_rot` = None Three numbers `rotx`, `rotz`, `roty` defining the starting rotation of the search model. Normally used along with `delta_phi=1000` or `max_rot=1` to generate exactly one defined rotation.
- `search_center` = None Optional coordinates in search model for centering search. Note this is different from `target_search_center` which is the location xyz in the map to look.
- `search_center_selection` = None Optional selection defining coordinates in search model for centering search.
- `target_search_center` = None Optional coordinates in reference map where `search_center` should be approximately located after superimposing maps. Used to eliminate possible superpositions that place the search center elsewhere. Overlap scores decreased based on distance to `target_search_center/density_radius`.
- `model_search_position` = None Optional coordinates (usually part of search model) for matching to `target_search_position` after transformation. These can be specified in addition to `target_search_center`. Can have multiple `model_search_position` and `target_search_position` pairs by specifying each multiple times in order. NOTE: Not compatible with map search
- `target_search_position` = None Target positions for `model_search_position` after transformation.
- `search_position_radius` = None Radius for comparison of `target_search_position` and transformed `search_position` values. Default is `density_radius`. If specified, must be a single value or the same number of values as entries in `model_search_position` and `target_search_position`
- `rot_id_n` = None Number of rotation groups. Along with `rot_id_group`, allows defining groups of rotations to be carried out in one run. `short_caption` = Number of rotation groups
- `rot_id_group` = None rotation group to include. See `rot_id_n`.
- `map_box` = True Run `map_box` to extract useful part of map before search
- `fix_search_position` = False You can choose to not move your model center of mass to the origin by fixing the search position
- `search_box_size` = None You can choose the size of the search box for local searches. Normally set by default corresponding to 3 times `density_radius`
- `lower_bounds` = None You can select a part of your map for analysis with `lower_bounds` and `upper_bounds`.
- `upper_bounds` = None You can select a part of your map for analysis with `lower_bounds` and `upper_bounds`.

- `keep_search_order = None` Keep search order as input
- `remove_water = False` Remove waters and other hetero atoms from input files
- `prediction_method = *alphafold` Prediction method to use
- `template_search_method = *mmseqs2` structure_search Method to identify templates from PDB and generate MSAs. If `structure_search` is set, you can specify what PDB databases to use (same as in `phenix.structure_search`). If `structure_search` is set, you must supply your own MSA file with the keyword `upload_msa_file=True` (structure_search does not generate MSAs).
- `starting_alphafold_model = None` Starting AlphaFold model. You can supply an AlphaFold model and skip the initial AlphaFold step. This is equivalent to setting up an `output_directory` with just an AlphaFold model named as the expected first AlphaFold model and specifying `carry_on=True`.
- `input_directory = ColabInputs` Input directory containing density map. The map filename must start with the same characters as the jobname (only including characters before the first underscore). If you are supplying an MSA file it goes here as well.
- `output_directory = ColabOutputs` Output directory. Copy outputs to `output_directory`. Used to restart with `carry on`.
- `save_outputs_in_google_drive = False` If run on Colab, copy outputs to `output_directory` in Google drive
- `content_dir = None` Content directory. Default is working directory
- `maxit_path = None` Path to maxit (pdb to cif) converter. Optional.
- `data_dir = /mnt` Data directory (location of AlphaFold parameters)
- `upload_file_with_jobname_resolution_sequence_lines = None` Upload a file with a set of jobs (Colab only). Each line in the file is a jobname, resolution, and sequence
- `maximum_cycles = 10` Maximum cycles to carry out
- `cycle_rmsd_to_resolution_ratio = 0.25` Stop iteration if rmsd between subsequent AlphaFold models is less than `cycle_rmsd_to_resolution_ratio` times the resolution for two cycles in a row
- `significant_increase_in_residues = 5` Significant increase in residues in model
- `password = None` Phenix download password (Colab only). The password used to download Phenix at your institution. Updated weekly, so you may need to request a new one frequently.
- `version = dev-4502` Version of Phenix to run (Colab only)
- `query_sequence = None` Sequence
- `resolution = None` Resolution of map (Å). Internal use only. Normally set instead `crystal_info.resolution`.
- `jobname = None` Name of this job. The first characters before any underscore must be unique and will define the first characters of the corresponding map file in the input directory. The job name will normally also be the name of the working directory. Used internally only
- `use_msa = True` Use multiple sequence alignments at some points
- `skip_all_msa_after_first_cycle = False` Skip multiple sequence alignments after first cycle
- `include_templates_from_pdb = True` Include templates from PDB. Note special behavior when running `predict_and_build`: applies only on first cycle, and if PhenixServer or Colab used prediction will be run with and without templates and PDB and highest plddt model will be kept
- `maximum_templates_from_pdb = 20` Maximum templates from PDB to include
- `release_date = None` release date for templates from pdb (only use up to this date. Format is 2020-05-14).
- `upload_msa_file = False` Supply MSA directly (.a3m format). You can supply the MSA as a file in your `input_directory` and it will be used instead of using `mmseqs2` to generate an MSA. Your file name must end in .a3m. The format is two lines per sequence, the first starts with a greater-than sign and is ignored, the second is a sequence of letters or minus signs. All sequence lines must have the same length. The first sequence must be the target.
- `upload_manual_templates = False` Supply templates for AlphaFold prediction. Used in the same way as templates from the PDB unless `uploaded_templates_are_map_to_model` is set. May be .cif or .pdb files. Templates must start with characters matching the first characters in the jobname (before the first underscore), the remainder of the file name can be anything but just end in .cif or .pdb. The files must be in the `input_directory` or be uploaded (in Colab only).
- `uploaded_tem...`
- `prediction_method = *alphafold` Prediction method to use
- `template_search_method = *mmseqs2` structure_search Method to identify templates from PDB and generate MSAs. If `structure_search` is set, you can specify what PDB databases to use (same as in `phenix.structure_search`). If `structure_search` is set, you must supply your own MSA file with the keyword

upload_msa_file=True (structure_search does not generate MSAs).

- starting_alphafold_model = None Starting AlphaFold model. You can supply an AlphaFold model and skip the initial AlphaFold step. This is equivalent to setting up an output_directory with just an AlphaFold model named as the expected first AlphaFold model and specifying carry_on=True.
- input_directory = ColabInputs Input directory containing density map. The map filename must start with the same characters as the jobname (only including characters before the first underscore). If you are supplying an MSA file it goes here as well.
- output_directory = ColabOutputs Output directory. Copy outputs to output_directory. Used to restart with carry on.
- save_outputs_in_google_drive = False If run on Colab, copy outputs to output_directory in Google drive
- content_dir = None Content directory. Default is working directory
- maxit_path = None Path to maxit (pdb to cif) converter. Optional.
- data_dir = /mnt Data directory (location of AlphaFold parameters)
- upload_file_with_jobname_resolution_sequence_lines = None Upload a file with a set of jobs (Colab only). Each line in the file is a jobname, resolution, and sequence
- maximum_cycles = 10 Maximum cycles to carry out
- cycle_rmsd_to_resolution_ratio = 0.25 Stop iteration if rmsd between subsequent AlphaFold models is less than cycle_rmsd_to_resolution_ratio times the resolution for two cycles in a row
- significant_increase_in_residues = 5 Significant increase in residues in model
- password = None Phenix download password (Colab only). The password used to download Phenix at your institution. Updated weekly, so you may need to request a new one frequently.
- version = dev-4502 Version of Phenix to run (Colab only)
- query_sequence = None Sequence
- resolution = None Resolution of map (Å). Internal use only. Normally set instead crystal_info.resolution.
- jobname = None Name of this job. The first characters before any underscore must be unique and will define the first characters of the corresponding map file in the input directory. The job name will normally also be the name of the working directory. Used internally only
- use_msa = True Use multiple sequence alignments at some point
- skip_all_msa_after_first_cycle = False Skip multiple sequence alignments after first cycle
- include_templates_from_pdb = True Include templates from PDB. Note special behavior when running predict_and_build: applies only on first cycle, and if PhenixServer or Colab used prediction will be run with and without templates and PDB and highest plddt model will be kept
- maximum_templates_from_pdb = 20 Maximum templates from PDB to include
- release_date = None release date for templates from pdb (only use up to this date. Format is 2020-05-14.
- upload_msa_file = False Supply MSA directly (.a3m format). You can supply the MSA as a file in your input_directory and it will be used instead of using mmseqs2 to generate an MSA. Your file name must end in .a3m. The format is two lines per sequence, the first starts with a greater-than sign and is ignored, the second is a sequence of letters or minus signs. All sequence lines must have the same length. The first sequence must be the target.

- `upload_manual_templates = False` Supply templates for AlphaFold prediction. Used in the same way as templates from the PDB unless `uploaded_templates_are_map_to_model` is set. May be .cif or .pdb files. Templates must start with characters matching the first characters in the jobname (before the first underscore), the remainder of the file name can be anything but just end in .cif or .pdb. The files must be in the `input_directory` or be uploaded (in Colab only).
- `uploaded_templates_are_map_to_model = False` The manual templates are models that may or may not have the sequence of the alphafold models. These are used only as suggestions for placement of the main chain.
- `upload_maps = True` Use maps (required to be True)
- `random_seed = 7231771` Random seed. Used internally
- `random_seed_iterations = 1` Random seed iterations of AlphaFold in first cycle. The model with the highest pLDDT will be used. If running `predict_and_build` and `include_templates_from_pdb` is True this many iterations will be carried out with and without templates . In `predict_and_build` and `predict_chain` set `number_of_models` instead.
- `minimum_random_seed_iterations = 1` Random seed iterations of AlphaFold after first cycle. The model with the highest pLDDT will be used
- `big_improvement = 10` How much improvement in pLDDT is worth going through all randomization cycles
- `good_enough_plddt = 80` Value of pLDDT that is good enough to not make any more models
- `nproc = 4` Number of processors to use. Internal use only. Normally set instead `control.nproc`
- `debug = False` Debugging run (print traceback error messages)
- `get_msa_only = False` Just get MSA and save it on server, no prediction. Does not return the MSA
- `carry_on = False` Carry on from where previous run ended. Used (usually in Colab) to go on after a crash or timeout. Requires that files are saved in the output directory
- `cif_dir = None` Location of templates (normally set automatically)
- `template_hit_list = None` List of templates (normally set automatically)
- `jobnames = None` List of jobnames (normally set automatically)
- `resolutions = None` List of resolutions (normally set automatically)
- `include_templates_from_pdb_list = None` List of `include_templates_from_pdb` ((normally set automatically)
- `include_side_in_templates_list = None` List of `include_side_in_templates_list` ((normally set automatically)
- `map_filename_dict = None` Map filename dict (used internally only)
- `msa_filename_dict = None` MSA filename dict (used internally only)
- `cif_filename_dict = None` CIF filename dict (used internally only)
- `query_sequences = None` List of query sequences (one per jobname) (used internally only)
- `manual_templates_uploaded = None` List of manual templates (used internally only)
- `upload_dir = None` Upload directory (used internally only)
- `working_directory = None` Working directory (used internally only)
- `maps_uploaded = None` List of maps (used internally only)
- `msas_uploaded = None` List of msas (used internally only)

- num_models = 1 Number of models (used internally only)
- homooligomer = 1 Number of copies (used internally only)
- cycle = None Cycle number (internal use only)
- host_url = https://api.colabfold.com Host url for mmseqs
- use_env = None Use env (internal use only)
- use_custom_msa = None Use custom msa (internal use only)
- use_templates = None Use templates (internal use only)
- template_paths = None Template paths (internal use only)
- mtm_file_name = None map_to_model file name (internal use only)
- cycle_model_file_name = None cycle_model_file_name (internal use only)
- previous_final_model_name = None previous_final_model_name (internal use only)
- msa = None MSA (internal use only)
- msa_is_msa_object = None MSA info (internal use only)
- deletion_matrix = None Deletion matrix (internal use only)
- structure_search_paramsstructure_searchpdb_file = None Enter a PDB file name sequence = None Optional Fasta sequence file. Only needed for a quick sequence search against RCSB without a PDB.
- output_prefix = 'output' Provide an output prefix if needed blastpath = None Enter path to blastall executable sequence_only = False Do a Blast search against PDBaa sequence instead of doing a Ramachandran-based structure searchstructure_only = False Do only a Ramachandran-based structure search. db_used = 'rcsb' structure database used in search. rcsb, scop95, or AF2db = 'rcsb' Database used in search. rcsb, scop95, or AF2.get_ligand = False Use get_ligand=True to retrieve ligands.get_ramcode_only = False Generate Rama code for input pdb/cif only. This is for developers only.get_xml_only = False Get BLAST XML output returned as a string object. No coordinate superposition will be performed. Developers only.use_pdb100aa = False Use PDB100 sequence database for sequence search.use_custom_db = False Use custom database specified by custom_db_files/custom_db_dir.custom_db_dir = None The directory of pdb/cif files to make custom database. Default is current directorycustom_db_files = None Filenames of the pdb/cif files separated by spaces for database. If none specified, all pdb/cif in the custom_db_dir will be collectedatom_selection = 'all' Choose part of the pdb used in the search (default=all). for example: chain B, resseq 113:219, ... etc. get_pdb = 10 get_pdb=N will collect and superpose the top N homologous pdbs. Use get_pdb=0 to disable this option.deposited_before = 0 Specify the latest year of matching structures to be considered for scoring. Pdb deposited after this year will be discarded.deposited_after = 0 Specify the earliest year of matching structures to be considered for scoring. Pdb deposited before this year will be discarded.batch_size = 0 Process the pdbs in batch of <batch_size> until <min_match> hits are identified or until all <get_pdb> pdbs are processedmin_match = 0 Finish structure_search when <min_match> matches are found. Usually uses with <trim_ends> to exit the search once find suitable pdbs.keep_all_pdb = False Keep all the PDB files, including full PDB, PDB_Chain and superposed PDB_Chain. Default is False which will keep only superposed PDB_Chain files in the directory specified in the output message.trim_ends = False Remove terminal residues of hit pdbs extending beyond those of the target pdb.write_pdb = True Set to False if no output pdb file is needed. Sometimes useful if use Structure_Search within another program and only want to pass pdb objects.write_results = True Set to False if no output results/log files is needed. Useful when calling Structure_Search within another program and only want to pass pdb objects.trim_hit_pdb = False Remove extra domains, extended loops, and unfit portions of hit pdbs after superposed to the target pdb.pickle_hits = False Pickle blast hit results

from xml output.coot_display = False (default) Display output pdb files in coot.ask_coot = True prompt for coot display optiosPDB_MIRRORDIR = None Enter the top directory of local RCSB PDB mirror. The program will try to retrieve PDBs and/or structure factors from this mirror first. Note this assumes the directory trees under it follows those in RCSB -- pdb files as 'pdb####.ent.gz' in PDB_MIRRORDIR/data/structures/divided/pdb directory. If you use PDB's rsync script, this variable would be the same as the \$MIRRORDIR set in the scriptPDB_MIRROR_MMCIF = None Enter the parent directory of the mmCIF files in the local PDB mirror. MMCIFs will be retrieved from subdirectory ## where ## are the second and third letters in the PDB id. This keyword should be \$PDB_MIRRORDIR/data/structures/divided/mmcif directory. PDB_MIRROR_PDB = None Enter the parent directory of the PDB files in the local PDB mirror. PDBs will be retrieved from subdirectory ## where ## are the second and...

- structure_searchpdb_file = None Enter a PDB file name sequence = None Optional Fasta sequence file. Only needed for a quick sequence search against RCSB without a PDB. output_prefix = 'output' Provide an output prefix if needed blastpath = None Enter path to blastall executable sequence_only = False Do a Blast search against PDBaa sequence instead of doing a Ramachandran-based structure searchstructure_only = False Do only a Ramachandran-based structure search. db_used = 'rcsb' structure database used in search. rcsb, scop95, or AF2db = 'rcsb' Database used in search. rcsb, scop95, or AF2.get_ligand = False Use get_ligand=True to retrieve ligands.get_rama_code_only = False Generate Rama code for input pdb/cif only. This is for developers only.get_xml_only = False Get BLAST XML output returned as a string object. No coordinate superposition will be performed. Developers only.use_pdb100aa = False Use PDB100 sequence database for sequence search.use_custom_db = False Use custom database specified by custom_db_files/custom_db_dir.custom_db_dir = None The directory of pdb/cif files to make custom database. Default is current directorycustom_db_files = None Filenames of the pdb/cif files separated by spaces for database. If none specified, all pdb/cif in the custom_db_dir will be collectedatom_selection = 'all' Choose part of the pdb used in the search (default=all). for example: chain B, resseq 113:219, ... etc. get_pdb = 10 get_pdb=N will collect and superpose the top N homologous pdbs. Use get_pdb=0 to disable this option.deposited_before = 0 Specify the latest year of matching structures to be considered for scoring. PDBs deposited after this year will be discarded.deposited_after = 0 Specify the earliest year of matching structures to be considered for scoring. PDBs deposited before this year will be discarded.batch_size = 0 Process the pdbs in batch of <batch_size> until <min_match> hits are identified or until all <get_pdb> pdbs are processedmin_match = 0 Finish structure_search when <min_match> matches are found. Usually uses with <trim_ends> to exit the search once find suitable pdbs.keep_all_pdb = False Keep all the PDB files, including full PDB, PDB_Chain and superposed PDB_Chain. Default is False which will keep only superposed PDB_Chain files in the directory specified in the output message.trim_ends = False Remove terminal residues of hit pdbs extending beyond those of the target pdb.write_pdb = True Set to False if no output pdb file is needed. Sometimes useful if use Structure_Search within another program and only want to pass pdb objects.write_results = True Set to False if no output results/log files is needed. Useful when calling Structure_Search within another program and only want to pass pdb objects.trim_hit_pdb = False Remove extra domains, extended loops, and unfit portions of hit pdbs after superposed to the target pdb.pickle_hits = False Pickle blast hit results from xml output.coot_display = False (default) Display output pdb files in coot.ask_coot = True prompt for coot display optiosPDB_MIRRORDIR = None Enter the top directory of local RCSB PDB mirror. The program will try to retrieve PDBs and/or structure factors from this mirror first. Note this assumes the directory trees under it follows those in RCSB -- pdb files as 'pdb####.ent.gz' in PDB_MIRRORDIR/data/structures/divided/pdb directory. If you use PDB's rsync script, this variable would be the same as the \$MIRRORDIR set in the scriptPDB_MIRROR_MMCIF = None Enter the parent directory of the mmCIF files in the local PDB mirror. MMCIFs will be retrieved from subdirectory ## where ## are the second and third letters in the PDB id. This keyword should be

\$PDB_MIRRORDIR/data/structures/divided/mmcif directory. PDB_MIRROR_PDB = None Enter the parent directory of the PDB files in the local PDB mirror. PDBs will be retrieved from subdirectory ## where ## are the second and third letters in the ...

- pdb_file = None Enter a PDB file name
- sequence = None Optional Fasta sequence file. Only needed for a quick sequence search against RCSB without a PDB.
- output_prefix = 'output' Provide an output prefix if needed
- blastpath = None Enter path to blastall executable
- sequence_only = False Do a Blast search against PDBaa sequence instead of doing a Ramachandran-based structure search
- structure_only = False Do only a Ramachandran-based structure search.
- db_used = 'rcsb' structure database used in search. rcsb, scop95, or AF2
- db = 'rcsb' Database used in search. rcsb, scop95, or AF2.
- get_ligand = False Use get_ligand=True to retrieve ligands.
- get_ramcode_only = False Generate Rama code for input pdb/cif only. This is for developers only.
- get_xml_only = False Get BLAST XML output returned as a string object. No coordinate superposition will be performed. Developers only.
- use_pdb100aa = False Use PDB100 sequence database for sequence search.
- use_custom_db = False Use custom database specified by custom_db_files/custom_db_dir.
- custom_db_dir = None The directory of pdb/cif files to make custom database. Default is current directory
- custom_db_files = None Filenames of the pdb/cif files separated by spaces for database. If none specified, all pdb/cif in the custom_db_dir will be collected
- atom_selection = 'all' Choose part of the pdb used in the search (default=all). for example: chain B, resseq 113:219, ... etc.
- get_pdb = 10 get_pdb=N will collect and superpose the top N homologous pdbs. Use get_pdb=0 to disable this option.
- deposited_before = 0 Specify the latest year of matching structures to be considered for scoring. Pdb's deposited after this year will be discarded.
- deposited_after = 0 Specify the earliest year of matching structures to be considered for scoring. Pdb's deposited before this year will be discarded.
- batch_size = 0 Process the pdbs in batch of <batch_size> until <min_match> hits are identified or until all <get_pdb> pdbs are processed
- min_match = 0 Finish structure_search when <min_match> matches are found. Usually uses with <trim_ends> to exit the search once find suitable pdbs.
- keep_all_pdb = False Keep all the PDB files, including full PDB, PDB_Chain and superposed PDB_Chain. Default is False which will keep only superposed PDB_Chain files in the directory specified in the output message.
- trim_ends = False Remove terminal residues of hit pdbs extending beyond those of the target pdb.
- write_pdb = True Set to False if no output pdb file is needed. Sometimes useful if use Structure_Search within another program and only want to pass pdb objects.

- `write_results = True` Set to False if no output results/log files is needed. Useful when calling `Structure_Search` within another program and only want to pass pdb objects.
- `trim_hit_pdb = False` Remove extra domains, extended loops, and unfit portions of hit pdbs after superposed to the target pdb.
- `pickle_hits = False` Pickle blast hit results from xml output.
- `coot_display = False` (default) Display output pdb files in coot.
- `ask_coot = True` prompt for coot display options
- `PDB_MIRRORDIR = None` Enter the top directory of local RCSB PDB mirror. The program will try to retrieve PDBs and/or structure factors from this mirror first. Note this assumes the directory trees under it follows those in RCSB -- pdb files as 'pdb####.ent.gz' in `PDB_MIRRORDIR/data/structures/divided/pdb` directory. If you use PDB's rsync script, this variable would be the same as the `$MIRRORDIR` set in the script
- `PDB_MIRROR_MMCIF = None` Enter the parent directory of the mmCIF files in the local PDB mirror. MMCIFs will be retrieved from subdirectory `##` where `##` are the second and third letters in the PDB id. This keyword should be `$PDB_MIRRORDIR/data/structures/divided/mmcif` directory.
- `PDB_MIRROR_PDB = None` Enter the parent directory of the PDB files in the local PDB mirror. PDBs will be retrieved from subdirectory `##` where `##` are the second and third letters in the PDB id. This keyword should be `$PDB_MIRRORDIR/data/structures/divided/pdb` directory. We recommend setting `PDB_MIRRORDIR` and it will take care of both `PDB_MIRROR_PDB` and others together. However, users may choose to specify `PDB_MIRROR_PDB` directly
- `PDB_MIRROR_STRUCTURE_FACTORS = None` Enter the parent directory of the PDB files in the local PDB mirror. structure factors s will be retrieved from subdirectory `##` where `##` are the second and third letters in the PDB id. This keyword should be the same as the `$PDB_MIRRORDIR/data/structures/divided/structure_factors` directory. We recommend setting `PDB_MIRRORDIR` and it will take care of both `PDB_MIRROR_PDB` and `PDB_MIRROR_STRUCTURE_FACTORS` together. However, users may choose to specify `PDB_MIRROR_STRUCTURE_FACTORS` directly
- `local_pdb_dir = None` Enter the path directly to your local PDB repository.
- `verbose = False` verbose output
- `debug = False` debugging output
- `job_title = None` Job title in PHENIX GUI, not used on command line
- guiGUI-specific parameter required for output directory `output_dir = None`
- `output_dir = None`
- `buildrebuild_strategy = Thorough` Standard Quick Rebuilding strategy. Standard is up to 3 cycles, refining and replacing poorly-fitting loops in predicted models each cycle, using autobuild for density modification (if Xray). Quick is refinement only, one cycle. Thorough is up to 10 cycles, extensive rebuilding.
`refine_only = None` Refine only, no rebuilding steps (set automatically with Quick)
`refine_only_resolution = 3.5` Default cutoff for using Refine only
`run_fit_loops = True` Run standard fit loops
`run_iterative_morph = True` Run iterative morphing
`run_trace_loops_through_density = True` Run loop fitting with trace_through_density algorithm
`run_refine = True` Refine morphed model. Required for `run_fit_loops` or `run_trace_loops_through_density`
`run_iterative_resolution_refine = True` Refine morphed model with iterative resolution method
`run_extend = True` Extend ends of morphed model
`extract_unique = True` Extract unique part of map. Applies after `density_select` (if set).
`use_symmetry_in_extract_unique = True` Use symmetry from symmetry file (if available) when extracting unique part of

map.acceptable_docking_cc = 0.5 Acceptable docking CC for a chain
 minimum_docking_cc = 0.15 Minimum docking CC for a chain
 minimum_cutoff = 0.1 Minimum cutoff for estimating density in better
 parts of model
 reasonable_cc_ratio = 0.80 Acceptable CC (ratio) for a chain relative to average. Used to
 make sure a chain entirely in very bad density is not kept
 reasonable_cc_diff = 0.15 Acceptable CC (difference) for a chain relative to average. Used to make sure a chain entirely in very bad density is not
 kept
 shift_field_distance = None Shift field characteristic distance (default = 10 Å) for morphing
 cc_sd_ratio = 3. Keep residues with CC at least within cc_sd_ratio of the mean CC for good residues
 cc_sd_ratio_end = 2. Keep residues with CC at least within cc_sd_ratio_end of the mean CC for good residues (applies to
 end of fragments)
 cc_sd_ratio_ok = 2. Residues with CC at least within cc_sd_ratio_ok of the mean
 CC for good residues are ok (do not delete on the basis of plddt)
 max_gap_ratio = 3. Allow CA-CA distance to be up to max_gap_ratio times expected
 maximum_connectivity_deviation = 15 Maximum connectivity deviation. Reject a solution with bigger deviation than this plus 2 *
 resolution.
 keep_fraction_of_best = 0.5 Acceptable CC to keep as ratio to best found. Applies if best found
 is at least acceptable_docking_cc.
 keep_maximum_entries = 10 Maximum dock positions to keep
 rmsd_for_similar_placement = None RMSD value indicating that two placements are similar. default
 is resolution of map
 rigid_body_refine_cycles = 1 Refinement cycles for rigid-body
 refinement
 overlap_ca_ca_distance = 3 Overlap distance for CA-CA atoms (or P-P)
 ca_ca_distance = 3.8 CA-CA distance (or P-P)
 allowed_fraction_overlapping = 0.10 Allowed fraction
 overlapping
 maximum_combinations = 100000 Maximum combinations of placements to
 consider
 proceed_with_any_symmetry = False Run even with symmetry that is not point-group or helical.
 Not recommended as symmetry may not work properly
 box_cushion = 20 Size of buffer around model when boxing
 refine_cycles = 3 Refinement cycles
 loop_refine_cycles = 5 Refinement cycles for loops
 loop_backup_residues = 3 Number of tries removing one residue at a time from each end of existing
 ends of loop if no loop is found with initial gap
 residues_to_trim = 5 Residues to trim on each end of all trimmed fragments
 find_ncs_from_model = True Find NCS (symmetry) from working models and apply in
 density modification. Also turns on finding NCS in any autobuild density modification (including
 density-based search)
 allow_split_more_than_one_chain_for_mr = False Allow splitting chains into domains for mr (X-ray only) even if there are multiple chains. Normally only split if a single chain as
 reassembly may not work with split multiple chains.
 acceptable_cc_ratio = 0.8 Keep segments with CC at least equal to average of base segments times minimum_cc_ratio minus difference between segment
 confidence...

- rebuild_strategy = Thorough Standard Quick Rebuilding strategy. Standard is up to 3 cycles, refining and replacing poorly-fitting loops in predicted models each cycle, using autobuild for density modification (if Xray). Quick is refinement only, one cycle. Thorough is up to 10 cycles, extensive rebuilding.
- refine_only = None Refine only, no rebuilding steps (set automatically with Quick)
- refine_only_resolution = 3.5 Default cutoff for using Refine only
- run_fit_loops = True Run standard fit_loops
- run_iterative_morph = True Run iterative morphing
- run_trace_loops_through_density = True Run loop fitting with trace_through_density algorithm
- run_refine = True Refine morphed model. Required for run_fit_loops or run_trace_loops_through_density
- run_iterative_resolution_refine = True Refine morphed model with iterative resolution method
- run_extend = True Extend ends of morphed model
- extract_unique = True Extract unique part of map. Applies after density_select (if set).

- `use_symmetry_in_extract_unique = True` Use symmetry from symmetry file (if available) when extracting unique part of map.
- `acceptable_docking_cc = 0.5` Acceptable docking CC for a chain
- `minimum_docking_cc = 0.15` Minimum docking CC for a chain
- `minimum_cutoff = 0.1` Minimum cutoff for estimating density in better parts of model
- `reasonable_cc_ratio = 0.80` Acceptable CC (ratio) for a chain relative to average. Used to make sure a chain entirely in very bad density is not kept
- `reasonable_cc_diff = 0.15` Acceptable CC (difference) for a chain relative to average. Used to make sure a chain entirely in very bad density is not kept
- `shift_field_distance = None` Shift field characteristic distance (default = 10 Å) for morphing
- `cc_sd_ratio = 3`. Keep residues with CC at least within `cc_sd_ratio` of the mean CC for good residues
- `cc_sd_ratio_end = 2`. Keep residues with CC at least within `cc_sd_ratio_end` of the mean CC for good residues (applies to end of fragments)
- `cc_sd_ratio_ok = 2`. Residues with CC at least within `cc_sd_ratio_ok` of the mean CC for good residues are ok (do not delete on the basis of plddt)
- `max_gap_ratio = 3`. Allow CA-CA distance to be up to `max_gap_ratio` times expected
- `maximum_connectivity_deviation = 15` Maximum connectivity deviation. Reject a solution with bigger deviation than this plus $2 * \text{resolution}$.
- `keep_fraction_of_best = 0.5` Acceptable CC to keep as ratio to best found. Applies if best found is at least `acceptable_docking_cc`.
- `keep_maximum_entries = 10` Maximum dock positions to keep
- `rmsd_for_similar_placement = None` RMSD value indicating that two placements are similar. default is resolution of map
- `rigid_body_refine_cycles = 1` Refinement cycles for rigid-body refinement
- `overlap_ca_ca_distance = 3` Overlap distance for CA-CA atoms (or P-P)
- `ca_ca_distance = 3.8` CA-CA distance (or P-P)
- `allowed_fraction_overlapping = 0.10` Allowed fraction overlapping
- `maximum_combinations = 100000` Maximum combinations of placements to consider
- `proceed_with_any_symmetry = False` Run even with symmetry that is not point-group or helical. Not recommended as symmetry may not work properly
- `box_cushion = 20` Size of buffer around model when boxing
- `refine_cycles = 3` Refinement cycles
- `loop_refine_cycles = 5` Refinement cycles for loops
- `loop_backup_residues = 3` Number of tries removing one residue at a time from each end of existing ends of loop if no loop is found with initial gap
- `residues_to_trim = 5` Residues to trim on each end of all trimmed fragments
- `find_ncs_from_model = True` Find NCS (symmetry) from working models and apply in density modification. Also turns on finding NCS in any autobuild density modification (including density-based search)

- `allow_split_more_than_one_chain_for_mr = False` Allow splitting chains into domains for mr (X-ray only) even if there are multiple chains. Normally only split if a single chain as reassembly may not work with split multiple chains.
- `acceptable_cc_ratio = 0.8` Keep segments with CC at least equal to average of base segments times `minimum_cc_ratio` minus difference between segment confidence (`plDDT`) and confidence cutoff (typically 0.7)
- `low_res_if_multiple_solutions = 3.5` Try phaser MR at increasing lower resolutions up to this value if multiple solutions are found
- `delta_low_res = 0.5` Resolution increment for `low_res_if_multiple_solutions`
- `controlstop_after_dock = None` Stop after docking step
`stop_after_morph = False` Stop after morphing step
`read_files = False` Read existing output files and use them if present
`write_files = True` Write output files
`nproc = 1` Number of processors to use on your local machine
`ignore_symmetry_conflicts = True` You can ignore the symmetry information (CRYST1) from coordinate files. This may be necessary if your model has been placed in a box with `box_map` for example.
`random_seed = 171731` Random seed
`max_dirs = 1000` Maximum number of directories (`dock_and_build_xxx`)
`verbose = False` Verbose output
`quick = True` Run quickly
- `gui` GUI-specific parameter required for output directory
`output_dir = None`

mmtbx.prepare_pdb_deposition

[prepare_pdb_deposition.html](#)

Description

This tool is primarily for adding sequence information to the mmCIF output from phenix.refine to make the file suitable for deposition into the Protein Data Bank. The minimum inputs for this use case are the model from phenix.refine and a sequence file.

If you are starting with a model in PDB format, we recommended that you run phenix.refine on your model and data to generate the necessary fields related to the data, then run this tool to add the sequence information. If your model contains ligands, additional restraints files for the ligands may be necessary.

The input sequence should just be the one letter canonical sequence. For non-standard amino acids or residues, this tool will replace the one letter code with the three letter code surrounded by the parentheses. For example, if there is a selenomethionine, your input sequence should represent that residue as "M". This tool will replace the "M" with "(MSE)" in the correct field in the output mmCIF file.

However, this feature is dependent on curated monomer libraries, so if there is an ambiguity in the parent residue (e.g. methionine is the parent residue for selenomethionine), the conversion will not be automatic. For this situation, you can manually specify which residues to convert from the one letter code to the three letter code with the "custom_residues" parameter. You just need to provide a space-separated sequence of residues. Two examples where this is necessary are SUI in PDB code 2jjj and OTY in PDB code 4jye.

Command line usage examples

```
% mmtbx.prepare_pdb_deposition model.cif sequence.fa % mmtbx.prepare_pdb_deposition 2jjj.cif  
sequence.fa custom_residues="SUI OTY" % mmtbx.prepare_pdb_deposition 4jye.cif sequence.fa  
custom_residues=OTY
```

```
% mmtbx.prepare_pdb_deposition model.cif sequence.fa
```

```
% mmtbx.prepare_pdb_deposition 2jjj.cif sequence.fa custom_residues="SUI OTY"
```

```
% mmtbx.prepare_pdb_deposition 4jye.cif sequence.fa custom_residues=OTY
```

GUI

This tool is available in the Phenix GUI in the "PDB deposition" section.

List of all available keywords

custom_residues = None Space-separated list of three letter codes to be included in the
entity_poly.pdbx_one_letter_code mmCIF loop (e.g. custom_residues='SUI PYL')keep_original_loops = True
Preserves mmCIF data from the input model file (if available) that is not overwritten by other
inputalign_columns = False When set to True, the columns are aligned. This will take longer because the
column widths need to be determined before outputting.job_title = None Job title in PHENIX GUI, not used
on command linevalidationsequencealignmentssimilarity_matrix = blosum50 dayhoff *identity
min_allowable_identity = 0.5 pdb_interpretationsort_atoms = True use_ncs_to_build_restraints = True
show_restraints_histograms = True flip_symmetric_amino_acids = True superpose_ideal_ligand = *None all
SF4 F3S DVT disable_uc_volume_vs_n_atoms_check = False allow_polymer_cross_special_position =

False correct_hydrogens = True c_beta_restraints = True use_neutron_distances = False Use neutron X-H distances (which are longer than X-ray ones) disulfide_bond_exclusions_selection_string = None exclusion_distance_cutoff = 3 If SG of CYS forming SS bond is closer than this distance to an atom that it may coordinate then this SG is excluded from SS bond. link_distance_cutoff = 3 Length of link between the linked residues disulfide_distance_cutoff = 3 add_angle_and_dihedral_restraints_for_disulfides = True dihedral_function_type = *determined_by_sign_of_periodicity all_sinusoidal all_harmonic chir_volume_esd = 0.2 max_reasonable_bond_distance = 50.0 nonbonded_distance_cutoff = None default_vdw_distance = 1 min_vdw_distance = 1 nonbonded_buffer = 1 **EXPERIMENTAL, developers only** nonbonded_weight = None Weighting of nonbonded restraints term. By default, this will be set to 16 if explicit hydrogens are used (this was the default in earlier versions of Phenix), or 100 if hydrogens are missing. const_shrink_donor_acceptor = 0 **EXPERIMENTAL, developers only** vdw_1_4_factor = 0.8 min_distance_sym_equiv = 0.5 custom_nonbonded_symmetry_exclusions = None translate_cns_dna_rna_residue_names = None proceed_with_excessive_length_bonds = False restraints_library_cdl = True Use Conformation Dependent Library (CDL) for geometry restraints include_modified_amino_acid_in_cdl = False mcl = True Use Metal Coordination Library (MCL) for tetrahedral Zn⁺⁺ and iron-sulfur clusters SF₄, FES, F₃S, ... cis_pro_eh99 = True omega_cdl = False Use Omega Conformation Dependent Library (omega-CDL) for geometry restraints cdl_svl = False rdl = False rdl_selection = *all TRP hpd1 = False cdl_nucleotides enable = False Use RestraintsLib for DNA and RNA for geometry restraints esd = *phenix csd factor = 2.0 user_supplied_path = None Root directory of user supplied restraints action = *pre post Choice to look here before GeoStd, the default, or after secondary structure ss_by_chain = True Only applies if search_method = from_ca. Find secondary structure only within individual chains. Alternative is to allow H-bonds between chains. Can be much slower with ss_by_chain=False. If your model is complete use ss_by_chain=True. If your model is many fragments, use ss_by_chain=False. from_ca_conservative = False various parameters changed to make from_ca method more conservative, hopefully to closer resemble ksdssp. max_rmsd = 1 Only applies if search_method = from_ca. Maximum rmsd to consider two chains with identical sequences as the same for ss identification use_representative_chains = True Only applies if search_method = from_ca. Use a representative of all chains with the same sequence. Alternative is to examine each chain individually. Can be much slower with use_representative_of_chain=False if there are many symmetry copies. Ignored unless ss_by_chain is True. max_representative_chains = 100 Only applies if search_method = from_ca. Maximum number of representative chains enabled = False Turn on secondary structure restraints (main switch) protein enabled = True Turn on secondary structure restraints fo...

- custom_residues = None Space-separated list of three letter codes to be included in the entity_poly.pdbx_one_letter_code mmCIF loop (e.g. custom_residues='SUI PYL')
- keep_original_loops = True Preserves mmCIF data from the input model file (if available) that is not overwritten by other input
- align_columns = False When set to True, the columns are aligned. This will take longer because the column widths need to be determined before outputting.
- job_title = None Job title in PHENIX GUI, not used on command line
- mmtbx_validation_sequences sequence_alignment_similarity_matrix = blosum50 dayhoff *identity min_allowable_identity = 0.5
- validation_sequences sequence_alignment_similarity_matrix = blosum50 dayhoff *identity min_allowable_identity = 0.5
- sequences sequence_alignment_similarity_matrix = blosum50 dayhoff *identity min_allowable_identity = 0.5
- sequence_alignments similarity_matrix = blosum50 dayhoff *identity min_allowable_identity = 0.5

- similarity_matrix = blosum50 dayhoff *identity
- min_allowable_identity = 0.5
- pdb_interpretationsort_atoms = True use_ncs_to_build_restraints = True show_restraints_histograms = True flip_symmetric_amino_acids = True superpose_ideal_ligand = *None all SF4 F3S DVT
 disable_uc_volume_vs_n_atoms_check = False allow_polymer_cross_special_position = False
 correct_hydrogens = True c_beta_restraints = True use_neutron_distances = False Use neutron X-H
 distances (which are longer than X-ray ones)disulfide_bond_exclusions_selection_string = None
 exclusion_distance_cutoff = 3 If SG of CYS forming SS bond is closer than this distance to an atom that it
 may coordinate then this SG is excluded from SS bond.link_distance_cutoff = 3 Length of link between
 the linked residuesdisulfide_distance_cutoff = 3 add_angle_and_dihedral_restraints_for_disulfides =
 True dihedral_function_type = *determined_by_sign_of_periodicity all_sinusoidal all_harmonic
 chir_volume_esd = 0.2 max_reasonable_bond_distance = 50.0 nonbonded_distance_cutoff = None
 default_vdw_distance = 1 min_vdw_distance = 1 nonbonded_buffer = 1 **EXPERIMENTAL, developers
 only**nonbonded_weight = None Weighting of nonbonded restraints term. By default, this will be set to 16
 if explicit hydrogens are used (this was the default in earlier versions of Phenix), or 100 if hydrogens are
 missing.const_shrink_donor_acceptor = 0 **EXPERIMENTAL, developers only**vdw_1_4_factor = 0.8
 min_distance_sym_equiv = 0.5 custom_nonbonded_symmetry_exclusions = None
 translate_cns_dna_rna_residue_names = None proceed_with_excessive_length_bonds = False
 restraints_librarycdl = True Use Conformation Dependent Library (CDL) for geometry
 restraintsinclude_modified_amino_acid_in_cdl = False mcl = True Use Metal Coordination Library (MCL)
 for tetrahedral Zn++ and iron-sulfur clusters SF4, FES, F3S, ...cis_pro_eh99 = True omega_cdl = False
 Use Omega Conformation Dependent Library (omega-CDL) for geometry restraintscdl_svl = False rdl =
 False rdl_selection = *all TRP hpdI = False cdl_nucleotidesenable = False Use RestraintsLib for DNA and
 RNA for geometry restraintsesd = *phenix csd factor = 2.0 user_suppliedpath = None Root directory of
 user supplied restraintsaction = *pre post Choice to look here before GeoStd, the default, or
 aftersecondary_structuress_by_chain = True Only applies if search_method = from_ca. Find secondary
 structure only within individual chains. Alternative is to allow H-bonds between chains. Can be much
 slower with ss_by_chain=False. If your model is complete use ss_by_chain=True. If your model is many
 fragments, use ss_by_chain=False.from_ca_conservative = False various parameters changed to make
 from_ca method more conservative, hopefully to closer resemble ksdssp.max_rmsd = 1 Only applies if
 search_method = from_ca. Maximum rmsd to consider two chains with identical sequences as the same
 for ss identificationuse_representative_chains = True Only applies if search_method = from_ca. Use a
 representative of all chains with the same sequence. Alternative is to examine each chain individually.
 Can be much slower with use_representative_of_chain=False if there are many symmetry copies.
 Ignored unless ss_by_chain is True.max_representative_chains = 100 Only applies if search_method =
 from_ca. Maximum number of representative chainsenabled = False Turn on secondary structure
 restraints (main switch)proteinenabled = True Turn on secondary structure restraints for
 proteinsearch_method = *ksdssp from_ca cablam Particular method to search protein secondary
 structure.distance_ideal_n_o = 2.9 Target length for N-O hydrogen bonddistance_cut_n_o = 3.5
 Hydrogen bond with length exceeding this value will not be establishedremove_outliers = True If true,
 h-bonds exceeding distance_cut_n_o length will not be establishedrestrain_hbond_angles = True
 helixserial_number = None helix_identifier = None enabled = True Restrained this particular helixselection =
 None helix_type = *alpha pi 3_10 unknown Type of helix, defaults to alpha. Only alpha, pi, and 3_10
 helices are used for hydrogen-bond res...
- sort_atoms = True
- use_ncs_to_build_restraints = True
- show_restraints_histograms = True

- flip_symmetric_amino_acids = True
- superpose_ideal_ligand = *None all SF4 F3S DVT
- disable_uc_volume_vs_n_atoms_check = False
- allow_polymer_cross_special_position = False
- correct_hydrogens = True
- c_beta_restraints = True
- use_neutron_distances = False Use neutron X-H distances (which are longer than X-ray ones)
- disulfide_bond_exclusions_selection_string = None
- exclusion_distance_cutoff = 3 If SG of CYS forming SS bond is closer than this distance to an atom that it may coordinate then this SG is excluded from SS bond.
- link_distance_cutoff = 3 Length of link between the linked residues
- disulfide_distance_cutoff = 3
- add_angle_and_dihedral_restraints_for_disulfides = True
- dihedral_function_type = *determined_by_sign_of_periodicity all_sinusoidal all_harmonic
- chir_volume_esd = 0.2
- max_reasonable_bond_distance = 50.0
- nonbonded_distance_cutoff = None
- default_vdw_distance = 1
- min_vdw_distance = 1
- nonbonded_buffer = 1 **EXPERIMENTAL, developers only**
- nonbonded_weight = None Weighting of nonbonded restraints term. By default, this will be set to 16 if explicit hydrogens are used (this was the default in earlier versions of Phenix), or 100 if hydrogens are missing.
- const_shrink_donor_acceptor = 0 **EXPERIMENTAL, developers only**
- vdw_1_4_factor = 0.8
- min_distance_sym_equiv = 0.5
- custom_nonbonded_symmetry_exclusions = None
- translate_cns_dna_rna_residue_names = None
- proceed_with_excessive_length_bonds = False
- restraints_librarycdl = True Use Conformation Dependent Library (CDL) for geometry
- restraintsinclude_modified_amino_acid_in_cdl = False mcl = True Use Metal Coordination Library (MCL) for tetrahedral Zn⁺⁺ and iron-sulfur clusters SF4, FES, F3S, ...cis_pro_eh99 = True omega_cdl = False Use Omega Conformation Dependent Library (omega-CDL) for geometry restraintscdl_svl = False rdl = False rdl_selection = *all TRP hpdI = False cdl_nucleotidesenable = False Use RestraintsLib for DNA and RNA for geometry restraintsesd = *phenix csd factor = 2.0 user_suppliedpath = None Root directory of user supplied restraintsaction = *pre post Choice to look here before GeoStd, the default, or after
- cdl = True Use Conformation Dependent Library (CDL) for geometry restraints
- include_modified_amino_acid_in_cdl = False
- mcl = True Use Metal Coordination Library (MCL) for tetrahedral Zn⁺⁺ and iron-sulfur clusters SF4, FES, F3S, ...

- cis_pro_eh99 = True
- omega_cdl = False Use Omega Conformation Dependent Library (omega-CDL) for geometry restraints
- cdl_svl = False
- rdl = False
- rdl_selection = *all TRP
- hpdl = False
- cdl_nucleotidesenable = False Use RestraintsLib for DNA and RNA for geometry restraintsesd = *phenix csd factor = 2.0
- enable = False Use RestraintsLib for DNA and RNA for geometry restraints
- esd = *phenix csd
- factor = 2.0
- user_suppliedpath = None Root directory of user supplied restraintsaction = *pre post Choice to look here before GeoStd, the default, or after
- path = None Root directory of user supplied restraints
- action = *pre post Choice to look here before GeoStd, the default, or after
- secondary_structuresss_by_chain = True Only applies if search_method = from_ca. Find secondary structure only within individual chains. Alternative is to allow H-bonds between chains. Can be much slower with ss_by_chain=False. If your model is complete use ss_by_chain=True. If your model is many fragments, use ss_by_chain=False.from_ca_conservative = False various parameters changed to make from_ca method more conservative, hopefully to closer resemble ksdssp.max_rmsd = 1 Only applies if search_method = from_ca. Maximum rmsd to consider two chains with identical sequences as the same for ss identificationuse_representative_chains = True Only applies if search_method = from_ca. Use a representative of all chains with the same sequence. Alternative is to examine each chain individually. Can be much slower with use_representative_of_chain=False if there are many symmetry copies. Ignored unless ss_by_chain is True.max_representative_chains = 100 Only applies if search_method = from_ca. Maximum number of representative chainsenabled = False Turn on secondary structure restraints (main switch)proteinenabled = True Turn on secondary structure restraints for proteinsearch_method = *ksdssp from_ca cablam Particular method to search protein secondary structure.distance_ideal_n_o = 2.9 Target length for N-O hydrogen bonddistance_cut_n_o = 3.5 Hydrogen bond with length exceeding this value will not be establishedremove_outliers = True If true, h-bonds exceeding distance_cut_n_o length will not be establishedrestrain_hbond_angles = True helixserial_number = None helix_identifier = None enabled = True Restraining this particular helixselection = None helix_type = *alpha pi 3_10 unknown Type of helix, defaults to alpha. Only alpha, pi, and 3_10 helices are used for hydrogen-bond restraints.sigma = 0.05 slack = 0 top_out = False angle_sigma_scale = 1 Multiply sigmas for h-bond angles by this value. Original sigmas range from 5 to 10.angle_sigma_set = None Use this parameter to set sigmas for h-bond angles to a particular valuehbonddonor = None acceptor = None sheetenabled = True Restraining this particular sheetfirst_strand = None sheet_id = None sigma = 0.05 slack = 0 top_out = False angle_sigma_scale = 1 Multiply sigmas for h-bond angles by this value. Original sigmas range from 5 to 10.angle_sigma_set = None Use this parameter to set sigmas for h-bond angles to a particular valuestrandselection = None sense = parallel antiparallel *unknown bond_start_current = None bond_start_previous = None hbonddonor = None acceptor = None nucleic_acidenabled = True Turn on secondary structure restraints for nucleic acidshbond_distance_cutoff = 3.4 Hydrogen bonds with length exceeding this limit will not be establishedscale_bonds_sigma = 1. All sigmas for h-bond length will be multiplied by this number. The

smaller number is tighter restraints.
base_pairenabled = True Restraint this particular base-pair
base1 = None Selection string selecting at least one atom in the desired residue
base2 = None Selection string selecting at least one atom in the desired residue
saenger_class = 0 Type of base-pair
restrain_planarity = False Apply planarity restraint to this base-pair
planarity_sigma = 0.176
restrain_hbonds = True Restrain hydrogen bonds length for this base-pair
restrain_hb_angles = True Restrain angles around hydrogen bonds for this base-pair
restrain_parallelity = True Apply parallelity restraint to this base-pair
parallelity_target = 0 parallelity_sigma = 0.0335
stacking_pairenabled = True Restraint this particular base-pair
base1 = None Selection string selecting at least one atom in the desired residue
base2 = None Selection string selecting at least one atom in the desired residue
angle = 0 sigma = 0.027

- ss_by_chain = True Only applies if search_method = from_ca. Find secondary structure only within individual chains. Alternative is to allow H-bonds between chains. Can be much slower with ss_by_chain=False. If your model is complete use ss_by_chain=True. If your model is many fragments, use ss_by_chain=False.
- from_ca_conservative = False various parameters changed to make from_ca method more conservative, hopefully to closer resemble ksdssp.
- max_rmsd = 1 Only applies if search_method = from_ca. Maximum rmsd to consider two chains with identical sequences as the same for ss identification
- use_representative_chains = True Only applies if search_method = from_ca. Use a representative of all chains with the same sequence. Alternative is to examine each chain individually. Can be much slower with use_representative_of_chain=False if there are many symmetry copies. Ignored unless ss_by_chain is True.
- max_representative_chains = 100 Only applies if search_method = from_ca. Maximum number of representative chains
- enabled = False Turn on secondary structure restraints (main switch)
- proteinenabled = True Turn on secondary structure restraints for protein
search_method = *ksdssp from_ca cablam Particular method to search protein secondary structure.
distance_ideal_n_o = 2.9 Target length for N-O hydrogen bond
distance_cut_n_o = 3.5 Hydrogen bond with length exceeding this value will not be established
remove_outliers = True If true, h-bonds exceeding distance_cut_n_o length will not be established
restrain_hbond_angles = True helixserial_number = None helix_identifier = None enabled = True Restrain this particular helix
selection = None helix_type = *alpha pi 3_10 unknown Type of helix, defaults to alpha. Only alpha, pi, and 3_10 helices are used for hydrogen-bond restraints.
sigma = 0.05 slack = 0 top_out = False angle_sigma_scale = 1 Multiply sigmas for h-bond angles by this value. Original sigmas range from 5 to 10.
angle_sigma_set = None Use this parameter to set sigmas for h-bond angles to a particular value
hbonddonor = None acceptor = None sheetenabled = True Restrain this particular sheet
first_strand = None sheet_id = None sigma = 0.05 slack = 0 top_out = False angle_sigma_scale = 1 Multiply sigmas for h-bond angles by this value. Original sigmas range from 5 to 10.
angle_sigma_set = None Use this parameter to set sigmas for h-bond angles to a particular value
strandselection = None sense = parallel antiparallel *unknown bond_start_current = None bond_start_previous = None
hbonddonor = None acceptor = None
- enabled = True Turn on secondary structure restraints for protein
- search_method = *ksdssp from_ca cablam Particular method to search protein secondary structure.
- distance_ideal_n_o = 2.9 Target length for N-O hydrogen bond
- distance_cut_n_o = 3.5 Hydrogen bond with length exceeding this value will not be established
- remove_outliers = True If true, h-bonds exceeding distance_cut_n_o length will not be established

- `restrain_hbond_angles = True`
- `helixserial_number = None` `helix_identifier = None` `enabled = True` Restrain this particular helix
`selection = None` `helix_type = *alpha pi 3_10 unknown` Type of helix, defaults to alpha. Only alpha, pi, and 3_10 helices are used for hydrogen-bond restraints.
`sigma = 0.05` `slack = 0` `top_out = False` `angle_sigma_scale = 1` Multiply sigmas for h-bond angles by this value. Original sigmas range from 5 to 10.
`angle_sigma_set = None` Use this parameter to set sigmas for h-bond angles to a particular value
`hbonddonor = None` `acceptor = None`
- `serial_number = None`
- `helix_identifier = None`
- `enabled = True` Restrain this particular helix
- `selection = None`
- `helix_type = *alpha pi 3_10 unknown` Type of helix, defaults to alpha. Only alpha, pi, and 3_10 helices are used for hydrogen-bond restraints.
- `sigma = 0.05`
- `slack = 0`
- `top_out = False`
- `angle_sigma_scale = 1` Multiply sigmas for h-bond angles by this value. Original sigmas range from 5 to 10.
- `angle_sigma_set = None` Use this parameter to set sigmas for h-bond angles to a particular value
- `hbonddonor = None` `acceptor = None`
- `donor = None`
- `acceptor = None`
- `sheetenabled = True` Restrain this particular sheet
`first_strand = None` `sheet_id = None` `sigma = 0.05` `slack = 0` `top_out = False` `angle_sigma_scale = 1` Multiply sigmas for h-bond angles by this value. Original sigmas range from 5 to 10.
`angle_sigma_set = None` Use this parameter to set sigmas for h-bond angles to a particular value
`strandselection = None` `sense = parallel antiparallel *unknown` `bond_start_current = None` `bond_start_previous = None` `hbonddonor = None` `acceptor = None`
- `enabled = True` Restrain this particular sheet
- `first_strand = None`
- `sheet_id = None`
- `sigma = 0.05`
- `slack = 0`
- `top_out = False`
- `angle_sigma_scale = 1` Multiply sigmas for h-bond angles by this value. Original sigmas range from 5 to 10.
- `angle_sigma_set = None` Use this parameter to set sigmas for h-bond angles to a particular value
- `strandselection = None` `sense = parallel antiparallel *unknown` `bond_start_current = None` `bond_start_previous = None`
- `selection = None`
- `sense = parallel antiparallel *unknown`

- `bond_start_current` = None
- `bond_start_previous` = None
- `hbonddonor` = None `acceptor` = None
- `donor` = None
- `acceptor` = None
- `nucleic_acidenabled` = True Turn on secondary structure restraints for nucleic acidshbond_distance_cutoff = 3.4 Hydrogen bonds with length exceeding this limit will not be establishedscale_bonds_sigma = 1. All sigmas for h-bond length will be multiplied by this number. The smaller number is tighter restraints.base_pairenabled = True Restraint this particular base-pairbase1 = None Selection string selecting at least one atom in the desired residuebase2 = None Selection string selecting at least one atom in the desired residuesaenger_class = 0 Type of base-pairingrestrain_planarity = False Apply planarity restraint to this base-pairplanarity_sigma = 0.176 restrain_hbonds = True Restrain hydrogen bonds length for this base-pairrestrain_hb_angles = True Restrain angles around hydrogen bonds for this base-pairrestrain_parallelity = True Apply parallelity restraint to this base-pairparallelity_target = 0 parallelity_sigma = 0.0335 stacking_pairenabled = True Restraint this particular base-pairbase1 = None Selection string selecting at least one atom in the desired residuebase2 = None Selection string selecting at least one atom in the desired residueangle = 0 sigma = 0.027
- `enabled` = True Turn on secondary structure restraints for nucleic acids
- `hbond_distance_cutoff` = 3.4 Hydrogen bonds with length exceeding this limit will not be established
- `scale_bonds_sigma` = 1. All sigmas for h-bond length will be multiplied by this number. The smaller number is tighter restraints.
- `base_pairenabled` = True Restraint this particular base-pairbase1 = None Selection string selecting at least one atom in the desired residuebase2 = None Selection string selecting at least one atom in the desired residuesaenger_class = 0 Type of base-pairingrestrain_planarity = False Apply planarity restraint to this base-pairplanarity_sigma = 0.176 restrain_hbonds = True Restrain hydrogen bonds length for this base-pairrestrain_hb_angles = True Restrain angles around hydrogen bonds for this base-pairrestrain_parallelity = True Apply parallelity restraint to this base-pairparallelity_target = 0 parallelity_sigma = 0.0335
- `enabled` = True Restraint this particular base-pair
- `base1` = None Selection string selecting at least one atom in the desired residue
- `base2` = None Selection string selecting at least one atom in the desired residue
- `saenger_class` = 0 Type of base-pairing
- `restrain_planarity` = False Apply planarity restraint to this base-pair
- `planarity_sigma` = 0.176
- `restrain_hbonds` = True Restrain hydrogen bonds length for this base-pair
- `restrain_hb_angles` = True Restrain angles around hydrogen bonds for this base-pair
- `restrain_parallelity` = True Apply parallelity restraint to this base-pair
- `parallelity_target` = 0
- `parallelity_sigma` = 0.0335
- `stacking_pairenabled` = True Restraint this particular base-pairbase1 = None Selection string selecting at least one atom in the desired residuebase2 = None Selection string selecting at least one atom in the

desired residueangle = 0 sigma = 0.027

- enabled = True Restraint this particular base-pair
- base1 = None Selection string selecting at least one atom in the desired residue
- base2 = None Selection string selecting at least one atom in the desired residue
- angle = 0
- sigma = 0.027
- reference_coordinate_restraintsRestrains coordinates in Cartesian space to stay near their starting positions. This is intended for use in generating simulated annealing omit maps, to prevent refined atoms from collapsing in on the region missing atoms. For conserving geometry quality at low resolution, the more flexible reference model restraints should be used instead.enabled = False exclude_outliers = True selection = all sigma = 0.2 limit = 1.0 top_out = False reference_is_average_alt_confs = False Sets the reference coordinate to the average of two alt. confs. Ignores selection option.
- enabled = False
- exclude_outliers = True
- selection = all
- sigma = 0.2
- limit = 1.0
- top_out = False
- reference_is_average_alt_confs = False Sets the reference coordinate to the average of two alt. confs. Ignores selection option.
- automatic_restraintsside_chain_parallelity = False
- side_chain_parallelity = False
- automatic_linkinglink_all = False If True, bond restraints will be generated for any appropriate ligand-protein or ligand-nucleic acid covalent bonds. This includes sugars, amino acid modifications, and other prosthetic groups.link_none = False link_metals = Auto link_residues = True link_amino_acid_rna_dna = False link_carbohydrates = True link_ligands = True link_small_molecules = False metal_coordination_cutoff = 3.0 amino_acid_bond_cutoff = 1.9 inter_residue_bond_cutoff = 2.2 buffer_for_second_row_elements = 0.5 carbohydrate_bond_cutoff = 1.99 ligand_bond_cutoff = 1.99 small_molecule_bond_cutoff = 1.98 exclude_hydrogens_from_bonding_decisions = False
- link_all = False If True, bond restraints will be generated for any appropriate ligand-protein or ligand-nucleic acid covalent bonds. This includes sugars, amino acid modifications, and other prosthetic groups.
- link_none = False
- link_metals = Auto
- link_residues = True
- link_amino_acid_rna_dna = False
- link_carbohydrates = True
- link_ligands = True
- link_small_molecules = False
- metal_coordination_cutoff = 3.0
- amino_acid_bond_cutoff = 1.9

- inter_residue_bond_cutoff = 2.2
- buffer_for_second_row_elements = 0.5
- carbohydrate_bond_cutoff = 1.99
- ligand_bond_cutoff = 1.99
- small_molecule_bond_cutoff = 1.98
- exclude_hydrogens_from_bonding_decisions = False
- include_in_automatic_linkingselection_1 = None selection_2 = None bond_cutoff = 4.5
- selection_1 = None
- selection_2 = None
- bond_cutoff = 4.5
- exclude_from_automatic_linkingselection_1 = None selection_2 = None
- selection_1 = None
- selection_2 = None
- apply_cis_trans_specificationcis_trans_mod = cis *trans residue_selection = None Residues containing C-alpha atom of omega dihedral
- cis_trans_mod = cis *trans
- residue_selection = None Residues containing C-alpha atom of omega dihedral
- apply_cif_restraintsrestraints_file_name = None residue_selection = None
- restraints_file_name = None
- residue_selection = None
- apply_cif_modificationdata_mod = None residue_selection = None
- data_mod = None
- residue_selection = None
- apply_cif_linkdata_link = None residue_selection_1 = None residue_selection_2 = None
- data_link = None
- residue_selection_1 = None
- residue_selection_2 = None
- peptide_linkramachandran_restraints = False !!! OBSOLETE. Kept for backward compatibility only !!!
 Restrains peptide backbone to fall within allowed regions of Ramachandran plot. Although it does not eliminate outliers, it can significantly improve the percent favored and percent outliers at low resolution. Probably not useful (and maybe even harmful) at resolutions much higher than 3.5Å.cis_threshold = 45
 apply_all_trans = False discard_omega = False discard_psi_phi = True apply_peptide_plane = False
 omega_esd_override_value = None rama_weight = 1.0 scale_allowed = 1.0 rama_potential = *oldfield
 emsley rama_selection = None Selection of part of the model for which Ramachandran restraints will be set up.restrain_rama_outliers = True Apply restraints to Ramachandran outliersrestrain_rama_allowed = True Apply restraints to residues in allowed region on Ramachandran
 plotrestrain_allowed_outliers_with_emsley = False In case of restrain_rama_outliers=True and/or restrain_rama_allowed=True still restrain these residues with emsley. Make sense only in case of using oldfield potential.oldfieldesd = 10.0 weight_scale = 1.0 dist_weight_max = 10.0 weight = None plot_cutoff = 0.027

- ramachandran_restraints = False !!! OBSOLETE. Kept for backward compatibility only !!! Restrains peptide backbone to fall within allowed regions of Ramachandran plot. Although it does not eliminate outliers, it can significantly improve the percent favored and percent outliers at low resolution. Probably not useful (and maybe even harmful) at resolutions much higher than 3.5Å.
- cis_threshold = 45
- apply_all_trans = False
- discard_omega = False
- discard_psi_phi = True
- apply_peptide_plane = False
- omega_esd_override_value = None
- rama_weight = 1.0
- scale_allowed = 1.0
- rama_potential = *oldfield emsley
- rama_selection = None Selection of part of the model for which Ramachandran restraints will be set up.
- restrain_rama_outliers = True Apply restraints to Ramachandran outliers
- restrain_rama_allowed = True Apply restraints to residues in allowed region on Ramachandran plot
- restrain_allowed_outliers_with_emsley = False In case of restrain_rama_outliers=True and/or restrain_rama_allowed=True still restrain these residues with emsley. Make sense only in case of using oldfield potential.
- oldfieldesd = 10.0 weight_scale = 1.0 dist_weight_max = 10.0 weight = None plot_cutoff = 0.027
- esd = 10.0
- weight_scale = 1.0
- dist_weight_max = 10.0
- weight = None
- plot_cutoff = 0.027
- ramachandran_plot_restraintsenabled = False favored = *oldfield emsley emsley8k phi_psi_2 allowed = *oldfield emsley emsley8k phi_psi_2 outlier = *oldfield emsley emsley8k phi_psi_2 selection = None Selection of part of the model for which Ramachandran restraints will be set up.inject_emsley8k_into_oldfield_favored = True Backdoor to disable temporary dirty hack to use botholdfieldweight = 0. Direct weight value. If 0 the weight will be calculated as following: $(w, op.esd, op.dist_weight_max, 2.0, op.weight_scale) \frac{1}{esd^2} * \max(2.0, \min(current_distance_to_allowed, dist_weight_max)) * weight_scale \max(2.0, current_distance_to_allowed) \frac{1}{esd^2} * weight_scale * \max(distance_to_allowed_cutoff, current_distance_to_allowed) weight_scale(=0.01) * \max(distance_weight_min(=2.), \min(distance_weight_max(=10.), current_distance_to_allowed))weight_scale = 0.01$ distance_weight_min = 2.0 minimum coefficient when scaling depending on how far the residue is from allowed region.distance_weight_max = 10.0 maximum coefficient when scaling depending on how far the residue is from allowed region.plot_cutoff = 0.027 emsleyweight = 1.0 scale_allowed = 1.0 emsley8kweight_favored = 5.0 weight_allowed = 10.0 weight_outlier = 10.0 phi_psi_2favored_strategy = *closest highest_probability random weighted_random allowed_strategy = *closest highest_probability random weighted_random outlier_strategy = *closest highest_probability random weighted_random
- enabled = False

- favored = *oldfield emsley emsley8k phi_psi_2
- allowed = *oldfield emsley emsley8k phi_psi_2
- outlier = *oldfield emsley emsley8k phi_psi_2
- selection = None Selection of part of the model for which Ramachandran restraints will be set up.
- inject_emsley8k_into_oldfield_favored = True Backdoor to disable temporary dirty hack to use both
- oldfieldweight = 0. Direct weight value. If 0 the weight will be calculated as following: (w, op.esd, op.dist_weight_max, 2.0, op.weight_scale) $1 / \text{esd}^2 * \max(2.0, \min(\text{current_distance_to_allowed}, \text{dist_weight_max})) * \text{weight_scale}$ $\max(2.0, \text{current_distance_to_allowed}) 1 / \text{esd}^2 * \text{weight_scale} * \max(\text{distance_to_allowed_cutoff}, \text{current_distance_to_allowed}) \text{weight_scale}(=0.01) *$ $\max(\text{distance_weight_min}(=2.), \min(\text{distance_weight_max}(=10.), \text{current_distance_to_allowed})) \text{weight_scale} = 0.01$ distance_weight_min = 2.0 minimum coefficient when scaling depending on how far the residue is from allowed region.distance_weight_max = 10.0 maximum coefficient when scaling depending on how far the residue is from allowed region.plot_cutoff = 0.027
- weight = 0. Direct weight value. If 0 the weight will be calculated as following: (w, op.esd, op.dist_weight_max, 2.0, op.weight_scale) $1 / \text{esd}^2 * \max(2.0, \min(\text{current_distance_to_allowed}, \text{dist_weight_max})) * \text{weight_scale}$ $\max(2.0, \text{current_distance_to_allowed}) 1 / \text{esd}^2 * \text{weight_scale} * \max(\text{distance_to_allowed_cutoff}, \text{current_distance_to_allowed}) \text{weight_scale}(=0.01) *$ $\max(\text{distance_weight_min}(=2.), \min(\text{distance_weight_max}(=10.), \text{current_distance_to_allowed}))$
- weight_scale = 0.01
- distance_weight_min = 2.0 minimum coefficient when scaling depending on how far the residue is from allowed region.
- distance_weight_max = 10.0 maximum coefficient when scaling depending on how far the residue is from allowed region.
- plot_cutoff = 0.027
- emsleyweight = 1.0 scale_allowed = 1.0
- weight = 1.0
- scale_allowed = 1.0
- emsley8kweight_favored = 5.0 weight_allowed = 10.0 weight_outlier = 10.0
- weight_favored = 5.0
- weight_allowed = 10.0
- weight_outlier = 10.0
- phi_psi_2favored_strategy = *closest highest_probability random weighted_random allowed_strategy = *closest highest_probability random weighted_random outlier_strategy = *closest highest_probability random weighted_random
- favored_strategy = *closest highest_probability random weighted_random
- allowed_strategy = *closest highest_probability random weighted_random
- outlier_strategy = *closest highest_probability random weighted_random
- rna_sugar_pucker_analysisbond_min_distance = 1.2 bond_max_distance = 1.8 epsilon_range_min = 155.0 epsilon_range_max = 310.0 delta_range_2p_min = 129.0 delta_range_2p_max = 162.0 delta_range_3p_min = 65.0 delta_range_3p_max = 104.0 p_distance_c1p_outbound_line_2p_max = 2.9 o3p_distance_c1p_outbound_line_2p_max = 2.4 bond_detection_distance_tolerance = 0.5 enable = True

- bond_min_distance = 1.2
- bond_max_distance = 1.8
- epsilon_range_min = 155.0
- epsilon_range_max = 310.0
- delta_range_2p_min = 129.0
- delta_range_2p_max = 162.0
- delta_range_3p_min = 65.0
- delta_range_3p_max = 104.0
- p_distance_c1p_outbound_line_2p_max = 2.9
- o3p_distance_c1p_outbound_line_2p_max = 2.4
- bond_detection_distance_tolerance = 0.5
- enable = True
- show_histogram_slotsbond_lengths = 5 nonbonded_interaction_distances = 5
bond_angle_deviations_from_ideal = 5 dihedral_angle_deviations_from_ideal = 5
chiral_volume_deviations_from_ideal = 5
- bond_lengths = 5
- nonbonded_interaction_distances = 5
- bond_angle_deviations_from_ideal = 5
- dihedral_angle_deviations_from_ideal = 5
- chiral_volume_deviations_from_ideal = 5
- show_max_itemsnot_linked = 5 bond_restraints_sorted_by_residual = 5
nonbonded_interactions_sorted_by_model_distance = 5 bond_angle_restraints_sorted_by_residual = 5
dihedral_angle_restraints_sorted_by_residual = 3 chirality_restraints_sorted_by_residual = 3
planarity_restraints_sorted_by_residual = 3 residues_with_excluded_nonbonded_symmetry_interactions
= 12 fatal_problem_max_lines = 10
- not_linked = 5
- bond_restraints_sorted_by_residual = 5
- nonbonded_interactions_sorted_by_model_distance = 5
- bond_angle_restraints_sorted_by_residual = 5
- dihedral_angle_restraints_sorted_by_residual = 3
- chirality_restraints_sorted_by_residual = 3
- planarity_restraints_sorted_by_residual = 3
- residues_with_excluded_nonbonded_symmetry_interactions = 12
- fatal_problem_max_lines = 10
- ncs_groupThe definition of one NCS group. Note, that almost always in refinement programs they will
be checked and filtered if needed.reference = None Residue selection string for the complete master
NCS copyselection = None Residue selection string for each NCS copy location in ASU
- reference = None Residue selection string for the complete master NCS copy
- selection = None Residue selection string for each NCS copy location in ASU

- `ncs_searchSet` of parameters for NCS search procedure. Some of them also used for filtering user-supplied `ncs_group.enabled = False` Enable NCS restraints or constraints in refinement (in some cases may be switched on inside refinement program). `exclude_selection = "element H or element D or water"` Atoms selected by this selection will be excluded from the model before any NCS search and/or filtering procedures. There is no way atoms defined by this selection will be in NCS. `chain_similarity_threshold = 0.85` Threshold for sequence similarity between matching chains. A smaller value may cause more chains to be grouped together and can lower the number of common residues. `chain_max_rmsd = 2.` limit of rms difference between chains to be considered as copies. `residue_match_radius = 4.0` Maximum allowed distance difference between pairs of matching atoms of two residues. `try_shortcuts = False` Try very quick check to speed up the search when chains are identical. If failed, regular search will be performed automatically. `minimum_number_of_atoms_in_copy = 3` Do not create ncs groups where master and copies would contain less than specified amount of atoms. `validate_user_supplied_groups = True` Enable validation of user-supplied `ncs_group`. Need to exercise a lot of caution turning this off. This option is for developers only.
- `enabled = False` Enable NCS restraints or constraints in refinement (in some cases may be switched on inside refinement program).
- `exclude_selection = "element H or element D or water"` Atoms selected by this selection will be excluded from the model before any NCS search and/or filtering procedures. There is no way atoms defined by this selection will be in NCS.
- `chain_similarity_threshold = 0.85` Threshold for sequence similarity between matching chains. A smaller value may cause more chains to be grouped together and can lower the number of common residues
- `chain_max_rmsd = 2.` limit of rms difference between chains to be considered as copies
- `residue_match_radius = 4.0` Maximum allowed distance difference between pairs of matching atoms of two residues
- `try_shortcuts = False` Try very quick check to speed up the search when chains are identical. If failed, regular search will be performed automatically.
- `minimum_number_of_atoms_in_copy = 3` Do not create ncs groups where master and copies would contain less than specified amount of atoms
- `validate_user_supplied_groups = True` Enable validation of user-supplied `ncs_group`. Need to exercise a lot of caution turning this off. This option is for developers only.
- `clash_guard.nonbonded_distance_threshold = 0.5` `max_number_of_distances_below_threshold = 100` `max_fraction_of_distances_below_threshold = 0.1`
- `nonbonded_distance_threshold = 0.5`
- `max_number_of_distances_below_threshold = 100`
- `max_fraction_of_distances_below_threshold = 0.1`
- `geometry_restraintseditsexcessive_bond_distance_limit = 10` `bondaction = *add delete change` `atom_selection_1 = None` `atom_selection_2 = None` `symmetry_operation = None` The bond is between `atom_1` and `symmetry_operation * atom_2`, with `atom_1` and `atom_2` given in fractional coordinates. Example: `symmetry_operation = -x-1,-y,z` `distance_ideal = None` `sigma = None` `slack = None` `limit = -1.0` `top_out = False` `angleaction = *add delete change` `atom_selection_1 = None` `atom_selection_2 = None` `atom_selection_3 = None` `angle_ideal = None` `sigma = None` `dihedralaction = *add delete change` `atom_selection_1 = None` `atom_selection_2 = None` `atom_selection_3 = None` `atom_selection_4 = None` `angle_ideal = None` `alt_angle_ideals = None` `sigma = None` `periodicity = 1` `planarityaction = *add delete change` `atom_selection = None` `sigma = None` `parallelityaction = *add delete change` `atom_selection_1 =`

None atom_selection_2 = None sigma = 0.027 target_angle_deg = 0 removeangles = None dihedrals = None chiralities = None planarities = None parallelities = None

- editsexcessive_bond_distance_limit = 10 bondaction = *add delete change atom_selection_1 = None atom_selection_2 = None symmetry_operation = None The bond is between atom_1 and symmetry_operation * atom_2, with atom_1 and atom_2 given in fractional coordinates. Example: symmetry_operation = -x-1,-y,z distance_ideal = None sigma = None slack = None limit = -1.0 top_out = False angleaction = *add delete change atom_selection_1 = None atom_selection_2 = None atom_selection_3 = None angle_ideal = None sigma = None dihedralaction = *add delete change atom_selection_1 = None atom_selection_2 = None atom_selection_3 = None atom_selection_4 = None angle_ideal = None alt_angle_ideals = None sigma = None periodicity = 1 planarityaction = *add delete change atom_selection = None sigma = None parallelityaction = *add delete change atom_selection_1 = None atom_selection_2 = None sigma = 0.027 target_angle_deg = 0

- excessive_bond_distance_limit = 10

- bondaction = *add delete change atom_selection_1 = None atom_selection_2 = None symmetry_operation = None The bond is between atom_1 and symmetry_operation * atom_2, with atom_1 and atom_2 given in fractional coordinates. Example: symmetry_operation = -x-1,-y,z distance_ideal = None sigma = None slack = None limit = -1.0 top_out = False

- action = *add delete change

- atom_selection_1 = None

- atom_selection_2 = None

- symmetry_operation = None The bond is between atom_1 and symmetry_operation * atom_2, with atom_1 and atom_2 given in fractional coordinates. Example: symmetry_operation = -x-1,-y,z

- distance_ideal = None

- sigma = None

- slack = None

- limit = -1.0

- top_out = False

- angleaction = *add delete change atom_selection_1 = None atom_selection_2 = None atom_selection_3 = None angle_ideal = None sigma = None

- action = *add delete change

- atom_selection_1 = None

- atom_selection_2 = None

- atom_selection_3 = None

- angle_ideal = None

- sigma = None

- dihedralaction = *add delete change atom_selection_1 = None atom_selection_2 = None atom_selection_3 = None atom_selection_4 = None angle_ideal = None alt_angle_ideals = None sigma = None periodicity = 1

- action = *add delete change

- atom_selection_1 = None

- atom_selection_2 = None

- atom_selection_3 = None
- atom_selection_4 = None
- angle_ideal = None
- alt_angle_ideals = None
- sigma = None
- periodicity = 1
- planarityaction = *add delete change atom_selection = None sigma = None
- action = *add delete change
- atom_selection = None
- sigma = None
- parallelityaction = *add delete change atom_selection_1 = None atom_selection_2 = None sigma = 0.027 target_angle_deg = 0
- action = *add delete change
- atom_selection_1 = None
- atom_selection_2 = None
- sigma = 0.027
- target_angle_deg = 0
- removeangles = None dihedrals = None chiralities = None planarities = None parallelities = None
- angles = None
- dihedrals = None
- chiralities = None
- planarities = None
- parallelities = None
- reference_modelenabled = False Restrains the dihedral angles to a high-resolution reference structure to reduce overfitting at low resolution. You will need to specify a reference PDB file (in the input list in the main window) to use this option.file = None use_starting_model_as_reference = False sigma = 1.0 limit = 15.0 hydrogens = False Include dihedrals with hydrogen atomsmain_chain = True Include dihedrals formed by main chain atomsside_chain = True Include dihedrals formed by side chain atomsfix_outliers = True Try to fix rotamer outliers in refined modelstrict_rotamer_matching = False Make sure that rotamers in refinement model matches those in reference model even when they are not outliersauto_shutoff_for_ncs = False Do not apply to parts of structure covered by NCS restraintssecondary_structure_only = False Only apply reference model restraints to secondary structure elements (helices and sheets)reference_groupreference = None selection = None file_name = None this is to used internally to disambiguate cases where multiple reference models contain the same chain ID. This normally does not need to be set by the usersearch_optionsSet of parameters for NCS search procedure. Some of them also used for filtering user-supplied ncs_group.exclude_selection = "element H or element D or water" Atoms selected by this selection will be excluded from the model before any NCS search and/or filtering procedures. There is no way atoms defined by this selection will be in NCS.chain_similarity_threshold = 0.85 Threshold for sequence similarity between matching chains. A smaller value may cause more chains to be grouped together and can lower the number of common residueschain_max_rmsd = 100. limit of rms difference between chains to be considered as copiesresidue_match_radius = 1000 Maximum allowed distance difference between pairs of matching

atoms of two residues `try_shortcuts = False` Try very quick check to speed up the search when chains are identical. If failed, regular search will be performed automatically. `minimum_number_of_atoms_in_copy = 3` Do not create ncs groups where master and copies would contain less than specified amount of atoms `validate_user_supplied_groups = True` Enable validation of user-supplied ncs_group. Need to exercise a lot of caution turning this off. This option is for developers only.

- `enabled = False` Restrains the dihedral angles to a high-resolution reference structure to reduce overfitting at low resolution. You will need to specify a reference PDB file (in the input list in the main window) to use this option.

- `file = None`

- `use_starting_model_as_reference = False`

- `sigma = 1.0`

- `limit = 15.0`

- `hydrogens = False` Include dihedrals with hydrogen atoms

- `main_chain = True` Include dihedrals formed by main chain atoms

- `side_chain = True` Include dihedrals formed by side chain atoms

- `fix_outliers = True` Try to fix rotamer outliers in refined model

- `strict_rotamer_matching = False` Make sure that rotamers in refinement model matches those in reference model even when they are not outliers

- `auto_shutoff_for_ncs = False` Do not apply to parts of structure covered by NCS restraints

- `secondary_structure_only = False` Only apply reference model restraints to secondary structure elements (helices and sheets)

- `reference_group` preference = None selection = None `file_name = None` this is to used internally to disambiguate cases where multiple reference models contain the same chain ID. This normally does not need to be set by the user

- `reference = None`

- `selection = None`

- `file_name = None` this is to used internally to disambiguate cases where multiple reference models contain the same chain ID. This normally does not need to be set by the user

- `search_options` Set of parameters for NCS search procedure. Some of them also used for filtering

user-supplied ncs_group. `exclude_selection = "element H or element D or water"` Atoms selected by this selection will be excluded from the model before any NCS search and/or filtering procedures. There is no way atoms defined by this selection will be in NCS. `chain_similarity_threshold = 0.85` Threshold for sequence similarity between matching chains. A smaller value may cause more chains to be grouped together and can lower the number of common residues. `chain_max_rmsd = 100.` limit of rms difference between chains to be considered as copies. `residue_match_radius = 1000` Maximum allowed distance difference between pairs of matching atoms of two residues. `try_shortcuts = False` Try very quick check to speed up the search when chains are identical. If failed, regular search will be performed automatically.

`minimum_number_of_atoms_in_copy = 3` Do not create ncs groups where master and copies would contain less than specified amount of atoms. `validate_user_supplied_groups = True` Enable validation of user-supplied ncs_group. Need to exercise a lot of caution turning this off. This option is for developers only.

- `exclude_selection = "element H or element D or water"` Atoms selected by this selection will be excluded from the model before any NCS search and/or filtering procedures. There is no way atoms

defined by this selection will be in NCS.

- `chain_similarity_threshold = 0.85` Threshold for sequence similarity between matching chains. A smaller value may cause more chains to be grouped together and can lower the number of common residues
- `chain_max_rmsd = 100` limit of rms difference between chains to be considered as copies
- `residue_match_radius = 1000` Maximum allowed distance difference between pairs of matching atoms of two residues
- `try_shortcuts = False` Try very quick check to speed up the search when chains are identical. If failed, regular search will be performed automatically.
- `minimum_number_of_atoms_in_copy = 3` Do not create ncs groups where master and copies would contain less than specified amount of atoms
- `validate_user_supplied_groups = True` Enable validation of user-supplied `ncs_group`. Need to exercise a lot of caution turning this off. This option is for developers only.
- `amberParameters` for using Amber in refinement.
 - `use_amber = False` Use Amber for all the gradients in refinement
 - `topology_file_name = None` A topology file needed by Amber. Can be generated using `phenix.AmberPrep`.
 - `coordinate_file_name = None` A coordinate file needed by Amber. Can be generated using `phenix.AmberPrep`.
 - `order_file_name = None` A file that maps amber atom numbers to phenix atom numbers.
 - `wxc_factor = 1`.
 - `restraint_wt = 0`.
 - `restraintmask = "`
 - `reference_file_name = "`
 - `bellymask = "` If given, turn on belly in sander
 - `qmmask = "` If given, turn on QM/MM with the given mask
 - `qmcharge = 0` Charge of the QM/MM region
 - `netcdf_trajectory_file_name = "` If given, turn on writing netcdf trajectory
 - `print_amber_energies = False` Print details of Amber energies during refinement
- `qiQMworking_directory = None` `qm_restraintsselection = None` selection for core of atoms to calculate new restraints via a QM geometry minimisation
 - `run_in_macro_cycles = *`first_only first_and_last all last_only test the steps of the refinement that the restraints generation is run
 - `buffer = 3.5` distance to include entire residues into the environment of the core
 - `calculate = *`in_situ_opt starting_energy final_energy starting_strain final_strain starting_bound final_bound starting_binding final_binding starting_higher_single_point final_higher_single_point Choose QM calculations to run
 - `write_files = *`restraints pdb_core pdb_buffer pdb_final_core pdb_final_buffer which ligand or cluster files to

writeprotein_optimisation_freeze = *all None main_chain main_chain_to_beta main_chain_to_delta
 torsions the parts of protein residues that are frozen when an amino acid is the main
 selectionremove_water = False restraints_filename = Auto restraints filename is based on model name if
 not specifiedcleanup = all *most None ignore_x_h_distance_protein = False skip check on transfer of
 proton during QM optimisationignore_lack_of_h_on_ligand = False skip check on protonation of ligand for
 entities such as MgF3capping_groups = True include_nearest_neighbours_in_optimisation = False
 include the side chains of protein in the QM optimisationinclude_inter_residue_restraints = False
 do_not_update_restraints = False For testing and maybe getting strain energy of standard
 restraintsbuffer_selection = None use this instead of distance from
 selectionsselectionspecific_atom_chargesatom_selection = None charge = None
 specific_atom_multiplicitiesatom_selection = None multiplicity = None
 freeze_specific_atomsatom_selection = None packageprogram = *mopac test charge = Auto multiplicity
 = Auto method = Auto basis_set = Auto solvent_model = None nproc = 1
 read_output_to_skip_opt_if_available = False ignore_input_differences = False view_output = None

- working_directory = None

- qm_restraintsselection = None selection for core of atoms to calculate new restraints via a QM
 geometry minimisationrun_in_macro_cycles = *first_only first_and_last all last_only test the steps of the
 refinement that the restraints generation is runbuffer = 3.5 distance to include entire residues into the
 enviroment of the corecalculate = *in_situ_opt starting_energy final_energy starting_strain final_strain
 starting_bound final_bound starting_binding final_binding starting_higher_single_point
 final_higher_single_point Choose QM calculations to runwrite_files = *restraints pdb_core pdb_buffer
 pdb_final_core pdb_final_buffer which ligand or cluster files to writeprotein_optimisation_freeze = *all
 None main_chain main_chain_to_beta main_chain_to_delta torsions the parts of protein residues that
 are frozen when an amino acid is the main selectionremove_water = False restraints_filename = Auto
 restraints filename is based on model name if not specifiedcleanup = all *most None
 ignore_x_h_distance_protein = False skip check on transfer of proton during QM
 optimisationignore_lack_of_h_on_ligand = False skip check on protonation of ligand for entities such as
 MgF3capping_groups = True include_nearest_neighbours_in_optimisation = False include the side
 chains of protein in the QM optimisationinclude_inter_residue_restraints = False
 do_not_update_restraints = False For testing and maybe getting strain energy of standard
 restraintsbuffer_selection = None use this instead of distance from
 selectionsselectionspecific_atom_chargesatom_selection = None charge = None
 specific_atom_multiplicitiesatom_selection = None multiplicity = None
 freeze_specific_atomsatom_selection = None packageprogram = *mopac test charge = Auto multiplicity
 = Auto method = Auto basis_set = Auto solvent_model = None nproc = 1
 read_output_to_skip_opt_if_available = False ignore_input_differences = False view_output = None

- selection = None selection for core of atoms to calculate new restraints via a QM geometry minimisation

- run_in_macro_cycles = *first_only first_and_last all last_only test the steps of the refinement that the
 restraints generation is run

- buffer = 3.5 distance to include entire residues into the enviroment of the core

- calculate = *in_situ_opt starting_energy final_energy starting_strain final_strain starting_bound
 final_bound starting_binding final_binding starting_higher_single_point final_higher_single_point Choose
 QM calculations to run

- write_files = *restraints pdb_core pdb_buffer pdb_final_core pdb_final_buffer which ligand or cluster
 files to write

- protein_optimisation_freeze = *all None main_chain main_chain_to_beta main_chain_to_delta torsions the parts of protein residues that are frozen when an amino acid is the main selection
- remove_water = False
- restraints_filename = Auto restraints filename is based on model name if not specified
- cleanup = all *most None
- ignore_x_h_distance_protein = False skip check on transfer of proton during QM optimisation
- ignore_lack_of_h_on_ligand = False skip check on protonation of ligand for entities such as MgF3
- capping_groups = True
- include_nearest_neighbours_in_optimisation = False include the side chains of protein in the QM optimisation
- include_inter_residue_restraints = False
- do_not_update_restraints = False For testing and maybe getting strain energy of standard restraints
- buffer_selection = None use this instead of distance from selection
- specific_atom_chargesatom_selection = None charge = None
- atom_selection = None
- charge = None
- specific_atom_multiplicitiesatom_selection = None multiplicity = None
- atom_selection = None
- multiplicity = None
- freeze_specific_atomsatom_selection = None
- atom_selection = None
- packageprogram = *mopac test charge = Auto multiplicity = Auto method = Auto basis_set = Auto solvent_model = None nproc = 1 read_output_to_skip_opt_if_available = False ignore_input_differences = False view_output = None
- program = *mopac test
- charge = Auto
- multiplicity = Auto
- method = Auto
- basis_set = Auto
- solvent_model = None
- nproc = 1
- read_output_to_skip_opt_if_available = False
- ignore_input_differences = False
- view_output = None
- outputsuffix = '.deposit' Suffix string added to automatically generated output filenames
- suffix = '.deposit' Suffix string added to automatically generated output filenames
- guiGUI-specific parameter required for output directoryoutput_dir = None
- output_dir = None

QI Energies

qi_energies.html

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- Authors
- Introduction
- Literature

Authors

Nigel W. Moriarty

Introduction

Under Construction

Literature

In situ ligand restraints from quantum-mechanical methods. D. Liebschner, N.W. Moriarty, B.K. Poon, and P.D. Adams. Acta Cryst. D79, 100-110 (2023).

- In situ ligand restraints from quantum-mechanical methods. D. Liebschner, N.W. Moriarty, B.K. Poon, and P.D. Adams. Acta Cryst. D79, 100-110 (2023).

QM Flipping (QMF)

qi_histidine.html

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- Authors
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Authors

Nigel W. Moriarty

Introduction

Histidine can be protonated on either or both of the two nitrogen atoms of the imidazole moiety. Each of the three possible forms occur as a result of the stereochemical environment of the histidine sidechain. In an atomic model, comparing the possible protonation states in situ, looking at possible H-bonding and metal coordination, it is possible to predict which is most likely correct. We describe a more direct method that uses quantum mechanical methods to calculate – also in situ – the minimum geometry and energy for comparison, and therefore more accurate identification of the most likely protonation state.

QMF is part of the Quantum Interface (QI) module that covers the interaction of Phenix with QM packages.

Running an example

QMF can only be run stand-alone in the `mmtbx.quantum_interface` procedure as it's not reasonable to do this during refinement. Using the same example as QMR of PDB:4w53, download from the PDB website using:

```
~> phenix.fetch_pdb 4w53 --mtz
```

Command to produce PHIL scope for QM minimisation using `mmtbx.quantum_interface` is:

```
~> mmtbx.quantum_interface 4w53.pdb iterate_NQH=HIS
```

which will fail because of the bulk test for hydrogen atoms.

```
Sorry: Model must have Hydrogen atoms
```

Also note that the `write_qmr_phil=True` option writes a PHIL file. In this example, the format of the PHIL file works for QI interface but the default works for `phenix.refine.Add Hydrogen atoms` to the model using:

```
~> phenix.ready_set 4w53.pdb
```

which produces a file with hydrogens atoms – `4w53.updated.pdb`. It also may produce a restraints file that needs to be used next but in this case because the restraints for BNZ (benzene) are shipped with Phenix it is not necessary to supply to the program unless you want a different list of restraints (not ideal values). This file can be created cheaply but is needed as the skeleton for the QMR values to be inserted in the bond, angle and torsion restraints. So, we can repeat the quantum interface command with the new files.:

```
~> mmtbx.quantum_interface 4w53.updated.pdb iterate_NQH=HIS
```

This will prompt for the selection of the ligand which in this case is “1”.


```
1 : "chain A and resid 31 and resname HIS"
2 : "chain A and resid 200 and resname MBN"
3 : "chain A and resid 201 and resname EPE"
```

Enter selection by choosing number or typing a new one ~>

The result is a PHIL scope for MBN QMR restraints:

```
4w53.updated_A_31_HIS.phil.
```

There is also a command given to run the six configurations.

```
~&gt; mmtbx.quantum_interface 4w53.updated.pdb iterate_NQH=HIS 4w53.updated_A_31_HIS.phil run_qmr=True qi.nproc
```

Note that you can edit the number of nproc to run all six configurations at once.

Once the jobs are done the important metrics are listed in the table:

```
Energies in units of kcal/mol
Proton energy used : 94.5100 kcal/mol

0. HD1 : rotamer "t70"
1. HD1, HE2 : -1272.239 kcal/mol ~&gt; 3.19 kcal/mol. H-Bonds : 14 rmsd : 0.03 rotamer "t70"
2. HD1 only : -1275.425 kcal/mol ~&gt; 0.00 kcal/mol. H-Bonds : 14 rmsd : 0.03 rotamer "t70"
3. HE2 only : -1249.172 kcal/mol ~&gt; 26.25 kcal/mol. H-Bonds : 13 rmsd : 0.10 rotamer "t70"
4. HD1, HE2 flipped : -1258.856 kcal/mol ~&gt; 16.57 kcal/mol. H-Bonds : 12 rmsd : 0.18 rotamer "t-90"
5. HD1 only flipped : -1265.615 kcal/mol ~&gt; 9.81 kcal/mol. H-Bonds : 13 rmsd : 0.22 rotamer "t-90"
6. HE2 only flipped : -1254.390 kcal/mol ~&gt; 21.03 kcal/mol. H-Bonds : 12 rmsd : 0.20 rotamer "t-90"
```

A Coot command is supplied:

```
phenix.start_coot qm_work_dir/iterate_HIS_01_cluster_final_A_31_HIS_3.5_C_D_T_PM6-D3H4_EPS_equals_78.4.pdb qm_w
```

And a list of PyMOL commands:

```
phenix.pymol qm_work_dir/iterate_HIS_01_cluster_final_A_31_HIS_3.5_C_D_T_PM6-D3H4_EPS_equals_78.4.pml &
phenix.pymol qm_work_dir/iterate_HIS_02_cluster_final_A_31_HIS_3.5_C_D_T_PM6-D3H4_EPS_equals_78.4.pml &
phenix.pymol qm_work_dir/iterate_HIS_03_cluster_final_A_31_HIS_3.5_C_D_T_PM6-D3H4_EPS_equals_78.4.pml &
phenix.pymol qm_work_dir/iterate_HIS_04_cluster_final_A_31_HIS_3.5_C_D_T_PM6-D3H4_EPS_equals_78.4.pml &
phenix.pymol qm_work_dir/iterate_HIS_05_cluster_final_A_31_HIS_3.5_C_D_T_PM6-D3H4_EPS_equals_78.4.pml &
phenix.pymol qm_work_dir/iterate_HIS_06_cluster_final_A_31_HIS_3.5_C_D_T_PM6-D3H4_EPS_equals_78.4.pml &
```

Finally the information about the lowest (and near) energy configuration:

```
!!! 2. HD1 only : -1275.425 kcal/mol ~&gt; 0.00 kcal/mol. H-Bonds : 14 rmsd : 0.03 rotamer "t70" SAME

Close
~&gt;&lt; 1. HD1, HE2 : -1272.239 kcal/mol ~&gt; 3.19 kcal/mol. H-Bonds : 14 rmsd : 0.03 rotamer "t70"
```

Literature

Protonation of histidine rings using quantum mechanical methods. N.W. Moriarty, J. Moussa, and P.D. Adams. Acta Cryst. (2024).

- Protonation of histidine rings using quantum mechanical methods. N.W. Moriarty, J. Moussa, and P.D. Adams. Acta Cryst. (2024).

In situ ligand restraints from quantum-mechanical methods. D. Liebschner, N.W. Moriarty, B.K. Poon, and P.D. Adams. Acta Cryst. D79, 100-110 (2023).

- In situ ligand restraints from quantum-mechanical methods. D. Liebschner, N.W. Moriarty, B.K. Poon, and P.D. Adams. Acta Cryst. D79, 100-110 (2023).

Model rebuilding keeping connectivity and allowing insertions and deletions

rebuild_model.html

Author(s)

- rebuild_model: Tom Terwilliger

Purpose

Rebuild a model, keeping all connectivity the same but allowing insertions and deletions.

How rebuild_model works:

Rebuild model rebuilds each chain in your model (optionally in segments of about 15-20 residues) and reassembles these into a new model with the same connectivity as the original.

Each segment is rebuilt with several different base methods (default base methods are refinement, fixing insertions and deletions, and rebuilding from CA positions).

For each base method, a sequence of optimization steps is carried out (typically refinement, the base method, replacement of side chains, regularization, and another replacement of side chains).

Rebuilt segments are scored based on map correlation. The highest-scoring segments are used to replace the corresponding segment in the original model.

If there are multiple chains in your model, each is rebuilt separately.

The quick option can be used to speed up the process. It sets the number of refinement cycles to 1 (otherwise there are 3 carried out) and skips all the optimization steps except the base method.

Examples

Standard run of rebuild_model:

Running rebuild_model is easy. From the command-line you can type:

```
phenix.rebuild_model my_model.pdb my_map.mrc resolution=3 nproc=8
```

This will rebuild my_model.pdb based on the map my_map.mrc at a resolution of 3 Å using 8 processors.

Possible Problems

Specific limitations and problems:

Literature

Additional information

List of all available keywords

job_title = None Job title in PHENIX GUI, not used on command line
input_filesseq_file = None Sequence file. Used to guess sequence if insertions and deletions are allowed. (Fasta format or sequences separated by blank lines)
selection = None If specified, use selected part of model (discard remainder)
map_modelfull_map = None Input full map file
half_map = None Input half map files
model = None Input model file
outputrebuilt_model = None Output file name
overwrite = True Overwrite files with same names
rebuild_modelfix_insertions_deletions = True Fix insertions/deletions
refine_cycles = None Refinement cycles (set to zero to not refine). Default is 3 for standard run and 1 for quickmacro_cycles = 1 Overall macro-cycles carrying out all steps
base_methods = *None fix_insertions_deletions ca_to_model refine Base methods to run. None will run all applicable methods
optimization_sequence = *None refine base replace_side regularize_model replace_side Optimization sequence to run. None will run all applicable
methodstake_anything_over_original = None True to not use original model
segment_length = None If set, break chains up into pieces about this size
crystal_inforesolution = None Nominal resolution of map
minimum_resolution = 2.5 If resolution is finer than minimum_resolution, use minimum_resolution. This speeds up model-building.
scattering_table = n_gaussian wk1995 it1992 *electron neutron Choice of scattering table for structure factor calculations. Standard for X-ray is n_gaussian, for cryoEM is electron.
wrapping = None You can specify whether the map is wrapped (can map values outside bounds to inside with cell translations).
sequence = None Sequences
controlnproc = 1 Number of processors (if None, use all available)
ignore_symmetry_conflicts = False You can ignore the symmetry information (CRYST1) from coordinate files. This may be necessary if your model has been placed in a box with box_map for example.
verbose = False Verbose output
quick = False Run quickly. Sets refinement_cycles to 1 and optimization_sequence to base_method
guiGUI-specific parameter required for output directory
output_dir = None

- job_title = None Job title in PHENIX GUI, not used on command line
- input_filesseq_file = None Sequence file. Used to guess sequence if insertions and deletions are allowed. (Fasta format or sequences separated by blank lines)
- selection = None If specified, use selected part of model (discard remainder)
- map_modelfull_map = None Input full map file
half_map = None Input half map files
model = None Input model file
- full_map = None Input full map file
- half_map = None Input half map files
- model = None Input model file
- outputrebuilt_model = None Output file name
overwrite = True Overwrite files with same names
- rebuilt_model = None Output file name
- overwrite = True Overwrite files with same names
- rebuild_modelfix_insertions_deletions = True Fix insertions/deletions
refine_cycles = None Refinement cycles (set to zero to not refine). Default is 3 for standard run and 1 for quickmacro_cycles = 1 Overall macro-cycles carrying out all steps
base_methods = *None fix_insertions_deletions ca_to_model refine Base methods to run. None will run all applicable methods
optimization_sequence = *None refine base replace_side regularize_model replace_side Optimization sequence to run. None will run all applicable

methodstake_anything_over_original = None True to not use original model
segment_length = None If set, break chains up into pieces about this size

- fix_insertions_deletions = True Fix insertions/deletions
- refine_cycles = None Refinement cycles (set to zero to not refine). Default is 3 for standard run and 1 for quick
- macro_cycles = 1 Overall macro-cycles carrying out all steps
- base_methods = *None fix_insertions_deletions ca_to_model refine Base methods to run. None will run all applicable methods
- optimization_sequence = *None refine base replace_side regularize_model replace_side Optimization sequence to run. None will run all applicable methods
- take_anything_over_original = None True to not use original model
- segment_length = None If set, break chains up into pieces about this size
- crystal_inforesolution = None Nominal resolution of map
minimum_resolution = 2.5 If resolution is finer than minimum_resolution, use minimum_resolution. This speeds up model-building.
scattering_table = n_gaussian wk1995 it1992 *electron neutron Choice of scattering table for structure factor calculations. Standard for X-ray is n_gaussian, for cryoEM is electron.
wrapping = None You can specify whether the map is wrapped (can map values outside bounds to inside with cell translations).
sequence = None Sequences
- resolution = None Nominal resolution of map
- minimum_resolution = 2.5 If resolution is finer than minimum_resolution, use minimum_resolution. This speeds up model-building.
- scattering_table = n_gaussian wk1995 it1992 *electron neutron Choice of scattering table for structure factor calculations. Standard for X-ray is n_gaussian, for cryoEM is electron.
- wrapping = None You can specify whether the map is wrapped (can map values outside bounds to inside with cell translations).
- sequence = None Sequences
- controlnproc = 1 Number of processors (if None, use all available)
ignore_symmetry_conflicts = False You can ignore the symmetry information (CRYST1) from coordinate files. This may be necessary if your model has been placed in a box with box_map for example.
verbose = False Verbose output
quick = False Run quickly. Sets refinement_cycles to 1 and optimization_sequence to base_method
- nproc = 1 Number of processors (if None, use all available)
- ignore_symmetry_conflicts = False You can ignore the symmetry information (CRYST1) from coordinate files. This may be necessary if your model has been placed in a box with box_map for example.
- verbose = False Verbose output
- quick = False Run quickly. Sets refinement_cycles to 1 and optimization_sequence to base_method
- guiGUI-specific parameter required for output directory
output_dir = None
- output_dir = None

<string>

reciprocal_space_arrays.html

phenix.reciprocal_space_arrays: compute various arrays such as Fcalc, Fmask, Fmodel, Fbulk, and more.

Inputs:

File with reflection data (Fobs or lobs), R-free flags, and optionally HL coefficients. It can be in most of known formats and spread across multiple files; label(s) selecting which reflection data arrays should be used (in case there are multiple choices in input file, there is no need to provide labels otherwise); Model file (PDB or mmCIF) with input model.

- File with reflection data (Fobs or lobs), R-free flags, and optionally HL coefficients. It can be in most of known formats and spread across multiple files;
- label(s) selecting which reflection data arrays should be used (in case there are multiple choices in input file, there is no need to provide labels otherwise);
- Model file (PDB or mmCIF) with input model.

Usage examples:

```
phenix.reciprocal_space_arrays model.pdb data.hkl f_obs_label="IOBS" phenix.reciprocal_space_arrays  
model.pdb data.hkl r_free_flags_label="FREE"
```

- phenix.reciprocal_space_arrays model.pdb data.hkl f_obs_label="IOBS"
- phenix.reciprocal_space_arrays model.pdb data.hkl r_free_flags_label="FREE"

Output:

MTZ file with data arrays. Depending on the input data, the following arrays may be present: FOBS : data from input reflection file SIGFOBS : corresponding sigmas R_FREE_FLAGS : R-free flags (0 - work, 1 - test) FMODEL : total model structure factor. See phenix.fmodel for details PHIFMODEL : corresponding phases FCALC : Fcalc from atomic model PHIFCALC : corresponding phases FMASK : Fmask from bulk-solvent mask. See phenix.fmodel for details PHIFMASK : corresponding phases FBULK : bulk-solvent contribution. See phenix.fmodel for details PHIFBULK : corresponding phases FB_CART : overall anisotropic scale factor FOM : figures of merit ALPHA : ML parameter used in m&D calculation for 2mFo-DFc maps BETA : ML parameter used in m&D calculation for 2mFo-DFc maps HLA : HL coefficients from from input reflection file HLB : HL coefficients from from input reflection file HLC : HL coefficients from from input reflection file HLD : HL coefficients from from input reflection file HLmodelA : HL coefficients from the model (C=D=0) HLmodelB : HL coefficients from the model (C=D=0) HLmodelC : HL coefficients from the model (C=D=0) HLmodelD : HL coefficients from the model (C=D=0) HLcombA : combined HL: model + input HLcombB : combined HL: model + input HLcombC : combined HL: model + input HLcombD : combined HL: model + input RESOLUTION : resolution per reflection

MTZ file with data arrays. Depending on the input data, the following arrays may be present:

- FOBS : data from input reflection file
- SIGFOBS : corresponding sigmas
- R_FREE_FLAGS : R-free flags (0 - work, 1 - test)
- FMODEL : total model structure factor. See phenix.fmodel for details
- PHIFMODEL : corresponding phases

- FCALC : Fcalc from atomic model
- PHIFCALC : corresponding phases
- FMASK : Fmask from bulk-solvent mask. See phenix.fmodel for details
- PHIFMASK : corresponding phases
- FBULK : bulk-solvent contribution. See phenix.fmodel for details
- PHIFBULK : corresponding phases
- FB_CART : overall anisotropic scale factor
- FOM : figures of merit
- ALPHA : ML parameter used in m&D calculation for 2mFo-DFc maps
- BETA : ML parameter used in m&D calculation for 2mFo-DFc maps
- HLA : HL coefficients from from input reflection file
- HLB : HL coefficients from from input reflection file
- HLC : HL coefficients from from input reflection file
- HLD : HL coefficients from from input reflection file
- HLmodelA : HL coefficients from the model (C=D=0)
- HLmodelB : HL coefficients from the model (C=D=0)
- HLmodelC : HL coefficients from the model (C=D=0)
- HLmodelD : HL coefficients from the model (C=D=0)
- HLcombA : combined HL: model + input
- HLcombB : combined HL: model + input
- HLcombC : combined HL: model + input
- HLcombD : combined HL: model + input
- RESOLUTION : resolution per reflection

Restraints Editor Especially Ligands (REEL)

reel.html

Contents

- Author
- Purpose
- Screen Shots
- General Procedure
- Input
- Editing
- Menus
- Examples
- References

Author

Nigel W. Moriarty

Purpose

Edit the geometry restraints of a ligand using a Graphical User Interface (GUI) including a 3D view of the ligand and a tabular view of the restraints.

Screen Shots

General Procedure

The general procedure is to load a restraints file (CIF) and manipulate the restraints via the table or molecule view. The geometry of the revised restraints can be tested using the Action->Geometry optimisation option. The final restraints can be saved to a CIF file for use with phenix.refine. The corresponding PDB file can also be saved.

Input

Restraints can be loaded into REEL using the command line, the pull-down menu to open a file and a pull-down menu to run eLBOW. Restraints files from eLBOW contain both the restraints and cartesian coordinates. For the purposes of REEL, the coordinates are generated by the restraints and can not be edited directly. The background colour is light steel blue to show that it is not used in the geometry editing actions.

REEL can load a molecule geometry from any format the eLBOW can read, including PDB, Mol2D and SDF. If the file does not contain the bonding information, the bonding is automatically determined using proximity. The limit on the size of molecule that the bonding is automatically determined is set at 200 atoms.

Molecule up to 2000 atoms can be loaded using the --view but only the bond connectivity is determined. The bond order is set to one but can be changed interactively.

Molecules can be loaded into REEL using the `--reel` option for eLBOW or using the eLBOW GUI dialog available in REEL.

The Chemical Components database can also be accessed using the `--chem` option so load the cartesian coordinates and atom names of a certain ligand.

The `--smiles` option will automatically run eLBOW on the SMILES string and load the CIF file in REEL.

It may be that a CIF without cartesian coordinates is the file to be loaded. REEL will load it and attempt but its better to provide the cartesian coordinates. A simple way is to use:

```
phenix.reel ligand.cif ligand.pdb
```

and to use the `--overlay` option to force it in the case of slightly different ligand details (ligand code).

Editing

The geometry restraints (bonds, angles, dihedrals, planes and chirals) are the driving coordinates in this editor. The cartesian coordinates are displayed in the atoms table only because there are generated for the viewer display. They are the driven coordinates and are therefore ignored if changed in the editor.

Many cells in the table view of the restraints can be changed but care must be taken to ensure that the changes are local or global. For example, changing an atom name in the bonds table view will only change it in that row. If you wish to change the name of the atom in all the restraints for should use the right mouse menu in the viewer window or change it in the atoms table view. The colour of the cell gives the impression as to whether the changes are propagated elsewhere. The cells, in some cases, have not been made read only to allow the user to make changes as desired.

Clicking an atom in the molecule view will highlight the various related topological elements in the table view and vice versa. Use the checkbox in the table view or a right-mouse pop-menu to remove a restraint from optimisation and output file. Chiral centres can be changed in the table view.

Right clicking in the viewer or in the table view provide pop-menu for various actions. Examples are below.

Double clicking the label headers will sort the rows.

Menus

File

REEL can open a large number of file formats but the best file is the CIF output from eLBOW. Using the menu or the shortcut will provide a dialog to choose an input file.

The template input allows for the transfer of atom names from a PDB to the molecule in the viewer

Two different cartesian options for overlaying onto the molecule in the viewer are in the "Cartesian coordinate inputs" sub-menu. The main difference is the adjustment of the restraints' ideal value from the input geometry.

The saving options include a data sheet which is a postscript file including a two-dimensional representation of the molecule.

The view defaults including the window size and position, the molecular representation (atom labels and bond view) and background can be managed using the "View defaults" menu items.

View

View options in include toggling the hydrogens, the background, the labels on the atoms ...

and showing the bond orders.

The listing of the geometry allows the ideal restraint values and the actual geometric values to be displayed. Any actual values more than three sigma the estimated standard deviations are highlighted.

The 2D options shows a 2-dimensional representation of the molecule.

Action

Geometry optimisation using a simple force field can be performed using the first option. This is also available in the toolbar as is the semi-empirical quantum method, AM1, optimisation using the eLBOW dialog.

The last three options will open the eLBOW dialog with either no defaults, the file initially loaded for the molecule or the current CIF restraints, respectively. Once the eLBOW job is finished the user is prompted to choose what to do with the results.

"Geometry transfer" will update the cartesian coordinates of the molecule while the "Geometry and restraints transfer" will update the ideal values of the restraints. The former can be combined with the View -> "List geometry values" menu option to see the differences.

Examples

To load a previously created restraints file

```
phenix.reel atp.cif
```

To load a restraints file into REEL from eLBOW

```
phenix.elbow --smiles 0 --reel
```

To load all the unknown ligands from a PDB file

```
phenix.reel model.pdb --do-all
```

or a single residue

```
phenix.reel model.pdb --residue ATP
```

References

An editor for the generation and customization of geometry restraints. N.W. Moriarty, E.J. Draizen, and P.D. Adams. Acta Crystallogr D Biol Crystallogr 73, 123-130 (2017).

- An editor for the generation and customization of geometry restraints. N.W. Moriarty, E.J. Draizen, and P.D. Adams. Acta Crystallogr D Biol Crystallogr 73, 123-130 (2017).

Running Phenix regression tests

regression_tests.html

Author(s)

- : All

Purpose

It is important to run regression tests any time a change is made in the Phenix code to make sure that the change does not break something that is already written.

Also regression tests can be used by users to make sure everything is working correctly.

Framework

Regression tests are located in many places in the Phenix and CCTBX codebase. They are typically either in subdirectories called regression or in the phenix_regression directory, or are specified in a run_tests.py file that can be anywhere in the CCTBX codebase.

Regression tests for GUI programs are created specifically for each program and can be run as a group in the GUI. These tests and their setup is described in Setting up GUI tests).

Running regression tests

You can run all the regression tests (usually on a machine with many processors) with a simple command:

```
phenix_regression.test_all_parallel nproc=96
```

There are some simple keywords that you might want to add to this:

```
sort: sort the tests by timings, run the longest ones first
git: identify all tests that use any files that are changed in the git
    database

tests_to_run="xx,yy": Run only tests that include the words "xx" or "yy"
tests_to_skip="aa bb": Skip any tests that include the words "aa" or "bb"
```

These keywords can be combined:

```
phenix_regression.test_all_parallel nproc=96 sort tests_to_run=refine
```

Remove test set reflections from a map file with `remove_free_from_map`

`remove_free_from_map.html`

Author(s)

- `remove_free_from_map`: Tom Terwilliger

Purpose

`remove_free_from_map` is a tool to remove test set reflections from a map coefficients file so that the map can be used in real-space refinement without including the test set reflections.

NOTE: Normally `remove_free_from_map` is called from `build_one_model` or `phase_and_build`, but you can run it interactively if you want.

Usage

How `remove_free_from_map` works:

`remove_free_from_map` uses `resolve` to set all test set reflections to zero.

Output files from `remove_free_from_map`

`map_coeffs_without_freer_set.mtz`: An MTZ file with test set reflections set to F=0

Examples

Standard run of `remove_free_from_map`:

Running `remove_free_from_map` is easy. From the command-line you can type:

```
phenix.remove_free_from_map map_coeffs.mtz free_r.mtz
```

where `free_r.mtz` has your `FreeR_flag` indicating which reflections are in the test set.

Removing protein chains built into RNA density with `remove_poor_fragments`

`remove_poor_fragments.html`

Author(s)

- `remove_poor_fragments`: Tom Terwilliger

Purpose

This routine takes a model that contains RNA and protein and tries to identify fragments of protein that were accidentally built into RNA density.

Usage

When carrying out model-building for complexes between protein and RNA it sometimes happens that protein can be built accidentally into regions that are really RNA. The `phenix.remove_poor_fragments` tool is used to try and identify such incorrectly-built regions.

How `remove_poor_fragments` works:

The key idea in this approach is that these incorrectly-built regions tend to be isolated (not in the middle of a protein domain) and to have lower map-model correlation than the correct parts of the protein chains. The `phenix.remove_poor_fragments` tool ranks all protein segments on these criteria and removes the worst ones. The threshold that is chosen based on the number of residues of RNA expected in the molecule that were not built, the fraction of protein that was built, and the quality of the model, using the formula:

$$\text{residues_to_remove} = A * \text{protein_built} * \text{rna_not_built} / (\text{cc} * \text{protein_present})$$

where `cc` is the map-model correlation for the protein part of the model, `protein_present` is the number of protein residues in the sequence file, `protein_built` are residues of protein built, and `rna_not_built` are residues of RNA in the sequence file minus the number built.

The logic of this formula is that there is a certain volume, proportional to `rna_not_built`, that is really RNA that might be accidentally built as protein. Furthermore the logic is that the number of residues of protein that might be built in that volume is higher if more residues of protein have been built and higher if the model-map correlation for the protein model is low. These relationships were seen in an analysis of ribosome structures. A scale factor `A` of 0.53 relating the optimal number of residues to remove to the other factors was found empirically.

Output files from `remove_poor_fragments`

`trimmed_pdb.pdb`: A PDB file with your trimmed model.

Examples

Standard run of `remove_poor_fragments`:

Running `remove_poor_fragments` is easy. From the command-line you can type:

```
phenix.remove_poor_fragments model_with_protein_and_rna.pdb \
seq.dat \
map.ccp4
```

Possible Problems

Specific limitations and problems:

Literature

Additional information

List of all available keywords

map_file_name = None Map file name
map_coeffs_file = None File with map coefficients (alternative to map file)
map_coeffs_labels = None Optional label specifying which columns of map coefficients to use
model_file_name = None Model file names. Normally only one specified.
output_pdb_file_name = trimmed_pdb.pdb Model file name
seq_file_name = None Sequence file name. Used to calculate n_protein_in_complex, n_rna_in_complex and rna_ratio
select_good_models = True Select good models from input models and write to output PDB file
cutoff_method = *volume_of_rna_not_built sigma_cutoff Choice of method for deciding how many residues to remove. Volume of RNA not built means calculate the volume in the structure that should be RNA, subtract the volume of RNA actually built, weight by 1/mean map CC and the scale factor rna_volume_ratio. Sigma cutoff means use a cutoff based on resolution and ratio of RNA:protein residues in the model with parameters. Default is Volume of RNA not built.
rna_volume_ratio = 0.53 Scale factor on RNA volume used to decide how many residues to remove.
n_protein_in_complex = None Number of protein residues in complex. Normally obtained from sequence file.
n_rna_in_complex = None Number of RNA residues in complex. Normally obtained from sequence file.
n_protein_in_model = None Number of protein residues in model. Normally obtained from the input model. Note distinction from n_protein_in_complex which is how many there are based on the sequence file.
n_rna_in_model = None Number of RNA residues in model. Normally obtained from the input model. Note distinction from n_protein_in_complex which is how many there are based on the sequence file.
rna_ratio = None Used to estimate optimal cutoff in sigma_cutoff method. Calculated automatically from sequence file. Note that this whole procedure is optimized for cases where the rna_ratio is greater than 0.2.
toss_sigma_cutoff_a = 1.4 For sigma_cutoff method, At the conclusion of the procedure, the worst portion of chains, with low z-scores, is removed. The cutoff is $\text{toss_sigma_cutoff} = \text{toss_sigma_cutoff_a} + \text{toss_sigma_cutoff_b} * (\text{resolution} - 3.0) + \text{toss_sigma_cutoff_c} * (\text{rna_ratio} - 0.5)$. You can just set the value by setting toss_sigma_cutoff_a and setting toss_sigma_cutoff_b and toss_sigma_cutoff_c to zero. To accept a few more chains, increase toss_sigma_cutoff_a by a little (try 0.1 higher).
toss_sigma_cutoff_b = -1.1 For sigma_cutoff method, At the conclusion of the procedure, the worst portion of chains, with low z-scores, is removed. The cutoff is $\text{toss_sigma_cutoff} = \text{toss_sigma_cutoff_a} + \text{toss_sigma_cutoff_b} * (\text{resolution} - 3.0) + \text{toss_sigma_cutoff_c} * (\text{rna_ratio} - 0.5)$
toss_sigma_cutoff_c = -3.3 For sigma_cutoff method, At the conclusion of the procedure, the worst portion of chains, with low z-scores, is removed. The cutoff is $\text{toss_sigma_cutoff} = \text{toss_sigma_cutoff_a} + \text{toss_sigma_cutoff_b} * (\text{resolution} - 3.0) + \text{toss_sigma_cutoff_c} * (\text{rna_ratio} - 0.5)$
resolution = None Resolution
metrics = *mean_delta_rama *map_value *cc fraction_with_neighbor n_residues fraction_favored Metrics for a chain (fragment) being likely to be correct. Mean_delta_rama is mean change in Ramachandran angles between neighboring residues. Map_value is maximum map value at coordinates of a CA atom in chain for low_resolution map (typically 30 Å) calculated from the model. Fraction_with_neighbor is fraction of residues in a chain near another chain.

N_residues is length of chain. Fraction_favored is fraction of residues with favored Ramachandran angles
 low_resolution_map_d_min = 30 Resolution for low-resolution map showing location of molecule
 distance_cutoff = 7 In Score for fraction_with_neighbor, a residue in the same chain has to be within distance_cutoff to count as being close to another chain
 any_in_series = 4 Score for fraction_with_neighbor, fraction of residues in a chain that are near another chain, counts a residue as close if any residue in a string of any_in_a_series is near another chain
 n_offset = 10 In Score for fraction_with_ne...

- map_file_name = None Map file name
- map_coeffs_file = None File with map coefficients (alternative to map file)
- map_coeffs_labels = None Optional label specifying which columns of map coefficients to use
- model_file_name = None Model file names. Normally only one specified.
- output_pdb_file_name = trimmed_pdb.pdb Model file name
- seq_file_name = None Sequence file name. Used to calculate n_protein_in_complex, n_rna_in_complex and rna_ratio
- select_good_models = True Select good models from input models and write to output PDB file
- cutoff_method = *volume_of_rna_not_built sigma_cutoff Choice of method for deciding how many residues to remove. Volume of RNA not built means calculate the volume in the structure that should be RNA, subtract the volume of RNA actually built, weight by 1/mean map CC and the scale factor rna_volume_ratio. Sigma cutoff means use a cutoff based on resolution and ratio of RNA:protein residues in the model with parameters. Default is Volume of RNA not built.
- rna_volume_ratio = 0.53 Scale factor on RNA volume used to decide how many residues to remove.
- n_protein_in_complex = None Number of protein residues in complex. Normally obtained from sequence file.
- n_rna_in_complex = None Number of RNA residues in complex. Normally obtained from sequence file.
- n_protein_in_model = None Number of protein residues in model. Normally obtained from the input model. Note distinction from n_protein_in_complex which is how many there are based on the sequence file.
- n_rna_in_model = None Number of RNA residues in model. Normally obtained from the input model. Note distinction from n_protein_in_complex which is how many there are based on the sequence file.
- rna_ratio = None Used to estimate optimal cutoff in sigma_cutoff method. Calculated automatically from sequence file. Note that this whole procedure is optimized for cases where the rna_ratio is greater than 0.2.
- toss_sigma_cutoff_a = 1.4 For sigma_cutoff method, At the conclusion of the procedure, the worst portion of chains, with low z-scores, is removed. The cutoff is $\text{toss_sigma_cutoff} = \text{toss_sigma_cutoff_a} + \text{toss_sigma_cutoff_b} * (\text{resolution} - 3.0) + \text{toss_sigma_cutoff_c} * (\text{rna_ratio} - 0.5)$. You can just set the value by setting toss_sigma_cutoff_a and setting toss_sigma_cutoff_b and toss_sigma_cutoff_c to zero. To accept a few more chains, increase toss_sigma_cutoff_a by a little (try 0.1 higher).
- toss_sigma_cutoff_b = -1.1 For sigma_cutoff method, At the conclusion of the procedure, the worst portion of chains, with low z-scores, is removed. The cutoff is $\text{toss_sigma_cutoff} = \text{toss_sigma_cutoff_a} + \text{toss_sigma_cutoff_b} * (\text{resolution} - 3.0) + \text{toss_sigma_cutoff_c} * (\text{rna_ratio} - 0.5)$
- toss_sigma_cutoff_c = -3.3 For sigma_cutoff method, At the conclusion of the procedure, the worst portion of chains, with low z-scores, is removed. The cutoff is $\text{toss_sigma_cutoff} = \text{toss_sigma_cutoff_a} + \text{toss_sigma_cutoff_b} * (\text{resolution} - 3.0) + \text{toss_sigma_cutoff_c} * (\text{rna_ratio} - 0.5)$
- resolution = None Resolution

- `metrics = *mean_delta_rama *map_value *cc fraction_with_neighbor n_residues fraction_favored`
Metrics for a chain (fragment) being likely to be correct. `Mean_delta_rama` is mean change in Ramachandran angles between neighboring residues. `Map_value` is maximum map value at coordinates of a CA atom in chain for low_resolution map (typically 30 Å) calculated from the model. `Fraction_with_neighbor` is fraction of residues in a chain near another chain. `N_residues` is length of chain. `Fraction_favored` is fraction of residues with favored Ramachandran angles
- `low_resolution_map_d_min = 30` Resolution for low-resolution map showing location of molecule
- `distance_cutoff = 7` In Score for `fraction_with_neighbor`, a residue in the same chain has to be within `distance_cutoff` to count as being close to another chain
- `any_in_series = 4` Score for `fraction_with_neighbor`, fraction of residues in a chain that are near another chain, counts a residue as close if any residue in a string of `any_in_a_series` is near another chain
- `n_offset = 10` In Score for `fraction_with_neighbor`, a residue in the same chain has to be separated by `n_offset` to count as being from another chain
- `atom_selection = "name ca and (not element Ca)"` Specification of atoms to use in `fraction_with_neighbor`. Normally CA only
- `top_half_value = 2`. The top set of chains (typically about one half) are used to estimate statistics for good chains. Value is $1/\text{top_half_value}$.
- `cross_validate = True` Cross-validate values of mean/sd for each metric.
- `optimize_weights = False` Optimize weights. Development only. Only applies if `model_file_name` specifies a model with good chains in a structure, `second_model_file_name` specifies one with poor chains in the same structure, and weights are to be optimized.
- `n_cycle = 2` Number of minor cycles of estimation of histograms.
- `n_big_cycle = 1` Number of big cycles of estimation of histograms.
- `weight_mean_delta_rama = 1` Weight on `mean_delta_rama` metric
- `weight_map_value = 1` Weight on `map_value` metric
- `weight_cc = 1` Weight on `cc` metric
- `weight_fraction_favored = 1` Weight on `fraction_favored` metric
- `weight_n_residues = 1` Weight on `n_residues` metric
- `weight_fraction_with_neighbor = 1` Weight on `fraction_with_neighbor` metric
- `minimum_number_of_fragments = 10` Minimum number of fragments evaluated with a metric to use that metric.
- `distance_cutoff_for_split_model = 5`. Distance cutoff for splitting chain into fragments.
- `job_title = None` Job title in PHENIX GUI, not used on command line
- `guiGUI-specific parameter required for output directory`
`output_dir = None`

phenix.renumber_fab

renumber_fab.html

Description

The renumber_fab tool is designed to streamline the renumbering process for FAB chains. It performs a search among a set of supplied sequences to find the most similar sequence. Once identified, this sequence is sent to the server to retrieve the appropriate numbering. The tool supports various renumbering schemes, such as Kabat, Chothia, and Martin.

Requirements: A model and its corresponding sequence.

Ensure that the sequence encompasses the entire FAB structure and adequately covers the entire chains in the model.

Additionally, the tool will update HELIX/SHEET records describing secondary structure. However, it will neither renumber nor output SSBOND, LINK, and certain other records.

You can customize the similarity threshold by adjusting the 'identity' parameter in case the sequences are not closely matched. The tool generates detailed logs, giving insights into similarity and matched chains.

!!! WARNING !!! This program sends supplied sequence to AbNum server at <http://www.bioinf.org.uk/abs/abnum/> Please acknowledge the server authors: Abhinandan KR, Martin AC. Mol Immunol. 2008 Aug; 45(14):3832-9. <https://doi.org/10.1016/j.molimm.2008.05.022>

Command line usage examples

```
% phenix.renumber_fab model.pdb sequence.fa % phenix.renumber_fab 2jjj.cif sequence.fa scheme=chothia identity=0.4
```

```
% phenix.renumber_fab model.pdb sequence.fa
```

```
% phenix.renumber_fab 2jjj.cif sequence.fa scheme=chothia identity=0.4
```

GUI

This tool is available in the Phenix GUI in the "PDB deposition" section.

List of all available keywords

scheme = *kabat chothia martin which scheme to use
verbose = True More log output
identity_cutoff = 0.5 chains with less identity between model and sequence file will be skipped. If you have a chain with a lot of deletions in sequence, lower this number.
job_title = None Job title in PHENIX GUI, not used on command line
outputsuffix = None Suffix string added to automatically generated output filenames. If None, _fab_renumbered and scheme name will be used.
gui GUI-specific parameter required for output directory
output_dir = None

- scheme = *kabat chothia martin which scheme to use
- verbose = True More log output
- identity_cutoff = 0.5 chains with less identity between model and sequence file will be skipped. If you have a chain with a lot of deletions in sequence, lower this number.

- `job_title = None` Job title in PHENIX GUI, not used on command line
- `outputsuffix = None` Suffix string added to automatically generated output filenames. If `None`, `_fab_renumbered` and scheme name will be used.
- `suffix = None` Suffix string added to automatically generated output filenames. If `None`, `_fab_renumbered` and scheme name will be used.
- `guiGUI-specific parameter required for output directory``output_dir = None`
- `output_dir = None`

Replace side chains in a model to match the sequence with `replace_side_chains`

`replace_side_chains.html`

Author(s)

- `replace_side_chains`: Tom Terwilliger

Purpose

`replace_side_chains` is a tool to take a main-chain model and to add side-chains from a rotamer library based on an electron density map.

NOTE: Normally this method is called from `phenix.phase_and_build` but you can use it interactively if you want.

You can also use it to identify side chains (sequence) from a model without side chains (or with) and a map using `sequence_from_density=True`.

Usage

How `replace_side_chains` works:

`replace_side_chains` uses `resolve` to replace the side chains in a model. You can choose to use the default `resolve` rotamer library or the Lovell et al penultimate rotamer library.

Output files from `replace_side_chains`

`replace_side_chains.pdb`: A PDB file with side chains replaced.

Examples

Standard run of `replace_side_chains`:

Running `replace_side_chains` is easy. From the command-line you can type:

```
phenix.replace_side_chains model.pdb map_coeffs.mtz sequence.dat
```

You should supply a sequence file that matches the sequence of the model.

Using `replace_side_chains` to sequence a protein from density:

From the command-line you can type:

```
phenix.replace_side_chains model.pdb map_coeffs.mtz sequence_from_density=True
```

You don't need a sequence file but if you supply one the residues allowed are those in your sequence file.

Possible Problems

Specific limitations and problems:

Literature

Additional information

Rebuild a model using fragments from the PDB

`replace_with_fragments_from_pdb.html`

Author(s)

- `replace_with_fragments_from_pdb`: Tom Terwilliger

Purpose

Create a composite model similar to a supplied model using fragments from the PDB.

This tool can be used in early stages of model-building to try and improve parts of a preliminary model by replacing them with similar fragments from a deposited structure.

Note that the composite model may not actually have good geometry because the fragments from the PDB are morphed (distorted) to match the supplied model and also because the junctions between fragments taken from the PDB are not refined.

Usage

`replace_with_fragments_from_pdb` can create composite models for protein chains

You can adjust the maximum number of insertions/deletions in each replacement fragment and whether to morph structures obtained by SSM matching.

You can supply a map file for refinement or calculation of map-model correlations.

You can also choose to exclude or include specific PDB entries in SSM matching and fragment searching, or exclude all PDB entries with sequences similar to a particular PDB entry.

You can choose to create a replacement for just a selected part of your model (by supplying a selection string that specifies what to select).

How `replace_with_fragments_from_pdb` works:

Search structures in the PDB (using SSM matching and fragment searching) for fragments similar to segments in a supplied model. Then choose the best matching segments for each part of the supplied model and create a composite model using the fragments from the PDB.

`replace_with_fragments_from_pdb` uses SSM-based (secondary structure matching) to find structures in the PDB that are similar to a supplied model. The SSM matching is the same as in `phenix.superpose_models`.

`replace_with_fragments_from_pdb` also uses fragment searching to find shorter fragments matching each loop (segment between two helices or strands) in the supplied model.

The fragments obtained with fragment searching are always morphed (distorted) to make the distance between starting and ending residues match the distance in the supplied structure.

Fragments obtained from SSM matching are optionally morphed to match the target structure as well. The distance over which morphing occurs (the shift-field distance) is adjustable (typically 10 Å), where structure is largely preserved over shorter distances and may be changed over longer distances.

The supplied model is then used as a template and matching parts of structures from the PDB are superimposed on the template. The best-matching segments (smallest rmsd and longest length) are chosen

and connected. Gaps are filled if possible by a targeted fragment search of the PDB.

Examples

Standard run of `replace_with_fragments_from_pdb`:

Running `replace_with_fragments_from_pdb` is easy. From the command-line you can type:

```
phenix.replace_with_fragments_from_pdb model.pdb nproc=8
```

This will find fragments in the PDB similar to `model.pdb`, put them together and write out a new model using these fragments.

Possible Problems

Specific limitations and problems:

The final model may have poor geometry at connections between fragments, so further refinement is normally required. Additionally the final model may have poor geometry within segments because morphing is carried out to match the target structure.

Literature

Additional information

List of all available keywords

`job_title` = None Job title in PHENIX GUI, not used on command line
`input_filesselection` = None If specified, use selected part of model
`local_pdb_dir` = None Local PDB mirror (something like `/User/files/pdb_mirror/`)
`map_model` = None Input full map
`full_map` = None Input half map
`half_map` = None Input model file
`outputrebuilt_model` = None Output file name. Default is to append `_rebuilt` to input file
`overwrite` = True Overwrite files with same names
`replace_with_fragments_from_pdb` = None Refinement cycles (set to zero to not refine)
`refine_cycles` = 3 Maximum insertions/deletions per fragment
`maximum_insertions_or_deletions` = None Database to search. `pdb100` is unique set of PDB. `pdb2018_90` is quality-filtered PDB at 90% identity
`database` = *pdb100
`number_of_sequence_neighbors` = 100 Number of sequence neighbors to search if using `pdb100`
`max_models_to_fetch` = 100 Maximum models to fetch from PDB
`allow_reverse_connectivity` = False Allow connectivity to be backwards or mixed
`morph` = False Morph replacement segments (NOTE: morphing of loops is intrinsic)
`shift_field_distance` = None Shift field distance. Distance over which morphing leaves model mostly the same. Over longer distances morphing changes model. Default is 10 Å
`pdb_include_list` = None List of PDB ID or pdb ID and chain_id to include (2DY1 or 2DY1_A)
`pdb_exclude_list` = None List of PDB ID or pdb ID and chain_id to exclude (2DY1 or 2DY1_A)
`exclude_similar_to_this_pdb_id` = None Exclude all PDB chains with sequence identity > `maximum_percent_identity_to_this_pdb_id` to
`exclude_similar_to_this_pdb_id` = 30 Maximum percent identity to supplied PDB chain to include. Exclude all PDB chains with sequence identity > `maximum_percent_identity_to_this_pdb_id` to
`exclude_similar_to_this_pdb_id` = 30 If `try_fragment_search_for_short_models` and model has no more than `maximum_residues_to_try_fragment_search`, try simple fragment search

firsttry_fragment_search_for_short_models = True If try_fragment_search_for_short_models and model has no more than maximum_residues_to_try_fragment_search, try simple fragment search
 firstcrystal_inforesolution = None Nominal resolution of mapscattering_table = n_gaussian wk1995 it1992
 *electron neutron Choice of scattering table for structure factor calculations. Standard for X-ray is n_gaussian, for cryoEM is electron.sequence = None Sequenceswrapping = None You can specify whether the map is wrapped (can map values outside bounds to inside with cell translations).controlnproc = 1 Number of processors (if None, use all available)ignore_symmetry_conflicts = False You can ignore the symmetry information (CRYST1) from coordinate files. This may be necessary if your model has been placed in a box with box_map for example.verbose = False Verbose outputquick = False Run quicklyguiGUI-specific parameter required for output directoryoutput_dir = None

- job_title = None Job title in PHENIX GUI, not used on command line
- input_filesselection = None If specified, use selected part of modellocal_pdb_dir = None Local PDB mirror (something like /User/files/pdb_mirror/)map_modelfull_map = None Input full map filehalf_map = None Input half map filesmodel = None Input model file
- selection = None If specified, use selected part of model
- local_pdb_dir = None Local PDB mirror (something like /User/files/pdb_mirror/)
- map_modelfull_map = None Input full map filehalf_map = None Input half map filesmodel = None Input model file
- full_map = None Input full map file
- half_map = None Input half map files
- model = None Input model file
- outputrebuilt_model = None Output file name. Default is to append _rebuilt to input fileoverwrite = True Overwrite files with same names
- rebuilt_model = None Output file name. Default is to append _rebuilt to input file
- overwrite = True Overwrite files with same names
- replace_with_fragments_from_pdbrefine_cycles = 3 Refinement cycles (set to zero to not refine)maximum_insertions_or_deletions = None Maximum insertions/deletions per fragmentdatabase = *pdb100 pdb2018_90 Database to search. pdb100 is unique set of PDB. pdb2018_90 is quality-filtered PDB at 90% identitynumber_of_sequence_neighbors = 100 Number of sequence neighbors to search if using pdb100max_models_to_fetch = 100 Maximum models to fetch from PDBallow_reverse_connectivity = False Allow connectivity to be backwards or mixedmorph = False Morph replacement segments (NOTE: morphing of loops is intrinsic)shift_field_distance = None Shift field distance. Distance over which morphing leaves model mostly the same. Over longer distances morphing changes model. Default is 10 A.pdb_include_list = None List of PDB ID or pdb ID and chain_id to include (2DY1 or 2DY1_A)pdb_exclude_list = None List of PDB ID or pdb ID and chain_id to exclude (2DY1 or 2DY1_A)exclude_similar_to_this_pdb_id = None Exclude all PDB chains with sequence identity > maximum_percent_identity_to_this_pdb_id to
 exclude_similar_to_this_pdb_idmaximum_percent_identity_to_this_pdb_id = 30 Maximum percent identity to supplied PDB chain to include. Exclude all PDB chains with sequence identity > maximum_percent_identity_to_this_pdb_id to
 exclude_similar_to_this_pdb_idmaximum_residues_to_try_fragment_search = 30 If try_fragment_search_for_short_models and model has no more than maximum_residues_to_try_fragment_search, try simple fragment search
 firsttry_fragment_search_for_short_models = True If try_fragment_search_for_short_models and model

has no more than maximum_residues_to_try_fragment_search, try simple fragment search first

- refine_cycles = 3 Refinement cycles (set to zero to not refine)
- maximum_insertions_or_deletions = None Maximum insertions/deletions per fragment
- database = *pdb100 pdb2018_90 Database to search. pdb100 is unique set of PDB. pdb2018_90 is quality-filtered PDB at 90% identity
- number_of_sequence_neighbors = 100 Number of sequence neighbors to search if using pdb100
- max_models_to_fetch = 100 Maximum models to fetch from PDB
- allow_reverse_connectivity = False Allow connectivity to be backwards or mixed
- morph = False Morph replacement segments (NOTE: morphing of loops is intrinsic)
- shift_field_distance = None Shift field distance. Distance over which morphing leaves model mostly the same. Over longer distances morphing changes model. Default is 10 Å.
- pdb_include_list = None List of PDB ID or pdb ID and chain_id to include (2DY1 or 2DY1_A)
- pdb_exclude_list = None List of PDB ID or pdb ID and chain_id to exclude (2DY1 or 2DY1_A)
- exclude_similar_to_this_pdb_id = None Exclude all PDB chains with sequence identity > maximum_percent_identity_to_this_pdb_id to exclude_similar_to_this_pdb_id
- maximum_percent_identity_to_this_pdb_id = 30 Maximum percent identity to supplied PDB chain to include. Exclude all PDB chains with sequence identity > maximum_percent_identity_to_this_pdb_id to exclude_similar_to_this_pdb_id
- maximum_residues_to_try_fragment_search = 30 If try_fragment_search_for_short_models and model has no more than maximum_residues_to_try_fragment_search, try simple fragment search first
- try_fragment_search_for_short_models = True If try_fragment_search_for_short_models and model has no more than maximum_residues_to_try_fragment_search, try simple fragment search first
- crystal_inforesolution = None Nominal resolution of mapscattering_table = n_gaussian wk1995 it1992 *electron neutron Choice of scattering table for structure factor calculations. Standard for X-ray is n_gaussian, for cryoEM is electron.sequence = None Sequenceswrapping = None You can specify whether the map is wrapped (can map values outside bounds to inside with cell translations).
- resolution = None Nominal resolution of map
- scattering_table = n_gaussian wk1995 it1992 *electron neutron Choice of scattering table for structure factor calculations. Standard for X-ray is n_gaussian, for cryoEM is electron.
- sequence = None Sequences
- wrapping = None You can specify whether the map is wrapped (can map values outside bounds to inside with cell translations).
- controlnproc = 1 Number of processors (if None, use all available)ignore_symmetry_conflicts = False You can ignore the symmetry information (CRYST1) from coordinate files. This may be necessary if your model has been placed in a box with box_map for example.verbose = False Verbose outputquick = False Run quickly
- nproc = 1 Number of processors (if None, use all available)
- ignore_symmetry_conflicts = False You can ignore the symmetry information (CRYST1) from coordinate files. This may be necessary if your model has been placed in a box with box_map for example.
- verbose = False Verbose output

- quick = False Run quickly
- guiGUI-specific parameter required for output directoryoutput_dir = None
- output_dir = None

Crystallographic refinement with Rosetta (phenix.rosetta_refine)

rosetta_refine.html

Contents

- Overview
- Installation
- Execution
- Limitations
- Troubleshooting
- References
- List of all available keywords

Overview

The program phenix.rosetta_refine provides a convenient wrapper for running Rosetta refinement of protein X-ray crystal structures (DiMaio et al. 2013), which integrates the Rosetta methods for conformational sampling with the X-ray targets, B-factor refinement, and map generation in phenix.refine (Afonine et al. 2012). This is intended for use in difficult cases, especially at low resolution where it combines a wide radius of convergence with excellent geometry, and for preparing crystal structures for further modelling in Rosetta.

Currently both the setup and execution of Rosetta refinement have some limitations; please read this entire document before attempting to use the program! Note that Rosetta does not currently run on Windows; you must have a Mac or Linux system to use this program.

Installation

First, you should install the appropriate version of Phenix. The binary installers should be sufficient; users of the Mac graphical package can set up the command-line environment with this command (replacing the version number as needed):

```
source /Applications/PHENIX-1.8.4-1496/Contents/phenix-1.8.4-1496/phenix_env.sh
```

To run phenix.rosetta_refine, you must have installed Rosetta, software developed from the Baker laboratory at the University of Washington.

See the central installation notes for Rosetta

Execution

Running phenix.rosetta_refine is superficially similar to running phenix.refine, but with a much more limited set of options. Required inputs are a PDB file, a data file (MTZ format), and CIF restraint files for any non-standard ligands. Other common parameters include the number of processors to use, the method for parallelizing Rosetta jobs, and the refinement protocol, for example:

```
phenix.rosetta_refine model.pdb data.mtz nproc=5 technology=sge protocol=hires
```

The default behavior is to run the full refinement protocol as described in the program citation, creating five unique models, and picking the model with the best free R-factor. This will then be passed through

phenix.refine with a null strategy to optimize hydrogen and bulk solvent parameters. If desired, the argument `postref=True` can be given to enable additional cycles of refinement in phenix.refine, including both coordinates and B-factors with weight optimization.

The main choice of refinement strategy is between the "full" and "hires" protocols, both of which are described using the RosettaScripts syntax (Fleishman et al. 2011). The "hires" protocol is intended both for high-resolution structures and in general any structure that is already near convergence. The XML scripts for these methods are included in the Rosetta distribution. You may also create your own XML script with a customized protocol and pass that to `phenix.rosetta_refine`.

You can ask the program to parallelize jobs across multiple CPUs or several different queuing systems; the `technology` parameter determines the method used, and `nproc` indicates the number of jobs to run at once. The actual number of processes will be limited by the `number_of_models` parameter, which defaults to 5. Requesting more models may provide incremental improvements in some cases but we have found it rarely makes a difference of more than 1-2% in R-free.

Output will be to a new directory with a name such as "rosetta_1", with the directory number increasing sequentially as additional runs are created. Because phenix.refine is used as the final step, the output files will be identical to those output by that program.

Limitations

The program has several restrictions compared to phenix.refine:

Molecules are limited to the standard amino acids and a small number of common ions and ligands. If you have additional molecules in your model, these will be removed prior to Rosetta refinement and replaced afterwards. You must still supply and necessary CIFs to the program, as these will be used later when running the final phenix.refine step. RNA is not supported. Alternate conformations are not supported. The B-factor refinement is limited to simply isotropic refinement at present; for more flexible parameterization you should run phenix.refine separately. Rosetta scales poorly above approximately 1000 residues; although larger structures may work, the performance tends to be worse. At present it is not possible to impose NCS restraints on structures in Rosetta.

- Molecules are limited to the standard amino acids and a small number of common ions and ligands. If you have additional molecules in your model, these will be removed prior to Rosetta refinement and replaced afterwards. You must still supply and necessary CIFs to the program, as these will be used later when running the final phenix.refine step.
- RNA is not supported.
- Alternate conformations are not supported.
- The B-factor refinement is limited to simply isotropic refinement at present; for more flexible parameterization you should run phenix.refine separately.
- Rosetta scales poorly above approximately 1000 residues; although larger structures may work, the performance tends to be worse.
- At present it is not possible to impose NCS restraints on structures in Rosetta.

Also note that the program is significantly more processor-intensive, although less so than MR-Rosetta (DiMaio et al. 2011). We recommend that you use a multiprocessor computer or queuing system for greatest efficiency.

Troubleshooting

This program is under development; please address any questions concerning its use to help@phenix-online.org. The following guidelines may be useful for best performance:

The default settings are optimized for difficult structures, where conventional refinement gets stuck quickly. The assumption is that fine details such as waters and ligand molecules have not yet been added. For now, the only alternative is the "high-resolution" protocol, which employs gentler optimization; additional refinement scripts will be provided in the future. For optimal results, especially when preparing a structure for further Rosetta modelling, you may wish to design your own Rosetta refinement protocol using RosettaScripts. Unlike methods such as simulated annealing with the default geometry restraints, Rosetta refinement will almost never improve the fit to the experimental data at the expense of model geometry. If your structure contains severe distortions as a result of aggressive optimization, Rosetta refinement may actually degrade the model. While Rosetta refinement can improve structures that are outside the radius of convergence of traditional crystallographic refinement, it still requires that the molecule(s) be correctly placed in the unit cell. If you are uncertain of your molecular replacement solution, we recommend trying MR-Rosetta instead. The tests performed in DiMaio et al. 2013 indicated that the best results may come from combining orthogonal refinement methods such as DEN (Schroder et al. 2010) with Rosetta.

- The default settings are optimized for difficult structures, where conventional refinement gets stuck quickly. The assumption is that fine details such as waters and ligand molecules have not yet been added. For now, the only alternative is the "high-resolution" protocol, which employs gentler optimization; additional refinement scripts will be provided in the future.
- For optimal results, especially when preparing a structure for further Rosetta modelling, you may wish to design your own Rosetta refinement protocol using RosettaScripts.
- Unlike methods such as simulated annealing with the default geometry restraints, Rosetta refinement will almost never improve the fit to the experimental data at the expense of model geometry. If your structure contains severe distortions as a result of aggressive optimization, Rosetta refinement may actually degrade the model.
- While Rosetta refinement can improve structures that are outside the radius of convergence of traditional crystallographic refinement, it still requires that the molecule(s) be correctly placed in the unit cell. If you are uncertain of your molecular replacement solution, we recommend trying MR-Rosetta instead.
- The tests performed in DiMaio et al. 2013 indicated that the best results may come from combining orthogonal refinement methods such as DEN (Schroder et al. 2010) with Rosetta.

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- Super-resolution biomolecular crystallography with low-resolution data. G.F. Schröder, M. Levitt, and A.T. Brunger. Nature 464, 1218-22 (2010).

List of all available keywords

inputsequence = None scattering_table = wk1995 it1992 *n_gaussian neutron electron wavelength = None energy = None use_experimental_phases = False twin_law = Auto Enter twin law if known.xray_dataScope of X-ray data and free-R flagsfile_name = None labels = None high_resolution = None low_resolution = None twin_law = None outliers_rejection = True Remove basic wilson outliers , extreme wilson outliers , and beamstop shadow outliersfrench_wilson_scale = True sigma_fobs_rejection_criterion = None sigma_iobs_rejection_criterion = None ignore_all_zeros = True force_anomalous_flag_to_be_equal_to = None convert_to_non_anomalous_if_ratio_pairs_lone_less_than_threshold = 0.5 french_wilsonmax_bins = 60 Maximum number of resolution binsmin_bin_size = 40 Minimum number of reflections per binr_free_flagsfile_name = None This is normally the same as the file containing Fobs and is usually selected automatically.label = None required = True Specify if free-r flags are must be present (or else generated)test_flag_value = None This value is usually selected automatically - do not change unless you really know what you're doing!ignore_r_free_flags = False Use all reflections in refinement (work and test)disable_suitability_test = False ignore_pdb_hexdigest = False If True, disables safety check based on MD5 hexdigests stored in PDB files produced by previous runs.generate = False Generate R-free flags (if not available in input files)fraction = 0.1 max_free = 2000 lattice_symmetry_max_delta = 5 use_lattice_symmetry = True use_dataman_shells = False Used to avoid biasing of the test set by certain types of non-crystallographic symmetry.n_shells = 20 pdbfile_name = None Model file(s) name (PDB)monomersfile_name = None Monomer file(s) name (CIF)mapsmap_file_name = None A CCP4-formatted mapd_min = None Resolution of mapmap_coefficients_file_name = None MTZ file containing mapmap_coefficients_label = None Data label for complex map coefficients in MTZfilerosettaprotocol = *full hires Refinement protocol to use, corresponding to XML scripts in the Rosetta distribution. The hires protocol is less aggressive and faster, but has a smaller radius of convergence.script = None XML file in RosettaScripts syntax describing a custom refinement protocol.rosetta_path = None Path to

Rosetta distribution. Normally this is set by the PHENIX_ROSETTA_PATH environment variable.
 database = None Path to the Rosetta database. Normally this is set by the ROSETTA3_DB environment variable.
 score_weights = *score12_full standard score13_env_hb number_of_models = 5 density_sampling = *fast thorough ignore_unrecognized = False keep_ligands = False adp_type = *individual group map_type = *Auto 2mFo-DFc mFo dm prime_and_switch feature_enhanced composite_omit This controls the procedure for generating a map for Rosetta real-space refinement. All map types other than mFo will essentially be similar to 2mFo-DFc, but with different processing steps.
 ligand_params_file = None phenixpost_refine = False Perform additional refinement (coordinates and B-factors) in phenix.refine after Rosetta is done.
 strip_hydrogens = False Remove hydrogen atoms after Rosetta refinement.
 cycles = 3 tls = False phil_file = None waters = False Add water molecules in phenix.refine after Rosetta refinement.
 outputprefix = None directory_number = None verbose = False debug = True gui_output_dir = None Output directory (for PHENIX GUI only)
 job_title = None Job title in PHENIX GUI, not used on command line
 runtimeproc = 1 technology = *multiprocessing sge lsf pbs condor slurm threading qsub_command = None

- inputsequence = None scattering_table = wk1995 it1992 *n_gaussian neutron electron wavelength = None energy = None use_experimental_phases = False twin_law = Auto Enter twin law if known.
 xray_dataScope of X-ray data and free-R flags
 file_name = None labels = None high_resolution = None low_resolution = None twin_law = None outliers_rejection = True Remove basic wilson outliers , extreme wilson outliers , and beamstop shadow outliers
 french_wilson_scale = True sigma_fobs_rejection_criterion = None sigma_iobs_rejection_criterion = None ignore_all_zeros = True force_anomalous_flag_to_be_equal_to = None
 convert_to_non_anomalous_if_ratio_pairs_lone_less_than_threshold = 0.5 french_wilsonmax_bins = 60 Maximum number of resolution bins
 min_bin_size = 40 Minimum number of reflections per bin
 free_flagsfile_name = None This is normally the same as the file containing Fobs and is usually selected automatically.
 label = None required = True Specify if free-r flags are must be present (or else generated)
 test_flag_value = None This value is usually selected automatically - do not change unless you really know what you're doing!
 ignore_r_free_flags = False Use all reflections in refinement (work and test)
 disable_suitability_test = False ignore_pdb_hexdigest = False If True, disables safety check based on MD5 hexdigests stored in PDB files produced by previous runs.
 generate = False Generate R-free flags (if not available in input files)
 fraction = 0.1 max_free = 2000 lattice_symmetry_max_delta = 5 use_lattice_symmetry = True use_dataman_shells = False Used to avoid biasing of the test set by certain types of non-crystallographic symmetry.
 n_shells = 20 pdbfile_name = None Model file(s) name (PDB)
 monomersfile_name = None Monomer file(s) name (CIF)
 mapsmap_file_name = None A CCP4-formatted map
 map_min = None Resolution of map
 map_coefficients_file_name = None MTZ file containing map
 map_coefficients_label = None Data label for complex map coefficients in MTZ file

- sequence = None

- scattering_table = wk1995 it1992 *n_gaussian neutron electron

- wavelength = None

- energy = None

- use_experimental_phases = False

- twin_law = Auto Enter twin law if known.

- xray_dataScope of X-ray data and free-R flags
 file_name = None labels = None high_resolution = None low_resolution = None twin_law = None outliers_rejection = True Remove basic wilson outliers , extreme wilson outliers , and beamstop shadow outliers
 french_wilson_scale = True sigma_fobs_rejection_criterion = None sigma_iobs_rejection_criterion = None ignore_all_zeros = True force_anomalous_flag_to_be_equal_to = None
 convert_to_non_anomalous_if_ratio_pairs_lone_less_than_threshold = 0.5 french_wilsonmax_bins = 60

Maximum number of resolution bins `min_bin_size = 40` Minimum number of reflections per bin `r_free_flagsfile_name = None` This is normally the same as the file containing Fobs and is usually selected automatically. `label = None` `required = True` Specify if free-r flags are must be present (or else generated) `test_flag_value = None` This value is usually selected automatically - do not change unless you really know what you're doing! `ignore_r_free_flags = False` Use all reflections in refinement (work and test) `disable_suitability_test = False` `ignore_pdb_hexdigest = False` If True, disables safety check based on MD5 hexdigests stored in PDB files produced by previous runs. `generate = False` Generate R-free flags (if not available in input files) `fraction = 0.1` `max_free = 2000` `lattice_symmetry_max_delta = 5` `use_lattice_symmetry = True` `use_dataman_shells = False` Used to avoid biasing of the test set by certain types of non-crystallographic symmetry. `n_shells = 20`

- `file_name = None`
- `labels = None`
- `high_resolution = None`
- `low_resolution = None`
- `twin_law = None`
- `outliers_rejection = True` Remove basic wilson outliers , extreme wilson outliers , and beamstop shadow outliers
- `french_wilson_scale = True`
- `sigma_fobs_rejection_criterion = None`
- `sigma_iobs_rejection_criterion = None`
- `ignore_all_zeros = True`
- `force_anomalous_flag_to_be_equal_to = None`
- `convert_to_non_anomalous_if_ratio_pairs_lone_less_than_threshold = 0.5`
- `french_wilsonmax_bins = 60` Maximum number of resolution bins `min_bin_size = 40` Minimum number of reflections per bin
- `max_bins = 60` Maximum number of resolution bins
- `min_bin_size = 40` Minimum number of reflections per bin
- `r_free_flagsfile_name = None` This is normally the same as the file containing Fobs and is usually selected automatically. `label = None` `required = True` Specify if free-r flags are must be present (or else generated) `test_flag_value = None` This value is usually selected automatically - do not change unless you really know what you're doing! `ignore_r_free_flags = False` Use all reflections in refinement (work and test) `disable_suitability_test = False` `ignore_pdb_hexdigest = False` If True, disables safety check based on MD5 hexdigests stored in PDB files produced by previous runs. `generate = False` Generate R-free flags (if not available in input files) `fraction = 0.1` `max_free = 2000` `lattice_symmetry_max_delta = 5` `use_lattice_symmetry = True` `use_dataman_shells = False` Used to avoid biasing of the test set by certain types of non-crystallographic symmetry. `n_shells = 20`
- `file_name = None` This is normally the same as the file containing Fobs and is usually selected automatically.
- `label = None`
- `required = True` Specify if free-r flags are must be present (or else generated)
- `test_flag_value = None` This value is usually selected automatically - do not change unless you really know what you're doing!

- ignore_r_free_flags = False Use all reflections in refinement (work and test)
- disable_suitability_test = False
- ignore_pdb_hexdigest = False If True, disables safety check based on MD5 hexdigests stored in PDB files produced by previous runs.
- generate = False Generate R-free flags (if not available in input files)
- fraction = 0.1
- max_free = 2000
- lattice_symmetry_max_delta = 5
- use_lattice_symmetry = True
- use_dataman_shells = False Used to avoid biasing of the test set by certain types of non-crystallographic symmetry.
- n_shells = 20
- pdbfile_name = None Model file(s) name (PDB)
- file_name = None Model file(s) name (PDB)
- monomersfile_name = None Monomer file(s) name (CIF)
- file_name = None Monomer file(s) name (CIF)
- mapsmap_file_name = None A CCP4-formatted mapd_min = None Resolution of mapmap_coefficients_file_name = None MTZ file containing mapmap_coefficients_label = None Data label for complex map coefficients in MTZ file
- map_file_name = None A CCP4-formatted map
- d_min = None Resolution of map
- map_coefficients_file_name = None MTZ file containing map
- map_coefficients_label = None Data label for complex map coefficients in MTZ file
- rosetta_protocol = *full hires Refinement protocol to use, corresponding to XML scripts in the Rosetta distribution. The hires protocol is less aggressive and faster, but has a smaller radius of convergence.script = None XML file in RosettaScripts syntax describing a custom refinement protocol.rosetta_path = None Path to Rosetta distribution. Normally this is set by the PHENIX_ROSETTA_PATH environment variable.database = None Path to the Rosetta database. Normally this is set by the ROSETTA3_DB environment variable.score_weights = *score12_full standard score13_env_hb number_of_models = 5 density_sampling = *fast thorough ignore_unrecognized = False keep_ligands = False adp_type = *individual group map_type = *Auto 2mFo-DFc mFo dm prime_and_switch feature_enhanced composite_omit This controls the procedure for generating a map for Rosetta real-space refinement. All map types other than mFo will essentially be similar to 2mFo-DFc, but with different processing steps.ligand_params_file = None
- protocol = *full hires Refinement protocol to use, corresponding to XML scripts in the Rosetta distribution. The hires protocol is less aggressive and faster, but has a smaller radius of convergence.
- script = None XML file in RosettaScripts syntax describing a custom refinement protocol.
- rosetta_path = None Path to Rosetta distribution. Normally this is set by the PHENIX_ROSETTA_PATH environment variable.
- database = None Path to the Rosetta database. Normally this is set by the ROSETTA3_DB environment variable.

- score_weights = *score12_full standard score13_env_hb
- number_of_models = 5
- density_sampling = *fast thorough
- ignore_unrecognized = False
- keep_ligands = False
- adp_type = *individual group
- map_type = *Auto 2mFo-DFc mFo dm prime_and_switch feature_enhanced composite_omit This controls the procedure for generating a map for Rosetta real-space refinement. All map types other than mFo will essentially be similar to 2mFo-DFc, but with different processing steps.
- ligand_params_file = None
- phenixpost_refine = False Perform additional refinement (coordinates and B-factors) in phenix.refine after Rosetta is done.strip_hydrogens = False Remove hydrogen atoms after Rosetta refinement.cycles = 3
- tls = False phil_file = None waters = False Add water molecules in phenix.refine after Rosetta refinement.
- post_refine = False Perform additional refinement (coordinates and B-factors) in phenix.refine after Rosetta is done.
- strip_hydrogens = False Remove hydrogen atoms after Rosetta refinement.
- cycles = 3
- tls = False
- phil_file = None
- waters = False Add water molecules in phenix.refine after Rosetta refinement.
- outputprefix = None directory_number = None verbose = False debug = True gui_output_dir = None Output directory (for PHENIX GUI only)job_title = None Job title in PHENIX GUI, not used on command line
- prefix = None
- directory_number = None
- verbose = False
- debug = True
- gui_output_dir = None Output directory (for PHENIX GUI only)
- job_title = None Job title in PHENIX GUI, not used on command line
- runtimenproc = 1 technology = *multiprocessing sge lsf pbs condor slurm threading qsub_command = None
- nproc = 1
- technology = *multiprocessing sge lsf pbs condor slurm threading
- qsub_command = None

Running SOLVE/RESOLVE in PHENIX

running-solve-resolve.html

Author(s)

- SOLVE/RESOLVE: Tom Terwilliger

Purpose

SOLVE and RESOLVE can be run directly in the PHENIX environment. This feature is normally only for advanced SOLVE/RESOLVE users who want to access the keywords in SOLVE/RESOLVE directly.

Usage

Running SOLVE/RESOLVE from the command-line or in a script.

- You can run solve with the command:

```
phenix.solve
```

This command will set the environmental variables CCP4_OPEN, SYMOP, SYMLIB, and SOLVEDIR and will run solve. If you want to run a different size of solve, then you can specify:

```
phenix.solve --giant
```

For a bigger version still, choose --huge, for biggest, --extra_huge.

- Running resolve or resolve_pattern is similar:

```
phenix.resolve  
phenix.resolve_pattern
```

- Running solve/resolve from a command file is simple. Here is a command file to run resolve:

```
phenix.resolve<<<EOD  
hklin solve.mtz  
labin FP=FP SIGFP=SIGFP PHIB=PHIB FOM=FOM HLA=HLA HLB=HLB HLC=HLC HLD=HLD  
solvent_content 0.43  
database 5  
EOD
```

Literature

Additional information

All the solve/resolve keywords are available in the PHENIX versions of solve and resolve. See the full documentation for solve/resolve at <https://solve.lanl.gov/> . A PDF of this documentation is available at https://phenix-online.org/phenix_data/terwilliger/solve_resolve_archive_doc/SOLVE_RESOLVE_213.pdf .

Downloading data from PDB and setting up for SAD structure determination with phenix.sad_data_from_pdb

sad_data_from_pdb.html

Author(s)

- phenix.sad_data_from_pdb: Tom Terwilliger

Purpose

phenix.sad_data_from_pdb is a command line tool for downloading coordinates and anomalous structure factor data for a PDB deposition and setting up for running one or more wavelengths of data as SAD datasets.

Usage

How phenix.sad_data_from_pdb works:

phenix.sad_data_from_pdb first uses phenix.fetch_pdb to download the data from the PDB.

Then it converts the structure factor data to Scalepack unmerged original index format, calculates model map coefficients, and identifies the anomalously-scattering atoms in the PDB file, the wavelengths of data collection, and the resolution of the data. If these are not present in the PDB deposition then the procedure will not work.

Examples

Standard run of phenix.sad_data_from_pdb:

Running phenix.sad_data_from_pdb is easy. You can just type:

```
phenix.sad_data_from_pdb pdb_ids="2etd"
```

and phenix.sad_data_from_pdb will download the data for 2etd and set up a directory with this data and with subdirectories for each wavelength of data present (w1, w2, and w3 in this case). In each subdirectory a file run_autosol_std.csh will be created that is suitable for running phenix.autosol using that anomalous data.

Possible Problems

Specific limitations and problems:

Literature

Additional information

List of all available keywords

base_directory = None Directory where files are to be located
 pdb_ids = None PDB ID to fetch and use to create SAD datasets
 wavelength_id = None Select only the wavelength specified by wavelength_id
 ha_list = None List of anomalously-scattering atoms (used instead of list extracted from PDB file)
 steps = *fetch *cif_to_sca *fmodel *ha *autosol Steps to carry out: fetch downloads files from the PDB, cif_to_sca converts PDB CIF files to Scalepack unmerged original index, fmodel calculates model structure factors, ha identifies anomalously-scattering atoms in the PDB file, autosol sets up for a run of autosol. To select several steps put them in quotes with the stars: steps='*ha *autosol'
 initialize = None If set, remove any existing data in the PDB directories
 best_only = True Keep only atom with strongest anomalous scattering and ignore the rest
 merge = None If set, merge multiple observations
 replace_full_directory = None You can change the full directory name for a single dataset

- base_directory = None Directory where files are to be located
- pdb_ids = None PDB ID to fetch and use to create SAD datasets
- wavelength_id = None Select only the wavelength specified by wavelength_id
- ha_list = None List of anomalously-scattering atoms (used instead of list extracted from PDB file)
- steps = *fetch *cif_to_sca *fmodel *ha *autosol Steps to carry out: fetch downloads files from the PDB, cif_to_sca converts PDB CIF files to Scalepack unmerged original index, fmodel calculates model structure factors, ha identifies anomalously-scattering atoms in the PDB file, autosol sets up for a run of autosol. To select several steps put them in quotes with the stars: steps='*ha *autosol'
- initialize = None If set, remove any existing data in the PDB directories
- best_only = True Keep only atom with strongest anomalous scattering and ignore the rest
- merge = None If set, merge multiple observations
- replace_full_directory = None You can change the full directory name for a single dataset

Scaling unmerged anomalous data or multiple datasets with `scale_and_merge`

`scale_and_merge.html`

Author(s)

- `scale_and_merge`: Tom Terwilliger

Purpose

`scale_and_merge` is a tool for scaling unmerged anomalous data or multiple data files and creating a scaled dataset and two scaled half-datasets. This tool normally is used in combination with `anomalous_signal` to create a scaled dataset and analyze anomalous signal in a SAD dataset. It can also be used to scale any other group of datasets.

Usage

How `scale_and_merge` works:

- You supply `scale_and_merge` with a directory containing datafiles or the name of a single unmerged datafile.
- You can optionally also specify a pair of labels that identifies datasets that are to be kept together. For example if you collected your data as pairs of datafiles with inverse beam geometry, you might have called the members of a pair `data_1_0_w1.HKL` and `data_1_0_w2.HKL`, related by `w1` and `w2`.
- `scale_and_merge` will first check the cell dimensions of all the datasets. Normally it will choose the largest set of similar crystals (you can have it keep all datasets with `only_similar_datasets=False`). It will also check the anisotropy in all the datafiles and calculate the average anisotropy (to be applied by default to all datafiles before scaling).
- `scale_and_merge` will then scale all the data together. It will split your datafiles into smaller files if your datafiles contain duplicate measurements of the same indices. The intensities in each datafile will be adjusted for anisotropy to match the average anisotropy of all the datafiles. In this way all the datafiles are matched but the overall character of the data is not changed.
- `scale_and_merge` will then scale each individual file with local scaling, producing files in the temporary directory with names ending in `_scale.mtz`. and then merge them all together.
- Merging of the individual datasets is done twice, and then optionally two additional times to optimize anomalous differences.
- In the first merging the individual datasets are simply averaged with weights based on the sigmas for each reflection. Then the individual datasets are compared with the overall mean dataset and dataset variances (estimates of systematic differences between datasets and the mean) are calculated based on the rms differences from the mean and the individual sigmas. In essence the dataset variance plus the individual variances are then to be used as estimates of the total variance for each reflection. The second merging uses these total variances in weighting rather than the original sigmas. The correlations of each individual dataset with the merged dataset are calculated.

- If anomalous differences are optimized (default with `optimize_anomalous=True`), merging is carried out another time in order to optimize the weighting of anomalous differences in the merging step. For each unique reflection in the asymmetric unit of each dataset, the mean I+ and mean I- are used to calculate anomalous differences. The anomalous differences from each individual dataset are then compared with the anomalous differences from the merged dataset to estimate individual dataset anomalous difference variances. Then the anomalous differences from each individual dataset are averaged, with weights based on the original sigmas and the dataset variances. These merged anomalous differences are then used to replace the anomalous differences in the merged dataset above (for example, a reflection in the merged dataset above that has I+,sigl+,I-,sigl- would get new values of I+ and I- that have a difference equal to the appropriate merged anomalous difference, but the same mean as before.) The correlation of the anomalous differences in the original merged dataset and after optimization is printed (this should be high, for example 0.80).
- The original datasets are then split into two parts for creation of two half-datasets. These half-datasets are useful for estimating the quality of the data and are used in `anomalous_signal` for this purpose.
- The splitting into half-datasets is done in one of three ways, and the method is chosen based on the number of anomalous differences available for comparison using each method. With each method the data in each half-dataset are scaled just as the entire dataset was scaled. The preferred method is to split by files. Half of the datafiles are used to create each half-dataset. The next preferred method is to split with the first half of each dataset in one half-dataset and the second half in the other. The third preferred method is splitting alternate reflections with each unique index (after mapping to the asymmetric unit) into the two half-datasets. The last method is to randomly assign reflections to the two half-datasets. The reason for this hierarchical approach is that reflections measured close in time, within the same dataset are better matched than those measured further in time or within different datasets. These approaches for splitting the datafiles attempt to pair anomalous differences measured close in time and in the same dataset.
- After half-datasets are obtained, the correlation of anomalous differences between half-datasets is estimated.

Output from `scale_and_merge`

`scale_and_merge` provides a summary half-dataset correlations in your dataset. Here is an example with very weak anomalous signal (but present):

```
Scale and merge...analysis of multi-dataset SAD data
input\_files {
  data = "dd"
  data\_labels = None
  paired\_group\_ids = None
}
output\_files {
  output\_file = "scaled\_data.mtz"
  output\_half\_dataset\_a = "half\_dataset\_a.mtz"
  output\_half\_dataset\_b = "half\_dataset\_b.mtz"
  output\_file\_format = *mtz sca
}
crystal\_info {
  resolution = None
  low\_resolution = None
  space\_group = None
  unit\_cell = None
}
data\_selection {
  minimum\_datafile\_fraction = 0.3
  require\_fpfm = True
}
```

```

only\_similar\_datasets = True
relative\_length\_tolerance = None
absolute\_angle\_tolerance = None
choose\_optimal\_datasets = False
sort\_datasets\_by\_anomalous\_cc = True
}
scaling {
  make\_anisotropy\_uniform = True
  overall\_scale = False
  skip\_scaling = False
  lowest\_resolution\_range = 6
}
merging {
  optimize\_anomalous = True
  use\_best\_group\_as\_target = False
  rescale\_sigmas = False
}
half\_dataset\_cc {
  get\_half\_dataset\_cc = True
  half\_dataset\_cc\_by\_files = True
  split\_as\_first\_second\_half = True
  split\_alternately = True
  split\_randomly = True
}
directories {
  temp\_dir = None
  output\_dir = ""
  gui\_output\_dir = None
}
control {
  verbose = False
  random\_seed = 714215
  comparison\_file = None
  clean\_up = True
}

```

Comparing crystal symmetries and noting anisotropy in data

Relative length tolerance: 0.010 Angle tolerance: 1.000 degrees

Taking initial symmetry from /Users/terwill/unix/misc/scale_and_merge/dd/WNV_NS1_205_w2_2.9_90.HKL
with 3 similar files

File SG A B C Alpha Beta Gamma

Symmetry group 1

```

WNV\_NS1\_202\_w1\_2.9\_90.HKL P 3 2 1 166.997 166.997 94.156 90.000 90.000 120.000
WNV\_NS1\_202\_w2\_2.9\_90.HKL P 3 2 1 166.704 166.704 94.113 90.000 90.000 120.000
WNV\_NS1\_205\_w1\_2.9\_90.HKL P 3 2 1 167.578 167.578 93.934 90.000 90.000 120.000
WNV\_NS1\_205\_w2\_2.9\_90.HKL P 3 2 1 167.639 167.639 93.859 90.000 90.000 120.000

```

Scaling 4 datasets

Scaling data files with local scaling

Files to scale:

```

/Users/terwill/unix/misc/scale\_and\_merge/dd/WNV\_NS1\_202\_w1\_2.9\_90.HKL
/Users/terwill/unix/misc/scale\_and\_merge/dd/WNV\_NS1\_202\_w2\_2.9\_90.HKL
/Users/terwill/unix/misc/scale\_and\_merge/dd/WNV\_NS1\_205\_w1\_2.9\_90.HKL
/Users/terwill/unix/misc/scale\_and\_merge/dd/WNV\_NS1\_205\_w2\_2.9\_90.HKL

```

Selecting just reflections that have both F+ and F- in the same
dataset or are centric

Splitting datafiles into sub-files with one copy of each unique hkl

File Refl (rejected) B1 B2 B3 B-avg

```

TEMP0/WNV\_NS1\_202\_w1\_2.9\_90\_1.sca: 138180 ( 6378) 63.9 63.9 55.00 60.9
TEMP0/WNV\_NS1\_202\_w1\_2.9\_90\_2.sca: 229 ( 0) 63.9 63.9 55.00 60.9

```

```

TEMP0\WNV\NS1\202\w2\2.9\90\1.sca: 136254 ( 6096) 59.4 59.4 49.78 56.2
TEMP0\WNV\NS1\202\w2\2.9\90\2.sca: 206 ( 0) 59.4 59.4 49.78 56.2
TEMP0\WNV\NS1\205\w1\2.9\90\1.sca: 148714 ( 4464) 73.7 73.7 63.92 70.4
TEMP0\WNV\NS1\205\w1\2.9\90\2.sca: 239 ( 0) 73.7 73.7 63.92 70.4
TEMP0\WNV\NS1\205\w2\2.9\90\1.sca: 147256 ( 4560) 70.3 70.3 60.50 67.0
TEMP0\WNV\NS1\205\w2\2.9\90\2.sca: 241 ( 0) 70.3 70.3 60.50 67.0

```

Notes:

```

Splitting /Users/terwill/unix/misc/scale\_and\_merge/dd/WNV\NS1\202\w1\2.9\90.HKL into 2 files:
Splitting /Users/terwill/unix/misc/scale\_and\_merge/dd/WNV\NS1\202\w2\2.9\90.HKL into 2 files:
Splitting /Users/terwill/unix/misc/scale\_and\_merge/dd/WNV\NS1\205\w1\2.9\90.HKL into 2 files:
Splitting /Users/terwill/unix/misc/scale\_and\_merge/dd/WNV\NS1\205\w2\2.9\90.HKL into 2 files:
Keeping split datafiles with at least 44614 reflections

```

High-resolution limit: 2.89

Scaling data in batches from individual data files

List of scaled data files:

```

TEMP0\WNV\NS1\205\w1\2.9\90\1\_scale.mtz (30844 refl)
TEMP0\WNV\NS1\205\w2\2.9\90\1\_scale.mtz (30828 refl)
TEMP0\WNV\NS1\202\w1\2.9\90\1\_scale.mtz (29641 refl)
TEMP0\WNV\NS1\202\w2\2.9\90\1\_scale.mtz (29527 refl)

```

Scaling and merging data files with overall scale factor

Files to scale:

```
TEMP0\WNV\NS1\205\w1\2.9\90\1\_scale.mt...
```

Possible Problems

Literature

List of all available keywords

input_filesdata = None Anomalous data (I+ and I- or F+ and F-). Any standard format is fine. You can have any number of input files. If you have multiple files, specify the directory that contains them and nothing else.data_labels = None Optional label specifying which columns of anomalous data to use. Not necessary if your input files have only one set of anomalous data. If you have multiple input files they need to have the same labels.assume_shelx_observation_type_is = *amplitudes intensities For shelx files you need to specify if they are amplitudes or intensitiespaired_group_ids = None Optional pair of identifiers that specify datasets that should be grouped together. The paired group ids must be the only characters that differ between datasets that should be grouped together, and the characters must uniquely define what datasets to group. For example paired_group_ids='w1 w2' would be the pair for the two datasets se_w1.sca and se_w2.sca. In this example you could also use paired_group_ids='w1.sca w2.sca' and that would have the same effect. Use this option when some of your datafiles are related to others by inverse-beam data collection for example. This matching of datasets can be very helpful.check_file_type = False You can request that the data type of input files be checked if a data directory is supplied. Default is to read everything in the data directory.reference_data = None Reference data for use in scaling. All datasets will be scaled to this reference dataset. Useful if you have very accurate native data.reference_data_labels = None Optional label specifying which columns of reference data to use.output_filesoutput_file = scaled_data.mtz Output data file name for merged scaled dataoutput_half_dataset_a = half_dataset_a.mtz Output data file name for half dataset A. Only written if get_half_datset_cc=True. This is the scaled merged data from half of the input data.output_half_dataset_b = half_dataset_b.mtz Output data file name for half dataset B. Only written if get_half_datset_cc=True. This is the scaled merged data from the other half of the input data.output_file_format = *mtz sca Output format for scaled data. Choices are MTZ and unmerged scalepack

formats. Note: unmerged scalepack format not normally used because indices may be changed from original values and that data from within one input file may be merged, so that this unmerged data is partially merged and not the same as the original data.

crystal_inforesolution = None high-resolution limit
 low_resolution = None Low-resolution limit
 space_group = None Space group (normally read from the data file)
 unit_cell = None Unit Cell (normally read from the data file)
 data_selectionanomalous = True Consider data as anomalous (otherwise average Bijvoet mates).
 minimum_datafile_fraction = 0.3 You can reject any datasets (after splitting) that have less than this fraction of the maximum number of reflections found in any datafile.
 require_fpfm = True If set, then when files are split into unique sets of indices only centric reflections and reflections with both F+ and F- present are written out. Note that in most space groups there may be more than one F+ or F- symmetry mate present. If any one is present for F+ and any one for F- the reflection is written out.
 only_similar_datasets = True Group data by lattice similarity and take biggest group only. If space_group and unit_cell are set, then take those only that match this space_group and unit_cell. Use relative_length_tolerance and absolute_angle_tolerance to decide if two datasets are similar.
 relative_length_tolerance = None Tolerance for considering two cells similar (fractional) You can allow anything with only_similar_datasets=False
 resolution_length_tolerance = None Tolerance for considering two cells similar (fraction of resolution. For example normalized_length_tolerance=0.25 means the unit cells must match within 1/4 of the resolution. You can allow anything with only_similar_datasets=False
 absolute_length_toler...

- input_filesdata = None Anomalous data (I+ and I- or F+ and F-). Any standard format is fine. You can have any number of input files. If you have multiple files, specify the directory that contains them and nothing else.
 data_labels = None Optional label specifying which columns of anomalous data to use. Not necessary if your input files have only one set of anomalous data. If you have multiple input files they need to have the same labels.
 assume_shelx_observation_type_is = *amplitudes intensities For shelx files you need to specify if they are amplitudes or intensities
 paired_group_ids = None Optional pair of identifiers that specify datasets that should be grouped together. The paired group ids must be the only characters that differ between datasets that should be grouped together, and the characters must uniquely define what datasets to group. For example paired_group_ids='w1 w2' would be the pair for the two datasets se_w1.sca and se_w2.sca. In this example you could also use paired_group_ids='w1.sca w2.sca' and that would have the same effect. Use this option when some of your datafiles are related to others by inverse-beam data collection for example. This matching of datasets can be very helpful.
 check_file_type = False You can request that the data type of input files be checked if a data directory is supplied. Default is to read everything in the data directory.
 reference_data = None Reference data for use in scaling. All datasets will be scaled to this reference dataset. Useful if you have very accurate native data.
 reference_data_labels = None Optional label specifying which columns of reference data to use.
- data = None Anomalous data (I+ and I- or F+ and F-). Any standard format is fine. You can have any number of input files. If you have multiple files, specify the directory that contains them and nothing else.
- data_labels = None Optional label specifying which columns of anomalous data to use. Not necessary if your input files have only one set of anomalous data. If you have multiple input files they need to have the same labels.
- assume_shelx_observation_type_is = *amplitudes intensities For shelx files you need to specify if they are amplitudes or intensities
- paired_group_ids = None Optional pair of identifiers that specify datasets that should be grouped together. The paired group ids must be the only characters that differ between datasets that should be grouped together, and the characters must uniquely define what datasets to group. For example paired_group_ids='w1 w2' would be the pair for the two datasets se_w1.sca and se_w2.sca. In this

example you could also use `paired_group_ids='w1.sca w2.sca'` and that would have the same effect. Use this option when some of your datafiles are related to others by inverse-beam data collection for example. This matching of datasets can be very helpful.

- `check_file_type = False` You can request that the data type of input files be checked if a data directory is supplied. Default is to read everything in the data directory.
- `reference_data = None` Reference data for use in scaling. All datasets will be scaled to this reference dataset. Useful if you have very accurate native data.
- `reference_data_labels = None` Optional label specifying which columns of reference data to use.
- `output_filesoutput_file = scaled_data.mtz` Output data file name for merged scaled data
- `dataoutput_half_dataset_a = half_dataset_a.mtz` Output data file name for half dataset A. Only written if `get_half_datset_cc=True`. This is the scaled merged data from half of the input data.
- `output_half_dataset_b = half_dataset_b.mtz` Output data file name for half dataset B. Only written if `get_half_datset_cc=True`. This is the scaled merged data from the other half of the input data.
- `data.output_file_format = *mtz sca` Output format for scaled data. Choices are MTZ and unmerged scalepack formats. Note: unmerged scalepack format not normally used because indices may be changed from original values and that data from within one input file may be merged, so that this unmerged data is partially merged and not the same as the original data.
- `output_file = scaled_data.mtz` Output data file name for merged scaled data
- `output_half_dataset_a = half_dataset_a.mtz` Output data file name for half dataset A. Only written if `get_half_datset_cc=True`. This is the scaled merged data from half of the input data.
- `output_half_dataset_b = half_dataset_b.mtz` Output data file name for half dataset B. Only written if `get_half_datset_cc=True`. This is the scaled merged data from the other half of the input data.
- `output_file_format = *mtz sca` Output format for scaled data. Choices are MTZ and unmerged scalepack formats. Note: unmerged scalepack format not normally used because indices may be changed from original values and that data from within one input file may be merged, so that this unmerged data is partially merged and not the same as the original data.
- `crystal_inforesolution = None` high-resolution limit `low_resolution = None` Low-resolution limit
`space_group = None` Space group (normally read from the data file)
`unit_cell = None` Unit Cell (normally read from the data file)
- `resolution = None` high-resolution limit
- `low_resolution = None` Low-resolution limit
- `space_group = None` Space group (normally read from the data file)
- `unit_cell = None` Unit Cell (normally read from the data file)
- `data_selectionanomalous = True` Consider data as anomalous (otherwise average Bijvoet mates).
- `minimum_datafile_fraction = 0.3` You can reject any datasets (after splitting) that have less than this fraction of the maximum number of reflections found in any datafile
- `require_fpm = True` If set, then when files are split into unique sets of indices only centric reflections and reflections with both F+ and F- present are written out. Note that in most space groups there may be more than one F+ or F- symmetry mate present. If any one is present for F+ and any one for F- the reflection is written out.
- `only_similar_datasets = True` Group data by lattice similarity and take biggest group only. If `space_group` and `unit_cell` are set, then take those only that match this `space_group` and `unit_cell`. Use `relative_length_tolerance` and `absolute_angle_tolerance` to decide if two datasets are similar.
- `relative_length_tolerance = None` Tolerance for considering two cells similar (fractional) You can allow anything with `only_similar_datasets=False`
- `resolution_length_tolerance = None` Tolerance for

considering two cells similar (fraction of resolution. For example `normalized_length_tolerance=0.25` means the unit cells must match within 1/4 of the resolution. You can allow anything with `only_similar_datasets=False`
`absolute_length_tolerance = None` `absolute_angle_tolerance = None`
 Tolerance for considering two cell angles similar (degrees) You can allow anything with `only_similar_datasets=False`
`choose_optimal_datasets = False` After initial merging, sort datasets by correlation to merged dataset, remove datasets until overall $I/\sigma I$ is maximized, including dataset systematic errors in $I/\sigma I$.
`sort_datasets_by_anomalous_cc = True` For choosing optimal datasets, sort the datasets by anomalous CC. Alternative is sorting by non-anomalous CC.

- `anomalous = True` Consider data as anomalous (otherwise average Bijvoet mates).
- `minimum_datafile_fraction = 0.3` You can reject any datasets (after splitting) that have less than this fraction of the maximum number of reflections found in any datafile
- `require_fpfm = True` If set, then when files are split into unique sets of indices only centric reflections and reflections with both F+ and F- present are written out. Note that in most space groups there may be more than one F+ or F- symmetry mate present. If any one is present for F+ and any one for F- the reflection is written out.
- `only_similar_datasets = True` Group data by lattice similarity and take biggest group only. If `space_group` and `unit_cell` are set, then take those only that match this `space_group` and `unit_cell`. Use `relative_length_tolerance` and `absolute_angle_tolerance` to decide if two datasets are similar.
- `relative_length_tolerance = None` Tolerance for considering two cells similar (fractional) You can allow anything with `only_similar_datasets=False`
- `resolution_length_tolerance = None` Tolerance for considering two cells similar (fraction of resolution. For example `normalized_length_tolerance=0.25` means the unit cells must match within 1/4 of the resolution. You can allow anything with `only_similar_datasets=False`
- `absolute_length_tolerance = None`
- `absolute_angle_tolerance = None` Tolerance for considering two cell angles similar (degrees) You can allow anything with `only_similar_datasets=False`
- `choose_optimal_datasets = False` After initial merging, sort datasets by correlation to merged dataset, remove datasets until overall $I/\sigma I$ is maximized, including dataset systematic errors in $I/\sigma I$.
- `sort_datasets_by_anomalous_cc = True` For choosing optimal datasets, sort the datasets by anomalous CC. Alternative is sorting by non-anomalous CC.
- `scalingmake_anisotropy_uniform = True` Calculate anisotropy for all datasets and adjust all of them to the average anisotropy. This is designed to improve scaling without changing the overall character of the data (i.e., not sharpening the data or removing anisotropy).
`overallscale = False` If set, only use overall scale and B (no local scaling)
`skip_scaling = False` You can skip the scaling step and just get half-dataset CC for a pair of existing files that you have put in a directory by themselves.
`lowest_resolution_range = 6`. Lowest value of the high-resolution cutoff for CC as a function of high-resolution cutoff with constant low-resolution cutoff
`min_common_refl = 200` Minimum number of reflections in common to scale datasets
`sort_on_batch_number = True` You can sort XDS and MTZ reflections by frame number. Otherwise scale and merge assumes reflections are collected in the order in the data file.
`scale_reference_data = True` Scale reference data to maximum of 99999 if writing intermediate scalepack data files
- `make_anisotropy_uniform = True` Calculate anisotropy for all datasets and adjust all of them to the average anisotropy. This is designed to improve scaling without changing the overall character of the data (i.e., not sharpening the data or removing anisotropy).

- `overall_scale = False` If set, only use overall scale and B (no local scaling)
- `skip_scaling = False` You can skip the scaling step and just get half-dataset CC for a pair of existing files that you have put in a directory by themselves.
- `lowest_resolution_range = 6` Lowest value of the high-resolution cutoff for CC as a function of high-resolution cutoff with constant low-resolution cutoff
- `min_common_refl = 200` Minimum number of reflections in common to scale datasets
- `sort_on_batch_number = True` You can sort XDS and MTZ reflections by frame number. Otherwise scale and merge assumes reflections are collected in the order in the data file.
- `scale_reference_data = True` Scale reference data to maximum of 99999 if writing intermediate scalepack data files
- `merging_include_dataset_variance = True` Scale datasets together including inter-dataset variance in in weighting. This takes into account the systematic differences among datasets and allows automatic down-weighting of datasets that are different than all the others. `optimize_anomalous = True` Scale datasets together including inter-dataset variance in anomalous differences in weighting. Involves separation of mean intensity of each reflection from anomalous differences and recombination after scaling to recreate I+ and I-. Alternative is to include inter-dataset variance in intensities. `use_best_group_as_target = False` Estimate inter-dataset variance by using the best dataset (highest I/sigma) as target for all datasets. This amounts to using all the other datasets to optimally estimate the data for this one crystal. `rescale_sigmas = False` Rescale sigmas assuming that at high resolution, there are no anomalous differences. This can sometimes be useful if the sigmas are all too high or low and the data are collected to sufficiently high resolution that there really are essentially no anomalous differences (only noise) at high resolution.
- `include_dataset_variance = True` Scale datasets together including inter-dataset variance in in weighting. This takes into account the systematic differences among datasets and allows automatic down-weighting of datasets that are different than all the others.
- `optimize_anomalous = True` Scale datasets together including inter-dataset variance in anomalous differences in weighting. Involves separation of mean intensity of each reflection from anomalous differences and recombination after scaling to recreate I+ and I-. Alternative is to include inter-dataset variance in intensities.
- `use_best_group_as_target = False` Estimate inter-dataset variance by using the best dataset (highest I/sigma) as target for all datasets. This amounts to using all the other datasets to optimally estimate the data for this one crystal.
- `rescale_sigmas = False` Rescale sigmas assuming that at high resolution, there are no anomalous differences. This can sometimes be useful if the sigmas are all too high or low and the data are collected to sufficiently high resolution that there really are essentially no anomalous differences (only noise) at high resolution.
- `half_dataset_cc_get_half_dataset_cc = True` Split datasets and get half-dataset CC if data are available. `half_dataset_cc_by_files = True` Split datasets into groups based on keeping all data in a file together (only applies if multiple data files are supplied). `half_dataset_cc_match_inverses = False` Split datasets into groups keeping (hkl, -h-k-l) together. Applies if only one file or `half_dataset_cc_by_files=False`. Does not apply if less than half of the reflections have (-h-k-l) pairs and other methods below have more matches. `split_as_first_second_half = True` split datasets into groups based on using the first or second half of each dataset in each half-dataset. Applies if `half_dataset_cc_by_files=False` or only one dataset is supplied and `half_dataset_cc_match_inverses=False`. Also applies if `half_dataset_cc_by_files` and

half_dataset_cc_match_inverses have failed to yield sufficient anomalous differences. split_alternately = True split datasets into groups based on sending observations of equivalent reflections alternately to the first or second half-dataset. Applies if split_as_first_second_half=False and half_dataset_cc_by_files=False and only one dataset is supplied. Also applies if other methods have failed to yield sufficient anomalous differences. split_randomly = True split datasets into groups based on sending observations of equivalent reflections randomly to the first or second half-dataset. Applies if split_as_first_second_half=False and half_dataset_cc_by_files=False and split_alternately is False and only one dataset is supplied. Also applies if other methods have failed to yield sufficient anomalous differences

- get_half_dataset_cc = True Split datasets and get half-dataset CC if data are available
- half_dataset_cc_by_files = True Split datasets into groups based on keeping all data in a file together (only applies if multiple data files are supplied.)
- half_dataset_cc_match_inverses = False Split datasets into groups keeping (hkl, -h-k-l) together. Applies if only one file or half_dataset_cc_by_files=False. Does not apply if less than half of the reflections have (-h-k-l) pairs and other methods below have more matches.
- split_as_first_second_half = True split datasets into groups based on using the first or second half of each dataset in each half-dataset. Applies if half_dataset_cc_by_files=False or only one dataset is supplied and half_dataset_cc_match_inverses=False. Also applies if half_dataset_cc_by_files and half_dataset_cc_match_inverses have failed to yield sufficient anomalous differences.
- split_alternately = True split datasets into groups based on sending observations of equivalent reflections alternately to the first or second half-dataset. Applies if split_as_first_second_half=False and half_dataset_cc_by_files=False and only one dataset is supplied. Also applies if other methods have failed to yield sufficient anomalous differences
- split_randomly = True split datasets into groups based on sending observations of equivalent reflections randomly to the first or second half-dataset. Applies if split_as_first_second_half=False and half_dataset_cc_by_files=False and split_alternately is False and only one dataset is supplied. Also applies if other methods have failed to yield sufficient anomalous differences
- directoriestemp_dir = None Optional temporary work directory. If specified, it will be used for temporary files. If clean_up=True and it did not exist before, it will be removed at the end of the run. If it did exist, it will be left.
- output_dir = "" Output directory where files are to be written
- gui_output_dir = None GUI use only - does not apply to command line version
- temp_dir = None Optional temporary work directory. If specified, it will be used for temporary files. If clean_up=True and it did not exist before, it will be removed at the end of the run. If it did exist, it will be left.
- controljob_title = None Job title in PHENIX GUI, not used on command line
- verbose = False Verbose
- output_random_seed = 714215 random seed
- comparison_file = None Data file with comparison
- anomalous_differences = *multiprocessing sge lsq pbs condor pbspro slurm Choices are multiprocessing (single machine) or queuing systems
- queue_run_command = None run command for queue jobs. For example qsub.nproc = 1 Number of processors to use in scaling
- solve_size = None You can set the size of Solve. Normally done automatically.
- clean_up = True Clean up and remove temp directory after finishing
- job_title = None Job title in PHENIX GUI, not used on command line

- verbose = False Verbose output
- random_seed = 714215 random seed
- comparison_file = None Data file with comparison anomalous differences
- multiprocessing = *multiprocessing sge lsf pbs condor pbspro slurm Choices are multiprocessing (single machine) or queuing systems
- queue_run_command = None run command for queue jobs. For example qsub.
- nproc = 1 Number of processors to use in scaling
- solve_size = None You can set the size of Solve. Normally done automatically.
- clean_up = True Clean up and remove temp directory after finishing

Simplified Carbohydrate Line-Entry System (SCaLES)

scales.html

There are many formats for specifying a polysaccharides as seen in this diagram from Herget et al. 2008.

The article introduces a format known as GlycoCT.

A common and compact molecule specification format is known as SMILES (ref) which stands for. It relies on the specification of the elements and the bonds (both implicit and explicit) of a molecule. Branches and rings can also be specified.

Because protein crystallographers are conversant with the concept of residues and polysaccharides are polymers of various carbohydrate units, the step to equating an element to a sugar residue and a chemical bond to a glycosidic link is easy. Polysaccharides are simpler because there are no polymeric rings (only rings within the residue unit). However, the linking is more complicated. Firstly, there are more types of links compared to the basic three in molecules (single, double, triple). Secondly, the type of the link relies on the attaching carbohydrate (α , β).

Conceptually, it is easiest to think of one sugar unit linking to the rest of the polysaccharide. For instance, this is the portion that attached to the protein. So, the most common polysaccharide attached to a protein is an N-linked chain that has three units in a linear chain linked to an ASN. The three-letter code for each of the units in order is NAG, NAG, MAN [1]. It is not a leap to imagine that this could be represented using:

NAG-NAG-MAN [2]

As with SMILES, there is a default link. In SMILES, it's a single bound while in SCaLES the default is $\beta(1-4)$. This linkage means that the attaching unit is connecting its anomeric carbon (C1) to the forth carbon (C4) on the other unit. It also specifies that the attaching unit is a β -isomer. In the example, MAN is a mannose saccharide but it is the α -isomer so it should:

NAG-NAG-BMA

However, the SCaLES parser automatically detects that the final residue is the incorrect isomer and changes the code from MAN to BMA: The linkage takes precedence. The specify an alternative to the default linkage, it can be indicated thus:

NAG-NAG- α -(1-4)-MAN

The hand of the linkage can be specified with "a" or "alpha" and the carbon atoms being linked can be specified using "(1-4)" or "(1,4)".

Branching is specified using square brackets (compared to parenthesis in SMILES) thus:

NAG-[α -(1-6)-FUC]-NAG-BMA

The branching can contain more than one saccharide:

NAG-NAG-BMA-[α -(1-3)-MAN- α -(1-2)-MAN- α -(1-2)-MAN]- α -(1-6)-MAN-[α -(1-3)-MAN- α -(1-2)-MAN]- α -(1-6)-MAN- α -(1-2)-MAN

References

Weininger, David (February 1988). "SMILES, a chemical language and information system. 1. Introduction to methodology and encoding rules". *Journal of Chemical Information and Modeling* 28 (1): 31–6. doi:10.1021/ci00057a005.

phenix.sceds: SCEDS - identify domains with normal mode analysis

sceds.html

Description

SCEDS is useful for identifying domains in a model to be used by molecular replacement when it is anticipated that there may be conformational motion of one or more domains relative to the domains in the target structure. Typically in such cases MR underperforms as it is not possible to match both domains onto the target structure. By partitioning a multidomain structure into distinct domains it is then possible to do MR searches with each separate domain and achieve a clearer solution than would otherwise have been possible. Domains are identified with a normal mode analysis. See literature reference below.

To use SCEDS supply it with the MR model and execute it. Depending on the size of the model it may take some time before it completes.

Output is one or more domains each saved into separate pdb files. These can then be used in molecular replacement. It is good practice to inspect the identified domains in a graphical viewer such as Coot or Pymol.

The structure with PDB id 1XHB before running SCEDS. Two distinct domains can be discerned in the structure.

After running SCEDS the two domains have been identified and saved into separate pdb files.

Literature

SCEDS: protein fragments for molecular replacement in Phaser McCoy AJ, Nicholls RA, Schneider TR Acta Cryst. D69, 2216-2225 (2013)

List of all available keywords

file_name = None root = None gui_output_dir = None job_title = None Job title in PHENIX GUI, not used on command line
validate_domains = True keywordsnmamode = 5 combinations = 2 enmoscillators = *rtb all
rtb_maxblocks = 125 rtb_nresidues = 0 radius = 5 force = 1 perturbationincrement = *rms dq dq = None rmsstep = 0.2 maximum = 0.3 direction = forward backward *tofro scedsenable = None ndom = 2 weightequality = 1
sphericity = 4 density = 1 continuity = 0 ddmslider = 0 cluster = *max first last distancestep = 50 min = 1 max = 5 separationmin = 7 max = 14 sequencemin = 0 max = 1 joinmin = 2 max = 12

- file_name = None
- root = None
- gui_output_dir = None
- job_title = None Job title in PHENIX GUI, not used on command line
- validate_domains = True
- keywordsnmamode = 5 combinations = 2 enmoscillators = *rtb all rtb_maxblocks = 125 rtb_nresidues = 0 radius = 5 force = 1 perturbationincrement = *rms dq dq = None rmsstep = 0.2 maximum = 0.3 direction = forward backward *tofro scedsenable = None ndom = 2 weightequality = 1 sphericity = 4 density = 1 continuity = 0 ddmslider = 0 cluster = *max first last distancestep = 50 min = 1 max = 5 separationmin = 7 max = 14 sequencemin = 0 max = 1 joinmin = 2 max = 12

- nmamode = 5 combinations = 2
- mode = 5
- combinations = 2
- enmoscillators = *rtb all rtb_maxblocks = 125 rtb_nresidues = 0 radius = 5 force = 1
- oscillators = *rtb all
- rtb_maxblocks = 125
- rtb_nresidues = 0
- radius = 5
- force = 1
- perturbincrement = *rms dq dq = None rmsstep = 0.2 maximum = 0.3 direction = forward backward *tofro
- increment = *rms dq
- dq = None
- rmsstep = 0.2 maximum = 0.3 direction = forward backward *tofro
- step = 0.2
- maximum = 0.3
- direction = forward backward *tofro
- scedsenable = None ndom = 2 weightequality = 1 sphericity = 4 density = 1 continuity = 0
- enable = None
- ndom = 2
- weightequality = 1 sphericity = 4 density = 1 continuity = 0
- equality = 1
- sphericity = 4
- density = 1
- continuity = 0
- ddmslider = 0 cluster = *max first last distancestep = 50 min = 1 max = 5 separationmin = 7 max = 14 sequencemin = 0 max = 1 joinmin = 2 max = 12
- slider = 0
- cluster = *max first last
- distancestep = 50 min = 1 max = 5
- step = 50
- min = 1
- max = 5
- separationmin = 7 max = 14
- min = 7
- max = 14
- sequencemin = 0 max = 1
- min = 0

- $\max = 1$
- $\text{joinmin} = 2 \quad \max = 12$
- $\min = 2$
- $\max = 12$

Segmenting cryo-EM maps with `segment_and_split_map`

`segment_and_split_map.html`

Author(s)

- `segment_and_split_map`: Tom Terwilliger

Purpose

The routine `segment_and_split_map` will identify the asymmetric unit of a map (typically cryo-EM) and contiguous regions of density within the asymmetric unit of the map.

Usage

How `segment_and_split_map` works:

If you have a CCP4-style (mrc, etc) map and a sequence file, you can use `segment_and_split_map` to split the map into smaller pieces suitable for model-building or viewing.

The `segment_and_split_map` tool works best at resolutions of about 4.5 Å or better.

The tool `segment_and_split_map` will find where your molecule is in the map and cut out and work with just that part of the density.

If your map has been averaged based on NCS symmetry and you supply a file with that NCS information (`.ncs_spec`, `biomtr.dat`, etc), `segment_and_split_map` will find the asymmetric unit of NCS and work with that.

Finally, the `segment_and_split_map` tool will cut the density in the asymmetric unit of your map into small pieces of connected density and write out a map for each one.

All the maps that are written by `segment_and_split_map` are superimposable on each other. They typically are all shifted from the original map to place the origin of the maps on the grid point (0,0,0).

Output files from `segment_and_split_map`

`shifted_map.ccp4`: Original map, shifted to place the origin on grid point (0,0,0)

`shifted_ncs.ncs_spec`: NCS operators (if any), shifted to match `shifted_map.ccp4`.

`shifted_pdb.pdb`: Input PDB file (if any), shifted to match `shifted_map.ccp4`.

`box_map_au.ccp4`: Same as `shifted_map.ccp4`, except that everything except the asymmetric unit of NCS is zeroed out (map shows the asymmetric unit only).

`box_mask_au.ccp4`: Mask showing location of NCS asymmetric unit. Superimposes on `box_map_au.ccp4` and `shifted_map.ccp4`.

`segment_and_split_map_info.pkl`: Pickled file with information about the segmentation. Used in `phenix.map_to_model` and to restore a shifted PDB file to original location.

Shifting the map to the origin

Most crystallographic maps have the origin at the corner of the map (grid point [0,0,0]), while most cryo-EM maps have the origin in the middle of the map. To make a consistent map, any maps with an origin not at the corner are shifted to put the origin at grid point [0,0,0]. This map is the shifted map that is used for further steps in model-building. At the conclusion of model-building, the model is shifted back to superimpose on the original map.

Finding the region containing the molecule

By default (`density_select=True`), the region of the map containing density is cut out of the entire map. This is particularly useful if the original map is very large and the molecule only takes up a small part of the map. This portion of the map is then shifted to place the origin at grid point [0,0,0]. (At the conclusion of model-building, the final model is shifted back to superimpose on the original map.) The region containing density is chosen as a box containing all the points above a threshold, typically 5% of the maximum in the map.

Finding the NCS asymmetric unit of the map

If you supply NCS matrices describing the NCS used to average the map (if any), then `segment_and_split_map` will try to define a region of the map that represents the NCS asymmetric unit. Application of the NCS operators to the NCS asymmetric unit will generate the entire map, and application to a model built into the asymmetric unit will generate the entire model. Normally identification of the NCS asymmetric unit and segmentation of the map (below) are done as a single step, yielding an asymmetric unit and a set of contiguous regions of density within that asymmetric unit. The asymmetric unit of NCS will be written out as a map to the `segmentation_dir` directory, superimposed on the shifted map (so that they can be viewed together in Coot).

Segmentation of the map

By default (`segment=True`) the map or NCS asymmetric unit of the map will be segmented (cut into small pieces) into regions of connected density. This is done by choosing a threshold of density and identifying contiguous regions where all grid points are above this threshold. The threshold is chosen to yield regions that have a size corresponding to about 50 residues. The regions of density are written out to the `segmentation_dir` directory and are superimposed on the shifted map (if you load the shifted map in Coot and a region map in Coot, they should superimpose.)

Special treatment of structures with very high symmetry

If your structure has very high symmetry (by default, 20 or more symmetry operators), then `segment_and_split_map` will try to cut out a piece of your map and work with that instead of working with the whole map. You can control this with the keyword `select_au_box=True` (or `None`, which will give the default behavior). If you use `select_au_box` then a box that contains about `n_au_box` asymmetric units of the map (default of 5) will be cut out of your map. That map will be worked with along with the symmetry operators that apply to it for the remainder of the analysis. placed in the original reference frame. This process can greatly save on memory.

Examples

Standard run of `segment_and_split_map`:

Running `segment_and_split_map` is easy. From the command-line you can type:

```
phenix.segment_and_split_map my_map.map seq.fa ncs_file=find_ncs.ncs_spec
```

where my_map.map is a CCP4, mrc or other related map format, seq.fa is a sequence file, and find_ncs.ncs_spec is an optional file specifying any NCS operators used in averaging the map. This can be in the form of BIOMTR records from a PDB file as well.

Possible Problems

Specific limitations and problems:

Literature

Additional information

List of all available keywords

input_filesseq_file = None Sequence file (unique chains only, 1-letter code, chains separated by blank line or greater-than sign.) Can have chains that are DNA/RNA/protein and all can be present in one file. If not supplied, must supply molecular mass or solvent content.map_file = None File with CCP4-style maphalf_map_file = None Half map (two should be supplied) for FSC calculation. Must have grid identical to map_fileexternal_map_file = None External map to be used to scale map_file (power vs resolution will be matched). Not used in segment_and_split_mapncs_file = None File with symmetry information (typically point-group NCS with the center specified). Typically in PDB format. Can also be a .ncs_spec file from phenix. Created automatically if symmetry is specified.pdb_file = None Optional PDB file matching map_file to be offsetpdb_to_restore = None Optional PDB file to restore to position matching original map_file. Used in combination with info_file = xxx.pkl and restored_pdb = yyyy.pdbinfo_file = None Optional pickle file with information from a previous run. Can be used with pdb_to_restore to restore a PDB file to to position matching original map_file.target_ncs_au_file = None Optional PDB file to partially define the ncs asymmetric unit of the map. The coordinates in this file will be used to mark part of the ncs au and all points nearby that are not part of another ncs au will be added.input_weight_map_pickle_file = None Weight map picklefileoutput_files magnification_map_file = magnification_map.ccp4 Input map file with magnification applied. Only written if magnification is applied.magnification_ncs_file = magnification_ncs.ncs_spec Input NCS with magnification applied. Only written if magnification is applied.shifted_map_file = shifted_map.ccp4 Input map file shifted to new origin.shifted_sharpened_map_file = shifted_sharpened_map.ccp4 Input map file shifted to new origin and sharpened.sharpened_map_file = sharpened_map.ccp4 Output sharpened map file, superimposed on the original map.shifted_pdb_file = shifted_pdb.pdb Input pdb file shifted to new origin.shifted_ncs_file = shifted_ncs.ncs_spec NCS information shifted to new origin.shifted_used_ncs_file = shifted_used_ncs.ncs_spec NCS information (just the part that is used) shifted to new origin.output_directory = segmented_maps Directory where output files are to be written applied.box_map_file = box_map_au.ccp4 Output map file with one NCS asymmetric unit, cut out boxbox_mask_file = box_mask_au.ccp4 Output mask file with one NCS asymmetric unit, cut out boxbox_buffer = 5 Buffer (grid units) around NCS asymmetric unit in box_mask and mapau_output_file_stem = shifted_au File stem for output map files with one NCS asymmetric unitwrite_intermediate_maps = False Write out intermediate maps and masks for visualizationwrite_output_maps = True Write out mapsremainder_map_file = remainder_map.ccp4 output map file with remainder after initial regions identifiedoutput_info_file = segment_and_split_map_info.pkl Output pickle file with information about map and masksrestored_pdb = None Output name of PDB restored to position matching original map_file. Used in combination with info_file = xxx.pkl and pdb_to_restore = xxxx.pdboutput_weight_map_pickle_file = weight_map_pickle_file.pkl Output weight map pickle

filecrystal_infochain_type = *None PROTEIN RNA DNA Chain type. Determined automatically from sequence file if not given. Mixed chain types are fine (leave blank if so).sequence = None Sequence as stringis_crystal = None Defines whether this is a crystal (or cryo-EM). Default is True if use_sg_symmetry = True and False otherwise.use_sg_symmetry = False If you set use_sg_symmetry = True then the symmetry of the space group will be used. For example in P1 a point at one end of the unit cell is next to a point on the other end. Normally for cryo-EM data this should be set to False and for crystal data it should be set to True. This will normally also set the value of is_crysta...

- input_filesseq_file = None Sequence file (unique chains only, 1-letter code, chains separated by blank line or greater-than sign.) Can have chains that are DNA/RNA/protein and all can be present in one file. If not supplied, must supply molecular mass or solvent content.map_file = None File with CCP4-style maphalf_map_file = None Half map (two should be supplied) for FSC calculation. Must have grid identical to map_fileexternal_map_file = None External map to be used to scale map_file (power vs resolution will be matched). Not used in segment_and_split_mapncs_file = None File with symmetry information (typically point-group NCS with the center specified). Typically in PDB format. Can also be a .ncs_spec file from phenix. Created automatically if symmetry is specified.pdb_file = None Optional PDB file matching map_file to be offsetpdb_to_restore = None Optional PDB file to restore to position matching original map_file. Used in combination with info_file = xxx.pkl and restored_pdb = yyyy.pdbinfo_file = None Optional pickle file with information from a previous run. Can be used with pdb_to_restore to restore a PDB file to to position matching original map_file.target_ncs_au_file = None Optional PDB file to partially define the ncs asymmetric unit of the map. The coordinates in this file will be used to mark part of the ncs au and all points nearby that are not part of another ncs au will be added.input_weight_map_pickle_file = None Weight map pickle file
- seq_file = None Sequence file (unique chains only, 1-letter code, chains separated by blank line or greater-than sign.) Can have chains that are DNA/RNA/protein and all can be present in one file. If not supplied, must supply molecular mass or solvent content.
- map_file = None File with CCP4-style map
- half_map_file = None Half map (two should be supplied) for FSC calculation. Must have grid identical to map_file
- external_map_file = None External map to be used to scale map_file (power vs resolution will be matched). Not used in segment_and_split_map
- ncs_file = None File with symmetry information (typically point-group NCS with the center specified). Typically in PDB format. Can also be a .ncs_spec file from phenix. Created automatically if symmetry is specified.
- pdb_file = None Optional PDB file matching map_file to be offset
- pdb_to_restore = None Optional PDB file to restore to position matching original map_file. Used in combination with info_file = xxx.pkl and restored_pdb = yyyy.pdb
- info_file = None Optional pickle file with information from a previous run. Can be used with pdb_to_restore to restore a PDB file to to position matching original map_file.
- target_ncs_au_file = None Optional PDB file to partially define the ncs asymmetric unit of the map. The coordinates in this file will be used to mark part of the ncs au and all points nearby that are not part of another ncs au will be added.
- input_weight_map_pickle_file = None Weight map pickle file
- output_files magnification_map_file = magnification_map.ccp4 Input map file with magnification applied. Only written if magnification is applied.magnification_ncs_file = magnification_ncs.ncs_spec Input NCS

with magnification applied. Only written if magnification is applied. `shifted_map_file = shifted_map.ccp4` Input map file shifted to new origin. `shifted_sharpened_map_file = shifted_sharpened_map.ccp4` Input map file shifted to new origin and sharpened. `sharpened_map_file = sharpened_map.ccp4` Output sharpened map file, superimposed on the original map. `shifted_pdb_file = shifted_pdb.pdb` Input pdb file shifted to new origin. `shifted_ncs_file = shifted_ncs.ncs_spec` NCS information shifted to new origin. `shifted_used_ncs_file = shifted_used_ncs.ncs_spec` NCS information (just the part that is used) shifted to new origin. `output_directory = segmented_maps` Directory where output files are to be written applied. `box_map_file = box_map_au.ccp4` Output map file with one NCS asymmetric unit, cut out box. `box_mask_file = box_mask_au.ccp4` Output mask file with one NCS asymmetric unit, cut out box. `box_buffer = 5` Buffer (grid units) around NCS asymmetric unit in `box_mask` and `map`. `au_output_file_stem = shifted_au` File stem for output map files with one NCS asymmetric unit. `write_intermediate_maps = False` Write out intermediate maps and masks for visualization. `write_output_maps = True` Write out maps. `remainder_map_file = remainder_map.ccp4` output map file with remainder after initial regions identified. `output_info_file = segment_and_split_map_info.pkl` Output pickle file with information about map and masks. `restored_pdb = None` Output name of PDB restored to position matching original `map_file`. Used in combination with `info_file = xxx.pkl` and `pdb_to_restore = xxxx.pdb`. `output_weight_map_pickle_file = weight_map_pickle_file.pkl` Output weight map pickle file

- `magnification_map_file = magnification_map.ccp4` Input map file with magnification applied. Only written if magnification is applied.
- `magnification_ncs_file = magnification_ncs.ncs_spec` Input NCS with magnification applied. Only written if magnification is applied.
- `shifted_map_file = shifted_map.ccp4` Input map file shifted to new origin.
- `shifted_sharpened_map_file = shifted_sharpened_map.ccp4` Input map file shifted to new origin and sharpened.
- `sharpened_map_file = sharpened_map.ccp4` Output sharpened map file, superimposed on the original map.
- `shifted_pdb_file = shifted_pdb.pdb` Input pdb file shifted to new origin.
- `shifted_ncs_file = shifted_ncs.ncs_spec` NCS information shifted to new origin.
- `shifted_used_ncs_file = shifted_used_ncs.ncs_spec` NCS information (just the part that is used) shifted to new origin.
- `output_directory = segmented_maps` Directory where output files are to be written applied.
- `box_map_file = box_map_au.ccp4` Output map file with one NCS asymmetric unit, cut out box
- `box_mask_file = box_mask_au.ccp4` Output mask file with one NCS asymmetric unit, cut out box
- `box_buffer = 5` Buffer (grid units) around NCS asymmetric unit in `box_mask` and `map`
- `au_output_file_stem = shifted_au` File stem for output map files with one NCS asymmetric unit
- `write_intermediate_maps = False` Write out intermediate maps and masks for visualization
- `write_output_maps = True` Write out maps
- `remainder_map_file = remainder_map.ccp4` output map file with remainder after initial regions identified
- `output_info_file = segment_and_split_map_info.pkl` Output pickle file with information about map and masks
- `restored_pdb = None` Output name of PDB restored to position matching original `map_file`. Used in combination with `info_file = xxx.pkl` and `pdb_to_restore = xxxx.pdb`

- `output_weight_map_pickle_file = weight_map_pickle_file.pkl` Output weight map pickle file
- `crystal_infchain_type = *None` PROTEIN RNA DNA Chain type. Determined automatically from sequence file if not given. Mixed chain types are fine (leave blank if so). `sequence = None` Sequence as string `is_crystal = None` Defines whether this is a crystal (or cryo-EM). Default is True if `use_sg_symmetry = True` and False otherwise. `use_sg_symmetry = False` If you set `use_sg_symmetry = True` then the symmetry of the space group will be used. For example in P1 a point at one end of the unit cell is next to a point on the other end. Normally for cryo-EM data this should be set to False and for crystal data it should be set to True. This will normally also set the value of `is_crystal` (same value as `use_sg_symmetry`) and `restrict_map_size` (False if `use_sg_symmetry = True`). `resolution = None` Nominal resolution of the map. This is used later to decide on resolution cutoffs for Fourier inversion of the map. Note: the resolution is not cut at this value, it is cut at $\text{resolution} * d_min_ratio$ if at all. `space_group = None` Space group (used for boxed maps) `unit_cell = None` Unit Cell (used for boxed maps) `original_unit_cell = None` Original unit cell (of input map). Used internally `original_unit_cell_grid = None` Original unit cell grid (of input map). Used internally `molecular_mass = None` Molecular mass of molecule in Da. Used as alternative method of specifying solvent content. `solvent_content = None` Solvent fraction of the cell. Used for ID of solvent content in boxed maps. `solvent_content_iterations = 3` Iterations of solvent fraction estimation. Used for ID of solvent content in boxed maps. `wang_radius = None` Wang radius for solvent identification. Default is $1.5 * \text{resolution}$ `buffer_radius = None` Buffer radius for mask smoothing. Default is `resolution` `pseudo_likelihood = None` Use pseudo-likelihood method for half-map sharpening. (In development)
- `chain_type = *None` PROTEIN RNA DNA Chain type. Determined automatically from sequence file if not given. Mixed chain types are fine (leave blank if so).
- `sequence = None` Sequence as string
- `is_crystal = None` Defines whether this is a crystal (or cryo-EM). Default is True if `use_sg_symmetry = True` and False otherwise.
- `use_sg_symmetry = False` If you set `use_sg_symmetry = True` then the symmetry of the space group will be used. For example in P1 a point at one end of the unit cell is next to a point on the other end. Normally for cryo-EM data this should be set to False and for crystal data it should be set to True. This will normally also set the value of `is_crystal` (same value as `use_sg_symmetry`) and `restrict_map_size` (False if `use_sg_symmetry = True`).
- `resolution = None` Nominal resolution of the map. This is used later to decide on resolution cutoffs for Fourier inversion of the map. Note: the resolution is not cut at this value, it is cut at $\text{resolution} * d_min_ratio$ if at all.
- `space_group = None` Space group (used for boxed maps)
- `unit_cell = None` Unit Cell (used for boxed maps)
- `original_unit_cell = None` Original unit cell (of input map). Used internally
- `original_unit_cell_grid = None` Original unit cell grid (of input map). Used internally
- `molecular_mass = None` Molecular mass of molecule in Da. Used as alternative method of specifying solvent content.
- `solvent_content = None` Solvent fraction of the cell. Used for ID of solvent content in boxed maps.
- `solvent_content_iterations = 3` Iterations of solvent fraction estimation. Used for ID of solvent content in boxed maps.
- `wang_radius = None` Wang radius for solvent identification. Default is $1.5 * \text{resolution}$
- `buffer_radius = None` Buffer radius for mask smoothing. Default is `resolution`

- `pseudo_likelihood = None` Use pseudo-likelihood method for half-map sharpening. (In development)
- `reconstruction_symmetry` = None Symmetry used in reconstruction. For example D7, C3, C2 I (icosahedral), T (tetrahedral), or ANY (try everything and use the highest symmetry found). Not needed if `ncs_file` is supplied.
- `include_helical_symmetry = True` You can include or exclude searches for helical symmetry
- `must_be_consistent_with_space_group_number = None` Searches for symmetry must be compatible with this space group number.
- `symmetry_center = None` Center (in Å) for symmetry operators (if symmetry is found automatically). If set to None, first guess is the center of the cell and then if that fails, found automatically as the center of the density in the map.
- `optimize_center = False` Optimize position of symmetry center. Also checks for center at (0, 0, 0) vs center of map
- `helical_rot_deg = None` helical rotation about z in degrees
- `helical_trans_z_angstrom = None` helical translation along z in Angstrom units
- `max_helical_optimizations = 2` Number of optimizations of helical parameters when finding symmetry
- `max_helical_ops_to_check = 5` Number of helical operations in each direction to check when finding symmetry
- `max_helical_rotations_to_check = None` Number of helical rotations to check when finding symmetry
- `two_fold_along_x = None` Specifies if D or I two-fold is along x (True) or y (False). If None, both are tried.
- `smallest_object = None` Dimension of smallest object to consider when finding symmetry. Default is 5 * resolution
- `score_basis = ncs_score cc * None` Symmetry score basis. Normally `ncs_score (sqrt(n) * cc)` is used except for identification of helical symmetry
- `scale_weight_fractional_translation = 1.05` Give slight increase in weighting in helical symmetry search to translations that are a fraction (1/2, 1/3) of the d-spacing of the peak of intensity in the fourier transform of the density.
- `random_points = 100` Number of random points in map to examine in finding symmetry
- `identify_ncs_id = True` If symmetry is not point-group symmetry, try each possible operator when evaluating symmetry and choose the one that results in the most uniform density at symmetry-related points.
- `min_ncs_cc = 0.75` Minimum symmetry CC to keep operators when identifying automatically
- `n_rescore = 5` Number of symmetry operators to rescore
- `op_max = 14` If symmetry is ANY, try up to op_max-fold symmetry
- `tol_r = 0.02` tolerance in rotations for point group or helical symmetry
- `tol_t = 2` tolerance in translations (Å) for point group or helical symmetry
- `max_helical_operators = None` Maximum helical operators (if extending existing helical operators)
- `rel_tol_t = .05` tolerance in translations (fractional) for point group or helical symmetry
- `require_helical_or_point_group_symmetry = False` normally helical or point-group symmetry (or none) is expected. However in some cases (helical + rotational symmetry for example) this is not needed and is not the case.
- `symmetry = None` Symmetry used in reconstruction. For example D7, C3, C2 I (icosahedral), T (tetrahedral), or ANY (try everything and use the highest symmetry found). Not needed if `ncs_file` is supplied.

- `max_helical_ops_to_check` = 5 Number of helical operations in each direction to check when finding symmetry
- `max_helical_rotations_to_check` = None Number of helical rotations to check when finding symmetry
- `two_fold_along_x` = None Specifies if D or I two-fold is along x (True) or y (False). If None, both are tried.
- `smallest_object` = None Dimension of smallest object to consider when finding symmetry. Default is 5 * resolution
- `score_basis` = `ncs_score cc` *None Symmetry score basis. Normally `ncs_score (sqrt(n)* cc)` is used except for identification of helical symmetry
- `scale_weight_fractional_translation` = 1.05 Give slight increase in weighting in helical symmetry search to translations that are a fraction (1/2, 1/3) of the d-spacing of the peak of intensity in the fourier transform of the density.
- `random_points` = 100 Number of random points in map to examine in finding symmetry
- `identify_ncs_id` = True If symmetry is not point-group symmetry, try each possible operator when evaluating symmetry and choose the one that results in the most uniform density at symmetry-related points.
- `min_ncs_cc` = 0.75 Minimum symmetry CC to keep operators when identifying automatically
- `n_rescore` = 5 Number of symmetry operators to rescore
- `op_max` = 14 If symmetry is ANY, try up to `op_max`-fold symmetries
- `tol_r` = 0.02 tolerance in rotations for point group or helical symmetry
- `abs_tol_t` = 2 tolerance in translations (A) for point group or helical symmetry
- `max_helical_operators` = None Maximum helical operators (if extending existing helical operators)
- `rel_tol_t` = .05 tolerance in translations (fractional) for point group or helical symmetry
- `require_helical_or_point_group_symmetry` = False normally helical or point-group symmetry (or none) is expected. However in some cases (helical + rotational symmetry for example) this is not needed and is not the case.
- `map_modificationmagnification` = None Magnification to apply to input map. Input map grid will be scaled by magnification factor before anything else is done.
`b_iso` = None Target B-value for map (sharpening will be applied to yield this value of `b_iso`). If sharpening method is not supplied, default is to use `b_iso_to_d_cut` sharpening.
`b_sharpen` = None Sharpen with this b-value. Contrast with `b_iso` that yield a targeted value of `b_iso`. B_sharpen greater than zero is sharpening. Less than zero is blurring.
`b_blur_hires` = 200 Blur high_resolution data (higher than `d_cut`) with this b-value. Contrast with `b_sharpen` applied to data up to `d_cut`. Note on defaults: If None and `b_sharpen` is positive (sharpening) then high-resolution data is left as is (not sharpened). If None and `b_sharpen` is negative (blurring) high-resolution data is also blurred.
`resolution_dependent_b` = None If set, apply resolution_dependent_b (`b0 b1 b2`). Log10(amplitudes) will start at 1, change to `b0` at half of resolution specified, changing linearly, change to `b1/2` at resolution specified, and change to `b1/2+b2` at `d_min_ratio*resolution`
`normalize_amplitudes_in_resdep` = False Normalize amplitudes in resolution-dependent sharpening
`d_min_ratio` = 0.833 Sharpening will be applied using `d_min` equal to `d_min_ratio` times resolution. Default is 0.833
`scale_max` = 100000 Scale amplitudes from inverse FFT to yield maximum of this value
`input_d_cut` = None High-resolution limit for sharpening
`rmsd` = None RMSD of model to true model (if supplied). Used to estimate expected fall-off with resolution of correct part of model-based map. If None, assumed to be resolution times

rmsd_resolution_factor.rmsd_resolution_factor = 0.25 default RMSD is resolution times resolution
 factorfraction_complete = None Completeness of model (if supplied). Used to estimate correct part of
 model-based map. If None, estimated from max(FSC).regions_to_keep = None You can specify a limit to
 the number of regions to keep when generating the asymmetric unit of density.auto_sharpen = True
 Automatically determine sharpening using kurtosis maximization or adjusted surface
 areaauto_sharpen_methods = no_sharpening b_iso *b_iso_to_d_cut resolution_dependent
 model_sharpening half_map_sharpening target_b_iso_to_d_cut None Methods to use in sharpening.
 b_iso searches for b_iso to maximize sharpening target (kurtosis or adjusted_sa). b_iso_to_d_cut applies
 b_iso only up to resolution specified, with fall-over of k_sharpen. Resolution dependent adjusts 3
 parameters to sharpen variably over resolution range. Default is b_iso_to_d_cut . target_b_iso_to_d_cut
 uses target_b_iso_ratio to set b_iso.box_in_auto_sharpen = False Use a representative box of density
 for initial auto-sharpening instead of the entire map.density_select_in_auto_sharpen = True Choose
 representative box of density for initial auto-sharpening with density_select method (choose region where
 there is high density). Normally use this as well as density_select = True which carries out density_select
 at start of segmentation.density_select_threshold_in_auto_sharpen = None Threshold for density select
 choice of box. Default is 0.05. If your map has low overall contrast you might need to make this bigger
 such as 0.2.allow_box_if_b_iso_set = False Allow box_in_auto_sharpen (if set to True) even if b_iso is
 set. Default is to set box_n_auto_sharpen = False if b_iso is set.soft_mask = True Use soft mask (smooth
 change from inside to outside with radius based on resolution of map). Required if you use half-map
 sharpening without a model, otherwise optional.use_weak_density = False When choosing box of
 representative density, use poor density (to get optimized map for weaker density)discard_if_worse =
 None Discard sharpening if worselocal_sharpening = None Sharpen locally using overlapping regions.
 NOTE: Best to turn off local_aniso_in_local_sharpening if symmetry is present. If
 local_aniso_in_local_sharpening is True and symmetry is present this can distort the map for s...

- magnification = None Magnification to apply to input map. Input map grid will be scaled by magnification factor before anything else is done.
- b_iso = None Target B-value for map (sharpening will be applied to yield this value of b_iso). If sharpening method is not supplied, default is to use b_iso_to_d_cut sharpening.
- b_sharpen = None Sharpen with this b-value. Contrast with b_iso that yield a targeted value of b_iso. B_sharpen greater than zero is sharpening. Less than zero is blurring.
- b_blur_hires = 200 Blur high_resolution data (higher than d_cut) with this b-value. Contrast with b_sharpen applied to data up to d_cut. Note on defaults: If None and b_sharpen is positive (sharpening) then high-resolution data is left as is (not sharpened). If None and b_sharpen is negative (blurring) high-resolution data is also blurred.
- resolution_dependent_b = None If set, apply resolution_dependent_b (b0 b1 b2). Log10(amplitudes) will start at 1, change to b0 at half of resolution specified, changing linearly, change to b1/2 at resolution specified, and change to b1/2+b2 at d_min_ratio*resolution
- normalize_amplitudes_in_resdep = False Normalize amplitudes in resolution-dependent sharpening
- d_min_ratio = 0.833 Sharpening will be applied using d_min equal to d_min_ratio times resolution. Default is 0.833
- scale_max = 100000 Scale amplitudes from inverse FFT to yield maximum of this value
- input_d_cut = None High-resolution limit for sharpening
- rmsd = None RMSD of model to true model (if supplied). Used to estimate expected fall-of with resolution of correct part of model-based map. If None, assumed to be resolution times rmsd_resolution_factor.

- `rmsd_resolution_factor` = 0.25 default RMSD is resolution times resolution factor
- `fraction_complete` = None Completeness of model (if supplied). Used to estimate correct part of model-based map. If None, estimated from max(FSC).
- `regions_to_keep` = None You can specify a limit to the number of regions to keep when generating the asymmetric unit of density.
- `auto_sharpen` = True Automatically determine sharpening using kurtosis maximization or adjusted surface area
- `auto_sharpen_methods` = no_sharpening b_iso *b_iso_to_d_cut resolution_dependent
`model_sharpening` half_map_sharpening target_b_iso_to_d_cut None Methods to use in sharpening.
`b_iso` searches for `b_iso` to maximize sharpening target (kurtosis or adjusted_sa). `b_iso_to_d_cut` applies `b_iso` only up to resolution specified, with fall-over of `k_sharpen`. Resolution dependent adjusts 3 parameters to sharpen variably over resolution range. Default is `b_iso_to_d_cut`. `target_b_iso_to_d_cut` uses `target_b_iso_ratio` to set `b_iso`.
- `box_in_auto_sharpen` = False Use a representative box of density for initial auto-sharpening instead of the entire map.
- `density_select_in_auto_sharpen` = True Choose representative box of density for initial auto-sharpening with `density_select` method (choose region where there is high density). Normally use this as well as `density_select` = True which carries out `density_select` at start of segmentation.
- `density_select_threshold_in_auto_sharpen` = None Threshold for density select choice of box. Default is 0.05. If your map has low overall contrast you might need to make this bigger such as 0.2.
- `allow_box_if_b_iso_set` = False Allow `box_in_auto_sharpen` (if set to True) even if `b_iso` is set. Default is to set `box_in_auto_sharpen` = False if `b_iso` is set.
- `soft_mask` = True Use soft mask (smooth change from inside to outside with radius based on resolution of map). Required if you use half-map sharpening without a model, otherwise optional.
- `use_weak_density` = False When choosing box of representative density, use poor density (to get optimized map for weaker density)
- `discard_if_worse` = None Discard sharpening if worse
- `local_sharpening` = None Sharpen locally using overlapping regions. NOTE: Best to turn off `local_aniso_in_local_sharpening` if symmetry is present. If `local_aniso_in_local_sharpening` is True and symmetry is present this can distort the map for some symmetry copies because an anisotropy correction is applied based on local density in one copy and is transferred without rotation to other copies.
- `local_aniso_in_local_sharpening` = None Use local anisotropy in local sharpening. Default is True unless symmetry is present.
- `overall_before_local` = True Apply overall scaling before local scaling
- `select_sharpened_map` = None Select a single sharpened map to use
- `read_sharpened_maps` = None Read in previously-calculated sharpened maps
- `write_sharpened_maps` = None Write out local sharpened maps
- `smoothing_radius` = None Sharpen locally using `smoothing_radius`. Default is 2/3 of mean distance between centers for sharpening
- `box_center` = None You can specify the center of the box (A units)
- `box_size` = 40 40 40 You can specify the size of the boxes to use (grid units)
- `target_n_overlap` = 10 You can specify the targeted overlap of boxes in local sharpening

- `restrict_map_size = None` Restrict box map to be inside full map (required for cryo-EM data). Default is True if `use_sg_symmetry = False`.
- `restrict_z_turns_for_helical_symmetry = 1` Restrict Z turns for helical symmetry. Number of turns of helix going each direction in Z is specified.
- `restrict_z_distance_for_helical_symmetry = None` Restrict Z distance (+/- this distance from center) for helical symmetry.
- `remove_aniso = True` You can remove anisotropy (overall and locally) during sharpening
- `cc_cut = 0.2` Estimate of minimum highly reliable CC in half-map FSC. Used to decide at what CC value to smooth the remaining CC values.
- `max_cc_for_rescale = 0.2` Min reliable CC in half-maps. Used along with `cc_cut` and `scale_using_last` to correct for small errors in FSC estimation at high resolution. If the value of FSC near the high-resolution limit is above `max_cc_for_rescale`, assume these values are correct and do not correct them. To keep all original values use `max_cc_for_rescale = 1`
- `scale_using_last = 3` If set, assume that the last `scale_using_last` bins in the FSC for half-map or model sharpening are about zero (corrects for errors in the half-map process).
- `max_box_fraction = 0.5` If box is greater than this fraction of entire map, use entire map.
- `density_select_max_box_fraction = 0.95` If box is greater than this fraction of entire map, use entire map for `density_select`. Default is 0.95
- `mask_atoms = True` Mask atoms when using model sharpening
- `mask_atoms_atom_radius = 3` Mask for `mask_atoms` will have `mask_atoms_atom_radius`
- `value_outside_atoms = None` Value of map outside atoms (set to mean to have mean value inside and outside mask be equal)
- `k_sharpen = 10` Steepness of transition between sharpening (up to resolution) and not sharpening ($d < \text{resolution}$). Note: for blurring, all data are blurred (regardless of resolution), while for sharpening, only data with d about resolution or lower are sharpened. This prevents making very high-resolution data too strong. Note 2: if `k_sharpen` is zero, then no transition is applied and all data is sharpened or blurred. Note 3: only used if `b_iso` is set.
- `iterate = False` You can iterate auto-sharpening. This is useful in cases where you do not specify the solvent content and it is not accurately estimated until sharpening is optimized.
- `optimize_b_blur_hires = False` Optimize value of `b_blur_hires`. Only applies for `auto_sharpen_methods` `b_iso_to_d_cut` and `b_iso`. This is normally carried out and helps prevent over-blurring at high resolution if the same map is sharpened more than once.
- `optimize_d_cut = None` Optimize value of `d_cut`. Only applies for `auto_sharpen_methods` `b_iso_to_d_cut` and `b_iso`. Not normally carried out.
- `adjust_region_weight = True` Adjust `region_weight` to make overall change in surface area equal to overall change in normalized regions over the range of `search_b_min` to `search_b_max` using `b_iso_to_d_cut`.
- `region_weight_method = initial_ratio * delta_ratio b_iso` Method for choosing `region_weights`. `Initial_ratio` uses ratio of surface area to regions at low B value. `Delta ratio` uses change in this ratio from low to high B. `B_iso` uses resolution-dependent `b_iso` (not weights) with the formula $b_iso = 5.9 * d_min^{**2}$
- `region_weight_factor = 1.0` Multiplies `region_weight` after calculation with `region_weight_method` above
- `region_weight_buffer = 0.1` `Region_weight` adjusted to be `region_weight_buffer` away from minimum or maximum values

- `region_weight_default = 30`. `Region_weight` adjusted to be `region_weight_default` if no information available
- `target_b_iso_ratio = 5.9` Target `b_iso` ratio : `b_iso` is estimated as `target_b_iso_ratio * resolution**2`
- `signal_min = 3.0` Minimum signal in estimation of optimal `b_iso`. If not achieved, use any other method chosen.
- `target_b_iso_model_scale = 0`. For model sharpening, the `target_biso` is scaled (normally zero).
- `search_b_min = -100` Low bound for `b_iso` search.
- `search_b_max = 300` High bound for `b_iso` search.
- `search_b_n = 21` Number of `b_iso` values to search.
- `residual_target = 'adjusted_sa'` Target for maximization steps in sharpening. Can be `kurtosis` or `adjusted_sa` (adjusted surface area)
- `sharpening_target = 'adjusted_sa'` Overall target for sharpening. Can be `kurtosis` or `adjusted_sa` (adjusted surface area) or `adjusted_path_length`. Used to decide which sharpening approach is used. Note that during optimization, `residual_target` is used (they can be the same.)
- `region_weight = 40` Region weighting in adjusted surface area calculation. Score is surface area minus `region_weight` times number of regions. Default is 40. A smaller value will give more sharpening.
- `sa_percent = 30`. Percent of target regions used in calculation of adjusted surface area. Default is 30.
- `fraction_occupied = 0.20` Fraction of molecular volume targeted to be inside contours. Used to set contour level. Default is 0.20
- `n_bins = 20` Number of resolution bins for sharpening. Default is 20.
- `max_regions_to_test = 30` Number of regions to test for surface area in `adjusted_sa` scoring of sharpening
- `eps = None`
- `k_sol = 0.35` `k_sol` value for model map calculation. IGNORED (Not applied)
- `b_sol = 50` `b_sol` value for model map calculation. IGNORED (Not applied)
- `segmentationselect_au_box = None` Select box containing at least one representative region of the map. Also select just symmetry operators relevant to that box. Default is true if number of operators is at least `n_ops_to_use_au_box`
`n_ops_to_use_au_box = 25` If number of operators is this big or more and `select_au_box` is None, set it to True.
`n_au_box = 5` Number of symmetry copies to try and get inside `au_box`
`lower_bounds = None` You can select a part of your map for analysis with `lower_bounds` and `upper_bounds`.
`upper_bounds = None` You can select a part of your map for analysis with `lower_bounds` and `upper_bounds`.
`density_select = True` Run `map_box` with `density_select = True` to cut out the region in the input map that contains density. Useful if the input map is much larger than the structure. Done before segmentation is carried out.
`density_select_threshold = 0.05` Choose region where density is this fraction of maximum or greater
`get_half_height_width = None` Use 4 times half-width at half-height as estimate of max size
`box_ncs_au = True` Box the map containing just the `au` of the map
`cell_cutoff_for_solvent_from_mask = 150` For cells with average edge over this cutoff, use the low resolution mask (backup) method for solvent estimation
`mask_padding_fraction = 0.025` Adjust threshold of standard deviation map in low resolution mask identification of solvent content to give this much more inside mask than would be obtained with the value of `fraction_of_max_mask_threshold`.
`fraction_of_max_mask_threshold = .05` threshold of standard deviation map in low resolution mask identification of solvent content.
`mask_threshold = None` threshold in identification of overall mask. If None, guess volume of molecule from sequence and symmetry

copies.grid_spacing_for_au = 3 Grid spacing for asymmetric unit when constructing asymmetric unit.radius = None Radius for constructing asymmetric unit.value_outside_mask = 0.0 Value to assign to density outside maskdensity_threshold = None Threshold density for identifying regions of density. Applied after normalizing the density in the region of the molecule to an rms of 1 and mean of zero.starting_density_threshold = None Optional guess of threshold densityiteration_fraction = 0.2 On iteration of finding regions, assume target volume is this fraction of the value on previous iterationmax_overlap_fraction = 0.05 Maximum fractional overlap allowed to density in another asymmetric unit. Definition of a bad region.remove_bad_regions_percent = 1 Remove the worst regions that are part of more than one NCS asymmetric unit, up to remove_bad_regions_percent of the totalrequire_complete = True Require all symmetry copies to be represented for a regionsplit_if_possible = True Split regions that are split in some symmetry copies. If None, split if most copies are split.write_all_regions = False Write all regions to ccp4 map files.max_per_au = None Maximum number of regions to be kept in the NCS asymmetric unitmax_per_au_ratio = 5. Maximum ratio of number of regions to be kept in the NCS asymmetric unit to those expectedmin_ratio_of_ncs_copy_to_first = 0.5 Minimum ratio of the last ncs_copy region size to maximummin_ratio = 0.1 Minimum ratio of region size to maximum to keep itmax_ratio_to_target = 3 Maximum ratio of grid points in top region to targetmin_ratio_to_target = 0.3 Minimum ratio of grid points in top region to targetmin_volume = 10 Minimum region size to consider (in grid points)residues_per_region = 50 Target number of residues per regionseeds_to_try = 10 Number of regions to try as centersiterate_with_remainder = True Iterate looking for regions based on remainder from first analysisweight_rad_gyr = 0.1 Weight on radius of gyration of group of regions in NCS AU relative to weight on closeness to neighbors. Normalized to largest cell dimension with weight = weight_rad_gyr*300/cell_maxexpand_size = None Grid points to expand size of regions when excluding for next round. If None, set to approx number of grid points to get expa...

- select_au_box = None Select box containing at least one representative region of the map. Also select just symmetry operators relevant to that box. Default is true if number of operators is at least n_ops_to_use_au_box
- n_ops_to_use_au_box = 25 If number of operators is this big or more and select_au_box is None, set it to True.
- n_au_box = 5 Number of symmetry copies to try and get inside au_box
- lower_bounds = None You can select a part of your map for analysis with lower_bounds and upper_bounds.
- upper_bounds = None You can select a part of your map for analysis with lower_bounds and upper_bounds.
- density_select = True Run map_box with density_select = True to cut out the region in the input map that contains density. Useful if the input map is much larger than the structure. Done before segmentation is carried out.
- density_select_threshold = 0.05 Choose region where density is this fraction of maximum or greater
- get_half_height_width = None Use 4 times half-width at half-height as estimate of max size
- box_ncs_au = True Box the map containing just the au of the map
- cell_cutoff_for_solvent_from_mask = 150 For cells with average edge over this cutoff, use the low resolution mask (backup) method for solvent estimation
- mask_padding_fraction = 0.025 Adjust threshold of standard deviation map in low resolution mask identification of solvent content to give this much more inside mask than would be obtained with the value of fraction_of_max_mask_threshold.

- `fraction_of_max_mask_threshold` = .05 threshold of standard deviation map in low resolution mask identification of solvent content.
- `mask_threshold` = None threshold in identification of overall mask. If None, guess volume of molecule from sequence and symmetry copies.
- `grid_spacing_for_au` = 3 Grid spacing for asymmetric unit when constructing asymmetric unit.
- `radius` = None Radius for constructing asymmetric unit.
- `value_outside_mask` = 0.0 Value to assign to density outside masks
- `density_threshold` = None Threshold density for identifying regions of density. Applied after normalizing the density in the region of the molecule to an rms of 1 and mean of zero.
- `starting_density_threshold` = None Optional guess of threshold density
- `iteration_fraction` = 0.2 On iteration of finding regions, assume target volume is this fraction of the value on previous iteration
- `max_overlap_fraction` = 0.05 Maximum fractional overlap allowed to density in another asymmetric unit. Definition of a bad region.
- `remove_bad_regions_percent` = 1 Remove the worst regions that are part of more than one NCS asymmetric unit, up to `remove_bad_regions_percent` of the total
- `require_complete` = True Require all symmetry copies to be represented for a region
- `split_if_possible` = True Split regions that are split in some symmetry copies. If None, split if most copies are split.
- `write_all_regions` = False Write all regions to ccp4 map files.
- `max_per_au` = None Maximum number of regions to be kept in the NCS asymmetric unit
- `max_per_au_ratio` = 5. Maximum ratio of number of regions to be kept in the NCS asymmetric unit to those expected
- `min_ratio_of_ncs_copy_to_first` = 0.5 Minimum ratio of the last ncs_copy region size to maximum
- `min_ratio` = 0.1 Minimum ratio of region size to maximum to keep it
- `max_ratio_to_target` = 3 Maximum ratio of grid points in top region to target
- `min_ratio_to_target` = 0.3 Minimum ratio of grid points in top region to target
- `min_volume` = 10 Minimum region size to consider (in grid points)
- `residues_per_region` = 50 Target number of residues per region
- `seeds_to_try` = 10 Number of regions to try as centers
- `iterate_with_remainder` = True Iterate looking for regions based on remainder from first analysis
- `weight_rad_gyr` = 0.1 Weight on radius of gyration of group of regions in NCS AU relative to weight on closeness to neighbors. Normalized to largest cell dimension with $\text{weight} = \text{weight_rad_gyr} \times 300 / \text{cell_max}$
- `expand_size` = None Grid points to expand size of regions when excluding for next round. If None, set to approx number of grid points to get `expand_target` below
- `expand_target` = 1.5 Target expansion of regions (A)
- `mask_additional_expand_size` = 1 Mask expansion in addition to `expand_size` for final map
- `mask_expand_ratio` = 1 Mask expansion relative to resolution for `save_box_map_ncs_au`
- `exclude_points_in_ncs_copies` = True Exclude points that are in symmetry copies when creating NCS au. Does not apply if `add_neighbors` = True

- `add_neighbors = True` Add neighboring regions around the au. Turns off `exclude_points_in_ncs_copies` also.
- `add_neighbors_dist = 1`. Max increase in radius of gyration by adding region to keep it.
- `controlverbose = False` Verbose output
- `shift_only = None` Shift map and half_maps and stop
- `sharpen_only = None` Sharpen map and stop
- `check_ncs = None` Check the NCS symmetry by estimating NCS correlation and stop
- `resolve_size = None` Size of resolve to use.
- `quick = True` Run quickly if possible
- `memory_check = True` Map-to-model checks to make sure you have enough memory on your machine to run. You can disable this by setting this keyword to False. The estimates are approximate so it is possible your job could run even if the check fails. Note the check does not take any other uses of the memory on your machine into account.
- `save_box_map_ncs_au = False` Controls whether the map_box ncs_au is saved. Internal use only
- `write_files = True` Controls whether files are written
- `multiprocessing = *multiprocessing sge lsf pbs condor pbspro slurm` Choices are multiprocessing (single machine) or queuing systems
- `queue_run_command = None` run command for queue jobs. For example qsub.
- `nproc = 1` Number of processors to use

Using the Selection Editor

selection_editor.html

Contents

- Overview
- Basic Usage
- Graphical Selection
- Known Issues

Overview

The selection editor is a uniform interface for graphically selecting groups of atoms. The editor generates the syntax that Phenix uses for atom selections and is a more advanced interface than the "View/Pick" button next to certain fields in the GUI. This interface is used to define

- Secondary structure annotations (helices, beta sheets, etc.)
- NCS groups
- TLS groups
- Refinement strategy options (occupancies, rigid bodies, etc.)
- Custom geometry restraints (bonds, angles, dihedrals, etc.)

Moreover, selections made in this editor can be saved as a file ("Export" button) and reloaded at a later time ("Load" button). It is important to note that selection files saved from one Phenix program GUI (e.g. phenix.refine) generally cannot be used with another Phenix program GUI (e.g. phenix.real_space_refine) because the organization for the parameters can be slightly different among the programs.

The "Validate Syntax" button can be used to check if the selections are valid for the given model. This only checks for syntactical correctness. This does not check if the selections are physically sensible or if there are overlapping/duplicated selections.

The "Sequence" button opens a new window showing the sequence for the given model and any secondary structure annotations for helices and beta sheets.

The "Model" button opens a new window for graphically selecting atoms.

It is important to note that selections can only be made for options that have been enabled in the Phenix program GUI. That is, if the "Use NCS" box is not checked in the phenix.refine GUI, there will be no panel for selecting NCS groups in the selection editor. Changing options in the Phenix program GUI will dynamically update the available panels in the selection editor. If selections were made for an option and that option is later disabled, those selections will not be passed to the Phenix program when the selection editor is closed.

Closing the selection editor will update the Phenix program GUI with the selections. This is required before running the program.

Basic Usage

The selection editor can generate two basic types of selections

- Groups composed of one selection (e.g. helix or rigid body)

- Groups composed of multiple selections (e.g. beta sheet or NCS group)

For both types, the general procedure for adding a new group is to click on the "Add ..." button (e.g. "Add Sheet"). This generates a new row and the selection can be directly typed in the row by double-clicking the appropriate column field. For the second type, you can add additional selections to the group (e.g. "Add Strand") by entering selections in the new row. Some selections have additional columns and they can be changed by double-clicking the appropriate column field and typing in the value. To expand or collapse the multiple selections within an group, you can click on the "Expand All" or "Collapse All" buttons, respectively. Some times, it is more sensible to toggle through a series of choices for a parameter (e.g. "Sense" for beta strands) and double-clicking that field will do that instead.

Deleting groups behaves similarly to adding groups. Multiple groups can be deleted at the same time by shift-clicking the range of groups or alt-clicking (command-clicking on macOS) a non-contiguous set. For the second type, be careful when deleting an entire group. That option is available if any selection within the group is highlighted. That is, an entire beta sheet can be deleted by clicking "Delete Sheet" when any strand within that sheet is highlighted.

Some options are shown in the left panel under "Other settings." These are usually options where it is sensible to apply them to many groups. After several groups are selected, either by shift-clicking a range or alt-clicking (command-clicking on macOS), changing a value in "Other settings" will update all the selected groups. If a parameter in "Other settings" is also shown as a column field, the selection editor will update that column to the new value for the selected groups.

An automatic detection option is available for

- Secondary structure annotations
- NCS groups
- TLS groups

by clicking the "Find ..." button in the left panel. If there are options for tweaking the automatic detection algorithm, they will show up in a new dialog window. These can be changed before running the algorithm. When automatic detection is finished, the selection editor will replace any existing selections with the automatically detected ones. Also, the automatic selections will be saved in your project directory.

Graphical Selection

In addition to manually typing in selections with the proper syntax, selections can also be made with the model viewer by clicking on "Model" in the top toolbar.

The model viewer will display the currently highlighted selections in the current panel in the selection editor. In order to use the model viewer to edit a selection

- Highlight the desired selection for editing in the selection editor
- Click on "Edit Selection" in the model viewer
- Make the changes in the model viewer
- In the model viewer, Click "Accept" to keep the changes, "Discard" to keep the original

Clicking atoms in the model viewer will toggle the highlight on them. To change between highlighting the entire residue or individual atoms, pick the appropriate option in the "Select" field.

The "Show" option in "View Controls" can toggle between showing all the atoms or just the backbone. This can simplify selecting residues.

When editing groups that have multiple selections in the same row (e.g. "Base Pairs" or "Stacking Pairs" under "Secondary Structure"), the specific selection can be changed by the "Field" widget in the model viewer.

A range of residues can be selected or deselected by shift-clicking or alt-clicking (option-clicking on macOS), respectively. The model viewer remembers the last residue that was clicked and uses that as the first residue in the range.

Known Issues

- With the model viewer open, highlighting many selections may be slow; close the model viewer for faster performance when highlighting many selections
- When an invalid type is entered (e.g. a letter is provided when the field expects a number), the selection editor will complain one more time after a valid type is entered.
- After clicking "Validate Syntax", the currently selected or most recently selected rows may not be highlighted in red if the syntax is incorrect.
- On macOS, the text in editing mode can be slightly offset in the row

Guessing sequence from a map with `sequence_from_map`

`sequence_from_map.html`

Author(s)

- `sequence_from_map`: Tom Terwilliger

Purpose

The routine `sequence_from_map` will use a map to guess the sequence of a supplied segment or of segments built automatically.

Usage

How `sequence_from_map` works:

Sequence-from-map examines the features in a map at the position where side chains in a model are expected to be located and uses these features to guess the identity of the side chain at each position.

The procedure used by `sequence-from-map` is has several steps:

If map (and no model) is supplied, a model is created by searching for regular secondary structure in the map. First the unique part of the map is identified and cut out to form a new map.

This is done first using the `map_symmetry` to find helical or point-group symmetry in the map. If any is found, a new map is created containing just the unique part of the map. Then (if `improper_symmetry` is set), the new map is examined to see if there is any local symmetry remaining. If so, the unique part is cut out and a new map is created. Then the tool `find_helices_strands` is used to analyze the new map to find regular secondary structure.

Once a map and model are available, each segment (a piece of chain without breaks) is compared to the map and sequences compatible with the map are identified.

The core of `sequence-from-map` is to generate a scoring matrix for each segment that reflects the relative probability that each possible side chain is located at each position in the segment. This probability is estimated from the map-model correlation for each side chain (after examining all rotamers of the side chain and picking the one with the highest correlation). A Z-score (value for this side chain minus mean for this side chain at all positions, divided by the standard deviation of this quantity) is calculated for each side chain at each position, where Z values less than zero are set to zero. These Z-scores are used as approximate minus-log-likelihood scores for side chain probabilities and make up the scoring matrix for the segment.

A best-fitting sequence is generated for each segment by simply picking the highest-scoring side chain at each position. Optionally, if `score_by_residue_groups` is set and residue groups are defined (`residue_groups = "VGASCTI P LDNEQM KR FHY W"`), one representative of each residue group is chosen to represent all the side chains in that group.

Optionally (if `minimum_discrimination` is defined and `trim_models` is set), the scoring matrix is used to cut the model up into pieces, removing any residues where the discrimination between the most probable and least probable side chain is less than `minimum_discrimination`. The segments are then cut up at these points and smaller segments are created.

Also optionally (if `trim_models` is set to some value N), the N residues on

each end of each segment are removed before analysis.

If one or more potential sequences representing the molecule in the map are provided, these sequences are scored, one at a time, by evaluating the optimal alignment of the sequence to the segments in the model.

The alignment of a sequence to a set of segments is done by considering all possible alignments for all segments, keeping the highest-scoring segment-sequence alignment, then repeating the process with all remaining segments. Optionally, the alignments can be restricted to a set that uses each residue in the sequence only once (`allow_duplicates` set to `False`). Optionally, sequences that are shorter than the number of residues in the model are rejected (if `skip_if_too_short` is set).

The raw score for one segment-sequence alignment is given by the sum of the position-dependent minus-log-likelihood scores for all the side chains in the alignment. This raw score is then adjusted by subtracting the mean value of raw segment-sequence alignment scores for random sequences of the same length as the sequence being considered. The random sequences are created using a residue frequency of eukaryotic amino acids (this can be adjusted with the `default_sequence` parameter). If `positive_only` is set, negative segment-sequence alignment scores are ignored.

Once the best-fitting sequence is identified, the side chains of the model are re-fit using this sequence and a model with the fitted side chains is written out.

Examples

Standard run of `sequence_from_map`:

You can use `sequence_from_map` to guess the sequence of a fragment you have built:

```
phenix.sequence_from_map my_fragment.pdb my_map.mrc resolution=2.8
```

You can also use it to try and pick out which sequence file matches your map:

```
phenix.sequence_from_map my_map.mrc resolution=3 seq_dir=sequences/
```

where the directory `sequences` has a set of fasta or similar files representing possible sequences that might match your map.

Re-sequencing and filling in a gap with `run_assign_sequence=True`:

You can use the special keyword `run_assign_sequence=True` in `sequence_from_map` to carry out a two-step process to try and fix a mistraced part of your model. If residues 154-162 of your model look bad and perhaps the sequence register is wrong after residue 162, you can try something like this:

First remove the bad residues:

```
phenix.pdbtools bad_model.pdb
```

```
remove="resseq 154:162" output.file_name=bad_model_del.pdb
```

Then run `sequence_from_map` to reassign the sequence and fill in the gap (you can supply a `map_file` or an `mtz_file`; either will be fine):

```
phenix.sequence_from_map run_assign_sequence=True
```

```
model_file=bad_model_del.pdb seq_file=sequence.dat map_file=map_file.ccp4
```

```
pdb_out=bad_model_fixed.pdb resolution=2
```

This will produce bad_model_fixed.pdb

Possible Problems

Specific limitations and problems:

Sequence_from_map does not recognize insertion codes, so if your model uses these, first run phenix.pdb_tools with the "renumber=True" keyword to remove them (actually to renumber the residues and leave the insertion codes in place).

Literature

Automated side-chain model building and sequence assignment by template matching. T.C. Terwilliger. Acta Crystallogr D Biol Crystallogr 59, 45-9 (2003). Rapid model building of alpha-helices in electron-density maps. T.C. Terwilliger. Acta Crystallogr D Biol Crystallogr 66, 268-75 (2010). Rapid model building of beta-sheets in electron-density maps. T.C. Terwilliger. Acta Crystallogr D Biol Crystallogr 66, 276-84 (2010). Protein identification from electron cryomicroscopy maps by automated model-building and side-chain matching. T.C. Terwilliger, O.V. Sobolev, Afonine, PV, P.D. Adams, Ho, CM, Li, X, Zhou, and ZH. Acta Cryst D. 77, 457-462 (2021).

- Automated side-chain model building and sequence assignment by template matching. T.C. Terwilliger. Acta Crystallogr D Biol Crystallogr 59, 45-9 (2003).
- Rapid model building of alpha-helices in electron-density maps. T.C. Terwilliger. Acta Crystallogr D Biol Crystallogr 66, 268-75 (2010).
- Rapid model building of beta-sheets in electron-density maps. T.C. Terwilliger. Acta Crystallogr D Biol Crystallogr 66, 276-84 (2010).
- Protein identification from electron cryomicroscopy maps by automated model-building and side-chain matching. T.C. Terwilliger, O.V. Sobolev, Afonine, PV, P.D. Adams, Ho, CM, Li, X, Zhou, and ZH. Acta Cryst D. 77, 457-462 (2021).

Additional information

List of all available keywords

job_title = None Job title in PHENIX GUI, not used on command line
input_filesmap_file = None File with CCP4-style map. May have origin in any location.
multiple_seq_file = None Optional sequence file containing multiple potential sequences. (Fasta format or sequences separated by blank lines)
seq_file = None Optional sequence file(s) containing potential sequences
seq_dir = None Optional directory containing sequence files
model_file = None Input PDB file with chains to be sequenced NOTE: if you supply model_file no model-building will be done. If you supply starting_model_file it will be used as a starting model for model-building.
starting_model_file = None Input PDB file to be used as starting model in model-building. NOTE: if you supply model_file no model-building will be done. If you supply starting_model_file it will be used as a starting model for model-building.
symmetry_file = None Optional symmetry file (.ncs_spec format or MATRX records) with reconstruction symmetry. Used in identification of unique part of map.
probability_matrix_file = None Optional .pkl file with probability matrices. Note: if build_type is all then a prefix of either helices_strands or full_model will be added to the file name (if not already there)
model_pickle_file = None Optional .pkl file with final fragments. Note: if build_type is all then a prefix of either helices_strands or full_model will be added to the file name (if not already there)
output_files
pdb_out =

sequence_from_map.pdb Model with sequence determined from map. Note: if build_type is all then a prefix of either helices_strands or full_model will be added to the file name (if not already there)output_probability_matrix_file = None Optional output .pkl file with probability matricesoutput_model_pickle_file = None Optional output .pkl file with final fragmentstemp_dir = None Temporary directory. Default is sequence_from_map_xx where xx creates new directorycrystal_inforesolution = None High-resolution limit for map analysis.scattering_table = n_gaussian wk1995 it1992 *electron neutron Choice of scattering table for structure factor calculations. Standard for X-ray is n_gaussian, for cryoEM is electron.sequences = None Sequences to be matched as text stringsdefault_sequence = " Sequence reflecting relative usage of residues/nucleotides. Default is amino acid composition for eukaryotes.chain_type = *PROTEIN DNA RNA Chain typemolecular_mass = None Molecular mass of the entire structure in the map (not the mass of one chain). Does not have to be accurate, but helpful if using symmetry to extract unique part of map.strategyfill_in_sequence = None Just fill in optimal sequence. Default if no sequence supplied.extract_best_parts = False Try to identify best-fitting parts of sequence and use thosecreate_gap_length = None Create gaps of this length if segments are nearly overlappingextract_length = 35 Target lengths of extracted best parts.extract_cc = 0.80 Target map CC of extracted best parts.extract_cc_delta = 0.01 Delta for target map CC of extracted best parts.extract_cc_min = 0.70 Minimum target map CC of extracted best parts.extract_n = 10 Maximum number of extracted best parts.extract_length_vs_cc = 10 Trade off between length and CCscore_with_optimal_sequence = True Identify optimal sequence with reduced residue set and use it to match sequenceunique_sequence = True Only allow one use of each part of sequence unless model is longer than sequencesplit_using_alignment = None At end, split fragments based on alignment. Default is True if score_with_optimal_sequence is True and sequence is suppliedkeep_segment_order = None Keep order of segments (only change their local positions). Only applies if unique_sequence=True and score_with_optimal_sequence=Trueextract_ss = False Extract secondary structure groups (usual alternative is extract regions)extract_ss_dist_max = 15 RMSD of ends of secondary structure must match within this dist.extract_ss_groups = 1 Number of secondary structure groups t...

- job_title = None Job title in PHENIX GUI, not used on command line
- input_filesmap_file = None File with CCP4-style map. May have origin in any location.multiple_seq_file = None Optional sequence file containing multiple potential sequences. (Fasta format or sequences separated by blank lines)seq_file = None Optional sequence file(s) containing potential sequencesseq_dir = None Optional directory containing sequence filesmodel_file = None Input PDB file with chains to be sequenced NOTE: if you supply model_file no model-building will be done. If you supply starting_model_file it will be used as a starting model for model-building.starting_model_file = None Input PDB file to be used as starting model in model-building. NOTE: if you supply model_file no model-building will be done. If you supply starting_model_file it will be used as a starting model for model-building.symmetry_file = None Optional symmetry file (.ncs_spec format or MATRX records) with reconstruction symmetry. Used in identification of unique part of map.probability_matrix_file = None Optional .pkl file with probability matrices. Note: if build_type is all then a prefix of either helices_strands or full_model will be added to the file name (if not already there)model_pickle_file = None Optional .pkl file with final fragments. Note: if build_type is all then a prefix of either helices_strands or full_model will be added to the file name (if not already there)
- map_file = None File with CCP4-style map. May have origin in any location.
- multiple_seq_file = None Optional sequence file containing multiple potential sequences. (Fasta format or sequences separated by blank lines)
- seq_file = None Optional sequence file(s) containing potential sequences
- seq_dir = None Optional directory containing sequence files

- `model_file` = None Input PDB file with chains to be sequenced NOTE: if you supply `model_file` no model-building will be done. If you supply `starting_model_file` it will be used as a starting model for model-building.
- `starting_model_file` = None Input PDB file to be used as starting model in model-building. NOTE: if you supply `model_file` no model-building will be done. If you supply `starting_model_file` it will be used as a starting model for model-building.
- `symmetry_file` = None Optional symmetry file (.ncs_spec format or MATRX records) with reconstruction symmetry. Used in identification of unique part of map.
- `probability_matrix_file` = None Optional .pkl file with probability matrices. Note: if `build_type` is all then a prefix of either `helices_strands` or `full_model` will be added to the file name (if not already there)
- `model_pickle_file` = None Optional .pkl file with final fragments. Note: if `build_type` is all then a prefix of either `helices_strands` or `full_model` will be added to the file name (if not already there)
- `output_files` `pdb_out` = `sequence_from_map.pdb` Model with sequence determined from map. Note: if `build_type` is all then a prefix of either `helices_strands` or `full_model` will be added to the file name (if not already there) `output_probability_matrix_file` = None Optional output .pkl file with probability matrices `output_model_pickle_file` = None Optional output .pkl file with final fragments `temp_dir` = None Temporary directory. Default is `sequence_from_map_xx` where `xx` creates new directory
- `pdb_out` = `sequence_from_map.pdb` Model with sequence determined from map. Note: if `build_type` is all then a prefix of either `helices_strands` or `full_model` will be added to the file name (if not already there)
- `output_probability_matrix_file` = None Optional output .pkl file with probability matrices
- `output_model_pickle_file` = None Optional output .pkl file with final fragments
- `temp_dir` = None Temporary directory. Default is `sequence_from_map_xx` where `xx` creates new directory
- `crystal_info` `resolution` = None High-resolution limit for map analysis. `scattering_table` = `n_gaussian` wk1995 it1992 *electron neutron Choice of scattering table for structure factor calculations. Standard for X-ray is `n_gaussian`, for cryoEM is `electron`. `sequences` = None Sequences to be matched as text strings `default_sequence` = " Sequence reflecting relative usage of residues/nucleotides. Default is amino acid composition for eukaryotes. `chain_type` = *PROTEIN DNA RNA Chain type `molecular_mass` = None Molecular mass of the entire structure in the map (not the mass of one chain). Does not have to be accurate, but helpful if using symmetry to extract unique part of map.
- `resolution` = None High-resolution limit for map analysis.
- `scattering_table` = `n_gaussian` wk1995 it1992 *electron neutron Choice of scattering table for structure factor calculations. Standard for X-ray is `n_gaussian`, for cryoEM is `electron`.
- `sequences` = None Sequences to be matched as text strings
- `default_sequence` = " Sequence reflecting relative usage of residues/nucleotides. Default is amino acid composition for eukaryotes.
- `chain_type` = *PROTEIN DNA RNA Chain type
- `molecular_mass` = None Molecular mass of the entire structure in the map (not the mass of one chain). Does not have to be accurate, but helpful if using symmetry to extract unique part of map.
- `strategy` `fill_in_sequence` = None Just fill in optimal sequence. Default if no sequence supplied. `extract_best_parts` = False Try to identify best-fitting parts of sequence and use those `create_gap_length` = None Create gaps of this length if segments are nearly overlapping `extract_length` = 35 Target lengths of extracted best parts. `extract_cc` = 0.80 Target map CC

of extracted best parts.
`extract_cc_delta = 0.01` Delta for target map CC of extracted best parts.
`extract_cc_min = 0.70` Minimum target map CC of extracted best parts.
`extract_n = 10` Maximum number of extracted best parts.
`extract_length_vs_cc = 10` Trade off between length and CC
`score_with_optimal_sequence = True` Identify optimal sequence with reduced residue set and use it to match sequence
`unique_sequence = True` Only allow one use of each part of sequence unless model is longer than sequence
`split_using_alignment = None` At end, split fragments based on alignment. Default is True if `score_with_optimal_sequence` is True and sequence is supplied
`keep_segment_order = None` Keep order of segments (only change their local positions). Only applies if `unique_sequence=True` and `score_with_optimal_sequence=True`
`extract_ss = False` Extract secondary structure groups (usual alternative is `extract_regions`)
`extract_ss_dist_max = 15` RMSD of ends of secondary structure must match within this dist.
`extract_ss_groups = 1` Number of secondary structure groups to generate
`build_only = False` Just set up and build model and stop
`retrace = None` Retrace chains and score with map correlation and sequence alignment in `trace_and_build` procedure. Default is False if `quick = True`
`block_gaps_in_secondary_structure = True` Block creation of gaps in alignments in the middle of secondary structure
`rerun_map_box = False` Run map box on input model with radius `atom_radius`
`map_box_radius = 10` RMSD of ends of secondary structure must match within this dist.
`rerun_with_avail_seq = True` Rerun sequence alignment with parts of the sequence already used removed. With a large structure this can take a long time.
`speed_up_ca_insertion = False` Speed up CA insertion by running on short segments first.
`chains_from_fixed_model = None` Use fixed model to mark initial chain positions.
`max_fraction_long_branches = None` Terminate trace if recent fraction of long branches is high
`split_and_join_iterations = None` Iterations of `split_and_join`
`split_and_join = None` Split fragments at low density and then rejoin. Criterion is `sd_ratio_pare_model`
`rejoin_split_fragments = False` Specifically rejoin the split fragments leaving out poor junctions
`test_threshold_fixed_segments = None` Test thresholds when creating fixed segments
`require_ends_in_region = None` Add ends of chain to region if necessary
`try_alternate_joins = None` try joining each end of pair of fragments, not just the first
`skip_path_if_missing = None` Skip path if points are missing

- `fill_in_sequence = None` Just fill in optimal sequence. Default if no sequence supplied.
- `extract_best_parts = False` Try to identify best-fitting parts of sequence and use those
- `create_gap_length = None` Create gaps of this length if segments are nearly overlapping
- `extract_length = 35` Target lengths of extracted best parts.
- `extract_cc = 0.80` Target map CC of extracted best parts.
- `extract_cc_delta = 0.01` Delta for target map CC of extracted best parts.
- `extract_cc_min = 0.70` Minimum target map CC of extracted best parts.
- `extract_n = 10` Maximum number of extracted best parts.
- `extract_length_vs_cc = 10` Trade off between length and CC
- `score_with_optimal_sequence = True` Identify optimal sequence with reduced residue set and use it to match sequence
- `unique_sequence = True` Only allow one use of each part of sequence unless model is longer than sequence
- `split_using_alignment = None` At end, split fragments based on alignment. Default is True if `score_with_optimal_sequence` is True and sequence is supplied
- `keep_segment_order = None` Keep order of segments (only change their local positions). Only applies if `unique_sequence=True` and `score_with_optimal_sequence=True`
- `extract_ss = False` Extract secondary structure groups (usual alternative is `extract_regions`)

- `extract_ss_dist_max` = 15 RMSD of ends of secondary structure must match within this dist.
- `extract_ss_groups` = 1 Number of secondary structure groups to generate
- `build_only` = False Just set up and build model and stop
- `retrace` = None Retrace chains and score with map correlation and sequence alignment in `trace_and_build` procedure. Default is False if `quick` = True
- `block_gaps_in_secondary_structure` = True Block creation of gaps in alignments in the middle of secondary structure
- `run_map_box` = False Run map box on input model with radius `atom_radius`
- `map_box_radius` = 10 RMSD of ends of secondary structure must match within this dist.
- `rerun_with_avail_seq` = True Rerun sequence alignment with parts of the sequence already used removed. With a large structure this can take a long time.
- `speed_up_ca_insertion` = False Speed up CA insertion by running on short segments first.
- `chains_from_fixed_model` = None Use fixed model to mark initial chain positions.
- `max_fraction_long_branches` = None Terminate trace if recent fraction of long branches is high
- `split_and_join_iterations` = None Iterations of `split_and_join`
- `split_and_join` = None Split fragments at low density and then rejoin. Criterion is `sd_ratio_pare_model`
- `rejoin_split_fragments` = False Specifically rejoin the split fragments leaving out poor junctions
- `test_threshold_fixed_segments` = None Test thresholds when creating fixed segments
- `require_ends_in_region` = None Add ends of chain to region if necessary
- `try_alternate_joins` = None try joining each end of pair of fragments, not just the first
- `skip_path_if_missing` = None Skip path if points are missing
- `sequencingcreate_final_model` = True Create final model with sequence
- `obtainedresidue_number_from_sequence` = True Start chains with best guess of residue number from sequence file
- `build_type` = *None helices_and_full helices_strands full_model trace_and_build trace_and_build_two_cycles Default is trace_and_build if `quick` = true and trace_and_build_two_cycles if `quick` = False. You can choose to build with trace_and_build, just helices/strands (quick) or a full model (takes more time). Note you can define how many regions to analyze with `regions_to_keep` (set it to 1 to run quickly). You can choose helices_and_full to run both helices_strands and full_model or trace_and_build_two_cycles to run trace and build two cycles and pick the best result.
- `tnb_build_cycles` = 1 Trace and build build cycles
- `regions_to_keep` = None You can specify a limit to the number of regions to keep when generating the asymmetric unit of density. Default is 1 for `build_type` = full_model or trace_and_build and None for `build_type` = helices_strands.
- `minimum_score` = -10. Minimum score. Scores less than this will be replaced with this value.
- `alignment_method` = *smith_waterman muscle Alignment method. Alternatives are muscle and smith_waterman
- `gap_ratio` = 0.5 Ratio of gap extension penalty to gap opening penalty
- `blocking_ratio` = 10 Ratio of gap extension penalty in regions to be blocked to to allowed regions
- `gap_opening_penalty_max` = None Gap opening penalty for sequence alignments. Goes from `gap_opening_penalty_min` to `gap_opening_penalty_max`. Default is 10 for muscle and 150 for phenix
- `gap_opening_penalty_min` = None Gap opening penalty for sequence alignments. Goes from `gap_opening_penalty_min` to `gap_opening_penalty_max`. Default is 2.5 for muscle and 35 for phenix
- `trim_ends_by` = 1 When blocking alignments with secondary structure, trim the secondary structure back by this amount to allow insertions and deletions at the end of the secondary structure.
- `convincing_delta_score` = 10. Convincing delta score for two solutions that nearly always means the higher-scoring one is correct.
- `average_z_scores` = False Use average Z-score for multiple builds.

Alternative is use highest score.
`minimum_final_z_score = 4` Skip additional builds if a solution with this Z-score or better is found.
`minimum_length_after_split = 5` Minimum length of a segment after splitting with `split_using_alignment`
`minimum_length_after_split_penalty = 2` Penalty per residue shorter than `minimum_length`
`minimum_length = None` Minimum length of a segment to consider it for sequencing. Default is 10 if a `multiple_sequence_file` or `sequence_dir` is supplied
`minimum_length_ratio = None` Reject segments that are less than this fraction of longest segment. Applies if `max_segments` is not set. Default is 0.15 if a `multiple_sequence_file` or `sequence_dir` is supplied.
`maximum_segments_in_sequencing = None` Maximum number of segments to consider (remainder ignored) in analyzing sequence placement. Default is 10 if a `multiple_sequence_file` or `sequence_dir` is supplied.
`max_segments = None` Maximum segments to consider in analyzing sequences. Default is to take all segments unless a `multiple_sequence_file` or `sequence_dir` is supplied, in which case the default is 3.
`minimum_residues = None` Try to choose enough segments to get this many residues. Applies if `max_segments` is not set. Default is 500 if multiple sequences are supplied.
`maximum_sequence_length = None` Maximum sequence length to consider. Default is 2000 if multiple sequences are supplied.
`subtract_expected = None` *single_try multiple_tries Subtract expected score based on random sequence for a single try or for the number of tries carried out.
`positive_only = False` Just sum scores that are positive and skip others
`allow_duplicates = True` Allow duplication of use of sequence.
`skip_if_too_short = False` Skip sequences that are shorter than the number of residues in the model.
`skip_if_too_short_for_segment = None` Skip sequences that are shorter than the number of residues in a segment, with tolerance of...

- `create_final_model = True` Create final model with sequence obtained
- `residue_number_from_sequence = True` Start chains with best guess of residue number from sequence file
- `build_type = *None helices_and_full helices_strands full_model trace_and_build trace_and_build_two_cycles` Default is `trace_and_build` if `quick = True` and `trace_and_build_two_cycles` if `quick = False`. You can choose to build with `trace_and_build`, just helices/strands (`quick`) or a full model (takes more time). Note you can define how many regions to analyze with `regions_to_keep` (set it to 1 to run quickly). You can choose `helices_and_full` to run both `helices_strands` and `full_model` or `trace_and_build_two_cycles` to run trace and build two cycles and pick the best result.
- `tnb_build_cycles = 1` Trace and build build cycles
- `regions_to_keep = None` You can specify a limit to the number of regions to keep when generating the asymmetric unit of density. Default is 1 for `build_type = full_model` or `trace_and_build` and `None` for `build_type = helices_strands`.
- `minimum_score = -10`. Minimum score. Scores less than this will be replaced with this value.
- `alignment_method = *smith_waterman muscle` Alignment method. Alternatives are `muscle` and `smith_waterman`
- `gap_ratio = 0.5` Ratio of gap extension penalty to gap opening penalty
- `blocking_ratio = 10` Ratio of gap extension penalty in regions to be blocked to to allowed regions
- `gap_opening_penalty_max = None` Gap opening penalty for sequence alignments. Goes from `gap_opening_penalty_min` to `gap_opening_penalty_max`. Default is 10 for `muscle` and 150 for `phenix`
- `gap_opening_penalty_min = None` Gap opening penalty for sequence alignments. Goes from `gap_opening_penalty_min` to `gap_opening_penalty_max`. Default is 2.5 for `muscle` and 35 for `phenix`
- `trim_ends_by = 1` When blocking alignments with secondary structure, trim the secondary structure back by this amount to allow insertions and deletions at the end of the secondary structure.

- `convincing_delta_score = 10`. Convincing delta score for two solutions that nearly always means the higher-scoring one is correct.
- `average_z_scores = False` Use average Z-score for multiple builds. Alternative is use highest score.
- `minimum_final_z_score = 4` Skip additional builds if a solution with this Z-score or better is found.
- `minimum_length_after_split = 5` Minimum length of a segment after splitting with `split_using_alignment`
- `minimum_length_after_split_penalty = 2` Penalty per residue shorter than minimum
- `minimum_length = None` Minimum length of a segment to consider it for sequencing. Default is 10 if a `multiple_sequence_file` or `sequence_dir` is supplied
- `minimum_length_ratio = None` Reject segments that are less than this fraction of longest segment. Applies if `max_segments` is not set. Default is 0.15 if a `multiple_sequence_file` or `sequence_dir` is supplied.
- `maximum_segments_in_sequencing = None` Maximum number of segments to consider (remainder ignored) in analyzing sequence placement. Default is 10 if a `multiple_sequence_file` or `sequence_dir` is supplied.
- `max_segments = None` Maximum segments to consider in analyzing sequences. Default is to take all segments unless a `multiple_sequence_file` or `sequence_dir` is supplied, in which case the default is 3.
- `minimum_residues = None` Try to choose enough segments to get this many residues. Applies if `max_segments` is not set. Default is 500 if multiple sequences are supplied.
- `maximum_sequence_length = None` Maximum sequence length to consider. Default is 2000 if multiple sequences are supplied.
- `subtract_expected = None` *`single_try` `multiple_tries` Subtract expected score based on random sequence for a single try or for the number of tries carried out.
- `positive_only = False` Just sum scores that are positive and skip others
- `allow_duplicates = True` Allow duplication of use of sequence.
- `skip_if_too_short = False` Skip sequences that are shorter than the number of residues in the model.
- `skip_if_too_short_for_segment = None` Skip sequences that are shorter than the number of residues in a segment, with tolerance of `short_tolerance_percent`.
- `short_tolerance_percent = 10` Tolerance for `skip_if_too_short_for_segment`
- `refinement = None` Run refinement after finding model. Default is True for `helices_strands` or `full_model` and False for `trace_and_build`.
- `simulated_annealing_tries = None` Run simulated annealing this many times on segments and weight sequence probability based on the resulting likelihood scores.
- `average_sa_by_residue = False` Average SA tries by residue (alternative is average by model)
- `split_then_fix = False` First use `split_at_unknown` to split up model, screen all input sequences and choose top `n_split_then_fix` sequences. Then run again on these sequences using `fix_insertions_deletions` on each of them for final scoring.
- `n_split_then_fix = None` Number of sequences to keep in play in `split_then_fix`
- `fix_insertions_deletions = False` Try to fix alignment allowing up to $n_split * n_sites / 10$ pieces but keep them linked to maximize sequence match
- `n_split = 1` Maximum number of pieces to cut segments up into is $n_sites * n_split / 10$

- `max_start_points_to_consider = 5` Maximum number of starting points for sequence to consider per 100 residues in sequence file.
- `positive_gap_penalty = 1` Penalty for missing residues is `positive_gap_penalty * gap**2`
- `negative_gap_penalty = 2` Gap penalty for extra residues is `negative_gap_penalty * gap**2`
- `max_gap_length = 2` Maximum gap length for adjacent alignments
- `end_overlap = 2` Maximum end overlap of fragments
- `minimum_alignment_length = 5` Minimum length of an alignment
- `test_lengths = 10 20` Lengths of segments to be tested for sequence matching.
- `macro_cycles = 5` Refinement macro-cycles
- `trim_from_ends = None` You can take final fragments and trim them N residues in from each end.
- `residue_groups = "GVASCTI P LDNEQM KR YFH W"` Optional groups of residues to score together. If RNA or DNA groups are not allowed.
- `score_by_residue_groups = None` Use residue groups in sequence alignment and listing of optimal sequences. Default is True unless no sequence is supplied.
- `trim_models = False` Trim models to break at residues with low discrimination. Default is False.
- `maximum_length = 50` If segment is longer than this, try to cut it into pieces, splitting at residues that are poorly-defined. Assumption is very long segments probably have register shifts. Only applies if `trim_models = True`
- `split_at_unknown = None` Split chains at positions where residue is unknown (X). Default is True unless `fix_insertions_deletions = True` or `score_with_optimal_sequence = True`
- `keep_split_together = True` Apply penalty if chains that are split at unknown are not close in sequence
- `vary_minimum_cc = False` Vary the minimum_cc in optimizing percent to reject. Alternative is `vary_minimum_z_score`.
- `percent_to_reject = 7.5` Adjust minimum_cc and minimum_z_score to reject this percentage of residues.
- `minimum_z_score = 0.5` Minimum residue Z-score for a residue (or residue groups if `score_by_residue_groups` is set) to keep the choice. Otherwise report residue as X.
- `minimum_cc = None` Minimum map-model correlation for a residue (or residue groups if `score_by_residue_groups` is set) to keep the choice. Otherwise report residue as X.
- `minimum_discrimination = 0.01` Minimum discrimination between lowest and highest probability residue (or residue groups if `score_by_residue_groups` is set) to keep the choice. Otherwise report residue as X.
- `random_sequences = 20` Number of random sequences of each length to use as baseline
- `use_default_sequence = False` Base the random sequences on the default sequence. Alternative is to base them on the actual sequence
- `remove_clashes = True` Truncate clashing side chains on output
- `symmetry = ANY` Optional symmetry used in reconstruction. For example D7, C3, C2 I (icosahedral), T (tetrahedral), or ANY (try everything and use the highest symmetry found). Not needed if symmetry is supplied.
- `include_helical_symmetry = False` Include search for helical symmetry if `symmetry = ALL`
- `run_assign_sequence = False` Run assign sequence to identify sequence of multiple fragments and connect them.

- `improper_symmetry = False` After any proper symmetry is found, take new unique part of map and search for any symmetry at all with `find_ncs_from_density`. Useful if your molecule has both proper reconstruction symmetry such as C3 and within the reconstruction unit there is local symmetry which can be of any kind. Final result is just the unique part of the map.
- `separate_chains = True` Write each segment as a separate chain with unique chain ID
- `null_score = -2` Score for a segment that is not placed
- `n_direct = 10` Extreme values for $N \leq n_direct$ calculated directly
- `distance_cutoff = None` Distance cutoff (CA-CA or P-P) for chain breaks. Large value used to prevent breaking chain due to big gaps.
- `min_ncs_cc = 0.80` Minimum Symmetry CC to keep operators when identifying automatically
- `rebuildingsplit_input_model = True` Input model will be split into segments
- `refine = True` Refine all-atom models at end of procedure
- `refine_b = None` Refine B-values
- `refine_cycles = 1` Refinement cycles (except final cycles). Typical is 1 for quick or superquick and 5 for thorough building
- `trace_and_build_min_segment_length = 15` Minimum segment length in `trace_and_build`
- `controlmultiprocessing = *` `multiprocessing` `sgs` `lsf` `pbs` `condor` `pbspro` `slurm` Choices are multiprocessing (single machine) or queuing systems
- `queue_run_command = None` run command for queue jobs. For example `qsub`.
- `nproc = 1` Number of processors to use
- `random_seed = 171731` Random seed
- `verbose = False` Verbose output
- `quick = True` Quick run. Suitable for most cases. Use `quick = False` to obtain maximum sensitivity by setting `build_type = trace_and_build_two_cycles`
- `simulated_annealing_tries = 32`
- `superquick = False` Very quick run
- `ignore_symmetry_conflicts = False` You can ignore the symmetry information (CRYST1) from coordinate files. This may be necessary if your model has been placed in a box with `box_map` for example.
- `max_dirs = 1000` Maximum number of directories (sequence_from_map_xxx)
- `em_side_density = False` Use EM side chain density. Alternative is to use standard x-ray side chain templates in sequence assignment
- `em_side_density_in_tnb = False` EM side chain density in `trace_and_build` (if `em_side_density` is True)
- `skip_temp_dir = False` Do not create or use a temp_dir
- `clean_up = False` Clean up and remove temporary sequence_from_map directory after finishing

- `ignore_symmetry_conflicts = False` You can ignore the symmetry information (CRYST1) from coordinate files. This may be necessary if your model has been placed in a box with `box_map` for example.
- `max_dirs = 1000` Maximum number of directories (`sequence_from_map_xxx`)
- `em_side_density = False` Use EM side chain density. Alternative is to use standard x-ray side chain templates in sequence assignment
- `em_side_density_in_tnb = False` EM side chain density in `trace_and_build` (if `em_side_density` is True)
- `skip_temp_dir = False` Do not create or use a `temp_dir`
- `clean_up = False` Clean up and remove temporary `sequence_from_map` directory after finishing
- guiGUI-specific parameter required for output directory `output_dir = None`
- `output_dir = None`

Setting up tests that can run automatically in the Phenix GUI

set_up_gui_tests.html

Author(s)

- Automatic GUI tests: Tom Terwilliger

Purpose

These tests can be used to make sure that programs in the GUI can run properly. They can also be used as an introduction to running in the GUI as they are all preset to run.

Framework

The GUI_TESTS directory has a set of subdirectories (CRYSTALS_A...) that can be imported as projects into the Phenix GUI. Once in the GUI, all the unique jobs in a project can be run sequentially with the "Auto-run all unique jobs" function.

Updating the GUI tests

New projects can be added to the GUI_TESTS directory by creating them in the Phenix GUI in an installation of GUI_TESTS. The existing and new projects can then be packaged into a new version of GUI_TESTS automatically. Additionally, new jobs can be run in an existing GUI_TESTS module and existing jobs can be deleted. On repackaging, the GUI_TESTS will reflect the state of these tests in the GUI that was worked on.

How to set up the tests

The tests are supplied with Phenix, but you have to install them and import them into the GUI to use them. Here is how:

Install the tests

Decide where you want to put the GUI_TESTS (in this example, ~/Downloads)

Go to the phenix_regression directory with GUI_TESTS (i.e.,

\$PHENIX/modules/phenix_regression/gui/auto_run). If \$PHENIX is not defined, you will need to type in the full path.

```
cd $PHENIX/modules/phenix_regression/gui/auto_run
```

To install in ~/Downloads, type:

```
./install_gui_tests.sh ~/Downloads
```

This will install the GUI_TESTS directory with the right path names in

~/Downloads.

Start the Phenix GUI. You might want to do this from the command line instead of clicking an icon so that the log goes to the terminal window.

In the Phenix GUI: select "Import Project". Navigate to your new GUI_TESTS directory, then to each subdirectory (CRYSTALS_A CRYSTALS_B, etc) and import each one individually.

- In the Phenix GUI: select "Import Project".
- Navigate to your new GUI_TESTS directory, then to each subdirectory (CRYSTALS_A CRYSTALS_B, etc) and import each one individually.

Navigate to your new GUI_TESTS directory, then to each subdirectory (CRYSTALS_A CRYSTALS_B, etc) and import each one individually.

Run a test

You are now ready to run one or all the tests. To run just one test:

In the Phenix GUI, select one of these projects (MODEL_A) and go to Job History. Select any job and hit "restore" or double click it The job will be restored and you can run it with "Run"

- In the Phenix GUI, select one of these projects (MODEL_A) and go to Job History.
- Select any job and hit "restore" or double click it
- The job will be restored and you can run it with "Run"

Run a group of tests

In the Phenix GUI, select one of these projects (MODEL_A) and go to Job History. Hit "Auto-run all unique jobs" The jobs will be run sequentially. Look for any errors. A few jobs will require clicking "OK" to go on. (It will stop and wait for your OK). It will also stop on most errors. Tests that cannot be run this way are in the ONE_BY_ONE project. For these, just reload the job and run it.

- In the Phenix GUI, select one of these projects (MODEL_A) and go to Job History.
- Hit "Auto-run all unique jobs"
- The jobs will be run sequentially. Look for any errors. A few jobs will require clicking "OK" to go on. (It will stop and wait for your OK). It will also stop on most errors. Tests that cannot be run this way are in the ONE_BY_ONE project. For these, just reload the job and run it.

The jobs will be run sequentially. Look for any errors.

A few jobs will require clicking "OK" to go on. (It will stop and wait for your OK). It will also stop on most errors. Tests that cannot be run this way are in the ONE_BY_ONE project. For these, just reload the job and run it.

Modifying the GUI_TESTS jobs

You can modify the GUI_TESTS jobs in any way and save the new version in phenix_regression. Details below. Basically: install the GUI_TESTS, create new jobs, delete any jobs that you do not want saved, create new projects (must be subdirectories of the directory GUI_TESTS), whatever. Then package up the new version you have created with the instructions below, and commit it to the repository.

Details of modifying the GUI_TESTS jobs by creating a new set of tests

When you have a new installation that you would like to save, you can use the create_new_gui_tests_package.csh script in your GUI_TESTS directory. NOTE: Script is c-shell so it won't work on bash for now.

- Get into your GUI_TESTS directory:

```
cd GUI_TESTS
```

2. Decide where your new installation package goes. The instructions here will put it in your ~/Downloads directory and you can copy it to your regression phenix_regression/gui/auto_run directory:

- Create a new installation package:

```
./create_new_gui_tests_package.csh ~/Downloads
```

```
cp ~/Downloads/GUI_TESTS.tgz $PHENIX/modules/phenix_regression/gui/auto_run/
```

This will overwrite the GUI_TESTS.tgz package in Downloads, and if you copy it, the one in your regression directory. You can put your new installation package anywhere you like.

Showing information about a map file with show_map_info

show_map_info.html

Author(s)

- show_map_info: Tom Terwilliger

Purpose

The routine show_map_info will show a summary of the data in a map file (ccp4/mrc/map).

Definitions of map specifications

A map file contains a box of data on a grid. This box of data may be the original box of data or it might be a smaller box cut out of the original box of data.

The map file also contains meta-data including a 'unit cell' and gridding for a unit cell. The unit cell usually corresponds to the full original box of data. The 'unit cell grid' is the grid corresponding to the unit cell. The unit cell has dimensions defined by 'unit cell parameters'.

The map that is present is represented on part or all of the unit cell grid. It starts at the grid point called the 'origin' and goes to one grid point in each direction before the grid point called 'last'. The map that is present has 'all' grid points in each direction. The map that is present has a 'map unit cell' and 'map grid' that correspond to the dimensions 'all'.

Example: a full map with a unit cell grid of (10, 10, 10) goes from (0, 0, 0) to (9, 9, 9) and has dimensions of (10, 10, 10).

Example: a partial map with a unit cell grid of (10,10,10) and an origin of (0, 0, 2) and a map grid or 'all' of (6, 6, 6), would go from (0, 0, 2) to (5, 5, 7).

A map file has a 'pixel size' in each direction (usually they are all the same). The pixel size is the dimension along x, y, or z divided by the number of grid points along that direction.

A map file has a 'map unit cell' which corresponds to the dimensions of the box of data that is present in the map. This will be the same as the unit cell of the full map if the entire map is present.

External origin (ORIGIN record in map file)

A map file can have an 'external_origin' defined. The external origin is an offset that is intended to be applied to a model file to superimpose the model on the map. If a map has an external origin defined, show_map_info will print out this value.

Phenix does not generally support the external_origin values. You can however shift a model file (PDB/mmCIF) based on a value of the external_origin with the tool phenix.shift_model_to_match_map

```
phenix.shift_model_to_match_map map_file.mrc model_file.pdb
```

Wrapping of a map

Maps can be periodic (wrapping = True) or non-periodic (wrapping = False). If you cut out a box from a periodic map, the new map is non-periodic.

If a map is periodic, then values at coordinates (x,y,z) outside the map are defined and can be obtained simply by translating (x,y,z) by integral numbers of the unit cell translations a,b,c ($N * a$, $M * b$, $O * c$) where N,M and O are integers. If a map is not periodic, values outside the supplied map are not defined.

In most Phenix methods, it is assumed that cryo-EM maps are non-periodic and X-ray (crystal) maps are periodic. You can specify in many cases whether your map is periodic or not, however, using the 'wrapping' keyword.

When you run `show_map_info` it will tell you whether the map supports wrapping.

Standard run of `show_map_info`:

You can use `show_map_info` to summarize a map file like this:

```
phenix.show_map_info my_map_file.mrc
```

You can get the value of the map at a particular grid point with:

```
phenix.show_map_info my_map_file.mrc grid_point_to_show=6,2,5
```

Possible Problems

Specific limitations and problems:

Literature

The MRC (ccp4/mrc/map) file format is defined at MRC/CCP4 file format for images and volumes

Additional information

List of all available keywords

`job_title` = None Job title in PHENIX GUI, not used on command line
`input_filesmap_file` = None File with CCP4-style map.
`map_infogrid_point_to_show` = None Display map value at this grid point if set. You can specify more than one.
`n_tries` = 23 Number of tries to find a common value
`relative_sd_tol` = 0.01 Tolerance for constant value at boundaries (relative_sd_tol). Used in identifying wrapping.
`check_wrapping` = None You can explicitly test whether this map appears suitable for wrapping (values outside map mapped inside with unit cell translations).
`gui` GUI-specific parameter required for output directory
`output_dir` = None

- `job_title` = None Job title in PHENIX GUI, not used on command line
- `input_filesmap_file` = None File with CCP4-style map.
- `map_file` = None File with CCP4-style map.
- `map_infogrid_point_to_show` = None Display map value at this grid point if set. You can specify more than one.
- `n_tries` = 23 Number of tries to find a common value
- `relative_sd_tol` = 0.01 Tolerance for constant value at boundaries (relative_sd_tol). Used in identifying wrapping.
- `check_wrapping` = None You can explicitly test whether this map appears suitable for wrapping (values outside map mapped inside with unit cell translations).
- `grid_point_to_show` = None Display map value at this grid point if set. You can specify more than one.
- `n_tries` = 23 Number of tries to find a common value

- `relative_sd_tol = 0.01` Tolerance for constant value at boundaries (`relative_sd_tol`). Used in identifying wrapping.
- `check_wrapping = None` You can explicitly test whether this map appears suitable for wrapping (values outside map mapped inside with unit cell translations).
- `guiGUI-specific parameter required for output directory``output_dir = None`
- `output_dir = None`

Homology modelling using simple_homology_model

simple_homology_model.html

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- Usage
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- Error messages
- Warnings
- Additional information List of all available keywords
- List of all available keywords

Purpose

This program can be used for building a homology model for a target sequence using a template structure and an alignment.

Installation

simple_homology_model uses Rosetta, software developed in the Baker laboratory at the University of Washington. See the central installation notes for Rosetta.

Usage

simple_homology_model is available from the command line.

Input files

- Structure (compulsory). The template structure. Accepted formats: PDB. Recognized extensions: .pdb, .ent.
- Alignment (compulsory). Alignment between the target and template sequences. Accepted formats: .aln: CLUSTAL format. .pir: PIR format. .ali: relaxed PIR-like format.

- .aln: CLUSTAL format.
- .pir: PIR format.
- .ali: relaxed PIR-like format.
- Sequence (optional). The complete target sequence if the alignment does not fully cover it. If not specified, the first sequence from the alignment will be used as target.

Output files

The homology model is written out as specified in output.model.

Description

The program first discard all residues from the template structure that are not present in the target and morphs the residue type when they are (this step is performed by Sculptor and is governed by the alignment). The missing segments are detected, a suitable loop modelling setup is devised, which is then fed into Rosetta to do the actual building. For a summary of all keywords with the corresponding defaults, see the Additional information section.

Chain-to-alignment-matching

This controls how chain-to-alignment matching is performed. For more information, see the Sculptor documentation.

Homology modelling setup

For each loop, the program backtracks min_residue_margin number of residues on both ends, then checks the distance between bridge residues. If this is shorter than residue_distance * number-of-residues-in-between, than backtracks one residues more on each side, and keeps doing this until the criterion is statisfied, or hits a terminus on either end. After this, it evaluates residue segments that are not assigned for loop modelling and will be kept fixed during the modelling step. Segments shorter than min_edge_segment_length will be discarded at the termini, and shorter than min_intenal_segment_length will be discarded internally, and the surrounding loops joined. It then checks the loop lengths, and stops with an error message if a loop is longer than max_loop_length.

Homology modelling

The program runs Rosetta with a kinematic loop closure algorithm (rosetta_loop_closure), as this does not require selected fragments. After the loop is closed, Rosetta can be instructed to run a refinement (rosetta_loop_refinement).

Command line

```
phaser.simple_homology_model \
[ command-line switches ] \
[ PHIL-format parameter files ] \
[ PHIL command-line assignments ] \
[ PDB file ] \
[ alignment file ] \
[ sequence file ]
```

Command-line switches:

```
-h, --help show this help message and exit
--show-defaults print PHIL and exit
-i, --stdin read PHIL from stdin as well
```

PHIL arguments:

Everything not starting with a dash('-') is interpreted as a PHIL argument. This can be a PHIL-format file containing parameters, command-line assignment or a file whose type is automatically recognized (based on file extension; structure files and alignment files are recognized automatically).

Warning and error messages

Error messages

- Structure file is missing: no PDB file specified.
- Chain .. is not recognised as protein: the chain specified for modelling is not protein.
- Alignment file is missing: no alignment file specified.
- Could not match chain with alignment: chain-to-alignment matching was not successful (reason indicated in previous line).
- ... loop longer than maximum allowed: loop .. is longer than the maximum allowed.
- Incomplete Rosetta environment setup: ... Rosetta environment is not setup.
- Cannot locate loopmodel: ..": the ``loopmodel Rosetta executable is not found.
- Cannot locate Rosetta database: the Rosetta database is not found.
- Rosetta finished with error code ...: Rosetta exited with an error code.
- No models produced: although Rosetta exited indicating success, no model is created.

Warnings

- No sequence file provided: no sequence file provided, the first sequence from the alignment will be used as target.
- `` Multiple models output - using only first``: Rosetta generated multiple models, and only the first one will be used.

Additional information

List of all available keywords

inputParameters controlling input
structure = None PDB file name
alignment = None Alignment file
name
sequence = None Target sequence
outputParameters controlling output
model = model.pdb Output
model file name
chain_to_alignment_matching Chain-to-alignment matching options
consecutivity = *geometry
numbering Consecutivity criterion to detect chain breaks
min_hss_length = 3 Minimum length of a sequence
fragment to be included in chain alignment
max_seed_hss_count = 12 Number of HSS to use in extensive
search
max_completion_hss_count = 6 Number of HSS to use in gap filling
min_sequence_overlap = 10
Minimum overlap between sequences to perform full alignment
min_sequence_identity = 0.80 Minimum
sequence identity of accepted chain alignment
homology_modellingParameters controlling homology
modelling
min_residue_margin = 2 Minimum number of residues to cut back on both sides of
loops
residue_distance = 2.5 Distance covered by a single residue (in Å)
min_edge_segment_length = 5

Discard edge segments if shorter (after considering gap margins)min_internal_segment_length = 2 Discard internal segments if shorter (after considering gap margins)max_loop_length = 60 Maximum loop lengthrosetta_max_build_attempts = 1000 Maximum build attempts to close looprosetta_bump_overlap_factor = 0.1 Allows some atomic overlap in initial loop closuresrosetta_loop_closure = kic *ngk Algorithm for Rosetta loop closurerosetta_loop_refinement = *no default quick fast test Algorithm for Rosetta loop refinement

- inputParameters controlling inputstructure = None PDB file namealignment = None Alignment file namesequence = None Target sequence
- structure = None PDB file name
- alignment = None Alignment file name
- sequence = None Target sequence
- outputParameters controlling outputmodel = model.pdb Output model file name
- model = model.pdb Output model file name
- chain_to_alignment_matchingChain-to-alignment matching optionsconsecutivity = *geometry numbering Consecutivity criterion to detect chain breaksmin_hss_length = 3 Minimum length of a sequence fragment to be included in chain alignmentmax_seed_hss_count = 12 Number of HSS to use in extensive searchmax_completion_hss_count = 6 Number of HSS to use in gap fillingmin_sequence_overlap = 10 Minimum overlap between sequences to perform full alignmentmin_sequence_identity = 0.80 Minimum sequence identity of accepted chain alignment
- consecutivity = *geometry numbering Consecutivity criterion to detect chain breaks
- min_hss_length = 3 Minimum length of a sequence fragment to be included in chain alignment
- max_seed_hss_count = 12 Number of HSS to use in extensive search
- max_completion_hss_count = 6 Number of HSS to use in gap filling
- min_sequence_overlap = 10 Minimum overlap between sequences to perform full alignment
- min_sequence_identity = 0.80 Minimum sequence identity of accepted chain alignment
- homology_modellingParameters controlling homology modellingmin_residue_margin = 2 Minimum number of residues to cut back on both sides of loopsresidue_distance = 2.5 Distance covered by a single residue (in A)min_edge_segment_length = 5 Discard edge segments if shorter (after considering gap margins)min_internal_segment_length = 2 Discard internal segments if shorter (after considering gap margins)max_loop_length = 60 Maximum loop lengthrosetta_max_build_attempts = 1000 Maximum build attempts to close looprosetta_bump_overlap_factor = 0.1 Allows some atomic overlap in initial loop closuresrosetta_loop_closure = kic *ngk Algorithm for Rosetta loop closurerosetta_loop_refinement = *no default quick fast test Algorithm for Rosetta loop refinement
- min_residue_margin = 2 Minimum number of residues to cut back on both sides of loops
- residue_distance = 2.5 Distance covered by a single residue (in A)
- min_edge_segment_length = 5 Discard edge segments if shorter (after considering gap margins)
- min_internal_segment_length = 2 Discard internal segments if shorter (after considering gap margins)
- max_loop_length = 60 Maximum loop length
- rosetta_max_build_attempts = 1000 Maximum build attempts to close loop
- rosetta_bump_overlap_factor = 0.1 Allows some atomic overlap in initial loop closures
- rosetta_loop_closure = kic *ngk Algorithm for Rosetta loop closure

- rosetta_loop_refinement = *no default quick fast test Algorithm for Rosetta loop refinement

Finding NCS in chains from a PDB file with `simple_ncs_from_pdb`

`simple_ncs_from_pdb.html`

Contents

- Author(s)
- Purpose
- Usage for searching How `simple_ncs_from_pdb` works: Additional notes on how `simple_ncs_from_pdb` works: Examples Other outputs Possible Problems Specific limitations and problems:
- How `simple_ncs_from_pdb` works:
- Additional notes on how `simple_ncs_from_pdb` works:
- Examples
- Other outputs
- Possible Problems
- Specific limitations and problems:
- Usage for validation Why we filter annotations How we filter annotations What you may see in the log file (example) General recommendations when you encounter this situation in refinement
- Why we filter annotations
- How we filter annotations
- What you may see in the log file (example)
- General recommendations when you encounter this situation in refinement
- List of all available keywords

Author(s)

- `simple_ncs_from_pdb` : Tom Terwilliger, Youval Dar, Oleg Sobolev
- Phil command interpreter: Ralf W. Grosse-Kunstleve
- `find_domain`: Peter Zwart

Purpose

- The `simple_ncs_from_pdb` method identifies NCS in the chains in a PDB file and writes out the NCS operators in forms suitable for `phenix.refine`, `resolve`, and the AutoSol and AutoBuild Wizards.
- Validate NCS annotations against a model

Usage for searching

How `simple_ncs_from_pdb` works:

The basic steps that the `simple_ncs_from_pdb` carries out are:

- Remove part of the structure specified in `exclude_selection` parameter.

- Identify sets of matching segments in chains in the PDB file by sequences (use value of `chain_similarity_threshold` as cutoff). These are potential NCS-related chains.
- Determine which matching segments have overall rms distance (RMSD) within the given tolerance (`chain_max_rmsd`, typically 2 Å)
- Remove residues from matching segments if locally spatially misaligned (`residue_match_radius`)
- Remove atoms from matching residues if not in both residues (for example, if a side chain is missing in one of the matching residues)
- Check atoms order residues

Additional notes on how `simple_ncs_from_pdb` works:

The chains matching is done using dynamic programming alignment of residues and atoms. The first pass contains restriction on minimal similarity, set by `min_percent` (number of matching residues)/(number of residues in longer chain)

From the matching chains list, we remove chain pairs where the matching segments exceed the RMSD limit.

Matching segments are scanned for local residue misalignment. Residues where (`max_atom_distance` - `min_atom_distance`) > `match_radius` are excluded from matching segment. This allow local differences in matching chains.

If matching residues have different number of atoms (For example if one containing the side chain while the other not), only the matching atoms will be included.

Grouping of chains is not performed.

Alternative conformation are excluded from matching segments.

The matching is done by the residues name strings, not by the residue numbers, this allows handling of insertions in PDB file.

The result of the NCS search is combination of NCS related groups and invariant or non-NCS related regions, to the atom level. In every NCS group all copies have the same number of atoms and can be reproduced by applying the appropriate rotation and translation to the master copy.

Examples

When running

```
phenix.simple_ncs_from_pdb 2h50.pdb
```

The following files will be produced

- NCS operators written in format for `phenix.refine`
`2h50_simple_ncs_from_pdb.phil`
- NCS operators written in format for the PHENIX Wizards (if `write_spec_files=True`)

```
2h50_simple_ncs_from_pdb.ncs_spec
2h50_simple_ncs_from_pdb.resolve
```

The file that should be used for refinement is `4boz_simple_ncs_from_pdb.phil`. This file can also be modified if a particular NCS relations need to be changed. The content of that file is the exact selection sting of the atoms in the NCS groups

The content of `2h50_simple_ncs_from_pdb.ncs` is

```

ncs_group {
  reference = chain 'A'
  selection = chain 'C'
  selection = chain 'E'
  selection = chain 'G'
  selection = chain 'I'
  selection = chain 'K'
  selection = chain 'M'
  selection = chain 'O'
  selection = chain 'Q'
  selection = chain 'S'
  selection = chain 'U'
  selection = chain 'W'
}

```

Other outputs

To get output in other format:

```
phenix.simple_ncs_from_pdb 4boz.pdb write_spec_files=True
```

Simple_ncs_from_pdb will analyze the chains in 4boz.pdb and identify any NCS that exists. For this sample run the following output is produced:

```

GROUP 1
Summary of NCS group with 2 operators:
ID of chain/residue where these apply: [['A', 'D'], [[147, 150], [152, 211],
[213, 275], [280, 305], [307, 308]], [[147, 150], [152, 211], [213, 275], [280, 305], [307, 308]]]
RMSD (A) from chain A: 0.0 0.72
Number of residues matching chain A:[155, 155]

OPERATOR 1
CENTER: 24.4880 -13.3177 -20.1848

ROTA 1: 1.0000 0.0000 0.0000
ROTA 2: 0.0000 1.0000 0.0000
ROTA 3: 0.0000 0.0000 1.0000
TRANS: 0.0000 0.0000 0.0000

OPERATOR 2
CENTER: 15.9430 11.8822 0.6609

ROTA 1: 0.7964 -0.5651 0.2152
ROTA 2: -0.5503 -0.8249 -0.1295
ROTA 3: 0.2507 -0.0152 -0.9680
TRANS: 18.3631 5.3425 -23.3604

GROUP 2
Summary of NCS group with 3 operators:
ID of chain/residue where these apply: [['B', 'C', 'E'], [[1, 41], [43, 71]],
[[1, 41], [43, 71]], [[1, 41], [43, 71]]]
RMSD (A) from chain B: 0.0 0.8 0.77
Number of residues matching chain B:[70, 70, 70]

OPERATOR 1
CENTER: 47.5581 -9.5652 -26.8403

ROTA 1: 1.0000 0.0000 0.0000
ROTA 2: 0.0000 1.0000 0.0000
ROTA 3: 0.0000 0.0000 1.0000
TRANS: 0.0000 0.0000 0.0000

OPERATOR 2
CENTER: 27.7410 -11.0237 -49.7361

ROTA 1: -0.6661 0.2615 -0.6986

```

ROTA 2: 0.2866 -0.7749 -0.5634
ROTA 3: -0.6886 -0.5755 0.4412
TRANS: 34.1743 -54.0788 7.8610

OPERATOR 3
CENTER: 29.3812 -4.6379 12.2671

ROTA 1: 0.7539 -0.5901 0.2888
ROTA 2: -0.5860 -0.8027 -0.1106
ROTA 3: 0.2971 -0.0858 -0.9510
TRANS: 19.1286 5.2864 -24.3021

Another way to view the results is

phenix.simple_ncs_from_pdb 4boz.pdb show_summary=true

Chains in model:

A B C D E

NCS summary:

Number of NCS groups : 2

Group # : 1

Number of copies : 2

Chains in master : 'A'

Chains in copies : 'D'

Group # : 2

Number of copies : 3

Chains in master : 'B'

Chains in copies : 'C', 'E'

Transforms:

Group # : 1

Transform # : 1

RMSD : 0

ROTA 0 1.0000 0.0000 0.0000

ROTA 1 0.0000 1.0000 0.0000

ROTA 2 0.0000 0.0000 1.0000

TRANS 0.0000 0.0000 0.0000

Transform # : 2

RMSD : 0.720792956817

ROTA 0 0.7964 -0.5503 0.2507

ROTA 1 -0.5651 -0.8249 -0.0152

ROTA 2 0.2152 -0.1295 -0.9680

TRANS -5.8295 14.4282 -25.8708

Group # : 2

Transform # : 1

RMSD : 0

ROTA 0 1.0000 0.0000 0.0000

ROTA 1 0.0000 1.0000 0.0000

ROTA 2 0.0000 0.0000 1.0000

TRANS 0.0000 0.0000 0.0000

Transform # : 2

RMSD : 0.796197645664

ROTA 0 -0.6661 0.2866 -0.6886

ROTA 1 0.2615 -0.7749 -0.5755

ROTA 2 -0.6986 -0.5634 0.4412

TRANS 43.6766 -46.3175 -10.0616

Transform # : 3

RMSD : 0.772963874518

```

ROTA 0 0.7539 -0.5860 0.2971
ROTA 1 -0.5901 -0.8027 -0.0858
ROTA 2 0.2888 -0.1106 -0.9510
TRANS -4.1022 13.4462 -28.0502

```

There are 5 chains in the PDB file (A,B,C,D,E). In the first group the master is A the the copy D and in the second group the master is B and the copy is C and E. Chain C is not in any group.

```
RMSD (A) from chain A: 0.0 0.72
```

shows the RMSD of matching atoms between the master and every other copy, The list of numbers is the list of matching residues by residue number. Note that this is not the exact selection, as it appears in: 4boz_simple_ncs_from_pdb.ncs.

ROTA and TRANS are the rotation and translation information. In the .ncs_spec file and the default on-screen representation of the results, the rotation and translation are:

```
Master = ROT x Copy + TRANS
```

While in the CCBTX implementation the common use is:

```
Copy = ROT x Master + TRANS
```

So the rotation/Translation are the inverse of each other in the two formats. Both formats are displayed in the summary.

A portion of the contents of the 4boz_simple_ncs_from_pdb.ncs_spec file, which you can edit if you want and which you can use in the AutoBuild Wizard, are shown below. NOTE: The ncs operators describe how to map the N'th ncs-related copy on to the first copy.

```

Summary of NCS information
Thu Apr 2 15:44:03 2015
/net/cci-filer2/raid1/home/...

new_ncs_group
new_operator

rota_matrix 1.0000 0.0000 0.0000
rota_matrix 0.0000 1.0000 0.0000
rota_matrix 0.0000 0.0000 1.0000
tran_orth 0.0000 0.0000 0.0000

center_orth 24.4880 -13.3177 -20.1848
CHAIN A
RMSD 0
MATCHING 155
  RESSEQ 147:150
  RESSEQ 152:211
  RESSEQ 213:275
  RESSEQ 280:305
  RESSEQ 307:308

new_operator

rota_matrix 0.7955 -0.5660 0.2164
rota_matrix -0.5511 -0.8242 -0.1300
rota_matrix 0.2520 -0.0159 -0.9676
tran_orth 18.4021 5.3569 -23.3621

center_orth 15.9430 11.8822 0.6609
CHAIN D
RMSD 0.817
MATCHING 155
  RESSEQ 147:150
  RESSEQ 152:211

```

RESSEQ 213:275
RESSEQ 280:305
RESSEQ 307:308

Possible Problems

- Simple_ncs_from_pdb cannot find NCS groups: try relaxing search parameters, e.g. set bigger chain_max_rmsd, smaller chain_similarity_threshold.

Simple_ncs_from_pdb cannot find NCS groups: try relaxing search parameters,

e.g. set bigger chain_max_rmsd, smaller chain_similarity_threshold.

e.g. set bigger chain_max_rmsd, smaller chain_similarity_threshold.

- Simple_ncs_from_pdb produces very complicated selections, like reference = (chain 'A' and (resid 147:150 or resid 152:194 or (resid 195 and (name N or name CA or name C or name O or name CB)) or resid 196:211 or resid 213:229 or (resid 230 and (name N or name CA or name C or name O or name CB or name CG)) or resid 231:298 or (resid 299 and (name N or name CA or name C or name O or name CB or name CG)) or resid 300:305 or resid 307:308)) This indicates that procedure is trying to exclude some residues or atoms from the selections. This could be caused by alternative conformations of residues (not supported), poor alignment of particular residues (try to increase residue_match_radius) or missing atoms in one of the chains.

Simple_ncs_from_pdb produces very complicated selections, like

```
reference = (chain 'A' and (resid 147:150 or resid 152:194 or (resid 195 and
(name N or name CA or name C or name O or name CB )) or resid 196:211 or
resid 213:229 or (resid 230 and (name N or name CA or name C or name O or name
CB or name CG )) or resid 231:298 or (resid 299 and (name N or name CA or name
C or name O or name CB or name CG )) or resid 300:305 or resid 307:308))
```

This indicates that procedure is trying to exclude some residues or atoms from the selections. This could be caused by alternative conformations of residues (not supported), poor alignment of particular residues (try to increase residue_match_radius) or missing atoms in one of the chains.

Specific limitations and problems:

- a large chain_similarity_threshold can prevent matching segments in chains that are of very different size.
- The value of chain_max_rmsd, the RMSD limit between chains should be set with care. Small value might exclude chains with problems in the model, even though they are NCS related. Large value might cause non-NCS related chains and to be constraint as related.
- The .ncs_spec files do not support representation of groups with multiple chains in master and copies. a group [(A,C),(B,D)] will be represented as [(A),(B)] and [(C),(D)], duplicating the rotation and translation operations.
- The .ncs_spec files do not support representation atoms, residue name and insertion selection, it only shows the numbers of the matching residues.
- The .ncs_spec files rotation and translation operation are

Master = ROT x Copy + TRANS

and not:

Copy = ROT x Master + TRANS

They are the inverse of the rotation and translation that are used in the implementation of the NCS relation.

Usage for validation

To run `simple_ncs_from_pdb` in validation mode one should supply NCS groups in form of phil file. The purpose of this mode is to illustrate how your NCS annotations will be filtered in `phenix.refine` and `phenix.real_space_refine`.

Why we filter annotations

We filter annotations because we need to be sure that:

- number of atoms in reference and selections are the same
- residues in reference and selections match each other
- matched residues have the same atoms in all copies

Without all these NCS restraints and constraints cannot work reliably. We understand that it could be tricky to manually create such selections, especially for poor models where some residues may be not modelled yet or lacking some atoms. Therefore we filter user-supplied annotations before refinement by default and there is no way to turn it off.

How we filter annotations

For every NCS group supplied by a user we:

- Select part of the model containing reference atoms and all selection atoms.

Select part of the model containing reference atoms and all selection atoms.

- Then we search for NCS in this sub-model using parameters from `ncs_search` scope with some modifications. Precisely: `chain_max_rmsd` or 10, whichever is greater `residue_match_radius` or 1000, whichever is greater `chain_similarity_threshold` or 0.5 whichever is smaller. These modifications are used to relax criteria for validation to a large extent to accomodate poor matches in poor models.

Then we search for NCS in this sub-model using parameters from `ncs_search` scope with some modifications. Precisely:

- `chain_max_rmsd` or 10, whichever is greater
- `residue_match_radius` or 1000, whichever is greater
- `chain_similarity_threshold` or 0.5 whichever is smaller.

These modifications are used to relax criteria for validation to a large extent to accomodate poor matches in poor models.

- If found NCS relations cover all atoms selected in 1, validation passed, the NCS group is used as was defined by a user. If found NCS relations do not cover all the initial atoms selected in 1, we create new NCS selections and use them further in refinement. If no NCS relations were found, the whole NCS group is discarded.

If found NCS relations cover all atoms selected in 1, validation passed, the NCS group is used as was defined by a user. If found NCS relations do not cover all the initial atoms selected in 1, we create new NCS selections and use them further in refinement. If no NCS relations were found, the whole NCS group is discarded.

Note, we only filter supplied selections. Nothing extra will be added.

What you may see in the log file (example)

One may find something like this in the log-file:

```
Validating user-supplied NCS groups...
Validating:
ncs_group {
  reference = "(chain A and resid 11:376 and not (resid 19:31 or resid 42:67 or resid 70:123 or resid 132:145 or
  selection = "(chain C and resid 11:376 and not (resid 19:31 or resid 42:67 or resid 70:123 or resid 132:145 or
}
MODIFIED. Some of the atoms were excluded from your selection.
The most common reasons are:
1. Missing residues in one or several copies in NCS group.
2. Presence of alternative conformations (they are excluded).
3. Residue mismatch in requested copies.
Please check the validated selection further down.
```

and further down something like:

```
Found NCS groups:
ncs_group {
  reference = (chain 'A' and (resid 11 through 18 or resid 32 through 41 or resid 68 through 6 \
9 or resid 124 through 125 or resid 129 through 131 or resid 146 through 147 or \
resid 185 through 189 or resid 250 through 338 or resid 358 through 364 or resid \
375 through 376))
  selection = (chain 'C' and (resid 11 through 18 or resid 32 through 41 or resid 68 through 6 \
9 or resid 124 through 131 or resid 146 through 147 or resid 185 through 189 or \
resid 250 through 338 or resid 358 through 376))
}
```

This is how NCS selections were modified for one of the reasons listed above. In this example, they are essentially the same with slightly different syntax. Let's look at the differences.

- In chain A residues 126-128 are excluded from selection. Indeed, there were no such residues in chain C.
- In chain A residues 365-374 are excluded. Same as #1, absent in chain C.

These two are the only differences from your selection and they don't change anything! They look much different because we have to construct new selections from scratch (internal representation of selections) automatically.

When one sees something like this in new selections:

```
(resid 28 and (name N or name CA or name C or name O or name CB or name CG or name CD ))
```

this is usually because another chain is lacking particular atom(s). Therefore we select particular atoms of the residue that present in all other copies. In this particular case, residue 28 in chain A has NE and CZ atoms, while chain B lacks them.

Other messages one can see is:

```
REJECTED because copies don't match good enough.
Try to revise selections or adjust chain_similarity_threshold or
chain_max_rmsd parameters.

OK. All atoms were included in
validated selection.
```

General recommendations when you encounter this situation in refinement

- If resulting selections (Found NCS groups) looks good to you - you should be fine, NCS is being applied and you can see exactly what selections are being used.
- Check the filtered selections. Maybe you will want to modify the model (put on missing atoms or remove extras).

- If you want your selections to be preserved - make them such that they satisfy all requirements in Why we filter annotations.

List of all available keywords

pdb_in = None Input PDB file to be used to identify ncstemp_dir = "" temporary directory (ncs_domain_pdb will be written there)ncs_domain_pdb_stem = None NCS domains will be written to ncs_domain_pdb_stem+"group_"+nnwrite_ncs_domain_pdb = False You can write out PDB files representing NCS domains for density modification if you wantwrite_ncs_phil = False Write out phil parameters for the NCS groups to file in working directoryshow_summary = False Show NCS information summarywrite_spec_files = False Write .ncs_spec and .resolve representation of NCS groupsncs_searchSet of parameters for NCS search procedure. Some of them also used for filtering user-supplied ncs_group.enabled = False Enable NCS restraints or constraints in refinement (in some cases may be switched on inside refinement program).exclude_selection = "element H or element D or water" Atoms selected by this selection will be excluded from the model before any NCS search and/or filtering procedures. There is no way atoms defined by this selection will be in NCS.chain_similarity_threshold = 0.85 Threshold for sequence similarity between matching chains. A smaller value may cause more chains to be grouped together and can lower the number of common residueschain_max_rmsd = 2. limit of rms difference between chains to be considered as copiesresidue_match_radius = 4.0 Maximum allowed distance difference between pairs of matching atoms of two residuesry_shortcuts = False Try very quick check to speed up the search when chains are identical. If failed, regular search will be performed automatically.minimum_number_of_atoms_in_copy = 3 Do not create ncs groups where master and copies would contain less than specified amount of atomsvalidate_user_supplied_groups = True Enable validation of user-supplied ncs_group. Need to exercise a lot of caution turning this off. This option is for developers only.ncs_groupThe definition of one NCS group. Note, that almost always in refinement programs they will be checked and filtered if needed.reference = None Residue selection string for the complete master NCS copyselection = None Residue selection string for each NCS copy location in ASU

- pdb_in = None Input PDB file to be used to identify ncs
- temp_dir = "" temporary directory (ncs_domain_pdb will be written there)
- ncs_domain_pdb_stem = None NCS domains will be written to ncs_domain_pdb_stem+"group_"+nn
- write_ncs_domain_pdb = False You can write out PDB files representing NCS domains for density modification if you want
- write_ncs_phil = False Write out phil parameters for the NCS groups to file in working directory
- show_summary = False Show NCS information summary
- write_spec_files = False Write .ncs_spec and .resolve representation of NCS groups
- ncs_searchSet of parameters for NCS search procedure. Some of them also used for filtering user-supplied ncs_group.enabled = False Enable NCS restraints or constraints in refinement (in some cases may be switched on inside refinement program).exclude_selection = "element H or element D or water" Atoms selected by this selection will be excluded from the model before any NCS search and/or filtering procedures. There is no way atoms defined by this selection will be in NCS.chain_similarity_threshold = 0.85 Threshold for sequence similarity between matching chains. A smaller value may cause more chains to be grouped together and can lower the number of common residueschain_max_rmsd = 2. limit of rms difference between chains to be considered as copiesresidue_match_radius = 4.0 Maximum allowed distance difference between pairs of matching atoms of two residuesry_shortcuts = False Try very quick check to speed up the search when chains are identical. If failed, regular search will be performed automatically.minimum_number_of_atoms_in_copy =

3 Do not create ncs groups where master and copies would contain less than specified amount of atoms
validate_user_supplied_groups = True Enable validation of user-supplied ncs_group. Need to exercise a lot of caution turning this off. This option is for developers only.

- enabled = False Enable NCS restraints or constraints in refinement (in some cases may be switched on inside refinement program).
- exclude_selection = "element H or element D or water" Atoms selected by this selection will be excluded from the model before any NCS search and/or filtering procedures. There is no way atoms defined by this selection will be in NCS.
- chain_similarity_threshold = 0.85 Threshold for sequence similarity between matching chains. A smaller value may cause more chains to be grouped together and can lower the number of common residues
- chain_max_rmsd = 2. limit of rms difference between chains to be considered as copies
- residue_match_radius = 4.0 Maximum allowed distance difference between pairs of matching atoms of two residues
- try_shortcuts = False Try very quick check to speed up the search when chains are identical. If failed, regular search will be performed automatically.
- minimum_number_of_atoms_in_copy = 3 Do not create ncs groups where master and copies would contain less than specified amount of atoms
- validate_user_supplied_groups = True Enable validation of user-supplied ncs_group. Need to exercise a lot of caution turning this off. This option is for developers only.
- ncs_group The definition of one NCS group. Note, that almost always in refinement programs they will be checked and filtered if needed.
reference = None Residue selection string for the complete master
NCS copy
selection = None Residue selection string for each NCS copy location in ASU
- reference = None Residue selection string for the complete master NCS copy
- selection = None Residue selection string for each NCS copy location in ASU

Sorting heteroatoms

sort_hetatms.html

The program `phenix.sort_hetatms` is a utility designed to re-group the non-polymeric molecules (heteroatoms) in a model in a roughly similar manner to the format used by the Protein Data Bank. This consists of matching all heteroatoms to the nearest polymer chain, resetting the chain IDs and residue numbers, converting ATOM labels to HETATM, and optionally sorting waters by B-factor. It is primarily intended for internal use, and is also run within `phenix.ligand_pipeline`. The only input required is a PDB file, since the program does not deal with molecular geometry or experimental data. Any REMARK records will be removed, but this can be disabled by setting `preserve_remarks=True`.

Note that the PDB will, inevitably, further modify the contents of the model according to its own rules. However, the program output is significantly closer to the PDB conventions than the output of `phenix.refine` or similar programs.

List of all available parameters

`file_name = None` Input file
`output_file = None` Output file
`unit_cell = None` space_group = None ignore_symmetry = False Don't take symmetry-related chains into account when determining the nearest macromolecule.
`preserve_remarks = False` Propagate all REMARK records to the output file.
`verbose = False`
`remove_hetatm_ter_records = True` The official PDB format only allows TER records at the end of polymer chains, whereas the CCTBX PDB-handling tools will insert TER after each chain of any type. If this parameter is True, the extra TER records will be removed.
`preserve_chain_id = False` The default behavior is to group heteroatoms with the nearest macromolecule chain, whose ID is inherited. This parameter disables the change of chain ID, and preserves the original chain ID.
`waters_only = False` Rearrange waters, but leave all other ligands alone.
`sort_waters_by = None` *b_iso Ordering of waters - by default it will sort them by the isotropic B-factor.
`set_hetatm_record = True` Convert ATOM to HETATM where appropriate.
`ignore_selection = None` Selection of atoms to skip. Any residue group which overlaps with this selection will be preserved with the original chain ID and numbering.
`renumber = True` Renumber heteroatoms once they are in new chains.
`sequential_numbering = True` If True, the heteroatoms will be renumbered starting from the next available residue number after the end of the associated macromolecule chain. Otherwise, numbering will start from 1.
`distance_cutoff = 6.0` Cutoff for identifying nearby macromolecule chains. This should be kept relatively small for speed reasons, but it may miss waters that are far out in solvent channels.
`remove_waters_outside_radius = False` Remove waters more than the specified distance cutoff to the nearest polymer chain (to avoid PDB complaints).
`loose_chain_id = X` Chain ID assigned to heteroatoms that can't be mapped to a nearby macromolecule chain.
`model_skip_expand_with_mtrix = False` If set to false then expand the model into NCS copies using matrix (MTRIX) records present in PDB file.
`job_title = None` Job title in PHENIX GUI, not used on command line

- `file_name = None` Input file
- `output_file = None`
- `unit_cell = None`
- `space_group = None`
- `ignore_symmetry = False` Don't take symmetry-related chains into account when determining the nearest macromolecule.
- `preserve_remarks = False` Propagate all REMARK records to the output file.

- `verbose = False`
- `remove_hetatm_ter_records = True` The official PDB format only allows TER records at the end of polymer chains, whereas the CCTBX PDB-handling tools will insert TER after each chain of any type. If this parameter is True, the extra TER records will be removed.
- `preserve_chain_id = False` The default behavior is to group heteroatoms with the nearest macromolecule chain, whose ID is inherited. This parameter disables the change of chain ID, and preserves the original chain ID.
- `waters_only = False` Rearrange waters, but leave all other ligands alone.
- `sort_waters_by = None` *`b_iso` Ordering of waters - by default it will sort them by the isotropic B-factor.
- `set_hetatm_record = True` Convert ATOM to HETATM where appropriate.
- `ignore_selection = None` Selection of atoms to skip. Any residue group which overlaps with this selection will be preserved with the original chain ID and numbering.
- `renumber = True` Renumber heteroatoms once they are in new chains.
- `sequential_numbering = True` If True, the heteroatoms will be renumbered starting from the next available residue number after the end of the associated macromolecule chain. Otherwise, numbering will start from 1.
- `distance_cutoff = 6.0` Cutoff for identifying nearby macromolecule chains. This should be kept relatively small for speed reasons, but it may miss waters that are far out in solvent channels.
- `remove_waters_outside_radius = False` Remove waters more than the specified distance cutoff to the nearest polymer chain (to avoid PDB complaints).
- `loose_chain_id = X` Chain ID assigned to heteroatoms that can't be mapped to a nearby macromolecule chain.
- `model_skip_expand_with_mtrix = False` If set to false then expand the model into NCS copies using matrix (MTRIX) records present in PDB file
- `job_title = None` Job title in PHENIX GUI, not used on command line

phenix.structural_domain_search

structural_domain_search.html

Contents

- Algorithm
- Usage Important parameters Output
- Important parameters
- Output
- References
- Command-line options

Finds domains by calculating low cuts in the interaction graph, and evaluates whether the cut would correspond to a viable domain decomposition.

Algorithm

Assuming that interaction within a domain are much stronger than between domains, a domain can be considered a low cut within a suitable constructed interaction graph.

The structure is divided up into chunks, and an interaction network is calculated. This is then turned into an interaction graph, and low cuts are enumerated. Cuts are then enumerated whether they would correspond to viable domains, based on size, compactness and interface strength, and viable domains are stored. The procedure is then repeated recursively until no viable cuts can be found.

Usage

The only required input to the program is a structure file (PDB/cif), which will then be analysed. The domain finding algorithm is then repeated for each protein chains, and results will be written to the logfile, and also into PHIL-syntax .eff-files.

The simplest command line:

```
phenix.structural_domain_search example.pdb
```

Runtime depends on the structure size, typically 1 min for structures between 200-300 residues, and 10 min for structures up to 1000 residues.

Important parameters

- `max_iterations`: controls how many low cuts to enumerate. Increasing this number will make the search more exhaustive, but also slower.
- `sufficient_count`: terminate the search early if the requested number of cuts have been found. It is probably a good idea to search for more than one, since the lowest cut does not always correspond to the highest domain quality score, but the default value (30) is perhaps slightly conservative, and good results can be achieved with `sufficient_count` = 10. On the other hand, the underlying clustering algorithm gets very slow when more than 100 values need to be clustered, and hence this is a practical upper limit.
- `min_size`: minimum size for an acceptable domain. Domains smaller than this will not be accepted. Default is 30.

- `prune_fraction`: controls whether to simplify the interaction graph by joining low-connected vertices. This can make the correct domain decomposition to appear earlier and hence make the search more efficient, at a slight loss of cut precision. A typical `prune_fraction` is 0.05-0.15.
- `algorithm`: the quick algorithm typically gives correct results, but for some tricky (especially multiple domain) structures the thorough algorithm may find a better decomposition. Running times up to 10 times of the quick algorithm are not uncommon.

Output

The program writes out the best domain decomposition for each chain, as a series of .eff-files containing the residue segments.

All considered decompositions are recorded in the logfile.

References

Efficient Algorithms for the Problems of Enumerating Cut by Non-decreasing Weights. L.-P. Yeh, B.-F. Wang, and H.-H. Su. *Algorithmica* 5626, 297-312 (2010).

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- An Experimental Comparison of Min-Cut/Max-Flow Algorithms for Energy Minimization in Vision. Y. Boykov, and V. Kolmogorov. *IEEE Transactions on Pattern Analysis and Machine Intelligence* 26, 1124-1137 (2004).

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Command-line options

inputInput filesstructure = None Structure fileoutputOutput filesroot = "domain" Root for domain
 fileswrite_pdb_file = False Write PDB file for each domainconfigurationCalculation
 parametersinteraction_strengthParameters to calculate interaction strength between two
 residuessignificance_threshold = 0.01 Discard interaction if below this levelprune_fraction = 0.00 Discard
 lowest connected segmentslong_range_contactParameters for defining a long-range
 contactmax_spatial_distance = 4.0 Maximum distance for atom-atom contactshort_range_contactParameters
 controlling the strength of short range contactsdefault = 10.0 Strength of a coil-to-coil and mixed type
 connectionshelix_to_helix_connection = 50.0 Strength of a helix-to-helix
 connectionstrand_to_strand_connection = 50.0 Strength of a strand-to-strand
 connectionassembly_membershipParameters rewarding membership in a higher order assemblysheet_bonus
 = 20.0 Extra strength if two strands are within the same sheetchunkingParameters controlling chunking of
 segmentscoil_max_size = 15 Minimum chunksize for a coilhelix_max_size = None Minimum chunksize for a
 helixstrand_max_size = None Minimum chunksize for a strandevaluationParameters controlling domain

acceptance_min_size = 30 Minimum domain size is residues_min_compactness = 0.6 Minimum compactness (internal interaction strength to atoms ratio)
 max_interface_strength = 0.4 Maximum interface strength to internal interaction strength ratio
 searchParameters controlling domain search
 algorithm = *quick thorough Algorithm to use
 max_macrocycles = 5 Maximum number of macrocycles
 max_iterations = 500 Maximum cut iterations to perform
 sufficient_count = 31 Quit search if sufficient number of domains has been located
 cluster_radius = 20 Radius for identifying quasi-equivalent domains
 max_residue_difference_for_equality = 10 Maximum residue differences between two domains
 max_residue_overlap_for_compatibility = 5 Max number of common residues between two domains
 strong_connection = 100 Do not separate multisegment domains that are at least as strongly connected

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Comparing related structures in the Phenix GUI

structure_comparison.html

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- Authors
- Introduction
- Limitations
- Video Tutorial
- Configuration and required inputs
- Handling heterogeneous sequences
- Results
- Interacting with Coot
- Reference
- Citations

Authors

Programming: Nigel W. Moriarty, Dorothee Liebschner, Nat Echols, Billy K. Poon, Pavel Afonine, Jeff Headd (LBL) Concept: Herb Klei (BMS)

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Introduction

The structure comparison GUI is a tool for parallel validation and analysis of near-identical protein structures, such as different crystal forms, various mutants, or NCS-related copies. It displays validation outliers and regions of difference for a set of individual chains on a spreadsheet-like grid, which is linked to the graphics windows (Coot and PyMOL). If desired, both the extracted chains and electron density maps can be superimposed into a common frame of reference. These chains may subsequently be edited in Coot to ensure consistency and/or fix errors, and recovered in their original orientations for further refinement and rebuilding.

We have included a set of representative structures of human Protein Kinase A (PKA) for comparison in the Phenix distribution. To automatically set up a new project with these data, click "New Project" in the main GUI toolbar and then "Set up tutorial data..." in the dialog which appears, and select "Protein Kinase A (Structure Comparison)" from the drop-down menu.

Limitations

Only protein chains are supported at this time. RNA/DNA are not supported. Only X-ray diffraction data are supported. It is planned to extend usage for cryo-EM and neutron data in the future. Only a single set of chains can be compared at a time; if you have a set of heterogeneous complexes, the components need to be dealt with separately. This tool is of limited use for comparing proteins which diverge significantly in sequence; the program will quit with a warning if the fractional identity drops below 0.8 (this can be adjusted if desired).

See below for additional details.

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- This tool is of limited use for comparing proteins which diverge significantly in sequence; the program will quit with a warning if the fractional identity drops below 0.8 (this can be adjusted if desired). See below for additional details.

Video Tutorial

The tutorial video is available on the Phenix YouTube channel and covers the following topics:

- Basic overview
- How to run Structure Comparison
- Example of a Structure Comparison run using the PKA tutorial data

Configuration and required inputs

At a minimum, two or more PDB files containing near-identical chains are needed. A sequence file containing a single amino-acid sequence can be used to define the chain to be compared. It is recommended to also provide either X-ray data or pre-calculated map coefficients (the latter are preferred since these calculations are the slowest part of the program). The files generated by phenix.refine or phenix.maps are suitable for the latter. Each PDB file may contain an unlimited number of matching chains; in the PKA example, the structure 1syk has two copies of the catalytic subunit, while the rest contain only one. Non-matching chains will be ignored.

There are several options to open files or file pairs:

- Files may be dragged into the "Input files" window in the center of the Structure Comparison GUI.

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- If you have a directory containing multiple allowed file types, you may simply drag the directory in, and it will automatically guess the appropriate use for each file. If the PDB and reflection files use similar root filenames, or occur in pairs in the same directories, the association between them will be guessed automatically. For instance, the following pair of files will automatically be recognized as model and experimental data: 1syk.pdb 1syk.mtz while these will be recognized as model and map coefficients: 1syk.pdb 1syk_map_coeffs.mtz

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The map calculations have been parallelized, and this part of the program scales extremely well over multiple processors. The GUI will automatically be set to use `ncpu=1` processes, but you may increase this if desired.

The most important configurable option is the choice of superposing the models and maps prior for viewing together in Coot or PyMOL. This is very advantageous for comparing regions of difference, instead of switching between views for each chain. However, it has the drawback of removing the chains being analyzed from their original context, possibly obscuring other features that account for the observed structures. If you choose to turn on the superpositioning, the extracted chains will be written out as separate files in the results directory (along with reoriented maps for each chain, if available). Otherwise, the original models will be used for viewing. (Note that the analyses which depend on local environment in the crystal will still be run on the original models, and the coordinates mapped to their new positions.)

In addition to the models actually being compared, you may also provide a "reference model" on which all other chains will be superposed. If a reference model is not explicitly provided, the first chain found in the list of input files will be used instead. However, it may be advantageous to explicitly define this model if you are concerned about the quality or consistency of the structural alignments.

Handling heterogeneous sequences

Homologous structures and insertion codes are supported. To ensure that the residue numbering is interpreted correctly, you will need to uncheck the box labeled "Assume identical chain numbering". The sequence provided as input will be used to pick out similar chains, which will then be aligned to the reference model. The numbering displayed in the output grids will reflect the residue IDs in the reference model; for residues present in other chains but not the reference model, the residue ID will be displayed as "--", but the residue name(s) will still be shown. The primary limitation at this point is the lack of any direct link back to the original residue IDs, but the models displayed in Coot and PyMOL preserve these IDs.

Results

All output files will be written to a directory named `StructureComparison_X` (where X is the job ID). If structure superpositioning is enabled, there will be a separate PDB file for each chain, plus CCP4 map files if map coefficients were available. If the map coefficients were calculated from input X-ray data (not pre-calculated) these will also be written out.

When the application is finished running, a summary tab will show a list of all chains used for the analysis. The summary tab also lists basic statistics (such as number of atoms and residues, mean isotropic B-factor, ...). Additional tabs are added for each analysis. Each tab contains a grid with chains organized one per row, and residues of interest in columns. Color coding indicates the relationship of the specific residue in each cell to the equivalent residues, and/or properties such as validation outlier or multiple conformations. Cells colored in

grey indicated residues missing from the input chain.

The criteria used to display results vary among result tabs, but as the primary purpose of the program is to highlight regions of difference, homogeneous regions will usually be omitted from the grid.

The following analyses are carried out. It should be noted that the result tabs only appear if differences were found.

Ligand Summary: If ligands are present in at least one of the chains, they are listed in the results table, along with basic properties such as residue name, number of atoms and mean isotropic B-factor. Note that columns are sortable, f.ex. the table can be sorted according to the number of atoms. **Rotamers:** Any set of matching residues containing more than one rotamer type will be included, plus all outliers. **Coloring:** red for outliers, orange for "minority" conformations, green for "majority" conformations (at least 50% of chains under consideration), blue for multiple conformations, grey for missing side-chains or residues. The rotamer IDs are taken from the Ultimate Rotamer Library, which is described in Hintze et al. (2016). In the bottom part of the window is a summary table which can serve as a legend for the cell colors. **Ramachandran angles:** Any set of matching residues with different phi,psi angles plus any outliers will be displayed. Additionally, residues which fall into more than one distinct "favored" region will be included (this is somewhat redundant with the secondary structure comparison). **Coloring:** red for outliers, orange for "allowed" angles, green for "favored", blue for multiple conformations, grey for missing residues. **Missing atoms:** Any set of matching residues with at least one atom missing will be displayed. **Coloring:** red for residues missing backbone atoms (C, O, N, CA, or CB), orange for residues missing sidechain atoms, blue for entire missing residues, green for complete residues. The cell width can be expanded to read the list of atom names in the cell. **Secondary structure:** Any set of matching residues with more than one type of secondary structure assignment will be included. **Coloring:** white for unstructured/loop, green for alpha helices, orange for beta-sheets, blue for 3₁₀ helices, grey for missing residues. **Water locations:** Any water molecules not conserved in all chains are listed in the results table. Likewise, ions are included in the analysis. It is of particular interest to investigate areas with waters in some chains and ions at equivalent positions in other chains. Mixed ion/water sites are therefore listed at the leftmost part of the results table. **Coloring:** orange for ions, green for water molecules, white for sites without corresponding water molecule. **Ligand Information:** If ligands are present, their centers of mass are compared. The table is useful to assess if ligands are at equivalent positions in all chains. **Coloring:** blue for ligands, white for sites without ligand at the equivalent position. **Histidine Protonation:** Histidine side chains can have three different protonation states. If H atoms are present in at least one input chain, the results tab lists all histidine residues with different protonation states. **Omega Angles:** Omega angles are categorized as trans, cis or twisted. Any matching residues with different assignments are displayed in the results table. **B-factors:** Unlike the other analyses, this result does not identify heterogeneity on a per-residue basis, but instead plots the mean B-factor for each residue by chain, with the option of viewing only the main-chain or sidechain average. In addition to the absolute mean values, it can also display the values normalized to the overall mean for each chain, which may be useful when the structures being compared differ significantly in the scale of their B-factors. Points denoting rotamer and Ramachandran outliers will also be displayed. (Note that this plot does not interact directly with Coot or PyMOL.)

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Interacting with Coot

To show controls for sending data to and from Coot, click the icon on the toolbar of the results window. A small panel will appear:

Click the button labeled "Load models and maps" to read all structures into Coot simultaneously. Once this is complete they will appear in a checklist in Phenix; unchecking any structure will hide the associated objects in Coot. As with other validation programs in the Phenix GUI, clicking on the field for any residue in the results grid will recenter Coot on that residue (in the appropriate frame of reference depending on whether models were superposed). You may now adjust the models to match a consensus if desired.

If the structures were superposed, you should not try to use the models in Coot for further refinement. Instead, click the button labeled "Fetch modified structures" to write out the models from Coot to the results directory. Phenix will automatically recover the original orientation and replace the unmodified chain in the input structure with the model from Coot, while leaving all other atoms in place. The complete files will be written out to the result directory, renamed with the extension "_modified.pdb".

Reference

Coming soon.

Citations

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structure_search

structure_search.html

Overview

Structure_search is a tool to quickly identify structural and/or sequence homologs of the input PDB file from the Protein Data Bank. It uses the SARST algorithm, and it's very fast. A typical search time against the whole PDB is usually less than one second. There is an option to allow users to obtain a list of ligands found in PDB structures of those homologs.

Usage

Identify and superpose homologous pdbs of mypdb.pdb
phenix.structure_search mypdb.pdb
obtain a list of homologs of mypdb.pdb and all ligands found in structures of those homologs
phenix.structure_search mypdb.pdb get_ligand=True
Use a local PDB mirror and obtain superposed homologs of mypdb.pdb
phenix.structure_search mypdb.pdb PDB_MIRRORDIR=/path/to/pdb_mirror/top-level

- Identify and superpose homologous pdbs of mypdb.pdb
phenix.structure_search mypdb.pdb

Identify and superpose homologous pdbs of mypdb.pdb

phenix.structure_search mypdb.pdb

- obtain a list of homologs of mypdb.pdb and all ligands found in structures of those homologs
phenix.structure_search mypdb.pdb get_ligand=True

obtain a list of homologs of mypdb.pdb and all ligands found in structures of those homologs

phenix.structure_search mypdb.pdb get_ligand=True

- Use a local PDB mirror and obtain superposed homologs of mypdb.pdb
phenix.structure_search mypdb.pdb PDB_MIRRORDIR=/path/to/pdb_mirror/top-level

Use a local PDB mirror and obtain superposed homologs of mypdb.pdb

phenix.structure_search mypdb.pdb PDB_MIRRORDIR=/path/to/pdb_mirror/top-level

More information can be found in input/Output files sections below:

Input files

required input:

pdb_file: the file containing the protein model of interest. By default, structure_search will search structure and sequence homologs of the input PDB file.

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pdb_file: the file containing the protein model of interest.

By default, structure_search will search structure and sequence homologs of the input PDB file.

Optional inputs :

get_ligand:"=True" if want a list of ligands found in homologous PDBs, Default = False. job_title: current job title output_prefix: prefix for output files if needed. get_pdb:Collect and superpose the top N homologous pdbs

(default=10). `coot_display`: Display superposed pdb files in coot. [default=(False/True) as E-value(>/<)1E-18].
`sequence_only`:"=False" Perform only sequence search against PDB database using Phenix' internal DB.
`structure_only`: Perform only structure search against PDB database using the SARST method.
`PDB_MIRRORDIR`: Set option to use local PDB mirror instead of using RCSB server. See 'Using Local PDB mirror' section.

- `get_ligand`:"=True" if want a list of ligands found in homologous PDBs, Default = False.
- `job_title`: current job title
- `output_prefix`: prefix for output files if needed.
- `get_pdb`:Collect and superpose the top N homologous pdbs (default=10).
- `coot_display`: Display superposed pdb files in coot. [default=(False/True) as E-value(>/<)1E-18].
- `sequence_only`:"=False" Perform only sequence search against PDB database using Phenix' internal DB.
- `structure_only`: Perform only structure search against PDB database using the SARST method.
- `PDB_MIRRORDIR`: Set option to use local PDB mirror instead of using RCSB server. See 'Using Local PDB mirror' section.

Output files

In addition to screen output, these files contains results of `structure_search`:

`output_(sequence/structure).txt`: files containing sequence/structure homologs of 'pdb_file' sorted by scores.
`MyBlast_(sequence/structure).log`: Standard BLAST output with selected pairwise alignments. NOTE: for structure alignment, the 'sequences' are structure-based Ramachandran codes (see reference), not 1-letter code for amino acids.
`pdb_ligand.txt` (if `get_ligand=True`): file containing all ligands found in all homologs from this search.
superposed PDB files: Can be found in `TEMPPDB_##` subdirectory as prompted in the program output.

- `output_(sequence/structure).txt`: files containing sequence/structure homologs of 'pdb_file' sorted by scores.
- `MyBlast_(sequence/structure).log`: Standard BLAST output with selected pairwise alignments. NOTE: for structure alignment, the 'sequences' are structure-based Ramachandran codes (see reference), not 1-letter code for amino acids.
- `pdb_ligand.txt` (if `get_ligand=True`): file containing all ligands found in all homologs from this search.
- superposed PDB files: Can be found in `TEMPPDB_##` subdirectory as prompted in the program output.

Using Local PDB mirror

By default the program retrieves homologous PDB mmCIF files from RCSB server for downstream processes. Users may choose to use their local PDB mirror if environmental variables "`PDB_MIRRORDIR`" has already been defined in the running shell of `phenix.structure_search`. Alternatively, users may define it in the command-line script or specify the path in the GUI. See more details below.

`PDB_MIRRORDIR`: Defines the top level of the local PDB mirror. The program will try to retrieve PDB mmCIF files from local mirror unless the path does not exist. Note this assumes the directory tree under it follows that in the RCSB server and will try to access `$PDB_MIRRORDIR/data/structures/divided/mmCIF`. The program will fall back to using RCSB server should the path contain errors.

- **PDB_MIRRORDIR**: Defines the top level of the local PDB mirror. The program will try to retrieve PDB mmCIF files from local mirror unless the path does not exist. Note this assumes the directory tree under it follows that in the RCSB server and will try to access \$PDB_MIRRORDIR/data/structures/divided/mmCIF. The program will fall back to using RCSB server should the path contain errors.

References

Lo WC, Huang PJ, Chang CH, Lyu PC. BMC Bioinformatics. 2007, 8:307

List of all available keywords

structure_searchpdb_file = None Enter a PDB file name sequence = None Optional Fasta sequence file. Only needed for a quick sequence search against RCSB without a PDB. output_prefix = 'output' Provide an output prefix if needed blastpath = None Enter path to blastall executable sequence_only = False Do a Blast search against PDBaa sequence instead of doing a Ramachandran-based structure search structure_only = False Do only a Ramachandran-based structure search. db_used = 'rcsb' structure database used in search. rcsb, scop95, or AF2db = 'rcsb' Database used in search. rcsb, scop95, or AF2.get_ligand = False Use get_ligand=True to retrieve ligands.get_ramcode_only = False Generate Rama code for input pdb/cif only. This is for developers only.get_xml_only = False Get BLAST XML output returned as a string object. No coordinate superposition will be performed. Developers only.use_pdb100aa = False Use PDB100 sequence database for sequence search.use_custom_db = False Use custom database specified by custom_db_files/custom_db_dir.custom_db_dir = None The directory of pdb/cif files to make custom database. Default is current directorycustom_db_files = None Filenames of the pdb/cif files separated by spaces for database. If none specified, all pdb/cif in the custom_db_dir will be collectedatom_selection = 'all' Choose part of the pdb used in the search (default=all). for example: chain B, resseq 113:219, ... etc. get_pdb = 10 get_pdb=N will collect and superpose the top N homologous pdbs. Use get_pdb=0 to disable this option.deposited_before = 0 Specify the latest year of matching structures to be considered for scoring. Pdb's deposited after this year will be discarded.deposited_after = 0 Specify the earliest year of matching structures to be considered for scoring. Pdb's deposited before this year will be discarded.batch_size = 0 Process the pdbs in batch of <batch_size> until <min_match> hits are identified or until all <get_pdb> pdbs are processedmin_match = 0 Finish structure_search when <min_match> matches are found. Usually uses with <trim_ends> to exit the search once find suitable pdbs.keep_all_pdb = False Keep all the PDB files, including full PDB, PDB_Chain and superposed PDB_Chain. Default is False which will keep only superposed PDB_Chain files in the directory specified in the output message.trim_ends = False Remove terminal residues of hit pdbs extending beyond those of the target pdb.write_pdb = True Set to False if no output pdb file is needed. Sometimes useful if use Structure_Search within another program and only want to pass pdb objects.write_results = True Set to False if no output results/log files is needed. Useful when calling Structure_Search within another program and only want to pass pdb objects.trim_hit_pdb = False Remove extra domains, extended loops, and unfit portions of hit pdbs after superposed to the target pdb.pickle_hits = False Pickle blast hit results from xml output.coot_display = False (default) Display output pdb files in coot.ask_coot = True prompt for coot display optionsPDB_MIRRORDIR = None Enter the top directory of local RCSB PDB mirror. The program will try to retrieve PDBs and/or structure factors from this mirror first. Note this assumes the directory trees under it follows those in RCSB -- pdb files as 'pdb####.ent.gz' in PDB_MIRRORDIR/data/structures/divided/pdb directory. If you use PDB's rsync script, this variable would be the same as the \$MIRRORDIR set in the scriptPDB_MIRROR_MMCIFF = None Enter the parent directory of the mmCIF files in the local PDB mirror. MMCIFFs will be retrieved from subdirectory ## where ## are the second and third letters in the PDB id. This keyword should be \$PDB_MIRRORDIR/data/structures/divided/mmCIF directory. PDB_MIRROR_PDB = None Enter the parent

directory of the PDB files in the local PDB mirror. PDBs will be retrieved from subdirectory ## where ## are the second and third letters in the ...

- `structure_searchpdb_file` = None Enter a PDB file name sequence = None Optional Fasta sequence file. Only needed for a quick sequence search against RCSB without a PDB. `output_prefix` = 'output' Provide an output prefix if needed `blastpath` = None Enter path to blastall executable `sequence_only` = False Do a Blast search against PDBaa sequence instead of doing a Ramachandran-based structure search `structure_only` = False Do only a Ramachandran-based structure search. `db_used` = 'rcsb' structure database used in search. `rcsb`, `scop95`, or `AF2db` = 'rcsb' Database used in search. `rcsb`, `scop95`, or `AF2`. `get_ligand` = False Use `get_ligand=True` to retrieve ligands. `get_ramcode_only` = False Generate Rama code for input pdb/cif only. This is for developers only. `get_xml_only` = False Get BLAST XML output returned as a string object. No coordinate superposition will be performed. Developers only. `use_pdb100aa` = False Use PDB100 sequence database for sequence search. `use_custom_db` = False Use custom database specified by `custom_db_files/custom_db_dir`. `custom_db_dir` = None The directory of pdb/cif files to make custom database. Default is current directory. `custom_db_files` = None Filenames of the pdb/cif files separated by spaces for database. If none specified, all pdb/cif in the `custom_db_dir` will be collected. `atom_selection` = 'all' Choose part of the pdb used in the search (default=all). for example: chain B, resseq 113:219, ... etc. `get_pdb` = 10 `get_pdb=N` will collect and superpose the top N homologous pdbs. Use `get_pdb=0` to disable this option. `deposited_before` = 0 Specify the latest year of matching structures to be considered for scoring. Pdb's deposited after this year will be discarded. `deposited_after` = 0 Specify the earliest year of matching structures to be considered for scoring. Pdb's deposited before this year will be discarded. `batch_size` = 0 Process the pdbs in batch of <batch_size> until <min_match> hits are identified or until all <get_pdb> pdbs are processed. `min_match` = 0 Finish structure_search when <min_match> matches are found. Usually uses with <trim_ends> to exit the search once find suitable pdbs. `keep_all_pdb` = False Keep all the PDB files, including full PDB, PDB_Chain and superposed PDB_Chain. Default is False which will keep only superposed PDB_Chain files in the directory specified in the output message. `trim_ends` = False Remove terminal residues of hit pdbs extending beyond those of the target pdb. `write_pdb` = True Set to False if no output pdb file is needed. Sometimes useful if use Structure_Search within another program and only want to pass pdb objects. `write_results` = True Set to False if no output results/log files is needed. Useful when calling Structure_Search within another program and only want to pass pdb objects. `trim_hit_pdb` = False Remove extra domains, extended loops, and unfit portions of hit pdbs after superposed to the target pdb. `pickle_hits` = False Pickle blast hit results from xml output. `coot_display` = False (default) Display output pdb files in coot. `ask_coot` = True prompt for coot display options. `PDB_MIRROR_DIR` = None Enter the top directory of local RCSB PDB mirror. The program will try to retrieve PDBs and/or structure factors from this mirror first. Note this assumes the directory trees under it follows those in RCSB -- pdb files as 'pdb####.ent.gz' in `PDB_MIRROR_DIR/data/structures/divided/pdb` directory. If you use PDB's rsync script, this variable would be the same as the `$MIRROR_DIR` set in the script. `PDB_MIRROR_MMCIF` = None Enter the parent directory of the mmCIF files in the local PDB mirror. MMCIFs will be retrieved from subdirectory ## where ## are the second and third letters in the PDB id. This keyword should be `$PDB_MIRROR_DIR/data/structures/divided/mmcif` directory. `PDB_MIRROR_PDB` = None Enter the parent directory of the PDB files in the local PDB mirror. PDBs will be retrieved from subdirectory ## where ## are the second and third letters in the ...

- `pdb_file` = None Enter a PDB file name

- `sequence` = None Optional Fasta sequence file. Only needed for a quick sequence search against RCSB without a PDB.

- `output_prefix` = 'output' Provide an output prefix if needed

- `blastpath` = None Enter path to blastall executable
- `sequence_only` = False Do a Blast search against PDBaa sequence instead of doing a Ramachandran-based structure search
- `structure_only` = False Do only a Ramachandran-based structure search.
- `db_used` = 'rcsb' structure database used in search. rcsb, scop95, or AF2
- `db` = 'rcsb' Database used in search. rcsb, scop95, or AF2.
- `get_ligand` = False Use `get_ligand=True` to retrieve ligands.
- `get_ramcode_only` = False Generate Rama code for input pdb/cif only. This is for developers only.
- `get_xml_only` = False Get BLAST XML output returned as a string object. No coordinate superposition will be performed. Developers only.
- `use_pdb100aa` = False Use PDB100 sequence database for sequence search.
- `use_custom_db` = False Use custom database specified by `custom_db_files/custom_db_dir`.
- `custom_db_dir` = None The directory of pdb/cif files to make custom database. Default is current directory
- `custom_db_files` = None Filenames of the pdb/cif files separated by spaces for database. If none specified, all pdb/cif in the `custom_db_dir` will be collected
- `atom_selection` = 'all' Choose part of the pdb used in the search (default=all). for example: chain B, resseq 113:219, ... etc.
- `get_pdb` = 10 `get_pdb=N` will collect and superpose the top N homologous pdbs. Use `get_pdb=0` to disable this option.
- `deposited_before` = 0 Specify the latest year of matching structures to be considered for scoring. Pdb's deposited after this year will be discarded.
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- `batch_size` = 0 Process the pdbs in batch of `<batch_size>` until `<min_match>` hits are identified or until all `<get_pdb>` pdbs are processed
- `min_match` = 0 Finish structure_search when `<min_match>` matches are found. Usually uses with `<trim_ends>` to exit the search once find suitable pdbs.
- `keep_all_pdb` = False Keep all the PDB files, including full PDB, PDB_Chain and superposed PDB_Chain. Default is False which will keep only superposed PDB_Chain files in the directory specified in the output message.
- `trim_ends` = False Remove terminal residues of hit pdbs extending beyond those of the target pdb.
- `write_pdb` = True Set to False if no output pdb file is needed. Sometimes useful if use Structure_Search within another program and only want to pass pdb objects.
- `write_results` = True Set to False if no output results/log files is needed. Useful when calling Structure_Search within another program and only want to pass pdb objects.
- `trim_hit_pdb` = False Remove extra domains, extended loops, and unfit portions of hit pdbs after superposed to the target pdb.
- `pickle_hits` = False Pickle blast hit results from xml output.
- `coot_display` = False (default) Display output pdb files in coot.
- `ask_coot` = True prompt for coot display options

- `PDB_MIRRORDIR = None` Enter the top directory of local RCSB PDB mirror. The program will try to retrieve PDBs and/or structure factors from this mirror first. Note this assumes the directory trees under it follows those in RCSB -- pdb files as 'pdb####.ent.gz' in `PDB_MIRRORDIR/data/structures/divided/pdb` directory. If you use PDB's rsync script, this variable would be the same as the `$MIRRORDIR` set in the script
- `PDB_MIRROR_MMCF = None` Enter the parent directory of the mmCIF files in the local PDB mirror. MMCFs will be retrieved from subdirectory `##` where `##` are the second and third letters in the PDB id. This keyword should be `$PDB_MIRRORDIR/data/structures/divided/mmCIF` directory.
- `PDB_MIRROR_PDB = None` Enter the parent directory of the PDB files in the local PDB mirror. PDBs will be retrieved from subdirectory `##` where `##` are the second and third letters in the PDB id. This keyword should be `$PDB_MIRRORDIR/data/structures/divided/pdb` directory. We recommend setting `PDB_MIRRORDIR` and it will take care of both `PDB_MIRROR_PDB` and others together. However, users may choose to specify `PDB_MIRROR_PDB` directly
- `PDB_MIRROR_STRUCTURE_FACTORS = None` Enter the parent directory of the PDB files in the local PDB mirror. structure factors s will be retrieved from subdirectory `##` where `##` are the second and third letters in the PDB id. This keyword should be the same as the `$PDB_MIRRORDIR/data/structures/divided/structure_factors` directory. We recommend setting `PDB_MIRRORDIR` and it will take care of both `PDB_MIRROR_PDB` and `PDB_MIRROR_STRUCTURE_FACTORS` together. However, users may choose to specify `PDB_MIRROR_STRUCTURE_FACTORS` directly
- `local_pdb_dir = None` Enter the path directly to your local PDB repository.
- `verbose = False` verbose output
- `debug = False` debugging output
- `job_title = None` Job title in PHENIX GUI, not used on command line
- `guiGUI-specific parameter required for output directory` `output_dir = None`
- `output_dir = None`

Welcome to AI-powered Phenix and CCTBX programming!

test.html

A new mindset for programming: Think about what you want to accomplish and ask your AI assistants to help you write the code to do it.

Why use AI: The Phenix developer AI chatbot has all the Phenix and CCTBX documentation and code in its sources. It can help you get going quickly. It can even write unit tests for you.

Be aware: AI chatbots usually do not get everything right on the first try. It is your job to tell them what is wrong (for starters, paste in a traceback!) and to make your code work just the way you want it to.

Use the right AI tool for the job: Use the Phenix developer chat to write Phenix and CCTBX code. It knows the code base and how to use the Program Template. Use Gemini or other general-purpose AI chatbots to write scripts that do not use Phenix or CCTBX tools. These chatbots are better at programming but they do not know about the Phenix code base.

Help chatbots

- Phenix developer chat. (Private, do not share link). Use this for Phenix or CCTBX programming questions or to write Python scripts in Phenix/CCTBX. Designed to allow easy creation of ProgramTemplate files.
- CCTBX developer chat. Use this for CCTBX programming questions or scripts.
- Phenix user help chat. Use this for questions on how to run Phenix or for general questions about crystallography and cryo-EM structure determination.

Also see some helpful hints on using AI chatbots.

Programming documentation

- CCTBX developer documentation Comprehensive guide to using CCTBX tools. High-level cctbx managers, maps, and low-level cctbx objects and tools. Includes tutorials, sample scripts, how to test your code, and many helpful hints on using CCTBX tools
- Phenix developer documentation How to use the Program Template, how to use the model-building tools in Phenix, handy restraint jiffies, Phenix/CCTBX high-level reference guide.
- PHENIX/CCTBX Programming API Fully indexed documentation of all CCTBX and Phenix Python classes and methods. Includes all code from CCTBX and from Phenix. You can update this on your machine with the command:

```
phenix.rebuild_phenix_api
```

- CCTBX Programming API Fully indexed documentation and code for all CCTBX Python classes and methods. You can update this on your machine with the command:

```
phenix.rebuild_cctbx_api
```

Analyzing unmerged data in Phenix

unmerged_data.html

Introduction

Most of the programs in Phenix assume that the input data, regardless of type or format, are already both scaled and merged to contain only unique reflections. The major exception is AutoSol, which can accept certain unmerged files (for a more limited set of formats, primarily Scalepack) and performs its own local scaling. In other situations, some programs will automatically merge the data themselves if necessary, discarding any information about the variance of individual observations.

A note on terminology: "unmerged" in this context means that the original individual observations are kept separate after scaling, including symmetry-related reflections as well as multiple observations of the same h,k,l . This is distinct from "anomalous" data, where the Friedel mates h,k,l and $-h,-k,-l$ are not merged, but each of these is (usually) still the average of multiple observations. Although many programs in Phenix deal with anomalous data, they are not useful for calculating merging statistics.

Several utilities are available in Phenix which use unmerged intensities to calculate various data quality-related metrics. These include Xtriage, which will display standard merging statistics; the standalone version of this routine, `phenix.merging_statistics`; a separate utility for calculating CC^* (Karplus & Diederichs 2012), `phenix.cc_star`; and the Table 1 program. Their use as it relates to unmerged data is described below. Note that due to implementation details, small numerical differences (up to a few tenths of a percent) may occur between the originally reported statistics in the processing logfiles, and the numbers reported by Phenix.

Important: we have provided a list of statistics with pointers to the relevant publications in the "Details" section, but this document does not attempt to explain the mathematics or logic behind the statistics calculated here. We strongly recommend that you read the listed references; the publication(s) associated with the data processing software used may also be useful (see for example Evans (2011)).

If you want to experiment with these programs and do not have any data of your own to use, the Phenix installation includes an unmerged data file in the `p9-sad` example:

```
$PHENIX/examples/p9-sad/p9_se_w2.sca
```

The format does not contain complete symmetry information (and the space group is specified incorrectly), but you can use the symmetry information in `p9.pdb` in the same directory.

Obtaining unmerged data

The procedure differs between various data-processing software; these are the methods we are familiar with:

xia2: the DataFiles directory will contain a file in Scalepack format, ending in `"_unmerged.sca"`. No additional steps are required. aimless (CCP4): the OUTPUT keyword has an UNMERGED option. XDS: the file named `XDS_ASCII.HKL` output by CORRECT will contain unmerged data. If you are using XSCALE to scale data from multiple crystals, add the keyword `MERGE=FALSE` to the input script. HKL2000: in the Macros tab, in the Scaling section type "no merge original index", then re-run scaling (with a different file name than you used for merged data). This is currently incompatible with the "automatic correction" option, which will always output a merged file.

- xia2: the DataFiles directory will contain a file in Scalepack format, ending in `"_unmerged.sca"`. No additional steps are required.

- aimless (CCP4): the OUTPUT keyword has an UNMERGED option.
- XDS: the file named XDS_ASCII.HKL output by CORRECT will contain unmerged data. If you are using XSCALE to scale data from multiple crystals, add the keyword MERGE=FALSE to the input script.
- HKL2000: in the Macros tab, in the Scaling section type "no merge original index", then re-run scaling (with a different file name than you used for merged data). This is currently incompatible with the "automatic correction" option, which will always output a merged file.

Note that unmerged Scalepack format does not include the unit cell, so symmetry needs to be supplied from a different source (this is handled automatically in many of the programs below). MTZ and XDS formats include complete symmetry. Not all files in these formats are necessarily suitable: in particular, note that MTZ files output by Mosflm have not had scaling performed yet.

Xtriage and phenix.merging_statistics

phenix.merging_statistics is a simple program to calculate the standard statistics output by most data processing software, such as mean(I/sigma), redundancy, R-merge, R-meas (Diederichs & Karplus 1997), R-pim (Weiss 2001), CC1/2, and others. The input may be in any format, including unmerged XDS, unmerged Scalepack, multi-batch MTZ, or SHELX (the latter will only work on the command line, however). The number of resolution bins is set to 10 by default, but is adjustable by the n_bins parameter. All data will be used unless the high_resolution or low_resolution parameters are set; the program does not attempt to identify an appropriate high-resolution cutoff on its own. Both a command-line tool and a GUI (pictured above) are available. The output in either case is a table of statistics by resolution bin:

```
Resolution: 28.53 - 1.75
Observations: 145453
Unique reflections: 21557
Redundancy: 6.7
Completeness: 99.92%
Mean intensity: 18672.0
Mean I/sigma(I): 14.9
R-merge: 0.073
R-meas: 0.079
R-pim: 0.030
```

Redundancies (non-anomalous):

```
1 : 50
2 : 163
3 : 597
4 : 1766
5 : 1869
6 : 3731
7 : 5512
8 : 5723
9 : 1526
10 : 490
11 : 74
12 : 53
13 : 3
```

Statistics by resolution bin:

d_max	d_min	#obs	#uniq	mult.	%comp	<I>	<I/sI>	r_mrg	r_meas	r_pim	cc1/2
28.53	3.77	15699	2254	6.96	99.87	78997.8	23.4	0.061	0.066	0.025	0.997
3.77	2.99	15703	2182	7.20	99.95	47400.1	23.1	0.061	0.066	0.024	0.997
2.99	2.61	15641	2172	7.20	100.00	17930.9	21.1	0.074	0.080	0.030	0.996
2.61	2.37	15309	2138	7.16	100.00	10520.1	18.6	0.090	0.097	0.036	0.995
2.37	2.21	15044	2146	7.01	99.95	9103.8	17.2	0.093	0.101	0.038	0.995
2.21	2.07	14571	2145	6.79	100.00	6560.2	13.5	0.108	0.117	0.045	0.993
2.07	1.97	13973	2135	6.54	100.00	5016.1	10.8	0.121	0.131	0.051	0.992
1.97	1.89	13540	2141	6.32	100.00	3620.6	8.6	0.145	0.158	0.062	0.984
1.89	1.81	13010	2104	6.18	99.95	2070.5	6.8	0.197	0.215	0.085	0.980

1.81 1.75 12963 2140 6.06 99.49 1477.4 5.6 0.247 0.270 0.108 0.970
28.53 1.75 145453 21557 6.75 99.92 18672.0 14.9 0.073 0.079 0.030 0.998

In the GUI version, plots are also available:

Xtrriage (Zwart et al. 2005) will also calculate these statistics if provided with unmerged intensities; in the GUI, these will appear in an extra tab before the data quality analyses.

phenix.cc_star

This is an implementation of the method outlined in Karplus & Diederichs (2012) for determining the optimal resolution cutoff, as an alternative to the traditional criteria such as limits on R-merge or I/sigma. It is essentially a combination of phenix.merging_statistics with the recalculation of R-factors for the model. The statistics CC* is an estimate of the "true" CC of the data under examination to the (unknown) true intensities. This places it on the same scale as CC(work) and CC(free) for the model and data, and the relationship between these statistics can be used as a guide for truncating the data.

Required inputs are unmerged intensities, merged data (either amplitudes or intensities - if the latter, French&Wilson treatment will be performed), R-free flags, and either a pre-calculated F(model) array, which is output by programs like phenix.refine, or the current refined model. The results should be essentially the same in either case, but supplying F(model) directly will run faster. The statistics overlap with those reported by the merging program, but with the addition of R-factors and CCs for model/data agreement.

phenix.table_one

This utility generates the standard Table 1 of crystallographic statistics found in most structure publications. If unmerged data are included along with the merged data and R-free flags, the intensity merging statistics will be calculated from these instead of parsing the data processing logfiles. We recommend that you use the unmerged data if available, as they are more reliably parsed than logfiles.

Details for individual statistics

Note that the programs listed above display slightly different subsets of these statistics.

Mean(I/signal): the signal-to-noise ratio. Traditionally the data are often cut off at the resolution where this statistic declines to 2, but this may exclude genuine measurements.

Redundancy or Multiplicity: the average number of observations of each reflection.

R-merge or R-symm: the simple merging R-factor for the multiple observations. Ideally this should be as low as possible, but for reasons discussed in the listed references, using it as a resolution cutoff is problematic. (Note that it will always increase significantly in the high-resolution shells.)

R-meas or R-rim: an alternative to R-merge which is intended to be independent of redundancy. See Diederichs & Karplus (1997) for explanation.

R-pim: another alternative to R-merge, intended to show the precision of the measurements. See Weiss (2001).

CC1/2: the correlation of one half of the observations (chosen at random, but with approximately even distribution for each reflection) to the other half. A similar statistic is used in electron microscopy; see Karplus & Diederichs (2012) for more information.

CC*: a modification of CC1/2, intended to show the true correlation of the observed data to the unknown "true" intensities. Used in conjunction with CC(work) and CC(free). See Karplus & Diederichs (2012).

CC(work) and CC(free): correlation of the observed intensities to the model-based intensities, for the work and test sets respectively. Should be on the same scale as CC*.

References

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- Xtriage and Fest: automatic assessment of X-ray data and substructure structure factor estimation. P.H. Zwart, R.W. Grosse-Kunstleve, and P.D. Adams. CCP4 Newsletter Winter, Contribution 7 (2005).

Using AI to help generate doc strings

using_ai_for_doc_strings.html

Documenting your code with doc strings or other helpful information makes it easier for you and others to understand, use, and modify your code later.

Additionally, doc strings can help AI tools identify what code would be useful in a new situation and to document the way to use your code.

libtbx.add_docstrings_with_ai: a tool to automatically add doc strings

There are many ways to add doc strings to your code, but even most AI tools will not add doc strings to large files. The CCTBX tool `libtbx.add_docstrings_with_ai` will break up a file into pieces, send each one to `gemini-1.5` for doc string generation, catch problems and resend, and assemble the result into a new file that has all the coding lines from the original file and a new set of documentation strings.

Here is an example. This is code supplied to `libtbx.add_docstrings_with_ai`:

```
class xh_connectivity_table2(object):
    def __init__(self, geometry, xray_structure):
        bond_proxies_simple, asu = geometry.geometry.get_all_bond_proxies(
            sites_cart=xray_structure.sites_cart())
        scatterers = xray_structure.scatterers()
```

The AI tool analyzes this, figures out something about what it is doing, and summarizes the arguments and any return values:

```
class xh_connectivity_table2(object):
    """
    Stores X-H connectivity information, including angle information.
    """
    def __init__(self, geometry, xray_structure):
        """
        Initialize the X-H connectivity table.

        Args:
        geometry (mmtbx.geometry_restraints.manager): Geometry restraints manager.
        xray_structure (cctbx.xray.structure.structure): X-ray structure object.
        """
        bond_proxies_simple, asu = geometry.geometry.get_all_bond_proxies(
            sites_cart=xray_structure.sites_cart())
        scatterers = xray_structure.scatterers()
        ...
```

Preparing to run libtbx.add_docstrings_with_ai

To run this tool, you will need a Google API key, as the tool talks to Google directly. You can get a free key like this:

Go to Google AI Studio: Navigate to makersuite.google.com/app/apikey. You may need to sign in with your Google account. Create a New Key: Click the button labeled "Create API key in new project". Copy Your Key: A pop-up window will appear displaying your new API key. Click the copy icon next to the key to copy it to your clipboard. Set the `GOOGLE_API_KEY` environment variable: `csh: setenv GOOGLE_API_KEY <your-api-key>` `bash: set GOOGLE_API_KEY=<your-api-key>`

Go to Google AI Studio: Navigate to makersuite.google.com/app/apikey. You may need to sign in with your Google account.

Create a New Key: Click the button labeled "Create API key in new project".

Copy Your Key: A pop-up window will appear displaying your new API key. Click the copy icon next to the key to copy it to your clipboard.

Set the GOOGLE_API_KEY environment variable:

csh: setenv GOOGLE_API_KEY <your-api-key> bash: set GOOGLE_API_KEY=<your-api-key>

How to run libtbx.add_docstrings_with_ai

You can run the tool with:

```
libtbx.add_docstrings_with_ai input_dir output_dir
```

You put the files you want documented in input_dir and it will write them to output_dir (creating output_dir if necessary).

Be safe: check the results

While this tool is designed to only change doc strings, you should carefully review the new version of each of your files. You can use

```
diff input_dir/my_file.py output_dir/my_file.py
```

to see what is changed, for example. Also just have a look at each new file.

Helpful hints for using AI in coding for Phenix and CCTBX

[using_ai_in_programming.html](#)

What if you could write pretty good code without knowing every detail of the tools you are going to use?

AI tools are great for creating scripts that do something pretty close to what you ask for. Then if you have a way of noticing what is not perfect about the results, you can iteratively ask the AI tools to fix your scripts. This can be an incredibly powerful, and relatively easy, process even if the AI is using tools that you do not know a lot about. The AI tools are great at explaining what the code is doing as well, so it can be great for learning how to use available tools.

How to be safe using an AI chatbot for coding

While AI chatbots are not malicious, they can write code that does not do what you intend.

Look through the code that a chatbot suggests before running it. It is your responsibility to be able to recognize whether the code is what you intend or not.

Make sure that the code does not accidentally delete or modify files or carry out procedures that you do not want.

Code from chatbots may require installation of new packages. Before installing any package, look it up and make sure that it is a mainstream package.

How to write code with an AI chatbot

First, the AI has to have been trained on programming with the tools that you need to use. For general scripting, tools such as Gemini and ChatGPT are already trained on most common tools, so you are all set. For writing scripts in a specialized environment such as Phenix or CCTBX, your AI tool may need to have imported the codebase that you are going to base your work on. And no worries: there are already AI tools all set up to work with CCTBX or with Phenix and CCTBX.

Second, you need to communicate clearly what you want the AI chatbot to create. If you want a script written in a certain way or with certain tools, you need to tell it these things. A good way to do this is to have a discussion with the AI tool, asking it how to go about carrying out the task you want, choosing a path that you like, and asking it to write a script to do the task in that way.

Third, and very importantly, you have to have a way of determining whether the code that the AI tool creates is actually doing what you want. There are several ways to do this. One is to specify just what it is supposed to do up front, and then take the script that the AI produces for you, run it, and see if it does what you want. Another is to ask the AI to create a set of unit tests for the script. Then you can look through the unit tests and make sure they are what you want and that they all pass.

Fourth, you have to expect that the AI will not get it right the first time. The amazing thing is that it is often quite close. If you run the script that the AI creates and it crashes, you can just paste the traceback text in to the AI chat and in most cases it will try to figure out what it did wrong and it will create an updated script fixing that problem. You can do this to fix anything that you don't like. If the formatting of your output is not to your liking, just ask the AI to change it. If you would like more input parameters, just ask it to add them. While this process is a little tedious, it is very often a lot faster than writing a script by yourself. Once the AI version of your script is pretty close to what you want, you can also just edit it yourself to make it do just what you want.

How to use the Phenix developer AI chatbot to write a ProgramTemplate script

While AI chatbots are quite amazing at writing code, they are (so far) not that great at following instructions to the letter. Consequently writing very specialized code that has to follow specific formatting is sometimes a little difficult with AI.

One way to get around this is to first specify everything that you want the AI chatbot to consider, then ask it to write code, then repeat instructions that the chatbot did not follow correctly and ask it to fix the code accordingly.

In the Phenix developer AI chatbot this process can be carried out when you want to write a script that uses the ProgramTemplate, the standard Phenix script for gathering information from a user and running a program. An important element of this is that the chatbot has already been pre-loaded with a source that tells it the rules for the ProgramTemplate. This source contains the text "Chatbot Program Template", and you can tell the chatbot to focus on the contents of this source.

Here is how you do this:

```
First chat with the AI bot about what you want to do and decide which
of the possibilities that the AI gives you that you want to follow.
```

```
Then, supposing you want to write a script to calculate MolProbity
scores, type a command like this:
```

```
1. Describe in detail all the instructions in the source that contains
the exact text: "Chatbot Program Template". 2. Following these
instructions, write a script that reads in model.pdb and prints
out its molprobity score."
```

Now the AI will try to follow the Program Template rules and write you a script. Although it may not be perfect, it is likely to be close. Then if it does not run, you can paste in the traceback or incorrect outputs and let it figure out what it did wrong. If there are specific things that you notice that it did wrong, you can point those out and ask for a new script.

Notes

When the AI chatbots write code, they print it out on the screen and usually also supply a button to copy the whole code snippet. Make sure you wait until the entire code snippet is written to the screen before hitting the copy button or you will get an incomplete script.

phenix.varref: a tool for refinement of a model against multiple maps

varref.html

Contents

- Description
- Availability
- Contact
- Command line examples
- Notes
- References

Description

phenix.varref stands for 'variability refinement'. The program refines a model into an ensemble of maps derived from cryo-EM. The ensemble of maps represents the sample's heterogeneity captured during the experiment. The inputs are typically the refined model against the consensus high-resolution map, and the map series derived from 3DVA (cryosparc), or multibody refinement (Relion) or equivalent. The result of refinement with phenix.varref is an ensemble of real-space refined models that adequately represents the ensemble of input maps. For more details refer to: <http://dx.doi.org/10.1016/j.bbamem.2023.184133>.

Availability

The program is available in the command line and in the GUI.

Contact

For questions, bug reports, feature requests: help@phenix-online.org

IMPORTANT: any problem report should be sent after trying the latest nightly build of Phenix:

https://www.phenix-online.org/download/nightly_builds.cgi

Bug or problem reports should include at least 1) all input files used to run the program, 2) list of parameters that were customized, 3) error message and 4) a brief description of steps that lead to the problem. Please use file sharing tools (such as Dropbox, Google Drive, etc) to send large files. All files sent to us are handled confidentially and only used to solve the reported issue, and discarded afterwards.

Command line examples

A typical run: `phenix.varref model.pdb map1.mrc map2.mrc... mapN.mrc resolution=4.2 \ nproc=10 models_per_map=50` The program takes one model and list of maps in any order. The resolution is a mandatory parameter. nproc specifies the number of processors available to the program to use. A run using a single processor can take a very long time! Use as many CPU as possible! models_per_map specifies the size of the ensemble to be refined against each individual map. The output ensemble accumulates overall best refined models taken from ensembles of models refined against each individual map. The model contains an unknown to Phenix ligand: `phenix.varref model.pdb ligand.cif map1.mrc map2.mrc... mapN.mrc \`

resolution=4.2 nproc=10 models_per_map=50 The ligand dictionary CIF file can be obtained using ReadySet! or eLBOW programs of Phenix and supplied to phenix.varref as an additional file. Anchoring solvent, ligands and other small molecules to the macromolecule: phenix.varref model.pdb map1.mrc map2.mrc... mapN.mrc resolution=4.2 \ nproc=10 models_per_map=50 parameters.eff phenix.varref uses a rather aggressive refinement technique such as morphing, simulated annealing and perturbing initial model by means of MD. This means that small molecule (non-macromolecules) entities of the model may drift away from their original position or binding sites. To prevent this from happening they can be anchored to the macromolecule using custom bonds (as well as angles and so on). Refer to phenix.refine documentation for more details. In the example above this is done by providing a parameter file containing the following lines: geometry_restraints { edits { bond { atom_selection_1 = chain B and resseq 87 and name N atom_selection_2 = chain S and resseq 1 and resname HOH and name O distance_ideal = 2.0 sigma = 0.01 } } } Here the water oxygen in chain S (residue number 1) is linked to the Nitrogen of residue 87 in chain B with the bond length of 2Å. Any number of bonds can be specified.

- A typical run: phenix.varref model.pdb map1.mrc map2.mrc... mapN.mrc resolution=4.2 \ nproc=10 models_per_map=50 The program takes one model and list of maps in any order. The resolution is a mandatory parameter. nproc specifies the number of processors available to the program to use. A run using a single processor can take a very long time! Use as many CPU as possible! models_per_map specifies the size of the ensemble to be refined against each individual map. The output ensemble accumulates overall best refined models taken from ensembles of models refined against each individual map.

A typical run:

```
phenix.varref model.pdb map1.mrc map2.mrc... mapN.mrc resolution=4.2 \
nproc=10 models_per_map=50
```

The program takes one model and list of maps in any order. The resolution is a mandatory parameter. nproc specifies the number of processors available to the program to use. A run using a single processor can take a very long time! Use as many CPU as possible! models_per_map specifies the size of the ensemble to be refined against each individual map. The output ensemble accumulates overall best refined models taken from ensembles of models refined against each individual map.

- The model contains an unknown to Phenix ligand: phenix.varref model.pdb ligand.cif map1.mrc map2.mrc... mapN.mrc \ resolution=4.2 nproc=10 models_per_map=50 The ligand dictionary CIF file can be obtained using ReadySet! or eLBOW programs of Phenix and supplied to phenix.varref as an additional file.

The model contains an unknown to Phenix ligand:

```
phenix.varref model.pdb ligand.cif map1.mrc map2.mrc... mapN.mrc \
resolution=4.2 nproc=10 models_per_map=50
```

The ligand dictionary CIF file can be obtained using ReadySet! or eLBOW programs of Phenix and supplied to phenix.varref as an additional file.

- Anchoring solvent, ligands and other small molecules to the macromolecule: phenix.varref model.pdb map1.mrc map2.mrc... mapN.mrc resolution=4.2 \ nproc=10 models_per_map=50 parameters.eff phenix.varref uses a rather aggressive refinement technique such as morphing, simulated annealing and perturbing initial model by means of MD. This means that small molecule (non-macromolecules) entities of the model may drift away from their original position or binding sites. To prevent this from happening they can be anchored to the macromolecule using custom bonds (as well as angles and so on). Refer to phenix.refine documentation for more details. In the example above this is done by providing a parameter file containing the following lines: geometry_restraints { edits { bond { atom_selection_1 = chain B and

resseq 87 and name N atom_selection_2 = chain S and resseq 1 and resname HOH and name O distance_ideal = 2.0 sigma = 0.01 } } } Here the water oxygen in chain S (residue number 1) is linked to the Nitrogen of residue 87 in chain B with the bond length of 2Å. Any number of bonds can be specified.

Anchoring solvent, ligands and other small molecules to the macromolecule:

```
phenix.varref model.pdb map1.mrc map2.mrc... mapN.mrc resolution=4.2 \
nproc=10 models_per_map=50 parameters.eff
```

phenix.varref uses a rather aggressive refinement technique such as morphing, simulated annealing and perturbing initial model by means of MD. This means that small molecule (non-macromolecules) entities of the model may drift away from their original position or binding sites. To prevent this from happening they can be anchored to the macromolecule using custom bonds (as well as angles and so on). Refer to phenix.refine documentation for more details. In the example above this is done by providing a parameter file containing the following lines:

```
geometry_restraints {
  edits {
    bond {
      atom_selection_1 = chain B and resseq 87 and name N
      atom_selection_2 = chain S and resseq 1 and resname HOH and name O
      distance_ideal = 2.0
      sigma = 0.01
    }
  }
}
```

Here the water oxygen in chain S (residue number 1) is linked to the Nitrogen of residue 87 in chain B with the bond length of 2Å. Any number of bonds can be specified.

Notes

The program can use multiple CPU. The program does not use GPU. Running on single processor can be very slow. Use as many CPUs as possible.

- The program can use multiple CPU. The program does not use GPU. Running on single processor can be very slow. Use as many CPUs as possible.

References

Afonine PV, Gobet A, Moissonnier L, Martin J, Poon BK, Chaptal V. Conformational space exploration of cryo-EM structures by variability refinement. *Biochim Biophys Acta Biomembr.* 2023 Apr;1865(4):184133. doi: 10.1016/j.bbamem.2023.184133. Epub 2023 Feb 3. PMID: 36738875.

- Afonine PV, Gobet A, Moissonnier L, Martin J, Poon BK, Chaptal V. Conformational space exploration of cryo-EM structures by variability refinement. *Biochim Biophys Acta Biomembr.* 2023 Apr;1865(4):184133. doi: 10.1016/j.bbamem.2023.184133. Epub 2023 Feb 3. PMID: 36738875.

Structure factor file manipulations with Xmanip

xmanip.html

Author(s)

- Xmanip: Peter Zwart
- Phil command interpreter: Ralf W. Grosse-Kunstleve

Purpose

Manipulation of reflection data and models

Usage

Command line interface

xmanip can be invoked via the command line interface with instructions given in a specific definition file:

```
phenix.xmanip params.def
```

The full set of definitions can be obtained by typing:

```
phenix.xmanip
```

which results in:

```
xmanip {
  input {
    unit_cell = None
    space_group = None
    xray_data {
      file_name = None
      labels = None
      label_appendix = None
      name = None
      write_out = None
    }
    model {
      file_name = None
    }
  }
  parameters {
    action = reindex manipulate_pdb *manipulate_miller
    reindex {
      standard_laws = niggli *reference_setting invert user_supplied
      user_supplied_law = "h,k,l"
    }
    manipulate_miller {
      task = get_dano get_diso lsq_scale sfcalc *custom None
      output_label_root = "FMODEL"
      get_dano {
        input_data = None
      }
      get_diso {
        native = None
        derivative = None
        use_intensities = True
        use_weights = True
        scale_weight = True
      }
    }
  }
}
```

```

lsq_scale {
input_data_1 = None
input_data_2 = None
use_intensities = True
use_weights = True
scale_weight = True
}
sfcalc {
fobs = None
output = *2mFo-DFc mFo-DFc complex_fcalc abs_fcalc intensities
use_bulk_and_scale = *as_estimated user_applied
bulk_and_scale_parameters {
d_min = 2
overall {
b_cart {
b_11 = 0
b_22 = 0
b_33 = 0
b_12 = 0
b_13 = 0
b_23 = 0
}
k_overall = 0.1
}
solvent {
k_sol = 0.3
b_sol = 56
}
}
}
custom{
code = print &gt;&gt; out, "hello world"
}
}

manipulate_pdb{
task = apply_operator *set_b
apply_operator{
operator = "x,y,z"
invert=False
concatenate_model=False
chain_id_increment=1
}
set_b{
b_iso = 30
}
}
output {
logfile = "xmanip.log"
hklout = "xmanip.mtz"
xyzout = "xmanip.pdb"
}
}

```

Detailed explanation of the scopes follow below.

Parameters and definitions

The xmanip.input scope defines which files and which data xmanip reads in:

```

input {
unit_cell = None # unit cell. Specify when not in reflection or pdb files
space_group = None # space group. Specify when not in reflection or pdb files
xray_data {
file_name = None # File from which data will be read
}
}

```

```

labels = None # Labels to read in.
label_appendix = None # Label appendix: when writing out the new mtz file, this
appendix will be added to the current label.
name = None # A data set name. Useful for manipulation
write_out = None # Determines if this data set will be written to the final
mtz file
}
model {
file_name = None # An input pdb file
}
}

```

One can define as many sub-scopes of xray_data as desired (see examples). The specific tasks of xmanip are controlled by the xmanip.parameters.action key. Possible options are:

- reindex
- manipulate_pdb
- manipulate_miller

Reindexing: reindexing of a data set (and a model) is controlled by the xmanip.parameters.reindex scope. Standard laws are available:

- niggli: Brings unit cell to the niggli setting.
- reference_setting: Brings space group to the reference setting
- invert: Inverts a data set
- user_supplied: A user supplied reindexing law is used, specified by reindex.user_supplied_law

manipulate_pdb: A pdb file can be modified by either applying a symmetry operator to the coordinates (select the apply_operator task from the manipulate_pdb.task list. The operator needs to be specified by apply_operator.operator. Setting apply_operator.invert to true will invert the supplied operator. One can choose to put out the newly generated chain with the original chain (set concatenate_model = True). The new chain ID can be controlled with the chain_id_increment parameter. manipulate_miller: Reflection data can be manipulate in various ways:

- get_dano: Get anomalous differences from the data set with name specified by manipulate_miller.get_dano.input_data.
- get_diso: Get isomorphous differences (derivative-native) from the data sets specified by the names manipulate_miller.get_diso.native and manipulate_miller.get_diso.derivative. Least squares scaling of the derivative to the native can be done on intensities (use_intensities=True), with or without using sigmas (use_weights) and by scaling the weights if desired (recommended).
- lsq_scale : As above, no isomorphous difference are computed, only input_data_2 is scaled and returned.
- sfcalc: Structure factor calculation. Requires a pdb file to be read in. Possible output coefficients are
- 2mFo-DFc (Fobs required. specify sfcalc.fobs).
- mFo-DFc (Fobs required. specify sfcalc.fobs).
- complex_fcalc (FC,PHIC)
- abs_fcalc (FC)
- intensities (FC^2)

bulk solvent and scaling parameters will be either estimated from observed data if supplied, or set by the user (using keywords in the bulk_and_scale_parameters scope)

- custom: If custom is selected, all data names for the xray data will become variable names accessible via the custom interface. The ``custom`` interface allows one to write a small piece of python code that directly works with the python objects them self. Basic knowledge of the cctbx and python are needed to bring this to a fruitful ending. Please contact the authors for detailed help if required. An example is given in the example section.

Examples

Reindexing a data set and model ::

```
xmanip {
  input {
    xray_data {
      file_name = mydata.mtz
      labels = FOBS,SIGFOBS
      write_out = True
    }
    xray_data {
      file_name = mydata.mtz
      labels = R_FREE_FLAG
      write_out = True
    }
    model {
      file_name = mymodel.pdb
    }
  }
  parameters {
    action = reindex
    reindex {
      standard_laws = *niggli
      user_supplied_law = "h,k,l"
    }
  }
  output {
    logfile = "xmanip.log"
    hklout = "reindex.mtz"
    xyzout = "reindex.pdb"
  }
}
```

Applying a symmetry operator to a pdb file ::

```
xmanip {
  input {
    model {
      file_name = mymodel.pdb
    }
  }
  parameters {
    action = manipulate_pdb
    manipulate_pdb {
      task = apply_operator
      apply_operator{
        operator = "x+1/3,y-2/3,z+1/8"
      }
    }
  }
  output {
    logfile = "xmanip.log"
    xyzout = "shifted.pdb"
  }
}
```

Printing out some useful information for an mtz file ::


```

xmanip {
  input {
    xray_data {
      file_name = mydata.mtz
      labels = FOBS,SIGFOBS
      name = fobs
    }
  }
  parameters {
    action = custom
    custom{
      code = """
      print &gt;&gt; out, "Printing d_spacings, epsilons and intensities"
      #change amplitude to intensities
      fobs = fobs.f_as_f_sq()
      #get epsilons
      epsilons = fobs.epsilons().data().as_double()
      #get d spacings
      d_hkl = fobs.d_spacings().data()
      #print the lot to a file
      output_file = open("jiffy_result.txt", 'w')
      for ii, eps, dd in zip( fobs.data(), epsilons, d_hkl):
        print &gt;&gt; output_file, ii, eps, dd
      print &gt;&gt; out, "Done"
      """
    }
  }
}

```

List of all available keywords

xmanip input unit_cell = None Unit cell parameters space_group = None space group xray_data Scope defining xray data. Multiple scopes are allowed file_name = None file name labels = None A unique label or unique substring of a label label_appendix = None Label appendix for output mtz filename = None An identifier of this particular miller array write_out = None Determines if this data is written to the output file model A model associated with the miller arrays. Only one model can be defined. file_name = None A model file parameters action = *reindex manipulate_pdb manipulate_miller Defines which action will be carried out. reindex Reindexing parameters. Acts on coordinates and miller arrays. standard_laws = niggli *reference_setting primitive_setting invert user_supplied Choices of reindexing operators. Will be applied on structure and miller arrays. user_supplied_law = 'h,k,l' User supplied operator. manipulate_miller Acts on a single miller array or a set of miller arrays. task = *get_dano get_diso lsq_scale sfcalc custom None Possible tasks output_label_root = None Output label root get_dano Get ||F+| - |F-| from input data. input_data = None get_diso Get |Fder| - |Fnat| native = None Name of native data derivative = None Name of derivative data use_intensities = True Scale on intensities use_weights = True Use experimental sigmas as weights in scaling scale_weight = True Whether or not to scale the sigmas during scaling lsq_scale input_data_1 = None Reference data input_data_2 = None Data to be scaled use_intensities = True Scale on intensities use_weights = True Use experimental sigmas as weights in scaling scale_weight = True Whether or not to scale the sigmas during scaling sfcalc fobs = None Data name of observed data output = 2mFo-DFc mFo-DFc *complex_fcalc abs_fcalc intensities Output coefficients use_bulk_and_scale = *as_estimated user_upplied estimate or use parameters given by user bulk_and_scale_parameters Parameters used in the structure factor calculation. Ignored if experimental data is given d_min = 2.0 resolution of the data to be calculated. overall Bulk solvent and scaling parameters k_overall = 0.1 Overall scalar b_cart Anisotropic B values b_11 = 0 b_22 = 0 b_33 = 0 b_12 = 0 b_13 = 0 b_23 = 0 solvent Solvent parameters k_sol = 0.3 Solvent scale b_sol = 56.0 Solvent B custom A custom script that uses miller_array data names as variables. code = None A piece of python code show_instructions = True Some instructions manipulate_pdb Manipulate elements of a pdb file task = set_b apply_operator *None How to manipulate a pdb file set_bb_iso = 30 new B value for all

atomsapply_operatorstandard_operators = *user_supplied_operator user_supplied_cartesian_rotation_matrix
Possible operatorsuser_supplied_operator = "x,y,z" Actually operator in x,y,z notationinvert = False Invert
operator given above before applying on coordinatesconcatenate_model = False Determines if new chain is
concatenated to old modelchain_id_increment = 1 Cain id
incrementuser_supplied_cartesian_rotation_matrixRotation,translation matrix in cartesian framer = None
Rotational part of operatorrot = None Translational part of operatoroutputOutput fileslogfile = xmanip.log
Logfilehklout = xmanip.mtz Ouptut miller indices and dataxyzout = xmanip.pdb output PDB file

- xmanipinputunit_cell = None Unit cell parametersspace_group = None space groupxray_dataScope defining xray data. Multiple scopes are allowedfile_name = None file namelabels = None A unique label or unique substring of a labellabel_appendix = None Label appendix for output mtz filename = None An identifier of this particular miller arraywrite_out = None Determines if this data is written to the output filemodelA model associated with the miller arrays. Only one model can be defined.file_name = None A model fileparametersaction = *reindex manipulate_pdb manipulate_miller Defines which action will be carried out.reindexReindexing parameters. Acts on coordinates and miller arrays.standard_laws = niggli *reference_setting primitive_setting invert user_supplied Choices of reindexing operators. Will be applied on structure and miller arrays.user_supplied_law = 'h,k,l' User supplied operator.manipulate_millerActs on a single miller array or a set of miller arrays.task = *get_dano get_diso lsq_scale sfcalc custom None Possible tasksooutput_label_root = None Output label rootget_danoGet ||F+| - |F-| from input data.input_data = None get_disoGet |Fder|-|Fnat|native = None Name of native dataderivative = None Name of derivative datause_intensities = True Scale on intensitiesuse_weights = True Use experimental sigmas as weights in scalingscale_weight = True Whether or not to scale the sigmas during scalingslsq_scaleinput_data_1 = None Reference datainput_data_2 = None Data to be scaleduse_intensities = True Scale on intensitiesuse_weights = True Use experimental sigmas as weights in scalingscale_weight = True Whether or not to scale the sigmas during scalingsfcalfobs = None Data name of observed dataoutput = 2mFo-DFc mFo-DFc *complex_fcalc abs_fcalc intensities Output coefficientsuse_bulk_and_scale = *as_estimated user_applied estimate or use parameters given by userbulk_and_scale_parametersParameters used in the structure factor calculation. Ignored if experimental data is givend_min = 2.0 resolution of the data to be calculated.overallBulk solvent and scaling parametersk_overall = 0.1 Overall scalarb_cartAnisotropic B valuesb_11 = 0 b_22 = 0 b_33 = 0 b_12 = 0 b_13 = 0 b_23 = 0 solventSolvent parametersk_sol = 0.3 Solvent scaleb_sol = 56.0 Solvent BcustomA custom script that uses miller_array data names as variables.code = None A piece of python codeshow_instructions = True Some instructionsmanipulate_pdbManipulate elements of a pdb filetask = set_b apply_operator *None How to manipulate a pdb filesset_bb_iso = 30 new B value for all atomsapply_operatorstandard_operators = *user_supplied_operator user_supplied_cartesian_rotation_matrix Possible operatorsuser_supplied_operator = "x,y,z" Actually operator in x,y,z notationinvert = False Invert operator given above before applying on coordinatesconcatenate_model = False Determines if new chain is concatenated to old modelchain_id_increment = 1 Cain id incrementuser_supplied_cartesian_rotation_matrixRotation,translation matrix in cartesian framer = None Rotational part of operatorrot = None Translational part of operatoroutputOutput fileslogfile = xmanip.log Logfilehklout = xmanip.mtz Ouptut miller indices and dataxyzout = xmanip.pdb output PDB file
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- action = *reindex manipulate_pdb manipulate_miller Defines which action will be carried out.
- reindexReindexing parameters. Acts on coordinates and miller arrays.standard_laws = niggli *reference_setting primitive_setting invert user_supplied Choices of reindexing operators. Will be applied on structure and miller arrays.user_supplied_law = 'h,k,l' User supplied operator.

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- output_label_root = None Output label root
- get_danoGet ||F+| - |F-|| from input data.input_data = None
- input_data = None
- get_disoGet |Fder|-|Fnat|native = None Name of native data derivative = None Name of derivative datause_intensities = True Scale on intensitiesuse_weights = True Use experimental sigmas as weights in scalingscale_weight = True Whether or not to scale the sigmas during scaling
- native = None Name of native data
- derivative = None Name of derivative data
- use_intensities = True Scale on intensities
- use_weights = True Use experimental sigmas as weights in scaling
- scale_weight = True Whether or not to scale the sigmas during scaling
- lsq_scaleinput_data_1 = None Reference datainput_data_2 = None Data to be scaleduse_intensities = True Scale on intensitiesuse_weights = True Use experimental sigmas as weights in scalingscale_weight = True Whether or not to scale the sigmas during scaling
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solvent and scaling parametersk_overall = 0.1 Overall scalarb_cartAnisotropic B valuesb_11 = 0 b_22 = 0 b_33 = 0 b_12 = 0 b_13 = 0 b_23 = 0 solventSolvent parametersk_sol = 0.3 Solvent scaleb_sol = 56.0 Solvent B

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- overallBulk solvent and scaling parametersk_overall = 0.1 Overall scalarb_cartAnisotropic B valuesb_11 = 0 b_22 = 0 b_33 = 0 b_12 = 0 b_13 = 0 b_23 = 0
- k_overall = 0.1 Overall scalar
- b_cartAnisotropic B valuesb_11 = 0 b_22 = 0 b_33 = 0 b_12 = 0 b_13 = 0 b_23 = 0
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- set_bb_iso = 30 new B value for all atoms
- b_iso = 30 new B value for all atoms
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concatenate_model = False Determines if new chain is concatenated to old model
chain_id_increment = 1 Chain id increment
user_supplied_cartesian_rotation_matrix Rotation, translation matrix in cartesian frame = None
Rotational part of operator = None Translational part of operator

- standard_operators = *user_supplied_operator user_supplied_cartesian_rotation_matrix Possible operators
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- invert = False Invert operator given above before applying on coordinates
- concatenate_model = False Determines if new chain is concatenated to old model
- chain_id_increment = 1 Chain id increment
- user_supplied_cartesian_rotation_matrix Rotation, translation matrix in cartesian frame = None
Rotational part of operator = None Translational part of operator
- r = None Rotational part of operator
- t = None Translational part of operator
- output Output files logfile = xmanip.log Logfile hklout = xmanip.mtz Output miller indices and data xyzout = xmanip.pdb output PDB file
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